

32-bit Arm<sup>®</sup> Cortex<sup>®</sup>-M7 480MHz MCUs, up to 2MB Flash,  
up to 1MB RAM, 46 com. and analog interfaces

Datasheet - production data

## Features

### Core

- 32-bit Arm<sup>®</sup> Cortex<sup>®</sup>-M7 core with double-precision FPU and L1 cache: 16 Kbytes of data and 16 Kbytes of instruction cache; frequency up to 480 MHz, MPU, 1027 DMIPS/2.14 DMIPS/MHz (Dhrystone 2.1), and DSP instructions

### Memories

- Up to 2 Mbytes of Flash memory with read-while-write support
- Up to 1 Mbyte of RAM: 192 Kbytes of TCM RAM (inc. 64 Kbytes of ITCM RAM + 128 Kbytes of DTCM RAM for time critical routines), Up to 864 Kbytes of user SRAM, and 4 Kbytes of SRAM in Backup domain
- Dual mode Quad-SPI memory interface running up to 133 MHz
- Flexible external memory controller with up to 32-bit data bus: SRAM, PSRAM, SDRAM/LPDDR SDRAM, NOR/NAND Flash memory clocked up to 100 MHz in Synchronous mode
- CRC calculation unit

### Security

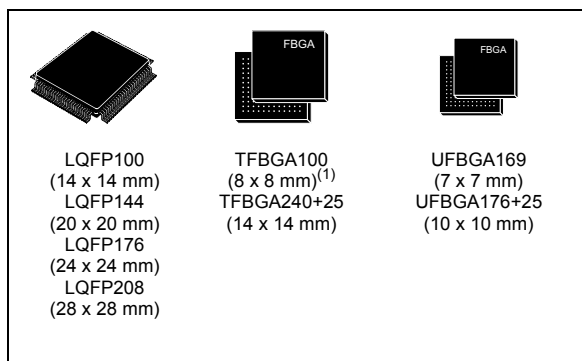
- ROP, PC-ROP, active tamper

### General-purpose input/outputs

- Up to 168 I/O ports with interrupt capability

### Reset and power management

- 3 separate power domains which can be independently clock-gated or switched off:
  - D1: high-performance capabilities
  - D2: communication peripherals and timers
  - D3: reset/clock control/power management



- 1.62 to 3.6 V application supply and I/Os
- POR, PDR, PVD and BOR
- Dedicated USB power embedding a 3.3 V internal regulator to supply the internal PHYs
- Embedded regulator (LDO) with configurable scalable output to supply the digital circuitry
- Voltage scaling in Run and Stop mode (6 configurable ranges)
- Backup regulator (~0.9 V)
- Voltage reference for analog peripheral/ $V_{REF+}$
- Low-power modes: Sleep, Stop, Standby and  $V_{BAT}$  supporting battery charging

### Low-power consumption

- $V_{BAT}$  battery operating mode with charging capability
- CPU and domain power state monitoring pins
- 2.95  $\mu$ A in Standby mode (Backup SRAM OFF, RTC/LSE ON)

### Clock management

- Internal oscillators: 64 MHz HSI, 48 MHz HSI48, 4 MHz CSI, 32 kHz LSI
- External oscillators: 4-48 MHz HSE, 32.768 kHz LSE
- 3 $\times$  PLLs (1 for the system clock, 2 for kernel clocks) with Fractional mode

**Interconnect matrix**

- 3 bus matrices (1 AXI and 2 AHB)
- Bridges (5× AHB2-APB, 2× AXI2-AHB)

**4 DMA controllers to unload the CPU**

- 1× high-speed master direct memory access controller (MDMA) with linked list support
- 2× dual-port DMAs with FIFO
- 1× basic DMA with request router capabilities

**Up to 35 communication peripherals**

- 4× I2Cs FM+ interfaces (SMBus/PMBus)
- 4× USARTs/4× UARTs (ISO7816 interface, LIN, IrDA, up to 12.5 Mbit/s) and 1× LPUART
- 6× SPIs, 3 with muxed duplex I2S audio class accuracy via internal audio PLL or external clock, 1× I2S in LP domain (up to 150 MHz)
- 4× SAls (serial audio interface)
- SPDIFRX interface
- SWPMI single-wire protocol master I/F
- MDIO Slave interface
- 2× SD/SDIO/MMC interfaces (up to 125 MHz)
- 2× CAN controllers: 2 with CAN FD, 1 with time-triggered CAN (TT-CAN)
- 2× USB OTG interfaces (1FS, 1HS/FS) crystal-less solution with LPM and BCD
- Ethernet MAC interface with DMA controller
- HDMI-CEC
- 8- to 14-bit camera interface (up to 80 MHz)

**11 analog peripherals**

- 3× ADCs with 16-bit max. resolution (up to 36 channels, up to 3.6 MSPS)
- 1× temperature sensor
- 2× 12-bit D/A converters (1 MHz)
- 2× ultra-low-power comparators
- 2× operational amplifiers (7.3 MHz bandwidth)
- 1× digital filters for sigma delta modulator (DFSDM) with 8 channels/4 filters

**Graphics**

- LCD-TFT controller up to XGA resolution

- Chrom-ART graphical hardware Accelerator™ (DMA2D) to reduce CPU load
- Hardware JPEG Codec

**Up to 22 timers and watchdogs**

- 1× high-resolution timer (2.1 ns max resolution)
- 2× 32-bit timers with up to 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input (up to 240 MHz)
- 2× 16-bit advanced motor control timers (up to 240 MHz)
- 10× 16-bit general-purpose timers (up to 240 MHz)
- 5× 16-bit low-power timers (up to 240 MHz)
- 2× watchdogs (independent and window)
- 1× SysTick timer
- RTC with sub-second accuracy and hardware calendar

**Debug mode**

- SWD & JTAG interfaces
- 4-Kbyte Embedded Trace Buffer

**True random number generators (3 oscillators each)****96-bit unique ID****All packages are ECOPACK®2 compliant****Table 1. Device summary**

| Reference     | Part number                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STM32H742xI/G | STM32H742VI, STM32H742ZI, STM32H742II, STM32H742BI, STM32H742XI, STM32H742AI, STM32H742VG, STM32H742ZG, STM32H742IG, STM32H742BG, STM32H742XG, STM32H742AG |
| STM32H743xI/G | STM32H743VI, STM32H743ZI, STM32H743II, STM32H743BI, STM32H743XI, STM32H743AI, STM32H743VG, STM32H743ZG, STM32H743IG, STM32H743BG, STM32H743XG, STM32H743AG |

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# 1 Introduction

This document provides information on STM32H742xI/G STM32H743xI/G microcontrollers, such as description, functional overview, pin assignment and definition, electrical characteristics, packaging, and ordering information.

This document should be read in conjunction with the STM32H742xI/G STM32H743xI/G reference manual (RM0433), available from the STMicroelectronics website [www.st.com](http://www.st.com).

For information on the Arm<sup>®(a)</sup> Cortex<sup>®</sup>-M7 core, please refer to the Cortex<sup>®</sup>-M7 Technical Reference Manual, available from the <http://www.arm.com> website.

**arm**

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a. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere.



## 2 Description

STM32H742xI/G and STM32H743xI/G devices are based on the high-performance Arm® Cortex®-M7 32-bit RISC core operating at up to 480 MHz. The Cortex®-M7 core features a floating point unit (FPU) which supports Arm® double-precision (IEEE 754 compliant) and single-precision data-processing instructions and data types. STM32H742xI/G and STM32H743xI/G devices support a full set of DSP instructions and a memory protection unit (MPU) to enhance application security.

STM32H742xI/G and STM32H743xI/G devices incorporate high-speed embedded memories with a dual-bank Flash memory of up to 2 Mbytes, up to 1 Mbyte of RAM (including 192 Kbytes of TCM RAM, up to 864 Kbytes of user SRAM and 4 Kbytes of backup SRAM), as well as an extensive range of enhanced I/Os and peripherals connected to APB buses, AHB buses, 2x32-bit multi-AHB bus matrix and a multi layer AXI interconnect supporting internal and external memory access.

All the devices offer three ADCs, two DACs, two ultra-low power comparators, a low-power RTC, a high-resolution timer, 12 general-purpose 16-bit timers, two PWM timers for motor control, five low-power timers, a true random number generator (RNG). The devices support four digital filters for external sigma-delta modulators (DFSDM). They also feature standard and advanced communication interfaces.

- Standard peripherals
  - Four I<sup>2</sup>Cs
  - Four USARTs, four UARTs and one LPUART
  - Six SPIs, three I<sup>2</sup>Ss in Half-duplex mode. To achieve audio class accuracy, the I<sup>2</sup>S peripherals can be clocked by a dedicated internal audio PLL or by an external clock to allow synchronization.
  - Four SAI serial audio interfaces
  - One SPDIFRX interface
  - One SWPMI (Single Wire Protocol Master Interface)
  - Management Data Input/Output (MDIO) slaves
  - Two SDMMC interfaces
  - A USB OTG full-speed and a USB OTG high-speed interface with full-speed capability (with the ULPI)
  - One FDCAN plus one TT-FDCAN interface
  - An Ethernet interface
  - Chrom-ART Accelerator™
  - HDMI-CEC
- Advanced peripherals including
  - A flexible memory control (FMC) interface
  - A Quad-SPI Flash memory interface
  - A camera interface for CMOS sensors
  - An LCD-TFT display controller (only available on STM32H743xI/G)
  - A JPEG hardware compressor/decompressor (only available on STM32H743xI/G)

Refer to [Table 2: STM32H742xI/G and STM32H743xI/G features and peripheral counts](#) for the list of peripherals available on each part number.

STM32H742xI/G and STM32H743xI/G devices operate in the  $-40$  to  $+85$  °C temperature range from a 1.62 to 3.6 V power supply. The supply voltage can drop down to 1.62 V by using an external power supervisor (see [Section 3.5.2: Power supply supervisor](#)) and connecting the PDR\_ON pin to  $V_{SS}$ . Otherwise the supply voltage must stay above 1.71 V with the embedded power voltage detector enabled.

Dedicated supply inputs for USB (OTG\_FS and OTG\_HS) are available on all packages except LQFP100 to allow a greater power supply choice.

A comprehensive set of power-saving modes allows the design of low-power applications.

STM32H742xI/G and STM32H743xI/G devices are offered in 8 packages ranging from 100 pins to 240 pins/balls. The set of included peripherals changes with the device chosen.

These features make STM32H742xI/G and STM32H743xI/G microcontrollers suitable for a wide range of applications:

- Motor drive and application control
- Medical equipment
- Industrial applications: PLC, inverters, circuit breakers
- Printers, and scanners
- Alarm systems, video intercom, and HVAC
- Home audio appliances
- Mobile applications, Internet of Things
- Wearable devices: smart watches.

[Figure 1](#) and [Figure 2](#) shows the device block diagrams.

**Table 2. STM32H742xI/G and STM32H743xI/G features and peripheral counts**

| Peripherals            |                          | STM32H 742VG   | STM32H742ZG | STM32H742AG | STM32H742IG | STM32H742BG | STM32H742XG | STM32H743VG | STM32H743ZG | STM32H743AG | STM32H743IG | STM32H743BG | STM32H743XG | STM32H 742VI | STM32H742ZI | STM32H742AI | STM32H742II | STM32H742BI | STM32H742XI | STM32H743VI | STM32H743ZI | STM32H743AI | STM32H743II | STM32H743BI | STM32H743XI |
|------------------------|--------------------------|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Flash memory in Kbytes |                          | 2 x 512 Kbytes |             |             |             |             |             |             |             |             |             |             |             | 2 x 1 Mbyte  |             |             |             |             |             |             |             |             |             |             |             |
| SRAM in Kbytes         | SRAM mapped onto AXI bus | 384            |             |             |             |             |             | 512         |             |             |             |             |             | 384          |             |             |             |             |             | 512         |             |             |             |             |             |
|                        | SRAM1 (D2 domain)        | 32             |             |             |             |             |             | 128         |             |             |             |             |             | 32           |             |             |             |             |             | 128         |             |             |             |             |             |
|                        | SRAM2 (D2 domain)        | 16             |             |             |             |             |             | 128         |             |             |             |             |             | 16           |             |             |             |             |             | 128         |             |             |             |             |             |
|                        | SRAM3 (D2 domain)        | -              |             |             |             |             |             | 32          |             |             |             |             |             | -            |             |             |             |             |             | 32          |             |             |             |             |             |
|                        | SRAM4 (D3 domain)        | 64             |             |             |             |             |             | 64          |             |             |             |             |             | 64           |             |             |             |             |             | 64          |             |             |             |             |             |
| TCM RAM in Kbytes      | ITCM RAM (instruction)   | 64             |             |             |             |             |             | 64          |             |             |             |             |             | 64           |             |             |             |             |             | 64          |             |             |             |             |             |
|                        | DTCM RAM (data)          | 128            |             |             |             |             |             | 128         |             |             |             |             |             | 128          |             |             |             |             |             | 128         |             |             |             |             |             |
| Backup SRAM (Kbytes)   |                          | 4              |             |             |             |             |             |             |             |             |             |             |             |              |             |             |             |             |             |             |             |             |             |             |             |
| FMC                    |                          | Yes            |             |             |             |             |             |             |             |             |             |             |             |              |             |             |             |             |             |             |             |             |             |             |             |
| GPIOs                  |                          | 82             | 114         | 131         | 140         | 168         | 82          | 114         | 131         | 140         | 168         | 82          | 114         | 131          | 140         | 168         | 82          | 114         | 131         | 140         | 168         | 82          | 114         | 131         | 140         |
| Quad-SPI               |                          | Yes            |             |             |             |             |             |             |             |             |             |             |             |              |             |             |             |             |             |             |             |             |             |             |             |
| Ethernet               |                          | Yes            |             |             |             |             |             |             |             |             |             |             |             |              |             |             |             |             |             |             |             |             |             |             |             |

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Table 2. STM32H742xI/G and STM32H743xI/G features and peripheral counts (continued)

| Peripherals                       | STM32H 742VG                                          | STM32H742ZG                  | STM32H742AG             | STM32H742IG            | STM32H742BG | STM32H742XG | STM32H743VG                           | STM32H743ZG                  | STM32H743AG             | STM32H743IG            | STM32H743BG | STM32H743XG | STM32H 742VI                          | STM32H742ZI                  | STM32H742AI             | STM32H742II            | STM32H742BI | STM32H742XI | STM32H743VI                           | STM32H743ZI                  | STM32H743AI             | STM32H743II            | STM32H743BI | STM32H743XI |
|-----------------------------------|-------------------------------------------------------|------------------------------|-------------------------|------------------------|-------------|-------------|---------------------------------------|------------------------------|-------------------------|------------------------|-------------|-------------|---------------------------------------|------------------------------|-------------------------|------------------------|-------------|-------------|---------------------------------------|------------------------------|-------------------------|------------------------|-------------|-------------|
| LCD-TFT                           | -                                                     |                              |                         |                        |             |             | Yes                                   |                              |                         |                        |             |             | -                                     |                              |                         |                        |             |             | Yes                                   |                              |                         |                        |             |             |
| JPEG Codec                        | -                                                     |                              |                         |                        |             |             | Yes                                   |                              |                         |                        |             |             | -                                     |                              |                         |                        |             |             | Yes                                   |                              |                         |                        |             |             |
| Chrom-ART Accelerator™ (DMA2D)    | Yes                                                   |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| 16-bit ADCs<br>Number of channels | 3                                                     |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
|                                   | Up to 36                                              |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| 12-bit DAC<br>Number of channels  | Yes<br>2                                              |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| Comparators                       | 2                                                     |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| Operational amplifiers            | 2                                                     |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| DFSDM                             | Yes                                                   |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| Maximum CPU frequency             | 480MHz <sup>(2)(3)</sup> /400 MHz                     |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| Operating voltage                 | 1.71<br>to<br>3.6<br>V <sup>(4)</sup>                 | 1.62 to 3.6 V <sup>(5)</sup> |                         |                        |             |             | 1.71<br>to<br>3.6<br>V <sup>(4)</sup> | 1.62 to 3.6 V <sup>(5)</sup> |                         |                        |             |             | 1.71<br>to<br>3.6<br>V <sup>(4)</sup> | 1.62 to 3.6 V <sup>(5)</sup> |                         |                        |             |             | 1.71<br>to<br>3.6<br>V <sup>(4)</sup> | 1.62 to 3.6 V <sup>(5)</sup> |                         |                        |             |             |
| Operating temperatures            | Ambient temperatures: –40 up to +85 °C <sup>(6)</sup> |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
|                                   | Junction temperature: –40 to + 125 °C                 |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |                                       |                              |                         |                        |             |             |
| Package                           | LQFP100<br>TFBGA100 <sup>(7)</sup>                    | LQFP144                      | UFBGA169 <sup>(7)</sup> | LQFP176<br>UFBGA176+25 | LQFP208     | TFBGA240+25 | LQFP100<br>TFBGA100 <sup>(7)</sup>    | LQFP144                      | UFBGA169 <sup>(7)</sup> | LQFP176<br>UFBGA176+25 | LQFP208     | TFBGA240+25 | LQFP100<br>TFBGA100 <sup>(7)</sup>    | LQFP144                      | UFBGA169 <sup>(7)</sup> | LQFP176<br>UFBGA176+25 | LQFP208     | TFBGA240+25 | LQFP100<br>TFBGA100 <sup>(7)</sup>    | LQFP144                      | UFBGA169 <sup>(7)</sup> | LQFP176<br>UFBGA176+25 | LQFP208     | TFBGA240+25 |

1. The SPI1, SPI2 and SPI3 interfaces give the flexibility to work in an exclusive way in either the SPI mode or the I2S audio mode.
2. The maximum CPU frequency of 480 MHz can be obtained on devices revision V.
3. The product junction temperature must be kept within the –40 to +105 °C temperature range.

4. Since the LQFP100 package does not feature the PDR\_ON pin (tied internally to  $V_{DD}$ ), the minimum  $V_{DD}$  value for this package is 1.71 V.
5.  $V_{DD}/V_{DDA}$  can drop down to 1.62 V by using an external power supervisor (see [Section 3.5.2: Power supply supervisor](#)) and connecting PDR\_ON pin to  $V_{SS}$ . Otherwise the supply voltage must stay above 1.71 V with the embedded power voltage detector enabled.
6. The product junction temperature must be kept within the  $-40$  to  $+125$  °C temperature range.
7. This package is under development. Please contact STMicroelectronics for details.

**Figure 1. STM32H742xI/G block diagram**

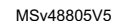
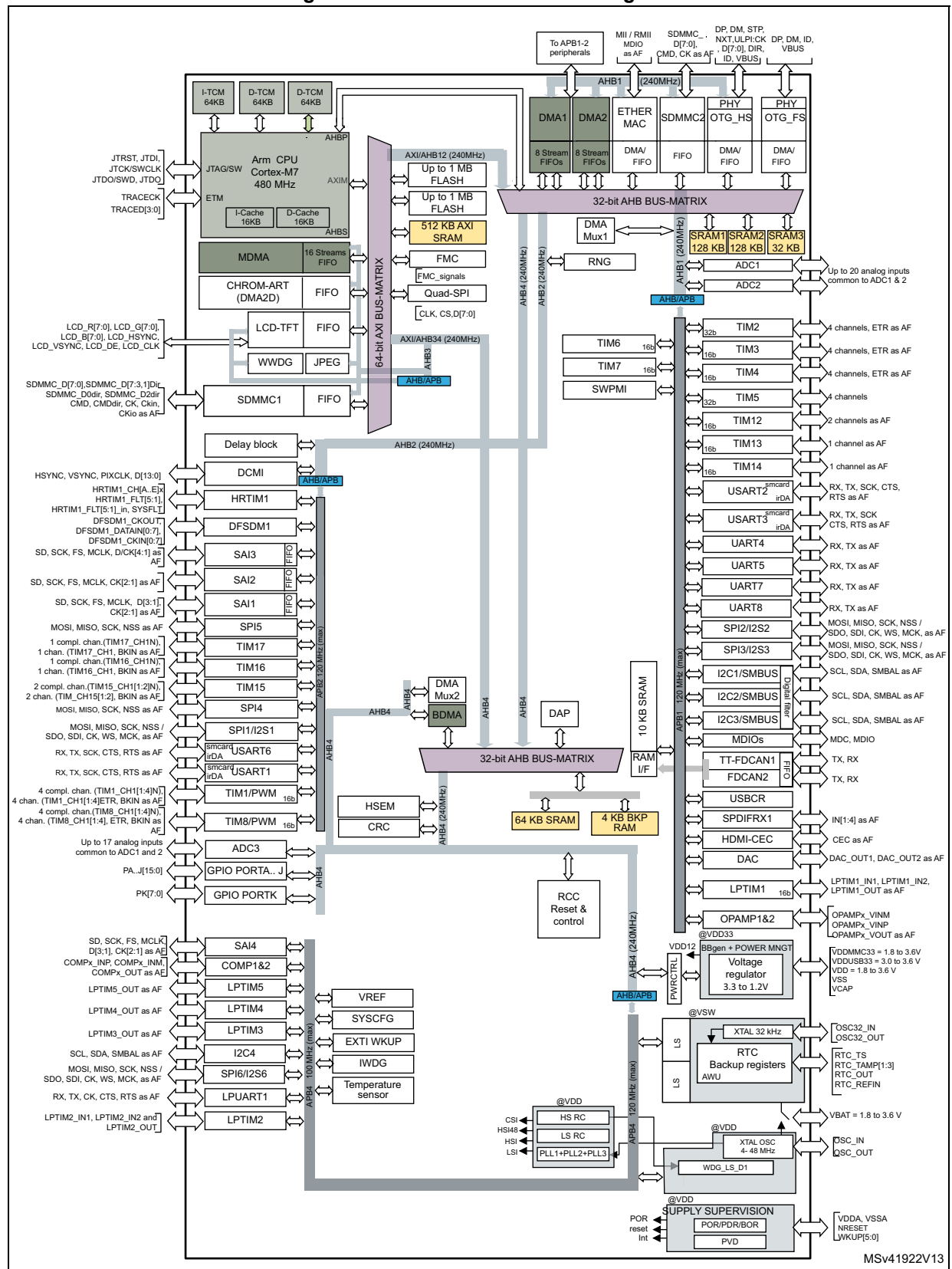




Figure 2. STM32H743xI/G block diagram



## 3 Functional overview

### 3.1 Arm® Cortex®-M7 with FPU

The Arm® Cortex®-M7 with double-precision FPU processor is the latest generation of Arm processors for embedded systems. It was developed to provide a low-cost platform that meets the needs of MCU implementation, with a reduced pin count and optimized power consumption, while delivering outstanding computational performance and low interrupt latency.

The Cortex®-M7 processor is a highly efficient high-performance featuring:

- Six-stage dual-issue pipeline
- Dynamic branch prediction
- Harvard architecture with L1 caches (16 Kbytes of I-cache and 16 Kbytes of D-cache)
- 64-bit AXI interface
- 64-bit ITCM interface
- 2x32-bit DTCM interfaces

The following memory interfaces are supported:

- Separate Instruction and Data buses (Harvard Architecture) to optimize CPU latency
- Tightly Coupled Memory (TCM) interface designed for fast and deterministic SRAM accesses
- AXI Bus interface to optimize Burst transfers
- Dedicated low-latency AHB-Lite peripheral bus (AHBP) to connect to peripherals.

The processor supports a set of DSP instructions which allow efficient signal processing and complex algorithm execution.

It also supports single and double precision FPU (floating point unit) speeds up software development by using metalanguage development tools, while avoiding saturation.

[Figure 1](#) and [Figure 2](#) shows the general block diagram of the STM32H742xI/G and STM32H743xI/G family.

*Note:* Cortex®-M7 with FPU core is binary compatible with the Cortex®-M4 core.

### 3.2 Memory protection unit (MPU)

The memory protection unit (MPU) manages the CPU access rights and the attributes of the system resources. It has to be programmed and enabled before use. Its main purposes are to prevent an untrusted user program to accidentally corrupt data used by the OS and/or by a privileged task, but also to protect data processes or read-protect memory regions.

The MPU defines access rules for privileged accesses and user program accesses. It allows defining up to 16 protected regions that can in turn be divided into up to 8 independent subregions, where region address, size, and attributes can be configured. The protection area ranges from 32 bytes to 4 Gbytes of addressable memory.

When an unauthorized access is performed, a memory management exception is generated.

### 3.3 Memories

#### 3.3.1 Embedded Flash memory

The STM32H742xI/G and STM32H743xI/G devices embed up to 2 Mbytes of Flash memory that can be used for storing programs and data.

The Flash memory is organized as 266-bit Flash words memory that can be used for storing both code and data constants. Each word consists of:

- One Flash word (8 words, 32 bytes or 256 bits)
- 10 ECC bits.

The Flash memory is divided into two independent banks. Each bank is organized as follows:

- A user Flash memory block of 512 Kbytes (STM32H7xxxG) or 1-Mbyte (STM32H7xxxI) containing eight user sectors of 128 Kbytes (4 K Flash memory words)
- 128 Kbytes of System Flash memory from which the device can boot
- 2 Kbytes (64 Flash words) of user option bytes for user configuration

#### 3.3.2 Embedded SRAM

All devices feature:

- 384 (STM32H742xI/G) or 512 Kbytes (STM32H743xI/G) of AXI-SRAM mapped onto AXI bus on D1 domain.
- SRAM1 mapped on D2 domain: 32 (STM32H742xI/G) or 128 Kbytes (STM32H743xI/G)
- SRAM2 mapped on D2 domain: 16 (STM32H742xI/G) or 128 Kbytes (STM32H743xI/G)
- SRAM3 mapped on D2 domain: 32 Kbytes (STM32H743xI/G only)
- SRAM4 mapped on D3 domain: 64 Kbytes
- 4 Kbytes of backup SRAM

The content of this area is protected against possible unwanted write accesses, and is retained in Standby or V<sub>BAT</sub> mode.

- RAM mapped to TCM interface (ITCM and DTCM):

Both ITCM and DTCM RAMs are 0 wait state memories. either They can be accessed either from the CPU or the MDMA (even in Sleep mode) through a specific AHB slave of the CPU(AHBP):

- 64 Kbytes of ITCM-RAM (instruction RAM)

This RAM is connected to ITCM 64-bit interface designed for execution of critical real-times routines by the CPU.

- 128 Kbytes of DTCM-RAM (2x 64-Kbyte DTCM-RAMs on 2x32-bit DTCM ports)

The DTCM-RAM could be used for critical real-time data, such as interrupt service routines or stack/heap memory. Both DTCM-RAMs can be used in parallel (for load/store operations) thanks to the Cortex<sup>®</sup>-M7 dual issue capability.

The MDMA can be used to load code or data in ITCM or DTCM RAMs.

### Error code correction (ECC)

Over the product lifetime, and/or due to external events such as radiations, invalid bits in memories may occur. They can be detected and corrected by ECC. This is an expected behavior that has to be managed at final-application software level in order to ensure data integrity through ECC algorithms implementation.

SRAM data are protected by ECC:

- 7 ECC bits are added per 32-bit word.
- 8 ECC bits are added per 64-bit word for AXI-SRAM and ITCM-RAM.

The ECC mechanism is based on the SECDED algorithm. It supports single-error correction and double-error detection.

## 3.4 Boot modes

At startup, the boot memory space is selected by the BOOT pin and BOOT\_ADDx option bytes, allowing to program any boot memory address from 0x0000 0000 to 0x3FFF FFFF which includes:

- All Flash address space
- All RAM address space: ITCM, DTCM RAMs and SRAMs
- The System memory bootloader

The boot loader is located in non-user System memory. It is used to reprogram the Flash memory through a serial interface (USART, I2C, SPI, USB-DFU). Refer to *STM32 microcontroller System memory Boot mode* application note (AN2606) for details.

## 3.5 Power supply management

### 3.5.1 Power supply scheme

STM32H742xI/G STM32H743xI/G power supply voltages are the following:

- $V_{DD}$  = 1.62 to 3.6 V: external power supply for I/Os, provided externally through  $V_{DD}$  pins.
- $V_{DDLDO}$  = 1.62 to 3.6 V: supply voltage for the internal regulator supplying  $V_{CORE}$
- $V_{DDA}$  = 1.62 to 3.6 V: external analog power supplies for ADC, DAC, COMP and OPAMP.
- $V_{DD33USB}$  and  $V_{DD50USB}$ :  
 $V_{DD50USB}$  can be supplied through the USB cable to generate the  $V_{DD33USB}$  via the USB internal regulator. This allows supporting a  $V_{DD}$  supply different from 3.3 V.  
 The USB regulator can be bypassed to supply directly  $V_{DD33USB}$  if  $V_{DD} = 3.3$  V.
- $V_{BAT}$  = 1.2 to 3.6 V: power supply for the  $V_{SW}$  domain when  $V_{DD}$  is not present.
- $V_{CAP}$ :  $V_{CORE}$  supply voltage, which values depend on voltage scaling (1.0 V, 1.1 V, 1.2 V or 1.35 V). They are configured through VOS bits in PWR\_D3CR register and

ODEN bit in the SYSCFG\_PWRCR register. The  $V_{\text{CORE}}$  domain is split into the following power domains that can be independently switch off.

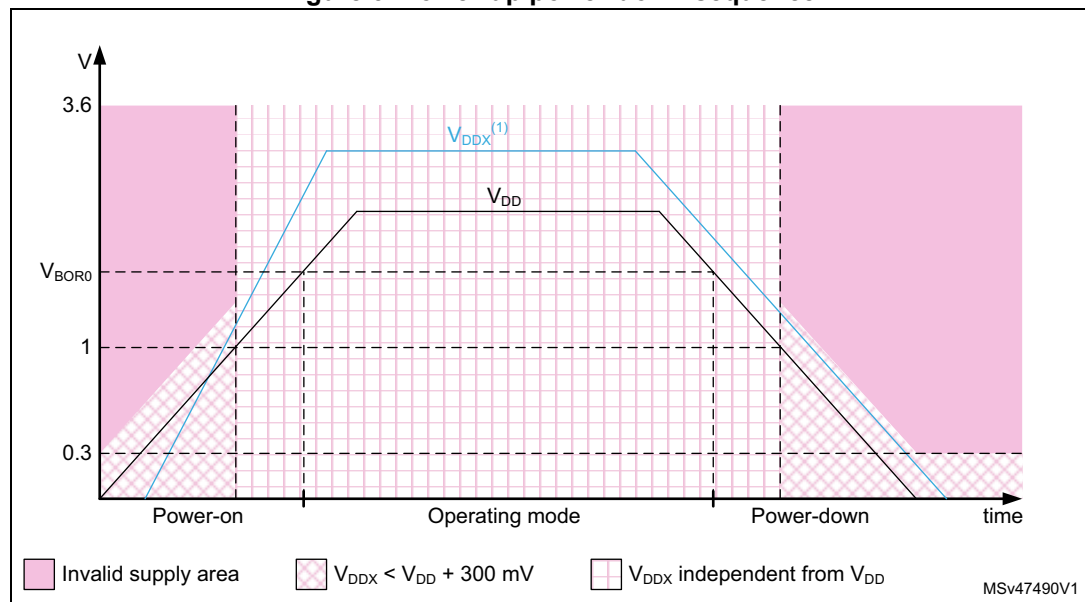
- D1 domain containing some peripherals and the Cortex<sup>®</sup>-M7 core.
- D2 domain containing a large part of the peripherals.
- D3 domain containing some peripherals and the system control.

During power-up and power-down phases, the following power sequence requirements must be respected (see [Figure 3](#)):

- When  $V_{\text{DD}}$  is below 1 V, other power supplies ( $V_{\text{DDA}}$ ,  $V_{\text{DD33USB}}$ ,  $V_{\text{DD50USB}}$ ) must remain below  $V_{\text{DD}} + 300$  mV.
- When  $V_{\text{DD}}$  is above 1 V, all power supplies are independent.

During the power-down phase,  $V_{\text{DD}}$  can temporarily become lower than other supplies only if the energy provided to the microcontroller remains below 1 mJ. This allows external decoupling capacitors to be discharged with different time constants during the power-down transient phase.

**Figure 3. Power-up/power-down sequence**



1.  $V_{\text{DDX}}$  refers to any power supply among  $V_{\text{DDA}}$ ,  $V_{\text{DD33USB}}$ ,  $V_{\text{DD50USB}}$ .

### 3.5.2 Power supply supervisor

The devices have an integrated power-on reset (POR)/ power-down reset (PDR) circuitry coupled with a Brownout reset (BOR) circuitry:

- Power-on reset (POR)  
The POR supervisor monitors  $V_{DD}$  power supply and compares it to a fixed threshold. The devices remain in Reset mode when  $V_{DD}$  is below this threshold,
- Power-down reset (PDR)  
The PDR supervisor monitors  $V_{DD}$  power supply. A reset is generated when  $V_{DD}$  drops below a fixed threshold.  
The PDR supervisor can be enabled/disabled through PDR\_ON pin.
- Brownout reset (BOR)  
The BOR supervisor monitors  $V_{DD}$  power supply. Three BOR thresholds (from 2.1 to 2.7 V) can be configured through option bytes. A reset is generated when  $V_{DD}$  drops below this threshold.

### 3.5.3 Voltage regulator

The same voltage regulator supplies the 3 power domains (D1, D2 and D3). D1 and D2 can be independently switched off.

Voltage regulator output can be adjusted according to application needs through 6 power supply levels:

- Run mode (VOS0 to VOS3)
  - Scale 0: boosted performance (available only with LDO regulator)
  - Scale 1: high performance
  - Scale 2: medium performance and consumption
  - Scale 3: optimized performance and low-power consumption
- Stop mode (SVOS3 to SVOS5)
  - Scale 3: peripheral with wakeup from Stop mode capabilities (UART, SPI, I2C, LPTIM) are operational
  - Scale 4 and 5 where the peripheral with wakeup from Stop mode is disabled  
The peripheral functionality is disabled but wakeup from Stop mode is possible through GPIO or asynchronous interrupt.

### 3.6 Low-power strategy

There are several ways to reduce power consumption on STM32H742xI/G and STM32H743xI/G:

- Decrease the dynamic power consumption by slowing down the system clocks even in Run mode and by individually clock gating the peripherals that are not used.
- Save power consumption when the CPU is idle, by selecting among the available low-power mode according to the user application needs. This allows achieving the best compromise between short startup time, low-power consumption, as well as available wakeup sources.

The devices feature several low-power modes:

- CSleep (CPU clock stopped)
- CStop (CPU sub-system clock stopped)
- DStop (Domain bus matrix clock stopped)
- Stop (System clock stopped)
- DStandby (Domain powered down)
- Standby (System powered down)

CSleep and CStop low-power modes are entered by the MCU when executing the WFI (Wait for Interrupt) or WFE (Wait for Event) instructions, or when the SLEEPONEXIT bit of the Cortex®-Mx core is set after returning from an interrupt service routine.

A domain can enter low-power mode (DStop or DStandby) when the processor, its subsystem and the peripherals allocated in the domain enter low-power mode.

If part of the domain is not in low-power mode, the domain remains in the current mode.

Finally the system can enter Stop or Standby when all EXTI wakeup sources are cleared and the power domains are in DStop or DStandby mode.

**Table 3. System vs domain low-power mode**

| System power mode | D1 domain power mode | D2 domain power mode | D3 domain power mode |
|-------------------|----------------------|----------------------|----------------------|
| Run               | DRun/DStop/DStandby  | DRun/DStop/DStandby  | DRun                 |
| Stop              | DStop/DStandby       | DStop/DStandby       | DStop                |
| Standby           | DStandby             | DStandby             | DStandby             |

## 3.7 Reset and clock controller (RCC)

The clock and reset controller is located in D3 domain. The RCC manages the generation of all the clocks, as well as the clock gating and the control of the system and peripheral resets. It provides a high flexibility in the choice of clock sources and allows to apply clock ratios to improve the power consumption. In addition, on some communication peripherals that are capable to work with two different clock domains (either a bus interface clock or a kernel peripheral clock), the system frequency can be changed without modifying the baudrate.

### 3.7.1 Clock management

The devices embed four internal oscillators, two oscillators with external crystal or resonator, two internal oscillators with fast startup time and three PLLs.

The RCC receives the following clock source inputs:

- Internal oscillators:
  - 64 MHz HSI clock
  - 48 MHz RC oscillator
  - 4 MHz CSI clock
  - 32 kHz LSI clock
- External oscillators:
  - HSE clock: 4-50 MHz (generated from an external source) or 4-48 MHz (generated from a crystal/ceramic resonator)
  - LSE clock: 32.768 kHz

The RCC provides three PLLs: one for system clock, two for kernel clocks.

The system starts on the HSI clock. The user application can then select the clock configuration.

### 3.7.2 System reset sources

Power-on reset initializes all registers while system reset reinitializes the system except for the debug, part of the RCC and power controller status registers, as well as the backup power domain.

A system reset is generated in the following cases:

- Power-on reset (pwr\_por\_rst)
- Brownout reset
- Low level on NRST pin (external reset)
- Window watchdog
- Independent watchdog
- Software reset
- Low-power mode security reset
- Exit from Standby



### 3.8 General-purpose input/outputs (GPIOs)

Each of the GPIO pins can be configured by software as output (push-pull or open-drain, with or without pull-up or pull-down), as input (floating, with or without pull-up or pull-down) or as peripheral alternate function. Most of the GPIO pins are shared with digital or analog alternate functions. All GPIOs are high-current-capable and have speed selection to better manage internal noise, power consumption and electromagnetic emission.

After reset, all GPIOs (except debug pins) are in Analog mode to reduce power consumption (refer to GPIOs register reset values in the device reference manual).

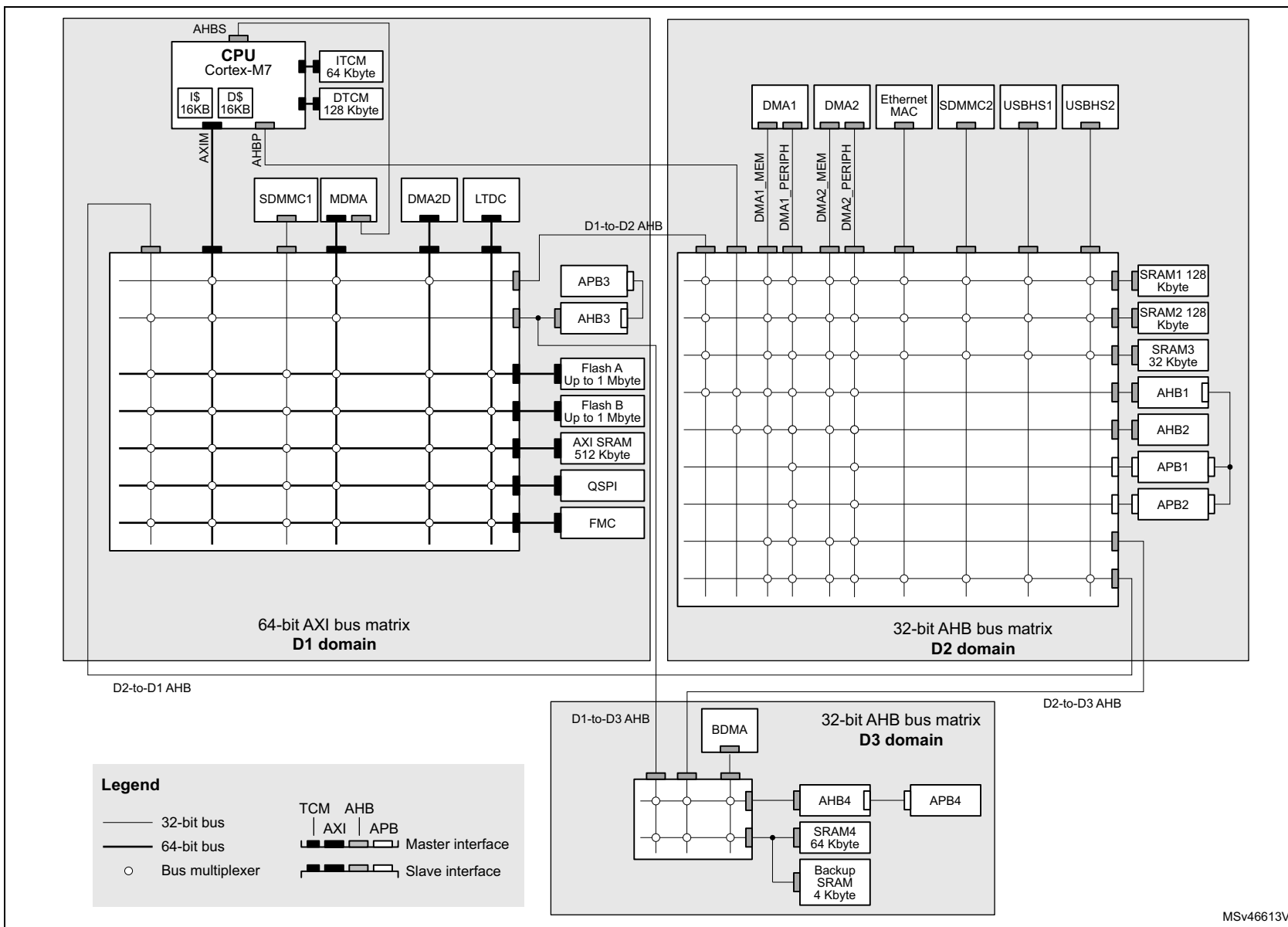
The I/O configuration can be locked if needed by following a specific sequence in order to avoid spurious writing to the I/Os registers.

### 3.9 Bus-interconnect matrix

The devices feature an AXI bus matrix, two AHB bus matrices and bus bridges that allow interconnecting bus masters with bus slaves (see [Figure 4](#)).

[Figure 4](#) shows STM32H743xI/G bus matrix. All peripherals may not be available for STM32H742xI/G (refer to [Table 2: STM32H742xI/G and STM32H743xI/G features and peripheral counts](#)).

Figure 4. STM32H743xI/G bus matrix



MSv46613V2

### 3.10 DMA controllers

The devices feature four DMA instances to unload CPU activity:

- A master direct memory access (MDMA)  
The MDMA is a high-speed DMA controller, which is in charge of all types of memory transfers (peripheral to memory, memory to memory, memory to peripheral), without any CPU action. It features a master AXI interface and a dedicated AHB interface to access Cortex<sup>®</sup>-M7 TCM memories.  
The MDMA is located in D1 domain. It is able to interface with the other DMA controllers located in D2 domain to extend the standard DMA capabilities, or can manage peripheral DMA requests directly.  
Each of the 16 channels can perform single block transfers, repeated block transfers and linked list transfers.
- Two dual-port DMAs (DMA1, DMA2) located in D2 domain, with FIFO and request router capabilities.
- One basic DMA (BDMA) located in D3 domain, with request router capabilities.

The DMA request router could be considered as an extension of the DMA controller. It routes the DMA peripheral requests to the DMA controller itself. This allowing managing the DMA requests with a high flexibility, maximizing the number of DMA requests that run concurrently, as well as generating DMA requests from peripheral output trigger or DMA event.

### 3.11 Chrom-ART Accelerator™ (DMA2D)

The Chrom-Art Accelerator™ (DMA2D) is a graphical accelerator which offers advanced bit blitting, row data copy and pixel format conversion. It supports the following functions:

- Rectangle filling with a fixed color
- Rectangle copy
- Rectangle copy with pixel format conversion
- Rectangle composition with blending and pixel format conversion

Various image format coding are supported, from indirect 4bpp color mode up to 32bpp direct color. It embeds dedicated memory to store color lookup tables. The DMA2D also supports block based YCbCr to handle JPEG decoder output.

An interrupt can be generated when an operation is complete or at a programmed watermark.

All the operations are fully automatized and are running independently from the CPU or the DMAs.

### 3.12 Nested vectored interrupt controller (NVIC)

The devices embed a nested vectored interrupt controller which is able to manage 16 priority levels, and handle up to 150 maskable interrupt channels plus the 16 interrupt lines of the Cortex<sup>®</sup>-M7 with FPU core.

- Closely coupled NVIC gives low-latency interrupt processing
- Interrupt entry vector table address passed directly to the core
- Allows early processing of interrupts
- Processing of late arriving, higher-priority interrupts
- Support tail chaining
- Processor context automatically saved on interrupt entry, and restored on interrupt exit with no instruction overhead

This hardware block provides flexible interrupt management features with minimum interrupt latency.

### 3.13 Extended interrupt and event controller (EXTI)

The EXTI controller performs interrupt and event management. In addition, it can wake up the processor, power domains and/or D3 domain from Stop mode.

The EXTI handles up to 89 independent event/interrupt lines split as 28 configurable events and 61 direct events .

Configurable events have dedicated pending flags, active edge selection, and software trigger capable.

Direct events provide interrupts or events from peripherals having a status flag.

### 3.14 Cyclic redundancy check calculation unit (CRC)

The CRC (cyclic redundancy check) calculation unit is used to get a CRC code using a programmable polynomial.

Among other applications, CRC-based techniques are used to verify data transmission or storage integrity. In the scope of the EN/IEC 60335-1 standard, they offer a means of verifying the Flash memory integrity. The CRC calculation unit helps compute a signature of the software during runtime, to be compared with a reference signature generated at link-time and stored at a given memory location.

### 3.15 Flexible memory controller (FMC)

The FMC controller main features are the following:

- Interface with static-memory mapped devices including:
  - Static random access memory (SRAM)
  - NOR Flash memory/OneNAND Flash memory
  - PSRAM (4 memory banks)
  - NAND Flash memory with ECC hardware to check up to 8 Kbytes of data
- Interface with synchronous DRAM (SDRAM/Mobile LPDDR SDRAM) memories
- 8-,16-,32-bit data bus width
- Independent Chip Select control for each memory bank
- Independent configuration for each memory bank
- Write FIFO
- Read FIFO for SDRAM controller
- The maximum FMC\_CLK/FMC\_SDCLK frequency for synchronous accesses is the FMC kernel clock divided by 2.

### 3.16 Quad-SPI memory interface (QUADSPI)

All devices embed a Quad-SPI memory interface, which is a specialized communication interface targeting Single, Dual or Quad-SPI Flash memories. It supports both single and double datarate operations.

It can operate in any of the following modes:

- Direct mode through registers
- External Flash status register polling mode
- Memory mapped mode.

Up to 256 Mbytes of external Flash memory can be mapped, and 8-, 16- and 32-bit data accesses are supported as well as code execution.

The opcode and the frame format are fully programmable.

### 3.17 Analog-to-digital converters (ADCs)

The STM32H742xI/G and STM32H743xI/G devices embed three analog-to-digital converters, which resolution can be configured to 16, 14, 12, 10 or 8 bits.

Each ADC shares up to 20 external channels, performing conversions in the Single-shot or Scan mode. In Scan mode, automatic conversion is performed on a selected group of analog inputs.

Additional logic functions embedded in the ADC interface allow:

- Simultaneous sample and hold
- Interleaved sample and hold

The ADC can be served by the DMA controller, thus allowing to automatically transfer ADC converted values to a destination location without any software action.

In addition, an analog watchdog feature can accurately monitor the converted voltage of one, some or all selected channels. An interrupt is generated when the converted voltage is outside the programmed thresholds.

To synchronize A/D conversion and timers, the ADCs could be triggered by any of TIM1, TIM2, TIM3, TIM4, TIM6, TIM8, TIM15, HRTIM1 and LPTIM1 timer.

### 3.18 Temperature sensor

STM32H742xI/G and STM32H743xI/G devices embed a temperature sensor that generates a voltage ( $V_{TS}$ ) that varies linearly with the temperature. This temperature sensor is internally connected to ADC3\_IN18. The conversion range is between 1.7 V and 3.6 V. It can measure the device junction temperature ranging from  $-40$  up to  $+125$  °C.

The temperature sensor have a good linearity, but it has to be calibrated to obtain a good overall accuracy of the temperature measurement. As the temperature sensor offset varies from chip to chip due to process variation, the uncalibrated internal temperature sensor is suitable for applications that detect temperature changes only. To improve the accuracy of the temperature sensor measurement, each device is individually factory-calibrated by ST. The temperature sensor factory calibration data are stored by ST in the System memory area, which is accessible in Read-only mode.

### 3.19 $V_{BAT}$ operation

The  $V_{BAT}$  power domain contains the RTC, the backup registers and the backup SRAM.

To optimize battery duration, this power domain is supplied by  $V_{DD}$  when available or by the voltage applied on VBAT pin (when  $V_{DD}$  supply is not present).  $V_{BAT}$  power is switched when the PDR detects that  $V_{DD}$  dropped below the PDR level.

The voltage on the VBAT pin could be provided by an external battery, a supercapacitor or directly by  $V_{DD}$ , in which case, the  $V_{BAT}$  mode is not functional.

$V_{BAT}$  operation is activated when  $V_{DD}$  is not present.

The  $V_{BAT}$  pin supplies the RTC, the backup registers and the backup SRAM.

*Note:* When the microcontroller is supplied from  $V_{BAT}$ , external interrupts and RTC alarm/events do not exit it from  $V_{BAT}$  operation.

When PDR\_ON pin is connected to  $V_{SS}$  (Internal Reset OFF), the  $V_{BAT}$  functionality is no more available and  $V_{BAT}$  pin should be connected to  $V_{DD}$ .

### 3.20 Digital-to-analog converters (DAC)

The two 12-bit buffered DAC channels can be used to convert two digital signals into two analog voltage signal outputs.

This dual digital Interface supports the following features:

- two DAC converters: one for each output channel
- 8-bit or 12-bit monotonic output
- left or right data alignment in 12-bit mode
- synchronized update capability
- noise-wave generation
- triangular-wave generation
- dual DAC channel independent or simultaneous conversions
- DMA capability for each channel including DMA underrun error detection
- external triggers for conversion
- input voltage reference  $V_{REF+}$  or internal VREFBUF reference.

The DAC channels are triggered through the timer update outputs that are also connected to different DMA streams.

### 3.21 Ultra-low-power comparators (COMP)

STM32H742xI/G and STM32H743xI/G devices embed two rail-to-rail comparators (COMP1 and COMP2). They feature programmable reference voltage (internal or external), hysteresis and speed (low speed for low-power) as well as selectable output polarity.

The reference voltage can be one of the following:

- An external I/O
- A DAC output channel
- An internal reference voltage or submultiple (1/4, 1/2, 3/4).

All comparators can wake up from Stop mode, generate interrupts and breaks for the timers, and be combined into a window comparator.

### 3.22 Operational amplifiers (OPAMP)

STM32H742xI/G and STM32H743xI/G devices embed two rail-to-rail operational amplifiers (OPAMP1 and OPAMP2) with external or internal follower routing and PGA capability.

The operational amplifier main features are:

- PGA with a non-inverting gain ranging of 2, 4, 8 or 16 or inverting gain ranging of -1, -3, -7 or -15
- One positive input connected to DAC
- Output connected to internal ADC
- Low input bias current down to 1 nA
- Low input offset voltage down to 1.5 mV
- Gain bandwidth up to 7.3 MHz

The devices embeds two operational amplifiers (OPAMP1 and OPAMP2) with two inputs and one output each. These three I/Os can be connected to the external pins, thus enabling any type of external interconnections. The operational amplifiers can be configured internally as a follower, as an amplifier with a non-inverting gain ranging from 2 to 16 or with inverting gain ranging from -1 to -15.

### 3.23 Digital filter for sigma-delta modulators (DFSDM)

The devices embed one DFSDM with 4 digital filters modules and 8 external input serial channels (transceivers) or alternately 8 internal parallel inputs support.

The DFSDM peripheral is dedicated to interface the external  $\Sigma\Delta$  modulators to microcontroller and then to perform digital filtering of the received data streams (which represent analog value on  $\Sigma\Delta$  modulators inputs). DFSDM can also interface PDM (Pulse Density Modulation) microphones and perform PDM to PCM conversion and filtering in hardware. DFSDM features optional parallel data stream inputs from internal ADC peripherals or microcontroller memory (through DMA/CPU transfers into DFSDM).

DFSDM transceivers support several serial interface formats (to support various  $\Sigma\Delta$  modulators). DFSDM digital filter modules perform digital processing according user selected filter parameters with up to 24-bit final ADC resolution.

The DFSDM peripheral supports:

- 8 multiplexed input digital serial channels:
  - configurable SPI interface to connect various SD modulator(s)
  - configurable Manchester coded 1 wire interface support
  - PDM (Pulse Density Modulation) microphone input support
  - maximum input clock frequency up to 20 MHz (10 MHz for Manchester coding)
  - clock output for SD modulator(s): 0..20 MHz
- alternative inputs from 8 internal digital parallel channels (up to 16 bit input resolution):
  - internal sources: ADC data or memory data streams (DMA)
- 4 digital filter modules with adjustable digital signal processing:
  - Sinc<sup>x</sup> filter: filter order/type (1..5), oversampling ratio (up to 1..1024)
  - integrator: oversampling ratio (1..256)
- up to 24-bit output data resolution, signed output data format
- automatic data offset correction (offset stored in register by user)
- continuous or single conversion
- start-of-conversion triggered by:
  - software trigger
  - internal timers
  - external events
  - start-of-conversion synchronously with first digital filter module (DFSDM0)
- analog watchdog feature:
  - low value and high value data threshold registers
  - dedicated configurable Sinc<sup>x</sup> digital filter (order = 1..3, oversampling ratio = 1..32)
  - input from final output data or from selected input digital serial channels
  - continuous monitoring independently from standard conversion



- short circuit detector to detect saturated analog input values (bottom and top range):
  - up to 8-bit counter to detect 1..256 consecutive 0's or 1's on serial data stream
  - monitoring continuously each input serial channel
- break signal generation on analog watchdog event or on short circuit detector event
- extremes detector:
  - storage of minimum and maximum values of final conversion data
  - refreshed by software
- DMA capability to read the final conversion data
- interrupts: end of conversion, overrun, analog watchdog, short circuit, input serial channel clock absence
- “regular” or “injected” conversions:
  - “regular” conversions can be requested at any time or even in Continuous mode without having any impact on the timing of “injected” conversions
  - “injected” conversions for precise timing and with high conversion priority

**Table 4. DFSDM implementation**

| DFSDM features                                         | DFSDM1 |
|--------------------------------------------------------|--------|
| Number of filters                                      | 4      |
| Number of input transceivers/channels                  | 8      |
| Internal ADC parallel input                            | X      |
| Number of external triggers                            | 16     |
| Regular channel information in identification register | X      |

### 3.24 Digital camera interface (DCMI)

The devices embed a camera interface that can connect with camera modules and CMOS sensors through an 8-bit to 14-bit parallel interface, to receive video data. The camera interface can achieve a data transfer rate up to 140 Mbyte/s using a 80 MHz pixel clock. It features:

- Programmable polarity for the input pixel clock and synchronization signals
- Parallel data communication can be 8-, 10-, 12- or 14-bit
- Supports 8-bit progressive video monochrome or raw bayer format, YCbCr 4:2:2 progressive video, RGB 565 progressive video or compressed data (like JPEG)
- Supports Continuous mode or Snapshot (a single frame) mode
- Capability to automatically crop the image

### 3.25 LCD-TFT controller

The LCD-TFT display controller (only available on STM32H743xI/G) provides a 24-bit parallel digital RGB (Red, Green, Blue) and delivers all signals to interface directly to a

broad range of LCD and TFT panels up to XGA (1024x768) resolution with the following features:

- 2 display layers with dedicated FIFO (64x64-bit)
- Color Look-Up table (CLUT) up to 256 colors (256x24-bit) per layer
- Up to 8 input color formats selectable per layer
- Flexible blending between two layers using alpha value (per pixel or constant)
- Flexible programmable parameters for each layer
- Color keying (transparency color)
- Up to 4 programmable interrupt events
- AXI master interface with burst of 16 words

### 3.26 JPEG Codec (JPEG)

The JPEG Codec (only available on STM32H743xI/G) can encode and decode a JPEG stream as defined in the *ISO/IEC 10918-1* specification. It provides an fast and simple hardware compressor and decompressor of JPEG images with full management of JPEG headers.

The JPEG codec main features are as follows:

- 8-bit/channel pixel depths
- Single clock per pixel encoding and decoding
- Support for JPEG header generation and parsing
- Up to four programmable quantization tables
- Fully programmable Huffman tables (two AC and two DC)
- Fully programmable minimum coded unit (MCU)
- Encode/decode support (non simultaneous)
- Single clock Huffman coding and decoding
- Two-channel interface: Pixel/Compress In, Pixel/Compressed Out
- Support for single greyscale component
- Ability to enable/disable header processing
- Fully synchronous design
- Configuration for High-speed decode mode

### 3.27 Random number generator (RNG)

All devices embed an RNG that delivers 32-bit random numbers generated by an integrated analog circuit.

### 3.28 Timers and watchdogs

The devices include one high-resolution timer, two advanced-control timers, ten general-purpose timers, two basic timers, five low-power timers, two watchdogs and a SysTick timer.

All timer counters can be frozen in Debug mode.

[Table 5](#) compares the features of the advanced-control, general-purpose and basic timers.

Table 5. Timer feature comparison

| Timer type            | Timer        | Counter resolution | Counter type      | Prescaler factor                         | DMA request generation | Capture/compare channels | Complementary output | Max interface clock (MHz) | Max timer clock (MHz) <sup>(1)</sup> |
|-----------------------|--------------|--------------------|-------------------|------------------------------------------|------------------------|--------------------------|----------------------|---------------------------|--------------------------------------|
| High-resolution timer | HRTIM1       | 16-bit             | Up                | /1 /2 /4<br>(x2 x4 x8 x16 x32, with DLL) | Yes                    | 10                       | Yes                  | 480                       | 480                                  |
| Advanced-control      | TIM1, TIM8   | 16-bit             | Up, Down, Up/down | Any integer between 1 and 65536          | Yes                    | 4                        | Yes                  | 120                       | 240                                  |
| General purpose       | TIM2, TIM5   | 32-bit             | Up, Down, Up/down | Any integer between 1 and 65536          | Yes                    | 4                        | No                   | 120                       | 240                                  |
|                       | TIM3, TIM4   | 16-bit             | Up, Down, Up/down | Any integer between 1 and 65536          | Yes                    | 4                        | No                   | 120                       | 240                                  |
|                       | TIM12        | 16-bit             | Up                | Any integer between 1 and 65536          | No                     | 2                        | No                   | 120                       | 240                                  |
|                       | TIM13, TIM14 | 16-bit             | Up                | Any integer between 1 and 65536          | No                     | 1                        | No                   | 120                       | 240                                  |
|                       | TIM15        | 16-bit             | Up                | Any integer between 1 and 65536          | Yes                    | 2                        | 1                    | 120                       | 240                                  |
|                       | TIM16, TIM17 | 16-bit             | Up                | Any integer between 1 and 65536          | Yes                    | 1                        | 1                    | 120                       | 240                                  |

Table 5. Timer feature comparison (continued)

| Timer type      | Timer                                  | Counter resolution | Counter type | Prescaler factor                | DMA request generation | Capture/compare channels | Complementary output | Max interface clock (MHz) | Max timer clock (MHz) <sup>(1)</sup> |
|-----------------|----------------------------------------|--------------------|--------------|---------------------------------|------------------------|--------------------------|----------------------|---------------------------|--------------------------------------|
| Basic           | TIM6, TIM7                             | 16-bit             | Up           | Any integer between 1 and 65536 | Yes                    | 0                        | No                   | 120                       | 240                                  |
| Low-power timer | LPTIM1, LPTIM2, LPTIM3, LPTIM4, LPTIM5 | 16-bit             | Up           | 1, 2, 4, 8, 16, 32, 64, 128     | No                     | 0                        | No                   | 120                       | 240                                  |

1. The maximum timer clock is up to 480 MHz depending on TIMPRE bit in the RCC\_CFGR register and D2PRE1/2 bits in RCC\_D2CFGR register.

### 3.28.1 High-resolution timer (HRTIM1)

The high-resolution timer (HRTIM1) allows generating digital signals with high-accuracy timings, such as PWM or phase-shifted pulses.

It consists of 6 timers, 1 master and 5 slaves, totaling 10 high-resolution outputs, which can be coupled by pairs for deadtime insertion. It also features 5 fault inputs for protection purposes and 10 inputs to handle external events such as current limitation, zero voltage or zero current switching.

The HRTIM1 timer is made of a digital kernel clocked at 480 MHz. The high-resolution is available on the 10 outputs in all operating modes: variable duty cycle, variable frequency, and constant ON time.

The slave timers can be combined to control multiswitch complex converters or operate independently to manage multiple independent converters.

The waveforms are defined by a combination of user-defined timings and external events such as analog or digital feedback signals.

HRTIM1 timer includes options for blanking and filtering out spurious events or faults. It also offers specific modes and features to offload the CPU: DMA requests, Burst mode controller, Push-pull and Resonant mode.

It supports many topologies including LLC, Full bridge phase shifted, buck or boost converters, either in voltage or current mode, as well as lighting application (fluorescent or LED). It can also be used as a general purpose timer, for instance to achieve high-resolution PWM-emulated DAC.

### 3.28.2 Advanced-control timers (TIM1, TIM8)

The advanced-control timers (TIM1, TIM8) can be seen as three-phase PWM generators multiplexed on 6 channels. They have complementary PWM outputs with programmable inserted dead times. They can also be considered as complete general-purpose timers. Their 4 independent channels can be used for:

- Input capture
- Output compare
- PWM generation (Edge- or Center-aligned modes)
- One-pulse mode output

If configured as standard 16-bit timers, they have the same features as the general-purpose TIMx timers. If configured as 16-bit PWM generators, they have full modulation capability (0-100%).

The advanced-control timer can work together with the TIMx timers via the Timer Link feature for synchronization or event chaining.

TIM1 and TIM8 support independent DMA request generation.

### 3.28.3 General-purpose timers (TIMx)

There are ten synchronizable general-purpose timers embedded in the STM32H742xI/G and STM32H743xI/G devices (see [Table 5](#) for differences).

- **TIM2, TIM3, TIM4, TIM5**

The devices include 4 full-featured general-purpose timers: TIM2, TIM3, TIM4 and TIM5. TIM2 and TIM5 are based on a 32-bit auto-reload up/downcounter and a 16-bit prescaler while TIM3 and TIM4 are based on a 16-bit auto-reload up/downcounter and a 16-bit prescaler. All timers feature 4 independent channels for input capture/output compare, PWM or One-pulse mode output. This gives up to 16 input capture/output compare/PWMs on the largest packages.

TIM2, TIM3, TIM4 and TIM5 general-purpose timers can work together, or with the other general-purpose timers and the advanced-control timers TIM1 and TIM8 via the Timer Link feature for synchronization or event chaining.

Any of these general-purpose timers can be used to generate PWM outputs.

TIM2, TIM3, TIM4, TIM5 all have independent DMA request generation. They are capable of handling quadrature (incremental) encoder signals and the digital outputs from 1 to 4 hall-effect sensors.

- **TIM12, TIM13, TIM14, TIM15, TIM16, TIM17**

These timers are based on a 16-bit auto-reload upcounter and a 16-bit prescaler. TIM13, TIM14, TIM16 and TIM17 feature one independent channel, whereas TIM12 and TIM15 have two independent channels for input capture/output compare, PWM or One-pulse mode output. They can be synchronized with the TIM2, TIM3, TIM4, TIM5 full-featured general-purpose timers or used as simple timebases.

### 3.28.4 Basic timers TIM6 and TIM7

These timers are mainly used for DAC trigger and waveform generation. They can also be used as a generic 16-bit time base.

TIM6 and TIM7 support independent DMA request generation.

### 3.28.5 Low-power timers (LPTIM1, LPTIM2, LPTIM3, LPTIM4, LPTIM5)

The low-power timers have an independent clock and is running also in Stop mode if it is clocked by LSE, LSI or an external clock. It is able to wakeup the devices from Stop mode.

This low-power timer supports the following features:

- 16-bit up counter with 16-bit autoreload register
- 16-bit compare register
- Configurable output: pulse, PWM
- Continuous / One-shot mode
- Selectable software / hardware input trigger
- Selectable clock source:
  - Internal clock source: LSE, LSI, HSI or APB clock
  - External clock source over LPTIM input (working even with no internal clock source running, used by the Pulse Counter Application)
- Programmable digital glitch filter
- Encoder mode

### 3.28.6 Independent watchdog

The independent watchdog is based on a 12-bit downcounter and 8-bit prescaler. It is clocked from an independent 32 kHz internal RC and as it operates independently from the main clock, it can operate in Stop and Standby modes. It can be used either as a watchdog to reset the device when a problem occurs, or as a free-running timer for application timeout management. It is hardware- or software-configurable through the option bytes.

### 3.28.7 Window watchdog

The window watchdog is based on a 7-bit downcounter that can be set as free-running. It can be used as a watchdog to reset the device when a problem occurs. It is clocked from the main clock. It has an early warning interrupt capability and the counter can be frozen in Debug mode.

### 3.28.8 SysTick timer

This timer is dedicated to real-time operating systems, but could also be used as a standard downcounter. It features:

- A 24-bit downcounter
- Autoreload capability
- Maskable system interrupt generation when the counter reaches 0
- Programmable clock source.

### 3.29 Real-time clock (RTC), backup SRAM and backup registers

The RTC is an independent BCD timer/counter. It supports the following features:

- Calendar with subsecond, seconds, minutes, hours (12 or 24 format), week day, date, month, year, in BCD (binary-coded decimal) format.
- Automatic correction for 28, 29 (leap year), 30, and 31 days of the month.
- Two programmable alarms.
- On-the-fly correction from 1 to 32767 RTC clock pulses. This can be used to synchronize it with a master clock.
- Reference clock detection: a more precise second source clock (50 or 60 Hz) can be used to enhance the calendar precision.
- Digital calibration circuit with 0.95 ppm resolution, to compensate for quartz crystal inaccuracy.
- Three anti-tamper detection pins with programmable filter.
- Timestamp feature which can be used to save the calendar content. This function can be triggered by an event on the timestamp pin, or by a tamper event, or by a switch to  $V_{BAT}$  mode.
- 17-bit auto-reload wakeup timer (WUT) for periodic events with programmable resolution and period.

The RTC and the 32 backup registers are supplied through a switch that takes power either from the  $V_{DD}$  supply when present or from the  $V_{BAT}$  pin.

The backup registers are 32-bit registers used to store 128 bytes of user application data when  $V_{DD}$  power is not present. They are not reset by a system or power reset, or when the device wakes up from Standby mode.

The RTC clock sources can be:

- A 32.768 kHz external crystal (LSE)
- An external resonator or oscillator (LSE)
- The internal low-power RC oscillator (LSI, with typical frequency of 32 kHz)
- The high-speed external clock (HSE) divided by 32.

The RTC is functional in  $V_{BAT}$  mode and in all low-power modes when it is clocked by the LSE. When clocked by the LSI, the RTC is not functional in  $V_{BAT}$  mode, but is functional in all low-power modes.

All RTC events (Alarm, Wakeup Timer, Timestamp or Tamper) can generate an interrupt and wakeup the device from the low-power modes.

### 3.30 Inter-integrated circuit interface (I2C)

STM32H742xI/G and STM32H743xI/G devices embed four I<sup>2</sup>C interfaces.

The I<sup>2</sup>C bus interface handles communications between the microcontroller and the serial I<sup>2</sup>C bus. It controls all I<sup>2</sup>C bus-specific sequencing, protocol, arbitration and timing.

The I2C peripheral supports:

- I<sup>2</sup>C-bus specification and user manual rev. 5 compatibility:
  - Slave and Master modes, multimaster capability
  - Standard-mode (Sm), with a bitrate up to 100 kbit/s
  - Fast-mode (Fm), with a bitrate up to 400 kbit/s
  - Fast-mode Plus (Fm+), with a bitrate up to 1 Mbit/s and 20 mA output drive I/Os
  - 7-bit and 10-bit addressing mode, multiple 7-bit slave addresses
  - Programmable setup and hold times
  - Optional clock stretching
- System Management Bus (SMBus) specification rev 2.0 compatibility:
  - Hardware PEC (Packet Error Checking) generation and verification with ACK control
  - Address resolution protocol (ARP) support
  - SMBus alert
- Power System Management Protocol (PMBus™) specification rev 1.1 compatibility
- Independent clock: a choice of independent clock sources allowing the I2C communication speed to be independent from the PCLK reprogramming.
- Wakeup from Stop mode on address match
- Programmable analog and digital noise filters
- 1-byte buffer with DMA capability

### 3.31 Universal synchronous/asynchronous receiver transmitter (USART)

STM32H742xI/G and STM32H743xI/G devices have four embedded universal synchronous receiver transmitters (USART1, USART2, USART3 and USART6) and four universal asynchronous receiver transmitters (UART4, UART5, UART7 and UART8). Refer to [Table 6](#) for a summary of USARTx and UARTx features.

These interfaces provide asynchronous communication, IrDA SIR ENDEC support, multiprocessor communication mode, single-wire Half-duplex communication mode and have LIN Master/Slave capability. They provide hardware management of the CTS and RTS signals, and RS485 Driver Enable. They are able to communicate at speeds of up to 12.5 Mbit/s.

USART1, USART2, USART3 and USART6 also provide Smartcard mode (ISO 7816 compliant) and SPI-like communication capability.

The USARTs embed a Transmit FIFO (TXFIFO) and a Receive FIFO (RXFIFO). FIFO mode is enabled by software and is disabled by default.



All USART have a clock domain independent from the CPU clock, allowing the USARTx to wake up the MCU from Stop mode. The wakeup from Stop mode is programmable and can be done on:

- Start bit detection
- Any received data frame
- A specific programmed data frame
- Specific TXFIFO/RXFIFO status when FIFO mode is enabled.

All USART interfaces can be served by the DMA controller.

**Table 6. USART features**

| USART modes/features <sup>(1)</sup>              | USART1/2/3/6    | UART4/5/7/8 |
|--------------------------------------------------|-----------------|-------------|
| Hardware flow control for modem                  | X               | X           |
| Continuous communication using DMA               | X               | X           |
| Multiprocessor communication                     | X               | X           |
| Synchronous mode (Master/Slave)                  | X               | -           |
| Smartcard mode                                   | X               | -           |
| Single-wire Half-duplex communication            | X               | X           |
| IrDA SIR ENDEC block                             | X               | X           |
| LIN mode                                         | X               | X           |
| Dual clock domain and wakeup from low power mode | X               | X           |
| Receiver timeout interrupt                       | X               | X           |
| Modbus communication                             | X               | X           |
| Auto baud rate detection                         | X               | X           |
| Driver Enable                                    | X               | X           |
| USART data length                                | 7, 8 and 9 bits |             |
| Tx/Rx FIFO                                       | X               | X           |
| Tx/Rx FIFO size                                  | 16              |             |

1. X = supported.

### 3.32 Low-power universal asynchronous receiver transmitter (LPUART)

The device embeds one Low-Power UART (LPUART1). The LPUART supports asynchronous serial communication with minimum power consumption. It supports half duplex single wire communication and modem operations (CTS/RTS). It allows multiprocessor communication.

The LPUARTs embed a Transmit FIFO (TXFIFO) and a Receive FIFO (RXFIFO). FIFO mode is enabled by software and is disabled by default.

The LPUART has a clock domain independent from the CPU clock, and can wakeup the system from Stop mode. The wakeup from Stop mode are programmable and can be done on:

- Start bit detection
- Any received data frame
- A specific programmed data frame
- Specific TXFIFO/RXFIFO status when FIFO mode is enabled.

Only a 32.768 kHz clock (LSE) is needed to allow LPUART communication up to 9600 baud. Therefore, even in Stop mode, the LPUART can wait for an incoming frame while having an extremely low energy consumption. Higher speed clock can be used to reach higher baudrates.

LPUART interface can be served by the DMA controller.

### 3.33 Serial peripheral interface (SPI)/inter- integrated sound interfaces (I2S)

The devices feature up to six SPIs (SPI2S1, SPI2S2, SPI2S3, SPI4, SPI5 and SPI6) that allow communicating up to 150 Mbits/s in Master and Slave modes, in Half-duplex, Full-duplex and Simplex modes. The 3-bit prescaler gives 8 master mode frequencies and the frame is configurable from 4 to 16 bits. All SPI interfaces support NSS pulse mode, TI mode, Hardware CRC calculation and 8x 8-bit embedded Rx and Tx FIFOs with DMA capability.

Three standard I<sup>2</sup>S interfaces (multiplexed with SPI1, SPI2 and SPI3) are available. They can be operated in Master or Slave mode, in Simplex communication modes, and can be configured to operate with a 16-/32-bit resolution as an input or output channel. Audio sampling frequencies from 8 kHz up to 192 kHz are supported. When either or both of the I<sup>2</sup>S interfaces is/are configured in Master mode, the master clock can be output to the external DAC/CODEC at 256 times the sampling frequency. All I<sup>2</sup>S interfaces support 16x 8-bit embedded Rx and Tx FIFOs with DMA capability.

### 3.34 Serial audio interfaces (SAI)

The devices embed 4 SAIs (SAI1, SAI2, SAI3 and SAI4) that allow designing many stereo or mono audio protocols such as I2S, LSB or MSB-justified, PCM/DSP, TDM or AC'97. An SPDIF output is available when the audio block is configured as a transmitter. To bring this level of flexibility and reconfigurability, the SAI contains two independent audio sub-blocks. Each block has its own clock generator and I/O line controller.

Audio sampling frequencies up to 192 kHz are supported.

In addition, up to 8 microphones can be supported thanks to an embedded PDM interface. The SAI can work in master or slave configuration. The audio sub-blocks can be either receiver or transmitter and can work synchronously or asynchronously (with respect to the other one). The SAI can be connected with other SAIs to work synchronously.

### 3.35 SPDIFRX Receiver Interface (SPDIFRX)

The SPDIFRX peripheral is designed to receive an S/PDIF flow compliant with IEC-60958 and IEC-61937. These standards support simple stereo streams up to high sample rate, and compressed multi-channel surround sound, such as those defined by Dolby or DTS (up to 5.1).

The main SPDIFRX features are the following:

- Up to 4 inputs available
- Automatic symbol rate detection
- Maximum symbol rate: 12.288 MHz
- Stereo stream from 32 to 192 kHz supported
- Supports Audio IEC-60958 and IEC-61937, consumer applications
- Parity bit management
- Communication using DMA for audio samples
- Communication using DMA for control and user channel information
- Interrupt capabilities

The SPDIFRX receiver provides all the necessary features to detect the symbol rate, and decode the incoming data stream. The user can select the wanted SPDIF input, and when a valid signal will be available, the SPDIFRX will re-sample the incoming signal, decode the Manchester stream, recognize frames, sub-frames and blocks elements. It delivers to the CPU decoded data, and associated status flags.

The SPDIFRX also offers a signal named `spdif_frame_sync`, which toggles at the S/PDIF sub-frame rate that will be used to compute the exact sample rate for clock drift algorithms.

### 3.36 Single wire protocol master interface (SWPMI)

The Single wire protocol master interface (SWPMI) is the master interface corresponding to the Contactless Frontend (CLF) defined in the ETSI TS 102 613 technical specification. The main features are:

- Full-duplex communication mode
- automatic SWP bus state management (active, suspend, resume)
- configurable bitrate up to 2 Mbit/s
- automatic SOF, EOF and CRC handling

SWPMI can be served by the DMA controller.

### 3.37 Management Data Input/Output (MDIO) slaves

The devices embed an MDIO slave interface it includes the following features:

- 32 MDIO Registers addresses, each of which is managed using separate input and output data registers:
  - 32 x 16-bit firmware read/write, MDIO read-only output data registers
  - 32 x 16-bit firmware read-only, MDIO write-only input data registers
- Configurable slave (port) address
- Independently maskable interrupts/events:
  - MDIO Register write
  - MDIO Register read
  - MDIO protocol error
- Able to operate in and wake up from Stop mode

### 3.38 SD/SDIO/MMC card host interfaces (SDMMC)

Two SDMMC host interfaces are available. They support *MultiMediaCard System Specification Version 4.51* in three different databus modes: 1 bit (default), 4 bits and 8 bits.

Both interfaces support the *SD memory card specifications version 4.1*. and the *SDIO card specification version 4.0*. in two different databus modes: 1 bit (default) and 4 bits.

Each SDMMC host interface supports only one SD/SDIO/MMC card at any one time and a stack of MMC Version 4.51 or previous.

The SDMMC host interface embeds a dedicated DMA controller allowing high-speed transfers between the interface and the SRAM.

### 3.39 Controller area network (FDCAN1, FDCAN2)

The controller area network (CAN) subsystem consists of two CAN modules, a shared message RAM memory and a clock calibration unit.

Both CAN modules (FDCAN1 and FDCAN2) are compliant with ISO 11898-1 (CAN protocol specification version 2.0 part A, B) and CAN FD protocol specification version 1.0.

FDCAN1 supports time triggered CAN (TT-FDCAN) specified in ISO 11898-4, including event synchronized time-triggered communication, global system time, and clock drift compensation. The FDCAN1 contains additional registers, specific to the time triggered feature. The CAN FD option can be used together with event-triggered and time-triggered CAN communication.

A 10-Kbyte message RAM memory implements filters, receive FIFOs, receive buffers, transmit event FIFOs, transmit buffers (and triggers for TT-FDCAN). This message RAM is shared between the two FDCAN1 and FDCAN2 modules.

The common clock calibration unit is optional. It can be used to generate a calibrated clock for both FDCAN1 and FDCAN2 from the HSI internal RC oscillator and the PLL, by evaluating CAN messages received by the FDCAN1.

### 3.40 Universal serial bus on-the-go high-speed (OTG\_HS)

The devices embed two USB OTG high-speed (up to 480 Mbit/s) device/host/OTG peripheral. OTG-HS1 supports both full-speed and high-speed operations, while OTG-HS2 supports only full-speed operations. They both integrate the transceivers for full-speed operation (12 Mbit/s) and are able to operate from the internal HSI48 oscillator. OTG-HS1 features a UTMI low-pin interface (ULPI) for high-speed operation (480 Mbit/s). When using the USB OTG-HS1 in HS mode, an external PHY device connected to the ULPI is required.

The USB OTG HS peripherals are compliant with the USB 2.0 specification and with the OTG 2.0 specification. They have software-configurable endpoint setting and supports suspend/resume. The USB OTG controllers require a dedicated 48 MHz clock that is generated by a PLL connected to the HSE oscillator.

The main features are:

- Combined Rx and Tx FIFO size of 4 Kbytes with dynamic FIFO sizing
- Supports the session request protocol (SRP) and host negotiation protocol (HNP)
- 9 bidirectional endpoints (including EP0)
- 16 host channels with periodic OUT support
- Software configurable to OTG1.3 and OTG2.0 modes of operation
- USB 2.0 LPM (Link Power Management) support
- Battery Charging Specification Revision 1.2 support
- Internal FS OTG PHY support
- External HS or HS OTG operation supporting ULPI in SDR mode (OTG\_HS1 only)  
The OTG PHY is connected to the microcontroller ULPI port through 12 signals. It can be clocked using the 60 MHz output.
- Internal USB DMA
- HNP/SNP/IP inside (no need for any external resistor)
- For OTG/Host modes, a power switch is needed in case bus-powered devices are connected

### 3.41 Ethernet MAC interface with dedicated DMA controller (ETH)

The devices provide an IEEE-802.3-2002-compliant media access controller (MAC) for ethernet LAN communications through an industry-standard medium-independent interface (MII) or a reduced medium-independent interface (RMII). The microcontroller requires an external physical interface device (PHY) to connect to the physical LAN bus (twisted-pair, fiber, etc.). The PHY is connected to the device MII port using 17 signals for MII or 9 signals for RMII, and can be clocked using the 25 MHz (MII) from the microcontroller.

The devices include the following features:

- Supports 10 and 100 Mbit/s rates
- Dedicated DMA controller allowing high-speed transfers between the dedicated SRAM and the descriptors
- Tagged MAC frame support (VLAN support)
- Half-duplex (CSMA/CD) and full-duplex operation
- MAC control sublayer (control frames) support
- 32-bit CRC generation and removal
- Several address filtering modes for physical and multicast address (multicast and group addresses)
- 32-bit status code for each transmitted or received frame
- Internal FIFOs to buffer transmit and receive frames. The transmit FIFO and the receive FIFO are both 2 Kbytes.
- Supports hardware PTP (precision time protocol) in accordance with IEEE 1588 2008 (PTP V2) with the time stamp comparator connected to the TIM2 input
- Triggers interrupt when system time becomes greater than target time

### 3.42 High-definition multimedia interface (HDMI) - consumer electronics control (CEC)

The devices embed a HDMI-CEC controller that provides hardware support for the Consumer Electronics Control (CEC) protocol (Supplement 1 to the HDMI standard).

This protocol provides high-level control functions between all audiovisual products in an environment. It is specified to operate at low speeds with minimum processing and memory overhead. It has a clock domain independent from the CPU clock, allowing the HDMI-CEC controller to wakeup the MCU from Stop mode on data reception.

### 3.43 Debug infrastructure

The devices offer a comprehensive set of debug and trace features to support software development and system integration.

- Breakpoint debugging
- Code execution tracing
- Software instrumentation
- JTAG debug port
- Serial-wire debug port
- Trigger input and output
- Serial-wire trace port
- Trace port
- Arm® CoreSight™ debug and trace components

The debug can be controlled via a JTAG/Serial-wire debug access port, using industry standard debugging tools.

The trace port performs data capture for logging and analysis.

## 4 Memory mapping

Refer to [Table 7](#) for details on STM32H742xI/G Flash and SRAM block memory mapping and to the product line reference manual for information on the boundary addresses for all STM32H742xI/G peripherals.

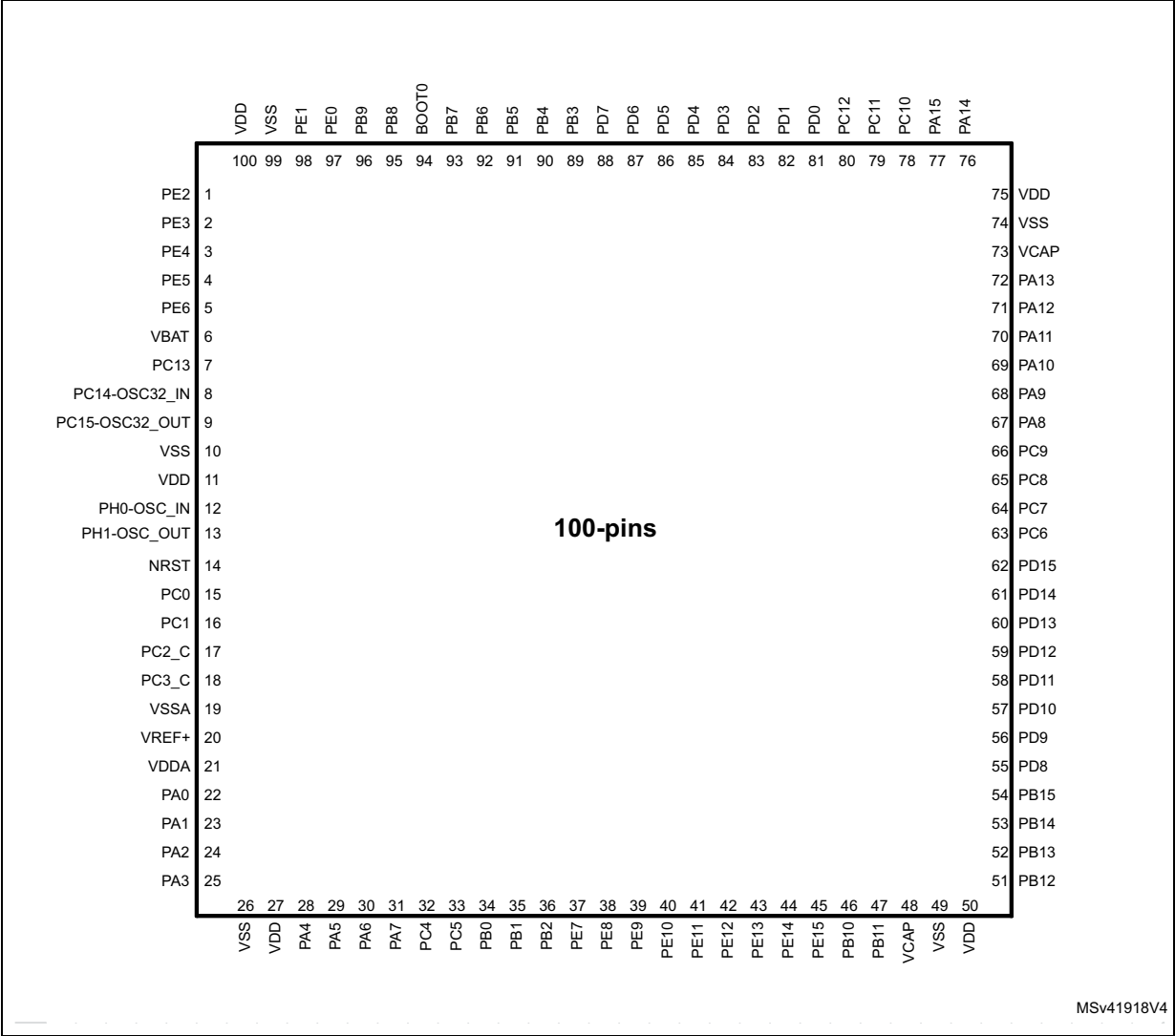
Details on STM32H743xGxI/G Flash and SRAM block memory mapping and boundary addresses for all STM32H743xI/G peripherals are given in the product line reference manual.

**Table 7. Flash memory and SRAM memory mapping for STM32H742xI/G**

|                  | Memory          | Size in Kbytes | Start address |
|------------------|-----------------|----------------|---------------|
| <b>RAM area</b>  | Backup SRAM     | 4              | 0x3880 0000   |
|                  | SRAM4           | 64             | 0x3800 0000   |
|                  | SRAM2           | 16             | 0x3002 0000   |
|                  | SRAM1           | 32             | 0x3000 0000   |
|                  | AXI-SRAM        | 384            | 0x2400 0000   |
|                  | DTCM            | 128            | 0x2000 0000   |
| <b>Code area</b> | System memory 2 |                | 0x1FF4 0000   |
|                  | System memory   |                | 0x1FF0 0000   |
|                  | Flash memory    | 2048           | 0x0800 0000   |
|                  | ITCM            | 64             | 0x0000 0000   |

5 Pin descriptions

Figure 5. LQFP100 pinout



1. The above figure shows the package top view.



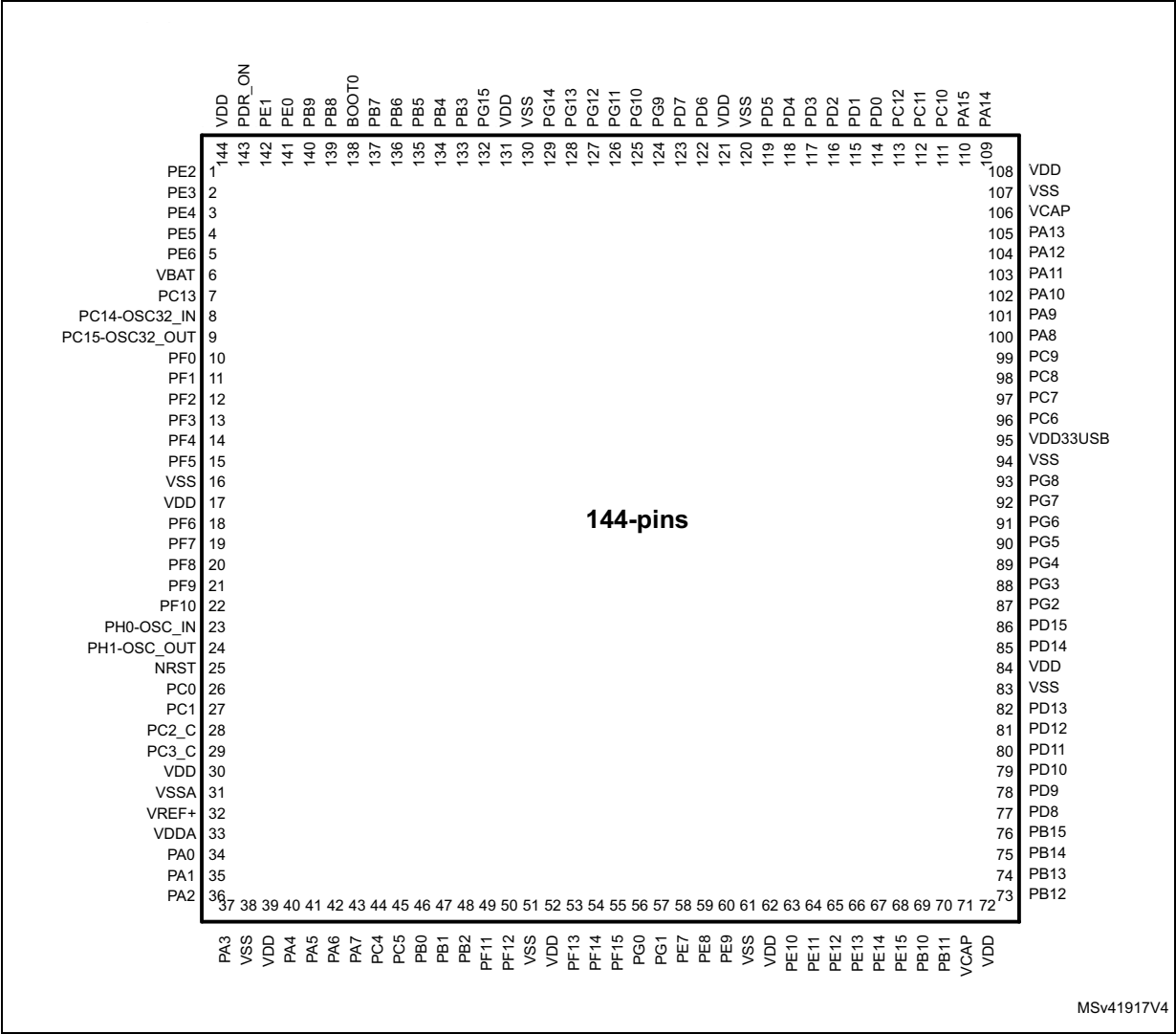
Figure 6. TFBGA100 pinout

|   | 1              | 2     | 3     | 4      | 5     | 6        | 7      | 8    | 9    | 10   |
|---|----------------|-------|-------|--------|-------|----------|--------|------|------|------|
| A | PC14-OSC32_IN  | PC13  | PE2   | PB9    | PB7   | PB4      | PB3    | PA15 | PA14 | PA13 |
| B | PC15-OSC32_OUT | VBAT  | PE3   | PB8    | PB6   | PD5      | PD2    | PC11 | PC10 | PA12 |
| C | PH0-OSC_IN     | VSS   | PE4   | PE1    | PB5   | PD6      | PD3    | PC12 | PA9  | PA11 |
| D | PH1-OSC_OUT    | VDD   | PE5   | PE0    | BOOT0 | PD7      | PD4    | PD0  | PA8  | PA10 |
| E | NRST           | PC2_C | PE6   | VSS    | VSS   | VSS      | VCAP   | PD1  | PC9  | PC7  |
| F | PC0            | PC1   | PC3_C | VDDLDO | VDD   | VDD33USB | PDR_ON | VCAP | PC8  | PC6  |
| G | VSSA           | PA0   | PA4   | PC4    | PB2   | PE10     | PE14   | PD15 | PD11 | PB15 |
| H | VDDA           | PA1   | PA5   | PC5    | PE7   | PE11     | PE15   | PD14 | PD10 | PB14 |
| J | VSS            | PA2   | PA6   | PB0    | PE8   | PE12     | PB10   | PB13 | PD9  | PD13 |
| K | VDD            | PA3   | PA7   | PB1    | PE9   | PE13     | PB11   | PB12 | PD8  | PD12 |

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1. The above figure shows the package top view.

Figure 7. LQFP144 pinout



1. The above figure shows the package top view.

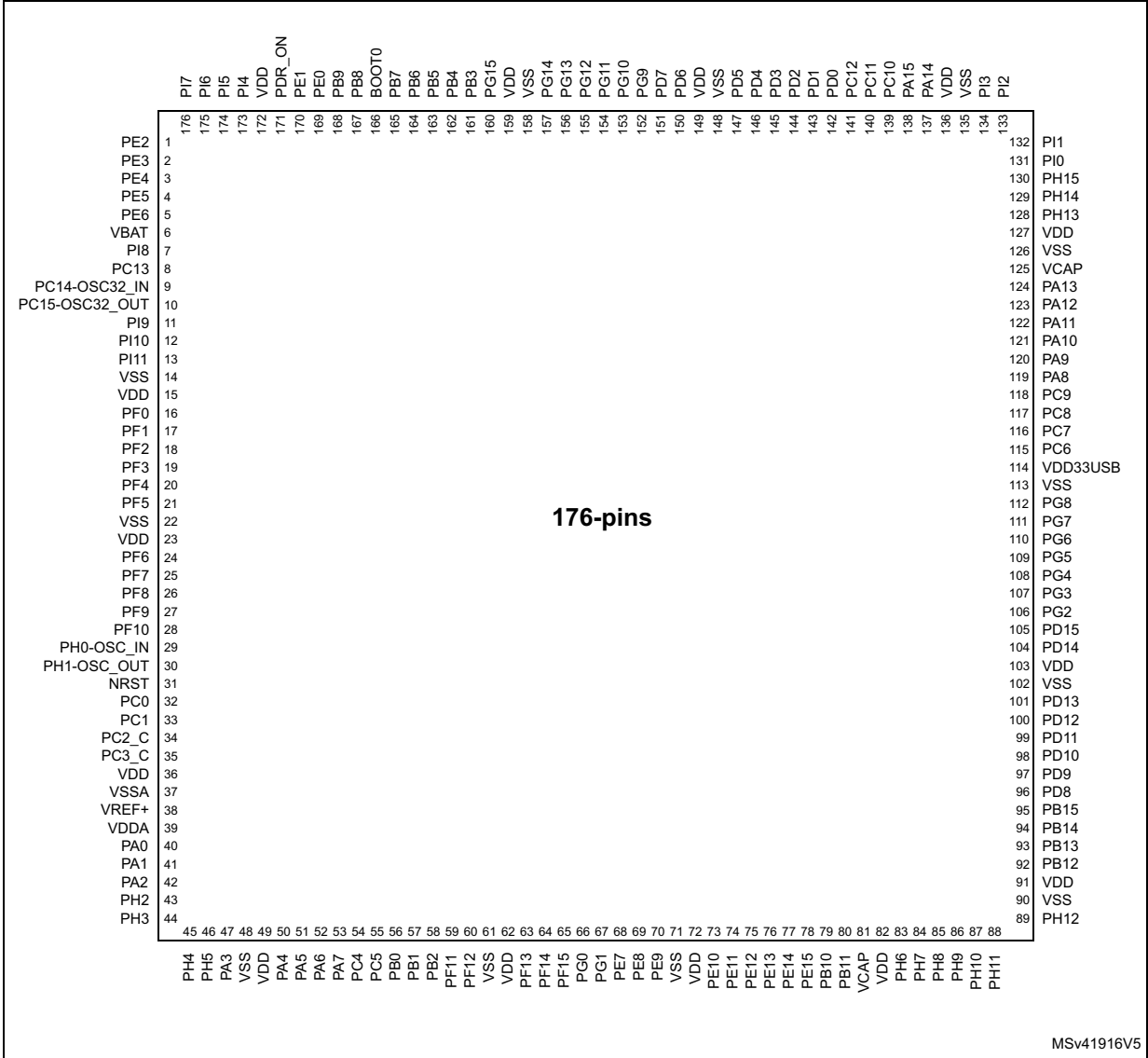
Figure 8. UFBGA169 ballout

|   | 1                      | 2                   | 3    | 4      | 5   | 6     | 7    | 8    | 9    | 10     | 11   | 12            | 13            |
|---|------------------------|---------------------|------|--------|-----|-------|------|------|------|--------|------|---------------|---------------|
| A | PE4                    | PE2                 | VDD  | PI6    | PB6 | PI2   | VDD  | PG10 | PD5  | VDD    | PC12 | PC10          | PI0           |
| B | PC15-<br>OSC32_<br>OUT | PE3                 | VSS  | VDDLDO | PB8 | PB4   | PI3  | PG11 | PD6  | VSS    | PC11 | PA14          | PI1           |
| C | PC14-<br>OSC32_<br>IN  | PE6                 | PE5  | PDR_ON | PB9 | PB5   | PG14 | PG9  | PD4  | PD1    | PA15 | VSS           | VDD           |
| D | VDD                    | VSS                 | PC13 | PE1    | PE0 | PB7   | PG13 | PD7  | PD3  | PD0    | PA13 | VDDLDO        | VCAP          |
| E | PI11                   | PI7                 | VBAT | PF1    | PF3 | BOOT0 | PG15 | PG12 | PD2  | PA10   | PA9  | PA8           | PA12          |
| F | PI13                   | PI12                | PF0  | PF2    | PF5 | PF7   | PB3  | PG4  | PC6  | PC7    | PC9  | PC8           | PA11          |
| G | VDD                    | VSS                 | PF4  | PF6    | PF9 | NRST  | PF13 | PE7  | PG6  | PG7    | PG8  | VDD50_<br>USB | VDD33_<br>USB |
| H | PH0-<br>OSC_<br>IN     | PH1-<br>OSC_<br>OUT | PF10 | PF8    | PJ1 | PA4   | PF14 | PE8  | PG2  | PG3    | PG5  | VSS           | VDD           |
| J | PC0                    | PC1                 | VSSA | PJ0    | PA0 | PA7   | PF15 | PE9  | PE14 | PD11   | PD13 | PD15          | PD14          |
| K | PC3_C                  | PC2_C               | PH4  | PA1    | PA6 | PC4   | PG0  | PE13 | PH10 | PH12   | PD9  | PD10          | PD12          |
| L | VDDA                   | VREF+               | PH5  | PA5    | PB1 | PB2   | PG1  | PE12 | PB10 | PH11   | PB13 | VSS           | VDD           |
| M | VDD                    | VSS                 | PH3  | VSS    | PB0 | PF11  | VSS  | PE10 | PB11 | VDDLDO | VSS  | PD8           | PB15          |
| N | PA2                    | PH2                 | PA3  | VDD    | PC5 | PF12  | VDD  | PE11 | PE15 | VCAP   | VDD  | PB12          | PB14          |

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1. The above figure shows the package top view.

Figure 9. LQFP176 pinout



1. The above figure shows the package top view.

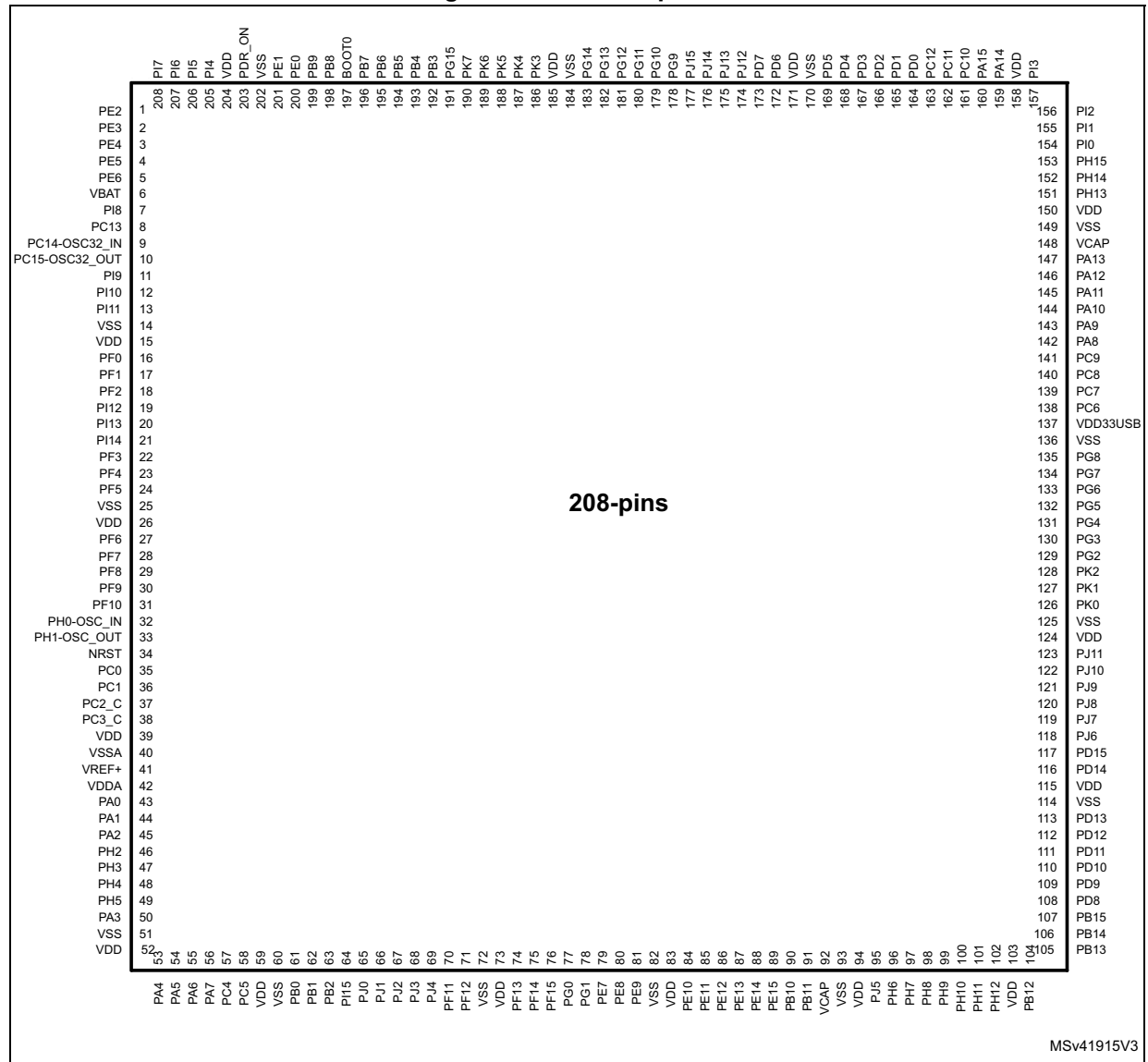
Figure 10. UFBGA176+25 ballout

|   | 1                      | 2   | 3    | 4     | 5     | 6      | 7    | 8    | 9    | 10   | 11   | 12   | 13           | 14   | 15   |
|---|------------------------|-----|------|-------|-------|--------|------|------|------|------|------|------|--------------|------|------|
| A | PE3                    | PE2 | PE1  | PE0   | PB8   | PB5    | PG14 | PG13 | PB4  | PB3  | PD7  | PC12 | PA15         | PA14 | PA13 |
| B | PE4                    | PE5 | PE6  | PB9   | PB7   | PB6    | PG15 | PG12 | PG11 | PG10 | PD6  | PD0  | PC11         | PC10 | PA12 |
| C | VBAT                   | PI7 | PI6  | PI5   | VDD   | PDR_ON | VDD  | VDD  | VDD  | PG9  | PD5  | PD1  | PI3          | PI2  | PA11 |
| D | PC13                   | PI8 | PI9  | PI4   | VSS   | BOOT0  | VSS  | VSS  | VSS  | PD4  | PD3  | PD2  | PH15         | PI1  | PA10 |
| E | PC14-<br>OSC32_<br>IN  | PF0 | PI10 | PI11  |       |        |      |      |      |      |      | PH13 | PH14         | PI0  | PA9  |
| F | PC15-<br>OSC32_<br>OUT | VSS | VDD  | PH2   |       | VSS    | VSS  | VSS  | VSS  | VSS  |      | VSS  | VCAP         | PC9  | PA8  |
| G | PH0-<br>OSC_IN         | VSS | VDD  | PH3   |       | VSS    | VSS  | VSS  | VSS  | VSS  |      | VSS  | VDD          | PC8  | PC7  |
| H | PH1-<br>OSC_<br>OUT    | PF2 | PF1  | PH4   |       | VSS    | VSS  | VSS  | VSS  | VSS  |      | VSS  | VDD<br>33USB | PG8  | PC6  |
| J | NRST                   | PF3 | PF4  | PH5   |       | VSS    | VSS  | VSS  | VSS  | VSS  |      | VDD  | VDD          | PG7  | PG6  |
| K | PF7                    | PF6 | PF5  | VDD   |       | VSS    | VSS  | VSS  | VSS  | VSS  |      | PH12 | PG5          | PG4  | PG3  |
| L | PF10                   | PF9 | PF8  | VSS   |       |        |      |      |      |      |      | PH11 | PH10         | PD15 | PG2  |
| M | VSSA                   | PC0 | PC1  | PC2_C | PC3_C | PB2    | PG1  | VSS  | VSS  | VCAP | PH6  | PH8  | PH9          | PD14 | PD13 |
| N | VREF-                  | PA1 | PA0  | PA4   | PC4   | PF13   | PG0  | VDD  | VDD  | VDD  | PE13 | PH7  | PD12         | PD11 | PD10 |
| P | VREF+                  | PA2 | PA6  | PA5   | PC5   | PF12   | PF15 | PE8  | PE9  | PE11 | PE14 | PB12 | PB13         | PD9  | PD8  |
| R | VDDA                   | PA3 | PA7  | PB1   | PB0   | PF11   | PF14 | PE7  | PE10 | PE12 | PE15 | PB10 | PB11         | PB14 | PB15 |

MSv41912V3

1. The above figure shows the package top view.

Figure 11. LQFP208 pinout



1. The above figure shows the package top view.

Figure 12. TFBGA240+25 ballout

|   | 1                     | 2                    | 3    | 4    | 5    | 6       | 7          | 8         | 9    | 10   | 11   | 12      | 13   | 14   | 15   | 16   | 17           |
|---|-----------------------|----------------------|------|------|------|---------|------------|-----------|------|------|------|---------|------|------|------|------|--------------|
| A | VSS                   | PI6                  | PI5  | PI4  | PB5  | VDD LDO | VCAP       | PK5       | PG10 | PG9  | PD5  | PD4     | PC10 | PA15 | PI1  | PI0  | VSS          |
| B | VBAT                  | VSS                  | PI7  | PE1  | PB6  | VSS     | PB4        | PK4       | PG11 | PJ15 | PD6  | PD3     | PC11 | PA14 | PI2  | PH15 | PH14         |
| C | PC15-<br>OSC32<br>OUT | PC14-<br>OSC32<br>IN | PE2  | PE0  | PB7  | PB3     | PK6        | PK3       | PG12 | VSS  | PD7  | PC12    | VSS  | PI3  | PA13 | VSS  | VDD LDO      |
| D | PE5                   | PE4                  | PE3  | PB9  | PB8  | PG15    | PK7        | PG14      | PG13 | PJ14 | PJ12 | PD2     | PD0  | PA10 | PA9  | PH13 | VCAP         |
| E | NC                    | PI9                  | PC13 | PI8  | PE6  | VDD     | PDR-<br>ON | BOO<br>T0 | VDD  | PJ13 | VDD  | PD1     | PC8  | PC9  | PA8  | PA12 | PA11         |
| F | NC                    | NC                   | PI10 | PI11 | VDD  |         |            |           |      |      |      |         | PC7  | PC6  | PG8  | PG7  | VDD33<br>USB |
| G | PF2                   | NC                   | PF1  | PF0  | VDD  |         | VSS        | VSS       | VSS  | VSS  | VSS  |         | VDD  | PG5  | PG6  | VSS  | VDD50<br>USB |
| H | PI12                  | PI13                 | PI14 | PF3  | VDD  |         | VSS        | VSS       | VSS  | VSS  | VSS  |         | VDD  | PG4  | PG3  | PG2  | PK2          |
| J | PH0-<br>OSC-<br>OUT   | PH0-<br>OSC-<br>IN   | VSS  | PF5  | PF4  |         | VSS        | VSS       | VSS  | VSS  | VSS  |         | VDD  | PK0  | PK1  | VSS  | VSS          |
| K | NRST                  | PF6                  | PF7  | PF8  | VDD  |         | VSS        | VSS       | VSS  | VSS  | VSS  |         | VDD  | PJ11 | VSS  | NC   | NC           |
| L | VDDA                  | PC0                  | PF10 | PF9  | VDD  |         | VSS        | VSS       | VSS  | VSS  | VSS  |         | VDD  | PJ10 | VSS  | NC   | NC           |
| M | VREF+                 | PC1                  | PC2  | PC3  | VDD  |         |            |           |      |      |      |         | VDD  | PJ9  | VSS  | NC   | NC           |
| N | VREF-                 | PH2                  | PA2  | PA1  | PA0  | PJ0     | VDD        | VDD       | PE10 | VDD  | VDD  | VDD     | PJ8  | PJ7  | PJ6  | VSS  | NC           |
| P | VSSA                  | PH3                  | PH4  | PH5  | PI15 | PJ1     | PF13       | PF14      | PE9  | PE11 | PB10 | PB11    | PH10 | PH11 | PD15 | PD14 | VDD          |
| R | PC2_C                 | PC3_C                | PA6  | VSS  | PA7  | PB2     | PF12       | VSS       | PF15 | PE12 | PE15 | PJ5     | PH9  | PH12 | PD11 | PD12 | PD13         |
| T | PA0_C                 | PA1_C                | PA5  | PC4  | PB1  | PJ2     | PF11       | PG0       | PE8  | PE13 | PH6  | VSS     | PH8  | PB12 | PB15 | PD10 | PD9          |
| U | VSS                   | PA3                  | PA4  | PC5  | PB0  | PJ3     | PJ4        | PG1       | PE7  | PE14 | VCAP | VDD LDO | PH7  | PB13 | PB14 | PD8  | VSS          |

MSv41911V2

1. The above figure shows the package top view.

Table 8. Legend/abbreviations used in the pinout table

| Name          |                      | Abbreviation                                                     | Definition                                                                                                                            |
|---------------|----------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Pin name      |                      |                                                                  | Unless otherwise specified in brackets below the pin name, the pin function during and after reset is the same as the actual pin name |
| Pin type      |                      | S                                                                | Supply pin                                                                                                                            |
|               |                      | I                                                                | Input only pin                                                                                                                        |
|               |                      | I/O                                                              | Input / output pin                                                                                                                    |
|               |                      | ANA                                                              | Analog-only Input                                                                                                                     |
| I/O structure |                      | FT                                                               | 5 V tolerant I/O                                                                                                                      |
|               |                      | TT                                                               | 3.3 V tolerant I/O                                                                                                                    |
|               |                      | B                                                                | Dedicated BOOT0 pin                                                                                                                   |
|               |                      | RST                                                              | Bidirectional reset pin with embedded weak pull-up resistor                                                                           |
|               |                      | <b>Option for TT and FT I/Os</b>                                 |                                                                                                                                       |
|               |                      | _f                                                               | I2C FM+ option                                                                                                                        |
|               |                      | _a                                                               | analog option (supplied by V <sub>DDA</sub> )                                                                                         |
|               |                      | _u                                                               | USB option (supplied by V <sub>DD33USB</sub> )                                                                                        |
|               |                      | _h                                                               | High-speed low-voltage I/O                                                                                                            |
| Notes         |                      |                                                                  | Unless otherwise specified by a note, all I/Os are set as floating inputs during and after reset.                                     |
| Pin functions | Alternate functions  | Functions selected through GPIOx_AFR registers                   |                                                                                                                                       |
|               | Additional functions | Functions directly selected/enabled through peripheral registers |                                                                                                                                       |

[Table 9](#) and [Table 10](#) to [Table 20](#) show STM32H743xI/G pin/ball definition and alternate functions, respectively. Refer to [Table 2](#) for the features and peripherals available on STM32H742xI/G devices.



Table 9. Pin/ball definition

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                        | Additional<br>functions        |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                            |                                |
| 1             | A3       | 1       | A2       | A2          | 1       | 1       | C3           | PE2                                      | I/O      | FT_h          | -     | TRACECLK, SAI1_CK1,<br>SPI4_SCK,<br>SAI1_MCLK_A,<br>SAI4_MCLK_A,<br>QUADSPI_BK1_IO2,<br>SAI4_CK1,<br>ETH_MII_TXD3,<br>FMC_A23, EVENTOUT                                    | -                              |
| 2             | B3       | 2       | B2       | A1          | 2       | 2       | D3           | PE3                                      | I/O      | FT_h          | -     | TRACED0, TIM15_BKIN,<br>SAI1_SD_B, SAI4_SD_B,<br>FMC_A19, EVENTOUT                                                                                                         | -                              |
| 3             | C3       | 3       | A1       | B1          | 3       | 3       | D2           | PE4                                      | I/O      | FT_h          | -     | TRACED1, SAI1_D2,<br>DFSDM1_DATIN3,<br>TIM15_CH1N,<br>SPI4_NSS, SAI1_FS_A,<br>SAI4_FS_A, SAI4_D2,<br>FMC_A20, DCMI_D4,<br>LCD_B0, EVENTOUT                                 | -                              |
| 4             | D3       | 4       | C3       | B2          | 4       | 4       | D1           | PE5                                      | I/O      | FT_h          | -     | TRACED2, SAI1_CK2,<br>DFSDM1_CKIN3,<br>TIM15_CH1, SPI4_MISO,<br>SAI1_SCK_A,<br>SAI4_SCK_A, SAI4_CK2,<br>FMC_A21, DCMI_D6,<br>LCD_G0, EVENTOUT                              | -                              |
| 5             | E3       | 5       | C2       | B3          | 5       | 5       | E5           | PE6                                      | I/O      | FT_h          | -     | TRACED3, TIM1_BKIN2,<br>SAI1_D1, TIM15_CH2,<br>SPI4_MOSI, SAI1_SD_A,<br>SAI4_SD_A, SAI4_D1,<br>SAI2_MCLK_B,<br>TIM1_BKIN2_COMP12,<br>FMC_A22, DCMI_D7,<br>LCD_G1, EVENTOUT | -                              |
| -             | -        | -       | M4       | H10         | -       | -       | A1           | VSS                                      | S        | -             | -     | -                                                                                                                                                                          | -                              |
| -             | -        | -       | A3       | -           | -       | -       | -            | VDD                                      | S        | -             | -     | -                                                                                                                                                                          | -                              |
| 6             | B2       | 6       | E3       | C1          | 6       | 6       | B1           | VBAT                                     | S        | -             | -     | -                                                                                                                                                                          | -                              |
| -             | -        | -       | -        | J6          | -       | -       | B2           | VSS                                      | S        | -             | -     | -                                                                                                                                                                          | -                              |
| -             | -        | -       | -        | D2          | 7       | 7       | E4           | PI8                                      | I/O      | FT            | -     | EVENTOUT                                                                                                                                                                   | RTC_TAMP2/<br>WKUP3            |
| 7             | A2       | 7       | D3       | D1          | 8       | 8       | E3           | PC13                                     | I/O      | FT            | -     | EVENTOUT                                                                                                                                                                   | RTC_TAMP1/<br>RTC_TS/<br>WKUP2 |
| -             | -        | -       | -        | J7          | -       | -       | B6           | VSS                                      | S        | -             | -     | -                                                                                                                                                                          | -                              |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |                   | Pin name<br>(function<br>after<br>reset)                 | Pin type | I/O structure | Notes | Alternate functions                                                    | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|-------------------|----------------------------------------------------------|----------|---------------|-------|------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25      |                                                          |          |               |       |                                                                        |                         |
| 8             | A1       | 8       | C1       | E1          | 9       | 9       | C2                | PC14-<br>OSC32_<br>IN<br>(OSC32_<br>IN) <sup>(1)</sup>   | I/O      | FT            | -     | EVENTOUT                                                               | OSC32_IN                |
| 9             | B1       | 9       | B1       | F1          | 10      | 10      | C1                | PC15-<br>OSC32_<br>OUT<br>(OSC32_<br>OUT) <sup>(1)</sup> | I/O      | FT            | -     | EVENTOUT                                                               | OSC32_<br>OUT           |
| -             | -        | -       | -        | D3          | 11      | 11      | E2                | PI9                                                      | I/O      | FT_h          | -     | UART4_RX,<br>FDCAN1_RX, FMC_D30,<br>LCD_VSYNC,<br>EVENTOUT             | -                       |
| -             | -        | -       | -        | E3          | 12      | 12      | F3                | PI10                                                     | I/O      | FT_h          | -     | FDCAN1_RXFD_MODE,<br>ETH_MII_RX_ER,<br>FMC_D31, LCD_HSYNC,<br>EVENTOUT | -                       |
| -             | -        | -       | E1       | E4          | 13      | 13      | F4                | PI11                                                     | I/O      | FT            | -     | LCD_G6,<br>OTG_HS_ULPI_DIR,<br>EVENTOUT                                | WKUP4                   |
| -             | C2       | -       | D2       | F2          | 14      | 14      | A17               | VSS                                                      | S        | -             | -     | -                                                                      | -                       |
| -             | D2       | -       | D1       | F3          | 15      | 15      | E6                | VDD                                                      | S        | -             | -     | -                                                                      | -                       |
| -             | -        | -       | -        | -           | -       | -       | E1 <sup>(2)</sup> | NC                                                       | -        | -             | -     | -                                                                      | -                       |
| -             | -        | -       | -        | -           | -       | -       | F1 <sup>(3)</sup> | NC                                                       | -        | -             | -     | -                                                                      | -                       |
| -             | -        | -       | -        | -           | -       | -       | G2 <sup>(4)</sup> | NC                                                       | -        | -             | -     | -                                                                      | -                       |
| -             | -        | 10      | F3       | E2          | 16      | 16      | G4                | PF0                                                      | I/O      | FT_f          | -     | I2C2_SDA, FMC_A0,<br>EVENTOUT                                          | -                       |
| -             | -        | 11      | E4       | H3          | 17      | 17      | G3                | PF1                                                      | I/O      | FT_f          | -     | I2C2_SCL, FMC_A1,<br>EVENTOUT                                          | -                       |
| -             | -        | 12      | F4       | H2          | 18      | 18      | G1                | PF2                                                      | I/O      | FT            | -     | I2C2_SMBA, FMC_A2,<br>EVENTOUT                                         | -                       |
| -             | -        | -       | F2       | -           | -       | 19      | H1                | PI12                                                     | I/O      | FT            | -     | LCD_HSYNC,<br>EVENTOUT                                                 | -                       |
| -             | -        | -       | F1       | -           | -       | 20      | H2                | PI13                                                     | I/O      | FT            | -     | LCD_VSYNC,<br>EVENTOUT                                                 | -                       |
| -             | -        | -       | -        | -           | -       | 21      | H3                | PI14                                                     | I/O      | FT_h          | -     | LCD_CLK, EVENTOUT                                                      | -                       |
| -             | -        | 13      | E5       | J2          | 19      | 22      | H4                | PF3                                                      | I/O      | FT_<br>ha     | -     | FMC_A3, EVENTOUT                                                       | ADC3_INP5               |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                           | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                               |                         |
| -             | -        | 14      | G3       | J3          | 20      | 23      | J5           | PF4                                      | I/O      | FT_<br>ha     | -     | FMC_A4, EVENTOUT                                                                                                              | ADC3_INN5,<br>ADC3_INP9 |
| -             | -        | 15      | F5       | K3          | 21      | 24      | J4           | PF5                                      | I/O      | FT_<br>ha     | -     | FMC_A5, EVENTOUT                                                                                                              | ADC3_INP4               |
| 10            | -        | 16      | B10      | G2          | 22      | 25      | C10          | VSS                                      | S        | -             | -     | -                                                                                                                             | -                       |
| 11            | -        | 17      | G1       | G3          | 23      | 26      | E9           | VDD                                      | S        | -             | -     | -                                                                                                                             | -                       |
| -             | -        | 18      | G4       | K2          | 24      | 27      | K2           | PF6                                      | I/O      | FT_<br>ha     | -     | TIM16_CH1, SPI5_NSS,<br>SAI1_SD_B, UART7_RX,<br>SAI4_SD_B,<br>QUADSPI_BK1_IO3,<br>EVENTOUT                                    | ADC3_INN4,<br>ADC3_INP8 |
| -             | -        | 19      | F6       | K1          | 25      | 28      | K3           | PF7                                      | I/O      | FT_<br>ha     | -     | TIM17_CH1, SPI5_SCK,<br>SAI1_MCLK_B,<br>UART7_TX,<br>SAI4_MCLK_B,<br>QUADSPI_BK1_IO2,<br>EVENTOUT                             | ADC3_INP3               |
| -             | -        | 20      | H4       | L3          | 26      | 29      | K4           | PF8                                      | I/O      | FT_<br>ha     | -     | TIM16_CH1N,<br>SPI5_MISO,<br>SAI1_SCK_B,<br>UART7_RTS/UART7_DE<br>, SAI4_SCK_B,<br>TIM13_CH1,<br>QUADSPI_BK1_IO0,<br>EVENTOUT | ADC3_INN3,<br>ADC3_INP7 |
| -             | -        | 21      | G5       | L2          | 27      | 30      | L4           | PF9                                      | I/O      | FT_<br>ha     | -     | TIM17_CH1N,<br>SPI5_MOSI, SAI1_FS_B,<br>UART7_CTS,<br>SAI4_FS_B, TIM14_CH1,<br>QUADSPI_BK1_IO1,<br>EVENTOUT                   | ADC3_INP2               |
| -             | -        | 22      | H3       | L1          | 28      | 31      | L3           | PF10                                     | I/O      | FT_<br>ha     | -     | TIM16_BKIN, SAI1_D3,<br>QUADSPI_CLK,<br>SAI4_D3, DCMI_D11,<br>LCD_DE, EVENTOUT                                                | ADC3_INN2,<br>ADC3_INP6 |
| 12            | C1       | 23      | H1       | G1          | 29      | 32      | J2           | PH0-<br>OSC_IN<br>(PH0)                  | I/O      | FT            | -     | EVENTOUT                                                                                                                      | OSC_IN                  |
| 13            | D1       | 24      | H2       | H1          | 30      | 33      | J1           | PH1-<br>OSC_OUT<br>(PH1)                 | I/O      | FT            | -     | EVENTOUT                                                                                                                      | OSC_OUT                 |
| 14            | E1       | 25      | G6       | J1          | 31      | 34      | K1           | NRST                                     | I/O      | RST           | -     | -                                                                                                                             | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name     |                   |                   |                   |                   |                   |                   |                   | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                               | Additional<br>functions                                       |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| LQFP100           | TFBGA100          | LQFP144           | UFBGA169          | UFBGA176+25       | LQFP176           | LQFP208           | TFBGA240 +25      |                                          |          |               |       |                                                                                                                                                                   |                                                               |
| 15                | F1                | 26                | J1                | M2                | 32                | 35                | L2                | PC0                                      | I/O      | FT_a          | -     | DFSDM1_CKIN0,<br>DFSDM1_DATIN4,<br>SAI2_FS_B,<br>OTG_HS_ULPI_STP,<br>FMC_SDNWE, LCD_R5,<br>EVENTOUT                                                               | ADC123_<br>INP10                                              |
| 16                | F2                | 27                | J2                | M3                | 33                | 36                | M2                | PC1                                      | I/O      | FT_ha         | -     | TRACED0, SAI1_D1,<br>DFSDM1_DATIN0,<br>DFSDM1_CKIN4,<br>SPI2_MOSI/I2S2_SDO,<br>SAI1_SD_A, SAI4_SD_A,<br>SDMMC2_CK, SAI4_D1,<br>ETH_MDC,<br>MDIOS_MDC,<br>EVENTOUT | ADC123_<br>INN10,<br>ADC123_<br>INP11,<br>RTC_TAMP3/W<br>KUP5 |
| -                 | -                 | -                 | -                 | -                 | -                 | -                 | M3 <sup>(5)</sup> | PC2                                      | I/O      | FT_a          | -     | CDSLEEP,<br>DFSDM1_CKIN1,<br>SPI2_MISO/I2S2_SDI,<br>DFSDM1_CKOUT,<br>OTG_HS_ULPI_DIR,<br>ETH_MII_TXD2,<br>FMC_SDNE0,<br>EVENTOUT                                  | ADC123_<br>INN11,<br>ADC123_<br>INP12                         |
| 17 <sup>(6)</sup> | E2 <sup>(6)</sup> | 28 <sup>(6)</sup> | K2 <sup>(6)</sup> | M4 <sup>(6)</sup> | 34 <sup>(6)</sup> | 37 <sup>(6)</sup> | R1 <sup>(5)</sup> | PC2_C                                    | ANA      | TT_a          | -     | CSLEEP,<br>DFSDM1_DATIN1,<br>SPI2_MOSI/I2S2_SDO,<br>OTG_HS_ULPI_NXT,<br>ETH_MII_TX_CLK,<br>FMC_SDCKE0,<br>EVENTOUT                                                | ADC3_INN1,<br>ADC3_INP0                                       |
| -                 | -                 | -                 | -                 | -                 | -                 | -                 | M4 <sup>(5)</sup> | PC3                                      | I/O      | FT_a          | -     | CSLEEP,<br>DFSDM1_DATIN1,<br>SPI2_MOSI/I2S2_SDO,<br>OTG_HS_ULPI_NXT,<br>ETH_MII_TX_CLK,<br>FMC_SDCKE0,<br>EVENTOUT                                                | ADC12_INN12,<br>ADC12_INP13                                   |
| 18 <sup>(6)</sup> | F3 <sup>(6)</sup> | 29 <sup>(6)</sup> | K1 <sup>(6)</sup> | M5 <sup>(6)</sup> | 35 <sup>(6)</sup> | 38 <sup>(6)</sup> | R2 <sup>(5)</sup> | PC3_C                                    | ANA      | TT_a          | -     | CSLEEP,<br>DFSDM1_DATIN1,<br>SPI2_MOSI/I2S2_SDO,<br>OTG_HS_ULPI_NXT,<br>ETH_MII_TX_CLK,<br>FMC_SDCKE0,<br>EVENTOUT                                                | ADC3_INP1                                                     |
| -                 | F5                | 30                | -                 | G3                | 36                | 39                | E11               | VDD                                      | S        | -             | -     | -                                                                                                                                                                 | -                                                             |
| -                 | E6                | -                 | B3                | J10               | -                 | -                 | C13               | VSS                                      | S        | -             | -     | -                                                                                                                                                                 | -                                                             |
| 19                | G1                | 31                | J3                | M1                | 37                | 40                | P1                | VSSA                                     | S        | -             | -     | -                                                                                                                                                                 | -                                                             |
| -                 | -                 | -                 | -                 | N1                | -                 | -                 | N1                | VREF-                                    | S        | -             | -     | -                                                                                                                                                                 | -                                                             |
| 20                | - <sup>(7)</sup>  | 32                | L2                | P1                | 38                | 41                | M1                | VREF+                                    | S        | -             | -     | -                                                                                                                                                                 | -                                                             |
| 21                | H1                | 33                | L1                | R1                | 39                | 42                | L1                | VDDA                                     | S        | -             | -     | -                                                                                                                                                                 | -                                                             |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |                   | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                | Additional<br>functions   |
|---------------|----------|---------|----------|-------------|---------|---------|-------------------|------------------------------------------|----------|---------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25      |                                          |          |               |       |                                                                                                                                                                                    |                           |
| 22            | G2       | 34      | J5       | N3          | 40      | 43      | N5 <sup>(5)</sup> | PA0                                      | I/O      | FT_a          | -     | TIM2_CH1/TIM2_ETR,<br>TIM5_CH1, TIM8_ETR,<br>TIM15_BKIN,<br>USART2_CTS/USART2_2_NSS, UART4_TX,<br>SDMMC2_CMD,<br>SAI2_SD_B,<br>ETH_MII_CRS,<br>EVENTOUT                            | ADC1_INP16,<br>WKUP0      |
| -             | -        | -       | -        | -           | -       | -       | T1 <sup>(5)</sup> | PA0_C                                    | ANA      | TT_a          | -     | USART2_CTS/USART2_2_NSS, UART4_TX,<br>SDMMC2_CMD,<br>SAI2_SD_B,<br>ETH_MII_CRS,<br>EVENTOUT                                                                                        | ADC12_INN1,<br>ADC12_INP0 |
| 23            | H2       | 35      | K4       | N2          | 41      | 44      | N4 <sup>(5)</sup> | PA1                                      | I/O      | FT_ha         | -     | TIM2_CH2, TIM5_CH2,<br>LPTIM3_OUT,<br>TIM15_CH1N,<br>USART2_RTS/USART2_2_DE, UART4_RX,<br>QUADSPI_BK1_IO3,<br>SAI2_MCLK_B,<br>ETH_MII_RX_CLK/ETH_RMII_REF_CLK,<br>LCD_R2, EVENTOUT | ADC1_INN16,<br>ADC1_INP17 |
| -             | -        | -       | -        | -           | -       | -       | T2 <sup>(5)</sup> | PA1_C                                    | ANA      | TT_a          | -     | USART2_RTS/USART2_2_DE, UART4_RX,<br>QUADSPI_BK1_IO3,<br>SAI2_MCLK_B,<br>ETH_MII_RX_CLK/ETH_RMII_REF_CLK,<br>LCD_R2, EVENTOUT                                                      | ADC12_INP1                |
| 24            | J2       | 36      | N1       | P2          | 42      | 45      | N3                | PA2                                      | I/O      | FT_a          | -     | TIM2_CH3, TIM5_CH3,<br>LPTIM4_OUT,<br>TIM15_CH1,<br>USART2_TX,<br>SAI2_SCK_B,<br>ETH_MDIO,<br>MDIOS_MDIO, LCD_R1,<br>EVENTOUT                                                      | ADC12_INP14,<br>WKUP1     |
| -             | -        | -       | N2       | F4          | 43      | 46      | N2                | PH2                                      | I/O      | FT_ha         | -     | LPTIM1_IN2,<br>QUADSPI_BK2_IO0,<br>SAI2_SCK_B,<br>ETH_MII_CRS,<br>FMC_SDCKE0, LCD_R0,<br>EVENTOUT                                                                                  | ADC3_INP13                |
| -             | K1       | -       | M1       | -           | -       | -       | F5                | VDD                                      | S        | -             | -     | -                                                                                                                                                                                  | -                         |
| -             | J1       | -       | M7       | J8          | -       | -       | C16               | VSS                                      | S        | -             | -     | -                                                                                                                                                                                  | -                         |
| -             | -        | -       | M3       | G4          | 44      | 47      | P2                | PH3                                      | I/O      | FT_ha         | -     | QUADSPI_BK2_IO1,<br>SAI2_MCLK_B,<br>ETH_MII_COL,<br>FMC_SDNE0, LCD_R1,<br>EVENTOUT                                                                                                 | ADC3_INN13,<br>ADC3_INP14 |
| -             | -        | -       | K3       | H4          | 45      | 48      | P3                | PH4                                      | I/O      | FT_fa         | -     | I2C2_SCL, LCD_G5,<br>OTG_HS_ULPI_NXT,<br>LCD_G4, EVENTOUT                                                                                                                          | ADC3_INN14,<br>ADC3_INP15 |
| -             | -        | -       | L3       | J4          | 46      | 49      | P4                | PH5                                      | I/O      | FT_fa         | -     | I2C2_SDA, SPI5_NSS,<br>FMC_SDNWE,<br>EVENTOUT                                                                                                                                      | ADC3_INN15,<br>ADC3_INP16 |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |                   | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                            | Additional<br>functions                   |
|---------------|----------|---------|----------|-------------|---------|---------|-------------------|------------------------------------------|----------|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25      |                                          |          |               |       |                                                                                                                                                                                |                                           |
| 25            | K2       | 37      | N3       | R2          | 47      | 50      | U2                | PA3                                      | I/O      | FT_<br>ha     | -     | TIM2_CH4, TIM5_CH4,<br>LPTIM5_OUT,<br>TIM15_CH2,<br>USART2_RX, LCD_B2,<br>OTG_HS_ULPI_D0,<br>ETH_MII_COL, LCD_B5,<br>EVENTOUT                                                  | ADC12_INP15                               |
| 26            | -        | 38      | G2       | K6          | -       | 51      | F2 <sup>(4)</sup> | VSS                                      | S        | -             | -     | -                                                                                                                                                                              | -                                         |
| -             | -        | -       | -        | L4          | 48      | -       | -                 | VSS                                      | S        | -             | -     | -                                                                                                                                                                              | -                                         |
| 27            | -        | 39      | -        | K4          | 49      | 52      | G5                | VDD                                      | S        | -             | -     | -                                                                                                                                                                              | -                                         |
| 28            | G3       | 40      | H6       | N4          | 50      | 53      | U3                | PA4                                      | I/O      | TT_a          | -     | D1PWREN, TIM5_ETR,<br>SPI1_NSS/I2S1_WS,<br>SPI3_NSS/I2S3_WS,<br>USART2_CK, SPI6_NSS,<br>OTG_HS_SOF,<br>DCMI_HSYNC,<br>LCD_VSYNC,<br>EVENTOUT                                   | ADC12_INP18,<br>DAC1_OUT1                 |
| 29            | H3       | 41      | L4       | P4          | 51      | 54      | T3                | PA5                                      | I/O      | TT_<br>ha     | -     | D2PWREN,<br>TIM2_CH1/TIM2_ETR,<br>TIM8_CH1N,<br>SPI1_SCK/I2S1_CK,<br>SPI6_SCK,<br>OTG_HS_ULPI_CK,<br>LCD_R4, EVENTOUT                                                          | ADC12_INN18,<br>ADC12_INP19,<br>DAC1_OUT2 |
| 30            | J3       | 42      | K5       | P3          | 52      | 55      | R3                | PA6                                      | I/O      | FT_a          | -     | TIM1_BKIN, TIM3_CH1,<br>TIM8_BKIN,<br>SPI1_MISO/I2S1_SDI,<br>SPI6_MISO, TIM13_CH1,<br>TIM8_BKIN_COMP12,<br>MDIOS_MDC,<br>TIM1_BKIN_COMP12,<br>DCMI_PIXCLK, LCD_G2,<br>EVENTOUT | ADC12_INP3                                |
| 31            | K3       | 43      | J6       | R3          | 53      | 56      | R5                | PA7                                      | I/O      | TT_a          | -     | TIM1_CH1N, TIM3_CH2,<br>TIM8_CH1N,<br>SPI1_MOSI/I2S1_SDO,<br>SPI6_MOSI, TIM14_CH1,<br>ETH_MII_RX_DV/ETH_R<br>MII_CRS_DV,<br>FMC_SDNWE,<br>EVENTOUT                             | ADC12_INN3,<br>ADC12_INP7,<br>OPAMP1_VINM |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                 | Additional<br>functions                                 |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                     |                                                         |
| 32            | G4       | 44      | K6       | N5          | 54      | 57      | T4           | PC4                                      | I/O      | TT_a          | -     | DFSDM1_CKIN2,<br>I2S1_MCK,<br>SPDIFRX1_IN3,<br>ETH_MII_RXD0/ETH_R<br>MII_RXD0, FMC_SDNE0,<br>EVENTOUT                               | ADC12_INP4,<br>OPAMP1_<br>VOUT,<br>COMP1_INM            |
| 33            | H4       | 45      | N5       | P5          | 55      | 58      | U4           | PC5                                      | I/O      | TT_a          | -     | SAI1_D3,<br>DFSDM1_DATIN2,<br>SPDIFRX1_IN4,<br>SAI4_D3,<br>ETH_MII_RXD1/ETH_R<br>MII_RXD1,<br>FMC_SDCKE0,<br>COMP1_OUT,<br>EVENTOUT | ADC12_INN4,<br>ADC12_INP8,<br>OPAMP1_<br>VINM           |
| -             | -        | -       | N4       | -           | -       | 59      | G13          | VDD                                      | S        | -             | -     | -                                                                                                                                   | -                                                       |
| -             | -        | -       | H12      | J9          | -       | 60      | R4           | VSS                                      | S        | -             | -     | -                                                                                                                                   | -                                                       |
| 34            | J4       | 46      | M5       | R5          | 56      | 61      | U5           | PB0                                      | I/O      | FT_a          | -     | TIM1_CH2N, TIM3_CH3,<br>TIM8_CH2N,<br>DFSDM1_CKOUT,<br>UART4_CTS, LCD_R3,<br>OTG_HS_ULPI_D1,<br>ETH_MII_RXD2,<br>LCD_G1, EVENTOUT   | ADC12_INN5,<br>ADC12_INP9,<br>OPAMP1_VINP,<br>COMP1_INP |
| 35            | K4       | 47      | L5       | R4          | 57      | 62      | T5           | PB1                                      | I/O      | TT_u          | -     | TIM1_CH3N, TIM3_CH4,<br>TIM8_CH3N,<br>DFSDM1_DATIN1,<br>LCD_R6,<br>OTG_HS_ULPI_D2,<br>ETH_MII_RXD3,<br>LCD_G0, EVENTOUT             | ADC12_INP5,<br>COMP1_INM                                |
| 36            | G5       | 48      | L6       | M6          | 58      | 63      | R6           | PB2                                      | I/O      | FT_<br>ha     | -     | RTC_OUT, SAI1_D1,<br>DFSDM1_CKIN1,<br>SAI1_SD_A,<br>SPI3_MOSI/I2S3_SDO,<br>SAI4_SD_A,<br>QUADSPI_CLK,<br>SAI4_D1, EVENTOUT          | COMP1_INP                                               |
| -             | -        | -       | -        | -           | -       | 64      | P5           | PI15                                     | I/O      | FT            | -     | LCD_G2, LCD_R0,<br>EVENTOUT                                                                                                         | -                                                       |
| -             | -        | -       | J4       | -           | -       | 65      | N6           | PJ0                                      | I/O      | FT            | -     | LCD_R7, LCD_R1,<br>EVENTOUT                                                                                                         | -                                                       |
| -             | -        | -       | H5       | -           | -       | 66      | P6           | PJ1                                      | I/O      | FT            | -     | LCD_R2, EVENTOUT                                                                                                                    | -                                                       |
| -             | -        | -       | -        | -           | -       | 67      | T6           | PJ2                                      | I/O      | FT            | -     | LCD_R3, EVENTOUT                                                                                                                    | -                                                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                       | Additional<br>functions       |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-----------------------------------------------------------------------------------------------------------|-------------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                           |                               |
| -             | -        | -       | -        | -           | -       | 68      | U6           | PJ3                                      | I/O      | FT            | -     | LCD_R4, EVENTOUT                                                                                          | -                             |
| -             | -        | -       | -        | -           | -       | 69      | U7           | PJ4                                      | I/O      | FT            | -     | LCD_R5, EVENTOUT                                                                                          | -                             |
| -             | -        | 49      | M6       | R6          | 59      | 70      | T7           | PF11                                     | I/O      | FT_a          | -     | SPI5_MOSI, SAI2_SD_B,<br>FMC_SDNRAS,<br>DCMI_D12, EVENTOUT                                                | ADC1_INP2                     |
| -             | -        | 50      | N6       | P6          | 60      | 71      | R7           | PF12                                     | I/O      | FT_ha         | -     | FMC_A6, EVENTOUT                                                                                          | ADC1_INN2,<br>ADC1_INP6       |
| -             | -        | 51      | M11      | M8          | 61      | 72      | J3           | VSS                                      | S        | -             | -     | -                                                                                                         | -                             |
| -             | -        | 52      | -        | N8          | 62      | 73      | H5           | VDD                                      | S        | -             | -     | -                                                                                                         | -                             |
| -             | -        | 53      | G7       | N6          | 63      | 74      | P7           | PF13                                     | I/O      | FT_ha         | -     | DFSDM1_DATIN6,<br>I2C4_SMBA, FMC_A7,<br>EVENTOUT                                                          | ADC2_INP2                     |
| -             | -        | 54      | H7       | R7          | 64      | 75      | P8           | PF14                                     | I/O      | FT_fha        | -     | DFSDM1_CKIN6,<br>I2C4_SCL, FMC_A8,<br>EVENTOUT                                                            | ADC2_INN2,<br>ADC2_INP6       |
| -             | -        | 55      | J7       | P7          | 65      | 76      | R9           | PF15                                     | I/O      | FT_fh         | -     | I2C4_SDA, FMC_A9,<br>EVENTOUT                                                                             | -                             |
| -             | -        | 56      | K7       | N7          | 66      | 77      | T8           | PG0                                      | I/O      | FT_h          | -     | FMC_A10, EVENTOUT                                                                                         | -                             |
| -             | -        | -       | M2       | F6          | -       | -       | J16          | VSS                                      | S        | -             | -     | -                                                                                                         | -                             |
| -             | -        | -       | A10      | -           | -       | -       | H13          | VDD                                      | S        | -             | -     | -                                                                                                         | -                             |
| -             | -        | 57      | L7       | M7          | 67      | 78      | U8           | PG1                                      | I/O      | TT_h          | -     | FMC_A11, EVENTOUT                                                                                         | OPAMP2_<br>VINM               |
| 37            | H5       | 58      | G8       | R8          | 68      | 79      | U9           | PE7                                      | I/O      | TT_ha         | -     | TIM1_ETR,<br>DFSDM1_DATIN2,<br>UART7_RX,<br>QUADSPI_BK2_IO0,<br>FMC_D4/FMC_DA4,<br>EVENTOUT               | OPAMP2_<br>VOUT,<br>COMP2_INM |
| 38            | J5       | 59      | H8       | P8          | 69      | 80      | T9           | PE8                                      | I/O      | TT_ha         | -     | TIM1_CH1N,<br>DFSDM1_CKIN2,<br>UART7_TX,<br>QUADSPI_BK2_IO1,<br>FMC_D5/FMC_DA5,<br>COMP2_OUT,<br>EVENTOUT | OPAMP2_<br>VINM               |
| 39            | K5       | 60      | J8       | P9          | 70      | 81      | P9           | PE9                                      | I/O      | TT_ha         | -     | TIM1_CH1,<br>DFSDM1_CKOUT,<br>UART7_RTS/UART7_DE<br>, QUADSPI_BK2_IO2,<br>FMC_D6/FMC_DA6,<br>EVENTOUT     | OPAMP2_VINP,<br>COMP2_INP     |



Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                           | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                               |                         |
| -             | -        | 61      | C12      | M9          | 71      | 82      | J17          | VSS                                      | S        | -             | -     | -                                                                                                             | -                       |
| -             | -        | 62      | C13      | N9          | 72      | 83      | J13          | VDD                                      | S        | -             | -     | -                                                                                                             | -                       |
| 40            | G6       | 63      | M8       | R9          | 73      | 84      | N9           | PE10                                     | I/O      | FT_ha         | -     | TIM1_CH2N,<br>DFSDM1_DATIN4,<br>UART7_CTS,<br>QUADSPI_BK2_IO3,<br>FMC_D7/FMC_DA7,<br>EVENTOUT                 | COMP2_INM               |
| 41            | H6       | 64      | N8       | P10         | 74      | 85      | P10          | PE11                                     | I/O      | FT_ha         | -     | TIM1_CH2,<br>DFSDM1_CKIN4,<br>SPI4_NSS, SAI2_SD_B,<br>FMC_D8/FMC_DA8,<br>LCD_G3, EVENTOUT                     | COMP2_INP               |
| 42            | J6       | 65      | L8       | R10         | 75      | 86      | R10          | PE12                                     | I/O      | FT_h          | -     | TIM1_CH3N,<br>DFSDM1_DATIN5,<br>SPI4_SCK,<br>SAI2_SCK_B,<br>FMC_D9/FMC_DA9,<br>COMP1_OUT, LCD_B4,<br>EVENTOUT | -                       |
| 43            | K6       | 66      | K8       | N11         | 76      | 87      | T10          | PE13                                     | I/O      | FT_h          | -     | TIM1_CH3,<br>DFSDM1_CKIN5,<br>SPI4_MISO, SAI2_FS_B,<br>FMC_D10/FMC_DA10,<br>COMP2_OUT, LCD_DE,<br>EVENTOUT    | -                       |
| -             | -        | -       | L12      | F7          | -       | -       | T12          | VSS                                      | S        | -             | -     | -                                                                                                             | -                       |
| -             | -        | -       | H13      | -           | -       | -       | K13          | VDD                                      | S        | -             | -     | -                                                                                                             | -                       |
| 44            | G7       | 67      | J9       | P11         | 77      | 88      | U10          | PE14                                     | I/O      | FT_h          | -     | TIM1_CH4, SPI4_MOSI,<br>SAI2_MCLK_B,<br>FMC_D11/FMC_DA11,<br>LCD_CLK, EVENTOUT                                | -                       |
| 45            | H7       | 68      | N9       | R11         | 78      | 89      | R11          | PE15                                     | I/O      | FT_h          | -     | TIM1_BKIN,<br>FMC_D12/FMC_DA12,<br>TIM1_BKIN_COMP12/<br>COMP_TIM1_BKIN,<br>LCD_R7, EVENTOUT                   | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                  | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                      |                         |
| 46            | J7       | 69      | L9       | R12         | 79      | 90      | P11          | PB10                                     | I/O      | FT_f          | -     | TIM2_CH3,<br>HRTIM_SCOUT,<br>LPTIM2_IN1, I2C2_SCL,<br>SPI2_SCK/I2S2_CK,<br>DFSDM1_DATIN7,<br>USART3_TX,<br>QUADSPI_BK1_NCS,<br>OTG_HS_ULPI_D3,<br>ETH_MII_RX_ER,<br>LCD_G4, EVENTOUT | -                       |
| 47            | K7       | 70      | M9       | R13         | 80      | 91      | P12          | PB11                                     | I/O      | FT_f          | -     | TIM2_CH4,<br>HRTIM_SCIN,<br>LPTIM2_ETR,<br>I2C2_SDA,<br>DFSDM1_CKIN7,<br>USART3_RX,<br>OTG_HS_ULPI_D4,<br>ETH_MII_TX_EN/ETH_R<br>MII_TX_EN, LCD_G5,<br>EVENTOUT                      | -                       |
| 48            | F8       | 71      | N10      | M10         | 81      | 92      | U11          | VCAP                                     | S        | -             | -     | -                                                                                                                                                                                    | -                       |
| 49            | E4       | -       | -        | K7          | -       | 93      | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                    | -                       |
| -             | -        | -       | M10      | -           | -       | -       | U12          | VDDLDO<br>(8)                            | S        | -             | -     | -                                                                                                                                                                                    | -                       |
| 50            | -        | 72      | M1       | N10         | 82      | 94      | L13          | VDD                                      | S        | -             | -     | -                                                                                                                                                                                    | -                       |
| -             | -        | -       | -        | -           | -       | 95      | R12          | PJ5                                      | I/O      | FT            | -     | LCD_R6, EVENTOUT                                                                                                                                                                     | -                       |
| -             | -        | -       | -        | M11         | 83      | 96      | T11          | PH6                                      | I/O      | FT            | -     | TIM12_CH1,<br>I2C2_SMBA, SPI5_SCK,<br>ETH_MII_RXD2,<br>FMC_SDNE1, DCMI_D8,<br>EVENTOUT                                                                                               | -                       |
| -             | -        | -       | -        | N12         | 84      | 97      | U13          | PH7                                      | I/O      | FT_fa         | -     | I2C3_SCL, SPI5_MISO,<br>ETH_MII_RXD3,<br>FMC_SDCKE1,<br>DCMI_D9, EVENTOUT                                                                                                            | -                       |
| -             | -        | -       | -        | M12         | 85      | 98      | T13          | PH8                                      | I/O      | FT_fa         | -     | TIM5_ETR, I2C3_SDA,<br>FMC_D16,<br>DCMI_HSYNC, LCD_R2,<br>EVENTOUT                                                                                                                   | -                       |
| -             | -        | -       | -        | F8          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                    | -                       |
| -             | -        | -       | L13      | -           | -       | -       | M13          | VDD                                      | S        | -             | -     | -                                                                                                                                                                                    | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                                  | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                                      |                         |
| -             | -        | -       | -        | M13         | 86      | 99      | R13          | PH9                                      | I/O      | FT_h          | -     | TIM12_CH2,<br>I2C3_SMBA, FMC_D17,<br>DCMI_D0, LCD_R3,<br>EVENTOUT                                                                                                                                    | -                       |
| -             | -        | -       | K9       | L13         | 87      | 100     | P13          | PH10                                     | I/O      | FT_h          | -     | TIM5_CH1, I2C4_SMBA,<br>FMC_D18, DCMI_D1,<br>LCD_R4, EVENTOUT                                                                                                                                        | -                       |
| -             | -        | -       | L10      | L12         | 88      | 101     | P14          | PH11                                     | I/O      | FT_fh         | -     | TIM5_CH2, I2C4_SCL,<br>FMC_D19, DCMI_D2,<br>LCD_R5, EVENTOUT                                                                                                                                         | -                       |
| -             | -        | -       | K10      | K12         | 89      | 102     | R14          | PH12                                     | I/O      | FT_fh         | -     | TIM5_CH3, I2C4_SDA,<br>FMC_D20, DCMI_D3,<br>LCD_R6, EVENTOUT                                                                                                                                         | -                       |
| -             | -        | -       | -        | H12         | 90      | -       | N16          | VSS                                      | S        | -             | -     | -                                                                                                                                                                                                    | -                       |
| -             | -        | -       | N11      | J12         | 91      | 103     | P17          | VDD                                      | S        | -             | -     | -                                                                                                                                                                                                    | -                       |
| 51            | K8       | 73      | N12      | P12         | 92      | 104     | T14          | PB12                                     | I/O      | FT_u          | -     | TIM1_BKIN, I2C2_SMBA,<br>SPI2_NSS/I2S2_WS,<br>DFSDM1_DATIN1,<br>USART3_CK,<br>FDCAN2_RX,<br>OTG_HS_ULPI_D5,<br>ETH_MII_TXD0/ETH_RM<br>II_TXD0, OTG_HS_ID,<br>TIM1_BKIN_COMP12,<br>UART5_RX, EVENTOUT |                         |
| 52            | J8       | 74      | L11      | P13         | 93      | 105     | U14          | PB13                                     | I/O      | FT_u          | -     | TIM1_CH1N,<br>LPTIM2_OUT,<br>SPI2_SCK/I2S2_CK,<br>DFSDM1_CKIN1,<br>USART3_CTS/USART3_<br>NSS, FDCAN2_TX,<br>OTG_HS_ULPI_D6,<br>ETH_MII_TXD1/ETH_RM<br>II_TXD1, UART5_TX,<br>EVENTOUT                 | OTG_HS_<br>VBUS         |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                       | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                           |                         |
| 53            | H10      | 75      | N13      | R14         | 94      | 106     | U15          | PB14                                     | I/O      | FT_u          | -     | TIM1_CH2N,<br>TIM12_CH1,<br>TIM8_CH2N,<br>USART1_TX,<br>SPI2_MISO/I2S2_SDI,<br>DFSDM1_DATIN2,<br>USART3_RTS/<br>USART3_DE,<br>UART4_RTS/UART4_DE,<br>SDMMC2_D0,<br>OTG_HS_DM,<br>EVENTOUT | -                       |
| 54            | G10      | 76      | M13      | R15         | 95      | 107     | T15          | PB15                                     | I/O      | FT_u          | -     | RTC_REFIN,<br>TIM1_CH3N,<br>TIM12_CH2,<br>TIM8_CH3N,<br>USART1_RX,<br>SPI2_MOSI/I2S2_SDO,<br>DFSDM1_CKIN2,<br>UART4_CTS,<br>SDMMC2_D1,<br>OTG_HS_DP,<br>EVENTOUT                          | -                       |
| 55            | K9       | 77      | M12      | P15         | 96      | 108     | U16          | PD8                                      | I/O      | FT_h          | -     | DFSDM1_CKIN3,<br>SAI3_SCK_B,<br>USART3_TX,<br>SPDIFRX1_IN2,<br>FMC_D13/FMC_DA13,<br>EVENTOUT                                                                                              | -                       |
| 56            | J9       | 78      | K11      | P14         | 97      | 109     | T17          | PD9                                      | I/O      | FT_h          | -     | DFSDM1_DATIN3,<br>SAI3_SD_B,<br>USART3_RX,<br>FDCAN2_RXFD_MODE,<br>FMC_D14/FMC_DA14,<br>EVENTOUT                                                                                          | -                       |
| 57            | H9       | 79      | K12      | N15         | 98      | 110     | T16          | PD10                                     | I/O      | FT_h          | -     | DFSDM1_CKOUT,<br>SAI3_FS_B,<br>USART3_CK,<br>FDCAN2_TXFD_MODE,<br>FMC_D15/FMC_DA15,<br>LCD_B3, EVENTOUT                                                                                   | -                       |
| -             | -        | -       | N7       | -           | -       | -       | N12          | VDD                                      | S        | -             | -     | -                                                                                                                                                                                         | -                       |
| -             | -        | -       | -        | F9          | -       | -       | U17          | VSS                                      | S        | -             | -     | -                                                                                                                                                                                         | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                             | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                 |                         |
| 58            | G9       | 80      | J10      | N14         | 99      | 111     | R15          | PD11                                     | I/O      | FT_h          | -     | LPTIM2_IN2,<br>I2C4_SMBA,<br>USART3_CTS/USART3_<br>NSS,<br>QUADSPI_BK1_IO0,<br>SAI2_SD_A, FMC_A16,<br>EVENTOUT                  | -                       |
| 59            | K10      | 81      | K13      | N13         | 100     | 112     | R16          | PD12                                     | I/O      | FT_fh         | -     | LPTIM1_IN1, TIM4_CH1,<br>LPTIM2_IN1, I2C4_SCL,<br>USART3_RTS/USART3_<br>DE, QUADSPI_BK1_IO1,<br>SAI2_FS_A, FMC_A17,<br>EVENTOUT | -                       |
| 60            | J10      | 82      | J11      | M15         | 101     | 113     | R17          | PD13                                     | I/O      | FT_fh         | -     | LPTIM1_OUT,<br>TIM4_CH2, I2C4_SDA,<br>QUADSPI_BK1_IO3,<br>SAI2_SCK_A, FMC_A18,<br>EVENTOUT                                      | -                       |
| -             | -        | 83      | -        | K8          | 102     | 114     | -            | VSS                                      | S        | -             | -     | -                                                                                                                               | -                       |
| -             | -        | 84      | -        | J13         | 103     | 115     | N11          | VDD                                      | S        | -             | -     | -                                                                                                                               | -                       |
| 61            | H8       | 85      | J13      | M14         | 104     | 116     | P16          | PD14                                     | I/O      | FT_h          | -     | TIM4_CH3,<br>SAI3_MCLK_B,<br>UART8_CTS,<br>FMC_D0/FMC_DA0,<br>EVENTOUT                                                          | -                       |
| 62            | G8       | 86      | J12      | L14         | 105     | 117     | P15          | PD15                                     | I/O      | FT_h          | -     | TIM4_CH4,<br>SAI3_MCLK_A,<br>UART8_RTS/UART8_DE<br>, FMC_D1/FMC_DA1,<br>EVENTOUT                                                | -                       |
| -             | -        | -       | -        | -           | -       | 118     | N15          | PJ6                                      | I/O      | FT            | -     | TIM8_CH2, LCD_R7,<br>EVENTOUT                                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | 119     | N14          | PJ7                                      | I/O      | FT            | -     | TRGIN, TIM8_CH2N,<br>LCD_G0, EVENTOUT                                                                                           | -                       |
| -             | -        | -       | -        | -           | -       | -       | N10          | VDD                                      | S        | -             | -     | -                                                                                                                               | -                       |
| -             | -        | -       | -        | F10         | -       | -       | R8           | VSS                                      | S        | -             | -     | -                                                                                                                               | -                       |
| -             | -        | -       | -        | -           | -       | 120     | N13          | PJ8                                      | I/O      | FT            | -     | TIM1_CH3N, TIM8_CH1,<br>UART8_TX, LCD_G1,<br>EVENTOUT                                                                           | -                       |
| -             | -        | -       | -        | -           | -       | 121     | M14          | PJ9                                      | I/O      | FT            | -     | TIM1_CH3, TIM8_CH1N,<br>UART8_RX, LCD_G2,<br>EVENTOUT                                                                           | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |                    | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                 | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25       |                                          |          |               |       |                                                                                     |                         |
| -             | -        | -       | -        | -           | -       | 122     | L14                | PJ10                                     | I/O      | FT            | -     | TIM1_CH2N, TIM8_CH2,<br>SPI5_MOSI, LCD_G3,<br>EVENTOUT                              | -                       |
| -             | -        | -       | -        | -           | -       | 123     | K14                | PJ11                                     | I/O      | FT            | -     | TIM1_CH2, TIM8_CH2N,<br>SPI5_MISO, LCD_G4,<br>EVENTOUT                              | -                       |
| -             | -        | -       | -        | -           | -       | 124     | N8                 | VDD                                      | S        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | G6          | -       | 125     | U1                 | VSS                                      | S        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | N17<br>(2)         | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | M16<br>(2)         | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | M17<br>(2)         | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | K15                | VSS                                      | S        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | L16 <sup>(2)</sup> | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | L17 <sup>(2)</sup> | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | K16<br>(2)         | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | K17<br>(2)         | NC                                       | -        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | -       | L15                | VSS                                      | S        | -             | -     | -                                                                                   | -                       |
| -             | -        | -       | -        | -           | -       | 126     | J14                | PK0                                      | I/O      | FT            | -     | TIM1_CH1N, TIM8_CH3,<br>SPI5_SCK, LCD_G5,<br>EVENTOUT                               | -                       |
| -             | -        | -       | -        | -           | -       | 127     | J15                | PK1                                      | I/O      | FT            | -     | TIM1_CH1, TIM8_CH3N,<br>SPI5_NSS, LCD_G6,<br>EVENTOUT                               | -                       |
| -             | -        | -       | -        | -           | -       | 128     | H17                | PK2                                      | I/O      | FT            | -     | TIM1_BKIN, TIM8_BKIN,<br>TIM8_BKIN_COMP12,<br>TIM1_BKIN_COMP12,<br>LCD_G7, EVENTOUT | -                       |
| -             | -        | 87      | H9       | L15         | 106     | 129     | H16                | PG2                                      | I/O      | FT_h          | -     | TIM8_BKIN,<br>TIM8_BKIN_COMP12,<br>FMC_A12, EVENTOUT                                | -                       |
| -             | -        | 88      | H10      | K15         | 107     | 130     | H15                | PG3                                      | I/O      | FT_h          | -     | TIM8_BKIN2,<br>TIM8_BKIN2_COMP12,<br>FMC_A13, EVENTOUT                              | -                       |
| -             | -        | -       | -        | G7          | -       | -       | -                  | VSS                                      | S        | -             | -     | -                                                                                   | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                       | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                           |                         |
| -             | -        | -       | -        | -           | -       | -       | N7           | VDD                                      | S        | -             | -     | -                                                                                                                                                                         | -                       |
| -             | -        | 89      | F8       | K14         | 108     | 131     | H14          | PG4                                      | I/O      | FT_h          | -     | TIM1_BKIN2,<br>TIM1_BKIN2_COMP12,<br>FMC_A14/FMC_BA0,<br>EVENTOUT                                                                                                         | -                       |
| -             | -        | 90      | H11      | K13         | 109     | 132     | G14          | PG5                                      | I/O      | FT_h          | -     | TIM1_ETR,<br>FMC_A15/FMC_BA1,<br>EVENTOUT                                                                                                                                 | -                       |
| -             | -        | 91      | G9       | J15         | 110     | 133     | G15          | PG6                                      | I/O      | FT_h          | -     | TIM17_BKIN,<br>HRTIM_CHE1,<br>QUADSPI_BK1_NCS,<br>FMC_NE3, DCMI_D12,<br>LCD_R7, EVENTOUT                                                                                  | -                       |
| -             | -        | 92      | G10      | J14         | 111     | 134     | F16          | PG7                                      | I/O      | FT_h          | -     | HRTIM_CHE2,<br>SAI1_MCLK_A,<br>USART6_CK, FMC_INT,<br>DCMI_D13, LCD_CLK,<br>EVENTOUT                                                                                      | -                       |
| -             | -        | 93      | G11      | H14         | 112     | 135     | F15          | PG8                                      | I/O      | FT_h          | -     | TIM8_ETR, SPI6_NSS,<br>USART6_RTS/USART6_<br>DE, SPDIFRX1_IN3,<br>ETH_PPS_OUT,<br>FMC_SDCLK, LCD_G7,<br>EVENTOUT                                                          | -                       |
| -             | -        | 94      | -        | G12         | 113     | 136     | G16          | VSS                                      | S        | -             | -     | -                                                                                                                                                                         | -                       |
| -             | -        | -       | G12      | -           | -       | -       | G17          | VDD50<br>USB                             | S        | -             | -     | -                                                                                                                                                                         | -                       |
| -             | F6       | 95      | G13      | H13         | 114     | 137     | F17          | VDD33<br>USB                             | S        | -             | -     | -                                                                                                                                                                         | -                       |
| -             | -        | -       | -        | -           | -       | -       | M5           | VDD                                      | S        | -             | -     | -                                                                                                                                                                         | -                       |
| 63            | F10      | 96      | F9       | H15         | 115     | 138     | F14          | PC6                                      | I/O      | FT_h          | -     | HRTIM_CHA1,<br>TIM3_CH1, TIM8_CH1,<br>DFSDM1_CKIN3,<br>I2S2_MCK, USART6_TX,<br>SDMMC1_D0DIR,<br>FMC_NWAIT,<br>SDMMC2_D6,<br>SDMMC1_D6, DCMI_D0,<br>LCD_HSYNC,<br>EVENTOUT | SWPMI_IO                |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                         | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                             |                         |
| 64            | E10      | 97      | F10      | G15         | 116     | 139     | F13          | PC7                                      | I/O      | FT_h          | -     | TRGIO, HRTIM_CHA2,<br>TIM3_CH2, TIM8_CH2,<br>DFSDM1_DATIN3,<br>I2S3_MCK,<br>USART6_RX,<br>SDMMC1_D123DIR,<br>FMC_NE1,<br>SDMMC2_D7,<br>SWPMI_TX,<br>SDMMC1_D7, DCMI_D1,<br>LCD_G6, EVENTOUT | -                       |
| 65            | F9       | 98      | F12      | G14         | 117     | 140     | E13          | PC8                                      | I/O      | FT_h          | -     | TRACED1,<br>HRTIM_CHB1,<br>TIM3_CH3, TIM8_CH3,<br>USART6_CK,<br>UART5_RTS/UART5_DE<br>, FMC_NE2/FMC_NCE,<br>SWPMI_RX,<br>SDMMC1_D0, DCMI_D2,<br>EVENTOUT                                    | -                       |
| 66            | E9       | 99      | F11      | F14         | 118     | 141     | E14          | PC9                                      | I/O      | FT_fh         | -     | MCO2, TIM3_CH4,<br>TIM8_CH4, I2C3_SDA,<br>I2S_CKIN, UART5_CTS,<br>QUADSPI_BK1_IO0,<br>LCD_G3,<br>SWPMI_SUSPEND,<br>SDMMC1_D1, DCMI_D3,<br>LCD_B2, EVENTOUT                                  | -                       |
| -             | -        | -       | -        | G8          | -       | -       | -            | VSS                                      | S        | -             | -     |                                                                                                                                                                                             | -                       |
| -             | -        | -       | -        | -           | -       | -       | L5           | VDD                                      | S        | -             | -     |                                                                                                                                                                                             | -                       |
| 67            | D9       | 100     | E12      | F15         | 119     | 142     | E15          | PA8                                      | I/O      | FT_fha        | -     | MCO1, TIM1_CH1,<br>HRTIM_CHB2,<br>TIM8_BKIN2, I2C3_SCL,<br>USART1_CK,<br>OTG_FS_SOF,<br>UART7_RX,<br>TIM8_BKIN2_COMP12,<br>LCD_B3, LCD_R6,<br>EVENTOUT                                      | -                       |



Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                               | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                   |                         |
| 68            | C9       | 101     | E11      | E15         | 120     | 143     | D15          | PA9                                      | I/O      | FT_u          | -     | TIM1_CH2,<br>HRTIM_CHC1,<br>LPUART1_TX,<br>I2C3_SMBA,<br>SPI2_SCK/I2S2_CK,<br>USART1_TX,<br>FDCAN1_RXFD_MODE,<br>DCMI_D0, LCD_R5,<br>EVENTOUT                                     | OTG_FS_VBUS             |
| 69            | D10      | 102     | E10      | D15         | 121     | 144     | D14          | PA10                                     | I/O      | FT_u          | -     | TIM1_CH3,<br>HRTIM_CHC2,<br>LPUART1_RX,<br>USART1_RX,<br>FDCAN1_TXFD_MODE,<br>OTG_FS_ID,<br>MDIOS_MDIO, LCD_B4,<br>DCMI_D1, LCD_B1,<br>EVENTOUT                                   | -                       |
| 70            | C10      | 103     | F13      | C15         | 122     | 145     | E17          | PA11                                     | I/O      | FT_u          | -     | TIM1_CH4,<br>HRTIM_CHD1,<br>LPUART1_CTS,<br>SPI2_NSS/I2S2_WS,<br>UART4_RX,<br>USART1_CTS/USART1_<br>NSS, FDCAN1_RX,<br>OTG_FS_DM, LCD_R4,<br>EVENTOUT                             | -                       |
| 71            | B10      | 104     | E13      | B15         | 123     | 146     | E16          | PA12                                     | I/O      | FT_u          | -     | TIM1_ETR,<br>HRTIM_CHD2,<br>LPUART1_RTS/<br>LPUART1_DE,<br>SPI2_SCK/I2S2_CK,<br>UART4_TX,<br>USART1_RTS/USART1_<br>DE, SAI2_FS_B,<br>FDCAN1_TX,<br>OTG_FS_DP, LCD_R5,<br>EVENTOUT | -                       |
| 72            | A10      | 105     | D11      | A15         | 124     | 147     | C15          | PA13<br>(JTMS/SW<br>DIO)                 | I/O      | FT            | -     | JTMS-SWDIO,<br>EVENTOUT                                                                                                                                                           | -                       |
| 73            | E7       | 106     | D13      | F13         | 125     | 148     | D17          | VCAP                                     | S        | -             | -     | -                                                                                                                                                                                 | -                       |
| 74            | E5       | 107     | -        | F12         | 126     | 149     | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                 | -                       |
| -             | -        | -       | D12      | -           | -       | -       | C17          | VDDLDO<br>(8)                            | -        | -             | -     | -                                                                                                                                                                                 | -                       |
| 75            | -        | 108     | -        | G13         | 127     | 150     | K5           | VDD                                      | S        | -             | -     | -                                                                                                                                                                                 | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                             | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                 |                         |
| -             | -        | -       | -        | E12         | 128     | 151     | D16          | PH13                                     | I/O      | FT_h          | -     | TIM8_CH1N, UART4_TX,<br>FDCAN1_TX, FMC_D21,<br>LCD_G2, EVENTOUT                                 | -                       |
| -             | -        | -       | -        | E13         | 129     | 152     | B17          | PH14                                     | I/O      | FT_h          | -     | TIM8_CH2N,<br>UART4_RX,<br>FDCAN1_RX, FMC_D22,<br>DCMI_D4, LCD_G3,<br>EVENTOUT                  | -                       |
| -             | -        | -       | -        | D13         | 130     | 153     | B16          | PH15                                     | I/O      | FT_h          | -     | TIM8_CH3N,<br>FDCAN1_TXFD_MODE,<br>FMC_D23, DCMI_D11,<br>LCD_G4, EVENTOUT                       | -                       |
| -             | -        | -       | A13      | E14         | 131     | 154     | A16          | PI0                                      | I/O      | FT_h          | -     | TIM5_CH4,<br>SPI2_NSS/I2S2_WS,<br>FDCAN1_RXFD_MODE,<br>FMC_D24, DCMI_D13,<br>LCD_G5, EVENTOUT   | -                       |
| -             | -        | -       | -        | G9          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                               | -                       |
| -             | -        | -       | B13      | D14         | 132     | 155     | A15          | PI1                                      | I/O      | FT_h          | -     | TIM8_BKIN2,<br>SPI2_SCK/I2S2_CK,<br>TIM8_BKIN2_COMP12,<br>FMC_D25, DCMI_D8,<br>LCD_G6, EVENTOUT | -                       |
| -             | -        | -       | A6       | C14         | 133     | 156     | B15          | PI2                                      | I/O      | FT_h          | -     | TIM8_CH4,<br>SPI2_MISO/I2S2_SDI,<br>FMC_D26, DCMI_D9,<br>LCD_G7, EVENTOUT                       | -                       |
| -             | -        | -       | B7       | C13         | 134     | 157     | C14          | PI3                                      | I/O      | FT_h          | -     | TIM8_ETR,<br>SPI2_MOSI/I2S2_SDO,<br>FMC_D27, DCMI_D10,<br>EVENTOUT                              | -                       |
| -             | -        | -       | -        | D9          | 135     | -       | -            | VSS                                      | S        | -             | -     | -                                                                                               | -                       |
| -             | -        | -       | -        | C9          | 136     | 158     | -            | VDD                                      | S        | -             | -     | -                                                                                               | -                       |
| 76            | A9       | 109     | B12      | A14         | 137     | 159     | B14          | PA14<br>(JTCK/SW<br>CLK)                 | I/O      | FT            | -     | JTCK-SWCLK,<br>EVENTOUT                                                                         | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                    | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                        |                         |
| 77            | A8       | 110     | C11      | A13         | 138     | 160     | A14          | PA15<br>(JTDI)                           | I/O      | FT            | -     | JTDI,<br>TIM2_CH1/TIM2_ETR,<br>HRTIM_FLT1, CEC,<br>SPI1_NSS/I2S1_WS,<br>SPI3_NSS/I2S3_WS,<br>SPI6_NSS,<br>UART4_RTS/UART4_DE,<br>UART7_TX,<br>EVENTOUT | -                       |
| 78            | B9       | 111     | A12      | B14         | 139     | 161     | A13          | PC10                                     | I/O      | FT_ha         | -     | HRTIM_EEV1,<br>DFSDM1_CKIN5,<br>SPI3_SCK/I2S3_CK,<br>USART3_TX,<br>UART4_TX,<br>QUADSPI_BK1_IO1,<br>SDMMC1_D2, DCMI_D8,<br>LCD_R2, EVENTOUT            | -                       |
| 79            | B8       | 112     | B11      | B13         | 140     | 162     | B13          | PC11                                     | I/O      | FT_h          | -     | HRTIM_FLT2,<br>DFSDM1_DATIN5,<br>SPI3_MISO/I2S3_SDI,<br>USART3_RX,<br>UART4_RX,<br>QUADSPI_BK2_NCS,<br>SDMMC1_D3, DCMI_D4,<br>EVENTOUT                 | -                       |
| 80            | C8       | 113     | A11      | A12         | 141     | 163     | C12          | PC12                                     | I/O      | FT_h          | -     | TRACED3,<br>HRTIM_EEV2,<br>SPI3_MOSI/I2S3_SDO,<br>USART3_CK,<br>UART5_TX,<br>SDMMC1_CK, DCMI_D9,<br>EVENTOUT                                           | -                       |
| -             | -        | -       | -        | G10         | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                      | -                       |
| 81            | D8       | 114     | D10      | B12         | 142     | 164     | D13          | PD0                                      | I/O      | FT_h          | -     | DFSDM1_CKIN6,<br>SAI3_SCK_A,<br>UART4_RX,<br>FDCAN1_RX,<br>FMC_D2/FMC_DA2,<br>EVENTOUT                                                                 | -                       |
| 82            | E8       | 115     | C10      | C12         | 143     | 165     | E12          | PD1                                      | I/O      | FT_h          | -     | DFSDM1_DATIN6,<br>SAI3_SD_A, UART4_TX,<br>FDCAN1_TX,<br>FMC_D3/FMC_DA3,<br>EVENTOUT                                                                    | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                                     | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                                         |                         |
| 83            | B7       | 116     | E9       | D12         | 144     | 166     | D12          | PD2                                      | I/O      | FT_h          | -     | TRACED2, TIM3_ETR,<br>UART5_RX,<br>SDMMC1_CMD,<br>DCMI_D11, EVENTOUT                                                                                                                                    | -                       |
| 84            | C7       | 117     | D9       | D11         | 145     | 167     | B12          | PD3                                      | I/O      | FT_h          | -     | DFSDM1_CKOUT,<br>SPI2_SCK/I2S2_CK,<br>USART2_CTS/USART2_<br>NSS, FMC_CLK,<br>DCMI_D5, LCD_G7,<br>EVENTOUT                                                                                               | -                       |
| 85            | D7       | 118     | C9       | D10         | 146     | 168     | A12          | PD4                                      | I/O      | FT_h          | -     | HRTIM_FLT3,<br>SAI3_FS_A,<br>USART2_RTS/USART2_<br>DE,<br>FDCAN1_RXFD_MODE,<br>FMC_NOE, EVENTOUT                                                                                                        | -                       |
| 86            | B6       | 119     | A9       | C11         | 147     | 169     | A11          | PD5                                      | I/O      | FT_h          | -     | HRTIM_EEV3,<br>USART2_TX,<br>FDCAN1_TXFD_MODE,<br>FMC_NWE, EVENTOUT                                                                                                                                     | -                       |
| -             | -        | 120     | -        | D8          | 148     | 170     | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                                       | -                       |
| -             | -        | 121     | -        | C8          | 149     | 171     | -            | VDD                                      | S        | -             | -     | -                                                                                                                                                                                                       | -                       |
| 87            | C6       | 122     | B9       | B11         | 150     | 172     | B11          | PD6                                      | I/O      | FT_h          | -     | SAI1_D1,<br>DFSDM1_CKIN4,<br>DFSDM1_DATIN1,<br>SPI3_MOSI/I2S3_SDO,<br>SAI1_SD_A,<br>USART2_RX,<br>SAI4_SD_A,<br>FDCAN2_RXFD_MODE,<br>SAI4_D1, SDMMC2_CK,<br>FMC_NWAIT,<br>DCMI_D10, LCD_B2,<br>EVENTOUT | -                       |
| 88            | D6       | 123     | D8       | A11         | 151     | 173     | C11          | PD7                                      | I/O      | FT_h          | -     | DFSDM1_DATIN4,<br>SPI1_MOSI/I2S1_SDO,<br>DFSDM1_CKIN1,<br>USART2_CK,<br>SPDIFRX1_IN1,<br>SDMMC2_CMD,<br>FMC_NE1, EVENTOUT                                                                               | -                       |
| -             | -        | -       | -        | -           | -       | 174     | D11          | PJ12                                     | I/O      | FT            | -     | TRGOUT, LCD_G3,<br>LCD_B0, EVENTOUT                                                                                                                                                                     | -                       |
| -             | -        | -       | -        | -           | -       | 175     | E10          | PJ13                                     | I/O      | FT            | -     | LCD_B4, LCD_B1,<br>EVENTOUT                                                                                                                                                                             | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                    | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                        |                         |
| -             | -        | -       | -        | -           | -       | 176     | D10          | PJ14                                     | I/O      | FT            | -     | LCD_B2, EVENTOUT                                                                                                                                       | -                       |
| -             | -        | -       | -        | -           | -       | 177     | B10          | PJ15                                     | I/O      | FT            | -     | LCD_B3, EVENTOUT                                                                                                                                       | -                       |
| -             | -        | -       | -        | H6          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                      | -                       |
| -             | -        | -       | A7       | -           | -       | -       | -            | VDD                                      | S        | -             | -     | -                                                                                                                                                      | -                       |
| -             | -        | 124     | C8       | C10         | 152     | 178     | A10          | PG9                                      | I/O      | FT_h          | -     | SPI1_MISO/I2S1_SDI,<br>USART6_RX,<br>SPDIFRX1_IN4,<br>QUADSPI_BK2_IO2,<br>SAI2_FS_B,<br>FMC_NE2/FMC_NCE,<br>DCMI_VSYNC,<br>EVENTOUT                    | -                       |
| -             | -        | 125     | A8       | B10         | 153     | 179     | A9           | PG10                                     | I/O      | FT_h          | -     | HRTIM_FLT5,<br>SPI1_NSS/I2S1_WS,<br>LCD_G3, SAI2_SD_B,<br>FMC_NE3, DCMI_D2,<br>LCD_B2, EVENTOUT                                                        | -                       |
| -             | -        | 126     | B8       | B9          | 154     | 180     | B9           | PG11                                     | I/O      | FT_h          | -     | LPTIM1_IN2,<br>HRTIM_EEV4,<br>SPI1_SCK/I2S1_CK,<br>SPDIFRX1_IN1,<br>SDMMC2_D2,<br>ETH_MII_TX_EN/ETH_RMII_TX_EN, DCMI_D3,<br>LCD_B3, EVENTOUT           | -                       |
| -             | -        | 127     | E8       | B8          | 155     | 181     | C9           | PG12                                     | I/O      | FT_h          | -     | LPTIM1_IN1,<br>HRTIM_EEV5,<br>SPI6_MISO,<br>USART6_RTS/USART6_DE, SPDIFRX1_IN2,<br>LCD_B4,<br>ETH_MII_TXD1/ETH_RMII_TXD1, FMC_NE4,<br>LCD_B1, EVENTOUT | -                       |
| -             | -        | 128     | D7       | A8          | 156     | 182     | D9           | PG13                                     | I/O      | FT_h          | -     | TRACED0,<br>LPTIM1_OUT,<br>HRTIM_EEV10,<br>SPI6_SCK,<br>USART6_CTS/USART6_NSS,<br>ETH_MII_TXD0/ETH_RMII_TXD0, FMC_A24,<br>LCD_R0, EVENTOUT             | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                  | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                      |                         |
| -             | -        | 129     | C7       | A7          | 157     | 183     | D8           | PG14                                     | I/O      | FT_h          | -     | TRACED1,<br>LPTIM1_ETR,<br>SPI6_MOSI,<br>USART6_TX,<br>QUADSPI_BK2_IO3,<br>ETH_MII_TXD1/ETH_RMII_TXD1, FMC_A25,<br>LCD_B0, EVENTOUT                                  | -                       |
| -             | -        | 130     | -        | D7          | 158     | 184     | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                    | -                       |
| -             | -        | 131     | -        | C7          | 159     | 185     | -            | VDD                                      | S        | -             | -     | -                                                                                                                                                                    | -                       |
| -             | -        | -       | -        | -           | -       | 186     | C8           | PK3                                      | I/O      | FT            | -     | LCD_B4, EVENTOUT                                                                                                                                                     | -                       |
| -             | -        | -       | -        | -           | -       | 187     | B8           | PK4                                      | I/O      | FT            | -     | LCD_B5, EVENTOUT                                                                                                                                                     | -                       |
| -             | -        | -       | -        | -           | -       | 188     | A8           | PK5                                      | I/O      | FT            | -     | LCD_B6, EVENTOUT                                                                                                                                                     | -                       |
| -             | -        | -       | -        | -           | -       | 189     | C7           | PK6                                      | I/O      | FT            | -     | LCD_B7, EVENTOUT                                                                                                                                                     | -                       |
| -             | -        | -       | -        | -           | -       | 190     | D7           | PK7                                      | I/O      | FT            | -     | LCD_DE, EVENTOUT                                                                                                                                                     | -                       |
| -             | -        | -       | -        | H7          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                    | -                       |
| -             | -        | 132     | E7       | B7          | 160     | 191     | D6           | PG15                                     | I/O      | FT_h          | -     | USART6_CTS/USART6_NSS, FMC_SDNCAS,<br>DCMI_D13, EVENTOUT                                                                                                             | -                       |
| 89            | A7       | 133     | F7       | A10         | 161     | 192     | C6           | PB3(JTDO<br>/TRACES<br>WO)               | I/O      | FT            | -     | JTDO/TRACESWO,<br>TIM2_CH2,<br>HRTIM_FLT4,<br>SPI1_SCK/I2S1_CK,<br>SPI3_SCK/I2S3_CK,<br>SPI6_SCK,<br>SDMMC2_D2,<br>CRS_SYNC, UART7_RX,<br>EVENTOUT                   | -                       |
| 90            | A6       | 134     | B6       | A9          | 162     | 193     | B7           | PB4(NJTR<br>ST)                          | I/O      | FT            | -     | NJTRST, TIM16_BKIN,<br>TIM3_CH1,<br>HRTIM_EEV6,<br>SPI1_MISO/I2S1_SDI,<br>SPI3_MISO/I2S3_SDI,<br>SPI2_NSS/I2S2_WS,<br>SPI6_MISO,<br>SDMMC2_D3,<br>UART7_TX, EVENTOUT | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                                                             | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                                                                 |                         |
| 91            | C5       | 135     | C6       | A6          | 163     | 194     | A5           | PB5                                      | I/O      | FT            | -     | TIM17_BKIN, TIM3_CH2,<br>HRTIM_EEV7,<br>I2C1_SMBA,<br>SPI1_MOSI/I2S1_SDO,<br>I2C4_SMBA,<br>SPI3_MOSI/I2S3_SDO,<br>SPI6_MOSI,<br>FDCAN2_RX,<br>OTG_HS_ULPI_D7,<br>ETH_PPS_OUT,<br>FMC_SDCKE1,<br>DCMI_D10, UART5_RX,<br>EVENTOUT | -                       |
| -             | -        | -       | -        | H8          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                                                               | -                       |
| 92            | B5       | 136     | A5       | B6          | 164     | 195     | B5           | PB6                                      | I/O      | FT_f          | -     | TIM16_CH1N,<br>TIM4_CH1,<br>HRTIM_EEV8,<br>I2C1_SCL, CEC,<br>I2C4_SCL, USART1_TX,<br>LPUART1_TX,<br>FDCAN2_TX,<br>QUADSPI_BK1_NCS,<br>DFSDM1_DATIN5,<br>FMC_SDNE1, DCMI_D5,<br>UART5_TX, EVENTOUT                               | -                       |
| 93            | A5       | 137     | D6       | B5          | 165     | 196     | C5           | PB7                                      | I/O      | FT_fa         | -     | TIM17_CH1N,<br>TIM4_CH2,<br>HRTIM_EEV9,<br>I2C1_SDA, I2C4_SDA,<br>USART1_RX,<br>LPUART1_RX,<br>FDCAN2_TXFD_MODE,<br>DFSDM1_CKIN5,<br>FMC_NL, DCMI_VSYNC,<br>EVENTOUT                                                            | PVD_IN                  |
| 94            | D5       | 138     | E6       | D6          | 166     | 197     | E8           | BOOT0                                    | I        | B             | -     | -                                                                                                                                                                                                                               | VPP                     |
| 95            | B4       | 139     | B5       | A5          | 167     | 198     | D5           | PB8                                      | I/O      | FT_fh         | -     | TIM16_CH1, TIM4_CH3,<br>DFSDM1_CKIN7,<br>I2C1_SCL, I2C4_SCL,<br>SDMMC1_CKIN,<br>UART4_RX,<br>FDCAN1_RX,<br>SDMMC2_D4,<br>ETH_MII_TXD3,<br>SDMMC1_D4, DCMI_D6,<br>LCD_B6, EVENTOUT                                               | -                       |

Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                                                                                                                                                                       | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                                                                                                                                                                           |                         |
| 96            | A4       | 140     | C5       | B4          | 168     | 199     | D4           | PB9                                      | I/O      | FT_fh         | -     | TIM17_CH1, TIM4_CH4,<br>DFSDM1_DATIN7,<br>I2C1_SDA,<br>SPI2_NSS/I2S2_WS,<br>I2C4_SDA,<br>SDMMC1_CDIO,<br>UART4_TX,<br>FDCAN1_TX,<br>SDMMC2_D5,<br>I2C4_SMB, A,<br>SDMMC1_D5, DCMI_D7,<br>LCD_B7, EVENTOUT | -                       |
| 97            | D4       | 141     | D5       | A4          | 169     | 200     | C4           | PE0                                      | I/O      | FT_h          | -     | LPTIM1_ETR,<br>TIM4_ETR,<br>HRTIM_SCIN,<br>LPTIM2_ETR,<br>UART8_RX,<br>FDCAN1_RXFD_MODE,<br>SAI2_MCLK_A,<br>FMC_NBL0, DCMI_D2,<br>EVENTOUT                                                                | -                       |
| 98            | C4       | 142     | D4       | A3          | 170     | 201     | B4           | PE1                                      | I/O      | FT_h          | -     | LPTIM1_IN2,<br>HRTIM_SCOUT,<br>UART8_TX,<br>FDCAN1_TXFD_MODE,<br>FMC_NBL1, DCMI_D3,<br>EVENTOUT                                                                                                           | -                       |
| -             | -        | -       | -        | -           | -       | -       | A7           | VCAP                                     | S        | -             | -     | -                                                                                                                                                                                                         | -                       |
| 99            | -        | -       | -        | D5          | -       | 202     | -            | VSS                                      | S        | -             | -     | -                                                                                                                                                                                                         | -                       |
| -             | F7       | 143     | C4       | C6          | 171     | 203     | E7           | PDR_ON                                   | I        | FT            | -     | -                                                                                                                                                                                                         | -                       |
| -             | F4       | -       | B4       | -           | -       | -       | A6           | VDDLDO<br>(8)                            | S        | -             | -     | -                                                                                                                                                                                                         | -                       |
| 100           | -        | 144     | -        | C5          | 172     | 204     | -            | VDD                                      | S        | -             | -     | -                                                                                                                                                                                                         | -                       |
| -             | -        | -       | -        | D4          | 173     | 205     | A4           | PI4                                      | I/O      | FT_h          | -     | TIM8_BKIN,<br>SAI2_MCLK_A,<br>TIM8_BKIN_COMP12,<br>FMC_NBL2, DCMI_D5,<br>LCD_B4, EVENTOUT                                                                                                                 | -                       |
| -             | -        | -       | -        | C4          | 174     | 206     | A3           | PI5                                      | I/O      | FT_h          | -     | TIM8_CH1,<br>SAI2_SCK_A,<br>FMC_NBL3,<br>DCMI_VSYNC, LCD_B5,<br>EVENTOUT                                                                                                                                  | -                       |



Table 9. Pin/ball definition (continued)

| Pin/ball name |          |         |          |             |         |         |              | Pin name<br>(function<br>after<br>reset) | Pin type | I/O structure | Notes | Alternate functions                                           | Additional<br>functions |
|---------------|----------|---------|----------|-------------|---------|---------|--------------|------------------------------------------|----------|---------------|-------|---------------------------------------------------------------|-------------------------|
| LQFP100       | TFBGA100 | LQFP144 | UFBGA169 | UFBGA176+25 | LQFP176 | LQFP208 | TFBGA240 +25 |                                          |          |               |       |                                                               |                         |
| -             | -        | -       | A4       | C3          | 175     | 207     | A2           | PI6                                      | I/O      | FT_h          | -     | TIM8_CH2, SAI2_SD_A,<br>FMC_D28, DCMI_D6,<br>LCD_B6, EVENTOUT | -                       |
| -             | -        | -       | E2       | C2          | 176     | 208     | B3           | PI7                                      | I/O      | FT_h          | -     | TIM8_CH3, SAI2_FS_A,<br>FMC_D29, DCMI_D7,<br>LCD_B7, EVENTOUT | -                       |
| -             | -        | -       | -        | H9          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                             | -                       |
| -             | -        | -       | -        | K9          | -       | -       | -            | VSS                                      | S        | -             | -     | -                                                             | -                       |
| -             | -        | -       | -        | K10         | -       | -       | M15          | VSS                                      | S        | -             | -     | -                                                             | -                       |

1. When this pin/ball was previously configured as an oscillator, the oscillator function is kept during and after a reset. This is valid for all resets except for power-on reset.
2. This ball should remain floating.
3. This ball should not remain floating. It can be connected to VSS or VDD. It is reserved for future use.
4. This ball should be connected to V<sub>SS</sub>.
5. Pxy\_C and Pxy pins/balls are two separate pads (analog switch open). The analog switch is configured through a SYSCFG register. Refer to the product reference manual for a detailed description of the switch configuration bits.
6. There is a direct path between Pxy\_C and Pxy pins/balls, through an analog switch. Pxy alternate functions are available on Pxy\_C when the analog switch is closed. The analog switch is configured through a SYSCFG register. Refer to the product reference manual for a detailed description of the switch configuration bits.
7. VREF+ pin, and consequently the internal voltage reference, are not available on the TFBGA100 package. On this package, this pin is double-bonded to VDDA which can be connected to an external reference. The internal voltage reference buffer is not available and must be kept disabled.
8. When it is not available on a package, the VDDLDO pin is internally tied to VDD.

Table 10. Port A alternate functions

| Port   |      | AF0         | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                 | AF13                       | AF14          | AF15      |
|--------|------|-------------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------|----------------------------|---------------|-----------|
|        |      | SYS         | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1/<br>3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC/<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS       |
| Port A | PA0  | -           | TIM2_CH1/<br>TIM2_ETR              | TIM5_CH1                        | TIM8_ETR                                               | TIM15_BKIN                                                    | -                      | -                                           | USART2_CTS/<br>USART2_NSS                      | UART4_TX                                                     | SDMMC2_CMD                                                                         | SAI2_SD_B                                                              | ETH_MII_CRS                                                                  | -                                                    | -                          | -             | EVENT-OUT |
|        | PA1  | -           | TIM2_CH2                           | TIM5_CH2                        | LPTIM3_OUT                                             | TIM15_CH1N                                                    | -                      | -                                           | USART2_RTS/<br>USART2_DE                       | UART4_RX                                                     | QUADSPI_BK1_IO3                                                                    | SAI2_MCLK_B                                                            | ETH_MII_RX_CLK/<br>ETH_RMII_REF_CLK                                          | -                                                    | -                          | LCD_R2        | EVENT-OUT |
|        | PA2  | -           | TIM2_CH3                           | TIM5_CH3                        | LPTIM4_OUT                                             | TIM15_CH1                                                     | -                      | -                                           | USART2_TX                                      | SAI2_SCK_B                                                   | -                                                                                  | -                                                                      | ETH_MDIO                                                                     | MDIOS_MDIO                                           | -                          | LCD_R1        | EVENT-OUT |
|        | PA3  | -           | TIM2_CH4                           | TIM5_CH4                        | LPTIM5_OUT                                             | TIM15_CH2                                                     | -                      | -                                           | USART2_RX                                      | -                                                            | LCD_B2                                                                             | OTG_HS_ULPI_D0                                                         | ETH_MII_COL                                                                  | -                                                    | -                          | LCD_B5        | EVENT-OUT |
|        | PA4  | D1<br>PWREN | -                                  | TIM5_ETR                        | -                                                      | -                                                             | SPI1_NSS/<br>I2S1_WS   | SPI3_NSS/<br>I2S3_WS                        | USART2_CK                                      | SPI6_NSS                                                     | -                                                                                  | -                                                                      | -                                                                            | OTG_HS_SOF                                           | DCMI_HSYNC                 | LCD_VSYNC     | EVENT-OUT |
|        | PA5  | D2<br>PWREN | TIM2_CH1/<br>TIM2_ETR              | -                               | TIM8_CH1N                                              | -                                                             | SPI1_SCK<br>/I2S1_CK   | -                                           | -                                              | SPI6_SCK                                                     | -                                                                                  | OTG_HS_ULPI_CK                                                         | -                                                                            | -                                                    | -                          | LCD_R4        | EVENT-OUT |
|        | PA6  | -           | TIM1_BKIN                          | TIM3_CH1                        | TIM8_BKIN                                              | -                                                             | SPI1_MISO<br>/I2S1_SDI | -                                           | -                                              | SPI6_MISO                                                    | TIM13_CH1                                                                          | TIM8_BKIN_COMP12                                                       | MDIOS_MDC                                                                    | TIM1_BKIN_COMP12                                     | DCMI_PIX_CLK               | LCD_G2        | EVENT-OUT |
|        | PA7  | -           | TIM1_CH1N                          | TIM3_CH2                        | TIM8_CH1_N                                             | -                                                             | SPI1_MOSI<br>/I2S1_SDO | -                                           | -                                              | SPI6_MOSI                                                    | TIM14_CH1                                                                          | -                                                                      | ETH_MII_RX_DV/<br>ETH_RMII_CRS_DV                                            | FMC_SDN_WE                                           | -                          | -             | EVENT-OUT |
|        | PA8  | MCO1        | TIM1_CH1                           | HRTIM_CH_B2                     | TIM8_BKIN_2                                            | I2C3_SCL                                                      | -                      | -                                           | USART1_CK                                      | -                                                            | -                                                                                  | OTG_FS_SOF                                                             | UART7_RX                                                                     | TIM8_BKIN_2_COMP12                                   | LCD_B3                     | LCD_R6        | EVENT-OUT |
|        | PA9  | -           | TIM1_CH2                           | HRTIM_CH_C1                     | LPUART1_TX                                             | I2C3_SMBA                                                     | SPI2_SCK/<br>I2S2_CK   | -                                           | USART1_TX                                      | -                                                            | FDCAN1_RXFD_MODE                                                                   | -                                                                      | -                                                                            | -                                                    | DCMI_D0                    | LCD_R5        | EVENT-OUT |
|        | PA10 | -           | TIM1_CH3                           | HRTIM_CH_C2                     | LPUART1_RX                                             | -                                                             | -                      | -                                           | USART1_RX                                      | -                                                            | FDCAN1_TXFD_MODE                                                                   | OTG_FS_ID                                                              | MDIOS_MDIO                                                                   | LCD_B4                                               | DCMI_D1                    | LCD_B1        | EVENT-OUT |
|        | PA11 | -           | TIM1_CH4                           | HRTIM_CH_D1                     | LPUART1_CTS                                            | -                                                             | SPI2_NSS<br>/I2S2_WS   | UART4_RX                                    | USART1_CTS/<br>USART1_NSS                      | -                                                            | FDCAN1_RX                                                                          | OTG_FS_DM                                                              | -                                                                            | -                                                    | -                          | LCD_R4        | EVENT-OUT |
|        | PA12 | -           | TIM1_ETR                           | HRTIM_CH_D2                     | LPUART1_RTS/<br>LPUART1_DE                             | -                                                             | SPI2_SCK/<br>I2S2_CK   | UART4_TX                                    | USART1_RTS/<br>USART1_DE                       | SAI2_FS_B                                                    | FDCAN1_TX                                                                          | OTG_FS_DP                                                              | -                                                                            | -                                                    | -                          | LCD_R5        | EVENT-OUT |



Table 10. Port A alternate functions (continued)

| Port   | AF0  | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        | SYS  | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1/<br>3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port A | PA13 | JTMS-<br>SWDIO                     | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PA14 | JTCK-<br>SWCLK                     | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PA15 | JTDI                               | TIM2_CH1/<br>TIM2_ETR           | HRTIM_<br>FLT1                                         | -                                                             | CEC                    | SPI1_NSS/<br>I2S1_WS                        | SPI3_NSS/<br>I2S3_WS                           | SPI6_NSS                                                     | UART4_<br>RTS/<br>UART4_<br>DE                                                     | -                                                                      | -                                                                            | UART7_TX                                            | -                          | -             | EVENT-<br>OUT |

Table 11. Port B alternate functions

| Port   |     | AF0           | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15      |
|--------|-----|---------------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|-----------|
|        |     | SYS           | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/5/<br>6/CEC | SPI2/3/SAI1/<br>3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/3<br>/6/UART7/S<br>DMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/<br>DCMI/LCD<br>/COMP | UART5/<br>LCD | SYS       |
| Port B | PB0 | -             | TIM1_CH2N                          | TIM3_CH3                        | TIM8_CH2N                                              | -                                                             | -                      | DFSDM1_CKOUT                                | -                                              | UART4_CTS                                                    | LCD_R3                                                                             | OTG_HS_ULPI_D1                                                         | ETH_MII_RXD2                                                                 | -                                                   | -                          | LCD_G1        | EVENT-OUT |
|        | PB1 | -             | TIM1_CH3N                          | TIM3_CH4                        | TIM8_CH3N                                              | -                                                             | -                      | DFSDM1_DATIN1                               | -                                              | -                                                            | LCD_R6                                                                             | OTG_HS_ULPI_D2                                                         | ETH_MII_RXD3                                                                 | -                                                   | -                          | LCD_G0        | EVENT-OUT |
|        | PB2 | RTC_OUT       | -                                  | SAI1_D1                         | -                                                      | DFSDM1_CKIN1                                                  | -                      | SAI1_SD_A                                   | SPI3_MOSI/I2S3_SDO                             | SAI4_SD_A                                                    | QUADSPI_CLK                                                                        | SAI4_D1                                                                | -                                                                            | -                                                   | -                          | -             | EVENT-OUT |
|        | PB3 | JTDO/TRACESWO | TIM2_CH2                           | HRTIM_FLT4                      | -                                                      | -                                                             | SPI1_SCK/I2S1_CK       | SPI3_SCK/I2S3_CK                            | -                                              | SPI6_SCK                                                     | SDMMC2_D2                                                                          | CRS_SYNC                                                               | UART7_RX                                                                     | -                                                   | -                          | -             | EVENT-OUT |
|        | PB4 | NJTRST        | TIM16_BKIN                         | TIM3_CH1                        | HRTIM_EEV6                                             | -                                                             | SPI1_MISO/I2S1_SDI     | SPI3_MISO/I2S3_SDI                          | SPI2_NSS/I2S2_WS                               | SPI6_MISO                                                    | SDMMC2_D3                                                                          | -                                                                      | UART7_TX                                                                     | -                                                   | -                          | -             | EVENT-OUT |
|        | PB5 | -             | TIM17_BKIN                         | TIM3_CH2                        | HRTIM_EEV7                                             | I2C1_SMBA                                                     | SPI1_MOSI/I2S1_SDO     | I2C4_SMBA                                   | SPI3_MOSI/I2S3_SDO                             | SPI6_MOSI                                                    | FDCAN2_RX                                                                          | OTG_HS_ULPI_D7                                                         | ETH_PPS_OUT                                                                  | FMC_SDCKE1                                          | DCMI_D10                   | UART5_RX      | EVENT-OUT |
|        | PB6 | -             | TIM16_CH1N                         | TIM4_CH1                        | HRTIM_EEV8                                             | I2C1_SCL                                                      | CEC                    | I2C4_SCL                                    | USART1_TX                                      | LPUART1_TX                                                   | FDCAN2_TX                                                                          | QUADSPI_BK1_NCS                                                        | DFSDM1_DATIN5                                                                | FMC_SDNE1                                           | DCMI_D5                    | UART5_TX      | EVENT-OUT |
|        | PB7 | -             | TIM17_CH1N                         | TIM4_CH2                        | HRTIM_EEV9                                             | I2C1_SDA                                                      | -                      | I2C4_SDA                                    | USART1_RX                                      | LPUART1_RX                                                   | FDCAN2_TXFD_MODE                                                                   | -                                                                      | DFSDM1_CKIN5                                                                 | FMC_NL                                              | DCMI_VSYNC                 | -             | EVENT-OUT |

**Table 11. Port B alternate functions (continued)**

| Port   |      | AF0       | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15      |
|--------|------|-----------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|-----------|
|        |      | SYS       | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/5/<br>6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/3<br>/6/UART7/S<br>DMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/<br>DCMI/LCD<br>/COMP | UART5/<br>LCD | SYS       |
| Port B | PB8  | -         | TIM16_CH1                          | TIM4_CH3                        | DFSDM1_CKIN7                                           | I2C1_SCL                                                      | -                      | I2C4_SCL                                    | SDMMC1_CKIN                                    | UART4_RX                                                     | FDCAN1_RX                                                                          | SDMMC2_D4                                                              | ETH_MII_TXD3                                                                 | SDMMC1_D4                                           | DCMI_D6                    | LCD_B6        | EVENT-OUT |
|        | PB9  | -         | TIM17_CH1                          | TIM4_CH4                        | DFSDM1_DATIN7                                          | I2C1_SDA                                                      | SPI2_NSS/I2S2_WS       | I2C4_SDA                                    | SDMMC1_CDIR                                    | UART4_TX                                                     | FDCAN1_TX                                                                          | SDMMC2_D5                                                              | I2C4_SMBA                                                                    | SDMMC1_D5                                           | DCMI_D7                    | LCD_B7        | EVENT-OUT |
|        | PB10 | -         | TIM2_CH3                           | HRTIM_SCOUT                     | LPTIM2_IN1                                             | I2C2_SCL                                                      | SPI2_SCK/I2S2_CK       | DFSDM1_DATIN7                               | USART3_TX                                      | -                                                            | QUADSPI_BK1_NCS                                                                    | OTG_HS_ULPI_D3                                                         | ETH_MII_RX_ER                                                                | -                                                   | -                          | LCD_G4        | EVENT-OUT |
|        | PB11 | -         | TIM2_CH4                           | HRTIM_SCIN                      | LPTIM2_ETR                                             | I2C2_SDA                                                      | -                      | DFSDM1_CKIN7                                | USART3_RX                                      | -                                                            | -                                                                                  | OTG_HS_ULPI_D4                                                         | ETH_MII_TX_EN/<br>ETH_RMII_TX_EN                                             | -                                                   | -                          | LCD_G5        | EVENT-OUT |
|        | PB12 | -         | TIM1_BKIN                          | -                               | -                                                      | I2C2_SMBA                                                     | SPI2_NSS/I2S2_WS       | DFSDM1_DATIN1                               | USART3_CK                                      | -                                                            | FDCAN2_RX                                                                          | OTG_HS_ULPI_D5                                                         | ETH_MII_TXD0/ETH_RMII_TXD0                                                   | OTG_HS_ID                                           | TIM1_BKIN_COMP12           | UART5_RX      | EVENT-OUT |
|        | PB13 | -         | TIM1_CH1N                          | -                               | LPTIM2_OUT                                             | -                                                             | SPI2_SCK/I2S2_CK       | DFSDM1_CKIN1                                | USART3_CTS/<br>USART3_NSS                      | -                                                            | FDCAN2_TX                                                                          | OTG_HS_ULPI_D6                                                         | ETH_MII_TXD1/ETH_RMII_TXD1                                                   | -                                                   | -                          | UART5_TX      | EVENT-OUT |
|        | PB14 | -         | TIM1_CH2N                          | TIM12_CH1                       | TIM8_CH2N                                              | USART1_TX                                                     | SPI2_MISO/I2S2_SDI     | DFSDM1_DATIN2                               | USART3_RTS/<br>USART3_DE                       | UART4_RTS/<br>UART4_DE                                       | SDMMC2_D0                                                                          | -                                                                      | -                                                                            | OTG_HS_DM                                           | -                          | -             | EVENT-OUT |
|        | PB15 | RTC_REFIN | TIM1_CH3N                          | TIM12_CH2                       | TIM8_CH3N                                              | USART1_RX                                                     | SPI2_MOSI/I2S2_SDO     | DFSDM1_CKIN2                                | -                                              | UART4_CTS                                                    | SDMMC2_D1                                                                          | -                                                                      | -                                                                            | OTG_HS_DP                                           | -                          | -             | EVENT-OUT |



Table 12. Port C alternate functions

| Port   |      | AF0     | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15      |
|--------|------|---------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|-----------|
|        |      | SYS     | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS       |
| Port C | PC0  | -       | -                                  | -                               | DFSDM1_CKIN0                                           | -                                                             | -                      | DFSDM1_DATIN4                               | -                                              | SAI2_FS_B                                                    | -                                                                                  | OTG_HS_ULPI_STP                                                        | -                                                                            | FMC_SDNWE                                           | -                          | LCD_R5        | EVENT-OUT |
|        | PC1  | TRACED0 | -                                  | SAI1_D1                         | DFSDM1_DATIN0                                          | DFSDM1_CKIN4                                                  | SPI2_MOSI/I2S2_SDO     | SAI1_SD_A                                   | -                                              | SAI4_SD_A                                                    | SDMMC2_CK                                                                          | SAI4_D1                                                                | ETH_MDC                                                                      | MDIOS_MDC                                           | -                          | -             | EVENT-OUT |
|        | PC2  | CDSLEEP | -                                  | -                               | DFSDM1_CKIN1                                           | -                                                             | SPI2_MISO/I2S2_SDI     | DFSDM1_CKOUT                                | -                                              | -                                                            | -                                                                                  | OTG_HS_ULPI_DIR                                                        | ETH_MII_TXD2                                                                 | FMC_SDNE0                                           | -                          | -             | EVENT-OUT |
|        | PC3  | CSLEEP  | -                                  | -                               | DFSDM1_DATIN1                                          | -                                                             | SPI2_MOSI/I2S2_SDO     | -                                           | -                                              | -                                                            | -                                                                                  | OTG_HS_ULPI_NXT                                                        | ETH_MII_TX_CLK                                                               | FMC_SDCKE0                                          | -                          | -             | EVENT-OUT |
|        | PC4  | -       | -                                  | -                               | DFSDM1_CKIN2                                           | -                                                             | I2S1_MCK               | -                                           | -                                              | -                                                            | SPDIFRX1_IN3                                                                       | -                                                                      | ETH_MII_RXD0/ETH_RMII_RXD0                                                   | FMC_SDNE0                                           | -                          | -             | EVENT-OUT |
|        | PC5  | -       | -                                  | SAI1_D3                         | DFSDM1_DATIN2                                          | -                                                             | -                      | -                                           | -                                              | SPDIFRX1_IN4                                                 | SAI4_D3                                                                            | ETH_MII_RXD1/ETH_RMII_RXD1                                             | FMC_SDCKE0                                                                   | COMP1_OUT                                           | -                          | -             | EVENT-OUT |
|        | PC6  | -       | HRTIM_CH_A1                        | TIM3_CH1                        | TIM8_CH1                                               | DFSDM1_CKIN3                                                  | I2S2_MCK               | -                                           | USART6_TX                                      | SDMMC1_D0DIR                                                 | FMC_NWAIT                                                                          | SDMMC2_D6                                                              | -                                                                            | SDMMC1_D6                                           | DCMI_D0                    | LCD_HSYNC     | EVENT-OUT |
|        | PC7  | TRGIO   | HRTIM_CH_A2                        | TIM3_CH2                        | TIM8_CH2                                               | DFSDM1_DATIN3                                                 | -                      | I2S3_MCK                                    | USART6_RX                                      | SDMMC1_D123DIR                                               | FMC_NE1                                                                            | SDMMC2_D7                                                              | SWPMI_TX                                                                     | SDMMC1_D7                                           | DCMI_D1                    | LCD_G6        | EVENT-OUT |
|        | PC8  | TRACED1 | HRTIM_CH_B1                        | TIM3_CH3                        | TIM8_CH3                                               | -                                                             | -                      | -                                           | USART6_CK                                      | UART5_RTS/UART5_DE                                           | FMC_NE2/FMC_NCE                                                                    | -                                                                      | SWPMI_RX                                                                     | SDMMC1_D0                                           | DCMI_D2                    | -             | EVENT-OUT |
|        | PC9  | MCO2    | -                                  | TIM3_CH4                        | TIM8_CH4                                               | I2C3_SDA                                                      | I2S_CKIN               | -                                           | -                                              | UART5_CTS                                                    | QUADSPI_BK1_IO0                                                                    | LCD_G3                                                                 | SWPMI_SUSPEND                                                                | SDMMC1_D1                                           | DCMI_D3                    | LCD_B2        | EVENT-OUT |
|        | PC10 | -       | -                                  | HRTIM_EEV1                      | DFSDM1_CKIN5                                           | -                                                             | -                      | SPI3_SCK/I2S3_CK                            | USART3_TX                                      | UART4_TX                                                     | QUADSPI_BK1_IO1                                                                    | -                                                                      | -                                                                            | SDMMC1_D2                                           | DCMI_D8                    | LCD_R2        | EVENT-OUT |
|        | PC11 | -       | -                                  | HRTIM_FLT2                      | DFSDM1_DATIN5                                          | -                                                             | -                      | SPI3_MISO/I2S3_SDI                          | USART3_RX                                      | UART4_RX                                                     | QUADSPI_BK2_NCS                                                                    | -                                                                      | -                                                                            | SDMMC1_D3                                           | DCMI_D4                    | -             | EVENT-OUT |
|        | PC12 | TRACED3 | -                                  | HRTIM_EEV2                      | -                                                      | -                                                             | -                      | SPI3_MOSI/I2S3_SDO                          | USART3_CK                                      | UART5_TX                                                     | -                                                                                  | -                                                                      | -                                                                            | SDMMC1_CK                                           | DCMI_D9                    | -             | EVENT-OUT |
|        | PC13 | -       | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | -         |

**Table 12. Port C alternate functions (continued)**

| Port   |      | AF0 | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|-----|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        |      | SYS | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port C | PC14 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PC15 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |

**Table 13. Port D alternate functions**

| Port   |     | AF0     | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                        | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|-----|---------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|----------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        |     | SYS     | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC     | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port D | PD0 | -       | -                                  | -                               | DFSDM1_<br>CKIN6                                       | -                                                             | -                          | SAI3_SCK_<br>A                              | -                                              | UART4_RX                                                     | FDCAN1_<br>RX                                                                      | -                                                                      | -                                                                            | FMC_D2/<br>FMC_DA2                                  | -                          | -             | EVENT-<br>OUT |
|        | PD1 | -       | -                                  | -                               | DFSDM1_<br>DATIN6                                      | -                                                             | -                          | SAI3_SD_A                                   | -                                              | UART4_TX                                                     | FDCAN1_<br>TX                                                                      | -                                                                      | -                                                                            | FMC_D3/<br>FMC_DA3                                  | -                          | -             | EVENT-<br>OUT |
|        | PD2 | TRACED2 | -                                  | TIM3_ETR                        | -                                                      | -                                                             | -                          | -                                           | -                                              | UART5_RX                                                     | -                                                                                  | -                                                                      | -                                                                            | SDMMC1_<br>CMD                                      | DCMI_D11                   | -             | EVENT-<br>OUT |
|        | PD3 | -       | -                                  | -                               | DFSDM1_<br>CKOUT                                       | -                                                             | SPI2_SCK/<br>I2S2_CK       | -                                           | USART2_<br>CTS/<br>USART2_<br>NSS              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_CLK                                             | DCMI_D5                    | LCD_G7        | EVENT-<br>OUT |
|        | PD4 | -       | -                                  | HRTIM_<br>FLT3                  | -                                                      | -                                                             | -                          | SAI3_FS_A                                   | USART2_<br>RTS/<br>USART2_<br>DE               | -                                                            | FDCAN1_R<br>XFD_MODE                                                               | -                                                                      | -                                                                            | FMC_NOE                                             | -                          | -             | EVENT-<br>OUT |
|        | PD5 | -       | -                                  | HRTIM_<br>EEV3                  | -                                                      | -                                                             | -                          | -                                           | USART2_<br>TX                                  | -                                                            | FDCAN1_T<br>XFD_MODE                                                               | -                                                                      | -                                                                            | FMC_NWE                                             | -                          | -             | EVENT-<br>OUT |
|        | PD6 | -       | -                                  | SAI1_D1                         | DFSDM1_<br>CKIN4                                       | DFSDM1_<br>DATIN1                                             | SPI3_<br>MOSI/I2S3_<br>SDO | SAI1_SD_A                                   | USART2_<br>RX                                  | SAI4_SD_<br>A                                                | FDCAN2_R<br>XFD_MODE                                                               | SAI4_D1                                                                | SDMMC2_<br>CK                                                                | FMC_NWAIT                                           | DCMI_D10                   | LCD_B2        | EVENT-<br>OUT |
|        | PD7 | -       | -                                  | -                               | DFSDM1_<br>DATIN4                                      | -                                                             | SPI1_<br>MOSI/I2S1_<br>SDO | DFSDM1_<br>CKIN1                            | USART2_<br>CK                                  | -                                                            | SPDIFRX1_<br>IN1                                                                   | -                                                                      | SDMMC2_<br>CMD                                                               | FMC_NE1                                             | -                          | -             | EVENT-<br>OUT |



Table 13. Port D alternate functions (continued)

| Port   |      | AF0 | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|-----|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        |      | SYS | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SP11/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port D | PD8  | -   | -                                  | -                               | DFSDM1_ CKIN3                                          | -                                                             | -                      | SAI3_SCK_ B                                 | USART3_ TX                                     | -                                                            | SPDIFRX1_ IN2                                                                      | -                                                                      | -                                                                            | FMC_D13/<br>FMC_DA13                                | -                          | -             | EVENT-<br>OUT |
|        | PD9  | -   | -                                  | -                               | DFSDM1_ DATIN3                                         | -                                                             | -                      | SAI3_SD_ B                                  | USART3_ RX                                     | -                                                            | FDCAN2_R XFD_MODE                                                                  | -                                                                      | -                                                                            | FMC_D14/<br>FMC_DA14                                | -                          | -             | EVENT-<br>OUT |
|        | PD10 | -   | -                                  | -                               | DFSDM1_ CKOUT                                          | -                                                             | -                      | SAI3_FS_ B                                  | USART3_ CK                                     | -                                                            | FDCAN2_T XFD_MODE                                                                  | -                                                                      | -                                                                            | FMC_D15/<br>FMC_DA15                                | -                          | LCD_B3        | EVENT-<br>OUT |
|        | PD11 | -   | -                                  | -                               | LPTIM2_ IN2                                            | I2C4_SMBA                                                     | -                      | -                                           | USART3_ CTS/<br>USART3_ N SS                   | -                                                            | QUADSPI_ BK1_IO0                                                                   | SAI2_SD_ A                                                             | -                                                                            | FMC_A16                                             | -                          | -             | EVENT-<br>OUT |
|        | PD12 | -   | LPTIM1_IN1                         | TIM4_CH1                        | LPTIM2_ IN1                                            | I2C4_SCL                                                      | -                      | -                                           | USART3_ RTS/<br>USART3_ DE                     | -                                                            | QUADSPI_ BK1_IO1                                                                   | SAI2_FS_ A                                                             | -                                                                            | FMC_A17                                             | -                          | -             | EVENT-<br>OUT |
|        | PD13 | -   | LPTIM1_ OUT                        | TIM4_CH2                        | -                                                      | I2C4_SDA                                                      | -                      | -                                           | -                                              | -                                                            | QUADSPI_ BK1_IO3                                                                   | SAI2_SCK_ A                                                            | -                                                                            | FMC_A18                                             | -                          | -             | EVENT-<br>OUT |
|        | PD14 | -   | -                                  | TIM4_CH3                        | -                                                      | -                                                             | -                      | SAI3_MCLK_ B                                | -                                              | UART8_ CTS                                                   | -                                                                                  | -                                                                      | -                                                                            | FMC_D0/<br>FMC_DA0                                  | -                          | -             | EVENT-<br>OUT |
|        | PD15 | -   | -                                  | TIM4_CH4                        | -                                                      | -                                                             | -                      | SAI3_MCLK_ A                                | -                                              | UART8_ RTS/<br>UART8_ DE                                     | -                                                                                  | -                                                                      | -                                                                            | FMC_D1/<br>FMC_DA1                                  | -                          | -             | EVENT-<br>OUT |

Table 14. Port E alternate functions

| Port   |      | AF0       | AF1                        | AF2                     | AF3                                    | AF4                                       | AF5                | AF6                             | AF7                                | AF8                                          | AF9                                                    | AF10                                           | AF11                                             | AF12                                | AF13               | AF14      | AF15      |
|--------|------|-----------|----------------------------|-------------------------|----------------------------------------|-------------------------------------------|--------------------|---------------------------------|------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------|--------------------------------------------------|-------------------------------------|--------------------|-----------|-----------|
|        |      | SYS       | TIM1/2/16/17/LPTIM1/HRTIM1 | SAI1/TIM3/4/5/12/HRTIM1 | LPUART/TIM8/LPTIM2/3/4/5/HRTIM1/DFSDM1 | I2C1/2/3/4/USART1/TIM15/LPTIM2/DFSDM1/CEC | SPI1/2/3/4/5/6/CEC | SPI2/3/SAI1/3/I2C4/UART4/DFSDM1 | SPI2/3/6/USART1/2/3/6/UART7/SDMMC1 | SPI6/SAI2/4/UART4/5/8/LPUART/SDMMC1/SPDIFRX1 | SAI4/FDCAN1/2/TIM13/14/QUADSPI/FMC/SDMMC2/LCD/SPDIFRX1 | SAI2/4/TIM8/QUADSPI/SDMMC2/OTG1_HS/OTG2_FS/LCD | I2C4/UART7/SWPMI1/TIM1/8/DFSDM1/SDMMC2/MDIOS/ETH | TIM1/8/FMC/SDMMC1/MDIOS/OTG1_FS/LCD | TIM1/DCMI/LCD/COMP | UART5/LCD | SYS       |
| Port E | PE0  | -         | LPTIM1_ETR                 | TIM4_ETR                | HRTIM_SCIN                             | LPTIM2_ETR                                | -                  | -                               | -                                  | UART8_RX                                     | FDCAN1_RXFD_MODE                                       | SAI2_MCLK_A                                    | -                                                | FMC_NBL0                            | DCMI_D2            | -         | EVENT-OUT |
|        | PE1  | -         | LPTIM1_IN2                 | -                       | HRTIM_SCOUT                            | -                                         | -                  | -                               | -                                  | UART8_TX                                     | FDCAN1_TXFD_MODE                                       | -                                              | -                                                | FMC_NBL1                            | DCMI_D3            | -         | EVENT-OUT |
|        | PE2  | TRACE_CLK | -                          | SAI1_CK1                | -                                      | -                                         | SPI4_SCK           | SAI1_MCLK_A                     | -                                  | SAI4_MCLK_A                                  | QUADSPI_BK1_IO2                                        | SAI4_CK1                                       | ETH_MII_TXD3                                     | FMC_A23                             | -                  | -         | EVENT-OUT |
|        | PE3  | TRACED0   | -                          | -                       | -                                      | TIM15_BKIN                                | -                  | SAI1_SD_B                       | -                                  | SAI4_SD_B                                    | -                                                      | -                                              | -                                                | FMC_A19                             | -                  | -         | EVENT-OUT |
|        | PE4  | TRACED1   | -                          | SAI1_D2                 | DFSDM1_DATIN3                          | TIM15_CH1_N                               | SPI4_NSS           | SAI1_FS_A                       | -                                  | SAI4_FS_A                                    | -                                                      | SAI4_D2                                        | -                                                | FMC_A20                             | DCMI_D4            | LCD_B0    | EVENT-OUT |
|        | PE5  | TRACED2   | -                          | SAI1_CK2                | DFSDM1_CKIN3                           | TIM15_CH1                                 | SPI4_MISO          | SAI1_SCK_A                      | -                                  | SAI4_SCK_A                                   | -                                                      | SAI4_CK2                                       | -                                                | FMC_A21                             | DCMI_D6            | LCD_G0    | EVENT-OUT |
|        | PE6  | TRACED3   | TIM1_BKIN2                 | SAI1_D1                 | -                                      | TIM15_CH2                                 | SPI4_MOSI          | SAI1_SD_A                       | -                                  | SAI4_SD_A                                    | SAI4_D1                                                | SAI2_MCLK_B                                    | TIM1_BKIN2_COMP12                                | FMC_A22                             | DCMI_D7            | LCD_G1    | EVENT-OUT |
|        | PE7  | -         | TIM1_ETR                   | -                       | DFSDM1_DATIN2                          | -                                         | -                  | -                               | UART7_RX                           | -                                            | -                                                      | QUADSPI_BK2_IO0                                | -                                                | FMC_D4/FMC_DA4                      | -                  | -         | EVENT-OUT |
|        | PE8  | -         | TIM1_CH1N                  | -                       | DFSDM1_CKIN2                           | -                                         | -                  | -                               | UART7_TX                           | -                                            | -                                                      | QUADSPI_BK2_IO1                                | -                                                | FMC_D5/FMC_DA5                      | COMP2_OUT          | -         | EVENT-OUT |
|        | PE9  | -         | TIM1_CH1                   | -                       | DFSDM1_CKOUT                           | -                                         | -                  | -                               | UART7_RTS/UART7_DE                 | -                                            | -                                                      | QUADSPI_BK2_IO2                                | -                                                | FMC_D6/FMC_DA6                      | -                  | -         | EVENT-OUT |
|        | PE10 | -         | TIM1_CH2N                  | -                       | DFSDM1_DATIN4                          | -                                         | -                  | -                               | UART7_CTS                          | -                                            | -                                                      | QUADSPI_BK2_IO3                                | -                                                | FMC_D7/FMC_DA7                      | -                  | -         | EVENT-OUT |
|        | PE11 | -         | TIM1_CH2                   | -                       | DFSDM1_CKIN4                           | -                                         | SPI4_NSS           | -                               | -                                  | -                                            | -                                                      | SAI2_SD_B                                      | -                                                | FMC_D8/FMC_DA8                      | -                  | LCD_G3    | EVENT-OUT |
|        | PE12 | -         | TIM1_CH3N                  | -                       | DFSDM1_DATIN5                          | -                                         | SPI4_SCK           | -                               | -                                  | -                                            | -                                                      | SAI2_SCK_B                                     | -                                                | FMC_D9/FMC_DA9                      | COMP1_OUT          | LCD_B4    | EVENT-OUT |
|        | PE13 | -         | TIM1_CH3                   | -                       | DFSDM1_CKIN5                           | -                                         | SPI4_MISO          | -                               | -                                  | -                                            | -                                                      | SAI2_FS_B                                      | -                                                | FMC_D10/FMC_DA10                    | COMP2_OUT          | LCD_DE    | EVENT-OUT |



**Table 14. Port E alternate functions (continued)**

| Port   |      | AF0 | AF1                        | AF2                     | AF3                                    | AF4                                       | AF5                | AF6                             | AF7                                | AF8                                          | AF9                                                    | AF10                                           | AF11                                             | AF12                                | AF13                            | AF14      | AF15      |
|--------|------|-----|----------------------------|-------------------------|----------------------------------------|-------------------------------------------|--------------------|---------------------------------|------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------|--------------------------------------------------|-------------------------------------|---------------------------------|-----------|-----------|
|        |      | SYS | TIM1/2/16/17/LPTIM1/HRTIM1 | SAI1/TIM3/4/5/12/HRTIM1 | LPUART/TIM8/LPTIM2/3/4/5/HRTIM1/DFSDM1 | I2C1/2/3/4/USART1/TIM15/LPTIM2/DFSDM1/CEC | SPI1/2/3/4/5/6/CEC | SPI2/3/SAI1/3/I2C4/UART4/DFSDM1 | SPI2/3/6/USART1/2/3/6/UART7/SDMMC1 | SPI6/SAI2/4/UART4/5/8/LPUART/SDMMC1/SPDIFRX1 | SAI4/FDCAN1/2/TIM13/14/QUADSPI/FMC/SDMMC2/LCD/SPDIFRX1 | SAI2/4/TIM8/QUADSPI/SDMMC2/OTG1_HS/OTG2_FS/LCD | I2C4/UART7/SWPMI1/TIM1/8/DFSDM1/SDMMC2/MDIOS/ETH | TIM1/8/FMC/SDMMC1/MDIOS/OTG1_FS/LCD | TIM1/DCMI/LCD/COMP              | UART5/LCD | SYS       |
| Port E | PE14 | -   | TIM1_CH4                   | -                       | -                                      | -                                         | SPI4_MOSI          | -                               | -                                  | -                                            | -                                                      | SAI2_MCLK_B                                    | -                                                | FMC_D11/FMC_DA11                    | -                               | LCD_CLK   | EVENT-OUT |
|        | PE15 | -   | TIM1_BKIN                  | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      |                                                | -                                                | FMC_D12/FMC_DA12                    | TIM1_BKIN_COMP12/COMP_TIM1_BKIN | LCD_R7    | EVENT-OUT |

**Table 15. Port F alternate functions**

| Port   |      | AF0 | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|-----|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        |      | SYS | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port F | PF0  | -   | -                                  | -                               | -                                                      | I2C2_SDA                                                      | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A0                                              | -                          | -             | EVENT-<br>OUT |
|        | PF1  | -   | -                                  | -                               | -                                                      | I2C2_SCL                                                      | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A1                                              | -                          | -             | EVENT-<br>OUT |
|        | PF2  | -   | -                                  | -                               | -                                                      | I2C2_SMBA                                                     | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A2                                              | -                          | -             | EVENT-<br>OUT |
|        | PF3  | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A3                                              | -                          | -             | EVENT-<br>OUT |
|        | PF4  | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A4                                              | -                          | -             | EVENT-<br>OUT |
|        | PF5  | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A5                                              | -                          | -             | EVENT-<br>OUT |
|        | PF6  | -   | TIM16_CH1                          | -                               | -                                                      | -                                                             | SPI5_NSS               | SAI1_SD_B                                   | UART7_RX                                       | SAI4_SD_B                                                    | QUADSPI_BK1_IO3                                                                    | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PF7  | -   | TIM17_CH1                          | -                               | -                                                      | -                                                             | SPI5_SCK               | SAI1_MCLK_B                                 | UART7_TX                                       | SAI4_MCLK_B                                                  | QUADSPI_BK1_IO2                                                                    | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PF8  | -   | TIM16_CH1N                         | -                               | -                                                      | -                                                             | SPI5_MISO              | SAI1_SCK_B                                  | UART7_RTS/<br>UART7_DE                         | SAI4_SCK_B                                                   | TIM13_CH1                                                                          | QUADSPI_BK1_IO0                                                        | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PF9  | -   | TIM17_CH1N                         | -                               | -                                                      | -                                                             | SPI5_MOSI              | SAI1_FS_B                                   | UART7_CTS                                      | SAI4_FS_B                                                    | TIM14_CH1                                                                          | QUADSPI_BK1_IO1                                                        | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PF10 | -   | TIM16_BKIN                         | SAI1_D3                         | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | QUADSPI_CLK                                                                        | SAI4_D3                                                                | -                                                                            | -                                                   | DCMI_D11                   | LCD_DE        | EVENT-<br>OUT |
|        | PF11 | -   | -                                  | -                               | -                                                      | -                                                             | SPI5_MOSI              | -                                           | -                                              | -                                                            | -                                                                                  | SAI2_SD_B                                                              | -                                                                            | FMC_SDNRAS                                          | DCMI_D12                   | -             | EVENT-<br>OUT |
|        | PF12 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A6                                              | -                          | -             | EVENT-<br>OUT |
|        | PF13 | -   | -                                  | -                               | DFSDM1_DATIN6                                          | I2C4_SMBA                                                     | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A7                                              | -                          | -             | EVENT-<br>OUT |
|        | PF14 | -   | -                                  | -                               | DFSDM1_CKIN6                                           | I2C4_SCL                                                      | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A8                                              | -                          | -             | EVENT-<br>OUT |
|        | PF15 | -   | -                                  | -                               | -                                                      | I2C4_SDA                                                      | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_A9                                              | -                          | -             | EVENT-<br>OUT |



Table 16. Port G alternate functions

| Port   |      | AF0     | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                        | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                               | AF11                                                                 | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|---------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|----------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        |      | SYS     | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC     | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/UART7<br>/SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/<br>DCMI/LCD<br>/COMP | UART5/<br>LCD | SYS           |
| Port G | PG0  | -       | -                                  | -                               | -                                                      | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | -                                                                    | FMC_A10                                             | -                          | -             | EVENT<br>-OUT |
|        | PG1  | -       | -                                  | -                               | -                                                      | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | -                                                                    | FMC_A11                                             | -                          | -             | EVENT<br>-OUT |
|        | PG2  | -       | -                                  | -                               | TIM8_BKIN                                              | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | TIM8_BKIN_<br>COMP12                                                 | FMC_A12                                             | -                          | -             | EVENT<br>-OUT |
|        | PG3  | -       | -                                  | -                               | TIM8_<br>BKIN2                                         | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | TIM8_BKIN2_<br>COMP12                                                | FMC_A13                                             | -                          | -             | EVENT<br>-OUT |
|        | PG4  | -       | TIM1_<br>BKIN2                     | -                               | -                                                      | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | TIM1_BKIN2_<br>COMP12                                                | FMC_A14/<br>FMC_BA0                                 | -                          | -             | EVENT<br>-OUT |
|        | PG5  | -       | TIM1_ETR                           | -                               | -                                                      | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                  | -                                                                    | FMC_A15/<br>FMC_BA1                                 | -                          | -             | EVENT<br>-OUT |
|        | PG6  | -       | TIM17_<br>BKIN                     | HRTIM_<br>CHE1                  | -                                                      | -                                                             | -                          | -                                           | -                                              | -                                                            | -                                                                                  | QUADSPI_<br>BK1_NCS                                                | -                                                                    | FMC_NE3                                             | DCMI_<br>D12               | LCD_<br>R7    | EVENT<br>-OUT |
|        | PG7  | -       | -                                  | HRTIM_<br>CHE2                  | -                                                      | -                                                             | -                          | SAI1_<br>MCLK_A                             | USART6_<br>CK                                  | -                                                            | -                                                                                  | -                                                                  | -                                                                    | FMC_INT                                             | DCMI_<br>D13               | LCD_<br>CLK   | EVENT<br>-OUT |
|        | PG8  | -       | -                                  | -                               | TIM8_ETR                                               | -                                                             | SPI6_NSS                   | -                                           | USART6_<br>RTS/<br>USART6_<br>DE               | SPDIFRX1_<br>IN3                                             | -                                                                                  | -                                                                  | ETH_PPS_<br>OUT                                                      | FMC_<br>SDCLK                                       | -                          | LCD_<br>G7    | EVENT<br>-OUT |
|        | PG9  | -       | -                                  | -                               | -                                                      | -                                                             | SPI1_<br>MISO/I2S1_<br>SDI | -                                           | USART6_<br>RX                                  | SPDIFRX1_<br>IN4                                             | QUADSPI_<br>BK2_IO2                                                                | SAI2_FS_B                                                          | -                                                                    | FMC_NE2/<br>FMC_NCE                                 | DCMI_<br>VSYNC             | -             | EVENT<br>-OUT |
|        | PG10 | -       | -                                  | HRTIM_<br>FLT5                  | -                                                      | -                                                             | SPI1_NSS/<br>I2S1_WS       | -                                           | -                                              | -                                                            | LCD_G3                                                                             | SAI2_SD_B                                                          | -                                                                    | FMC_NE3                                             | DCMI_D2                    | LCD_<br>B2    | EVENT<br>-OUT |
|        | PG11 | -       | LPTIM1_IN2                         | HRTIM_<br>EEV4                  | -                                                      | -                                                             | SPI1_SCK/<br>I2S1_CK       | -                                           | -                                              | SPDIFRX1_<br>IN1                                             | -                                                                                  | SDMMC2_D2                                                          | ETH_MII_<br>TX_EN/<br>ETH_RMII_<br>TX_EN                             | -                                                   | DCMI_D3                    | LCD_<br>B3    | EVENT<br>-OUT |
|        | PG12 | -       | LPTIM1_IN1                         | HRTIM_<br>EEV5                  | -                                                      | -                                                             | SPI6_MISO                  | -                                           | USART6_<br>RTS/<br>USART6_<br>DE               | SPDIFRX1_<br>IN2                                             | LCD_B4                                                                             | -                                                                  | ETH_MII_<br>TXD0/ETH_<br>RMII_TXD1                                   | FMC_NE4                                             | -                          | LCD_<br>B1    | EVENT<br>-OUT |
|        | PG13 | TRACED0 | LPTIM1_<br>OUT                     | HRTIM_<br>EEV10                 | -                                                      | -                                                             | SPI6_SCK                   | -                                           | USART6_<br>CTS/<br>USART6_<br>NSS              | -                                                            | -                                                                                  | -                                                                  | ETH_MII_<br>TXD0/ETH_<br>RMII_TXD0                                   | FMC_A24                                             | -                          | LCD_<br>R0    | EVENT<br>-OUT |

**Table 16. Port G alternate functions (continued)**

| Port   |      | AF0     | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                               | AF11                                                                 | AF12                                                | AF13                       | AF14          | AF15      |
|--------|------|---------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|-----------|
|        |      | SYS     | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4/<br>5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/UART7<br>/SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/<br>DCMI/LCD<br>/COMP | UART5/<br>LCD | SYS       |
| Port G | PG14 | TRACED1 | LPTIM1_ETR                         | -                               | -                                                      | -                                                             | SPI6_MOSI              | -                                           | USART6_TX                                      |                                                              | QUADSPI_BK2_IO3                                                                    | -                                                                  | ETH_MII_TXD1/ETH_RMII_TXD1                                           | FMC_A25                                             | -                          | LCD_B0        | EVENT-OUT |
|        | PG15 | -       | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | USART6_CTS/<br>USART6_NSS                      | -                                                            | -                                                                                  | -                                                                  | -                                                                    | FMC_SDNCAS                                          | DCMI_D13                   | -             | EVENT-OUT |



Table 17. Port H alternate functions

| Port   | AF0 AF1 AF2 AF3 AF4 AF5 AF6 AF7 AF8 AF9 AF10 AF11 AF12 AF13 AF14 AF15 |                                    |                                 |                                                        |                                                               |                        |                                             |                                                |                                                              |                                                                                    |                                                                        |                                                                              |                                                     |                            |               |               |
|--------|-----------------------------------------------------------------------|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        | SYS                                                                   | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port H | PH0                                                                   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PH1                                                                   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | EVENT-<br>OUT |
|        | PH2                                                                   | -                                  | LPTIM1_IN2                      | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | QUADSPI_<br>BK2_IO0                                                                | SAI2_SCK_<br>B                                                         | ETH_MII_<br>CRS                                                              | FMC_<br>SDCKE0                                      | -                          | LCD_R0        | EVENT-<br>OUT |
|        | PH3                                                                   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | QUADSPI_<br>BK2_IO1                                                                | SAI2_<br>MCLK_B                                                        | ETH_MII_<br>COL                                                              | FMC_<br>SDNE0                                       | -                          | LCD_R1        | EVENT-<br>OUT |
|        | PH4                                                                   | -                                  | -                               | -                                                      | -                                                             | I2C2_SCL               | -                                           | -                                              | -                                                            | LCD_G5                                                                             | OTG_HS_<br>ULPI_NXT                                                    | -                                                                            | -                                                   | -                          | LCD_G4        | EVENT-<br>OUT |
|        | PH5                                                                   | -                                  | -                               | -                                                      | -                                                             | I2C2_SDA               | SPI5_NSS                                    | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_<br>SDNWE                                       | -                          | -             | EVENT-<br>OUT |
|        | PH6                                                                   | -                                  | -                               | TIM12_<br>CH1                                          | -                                                             | I2C2_SMBA              | SPI5_SCK                                    | -                                              | -                                                            | -                                                                                  | -                                                                      | ETH_MII_<br>RXD2                                                             | FMC_<br>SDNE1                                       | DCMI_D8                    | -             | EVENT-<br>OUT |
|        | PH7                                                                   | -                                  | -                               | -                                                      | -                                                             | I2C3_SCL               | SPI5_<br>MISO                               | -                                              | -                                                            | -                                                                                  | -                                                                      | ETH_MII_<br>RXD3                                                             | FMC_<br>SDCKE1                                      | DCMI_D9                    | -             | EVENT-<br>OUT |
|        | PH8                                                                   | -                                  | -                               | TIM5_ETR                                               | -                                                             | I2C3_SDA               | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D16                                             | DCMI_<br>HSYNC             | LCD_R2        | EVENT-<br>OUT |
|        | PH9                                                                   | -                                  | -                               | TIM12_<br>CH2                                          | -                                                             | I2C3_SMBA              | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D17                                             | DCMI_D0                    | LCD_R3        | EVENT-<br>OUT |
|        | PH10                                                                  | -                                  | -                               | TIM5_CH1                                               | -                                                             | I2C4_SMBA              | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D18                                             | DCMI_D1                    | LCD_R4        | EVENT-<br>OUT |
|        | PH11                                                                  | -                                  | -                               | TIM5_CH2                                               | -                                                             | I2C4_SCL               | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D19                                             | DCMI_D2                    | LCD_R5        | EVENT-<br>OUT |
|        | PH12                                                                  | -                                  | -                               | TIM5_CH3                                               | -                                                             | I2C4_SDA               | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D20                                             | DCMI_D3                    | LCD_R6        | EVENT-<br>OUT |
|        | PH13                                                                  | -                                  | -                               | -                                                      | TIM8_<br>CH1N                                                 | -                      | -                                           | -                                              | -                                                            | UART4_TX                                                                           | FDCAN1_<br>TX                                                          | -                                                                            | FMC_D21                                             | -                          | LCD_G2        | EVENT-<br>OUT |
|        | PH14                                                                  | -                                  | -                               | -                                                      | TIM8_<br>CH2N                                                 | -                      | -                                           | -                                              | -                                                            | UART4_RX                                                                           | FDCAN1_<br>RX                                                          | -                                                                            | FMC_D22                                             | DCMI_D4                    | LCD_G3        | EVENT-<br>OUT |
|        | PH15                                                                  | -                                  | -                               | -                                                      | TIM8_<br>CH3N                                                 | -                      | -                                           | -                                              | -                                                            | -                                                                                  | FDCAN1_<br>TXFD_<br>MODE                                               | -                                                                            | FMC_D23                                             | DCMI_D11                   | LCD_G4        | EVENT-<br>OUT |

Table 18. Port I alternate functions

| Port   |      | AF0 | AF1                                | AF2                             | AF3                                                    | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |               |
|--------|------|-----|------------------------------------|---------------------------------|--------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|---------------|
|        |      | SYS | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1 | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |               |
| Port I | PI0  | -   | -                                  | TIM5_CH4                        | -                                                      | -                                                             | SPI2_NSS/<br>I2S2_WS   | -                                           | -                                              | -                                                            | FDCAN1_<br>RXFD_<br>MODE                                                           | -                                                                      | -                                                                            | FMC_D24                                             | DCMI_D13                   | LCD_G5        | EVENT-<br>OUT |               |
|        | PI1  | -   | -                                  | -                               | TIM8_BKIN2                                             | -                                                             | SPI2_SCK/<br>I2S2_CK   | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | TIM8_BKIN<br>2_COMP12                                                        | FMC_D25                                             | DCMI_D8                    | LCD_G6        | EVENT-<br>OUT |               |
|        | PI2  | -   | -                                  | -                               | TIM8_CH4                                               | -                                                             | SPI2_MISO/I2S2<br>_SDI | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D26                                             | DCMI_D9                    | LCD_G7        | EVENT-<br>OUT |               |
|        | PI3  | -   | -                                  | -                               | TIM8_ETR                                               | -                                                             | SPI2_MOSI/I2S2<br>_SDO | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | FMC_D27                                             | DCMI_D10                   | -             | EVENT-<br>OUT |               |
|        | PI4  | -   | -                                  | -                               | TIM8_BKIN                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | SAI2_MCLK_A                                                            | TIM8_BKIN<br>_COMP12                                                         | FMC_NBL2                                            | DCMI_D5                    | LCD_B4        | EVENT-<br>OUT |               |
|        | PI5  | -   | -                                  | -                               | TIM8_CH1                                               | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | SAI2_SCK_A                                                             | -                                                                            | FMC_NBL3                                            | DCMI_VSYNC                 | LCD_B5        | EVENT-<br>OUT |               |
|        | PI6  | -   | -                                  | -                               | TIM8_CH2                                               | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | SAI2_SD_A                                                              | -                                                                            | FMC_D28                                             | DCMI_D6                    | LCD_B6        | EVENT-<br>OUT |               |
|        | PI7  | -   | -                                  | -                               | TIM8_CH3                                               | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | SAI2_FS_A                                                              | -                                                                            | FMC_D29                                             | DCMI_D7                    | LCD_B7        | EVENT-<br>OUT |               |
|        | PI8  | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | -             | EVENT-<br>OUT |
|        | PI9  | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | UART4_RX                                                                           | FDCAN1_<br>RX                                                          | -                                                                            | -                                                   | FMC_D30                    | -             | LCD_VSYNC     | EVENT-<br>OUT |
|        | PI10 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | FDCAN1_<br>RXFD_<br>MODE                                                           | -                                                                      | ETH_MII_<br>RX_ER                                                            | FMC_D31                                             | -                          | LCD_HSYNC     | EVENT-<br>OUT |               |
|        | PI11 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | LCD_G6                                                                             | OTG_HS_<br>ULPI_DIR                                                    | -                                                                            | -                                                   | -                          | -             | -             | EVENT-<br>OUT |
|        | PI12 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | LCD_HSYNC     | EVENT-<br>OUT |
|        | PI13 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | LCD_VSYNC     | EVENT-<br>OUT |
|        | PI14 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | -             | LCD_CLK       | EVENT-<br>OUT |
|        | PI15 | -   | -                                  | -                               | -                                                      | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | LCD_G2                                                                 | -                                                                            | -                                                   | -                          | -             | LCD_R0        | EVENT-<br>OUT |



Table 19. Port J alternate functions

| Port   | AF0  | AF1                                | AF2                             | AF3                                                            | AF4                                                           | AF5                    | AF6                                         | AF7                                            | AF8                                                          | AF9                                                                                | AF10                                                                   | AF11                                                                         | AF12                                                | AF13                       | AF14          | AF15          |
|--------|------|------------------------------------|---------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|------------------------|---------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|----------------------------|---------------|---------------|
|        | SYS  | TIM1/2/16/<br>17/LPTIM1/<br>HRTIM1 | SAI1/TIM3/<br>4/5/12/<br>HRTIM1 | LPUART/<br>TIM8/<br>LPTIM2/3/4<br>/5/HRTIM1/<br>DFSDM1/<br>CEC | I2C1/2/3/4/<br>USART1/<br>TIM15/<br>LPTIM2/<br>DFSDM1/<br>CEC | SPI1/2/3/4/<br>5/6/CEC | SPI2/3/SAI1<br>/3/I2C4/<br>UART4/<br>DFSDM1 | SPI2/3/6/<br>USART1/2/<br>3/6/UART7/<br>SDMMC1 | SPI6/SAI2/<br>4/UART4/5/<br>8/LPUART/<br>SDMMC1/<br>SPDIFRX1 | SAI4/<br>FDCAN1/2/<br>TIM13/14/<br>QUADSPI/<br>FMC/<br>SDMMC2/<br>LCD/<br>SPDIFRX1 | SAI2/4/<br>TIM8/<br>QUADSPI/<br>SDMMC2/<br>OTG1_HS/<br>OTG2_FS/<br>LCD | I2C4/<br>UART7/<br>SWPMI1/<br>TIM1/8/<br>DFSDM1/<br>SDMMC2/<br>MDIOS/<br>ETH | TIM1/8/FMC<br>/SDMMC1/<br>MDIOS/<br>OTG1_FS/<br>LCD | TIM1/DCMI<br>/LCD/<br>COMP | UART5/<br>LCD | SYS           |
| Port J | PJ0  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | LCD_R7                                                                             | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R1        | EVENT-<br>OUT |
|        | PJ1  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R2        | EVENT-<br>OUT |
|        | PJ2  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R3        | EVENT-<br>OUT |
|        | PJ3  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R4        | EVENT-<br>OUT |
|        | PJ4  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R5        | EVENT-<br>OUT |
|        | PJ5  | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R6        | EVENT-<br>OUT |
|        | PJ6  | -                                  | -                               | -                                                              | TIM8_CH2                                                      | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_R7        | EVENT-<br>OUT |
|        | PJ7  | TRGIN                              | -                               | -                                                              | TIM8_CH2N                                                     | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_G0        | EVENT-<br>OUT |
|        | PJ8  | -                                  | TIM1_CH3N                       | -                                                              | TIM8_CH1                                                      | -                      | -                                           | -                                              | UART8_TX                                                     | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_G1        | EVENT-<br>OUT |
|        | PJ9  | -                                  | TIM1_CH3                        | -                                                              | TIM8_CH1N                                                     | -                      | -                                           | -                                              | UART8_RX                                                     | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_G2        | EVENT-<br>OUT |
|        | PJ10 | -                                  | TIM1_CH2N                       | -                                                              | TIM8_CH2                                                      | -                      | SPI5_MOSI                                   | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_G3        | EVENT-<br>OUT |
|        | PJ11 | -                                  | TIM1_CH2                        | -                                                              | TIM8_CH2N                                                     | -                      | SPI5_MISO                                   | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_G4        | EVENT-<br>OUT |
|        | PJ12 | TRGOUT                             | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | LCD_G3                                                                             | -                                                                      | -                                                                            | -                                                   | -                          | LCD_B0        | EVENT-<br>OUT |
|        | PJ13 | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | LCD_B4                                                                             | -                                                                      | -                                                                            | -                                                   | -                          | LCD_B1        | EVENT-<br>OUT |
|        | PJ14 | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_B2        | EVENT-<br>OUT |
|        | PJ15 | -                                  | -                               | -                                                              | -                                                             | -                      | -                                           | -                                              | -                                                            | -                                                                                  | -                                                                      | -                                                                            | -                                                   | -                          | LCD_B3        | EVENT-<br>OUT |

**Table 20. Port K alternate functions**

| Port   | SYS | AF0 | AF1                        | AF2                     | AF3                                    | AF4                                       | AF5                | AF6                             | AF7                                | AF8                                          | AF9                                                    | AF10                                           | AF11                                             | AF12                                | AF13               | AF14      | AF15      |
|--------|-----|-----|----------------------------|-------------------------|----------------------------------------|-------------------------------------------|--------------------|---------------------------------|------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------|--------------------------------------------------|-------------------------------------|--------------------|-----------|-----------|
|        |     |     | TIM1/2/16/17/LPTIM1/HRTIM1 | SAI1/TIM3/4/5/12/HRTIM1 | LPUART/TIM8/LPTIM2/3/4/5/HRTIM1/DFSDM1 | I2C1/2/3/4/USART1/TIM15/LPTIM2/DFSDM1/CEC | SPI1/2/3/4/5/6/CEC | SPI2/3/SAI1/3/I2C4/UART4/DFSDM1 | SPI2/3/6/USART1/2/3/6/UART7/SDMMC1 | SPI6/SAI2/4/UART4/5/8/LPUART/SDMMC1/SPDIFRX1 | SAI4/FDCAN1/2/TIM13/14/QUADSPI/FMC/SDMMC2/LCD/SPDIFRX1 | SAI2/4/TIM8/QUADSPI/SDMMC2/OTG1_HS/OTG2_FS/LCD | I2C4/UART7/SWPMI1/TIM1/8/DFSDM1/SDMMC2/MDIOS/ETH | TIM1/8/FMC/SDMMC1/MDIOS/OTG1_FS/LCD | TIM1/DCMI/LCD/COMP | UART5/LCD | SYS       |
| Port K | PK0 | -   | TIM1_CH1N                  | -                       | TIM8_CH3                               | -                                         | SPI5_SCK           | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_G5    | EVENT-OUT |
|        | PK1 | -   | TIM1_CH1                   | -                       | TIM8_CH3N                              | -                                         | SPI5_NSS           | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_G6    | EVENT-OUT |
|        | PK2 | -   | TIM1_BKIN                  | -                       | TIM8_BKIN                              | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | TIM8_BKIN_COMP12                               | TIM1_BKIN_COMP12                                 | -                                   | -                  | LCD_G7    | EVENT-OUT |
|        | PK3 | -   | -                          | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_B4    | EVENT-OUT |
|        | PK4 | -   | -                          | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_B5    | EVENT-OUT |
|        | PK5 | -   | -                          | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_B6    | EVENT-OUT |
|        | PK6 | -   | -                          | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_B7    | EVENT-OUT |
|        | PK7 | -   | -                          | -                       | -                                      | -                                         | -                  | -                               | -                                  | -                                            | -                                                      | -                                              | -                                                | -                                   | -                  | LCD_DE    | EVENT-OUT |



## 6 Electrical characteristics (rev Y)

### 6.1 Parameter conditions

Unless otherwise specified, all voltages are referenced to  $V_{SS}$ .

#### 6.1.1 Minimum and maximum values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of junction temperature, supply voltage and frequencies by tests in production on 100% of the devices with an junction temperature at  $T_J = 25\text{ }^{\circ}\text{C}$  and  $T_J = T_{Jmax}$  (given by the selected temperature range).

Data based on characterization results, design simulation and/or technology characteristics are indicated in the table footnotes. Based on characterization, the minimum and maximum values refer to sample tests and represent the mean value plus or minus three times the standard deviation ( $\text{mean} \pm 3\sigma$ ).

#### 6.1.2 Typical values

Unless otherwise specified, typical data are based on  $T_J = 25\text{ }^{\circ}\text{C}$ ,  $V_{DD} = 3.3\text{ V}$  (for the  $1.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  voltage range). They are given only as design guidelines and are not tested.

Typical ADC accuracy values are determined by characterization of a batch of samples from a standard diffusion lot over the full temperature range, where 95% of the devices have an error less than or equal to the value indicated ( $\text{mean} \pm 2\sigma$ ).

#### 6.1.3 Typical curves

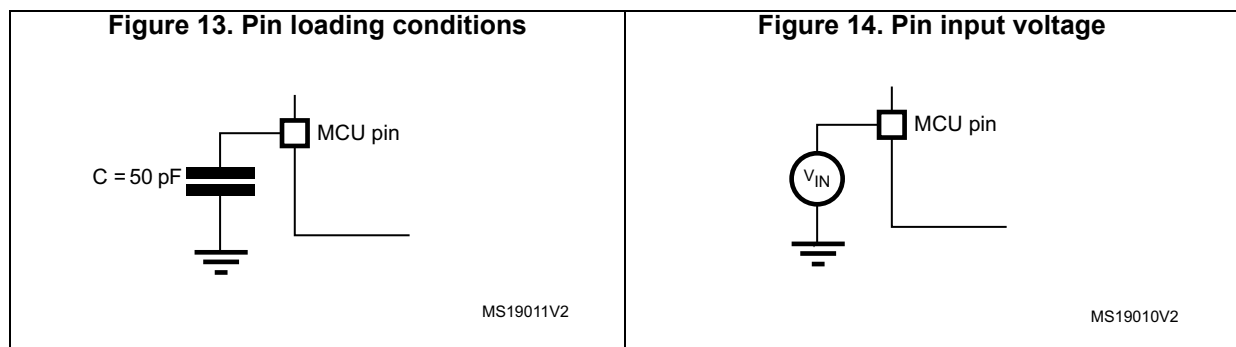
Unless otherwise specified, all typical curves are given only as design guidelines and are not tested.

#### 6.1.4 Loading capacitor

The loading conditions used for pin parameter measurement are shown in [Figure 13](#).

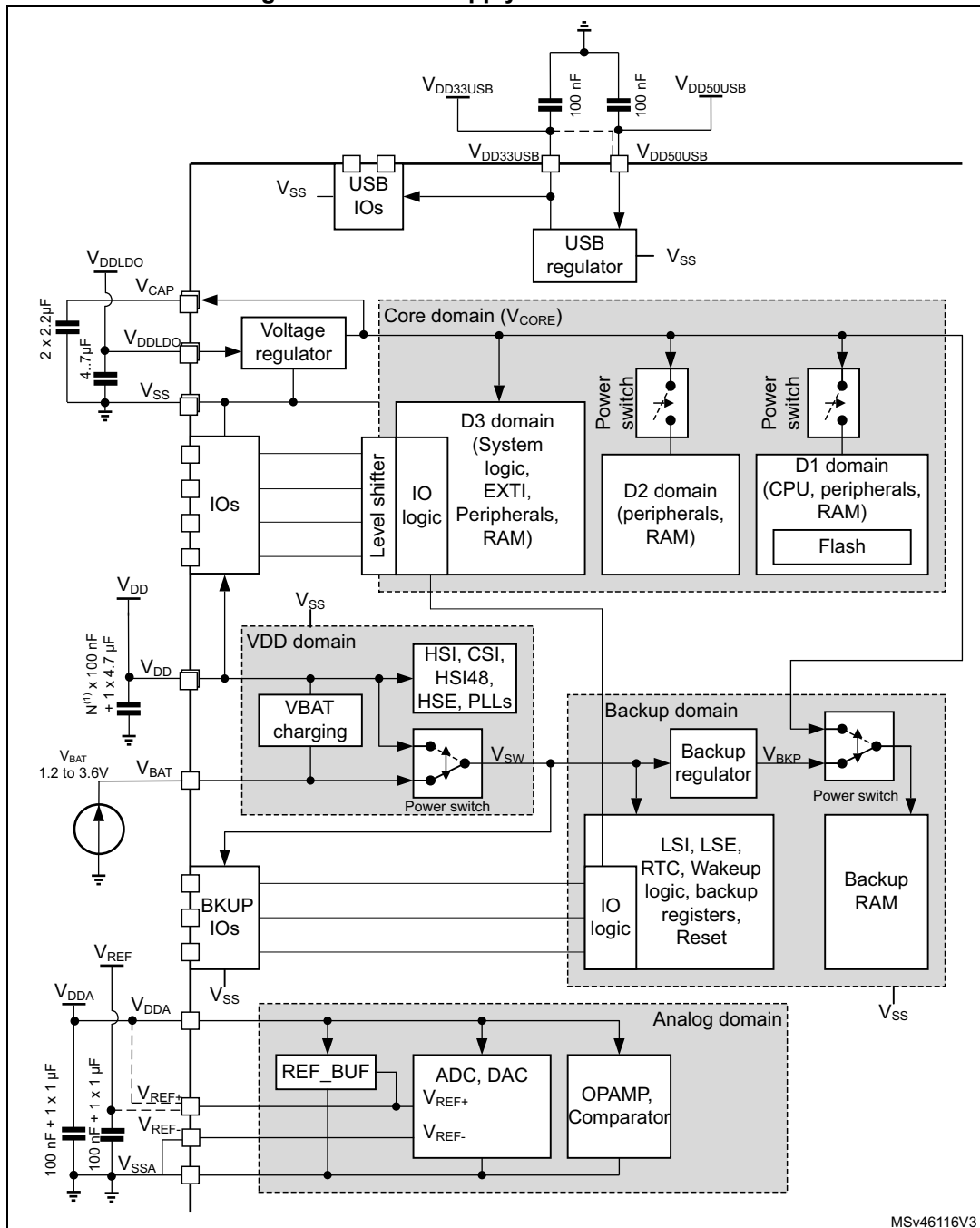
#### 6.1.5 Pin input voltage

The input voltage measurement on a pin of the device is described in [Figure 14](#).



## 6.1.6 Power supply scheme

Figure 15. Power supply scheme



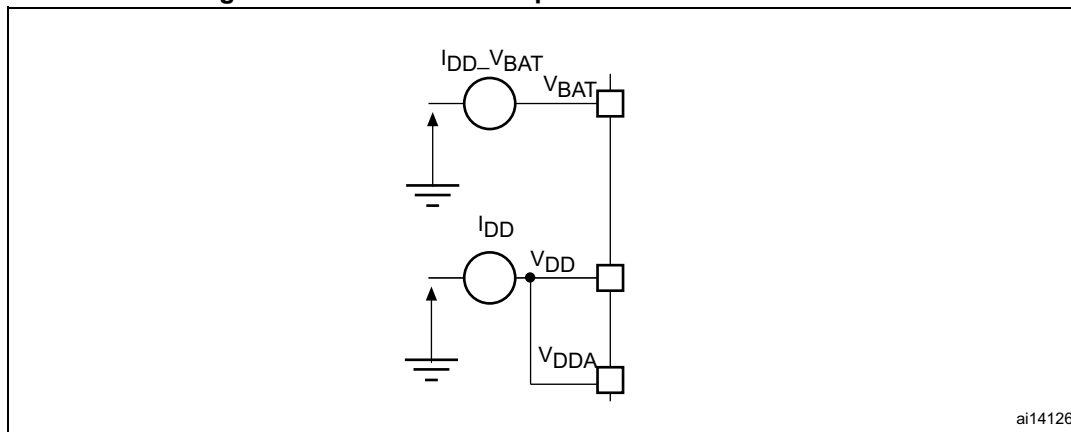
1. N corresponds to the number of VDD pins available on the package.
2. A tolerance of  $\pm 20\%$  is acceptable on decoupling capacitors.

**Caution:** Each power supply pair ( $V_{DD}/V_{SS}$ ,  $V_{DDA}/V_{SSA}$  ...) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure good operation of the

device. It is not recommended to remove filtering capacitors to reduce PCB size or cost. This might cause incorrect operation of the device.

### 6.1.7 Current consumption measurement

Figure 16. Current consumption measurement scheme



## 6.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in [Table 21: Voltage characteristics](#), [Table 22: Current characteristics](#), and [Table 23: Thermal characteristics](#) may cause permanent damage to the device. These are stress ratings only and the functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 21. Voltage characteristics <sup>(1)</sup>

| Symbols            | Ratings                                                                                                  | Min          | Max                                                                | Unit |
|--------------------|----------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------|------|
| $V_{DDX} - V_{SS}$ | External main supply voltage (including $V_{DD}$ , $V_{DDLDO}$ , $V_{DDA}$ , $V_{DD33USB}$ , $V_{BAT}$ ) | -0.3         | 4.0                                                                | V    |
| $V_{IN}^{(2)}$     | Input voltage on FT_XXX pins                                                                             | $V_{SS}-0.3$ | $\text{Min}(V_{DD}, V_{DDA}, V_{DD33USB}, V_{BAT}) + 4.0^{(3)(4)}$ | V    |
|                    | Input voltage on TT_XX pins                                                                              | $V_{SS}-0.3$ | 4.0                                                                | V    |
|                    | Input voltage on BOOT0 pin                                                                               | $V_{SS}$     | 9.0                                                                | V    |
|                    | Input voltage on any other pins                                                                          | $V_{SS}-0.3$ | 4.0                                                                | V    |
| $ \Delta V_{DDX} $ | Variations between different $V_{DDX}$ power pins of the same domain                                     | -            | 50                                                                 | mV   |
| $ V_{SSx}-V_{SS} $ | Variations between all the different ground pins                                                         | -            | 50                                                                 | mV   |

1. All main power ( $V_{DD}$ ,  $V_{DDA}$ ,  $V_{DD33USB}$ ,  $V_{BAT}$ ) and ground ( $V_{SS}$ ,  $V_{SSA}$ ) pins must always be connected to the external power supply, in the permitted range.
2.  $V_{IN}$  maximum must always be respected. Refer to [Table 59](#) for the maximum allowed injected current values.
3. This formula has to be applied on power supplies related to the IO structure described by the pin definition table.
4. To sustain a voltage higher than 4V the internal pull-up/pull-down resistors must be disabled.

Table 22. Current characteristics

| Symbols                 | Ratings                                                                         | Max   | Unit |
|-------------------------|---------------------------------------------------------------------------------|-------|------|
| $\Sigma I_{V_{DD}}$     | Total current into sum of all $V_{DD}$ power lines (source) <sup>(1)</sup>      | 620   | mA   |
| $\Sigma I_{V_{SS}}$     | Total current out of sum of all $V_{SS}$ ground lines (sink) <sup>(1)</sup>     | 620   |      |
| $I_{V_{DD}}$            | Maximum current into each $V_{DD}$ power pin (source) <sup>(1)</sup>            | 100   |      |
| $I_{V_{SS}}$            | Maximum current out of each $V_{SS}$ ground pin (sink) <sup>(1)</sup>           | 100   |      |
| $I_{IO}$                | Output current sunk by any I/O and control pin                                  | 20    |      |
| $\Sigma I_{(PIN)}$      | Total output current sunk by sum of all I/Os and control pins <sup>(2)</sup>    | 140   |      |
|                         | Total output current sourced by sum of all I/Os and control pins <sup>(2)</sup> | 140   |      |
| $I_{INJ(PIN)}^{(3)(4)}$ | Injected current on FT_xxx, TT_xx, RST and B pins except PA4, PA5               | -5/+0 |      |
|                         | Injected current on PA4, PA5                                                    | -0/0  |      |
| $\Sigma I_{INJ(PIN)}$   | Total injected current (sum of all I/Os and control pins) <sup>(5)</sup>        | ±25   |      |

1. All main power ( $V_{DD}$ ,  $V_{DDA}$ ,  $V_{DD33USB}$ ) and ground ( $V_{SS}$ ,  $V_{SSA}$ ) pins must always be connected to the external power supplies, in the permitted range.
2. This current consumption must be correctly distributed over all I/Os and control pins. The total output current must not be sunk/sourced between two consecutive power supply pins referring to high pin count QFP packages.
3. Positive injection is not possible on these I/Os and does not occur for input voltages lower than the specified maximum value.
4. A positive injection is induced by  $V_{IN} > V_{DD}$  while a negative injection is induced by  $V_{IN} < V_{SS}$ .  $I_{INJ(PIN)}$  must never be exceeded. Refer also to [Table 21: Voltage characteristics](#) for the maximum allowed input voltage values.
5. When several inputs are submitted to a current injection, the maximum  $\Sigma I_{INJ(PIN)}$  is the absolute sum of the positive and negative injected currents (instantaneous values).

Table 23. Thermal characteristics

| Symbol    | Ratings                      | Value        | Unit |
|-----------|------------------------------|--------------|------|
| $T_{STG}$ | Storage temperature range    | - 65 to +150 | °C   |
| $T_J$     | Maximum junction temperature | 125          |      |

## 6.3 Operating conditions

### 6.3.1 General operating conditions

Table 24. General operating conditions

| Symbol               | Parameter                                                               |                                      | Operating conditions                    | Min                 | Max                                                                                           | Unit |
|----------------------|-------------------------------------------------------------------------|--------------------------------------|-----------------------------------------|---------------------|-----------------------------------------------------------------------------------------------|------|
| V <sub>DD</sub>      | Standard operating voltage                                              |                                      | -                                       | 1.62 <sup>(1)</sup> | 3.6                                                                                           | V    |
| V <sub>DDLDO</sub>   | Supply voltage for the internal regulator                               |                                      | V <sub>DDLDO</sub> ≤ V <sub>DD</sub>    | 1.62 <sup>(1)</sup> | 3.6                                                                                           |      |
| V <sub>DD33USB</sub> | Standard operating voltage, USB domain                                  |                                      | USB used                                | 3.0                 | 3.6                                                                                           |      |
|                      |                                                                         |                                      | USB not used                            | 0                   | 3.6                                                                                           |      |
| V <sub>DDA</sub>     | Analog operating voltage                                                |                                      | ADC or COMP used                        | 1.62                | 3.6                                                                                           |      |
|                      |                                                                         |                                      | DAC used                                | 1.8                 |                                                                                               |      |
|                      |                                                                         |                                      | OPAMP used                              | 2.0                 |                                                                                               |      |
|                      |                                                                         |                                      | VREFBUF used                            | 1.8                 |                                                                                               |      |
|                      |                                                                         |                                      | ADC, DAC, OPAMP, COMP, VREFBUF not used | 0                   |                                                                                               |      |
| V <sub>IN</sub>      | I/O Input voltage                                                       |                                      | TT_xx I/O                               | −0.3                | V <sub>DD</sub> +0.3                                                                          |      |
|                      |                                                                         |                                      | BOOT0                                   | 0                   | 9                                                                                             |      |
|                      |                                                                         |                                      | All I/O except BOOT0 and TT_xx          | −0.3                | Min(V <sub>DD</sub> , V <sub>DDA</sub> , V <sub>DD33USB</sub> )+3.6V < 5.5V <sup>(2)(3)</sup> |      |
| P <sub>D</sub>       | Power dissipation at T <sub>A</sub> = 85 °C for suffix 6 <sup>(4)</sup> | TFBGA240+25                          | -                                       | -                   | 1093                                                                                          | mW   |
|                      |                                                                         | LQFP208                              | -                                       | -                   | 943                                                                                           |      |
|                      |                                                                         | LQFP176                              | -                                       | -                   | 930                                                                                           |      |
|                      |                                                                         | UFBGA176+25                          | -                                       | -                   | 1070                                                                                          |      |
|                      |                                                                         | UFBGA169                             | -                                       | -                   | 1061                                                                                          |      |
|                      |                                                                         | LQFP144                              | -                                       | -                   | 915                                                                                           |      |
|                      |                                                                         | LQFP100                              | -                                       | -                   | 889                                                                                           |      |
|                      |                                                                         | TFBGA100                             | -                                       | -                   | 1018                                                                                          |      |
| T <sub>A</sub>       | Ambient temperature for the suffix 6 version                            | Maximum power dissipation            |                                         | −40                 | 85                                                                                            | °C   |
|                      |                                                                         | Low-power dissipation <sup>(5)</sup> |                                         | −40                 | 105                                                                                           |      |
|                      | Ambient temperature for the suffix 3 version                            | Maximum power dissipation            |                                         | −40                 | 125                                                                                           |      |
|                      |                                                                         | Low-power dissipation <sup>(5)</sup> |                                         | −40                 | 130                                                                                           |      |
| T <sub>J</sub>       | Junction temperature range                                              | Suffix 6 version                     |                                         | −40                 | 125                                                                                           | °C   |

1. When RESET is released functionality is guaranteed down to  $V_{BOR0}$  min

2. This formula has to be applied on power supplies related to the IO structure described by the pin definition table.

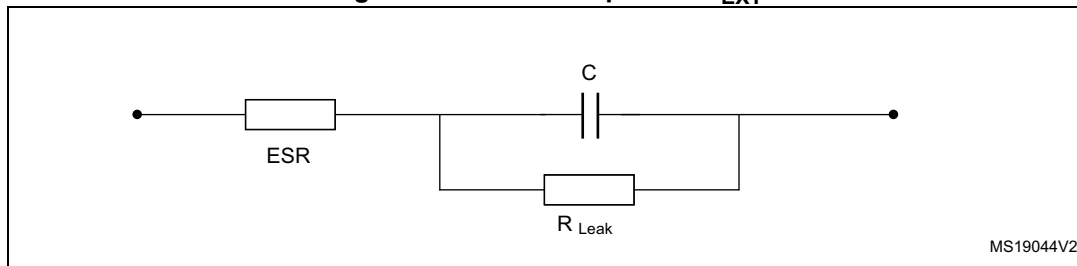
3. For operation with voltage higher than  $\text{Min}(V_{DD}, V_{DDA}, V_{DD33USB}) + 0.3V$ , the internal Pull-up and Pull-Down resistors must be disabled.

4. If  $T_A$  is lower, higher  $P_D$  values are allowed as long as  $T_J$  does not exceed  $T_{Jmax}$  (see [Section 8.9: Thermal characteristics](#)).
5. In low-power dissipation state,  $T_A$  can be extended to this range as long as  $T_J$  does not exceed  $T_{Jmax}$  (see [Section 8.9: Thermal characteristics](#)).

### 6.3.2 VCAP external capacitor

Stabilization for the main regulator is achieved by connecting an external capacitor  $C_{EXT}$  to the VCAP pin.  $C_{EXT}$  is specified in [Table 25](#). Two external capacitors can be connected to VCAP pins.

**Figure 17. External capacitor  $C_{EXT}$**



1. Legend: ESR is the equivalent series resistance.

**Table 25. VCAP operating conditions<sup>(1)</sup>**

| Symbol | Parameter                         | Conditions                 |
|--------|-----------------------------------|----------------------------|
| CEXT   | Capacitance of external capacitor | 2.2 $\mu F$ <sup>(2)</sup> |
| ESR    | ESR of external capacitor         | < 100 m $\Omega$           |

1. When bypassing the voltage regulator, the two 2.2  $\mu F$   $V_{CAP}$  capacitors are not required and should be replaced by two 100 nF decoupling capacitors.
2. This value corresponds to CEXT typical value. A variation of +/-20% is tolerated.

### 6.3.3 Operating conditions at power-up / power-down

Subject to general operating conditions for  $T_A$ .

**Table 26. Operating conditions at power-up / power-down (regulator ON)**

| Symbol       | Parameter                  | Min | Max      | Unit      |
|--------------|----------------------------|-----|----------|-----------|
| $t_{VDD}$    | $V_{DD}$ rise time rate    | 0   | $\infty$ | $\mu s/V$ |
|              | $V_{DD}$ fall time rate    | 10  | $\infty$ |           |
| $t_{VDDA}$   | $V_{DDA}$ rise time rate   | 0   | $\infty$ |           |
|              | $V_{DDA}$ fall time rate   | 10  | $\infty$ |           |
| $t_{VDDUSB}$ | $V_{DDUSB}$ rise time rate | 0   | $\infty$ |           |
|              | $V_{DDUSB}$ fall time rate | 10  | $\infty$ |           |

### 6.3.4 Embedded reset and power control block characteristics

The parameters given in [Table 27](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

**Table 27. Reset and power control block characteristics**

| Symbol                   | Parameter                                                          | Conditions                 | Min  | Typ  | Max   | Unit    |
|--------------------------|--------------------------------------------------------------------|----------------------------|------|------|-------|---------|
| $t_{RSTEMPO}^{(1)}$      | Reset temporization after BOR0 released                            | -                          | -    | 377  | -     | $\mu s$ |
| $V_{BOR0}$               | Brown-out reset threshold 0                                        | Rising edge <sup>(1)</sup> | 1.62 | 1.67 | 1.71  | V       |
|                          |                                                                    | Falling edge               | 1.58 | 1.62 | 1.68  |         |
| $V_{BOR1}$               | Brown-out reset threshold 1                                        | Rising edge                | 2.04 | 2.10 | 2.15  |         |
|                          |                                                                    | Falling edge               | 1.95 | 2.00 | 2.06  |         |
| $V_{BOR2}$               | Brown-out reset threshold 2                                        | Rising edge                | 2.34 | 2.41 | 2.47  |         |
|                          |                                                                    | Falling edge               | 2.25 | 2.31 | 2.37  |         |
| $V_{BOR3}$               | Brown-out reset threshold 3                                        | Rising edge                | 2.63 | 2.70 | 2.78  |         |
|                          |                                                                    | Falling edge               | 2.54 | 2.61 | 2.68  |         |
| $V_{PVD0}$               | Programmable Voltage Detector threshold 0                          | Rising edge                | 1.90 | 1.96 | 2.01  |         |
|                          |                                                                    | Falling edge               | 1.81 | 1.86 | 1.91  |         |
| $V_{PVD1}$               | Programmable Voltage Detector threshold 1                          | Rising edge                | 2.05 | 2.10 | 2.16  |         |
|                          |                                                                    | Falling edge               | 1.96 | 2.01 | 2.06  |         |
| $V_{PVD2}$               | Programmable Voltage Detector threshold 2                          | Rising edge                | 2.19 | 2.26 | 2.32  |         |
|                          |                                                                    | Falling edge               | 2.10 | 2.15 | 2.21  |         |
| $V_{PVD3}$               | Programmable Voltage Detector threshold 3                          | Rising edge                | 2.35 | 2.41 | 2.47  |         |
|                          |                                                                    | Falling edge               | 2.25 | 2.31 | 2.37  |         |
| $V_{PVD4}$               | Programmable Voltage Detector threshold 4                          | Rising edge                | 2.49 | 2.56 | 2.62  |         |
|                          |                                                                    | Falling edge               | 2.39 | 2.45 | 2.51  |         |
| $V_{PVD5}$               | Programmable Voltage Detector threshold 5                          | Rising edge                | 2.64 | 2.71 | 2.78  |         |
|                          |                                                                    | Falling edge               | 2.55 | 2.61 | 2.68  |         |
| $V_{PVD6}$               | Programmable Voltage Detector threshold 6                          | Rising edge                | 2.78 | 2.86 | 2.94  |         |
|                          |                                                                    | Falling edge in Run mode   | 2.69 | 2.76 | 2.83  |         |
| $V_{hyst\_BOR\_PVD}$     | Hysteresis voltage of BOR (unless BOR0) and PVD                    | Hysteresis in Run mode     | -    | 100  | -     | mV      |
| $I_{DD\_BOR\_PVD}^{(1)}$ | BOR <sup>(2)</sup> (unless BOR0) and PVD consumption from $V_{DD}$ | -                          | -    |      | 0.630 | $\mu A$ |

Table 27. Reset and power control block characteristics (continued)

| Symbol           | Parameter                                         | Conditions      | Min  | Typ  | Max  | Unit    |
|------------------|---------------------------------------------------|-----------------|------|------|------|---------|
| $V_{AVM\_0}$     | Analog voltage detector for $V_{DDA}$ threshold 0 | Rising edge     | 1.66 | 1.71 | 1.76 | V       |
|                  |                                                   | Falling edge    | 1.56 | 1.61 | 1.66 |         |
| $V_{AVM\_1}$     | Analog voltage detector for $V_{DDA}$ threshold 1 | Rising edge     | 2.06 | 2.12 | 2.19 |         |
|                  |                                                   | Falling edge    | 1.96 | 2.02 | 2.08 |         |
| $V_{AVM\_2}$     | Analog voltage detector for $V_{DDA}$ threshold 2 | Rising edge     | 2.42 | 2.50 | 2.58 |         |
|                  |                                                   | Falling edge    | 2.35 | 2.42 | 2.49 |         |
| $V_{AVM\_3}$     | Analog voltage detector for $V_{DDA}$ threshold 3 | Rising edge     | 2.74 | 2.83 | 2.91 |         |
|                  |                                                   | Falling edge    | 2.64 | 2.72 | 2.80 |         |
| $V_{hyst\_VDDA}$ | Hysteresis of $V_{DDA}$ voltage detector          | -               | -    | 100  | -    | mV      |
| $I_{DD\_PVM}$    | PVM consumption from $V_{DD(1)}$                  | -               | -    | -    | 0.25 | $\mu$ A |
| $I_{DD\_VDDA}$   | Voltage detector consumption on $V_{DDA}^{(1)}$   | Resistor bridge | -    | -    | 2.5  | $\mu$ A |

1. Guaranteed by design.

2. BOR0 is enabled in all modes and its consumption is therefore included in the supply current characteristics tables (refer to [Section 6.3.6: Supply current characteristics](#)).

### 6.3.5 Embedded reference voltage

The parameters given in [Table 28](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

Table 28. Embedded reference voltage

| Symbol                    | Parameter                                                           | Conditions                                                                     | Min   | Typ   | Max   | Unit                    |
|---------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|-------|-------|-------|-------------------------|
| $V_{REFINT}$              | Internal reference voltages                                         | $-40^{\circ}\text{C} < T_J < 105^{\circ}\text{C}$ ,<br>$V_{DD} = 3.3\text{ V}$ | 1.180 | 1.216 | 1.255 | V                       |
| $t_{S\_vrefint}^{(1)(2)}$ | ADC sampling time when reading the internal reference voltage       | -                                                                              | 4.3   | -     | -     | $\mu$ s                 |
| $t_{S\_vbat}^{(1)(2)}$    | VBAT sampling time when reading the internal VBAT reference voltage | -                                                                              | 9     | -     | -     |                         |
| $I_{refbuf}^{(2)}$        | Reference Buffer consumption for ADC                                | $V_{DDA}=3.3\text{ V}$                                                         | 9     | 13.5  | 23    | $\mu$ A                 |
| $\Delta V_{REFINT}^{(2)}$ | Internal reference voltage spread over the temperature range        | $-40^{\circ}\text{C} < T_J < 105^{\circ}\text{C}$                              | -     | 5     | 15    | mV                      |
| $T_{coeff}^{(2)}$         | Average temperature coefficient                                     | Average temperature coefficient                                                | -     | 20    | 70    | ppm/ $^{\circ}\text{C}$ |
| $V_{DDcoeff}^{(2)}$       | Average Voltage coefficient                                         | $3.0\text{V} < V_{DD} < 3.6\text{V}$                                           | -     | 10    | 1370  | ppm/V                   |



Table 28. Embedded reference voltage (continued)

| Symbol                   | Parameter             | Conditions | Min | Typ | Max | Unit                     |
|--------------------------|-----------------------|------------|-----|-----|-----|--------------------------|
| V <sub>REFINT_DIV1</sub> | 1/4 reference voltage | -          | -   | 25  | -   | %<br>V <sub>REFINT</sub> |
| V <sub>REFINT_DIV2</sub> | 1/2 reference voltage | -          | -   | 50  | -   |                          |
| V <sub>REFINT_DIV3</sub> | 3/4 reference voltage | -          | -   | 75  | -   |                          |

1. The shortest sampling time for the application can be determined by multiple iterations.
2. Guaranteed by design.

Table 29. Internal reference voltage calibration values

| Symbol                 | Parameter                                                           | Memory address      |
|------------------------|---------------------------------------------------------------------|---------------------|
| V <sub>REFIN_CAL</sub> | Raw data acquired at temperature of 30 °C, V <sub>DDA</sub> = 3.3 V | 1FF1E860 - 1FF1E861 |

### 6.3.6 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, program location in memory and executed binary code.

The current consumption is measured as described in [Figure 16: Current consumption measurement scheme](#).

All the run-mode current consumption measurements given in this section are performed with a CoreMark code.

#### Typical and maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog input mode.
- All peripherals are disabled except when explicitly mentioned.
- The Flash memory access time is adjusted with the minimum wait states number, depending on the f<sub>ACLK</sub> frequency (refer to the table “Number of wait states according to CPU clock (f<sub>RC\_C\_ck</sub>) frequency and V<sub>CORE</sub> range” available in the reference manual).
- When the peripherals are enabled, the AHB clock frequency is the CPU frequency divided by 2 and the APB clock frequency is AHB clock frequency divided by 2.

The parameters given in [Table 30](#) to [Table 38](#) are derived from tests performed under ambient temperature and supply voltage conditions summarized in [Table 24: General operating conditions](#).

**Table 30. Typical and maximum current consumption in Run mode, code with data processing running from ITCM, regulator ON<sup>(1)</sup>**

| Symbol          | Parameter                        | Conditions                     |      | f <sub>rcc_c_ck</sub><br>(MHz) | Typ  | Max <sup>(2)</sup>       |                          |                           |                           | unit |
|-----------------|----------------------------------|--------------------------------|------|--------------------------------|------|--------------------------|--------------------------|---------------------------|---------------------------|------|
|                 |                                  |                                |      |                                |      | T <sub>J</sub> =<br>25°C | T <sub>J</sub> =<br>85°C | T <sub>J</sub> =<br>105°C | T <sub>J</sub> =<br>125°C |      |
| I <sub>DD</sub> | Supply<br>current in Run<br>mode | All<br>peripherals<br>disabled | VOS1 | 400                            | 71   | 110                      | 210                      | 290                       | 540                       | mA   |
|                 |                                  |                                |      | 300                            | 56   | -                        | -                        | -                         | -                         |      |
|                 |                                  |                                | VOS2 | 300                            | 50   | 72                       | 170                      | 230                       | 370                       |      |
|                 |                                  |                                |      | 216                            | 37   | 58                       | 150                      | 210                       | 380                       |      |
|                 |                                  |                                |      | 200                            | 35.5 | -                        | -                        | -                         | -                         |      |
|                 |                                  |                                | VOS3 | 200                            | 33   | 50                       | 130                      | 190                       | 300                       |      |
|                 |                                  |                                |      | 180                            | 30   | 47                       | 130                      | 180                       | 290                       |      |
|                 |                                  |                                |      | 168                            | 28   | 45                       | 130                      | 180                       | 290                       |      |
|                 |                                  |                                |      | 144                            | 25   | 41                       | 120                      | 180                       | 290                       |      |
|                 |                                  |                                |      | 60                             | 13   | 28                       | 110                      | 160                       | 280                       |      |
|                 |                                  |                                |      | 25                             | 10   | 24                       | 99                       | 160                       | 270                       |      |
|                 |                                  | All<br>peripherals<br>enabled  | VOS1 | 400                            | 165  | 220 <sup>(3)</sup>       | 400                      | 500 <sup>(3)</sup>        | 840                       |      |
|                 |                                  |                                |      | 300                            | 130  | -                        | -                        | -                         | -                         |      |
|                 |                                  |                                | VOS2 | 300                            | 120  | 170                      | 300                      | 390                       | 570                       |      |
|                 |                                  |                                |      | 200                            | 83   | -                        | -                        | -                         | -                         |      |
|                 |                                  |                                | VOS3 | 200                            | 78   | 110                      | 220                      | 300                       | 470                       |      |

1. Data are in DTCM for best computation performance, cache has no influence on consumption in this case.

2. Guaranteed by characterization results unless otherwise specified.

3. Guaranteed by test in production.

**Table 31. Typical and maximum current consumption in Run mode, code with data processing running from Flash memory, cache ON, regulator ON**

| Symbol          | Parameter                  | Conditions               |      | $f_{\text{rcc\_c\_ck}}$<br>(MHz) | Typ | Max <sup>(1)</sup>       |                          |                           |                           | unit |
|-----------------|----------------------------|--------------------------|------|----------------------------------|-----|--------------------------|--------------------------|---------------------------|---------------------------|------|
|                 |                            |                          |      |                                  |     | $T_J = 25^\circ\text{C}$ | $T_J = 85^\circ\text{C}$ | $T_J = 105^\circ\text{C}$ | $T_J = 125^\circ\text{C}$ |      |
| $I_{\text{DD}}$ | Supply current in Run mode | All peripherals disabled | VOS1 | 400                              | 105 | 160                      | 310                      | 420                       | 750                       | mA   |
|                 |                            |                          |      | 300                              | 55  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS2 | 300                              | 50  | 72                       | 160                      | 230                       | 370                       |      |
|                 |                            |                          |      | 216                              | 38  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS3 | 200                              | 36  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          |      | 200                              | 33  | 50                       | 130                      | 190                       | 300                       |      |
|                 |                            |                          |      | 180                              | 30  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          |      | 168                              | 29  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          |      | 144                              | 26  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          |      | 60                               | 14  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          |      | 25                               | 14  | -                        | -                        | -                         | -                         |      |
|                 |                            | All peripherals enabled  | VOS1 | 400                              | 160 | 220                      | 400                      | 500                       | 750                       |      |
|                 |                            |                          |      | 300                              | 130 | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS2 | 300                              | 120 | 160                      | 300                      | 390                       | 560                       |      |
|                 |                            |                          |      | 200                              | 81  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS3 | 200                              | 77  | 110                      | 220                      | 300                       | 460                       |      |
|                 |                            |                          |      |                                  |     |                          |                          |                           |                           |      |

1. Guaranteed by characterization results unless otherwise specified.

**Table 32. Typical and maximum current consumption in Run mode, code with data processing running from Flash memory, cache OFF, regulator ON**

| Symbol          | Parameter                  | Conditions               |      | $f_{\text{rcc\_c\_ck}}$<br>(MHz) | Typ | Max <sup>(1)</sup>       |                          |                           |                           | unit |
|-----------------|----------------------------|--------------------------|------|----------------------------------|-----|--------------------------|--------------------------|---------------------------|---------------------------|------|
|                 |                            |                          |      |                                  |     | $T_J = 25^\circ\text{C}$ | $T_J = 85^\circ\text{C}$ | $T_J = 105^\circ\text{C}$ | $T_J = 125^\circ\text{C}$ |      |
| $I_{\text{DD}}$ | Supply current in Run mode | All peripherals disabled | VOS1 | 400                              | 73  | 110                      | 220                      | 290                       | 540                       | mA   |
|                 |                            |                          | VOS2 | 300                              | 52  | 75                       | 170                      | 230                       | 370                       |      |
|                 |                            |                          | VOS3 | 200                              | 34  | 52                       | 130                      | 190                       | 300                       |      |
|                 |                            | All peripherals enabled  | VOS1 | 400                              | 135 | 190                      | 360                      | 470                       | 730                       |      |
|                 |                            |                          | VOS2 | 300                              | 100 | 150                      | 270                      | 370                       | 550                       |      |
|                 |                            |                          | VOS3 | 200                              | 70  | 100                      | 210                      | 300                       | 460                       |      |
|                 |                            |                          |      |                                  |     |                          |                          |                           |                           |      |

1. Guaranteed by characterization results.

**Table 33. Typical consumption in Run mode and corresponding performance versus code position**

| Symbol          | Parameter                  | Conditions                          |          | $f_{\text{rcc\_c\_ck}}$<br>(MHz) | CoreMark | Typ  | Unit | IDD/<br>CoreMark | Unit                          |
|-----------------|----------------------------|-------------------------------------|----------|----------------------------------|----------|------|------|------------------|-------------------------------|
|                 |                            | Peripheral                          | Code     |                                  |          |      |      |                  |                               |
| $I_{\text{DD}}$ | Supply current in Run mode | All peripherals disabled, cache ON  | ITCM     | 400                              | 2012     | 71   | mA   | 35               | $\mu\text{A}/\text{CoreMark}$ |
|                 |                            |                                     | FLASH A  | 400                              | 2012     | 105  |      | 52               |                               |
|                 |                            |                                     | AXI SRAM | 400                              | 2012     | 105  |      | 52               |                               |
|                 |                            |                                     | SRAM1    | 400                              | 2012     | 105  |      | 52               |                               |
|                 |                            |                                     | SRAM4    | 400                              | 2012     | 105  |      | 52               |                               |
|                 |                            | All peripherals disabled, cache OFF | ITCM     | 400                              | 2012     | 71   |      | 35               |                               |
|                 |                            |                                     | FLASH A  | 400                              | 593      | 70.5 |      | 119              |                               |
|                 |                            |                                     | AXI SRAM | 400                              | 344      | 70.5 |      | 205              |                               |
|                 |                            |                                     | SRAM1    | 400                              | 472      | 74.5 |      | 158              |                               |
|                 |                            |                                     | SRAM4    | 400                              | 432      | 72   |      | 167              |                               |

**Table 34. Typical current consumption batch acquisition mode**

| Symbol          | Parameter                                | Conditions                  |      | $f_{\text{rcc\_ahb\_ck(AHB4)}}$<br>(MHz) | Typ | unit |
|-----------------|------------------------------------------|-----------------------------|------|------------------------------------------|-----|------|
| $I_{\text{DD}}$ | Supply current in batch acquisition mode | D1Standby, D2Standby, D3Run | VOS3 | 64                                       | 6.5 | mA   |
|                 |                                          | D1Stop, D2Stop, D3Run       | VOS3 | 64                                       | 12  |      |

**Table 35. Typical and maximum current consumption in Sleep mode, regulator ON**

| Symbol                 | Parameter                    | Conditions               |      | $f_{\text{rcc\_c\_ck}}$<br>(MHz) | Typ  | Max <sup>(1)</sup>       |                          |                           |                           | unit |
|------------------------|------------------------------|--------------------------|------|----------------------------------|------|--------------------------|--------------------------|---------------------------|---------------------------|------|
|                        |                              |                          |      |                                  |      | $T_J = 25^\circ\text{C}$ | $T_J = 85^\circ\text{C}$ | $T_J = 105^\circ\text{C}$ | $T_J = 125^\circ\text{C}$ |      |
| $I_{\text{DD(Sleep)}}$ | Supply current in Sleep mode | All peripherals disabled | VOS1 | 400                              | 31.0 | 64                       | 220                      | 330                       | 660                       | mA   |
|                        |                              |                          |      | 300                              | 24.5 | 57                       | 210                      | 330                       | 650                       |      |
|                        |                              |                          | VOS2 | 300                              | 22.0 | 48                       | 180                      | 270                       | 500                       |      |
|                        |                              |                          |      | 200                              | 17.0 | 42                       | 170                      | 270                       | 490                       |      |
|                        |                              |                          | VOS3 | 200                              | 15.5 | 37                       | 150                      | 230                       | 400                       |      |

1. Guaranteed by characterization results.

Table 36. Typical and maximum current consumption in Stop mode, regulator ON

| Symbol                | Parameter                    | Conditions                              |       | Typ   | Max <sup>(1)</sup>    |                       |                        |                        | unit |
|-----------------------|------------------------------|-----------------------------------------|-------|-------|-----------------------|-----------------------|------------------------|------------------------|------|
|                       |                              |                                         |       |       | T <sub>J</sub> = 25°C | T <sub>J</sub> = 85°C | T <sub>J</sub> = 105°C | T <sub>J</sub> = 125°C |      |
| I <sub>DD(Stop)</sub> | D1Stop, D2Stop, D3Stop       | Flash memory in low-power mode, no IWDG | SVOS5 | 1.4   | 7.2 <sup>(2)</sup>    | 49                    | 75 <sup>(2)</sup>      | 140                    | mA   |
|                       |                              |                                         | SVOS4 | 1.95  | 11                    | 66                    | 110                    | 200                    |      |
|                       |                              |                                         | SVOS3 | 2.85  | 16 <sup>(2)</sup>     | 91                    | 150 <sup>(2)</sup>     | 240                    |      |
|                       |                              | Flash memory ON, no IWDG                | SVOS5 | 1.65  | 7.2                   | 49                    | 75                     | 140                    |      |
|                       |                              |                                         | SVOS4 | 2.2   | 11                    | 66                    | 110                    | 180                    |      |
|                       |                              |                                         | SVOS3 | 3.15  | 16                    | 91                    | 150                    | 300                    |      |
|                       | D1Stop, D2Standby, D3Stop    | Flash memory OFF, no IWDG               | SVOS5 | 0.99  | 5.1                   | 35                    | 60                     | 97                     |      |
|                       |                              |                                         | SVOS4 | 1.4   | 7.5                   | 47                    | 79                     | 130                    |      |
|                       |                              |                                         | SVOS3 | 2.05  | 12                    | 64                    | 110                    | 170                    |      |
|                       |                              | Flash memory ON, no IWDG                | SVOS5 | 1.25  | 5.5                   | 35                    | 61                     | 98                     |      |
|                       |                              |                                         | SVOS4 | 1.65  | 7.8                   | 47                    | 80                     | 130                    |      |
|                       |                              |                                         | SVOS3 | 2.3   | 12                    | 65                    | 110                    | 170                    |      |
|                       | D1Standby, D2Stop, D3Stop    | Flash OFF, no IWDG                      | SVOS5 | 0.57  | 3                     | 21                    | 36                     | 57                     |      |
|                       |                              |                                         | SVOS4 | 0.805 | 4.5                   | 27                    | 47                     | 74                     |      |
|                       |                              |                                         | SVOS3 | 1.2   | 6.7                   | 37                    | 63                     | 99                     |      |
|                       | D1Standby, D2Standby, D3Stop |                                         | SVOS5 | 0.17  | 1.1 <sup>(2)</sup>    | 8                     | 13 <sup>(2)</sup>      | 20                     |      |
|                       |                              |                                         | SVOS4 | 0.245 | 1.5                   | 11                    | 17                     | 26                     |      |
|                       |                              |                                         | SVOS3 | 0.405 | 2.4 <sup>(2)</sup>    | 15                    | 23 <sup>(2)</sup>      | 35                     |      |

1. Guaranteed by characterization results.

2. Guaranteed by test in production.

Table 37. Typical and maximum current consumption in Standby mode

| Symbol                       | Parameter                      | Conditions  |           | Typ <sup>(3)</sup> |       |      |       | Max (3 V) <sup>(1)</sup> |                       |                        |                        | Unit |
|------------------------------|--------------------------------|-------------|-----------|--------------------|-------|------|-------|--------------------------|-----------------------|------------------------|------------------------|------|
|                              |                                | Backup SRAM | RTC & LSE | 1.62 V             | 2.4 V | 3 V  | 3.3 V | T <sub>J</sub> = 25°C    | T <sub>J</sub> = 85°C | T <sub>J</sub> = 105°C | T <sub>J</sub> = 125°C |      |
| I <sub>DD</sub><br>(Standby) | Supply current in Standby mode | OFF         | OFF       | 1.8                | 1.9   | 1.95 | 2.05  | 4 <sup>(2)</sup>         | 18 <sup>(3)</sup>     | 40 <sup>(2)</sup>      | 90 <sup>(3)</sup>      | μA   |
|                              |                                | ON          | OFF       | 3.4                | 3.4   | 3.5  | 3.7   | 8.2 <sup>(3)</sup>       | 47 <sup>(3)</sup>     | 83 <sup>(3)</sup>      | 141 <sup>(3)</sup>     |      |
|                              |                                | OFF         | ON        | 2.4                | 3.5   | 3.86 | 4.12  | -                        | -                     | -                      | -                      |      |
|                              |                                | ON          | ON        | 3.95               | 5.1   | 5.46 | 5.97  | -                        | -                     | -                      | -                      |      |

1. The maximum current consumption values are given for PDR OFF (internal reset OFF). When the PDR is OFF (internal reset OFF), the current consumption is reduced by 1.2 μA compared to PDR ON.

2. Guaranteed by test in production.

3. Guaranteed by characterization results.

Table 38. Typical and maximum current consumption in VBAT mode

| Symbol                    | Parameter                      | Conditions  |           | Typ <sup>(1)</sup> |       |       |       | Max (3 V)             |                       |                        |                        | Unit |
|---------------------------|--------------------------------|-------------|-----------|--------------------|-------|-------|-------|-----------------------|-----------------------|------------------------|------------------------|------|
|                           |                                | Backup SRAM | RTC & LSE | 1.2 V              | 2 V   | 3 V   | 3.4 V | T <sub>J</sub> = 25°C | T <sub>J</sub> = 85°C | T <sub>J</sub> = 105°C | T <sub>J</sub> = 125°C |      |
| I <sub>DD</sub><br>(VBAT) | Supply current in standby mode | OFF         | OFF       | 0.024              | 0.035 | 0.062 | 0.096 | 0.5 <sup>(1)</sup>    | 4.1 <sup>(1)</sup>    | 10 <sup>(1)</sup>      | 24 <sup>(1)</sup>      | μA   |
|                           |                                | ON          | OFF       | 1.4                | 1.6   | 1.8   | 1.8   | 4.4 <sup>(1)</sup>    | 22 <sup>(1)</sup>     | 48 <sup>(1)</sup>      | 87 <sup>(1)</sup>      |      |
|                           |                                | OFF         | ON        | 0.24               | 0.45  | 0.62  | 0.73  | -                     | -                     | -                      | -                      |      |
|                           |                                | ON          | ON        | 1.97               | 2.37  | 2.57  | 2.77  | -                     | -                     | -                      | -                      |      |

1. Guaranteed by characterization results.

### I/O system current consumption

The current consumption of the I/O system has two components: static and dynamic.

#### I/O static current consumption

All the I/Os used as inputs with pull-up generate a current consumption when the pin is externally held low. The value of this current consumption can be simply computed by using the pull-up/pull-down resistors values given in [Table 60: I/O static characteristics](#).

For the output pins, any external pull-down or external load must also be considered to estimate the current consumption.

An additional I/O current consumption is due to I/Os configured as inputs if an intermediate voltage level is externally applied. This current consumption is caused by the input Schmitt trigger circuits used to discriminate the input value. Unless this specific configuration is required by the application, this supply current consumption can be avoided by configuring these I/Os in analog mode. This is notably the case of ADC input pins which should be configured as analog inputs.

**Caution:** Any floating input pin can also settle to an intermediate voltage level or switch inadvertently, as a result of external electromagnetic noise. To avoid a current consumption related to floating pins, they must either be configured in analog mode, or forced internally to a definite digital value. This can be done either by using pull-up/down resistors or by configuring the pins in output mode.

#### I/O dynamic current consumption

In addition to the internal peripheral current consumption (see [Table 39: Peripheral current consumption in Run mode](#)), the I/Os used by an application also contribute to the current consumption. When an I/O pin switches, it uses the current from the MCU supply voltage to supply the I/O pin circuitry and to charge/discharge the capacitive load (internal or external) connected to the pin:

$$I_{SW} = V_{DDX} \times f_{SW} \times C_L$$

where

$I_{SW}$  is the current sunk by a switching I/O to charge/discharge the capacitive load

$V_{DDX}$  is the MCU supply voltage

$f_{SW}$  is the I/O switching frequency

$C_L$  is the total capacitance seen by the I/O pin:  $C = C_{INT} + C_{EXT}$

The test pin is configured in push-pull output mode and is toggled by software at a fixed frequency.

### On-chip peripheral current consumption

The MCU is placed under the following conditions:

- At startup, all I/O pins are in analog input configuration.
- All peripherals are disabled unless otherwise mentioned.
- The I/O compensation cell is enabled.
- $f_{\text{rcc\_c\_ck}}$  is the CPU clock.  $f_{\text{PCLK}} = f_{\text{rcc\_c\_ck}}/4$ , and  $f_{\text{HCLK}} = f_{\text{rcc\_c\_ck}}/2$ .  
The given value is calculated by measuring the difference of current consumption
  - with all peripherals clocked off
  - with only one peripheral clocked on
  - $f_{\text{rcc\_c\_ck}} = 400 \text{ MHz}$  (Scale 1),  $f_{\text{rcc\_c\_ck}} = 300 \text{ MHz}$  (Scale 2),  
 $f_{\text{rcc\_c\_ck}} = 200 \text{ MHz}$  (Scale 3)
- The ambient operating temperature is 25 °C and  $V_{\text{DD}}=3.3 \text{ V}$ .

Table 39. Peripheral current consumption in Run mode

| Peripheral |                    | I <sub>DD</sub> (Typ) |      |      | Unit   |
|------------|--------------------|-----------------------|------|------|--------|
|            |                    | VOS1                  | VOS2 | VOS3 |        |
| AHB3       | MDMA               | 8.3                   | 7.6  | 7    | μA/MHz |
|            | DMA2D              | 21                    | 20   | 18   |        |
|            | JPEG               | 24                    | 23   | 21   |        |
|            | FLASH              | 9.9                   | 9    | 8.3  |        |
|            | FMC registers      | 0.9                   | 0.9  | 0.8  |        |
|            | FMC kernel         | 6.1                   | 5.5  | 5.3  |        |
|            | QUADSPI registers  | 1.5                   | 1.4  | 1.3  |        |
|            | QUADSPI kernel     | 0.9                   | 0.8  | 0.7  |        |
|            | SDMMC1 registers   | 8                     | 7.2  | 6.8  |        |
|            | SDMMC1 kernel      | 2.4                   | 2    | 1.8  |        |
|            | DTCM1              | 5.7                   | 5    | 4.5  |        |
|            | DTCM2              | 5.5                   | 4.8  | 4.3  |        |
|            | ITCM               | 3.2                   | 2.9  | 2.6  |        |
|            | D1SRAM1            | 7.6                   | 6.8  | 6.1  |        |
|            | AHB3 bridge        | 7.5                   | 6.8  | 6.3  |        |
| AHB1       | DMA1               | 1.1                   | 1    | 1    |        |
|            | DMA2               | 1.7                   | 1.4  | 1.1  |        |
|            | ADC1/2 registers   | 3.9                   | 3.2  | 3.1  |        |
|            | ADC1/2 kernel      | 0.9                   | 0.8  | 0.7  |        |
|            | ART accelerator    | 5.5                   | 4.5  | 4.2  |        |
|            | ETH1MAC            | 16                    | 14   | 13   |        |
|            | ETH1TX             |                       |      |      |        |
|            | ETH1RX             |                       |      |      |        |
|            | USB1 OTG registers | 15                    | 14   | 13   |        |
|            | USB1 OTG kernel    | -                     | 8.5  | 8.5  |        |
|            | USB1 ULPI          | 0.3                   | 0.3  | 0.1  |        |
|            | USB2 OTG registers | 15                    | 13   | 12   |        |
|            | USB2 OTG kernel    | -                     | 8.6  | 8.6  |        |
|            | USB2 ULPI          | 16                    | 16   | 16   |        |
|            | AHB1 Bridge        | 10                    | 9.6  | 8.6  |        |



Table 39. Peripheral current consumption in Run mode (continued)

| Peripheral |                  | I <sub>DD</sub> (Typ) |      |      | Unit   |
|------------|------------------|-----------------------|------|------|--------|
|            |                  | VOS1                  | VOS2 | VOS3 |        |
| AHB2       | DCMI             | 1.7                   | 1.7  | 1.7  | μA/MHz |
|            | RNG registers    | 1.8                   | 1.4  | 1.2  |        |
|            | RNG kernel       | -                     | 9.6  | 9.6  |        |
|            | SDMMC2 registers | 13                    | 12   | 11   |        |
|            | SDMMC2 kernel    | 2.7                   | 2.5  | 2.4  |        |
|            | D2SRAM1          | 3.3                   | 3.1  | 2.8  |        |
|            | D2SRAM2          | 2.9                   | 2.7  | 2.5  |        |
|            | D2SRAM3          | 1.9                   | 1.8  | 1.7  |        |
|            | AHB2 bridge      | 0.1                   | 0.1  | 0.1  |        |
| AHB4       | GPIOA            | 1.1                   | 1    | 0.9  |        |
|            | GPIOB            | 1                     | 0.9  | 0.9  |        |
|            | GPIOC            | 1.4                   | 1.3  | 1.3  |        |
|            | GPIOD            | 1.1                   | 1    | 0.9  |        |
|            | GPIOE            | 1                     | 0.9  | 0.8  |        |
|            | GPIOF            | 0.9                   | 0.8  | 0.8  |        |
|            | GPIOG            | 0.9                   | 0.7  | 0.7  |        |
|            | GPIOH            | 1                     | 0.9  | 0.9  |        |
|            | GPIOI            | 0.9                   | 0.9  | 0.8  |        |
|            | GPIOJ            | 0.9                   | 0.8  | 0.8  |        |
|            | GPIOK            | 0.9                   | 0.8  | 0.7  |        |
|            | CRC              | 0.5                   | 0.4  | 0.4  |        |
|            | BDMA             | 6.2                   | 5.8  | 5.5  |        |
|            | ADC3 registers   | 1.8                   | 1.7  | 1.7  |        |
|            | ADC3 kernel      | 0.1                   | 0.1  | 0.1  |        |
|            | Backup SRAM      | 1.9                   | 1.8  | 1.8  |        |
|            | Bridge AHB4      | 0.1                   | 0.1  | 0.1  |        |
| APB3       | LCD-TFT          | 12                    | 11   | 10   | μA/MHz |
|            | WWDG1            | 0.5                   | 0.4  | 0.3  |        |
|            | APB3 bridge      | 0.5                   | 0.2  | 0.1  |        |

Table 39. Peripheral current consumption in Run mode (continued)

| Peripheral |                    | I <sub>DD</sub> (Typ) |      |      | Unit   |
|------------|--------------------|-----------------------|------|------|--------|
|            |                    | VOS1                  | VOS2 | VOS3 |        |
| APB1       | TIM2               | 3.5                   | 3.2  | 2.9  | μA/MHz |
|            | TIM3               | 3.4                   | 3.1  | 2.7  |        |
|            | TIM4               | 2.7                   | 2.5  | 1.9  |        |
|            | TIM5               | 3.2                   | 2.9  | 2.5  |        |
|            | TIM6               | 1                     | 0.8  | 0.7  |        |
|            | TIM7               | 1                     | 0.9  | 0.7  |        |
|            | TIM12              | 1.7                   | 1.5  | 1.2  |        |
|            | TIM13              | 1.5                   | 1.3  | 1    |        |
|            | TIM14              | 1.4                   | 1.3  | 0.9  |        |
|            | LPTIM1 registers   | 0.7                   | 0.6  | 0.5  |        |
|            | LPTIM1 kernel      | 2.3                   | 2.1  | 1.9  |        |
|            | WWDG2              | 0.6                   | 0.4  | 0.4  |        |
|            | SPI2 registers     | 1.8                   | 1.5  | 1.2  |        |
|            | SPI2 kernel        | 0.6                   | 0.5  | 0.5  |        |
|            | SPI3 registers     | 1.5                   | 1.3  | 1.1  |        |
|            | SPI3 kernel        | 0.6                   | 0.5  | 0.5  |        |
|            | SPDIFRX1 registers | 0.6                   | 0.5  | 0.3  |        |
|            | SPDIFRX1 kernel    | 2.9                   | 2.4  | 2.4  |        |
|            | USART2 registers   | 1.4                   | 1.3  | 1    |        |
|            | USART2 kernel      | 4.7                   | 4.1  | 4    |        |
|            | USART3 registers   | 1.4                   | 1.3  | 1    |        |
|            | USART3 kernel      | 4.2                   | 3.8  | 3.5  |        |
|            | UART4 registers    | 1.5                   | 1.1  | 1    |        |
|            | UART4 kernel       | 3.7                   | 3.6  | 3.2  |        |

Table 39. Peripheral current consumption in Run mode (continued)

| Peripheral          |                    | I <sub>DD</sub> (Typ) |      |      | Unit   |
|---------------------|--------------------|-----------------------|------|------|--------|
|                     |                    | VOS1                  | VOS2 | VOS3 |        |
| APB1<br>(continued) | UART5 registers    | 1.4                   | 1.4  | 1    | μA/MHz |
|                     | UART5 kernel       | 3.6                   | 3.2  | 3.1  |        |
|                     | I2C1 registers     | 0.8                   | 0.8  | 0.6  |        |
|                     | I2C1 kernel        | 2                     | 1.8  | 1.7  |        |
|                     | I2C2 registers     | 0.7                   | 0.7  | 0.4  |        |
|                     | I2C2 kernel        | 1.9                   | 1.7  | 1.6  |        |
|                     | I2C3 registers     | 0.9                   | 0.7  | 0.6  |        |
|                     | I2C3 kernel        | 2.1                   | 1.9  | 1.9  |        |
|                     | HDMI-CEC registers | 0.5                   | 0.3  | 0.3  |        |
|                     | DAC1/2             | 1.4                   | 1.1  | 0.9  |        |
|                     | USART7 registers   | 1.9                   | 1.8  | 1.3  |        |
|                     | USART7 kernel      | 4                     | 3.5  | 3.3  |        |
|                     | USART8 registers   | 1.6                   | 1.5  | 1.2  |        |
|                     | USART8 kernel      | 4                     | 3.6  | 3.3  |        |
|                     | CRS                | 3.4                   | 3.1  | 2.9  |        |
|                     | SWPMI registers    | 2.3                   | 2    | 2    |        |
|                     | SWPMI kernel       | 0.1                   | 0.1  | 0.1  |        |
|                     | OPAMP              | 0.5                   | 0.4  | 0.4  |        |
|                     | MDIO               | 2.7                   | 2.4  | 2.3  |        |
|                     | FDCAN registers    | 16                    | 15   | 14   |        |
|                     | FDCAN kernel       | 7.8                   | 7.6  | 7.1  |        |
|                     | Bridge APB1        | 0.1                   | 0.1  | 0.1  |        |

Table 39. Peripheral current consumption in Run mode (continued)

| Peripheral |                  | I <sub>DD</sub> (Typ) |      |      | Unit   |
|------------|------------------|-----------------------|------|------|--------|
|            |                  | VOS1                  | VOS2 | VOS3 |        |
| APB2       | TIM1             | 5.1                   | 4.8  | 4.3  | μA/MHz |
|            | TIM8             | 5.4                   | 4.9  | 4.6  |        |
|            | USART1 registers | 2.7                   | 2.6  | 2.5  |        |
|            | USART1 kernel    | 0.1                   | 0.1  | 0.1  |        |
|            | USART6 registers | 2.6                   | 2.5  | 2.5  |        |
|            | USART6 kernel    | 0.1                   | 0.1  | 0.1  |        |
|            | SPI1 registers   | 1.8                   | 1.6  | 1.6  |        |
|            | SPI1 kernel      | 1                     | 0.8  | 0.6  |        |
|            | SPI4 registers   | 1.6                   | 1.5  | 1.5  |        |
|            | SPI4 kernel      | 0.5                   | 0.4  | 0.4  |        |
|            | TIM15            | 3.1                   | 2.8  | 2.7  |        |
|            | TIM16            | 2.4                   | 2.1  | 2.1  |        |
|            | TIM17            | 2.2                   | 2    | 1.9  |        |
|            | SPI5 registers   | 1.8                   | 1.7  | 1.7  |        |
|            | SPI5 kernel      | 0.6                   | 0.5  | 0.3  |        |
|            | SAI1 registers   | 1.5                   | 1.4  | 1.4  |        |
|            | SAI1 kernel      | 2                     | 1.7  | 1.5  |        |
|            | SAI2 registers   | 1.5                   | 1.5  | 1.3  |        |
|            | SAI2 kernel      | 2.2                   | 1.9  | 1.8  |        |
|            | SAI3 registers   | 1.8                   | 1.6  | 1.6  |        |
|            | SAI3 kernel      | 2.5                   | 2.3  | 2.1  |        |
|            | DFSDM1 registers | 6                     | 5.4  | 5.2  |        |
|            | DFSDM1 kernel    | 0.9                   | 0.8  | 0.7  |        |
|            | HRTIM            | 40                    | 37   | 35   |        |
|            | Bridge APB2      | 0.1                   | 0.1  | 0.1  |        |

Table 39. Peripheral current consumption in Run mode (continued)

| Peripheral |                   | I <sub>DD</sub> (Typ) |      |      | Unit   |
|------------|-------------------|-----------------------|------|------|--------|
|            |                   | VOS1                  | VOS2 | VOS3 |        |
| APB4       | SYSCFG            | 1                     | 0.7  | 0.7  | μA/MHz |
|            | LPUART1 registers | 1.1                   | 1.1  | 1.1  |        |
|            | LPUART1 kernel    | 2.6                   | 2.4  | 2.1  |        |
|            | SPI6 registers    | 1.6                   | 1.5  | 1.4  |        |
|            | SPI6 kernel       | 0.2                   | 0.2  | 0.2  |        |
|            | I2C4 registers    | 0.1                   | 0.1  | 0.1  |        |
|            | I2C4 kernel       | 2.4                   | 2.1  | 2    |        |
|            | LPTIM2 registers  | 0.5                   | 0.5  | 0.5  |        |
|            | LPTIM2 kernel     | 2.3                   | 2.1  | 1.8  |        |
|            | LPTIM3 registers  | 0.5                   | 0.5  | 0.5  |        |
|            | LPTIM3 kernel     | 2                     | 2.1  | 1.5  |        |
|            | LPTIM4 registers  | 0.5                   | 0.5  | 0.5  |        |
|            | LPTIM4 kernel     | 2                     | 2    | 1.9  |        |
|            | LPTIM5 registers  | 0.5                   | 0.5  | 0.5  |        |
|            | LPTIM5 kernel     | 2                     | 1.8  | 1.5  |        |
|            | COMP1/2           | 0.7                   | 0.5  | 0.5  |        |
|            | VREFBUF           | 0.6                   | 0.4  | 0.4  |        |
|            | RTC               | 1.2                   | 1.1  | 1.1  |        |
|            | SAI4 registers    | 1.6                   | 1.5  | 1.4  |        |
|            | SAI4 kernel       | 1.3                   | 1.3  | 1.2  |        |
|            | Bridge APB4       | 0.1                   | 0.1  | 0.1  |        |

Table 40. Peripheral current consumption in Stop, Standby and VBAT mode

| Symbol          | Parameter                 | Conditions | Typ  | Unit |
|-----------------|---------------------------|------------|------|------|
|                 |                           |            | 3 V  |      |
| I <sub>DD</sub> | RTC+LSE low drive         | -          | 2.32 | μA   |
|                 | RTC+LSE medium-low drive  | -          | 2.4  |      |
|                 | RTC+LSE medium-high drive | -          | 2.7  |      |
|                 | RTC+LSE High drive        | -          | 3    |      |

### 6.3.7 Wakeup time from low-power modes

The wakeup times given in [Table 41](#) are measured starting from the wakeup event trigger up to the first instruction executed by the CPU:

- For Stop or Sleep modes: the wakeup event is WFE.
- WKUP (PC1) pin is used to wakeup from Standby, Stop and Sleep modes.

All timings are derived from tests performed under ambient temperature and  $V_{DD}=3.3$  V.

**Table 41. Low-power mode wakeup timings**

| Symbol              | Parameter                            | Conditions                                | Typ <sup>(1)</sup> | Max <sup>(1)</sup> | Unit             |
|---------------------|--------------------------------------|-------------------------------------------|--------------------|--------------------|------------------|
| $t_{WUSLEEP}^{(2)}$ | Wakeup from Sleep                    | -                                         | 9                  | 10                 | CPU clock cycles |
| $t_{WUSTOP}^{(2)}$  | Wakeup from Stop                     | VOS3, HSI, Flash memory in normal mode    | 4.4                | 5.6                | $\mu s$          |
|                     |                                      | VOS3, HSI, Flash memory in low-power mode | 12                 | 15                 |                  |
|                     |                                      | VOS4, HSI, Flash memory in normal mode    | 15                 | 20                 |                  |
|                     |                                      | VOS4, HSI, Flash memory in low-power mode | 23                 | 28                 |                  |
|                     |                                      | VOS5, HSI, Flash memory in normal mode    | 30                 | 71                 |                  |
|                     |                                      | VOS5, HSI, Flash memory in low-power mode | 38                 | 47                 |                  |
|                     |                                      | VOS3, CSI, Flash memory in normal mode    | 27                 | 37                 |                  |
|                     |                                      | VOS3, CSI, Flash memory in low power mode | 36                 | 50                 |                  |
|                     |                                      | VOS4, CSI, Flash memory in normal mode    | 38                 | 48                 |                  |
|                     |                                      | VOS4, CSI, Flash memory in low-power mode | 47                 | 61                 |                  |
|                     |                                      | VOS5, CSI, Flash memory in normal mode    | 52                 | 64                 |                  |
|                     |                                      | VOS5, CSI, Flash memory in low-power mode | 62                 | 77                 |                  |
| $t_{WUSTOP2}^{(2)}$ | Wakeup from Stop, clock kept running | VOS3, HSI, Flash memory in normal mode    | 2.6                | 3.4                |                  |
|                     |                                      | VOS3, CSI, Flash memory in normal mode    | 26                 | 36                 |                  |
| $t_{WUSTDBY}^{(2)}$ | Wakeup from Standby mode             | -                                         | 390                | 500                |                  |

1. Guaranteed by characterization results.

2. The wakeup times are measured from the wakeup event to the point in which the application code reads the first instruction.

### 6.3.8 External clock source characteristics

#### High-speed external user clock generated from an external source

In bypass mode the HSE oscillator is switched off and the input pin is a standard I/O.

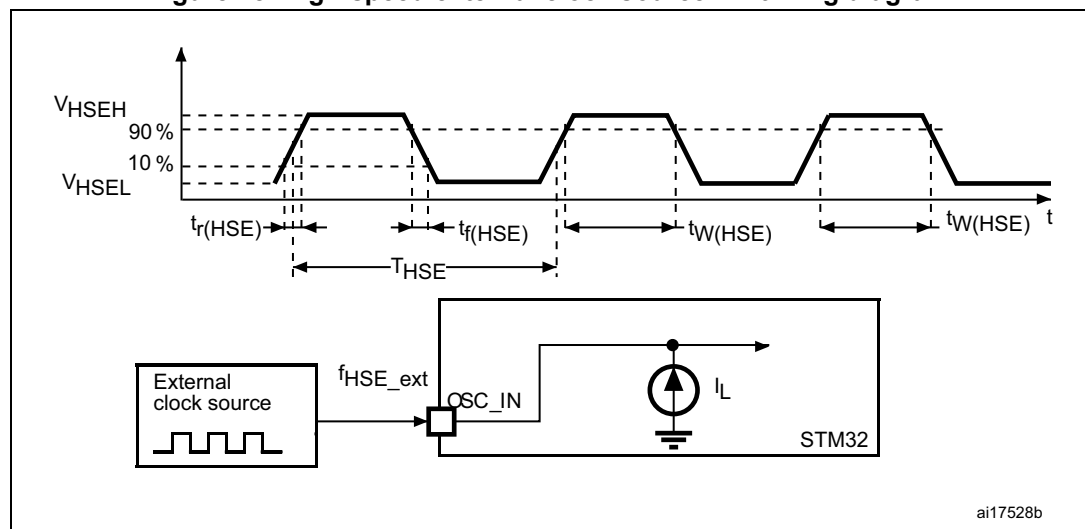
The external clock signal has to respect the [Table 60: I/O static characteristics](#). However, the recommended clock input waveform is shown in [Figure 18](#).

**Table 42. High-speed external user clock characteristics<sup>(1)</sup>**

| Symbol                                | Parameter                            | Min         | Typ | Max         | Unit |
|---------------------------------------|--------------------------------------|-------------|-----|-------------|------|
| $f_{HSE\_ext}$                        | User external clock source frequency | 4           | 25  | 50          | MHz  |
| $V_{SW}$<br>( $V_{HSEH} - V_{HSEL}$ ) | OSC_IN amplitude                     | $0.7V_{DD}$ | -   | $V_{DD}$    | V    |
| $V_{DC}$                              | OSC_IN input voltage                 | $V_{SS}$    | -   | $0.3V_{SS}$ |      |
| $t_{W(HSE)}$                          | OSC_IN high or low time              | 7           | -   | -           | ns   |

1. Guaranteed by design.

**Figure 18. High-speed external clock source AC timing diagram**



### Low-speed external user clock generated from an external source

In bypass mode the LSE oscillator is switched off and the input pin is a standard I/O. The external clock signal has to respect the [Table 60: I/O static characteristics](#). However, the recommended clock input waveform is shown in [Figure 19](#).

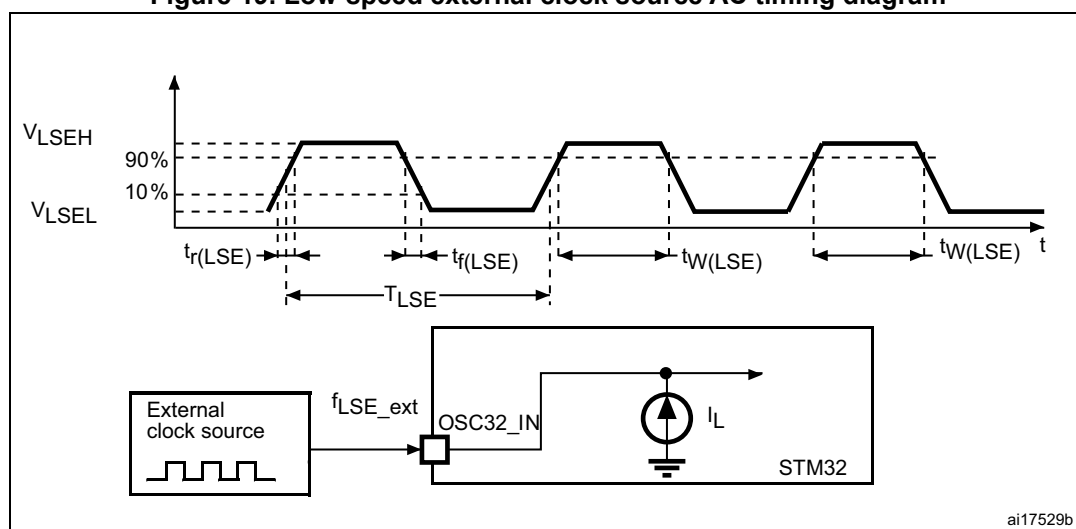
**Table 43. Low-speed external user clock characteristics<sup>(1)</sup>**

| Symbol                         | Parameter                             | Conditions | Min             | Typ    | Max             | Unit |
|--------------------------------|---------------------------------------|------------|-----------------|--------|-----------------|------|
| $f_{LSE\_ext}$                 | User external clock source frequency  | -          | -               | 32.768 | 1000            | kHz  |
| $V_{LSEH}$                     | OSC32_IN input pin high level voltage | -          | $0.7 V_{DDIOx}$ | -      | $V_{DDIOx}$     | V    |
| $V_{LSEL}$                     | OSC32_IN input pin low level voltage  | -          | $V_{SS}$        | -      | $0.3 V_{DDIOx}$ |      |
| $t_{w(LSEH)}$<br>$t_{w(LSEL)}$ | OSC32_IN high or low time             | -          | 250             | -      | -               | ns   |

1. Guaranteed by design.

**Note:** For information on selecting the crystal, refer to the application note AN2867 “Oscillator design guide for ST microcontrollers” available from the ST website [www.st.com](http://www.st.com).

**Figure 19. Low-speed external clock source AC timing diagram**



ai17529b



### High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 48 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in [Table 44](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

**Table 44. 4-48 MHz HSE oscillator characteristics<sup>(1)</sup>**

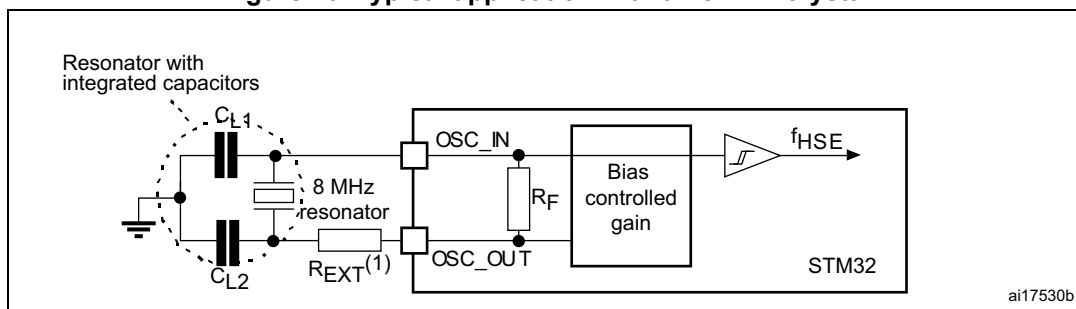
| Symbol                            | Parameter                   | Operating conditions <sup>(2)</sup>                                           | Min | Typ  | Max | Unit |
|-----------------------------------|-----------------------------|-------------------------------------------------------------------------------|-----|------|-----|------|
| F                                 | Oscillator frequency        | -                                                                             | 4   | -    | 48  | MHz  |
| R <sub>F</sub>                    | Feedback resistor           | -                                                                             | -   | 200  | -   | kΩ   |
| I <sub>DD(HSE)</sub>              | HSE current consumption     | During startup <sup>(3)</sup>                                                 | -   | -    | 4   | mA   |
|                                   |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF@4MHz      | -   | 0.35 | -   |      |
|                                   |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 8 MHz  | -   | 0.40 | -   |      |
|                                   |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 16 MHz | -   | 0.45 | -   |      |
|                                   |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 32 MHz | -   | 0.65 | -   |      |
|                                   |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 48 MHz | -   | 0.95 | -   |      |
| G <sub>m</sub> <sub>critmax</sub> | Maximum critical crystal gm | Startup                                                                       | -   | -    | 1.5 | mA/V |
| t <sub>SU</sub> <sup>(4)</sup>    | Start-up time               | V <sub>DD</sub> is stabilized                                                 | -   | 2    | -   | ms   |

1. Guaranteed by design.
2. Resonator characteristics given by the crystal/ceramic resonator manufacturer.
3. This consumption level occurs during the first 2/3 of the t<sub>SU(HSE)</sub> startup time.
4. t<sub>SU(HSE)</sub> is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

For C<sub>L1</sub> and C<sub>L2</sub>, it is recommended to use high-quality external ceramic capacitors in the 5 pF to 25 pF range (typical), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator (see [Figure 20](#)). C<sub>L1</sub> and C<sub>L2</sub> are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C<sub>L1</sub> and C<sub>L2</sub>. The PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing C<sub>L1</sub> and C<sub>L2</sub>.

**Note:** For information on selecting the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website [www.st.com](http://www.st.com).

Figure 20. Typical application with an 8 MHz crystal



1.  $R_{EXT}$  value depends on the crystal characteristics.

### Low-speed external clock generated from a crystal/ceramic resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in [Table 45](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 45. Low-speed external user clock characteristics<sup>(1)</sup>

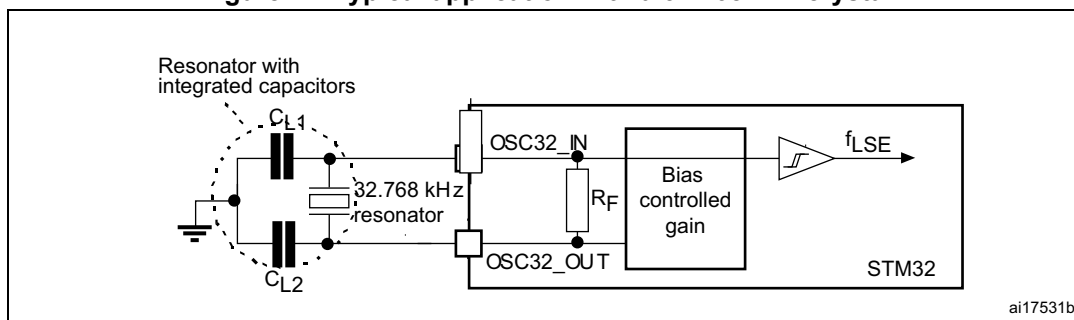
| Symbol            | Parameter                   | Operating conditions <sup>(2)</sup>               | Min | Typ    | Max  | Unit      |
|-------------------|-----------------------------|---------------------------------------------------|-----|--------|------|-----------|
| F                 | Oscillator frequency        | -                                                 | -   | 32.768 | -    | kHz       |
| $I_{DD}$          | LSE current consumption     | LSEDRV[1:0] = 00,<br>Low drive capability         | -   | 290    | -    | nA        |
|                   |                             | LSEDRV[1:0] = 01,<br>Medium Low drive capability  | -   | 390    | -    |           |
|                   |                             | LSEDRV[1:0] = 10,<br>Medium high drive capability | -   | 550    | -    |           |
|                   |                             | LSEDRV[1:0] = 11,<br>High drive capability        | -   | 900    | -    |           |
| $G_{m_{critmax}}$ | Maximum critical crystal gm | LSEDRV[1:0] = 00,<br>Low drive capability         | -   | -      | 0.5  | $\mu A/V$ |
|                   |                             | LSEDRV[1:0] = 01,<br>Medium Low drive capability  | -   | -      | 0.75 |           |
|                   |                             | LSEDRV[1:0] = 10,<br>Medium high drive capability | -   | -      | 1.7  |           |
|                   |                             | LSEDRV[1:0] = 11,<br>High drive capability        | -   | -      | 2.7  |           |
| $t_{SU}^{(3)}$    | Startup time                | VDD is stabilized                                 | -   | 2      | -    | s         |

1. Guaranteed by design.

2. Refer to the note and caution paragraphs below the table, and to the application note AN2867 "Oscillator design guide for ST microcontrollers.
3.  $t_{SU}$  is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768 kHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

**Note:** For information on selecting the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website [www.st.com](http://www.st.com).

**Figure 21. Typical application with a 32.768 kHz crystal**



1. An external resistor is not required between OSC32\_IN and OSC32\_OUT and it is forbidden to add one.

### 6.3.9 Internal clock source characteristics

The parameters given in [Table 46](#) and [Table 49](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

#### 48 MHz high-speed internal RC oscillator (HSI48)

**Table 46. HSI48 oscillator characteristics**

| Symbol                             | Parameter                                                                  | Conditions                                                                  | Min                 | Typ        | Max                 | Unit          |
|------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------|------------|---------------------|---------------|
| $f_{HSI48}$                        | HSI48 frequency                                                            | $V_{DD}=3.3\text{ V}$ , $T_J=30\text{ °C}$                                  | 47.5 <sup>(1)</sup> | 48         | 48.5 <sup>(1)</sup> | MHz           |
| TRIM <sup>(2)</sup>                | USER trimming step                                                         | -                                                                           | -                   | 0.17       | -                   | %             |
| USER TRIM COVERAGE <sup>(3)</sup>  | USER TRIMMING Coverage                                                     | $\pm 32$ steps                                                              | -                   | $\pm 5.45$ | -                   | %             |
| DuCy(HSI48) <sup>(2)</sup>         | Duty Cycle                                                                 | -                                                                           | 45                  | -          | 55                  | %             |
| ACCHSI48_REL <sup>(3)</sup>        | Accuracy of the HSI48 oscillator over temperature (factory calibrated)     | $V_{DD}=1.62\text{ to }3.6\text{ V}$ ,<br>$T_J=-40\text{ to }125\text{ °C}$ | -4.5                | -          | 3.5                 | %             |
| $\Delta VDD(HSI48)$ <sup>(3)</sup> | HSI48 oscillator frequency drift with $V_{DD}$ <sup>(4)</sup>              | $V_{DD}=3\text{ to }3.6\text{ V}$                                           | -                   | 0.025      | 0.05                | %             |
|                                    |                                                                            | $V_{DD}=1.62\text{ V to }3.6\text{ V}$                                      | -                   | 0.05       | 0.1                 |               |
| $t_{su}(HSI48)$ <sup>(2)</sup>     | HSI48 oscillator start-up time                                             | -                                                                           | -                   | 2.1        | 3.5                 | $\mu\text{s}$ |
| $I_{DD}(HSI48)$ <sup>(2)</sup>     | HSI48 oscillator power consumption                                         | -                                                                           | -                   | 350        | 400                 | $\mu\text{A}$ |
| $N_T$ jitter                       | Next transition jitter<br>Accumulated jitter on 28 cycles <sup>(5)</sup>   | -                                                                           | -                   | $\pm 0.15$ | -                   | ns            |
| $P_T$ jitter                       | Paired transition jitter<br>Accumulated jitter on 56 cycles <sup>(5)</sup> | -                                                                           | -                   | $\pm 0.25$ | -                   | ns            |

1. Guaranteed by test in production.
2. Guaranteed by design.
3. Guaranteed by characterization.
4. These values are obtained by using the formula:  
(Freq(3.6V) - Freq(3.0V)) / Freq(3.0V) or (Freq(3.6V) - Freq(1.62V)) / Freq(1.62V).
5. Jitter measurements are performed without clock source activated in parallel.

### 64 MHz high-speed internal RC oscillator (HSI)

Table 47. HSI oscillator characteristics<sup>(1)</sup>

| Symbol                  | Parameter                                                                | Conditions                                                                 | Min                 | Typ   | Max                 | Unit |
|-------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------|-------|---------------------|------|
| f <sub>HSI</sub>        | HSI frequency                                                            | V <sub>DD</sub> =3.3 V, T <sub>J</sub> =30 °C                              | 63.7 <sup>(2)</sup> | 64    | 64.3 <sup>(2)</sup> | MHz  |
| TRIM                    | HSI user trimming step                                                   | Trimming is not a multiple of 32                                           | -                   | 0.24  | 0.32                | %    |
|                         |                                                                          | Trimming is 128, 256 and 384                                               | -5.2                | -1.8  | -                   |      |
|                         |                                                                          | Trimming is 64, 192, 320 and 448                                           | -1.4                | -0.8  | -                   |      |
|                         |                                                                          | Other trimming are a multiple of 32 (not including multiple of 64 and 128) | -0.6                | -0.25 | -                   |      |
| DuCy(HSI)               | Duty Cycle                                                               | -                                                                          | 45                  | -     | 55                  | %    |
| ΔV <sub>DD</sub> (HSI)  | HSI oscillator frequency drift over V <sub>DD</sub> (reference is 3.3 V) | V <sub>DD</sub> =1.62 to 3.6 V                                             | -0.12               | -     | 0.03                | %    |
| ΔT <sub>EMP</sub> (HSI) | HSI oscillator frequency drift over temperature (reference is 64 MHz)    | T <sub>J</sub> = -20 to 105 °C                                             | -1 <sup>(3)</sup>   | -     | 1 <sup>(3)</sup>    | %    |
|                         |                                                                          | T <sub>J</sub> = -40 to T <sub>Jmax</sub> °C                               | -2 <sup>(3)</sup>   | -     | 1 <sup>(3)</sup>    |      |
| t <sub>su</sub> (HSI)   | HSI oscillator start-up time                                             | -                                                                          | -                   | 1.4   | 2                   | μs   |
| t <sub>stab</sub> (HSI) | HSI oscillator stabilization time                                        | at 1% of target frequency                                                  | -                   | 4     | 8                   | μs   |
| I <sub>DD</sub> (HSI)   | HSI oscillator power consumption                                         | -                                                                          | -                   | 300   | 400                 | μA   |

1. Guaranteed by design unless otherwise specified.
2. Guaranteed by test in production.
3. Guaranteed by characterization.

### 4 MHz low-power internal RC oscillator (CSI)

Table 48. CSI oscillator characteristics<sup>(1)</sup>

| Symbol           | Parameter     | Conditions                                    | Min                 | Typ  | Max                 | Unit |
|------------------|---------------|-----------------------------------------------|---------------------|------|---------------------|------|
| f <sub>CSI</sub> | CSI frequency | V <sub>DD</sub> =3.3 V, T <sub>J</sub> =30 °C | 3.96 <sup>(2)</sup> | 4    | 4.04 <sup>(2)</sup> | MHz  |
| TRIM             | Trimming step | -                                             | -                   | 0.35 | -                   | %    |
| DuCy(CSI)        | Duty Cycle    | -                                             | 45                  | -    | 55                  | %    |

Table 48. CSI oscillator characteristics<sup>(1)</sup> (continued)

| Symbol                       | Parameter                                                                   | Conditions                                         | Min | Typ                 | Max                | Unit          |
|------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------|-----|---------------------|--------------------|---------------|
| $\Delta_{\text{TEMP}}$ (CSI) | CSI oscillator frequency drift over temperature                             | $T_J = 0 \text{ to } 85 \text{ }^\circ\text{C}$    | -   | -3.7 <sup>(3)</sup> | 4.5 <sup>(3)</sup> | %             |
|                              |                                                                             | $T_J = -40 \text{ to } 125 \text{ }^\circ\text{C}$ | -   | -11 <sup>(3)</sup>  | 7.5 <sup>(3)</sup> |               |
| $D_{\text{VDD}}$ (CSI)       | CSI oscillator frequency drift over $V_{\text{DD}}$                         | $V_{\text{DD}} = 1.62 \text{ to } 3.6 \text{ V}$   | -   | -0.06               | 0.06               | %             |
| $t_{\text{su}}$ (CSI)        | CSI oscillator startup time                                                 | -                                                  | -   | 1                   | 2                  | $\mu\text{s}$ |
| $t_{\text{stab}}$ (CSI)      | CSI oscillator stabilization time (to reach $\pm 3\%$ of $f_{\text{CSI}}$ ) | -                                                  | -   | 4                   | 8                  | cycle         |
| $I_{\text{DD}}$ (CSI)        | CSI oscillator power consumption                                            | -                                                  | -   | 23                  | 30                 | $\mu\text{A}$ |

1. Guaranteed by design.
2. Guaranteed by test in production.
3. Guaranteed by characterization.

### Low-speed internal (LSI) RC oscillator

Table 49. LSI oscillator characteristics

| Symbol                                 | Parameter                                             | Conditions                                                                                         | Min   | Typ | Max   | Unit          |
|----------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------|-----|-------|---------------|
| $f_{\text{LSI}}$ <sup>(1)</sup>        | LSI frequency                                         | $V_{\text{DD}} = 3.3 \text{ V}, T_J = 25 \text{ }^\circ\text{C}$                                   | 31.4  | 32  | 32.6  | kHz           |
|                                        |                                                       | $T_J = -40 \text{ to } 105 \text{ }^\circ\text{C}, V_{\text{DD}} = 1.62 \text{ to } 3.6 \text{ V}$ | 29.76 | -   | 33.60 |               |
| $t_{\text{su}}$ (LSI) <sup>(2)</sup>   | LSI oscillator startup time                           | -                                                                                                  | -     | 80  | 130   | $\mu\text{s}$ |
| $t_{\text{stab}}$ (LSI) <sup>(2)</sup> | LSI oscillator stabilization time (5% of final value) | -                                                                                                  | -     | 120 | 170   |               |
| $I_{\text{DD}}$ (LSI) <sup>(2)</sup>   | LSI oscillator power consumption                      | -                                                                                                  | -     | 130 | 280   | nA            |

1. Guaranteed by characterization results.
2. Guaranteed by design.

### 6.3.10 PLL characteristics

The parameters given in [Table 50](#) are derived from tests performed under temperature and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

Table 50. PLL characteristics (wide VCO frequency range)<sup>(1)</sup>

| Symbol               | Parameter                  | Conditions | Min | Typ | Max | Unit |
|----------------------|----------------------------|------------|-----|-----|-----|------|
| $f_{\text{PLL\_IN}}$ | PLL input clock            | -          | 2   | -   | 16  | MHz  |
|                      | PLL input clock duty cycle | -          | 10  | -   | 90  | %    |

Table 50. PLL characteristics (wide VCO frequency range)<sup>(1)</sup> (continued)

| Symbol                              | Parameter                                | Conditions                          | Min               | Typ               | Max                | Unit |
|-------------------------------------|------------------------------------------|-------------------------------------|-------------------|-------------------|--------------------|------|
| f <sub>PLL_P_OUT</sub>              | PLL multiplier output clock P            | VOS1                                | 1.5               | -                 | 400 <sup>(2)</sup> | MHz  |
|                                     |                                          | VOS2                                | 1.5               | -                 | 300                |      |
|                                     |                                          | VOS3                                | 1.5               | -                 | 200                |      |
| f <sub>PLL_Q_OUT</sub>              | PLL multiplier output clock Q/R          | VOS1                                | 1.5               | -                 | 400 <sup>(2)</sup> |      |
|                                     |                                          | VOS2                                | 1.5               | -                 | 300                |      |
|                                     |                                          | VOS3                                | 1.5               | -                 | 200                |      |
| f <sub>VCO_OUT</sub>                | PLL VCO output                           | -                                   | 192               | -                 | 836                |      |
| t <sub>LOCK</sub>                   | PLL lock time                            | Normal mode                         | -                 | 50 <sup>(3)</sup> | 150 <sup>(3)</sup> | μs   |
|                                     |                                          | Sigma-delta mode<br>(CKIN ≥ 8 MHz)  | -                 | 58 <sup>(3)</sup> | 166 <sup>(3)</sup> |      |
| Jitter                              | Cycle-to-cycle jitter <sup>(4)</sup>     | VCO = 192 MHz                       | -                 | 134               | -                  | ±ps  |
|                                     |                                          | VCO = 200 MHz                       | -                 | 134               | -                  |      |
|                                     |                                          | VCO = 400 MHz                       | -                 | 76                | -                  |      |
|                                     |                                          | VCO = 800 MHz                       | -                 | 39                | -                  |      |
|                                     | Long term jitter                         | Normal mode                         | -                 | ±0.7              | -                  | %    |
|                                     |                                          | Sigma-delta mode<br>(CKIN = 16 MHz) | -                 | ±0.8              | -                  |      |
| I <sub>DD(PLL)</sub> <sup>(3)</sup> | PLL power consumption on V <sub>DD</sub> | VCO freq =<br>420 MHz               | V <sub>DDA</sub>  | -                 | 440                | μA   |
|                                     |                                          |                                     | V <sub>CORE</sub> | -                 | 530                |      |
|                                     |                                          | VCO freq =<br>150 MHz               | V <sub>DDA</sub>  | -                 | 180                |      |
|                                     |                                          |                                     | V <sub>CORE</sub> | -                 | 200                |      |

1. Guaranteed by design unless otherwise specified.

2. This value must be limited to the maximum frequency due to the product limitation (400 MHz for VOS1, 300 MHz for VOS2, 200 MHz for VOS3).

3. Guaranteed by characterization results.

4. Integer mode only.

Table 51. PLL characteristics (medium VCO frequency range)<sup>(1)</sup>

| Symbol               | Parameter                           | Conditions       | Min       | Typ               | Max                | Unit |
|----------------------|-------------------------------------|------------------|-----------|-------------------|--------------------|------|
| f <sub>PLL_IN</sub>  | PLL input clock                     | -                | 1         | -                 | 2                  | MHz  |
|                      | PLL input clock duty cycle          | -                | 10        | -                 | 90                 | %    |
| f <sub>PLL_OUT</sub> | PLL multiplier output clock P, Q, R | VOS1             | 1.17      | -                 | 210                | MHz  |
|                      |                                     | VOS2             | 1.17      | -                 | 210                |      |
|                      |                                     | VOS3             | 1.17      | -                 | 200                |      |
| f <sub>VCO_OUT</sub> | PLL VCO output                      | -                | 150       | -                 | 420                | MHz  |
| t <sub>LOCK</sub>    | PLL lock time                       | Normal mode      | -         | 60 <sup>(2)</sup> | 100 <sup>(2)</sup> | μs   |
|                      |                                     | Sigma-delta mode | forbidden | -                 | -                  | μs   |

Table 51. PLL characteristics (medium VCO frequency range)<sup>(1)</sup> (continued)

| Symbol                | Parameter                                | Conditions                             |               | Min | Typ    | Max  | Unit          |
|-----------------------|------------------------------------------|----------------------------------------|---------------|-----|--------|------|---------------|
| Jitter                | Cycle-to-cycle jitter <sup>(3)</sup>     | -                                      | VCO = 150 MHz | -   | 145    | -    | +/- ps        |
|                       |                                          |                                        | VCO = 300 MHz | -   | 91     | -    |               |
|                       |                                          |                                        | VCO = 400 MHz | -   | 64     | -    |               |
|                       |                                          |                                        | VCO = 420 MHz | -   | 63     | -    |               |
|                       | Period jitter                            | $f_{\text{PLL\_OUT}} = 50 \text{ MHz}$ | VCO = 150 MHz | -   | 55     | -    | +/- ps        |
|                       |                                          |                                        | VCO = 400 MHz | -   | 30     | -    |               |
|                       | Long term jitter                         | Normal mode                            | VCO = 150 MHz | -   | -      | -    | %             |
|                       |                                          |                                        | VCO = 300 MHz | -   | -      | -    |               |
|                       |                                          |                                        | VCO = 400 MHz | -   | +/-0.3 | -    |               |
| $I(\text{PLL})^{(2)}$ | PLL power consumption on $V_{\text{DD}}$ | VCO freq = 420MHz                      | VDD           | -   | 440    | 1150 | $\mu\text{A}$ |
|                       |                                          |                                        | VCORE         | -   | 530    | -    |               |
|                       |                                          | VCO freq = 150MHz                      | VDD           | -   | 180    | 500  |               |
|                       |                                          |                                        | VCORE         | -   | 200    | -    |               |

1. Guaranteed by design unless otherwise specified.

2. Guaranteed by characterization results.

3. Integer mode only.

### 6.3.11 Memory characteristics

#### Flash memory

The characteristics are given at  $T_J = -40$  to  $125^\circ\text{C}$  unless otherwise specified.

The devices are shipped to customers with the Flash memory erased.

Table 52. Flash memory characteristics

| Symbol          | Parameter      | Conditions                | Min | Typ  | Max | Unit |
|-----------------|----------------|---------------------------|-----|------|-----|------|
| $I_{\text{DD}}$ | Supply current | Write / Erase 8-bit mode  | -   | 6.5  | -   | mA   |
|                 |                | Write / Erase 16-bit mode | -   | 11.5 | -   |      |
|                 |                | Write / Erase 32-bit mode | -   | 20   | -   |      |
|                 |                | Write / Erase 64-bit mode | -   | 35   | -   |      |

Table 53. Flash memory programming (single bank configuration nDBANK=1)

| Symbol                  | Parameter                        | Conditions                     | Min <sup>(1)</sup> | Typ | Max <sup>(1)</sup> | Unit          |
|-------------------------|----------------------------------|--------------------------------|--------------------|-----|--------------------|---------------|
| $t_{\text{prog}}$       | Word (266 bits) programming time | Program/erase parallelism x 8  | -                  | 290 | 580 <sup>(2)</sup> | $\mu\text{s}$ |
|                         |                                  | Program/erase parallelism x 16 | -                  | 180 | 360                |               |
|                         |                                  | Program/erase parallelism x 32 | -                  | 130 | 260                |               |
|                         |                                  | Program/erase parallelism x 64 | -                  | 100 | 200                |               |
| $t_{\text{ERASE128KB}}$ | Sector (128 KB) erase time       | Program/erase parallelism x 8  | -                  | 2   | 4                  | s             |
|                         |                                  | Program/erase parallelism x 16 | -                  | 1.8 | 3.6                |               |
|                         |                                  | Program/erase parallelism x 32 | -                  |     |                    |               |
| $t_{\text{ME}}$         | Mass erase time                  | Program/erase parallelism x 8  | -                  | 13  | 26                 |               |
|                         |                                  | Program/erase parallelism x 16 | -                  | 8   | 16                 |               |
|                         |                                  | Program/erase parallelism x 32 | -                  | 6   | 12                 |               |
|                         |                                  | Program/erase parallelism x 64 | -                  | 5   | 10                 |               |
| $V_{\text{prog}}$       | Programming voltage              | Program parallelism x 8        | 1.62               | -   | 3.6                | V             |
|                         |                                  | Program parallelism x 16       |                    |     |                    |               |
|                         |                                  | Program parallelism x 32       |                    |     |                    |               |
|                         |                                  | Program parallelism x 64       | 1.8                | -   | 3.6                |               |

1. Guaranteed by characterization results.

2. The maximum programming time is measured after 10K erase operations.

Table 54. Flash memory endurance and data retention

| Symbol           | Parameter      | Conditions                                   | Value              | Unit    |
|------------------|----------------|----------------------------------------------|--------------------|---------|
|                  |                |                                              | Min <sup>(1)</sup> |         |
| $N_{\text{END}}$ | Endurance      | $T_J = -40$ to $+125$ °C (6 suffix versions) | 10                 | kcycles |
| $t_{\text{RET}}$ | Data retention | 1 kcycle at $T_A = 85$ °C                    | 30                 | Years   |
|                  |                | 10 kcycles at $T_A = 55$ °C                  | 20                 |         |

1. Guaranteed by characterization results.



### 6.3.12 EMC characteristics

Susceptibility tests are performed on a sample basis during device characterization.

#### Functional EMS (electromagnetic susceptibility)

While a simple application is executed on the device (toggling 2 LEDs through I/O ports), the device is stressed by two electromagnetic events until a failure occurs. The failure is indicated by the LEDs:

- **Electrostatic discharge (ESD)** (positive and negative) is applied to all device pins until a functional disturbance occurs. This test is compliant with the IEC 61000-4-2 standard.
- **FTB**: A burst of fast transient voltage (positive and negative) is applied to  $V_{DD}$  and  $V_{SS}$  through a 100 pF capacitor, until a functional disturbance occurs. This test is compliant with the IEC 61000-4-4 standard.

A device reset allows normal operations to be resumed.

The test results are given in [Table 55](#). They are based on the EMS levels and classes defined in application note AN1709.

**Table 55. EMS characteristics**

| Symbol     | Parameter                                                                                                                         | Conditions                                                                                                                | Level/<br>Class |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|
| $V_{FESD}$ | Voltage limits to be applied on any I/O pin to induce a functional disturbance                                                    | $V_{DD} = 3.3\text{ V}$ , $T_A = +25\text{ °C}$ , UFBGA240, $f_{rcc\_c\_ck} = 400\text{ MHz}$ , conforms to IEC 61000-4-2 | 3B              |
| $V_{FTB}$  | Fast transient voltage burst limits to be applied through 100 pF on $V_{DD}$ and $V_{SS}$ pins to induce a functional disturbance |                                                                                                                           | 4B              |

As a consequence, it is recommended to add a serial resistor (1 kΩ) located as close as possible to the MCU to the pins exposed to noise (connected to tracks longer than 50 mm on PCB).

#### Designing hardened software to avoid noise problems

EMC characterization and optimization are performed at component level with a typical application environment and simplified MCU software. It should be noted that good EMC performance is highly dependent on the user application and the software in particular.

Therefore it is recommended that the user applies EMC software optimization and prequalification tests in relation with the EMC level requested for his application.

#### Software recommendations

The software flowchart must include the management of runaway conditions such as:

- Corrupted program counter
- Unexpected reset
- Critical Data corruption (control registers...)

### Prequalification trials

Most of the common failures (unexpected reset and program counter corruption) can be reproduced by manually forcing a low state on the NRST pin or the Oscillator pins for 1 second.

To complete these trials, ESD stress can be applied directly on the device, over the range of specification values. When unexpected behavior is detected, the software can be hardened to prevent unrecoverable errors occurring (see application note AN1015).

### Electromagnetic Interference (EMI)

The electromagnetic field emitted by the device are monitored while a simple application, executing EEMBC code, is running. This emission test is compliant with SAE IEC61967-2 standard which specifies the test board and the pin loading.

**Table 56. EMI characteristics**

| Symbol           | Parameter  | Conditions                                                                                  | Monitored frequency band | Max vs. [f <sub>HSE</sub> /f <sub>CPU</sub> ] | Unit |
|------------------|------------|---------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------|------|
|                  |            |                                                                                             |                          | 8/400 MHz                                     |      |
| S <sub>EMI</sub> | Peak level | V <sub>DD</sub> = 3.6 V, T <sub>A</sub> = 25 °C, UFBGA240 package, conforming to IEC61967-2 | 0.1 to 30 MHz            | 6                                             | dBμV |
|                  |            |                                                                                             | 30 to 130 MHz            | 5                                             |      |
|                  |            |                                                                                             | 130 MHz to 1 GHz         | 13                                            |      |
|                  |            |                                                                                             | 1 GHz to 2 GHz           | 7                                             |      |
|                  |            |                                                                                             | EMI Level                | 2.5                                           | -    |

### 6.3.13 Absolute maximum ratings (electrical sensitivity)

Based on three different tests (ESD, LU) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

#### Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse) are applied to the pins of each sample according to each pin combination. This test conforms to the ANSI/ESDA/JEDEC JS-001 and ANSI/ESDA/JEDEC JS-002 standards.

**Table 57. ESD absolute maximum ratings**

| Symbol                | Ratings                                               | Conditions                                                   | Packages | Class | Maximum value <sup>(1)</sup> | Unit |
|-----------------------|-------------------------------------------------------|--------------------------------------------------------------|----------|-------|------------------------------|------|
| V <sub>ESD(HBM)</sub> | Electrostatic discharge voltage (human body model)    | T <sub>A</sub> = +25 °C conforming to ANSI/ESDA/JEDEC JS-001 | All      | 1C    | 1000                         | V    |
| V <sub>ESD(CDM)</sub> | Electrostatic discharge voltage (charge device model) | T <sub>A</sub> = +25 °C conforming to ANSI/ESDA/JEDEC JS-002 | All      | C1    | 250                          |      |

1. Guaranteed by characterization results.

### Static latchup

Two complementary static tests are required on six parts to assess the latchup performance:

- A supply overvoltage is applied to each power supply pin
- A current injection is applied to each input, output and configurable I/O pin

These tests are compliant with JESD78 IC latchup standard.

**Table 58. Electrical sensitivities**

| Symbol | Parameter            | Conditions                                               | Class      |
|--------|----------------------|----------------------------------------------------------|------------|
| LU     | Static latchup class | $T_A = +25\text{ }^{\circ}\text{C}$ conforming to JESD78 | II level A |

### 6.3.14 I/O current injection characteristics

As a general rule, a current injection to the I/O pins, due to external voltage below  $V_{SS}$  or above  $V_{DD}$  (for standard, 3.3 V-capable I/O pins) should be avoided during the normal product operation. However, in order to give an indication of the robustness of the microcontroller in cases when an abnormal injection accidentally happens, susceptibility tests are performed on a sample basis during the device characterization.

#### Functional susceptibility to I/O current injection

While a simple application is executed on the device, the device is stressed by injecting current into the I/O pins programmed in floating input mode. While current is injected into the I/O pin, one at a time, the device is checked for functional failures.

The failure is indicated by an out of range parameter: ADC error above a certain limit (higher than 5 LSB TUE), out of conventional limits of induced leakage current on adjacent pins (out of  $-5\text{ }\mu\text{A}/+0\text{ }\mu\text{A}$  range), or other functional failure (for example reset, oscillator frequency deviation).

The following tables are the compilation of the SIC1/SIC2 and functional ESD results.

Negative induced A negative induced leakage current is caused by negative injection and positive induced leakage current by positive injection.

**Table 59. I/O current injection susceptibility<sup>(1)</sup>**

| Symbol    | Description                                                              | Functional susceptibility |                    | Unit |
|-----------|--------------------------------------------------------------------------|---------------------------|--------------------|------|
|           |                                                                          | Negative injection        | Positive injection |      |
| $I_{INJ}$ | PA7, PC5, PG1, PB14, PJ7, PA11, PA12, PA13, PA14, PA15, PJ12, PB4        | 5                         | 0                  | mA   |
|           | PA2, PH2, PH3, PE8, PA6, PA7, PC4, PE7, PE10, PE11                       | 0                         | NA                 |      |
|           | PA0, PA_C, PA1, PA1_C, PC2, PC2_C, PC3, PC3_C, PA4, PA5, PH4, PH5, BOOT0 | 0                         | 0                  |      |
|           | All other I/Os                                                           | 5                         | NA                 |      |

1. Guaranteed by characterization.

### 6.3.15 I/O port characteristics

#### General input/output characteristics

Unless otherwise specified, the parameters given in [Table 60: I/O static characteristics](#) are derived from tests performed under the conditions summarized in [Table 24: General operating conditions](#). All I/Os are CMOS and TTL compliant (except for BOOT0).

**Table 60. I/O static characteristics**

| Symbol           | Parameter                                                | Condition                                                                  | Min                                | Typ | Max                               | Unit |
|------------------|----------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------|-----|-----------------------------------|------|
| $V_{IL}$         | I/O input low level voltage except BOOT0                 | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                                 | -                                  | -   | $0.3V_{DD}^{(1)}$                 | V    |
|                  | I/O input low level voltage except BOOT0                 |                                                                            | -                                  | -   | $0.4V_{DD}^{(2)}$<br>$0.1^{(2)}$  |      |
|                  | BOOT0 I/O input low level voltage                        |                                                                            | -                                  | -   | $0.19V_{DD}^{(2)}$<br>$0.1^{(2)}$ |      |
| $V_{IH}$         | I/O input high level voltage except BOOT0                | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                                 | $0.7V_{DD}^{(1)}$                  | -   | -                                 | V    |
|                  | I/O input high level voltage except BOOT0 <sup>(3)</sup> |                                                                            | $0.47V_{DD}^{(2)}$<br>$0.25^{(2)}$ | -   | -                                 |      |
|                  | BOOT0 I/O input high level voltage <sup>(3)</sup>        |                                                                            | $0.17V_{DD}^{(2)}$<br>$0.6^{(2)}$  | -   | -                                 |      |
| $V_{HYS}^{(2)}$  | TT_xx, FT_xxx and NRST I/O input hysteresis              | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                                 | -                                  | 250 | -                                 | mV   |
|                  | BOOT0 I/O input hysteresis                               |                                                                            | -                                  | 200 | -                                 |      |
| $I_{leak}^{(4)}$ | FT_xx Input leakage current <sup>(2)</sup>               | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                              | -                                  | -   | +/-250                            | nA   |
|                  |                                                          | $\text{Max}(V_{DDXXX}) < V_{IN} \leq 5.5\text{ V}$<br><sup>(5)(6)(9)</sup> | -                                  | -   | 1500                              |      |
|                  | FT_u IO                                                  | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                              | -                                  | -   | +/- 350                           |      |
|                  |                                                          | $\text{Max}(V_{DDXXX}) < V_{IN} \leq 5.5\text{ V}$<br><sup>(5)(6)(9)</sup> | -                                  | -   | 5000 <sup>(7)</sup>               |      |
|                  | TT_xx Input leakage current                              | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                              | -                                  | -   | +/-250                            |      |
|                  | VPP (BOOT0 alternate function)                           | $0 < V_{IN} \leq V_{DDIOx}$                                                | -                                  | -   | 15                                |      |
|                  |                                                          | $V_{DDIOx} < V_{IN} \leq 9\text{ V}$                                       | -                                  | -   | 35                                |      |
| $R_{PU}$         | Weak pull-up equivalent resistor <sup>(8)</sup>          | $V_{IN}=V_{SS}$                                                            | 30                                 | 40  | 50                                | kΩ   |
| $R_{PD}$         | Weak pull-down equivalent resistor <sup>(8)</sup>        | $V_{IN}=V_{DD}^{(9)}$                                                      | 30                                 | 40  | 50                                |      |
| $C_{IO}$         | I/O pin capacitance                                      | -                                                                          | -                                  | 5   | -                                 | pF   |

1. Compliant with CMOS requirement.

2. Guaranteed by design.

3.  $V_{DDIOx}$  represents  $V_{DDIO1}$ ,  $V_{DDIO2}$  or  $V_{DDIO3}$ .  $V_{DDIOx} = V_{DD}$ .

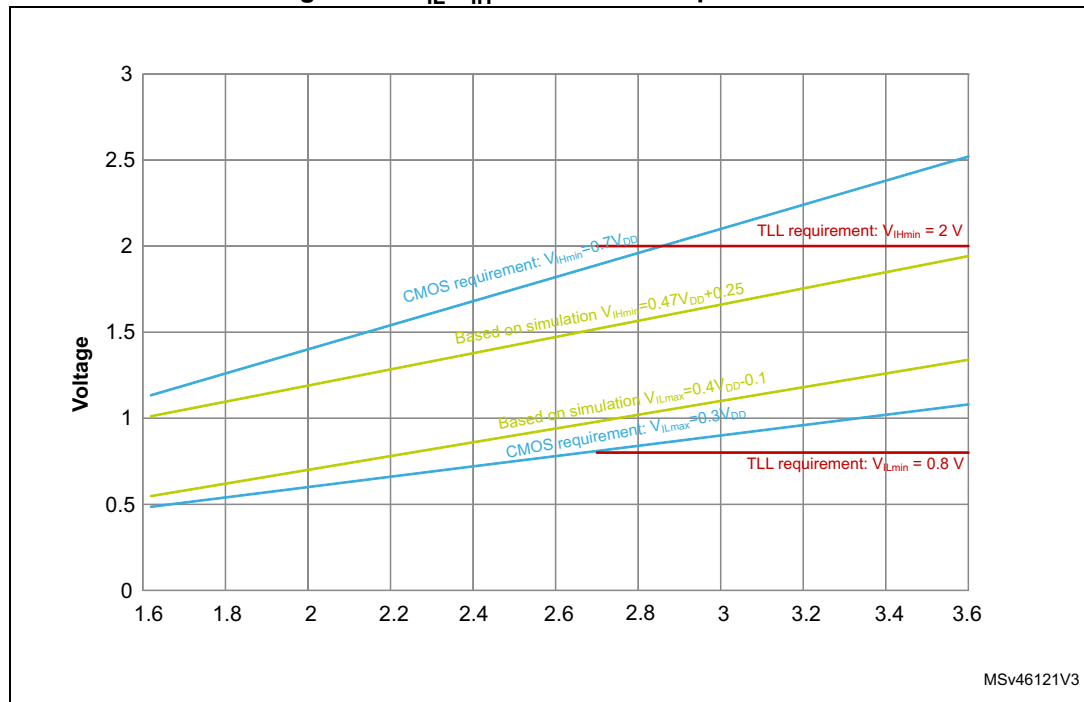
4. This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula:  $I_{Total\_leak\_max} = 10\text{ }\mu\text{A} + [\text{number of I/Os where } V_{IN} \text{ is applied on the pad}] \times I_{lk}(Max)$ .

5. All FT\_xx IO except FT\_lu, FT\_u and PC3.

6.  $V_{IN}$  must be less than  $\text{Max}(V_{DDXXX}) + 3.6 \text{ V}$ .
7. To sustain a voltage higher than  $\text{MIN}(V_{DD}, V_{DDA}, V_{DD33USB}) + 0.3 \text{ V}$ , the internal pull-up and pull-down resistors must be disabled.
8. The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).
9.  $\text{Max}(V_{DDXXX})$  is the maximum value of all the I/O supplies.

All I/Os are CMOS and TTL compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements for FT I/Os is shown in [Figure 22](#).

**Figure 22.  $V_{IL}/V_{IH}$  for all I/Os except BOOT0**



### Output driving current

The GPIOs (general purpose input/outputs) can sink or source up to  $\pm 8 \text{ mA}$ , and sink or source up to  $\pm 20 \text{ mA}$  (with a relaxed  $V_{OL}/V_{OH}$ ).

In the user application, the number of I/O pins which can drive current must be limited to respect the absolute maximum rating specified in [Section 6.2](#). In particular:

- The sum of the currents sourced by all the I/Os on  $V_{DD}$ , plus the maximum Run consumption of the MCU sourced on  $V_{DD}$ , cannot exceed the absolute maximum rating  $\Sigma I_{VDD}$  (see [Table 22](#)).
- The sum of the currents sunk by all the I/Os on  $V_{SS}$  plus the maximum Run consumption of the MCU sunk on  $V_{SS}$  cannot exceed the absolute maximum rating  $\Sigma I_{VSS}$  (see [Table 22](#)).

### Output voltage levels

Unless otherwise specified, the parameters given in [Table 61](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#). All I/Os are CMOS and TTL compliant.

**Table 61. Output voltage characteristics for all I/Os except PC13, PC14, PC15 and PI8<sup>(1)</sup>**

| Symbol            | Parameter                                               | Conditions <sup>(3)</sup>                                                                         | Min          | Max | Unit |
|-------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------|-----|------|
| $V_{OL}$          | Output low level voltage                                | CMOS port <sup>(2)</sup><br>$I_{IO}=8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -            | 0.4 | V    |
| $V_{OH}$          | Output high level voltage                               | CMOS port <sup>(2)</sup><br>$I_{IO}=-8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ | $V_{DD}-0.4$ | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | TTL port <sup>(2)</sup><br>$I_{IO}=8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$   | -            | 0.4 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | TTL port <sup>(2)</sup><br>$I_{IO}=-8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | 2.4          | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | $I_{IO}=20\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 1.3 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | $I_{IO}=-20\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                            | $V_{DD}-1.3$ | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | $I_{IO}=4\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 0.4 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | $I_{IO}=-4\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} < 3.6\text{ V}$                               | $V_{DD}-0.4$ | -   |      |
| $V_{OLFM+}^{(3)}$ | Output low level voltage for an FTf I/O pin in FM+ mode | $I_{IO}=20\text{ mA}$<br>$2.3\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 0.4 |      |
|                   |                                                         | $I_{IO}=10\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                            | -            | 0.4 |      |

1. The I/O current sourced or sunk by the device must always respect the absolute maximum rating specified in [Table 21: Voltage characteristics](#), and the sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings  $\Sigma I_{IO}$ .
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Guaranteed by design.

Table 62. Output voltage characteristics for PC13, PC14, PC15 and PI8<sup>(1)</sup>

| Symbol         | Parameter                 | Conditions <sup>(3)</sup>                                                                         | Min          | Max | Unit |
|----------------|---------------------------|---------------------------------------------------------------------------------------------------|--------------|-----|------|
| $V_{OL}$       | Output low level voltage  | CMOS port <sup>(2)</sup><br>$I_{IO}=3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -            | 0.4 | V    |
| $V_{OH}$       | Output high level voltage | CMOS port <sup>(2)</sup><br>$I_{IO}=-3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ | $V_{DD}-0.4$ | -   |      |
| $V_{OL}^{(3)}$ | Output low level voltage  | TTL port <sup>(2)</sup><br>$I_{IO}=3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$   | -            | 0.4 |      |
| $V_{OH}^{(3)}$ | Output high level voltage | TTL port <sup>(2)</sup><br>$I_{IO}=-3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | 2.4          | -   |      |
| $V_{OL}^{(3)}$ | Output low level voltage  | $I_{IO}=1.5\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                           | -            | 0.4 |      |
| $V_{OH}^{(3)}$ | Output high level voltage | $I_{IO}=-1.5\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                          | $V_{DD}-0.4$ | -   |      |

1. The I/O current sourced or sunk by the device must always respect the absolute maximum rating specified in [Table 21: Voltage characteristics](#), and the sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings  $\Sigma I/O$ .
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Guaranteed by design.

**Output buffer timing characteristics (HSLV option disabled)**

The HSLV bit of SYSCFG\_CCCSR register can be used to optimize the I/O speed when the product voltage is below 2.5 V.

**Table 63. Output timing characteristics (HSLV OFF)<sup>(1)(2)</sup>**

| Speed | Symbol           | Parameter                                                                 | conditions                                | Min | Max  | Unit |
|-------|------------------|---------------------------------------------------------------------------|-------------------------------------------|-----|------|------|
| 00    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 12   | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 3    |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 12   |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 3    |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 16   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 4    |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 16.6 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 33.3 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 13.3 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 25   |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 10   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 20   |      |
| 01    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 60   | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 15   |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 80   |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 15   |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 110  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 20   |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 5.2  | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 10   |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 4.2  |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 7.5  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 2.8  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 5.2  |      |



Table 63. Output timing characteristics (HSLV OFF)<sup>(1)(2)</sup> (continued)

| Speed | Symbol           | Parameter                                                                 | conditions                                               | Min | Max | Unit |
|-------|------------------|---------------------------------------------------------------------------|----------------------------------------------------------|-----|-----|------|
| 10    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 85  | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 35  |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 110 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 40  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 166 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 100 |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 3.8 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 6.9 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 2.8 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 5.2 |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 1.8 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 3.3 |      |
| 11    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 100 | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 50  |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 133 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 66  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 220 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 85  |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 3.3 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 6.6 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 2.4 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 4.5 |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 1.5 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 2.7 |      |

1. Guaranteed by design.
2. The frequency of the GPIOs that can be supplied in V<sub>BAT</sub> mode (PC13, PC14, PC15 and PI8) is limited to 2 MHz
3. The maximum frequency is defined with the following conditions:  
 $(t_r + t_f) \leq 2/3 T$   
Skew ≤ 1/20 T  
45% < Duty cycle < 55%
4. The fall and rise times are defined between 90% and 10% and between 10% and 90% of the output waveform, respectively.
5. Compensation system enabled.

## Output buffer timing characteristics (HSLV option enabled)

Table 64. Output timing characteristics (HSLV ON)<sup>(1)</sup>

| Speed | Symbol           | Parameter                                                                 | conditions                                               | Min | Max | Unit |
|-------|------------------|---------------------------------------------------------------------------|----------------------------------------------------------|-----|-----|------|
| 00    | $F_{\max}^{(2)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  |      |
|       | $t_r/t_f^{(3)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 11  | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 9   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 6.6 |      |
| 01    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 50  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 58  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 66  |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 6.6 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 4.8 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 3   |      |
| 10    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 55  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 80  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 133 |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 5.8 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 4   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 2.4 |      |
| 11    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 60  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 90  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 175 |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 5.3 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 3.6 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 1.9 |      |

1. Guaranteed by design.

2. The maximum frequency is defined with the following conditions:  
 $(t_r + t_f) \leq 2/3 T$   
 Skew ≤ 1/20 T  
 45% < Duty cycle < 55%

3. The fall and rise times are defined between 90% and 10% and between 10% and 90% of the output waveform, respectively.

4. Compensation system enabled.

### 6.3.16 NRST pin characteristics

The NRST pin input driver uses CMOS technology. It is connected to a permanent pull-up resistor,  $R_{PU}$  (see [Table 60: I/O static characteristics](#)).

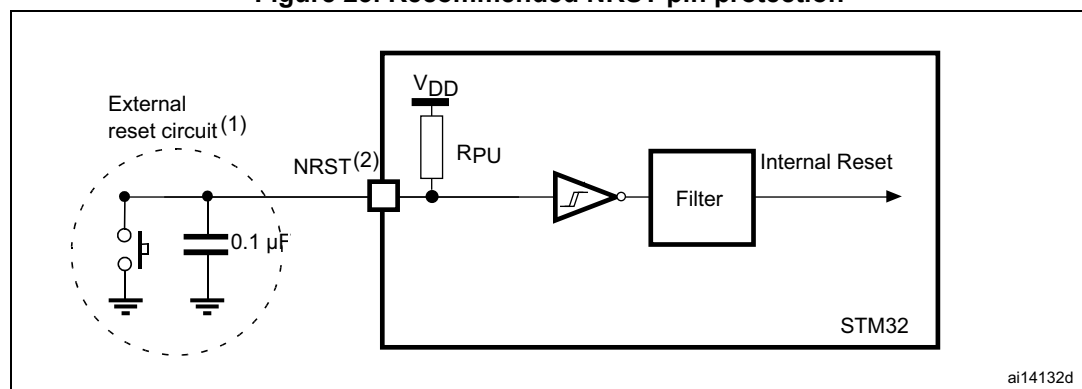
Unless otherwise specified, the parameters given in [Table 65](#) are derived from tests performed under the ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

**Table 65. NRST pin characteristics**

| Symbol               | Parameter                                       | Conditions                              | Min  | Typ | Max | Unit       |
|----------------------|-------------------------------------------------|-----------------------------------------|------|-----|-----|------------|
| $R_{PU}^{(2)}$       | Weak pull-up equivalent resistor <sup>(1)</sup> | $V_{IN} = V_{SS}$                       | 30   | 40  | 50  | k $\Omega$ |
| $V_{F(NRST)}^{(2)}$  | NRST Input filtered pulse                       | $1.71\text{ V} < V_{DD} < 3.6\text{ V}$ | -    | -   | 50  | ns         |
| $V_{NF(NRST)}^{(2)}$ | NRST Input not filtered pulse                   | $1.71\text{ V} < V_{DD} < 3.6\text{ V}$ | 300  | -   | -   |            |
|                      |                                                 | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | 1000 | -   | -   |            |

1. The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance must be minimum (~10% order).
2. Guaranteed by design.

**Figure 23. Recommended NRST pin protection**



1. The reset network protects the device against parasitic resets.
2. The user must ensure that the level on the NRST pin can go below the  $V_{IL(NRST)}$  max level specified in [Table 60](#). Otherwise the reset is not taken into account by the device.

### 6.3.17 FMC characteristics

Unless otherwise specified, the parameters given in [Table 66](#) to [Table 79](#) for the FMC interface are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $\text{OSPEEDRy}[1:0] = 11$
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$

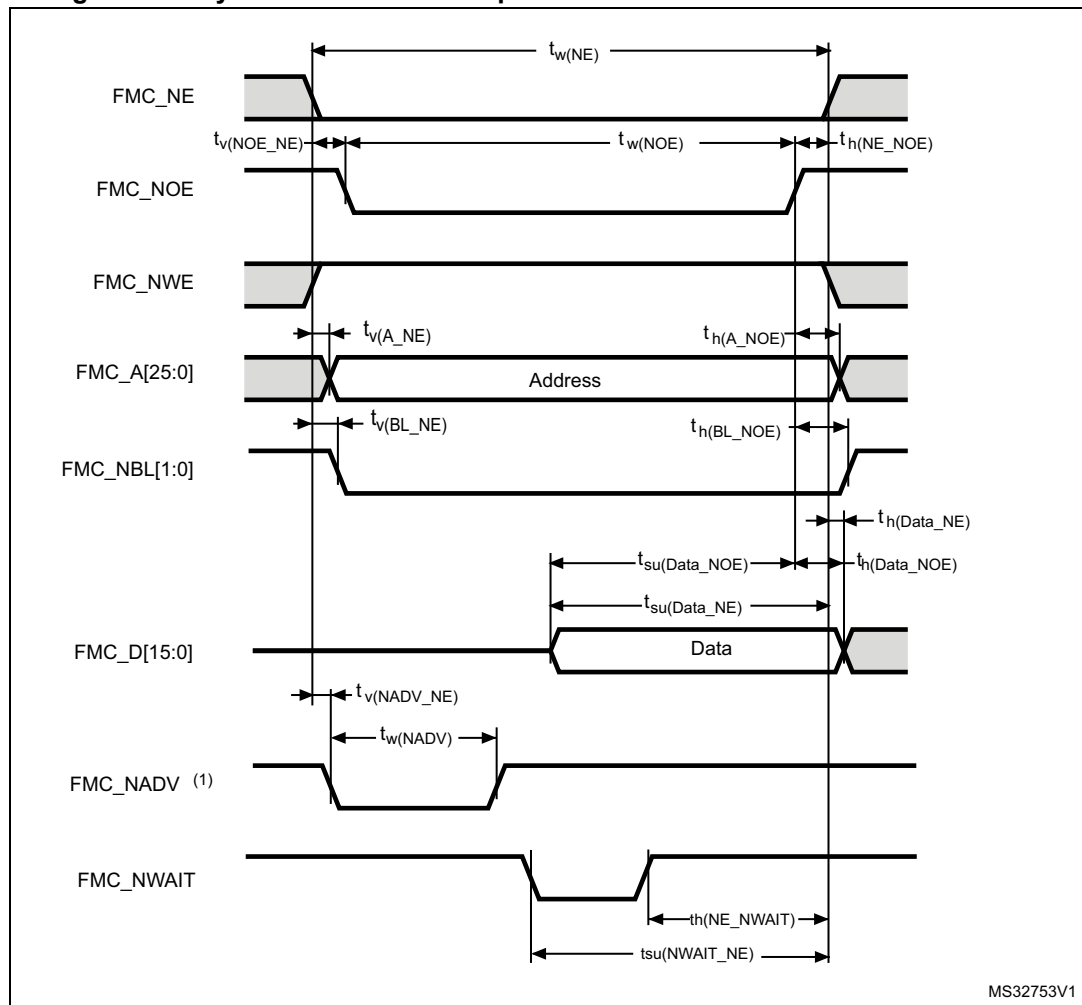
Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

#### Asynchronous waveforms and timings

[Figure 24](#) through [Figure 27](#) represent asynchronous waveforms and [Table 66](#) through [Table 73](#) provide the corresponding timings. The results shown in these tables are obtained with the following FMC configuration:

- AddressSetupTime = 0x1
- AddressHoldTime = 0x1
- DataSetupTime = 0x1 (except for asynchronous NWAIT mode, DataSetupTime = 0x5)
- BusTurnAroundDuration = 0x0
- Capacitive load  $C_L = 30 \text{ pF}$

In all timing tables, the  $T_{\text{KERCK}}$  is the  $f_{\text{mc\_ker\_ck}}$  clock period.

**Figure 24. Asynchronous non-multiplexed SRAM/PSRAM/NOR read waveforms**

1. Mode 2/B, C and D only. In Mode 1, FMC\_NADV is not used.

Table 66. Asynchronous non-multiplexed SRAM/PSRAM/NOR read timings<sup>(1)</sup>

| Symbol              | Parameter                             | Min                     | Max                     | Unit |
|---------------------|---------------------------------------|-------------------------|-------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 1$ | ns   |
| $t_{v(NOE\_NE)}$    | FMC_NEx low to FMC_NOE low            | 0                       | 0.5                     |      |
| $t_{w(NOE)}$        | FMC_NOE low time                      | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 1$ |      |
| $t_{h(NE\_NOE)}$    | FMC_NOE high to FMC_NE high hold time | 0                       | -                       |      |
| $t_{v(A\_NE)}$      | FMC_NEx low to FMC_A valid            | -                       | 0.5                     |      |
| $t_{h(A\_NOE)}$     | Address hold time after FMC_NOE high  | 0                       | -                       |      |
| $t_{v(BL\_NE)}$     | FMC_NEx low to FMC_BL valid           | -                       | 0.5                     |      |
| $t_{h(BL\_NOE)}$    | FMC_BL hold time after FMC_NOE high   | 0                       | -                       |      |
| $t_{su(Data\_NE)}$  | Data to FMC_NEx high setup time       | 11                      | -                       |      |
| $t_{su(Data\_NOE)}$ | Data to FMC_NOEx high setup time      | 11                      | -                       |      |
| $t_{h(Data\_NOE)}$  | Data hold time after FMC_NOE high     | 0                       | -                       |      |
| $t_{h(Data\_NE)}$   | Data hold time after FMC_NEx high     | 0                       | -                       |      |
| $t_{v(NADV\_NE)}$   | FMC_NEx low to FMC_NADV low           | -                       | 0                       |      |
| $t_{w(NADV)}$       | FMC_NADV low time                     | -                       | $T_{fmc\_ker\_ck} + 1$  |      |

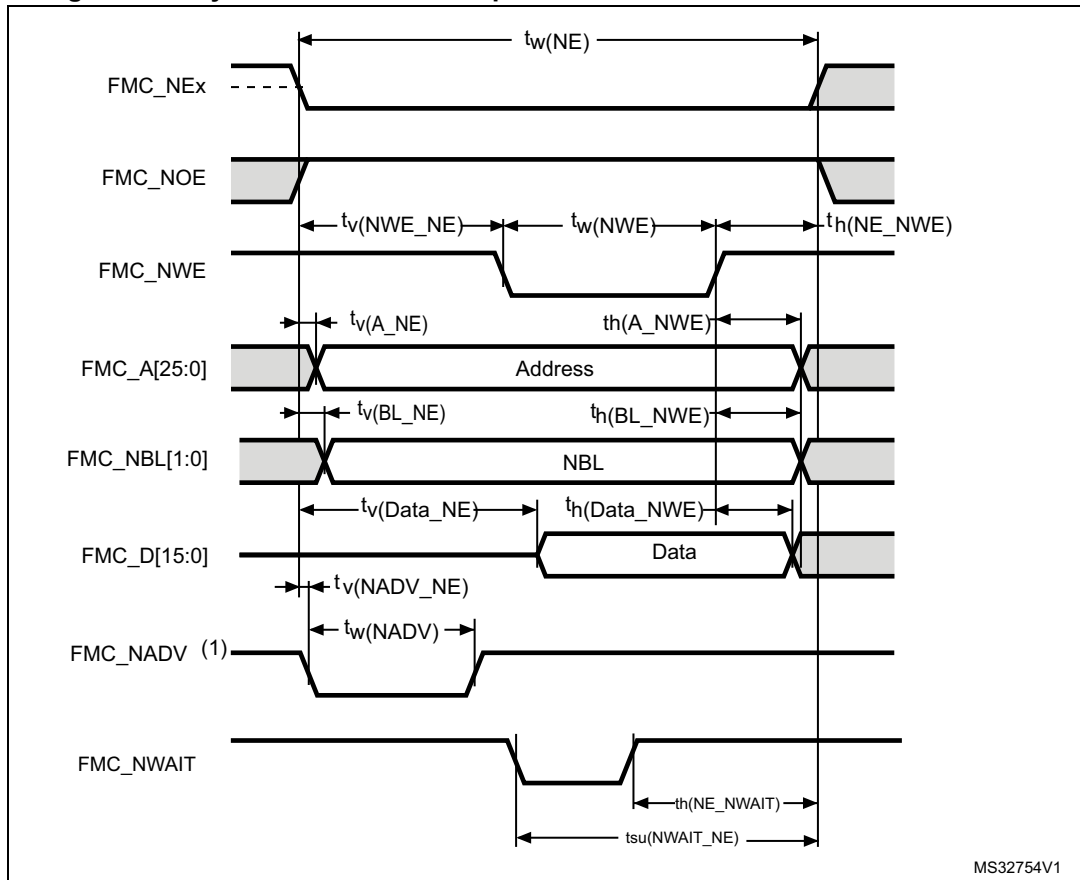
1. Guaranteed by characterization results.

Table 67. Asynchronous non-multiplexed SRAM/PSRAM/NOR read - NWAIT timings<sup>(1)(2)</sup>

| Symbol              | Parameter                                 | Min                        | Max                     | Unit |
|---------------------|-------------------------------------------|----------------------------|-------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $7T_{fmc\_ker\_ck} + 1$    | $7T_{fmc\_ker\_ck} + 1$ | ns   |
| $t_{w(NOE)}$        | FMC_NWE low time                          | $5T_{fmc\_ker\_ck} - 1$    | $5T_{fmc\_ker\_ck} + 1$ |      |
| $t_{w(NWAIT)}$      | FMC_NWAIT low time                        | $T_{fmc\_ker\_ck} - 0.5$   | -                       |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $4T_{fmc\_ker\_ck} + 11$   | -                       |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $3T_{fmc\_ker\_ck} + 11.5$ | -                       |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

**Figure 25. Asynchronous non-multiplexed SRAM/PSRAM/NOR write waveforms**

1. Mode 2/B, C and D only. In Mode 1, FMC\_NADV is not used.

**Table 68. Asynchronous non-multiplexed SRAM/PSRAM/NOR write timings<sup>(1)</sup>**

| Symbol             | Parameter                             | Min                      | Max                      | Unit |
|--------------------|---------------------------------------|--------------------------|--------------------------|------|
| $t_{w(NE)}$        | FMC_NE low time                       | $3T_{fmc\_ker\_ck} - 1$  | $3T_{fmc\_ker\_ck}$      | ns   |
| $t_{v(NWE\_NE)}$   | FMC_NEx low to FMC_NWE low            | $T_{fmc\_ker\_ck}$       | $T_{fmc\_ker\_ck} + 1$   |      |
| $t_{w(NWE)}$       | FMC_NWE low time                      | $T_{fmc\_ker\_ck} - 0.5$ | $T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{h(NE\_NWE)}$   | FMC_NWE high to FMC_NE high hold time | $T_{fmc\_ker\_ck}$       | -                        |      |
| $t_{v(A\_NE)}$     | FMC_NEx low to FMC_A valid            | -                        | 2                        |      |
| $t_{h(A\_NWE)}$    | Address hold time after FMC_NWE high  | $T_{fmc\_ker\_ck} - 0.5$ | -                        |      |
| $t_{v(BL\_NE)}$    | FMC_NEx low to FMC_BL valid           | -                        | 0.5                      |      |
| $t_{h(BL\_NWE)}$   | FMC_BL hold time after FMC_NWE high   | $T_{fmc\_ker\_ck} - 0.5$ | -                        |      |
| $t_{v(Data\_NE)}$  | Data to FMC_NEx low to Data valid     | -                        | $T_{fmc\_ker\_ck} + 2.5$ |      |
| $t_{h(Data\_NWE)}$ | Data hold time after FMC_NWE high     | $T_{fmc\_ker\_ck} + 0.5$ | -                        |      |
| $t_{v(NADV\_NE)}$  | FMC_NEx low to FMC_NADV low           | -                        | 0                        |      |
| $t_{w(NADV)}$      | FMC_NADV low time                     | -                        | $T_{fmc\_ker\_ck} + 1$   |      |

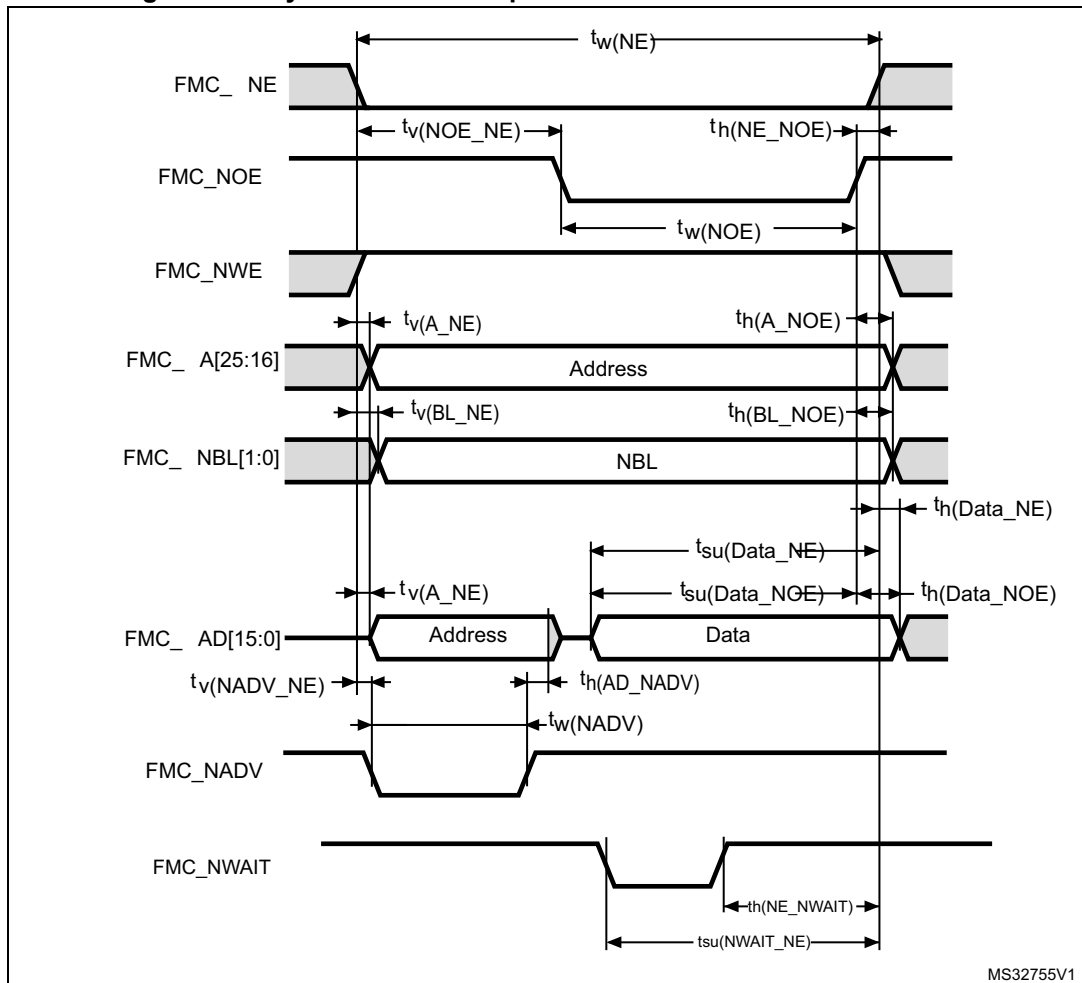
1. Guaranteed by characterization results.

**Table 69. Asynchronous non-multiplexed SRAM/PSRAM/NOR write - NWAIT timings<sup>(1)(2)</sup>**

| Symbol              | Parameter                                 | Min                       | Max                       | Unit |
|---------------------|-------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $8T_{fmc\_ker\_ck} - 1$   | $8T_{fmc\_ker\_ck} + 1$   | ns   |
| $t_{w(NWE)}$        | FMC_NWE low time                          | $6T_{fmc\_ker\_ck} - 1.5$ | $6T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $5T_{fmc\_ker\_ck} + 13$  | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $4T_{fmc\_ker\_ck} + 13$  | -                         |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

**Figure 26. Asynchronous multiplexed PSRAM/NOR read waveforms**



**Table 70. Asynchronous multiplexed PSRAM/NOR read timings<sup>(1)</sup>**

| Symbol              | Parameter                                           | Min                      | Max                       | Unit |
|---------------------|-----------------------------------------------------|--------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                                     | $3T_{fmc\_ker\_ck} - 1$  | $3T_{fmc\_ker\_ck} + 1$   | ns   |
| $t_{v(NOE\_NE)}$    | FMC_NEx low to FMC_NOE low                          | $2T_{fmc\_ker\_ck}$      | $2T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{w(NOE)}$        | FMC_NOE low time                                    | $T_{fmc\_ker\_ck} - 1$   | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_{h(NE\_NOE)}$    | FMC_NOE high to FMC_NE high hold time               | 0                        | -                         |      |
| $t_{v(A\_NE)}$      | FMC_NEx low to FMC_A valid                          | -                        | 0.5                       |      |
| $t_{v(NADV\_NE)}$   | FMC_NEx low to FMC_NADV low                         | 0                        | 0.5                       |      |
| $t_{w(NADV)}$       | FMC_NADV low time                                   | $T_{fmc\_ker\_ck} - 0.5$ | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_{h(AD\_NADV)}$   | FMC_AD(address) valid hold time after FMC_NADV high | $T_{fmc\_ker\_ck} + 0.5$ | -                         |      |
| $t_{h(A\_NOE)}$     | Address hold time after FMC_NOE high                | $T_{fmc\_ker\_ck} - 0.5$ | -                         |      |
| $t_{h(BL\_NOE)}$    | FMC_BL time after FMC_NOE high                      | 0                        | -                         |      |
| $t_{v(BL\_NE)}$     | FMC_NEx low to FMC_BL valid                         | -                        | 0.5                       |      |
| $t_{su(Data\_NE)}$  | Data to FMC_NEx high setup time                     | $T_{fmc\_ker\_ck} - 2$   | -                         |      |
| $t_{su(Data\_NOE)}$ | Data to FMC_NOE high setup time                     | $T_{fmc\_ker\_ck} - 2$   | -                         |      |
| $t_{h(Data\_NE)}$   | Data hold time after FMC_NEx high                   | 0                        | -                         |      |
| $t_{h(Data\_NOE)}$  | Data hold time after FMC_NOE high                   | 0                        | -                         |      |

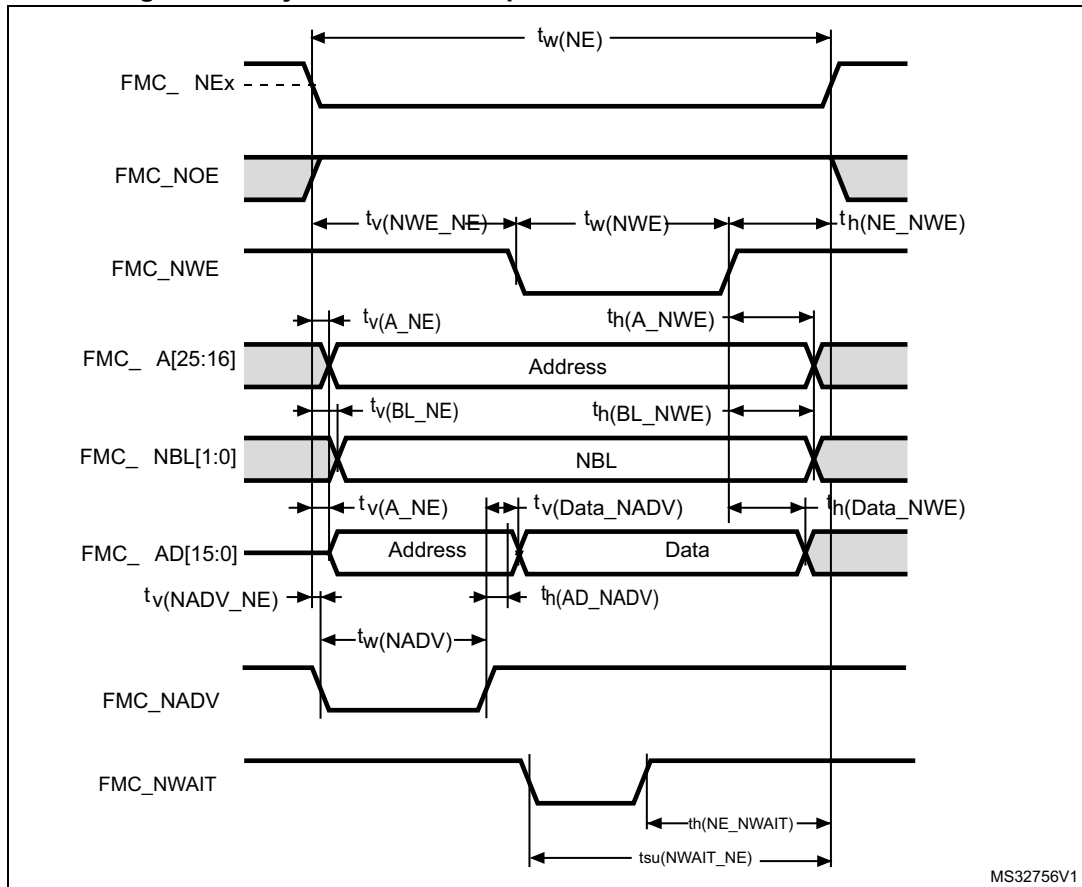
1. Guaranteed by characterization results.

**Table 71. Asynchronous multiplexed PSRAM/NOR read-NWAIT timings<sup>(1)</sup>**

| Symbol              | Parameter                                 | Min                       | Max                       | Unit |
|---------------------|-------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $8T_{fmc\_ker\_ck} - 1$   | $8T_{fmc\_ker\_ck}$       | ns   |
| $t_{w(NOE)}$        | FMC_NWE low time                          | $5T_{fmc\_ker\_ck} - 1.5$ | $5T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $5T_{fmc\_ker\_ck} + 3$   | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $4T_{fmc\_ker\_ck}$       | -                         |      |

1. Guaranteed by characterization results.

Figure 27. Asynchronous multiplexed PSRAM/NOR write waveforms

Table 72. Asynchronous multiplexed PSRAM/NOR write timings<sup>(1)</sup>

| Symbol            | Parameter                                           | Min                       | Max                       | Unit |
|-------------------|-----------------------------------------------------|---------------------------|---------------------------|------|
| $t_w(NE)$         | FMC_NE low time                                     | $4T_{fmc\_ker\_c} - 1$    | $4T_{fmc\_ker\_ck}$       | ns   |
| $t_v(NWE\_NE)$    | FMC_NEx low to FMC_NWE low                          | $T_{fmc\_ker\_c} - 1$     | $T_{fmc\_ker\_ck} + 0.5$  |      |
| $t_w(NWE)$        | FMC_NWE low time                                    | $2T_{fmc\_ker\_ck} - 0.5$ | $2T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_h(NE\_NWE)$    | FMC_NWE high to FMC_NE high hold time               | $T_{fmc\_ker\_ck} - 0.5$  | -                         |      |
| $t_v(A\_NE)$      | FMC_NEx low to FMC_A valid                          | -                         | 0                         |      |
| $t_v(NADV\_NE)$   | FMC_NEx low to FMC_NADV low                         | 0                         | 0.5                       |      |
| $t_w(NADV)$       | FMC_NADV low time                                   | $T_{fmc\_ker\_ck}$        | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_h(AD\_NADV)$   | FMC_AD(address) valid hold time after FMC_NADV high | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |
| $t_h(A\_NWE)$     | Address hold time after FMC_NWE high                | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |
| $t_h(BL\_NWE)$    | FMC_BL hold time after FMC_NWE high                 | $T_{fmc\_ker\_ck} - 0.5$  | -                         |      |
| $t_v(BL\_NE)$     | FMC_NEx low to FMC_BL valid                         | -                         | 0.5                       |      |
| $t_v(Data\_NADV)$ | FMC_NADV high to Data valid                         | -                         | $T_{fmc\_ker\_ck} + 2$    |      |
| $t_h(Data\_NWE)$  | Data hold time after FMC_NWE high                   | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |

1. Guaranteed by characterization results.

**Table 73. Asynchronous multiplexed PSRAM/NOR write-NWAIT timings<sup>(1)</sup>**

| Symbol              | Parameter                                 | Min                       | Max                       | Unit |
|---------------------|-------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $9T_{fmc\_ker\_ck} - 1$   | $9T_{fmc\_ker\_ck}$       | ns   |
| $t_{w(NWE)}$        | FMC_NWE low time                          | $7T_{fmc\_ker\_ck} - 0.5$ | $7T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $6T_{fmc\_ker\_ck} + 3$   | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $4T_{fmc\_ker\_ck}$       | -                         |      |

1. Guaranteed by characterization results.

### Synchronous waveforms and timings

[Figure 28](#) through [Figure 31](#) represent synchronous waveforms and [Table 74](#) through [Table 77](#) provide the corresponding timings. The results shown in these tables are obtained with the following FMC configuration:

- BurstAccessMode = FMC\_BurstAccessMode\_Enable
- MemoryType = FMC\_MemoryType\_CRAM
- WriteBurst = FMC\_WriteBurst\_Enable
- CLKDivision = 1
- DataLatency = 1 for NOR Flash; DataLatency = 0 for PSRAM

In all the timing tables, the  $T_{fmc\_ker\_ck}$  is the  $f_{mc\_ker\_ck}$  clock period, with the following FMC\_CLK maximum values:

- For  $2.7\text{ V} < V_{DD} < 3.6\text{ V}$ , FMC\_CLK = 100 MHz at 20 pF
- For  $1.8\text{ V} < V_{DD} < 1.9\text{ V}$ , FMC\_CLK = 100 MHz at 20 pF
- For  $1.62\text{ V} < V_{DD} < 1.8\text{ V}$ , FMC\_CLK = 100 MHz at 15 pF

### Figure 28. Synchronous multiplexed NOR/PSRAM read timings

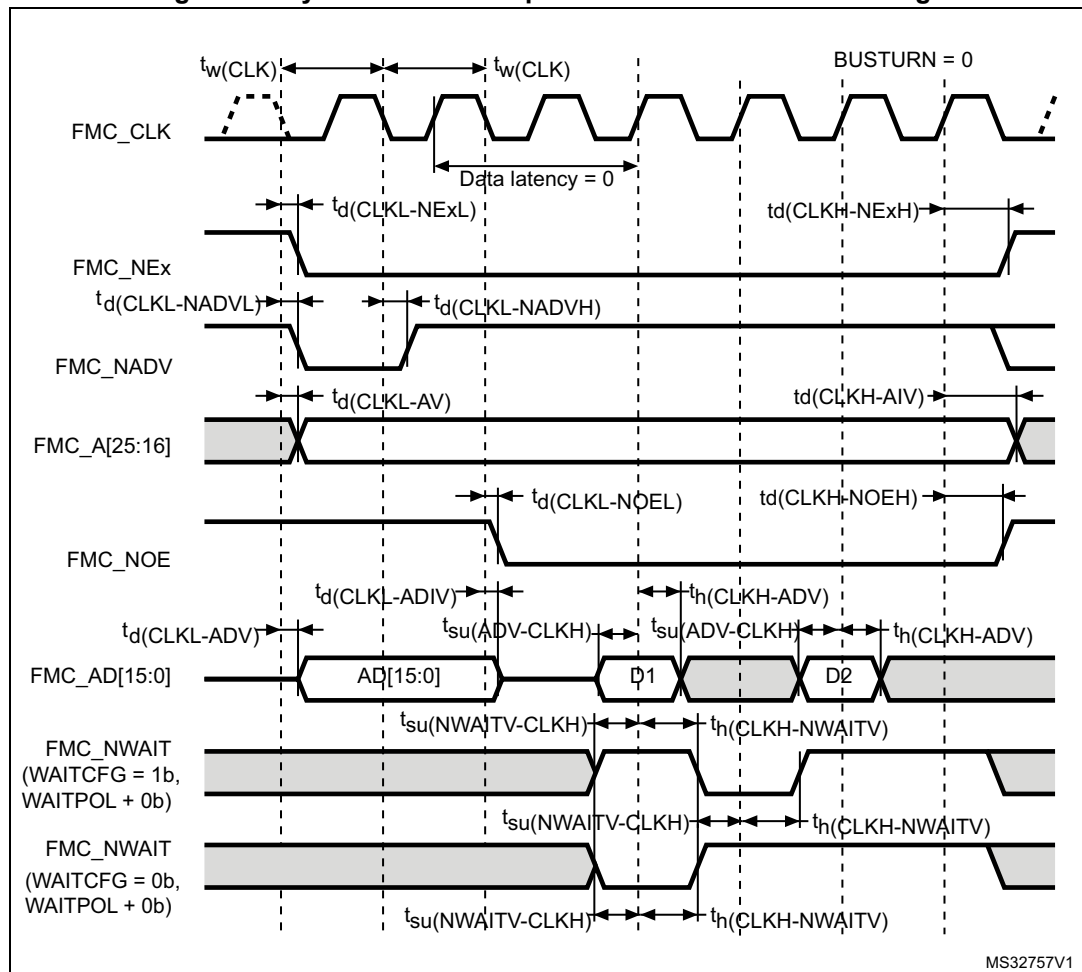


Table 74. Synchronous multiplexed NOR/PSRAM read timings<sup>(1)</sup>

| Symbol               | Parameter                                    | Min                      | Max | Unit |
|----------------------|----------------------------------------------|--------------------------|-----|------|
| $t_{w(CLK)}$         | FMC_CLK period                               | $2T_{fmc\_ker\_ck} - 1$  | -   | ns   |
| $t_{d(CLKL-NExL)}$   | FMC_CLK low to FMC_NEx low (x=0..2)          | -                        | 1   |      |
| $t_{d(CLKH-NExH)}$   | FMC_CLK high to FMC_NEx high (x= 0...2)      | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_{d(CLKL-NADVL)}$  | FMC_CLK low to FMC_NADV low                  | -                        | 1.  |      |
| $t_{d(CLKL-NADVH)}$  | FMC_CLK low to FMC_NADV high                 | 0                        | -   |      |
| $t_{d(CLKL-AV)}$     | FMC_CLK low to FMC_Ax valid (x=16...25)      | -                        | 2.5 |      |
| $t_{d(CLKH-AIV)}$    | FMC_CLK high to FMC_Ax invalid (x=16...25)   | $T_{fmc\_ker\_ck}$       | -   |      |
| $t_{d(CLKL-NOEL)}$   | FMC_CLK low to FMC_NOE low                   | -                        | 1.5 |      |
| $t_{d(CLKH-NOEH)}$   | FMC_CLK high to FMC_NOE high                 | $T_{fmc\_ker\_ck} - 0.5$ | -   |      |
| $t_{d(CLKL-ADV)}$    | FMC_CLK low to FMC_AD[15:0] valid            | -                        | 3   |      |
| $t_{d(CLKL-ADIV)}$   | FMC_CLK low to FMC_AD[15:0] invalid          | 0                        | -   |      |
| $t_{su(ADV-CLKH)}$   | FMC_A/D[15:0] valid data before FMC_CLK high | 2                        | -   |      |
| $t_h(CLKH-ADV)$      | FMC_A/D[15:0] valid data after FMC_CLK high  | 1                        | -   |      |
| $t_{su(NWAIT-CLKH)}$ | FMC_NWAIT valid before FMC_CLK high          | 2                        | -   |      |
| $t_h(CLKH-NWAIT)$    | FMC_NWAIT valid after FMC_CLK high           | 2                        | -   |      |

1. Guaranteed by characterization results.

Figure 29. Synchronous multiplexed PSRAM write timings

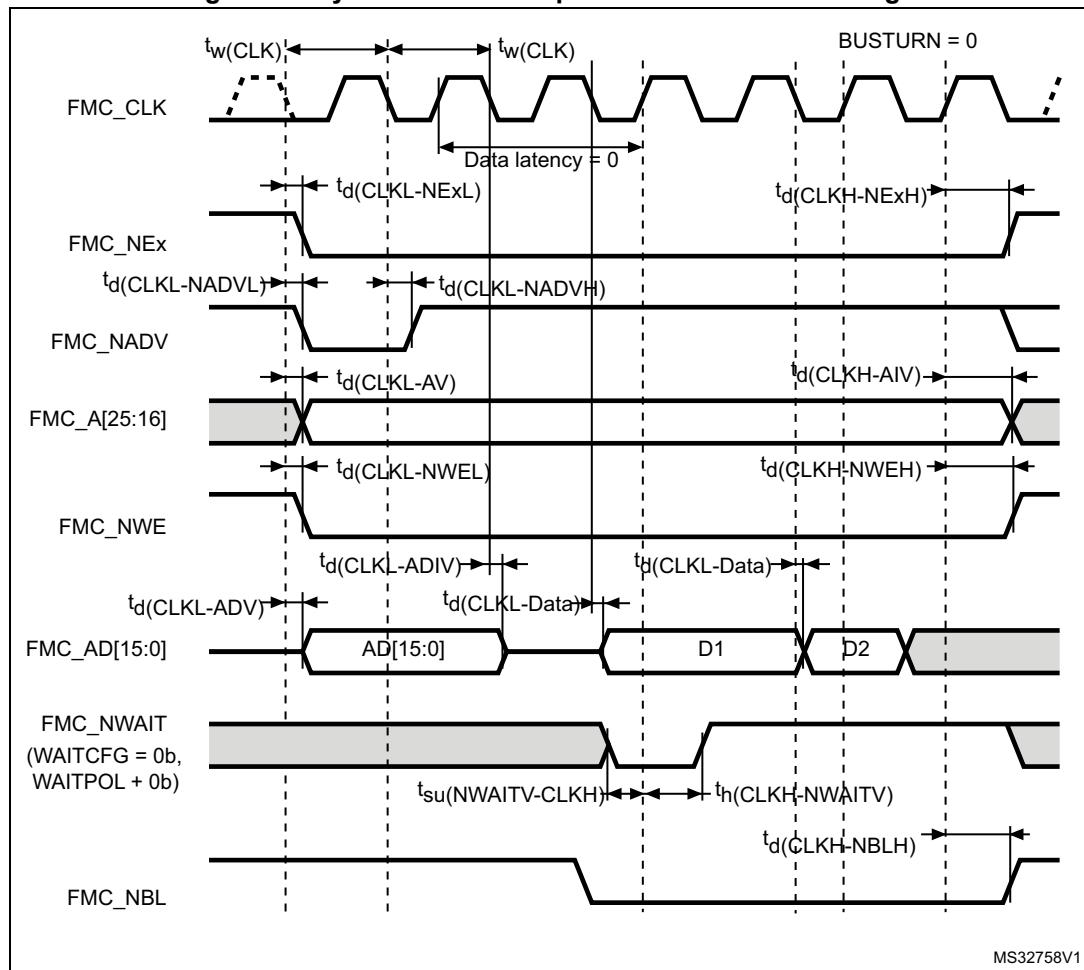
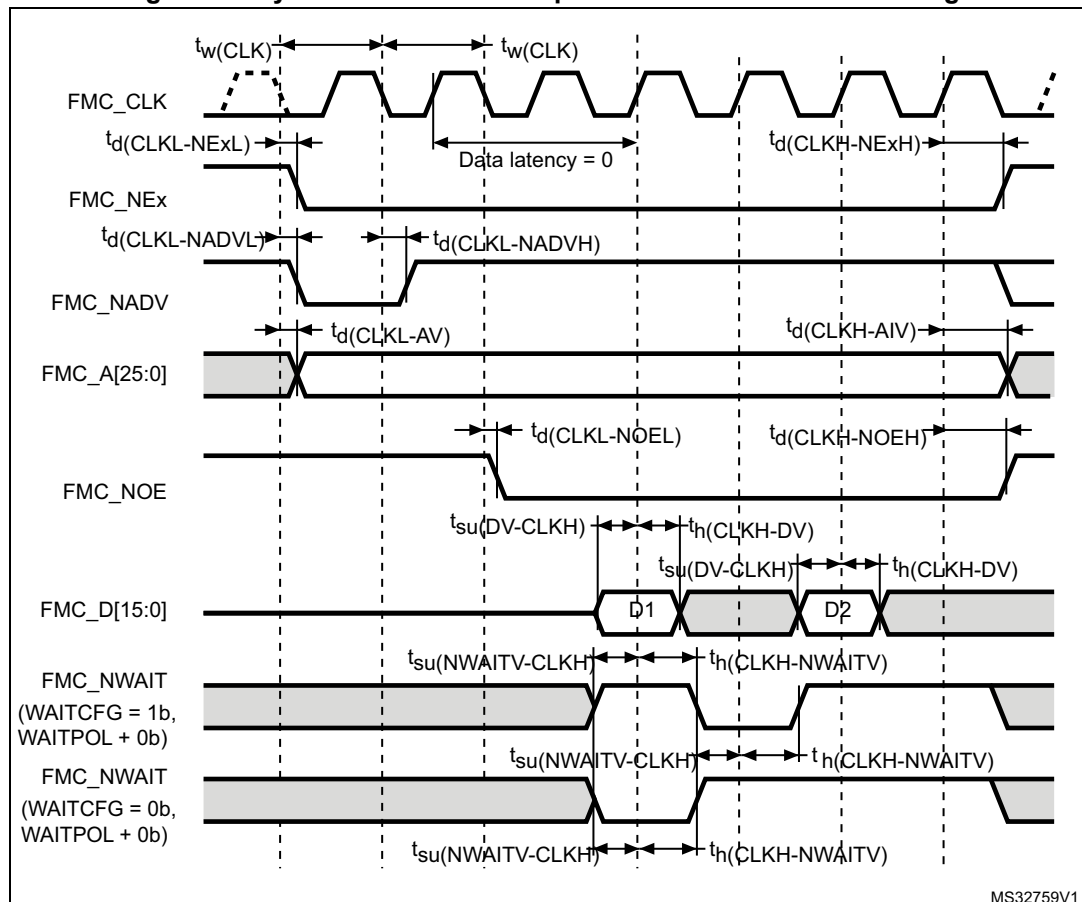


Table 75. Synchronous multiplexed PSRAM write timings<sup>(1)</sup>

| Symbol               | Parameter                                     | Min                      | Max | Unit |
|----------------------|-----------------------------------------------|--------------------------|-----|------|
| $t_{w(CLK)}$         | FMC_CLK period                                | $2T_{fmc\_ker\_ck} - 1$  | -   | ns   |
| $t_{d(CLKL-NExL)}$   | FMC_CLK low to FMC_NEx low ( $x=0..2$ )       | -                        | 1   |      |
| $t_{d(CLKH-NExH)}$   | FMC_CLK high to FMC_NEx high ( $x=0..2$ )     | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_{d(CLKL-NADV L)}$ | FMC_CLK low to FMC_NADV low                   | -                        | 1.5 |      |
| $t_{d(CLKL-NADV H)}$ | FMC_CLK low to FMC_NADV high                  | 0                        | -   |      |
| $t_{d(CLKL-AV)}$     | FMC_CLK low to FMC_Ax valid ( $x=16..25$ )    | -                        | 2   |      |
| $t_{d(CLKH-AIV)}$    | FMC_CLK high to FMC_Ax invalid ( $x=16..25$ ) | $T_{fmc\_ker\_ck}$       | -   |      |
| $t_{d(CLKL-NWEL)}$   | FMC_CLK low to FMC_NWE low                    | -                        | 1.5 |      |
| $t_{d(CLKH-NWEH)}$   | FMC_CLK high to FMC_NWE high                  | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_{d(CLKL-ADV)}$    | FMC_CLK low to FMC_AD[15:0] valid             | -                        | 2.5 |      |
| $t_{d(CLKL-ADIV)}$   | FMC_CLK low to FMC_AD[15:0] invalid           | 0                        | -   |      |
| $t_{d(CLKL-DATA)}$   | FMC_A/D[15:0] valid data after FMC_CLK low    | -                        | 2.5 |      |
| $t_{d(CLKL-NBL L)}$  | FMC_CLK low to FMC_NBL low                    | -                        | 2   |      |
| $t_{d(CLKH-NBL H)}$  | FMC_CLK high to FMC_NBL high                  | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_{su(NWAIT-CLKH)}$ | FMC_NWAIT valid before FMC_CLK high           | 2                        | -   |      |
| $t_h(CLKH-NWAIT)$    | FMC_NWAIT valid after FMC_CLK high            | 2                        | -   |      |

1. Guaranteed by characterization results.

Figure 30. Synchronous non-multiplexed NOR/PSRAM read timings

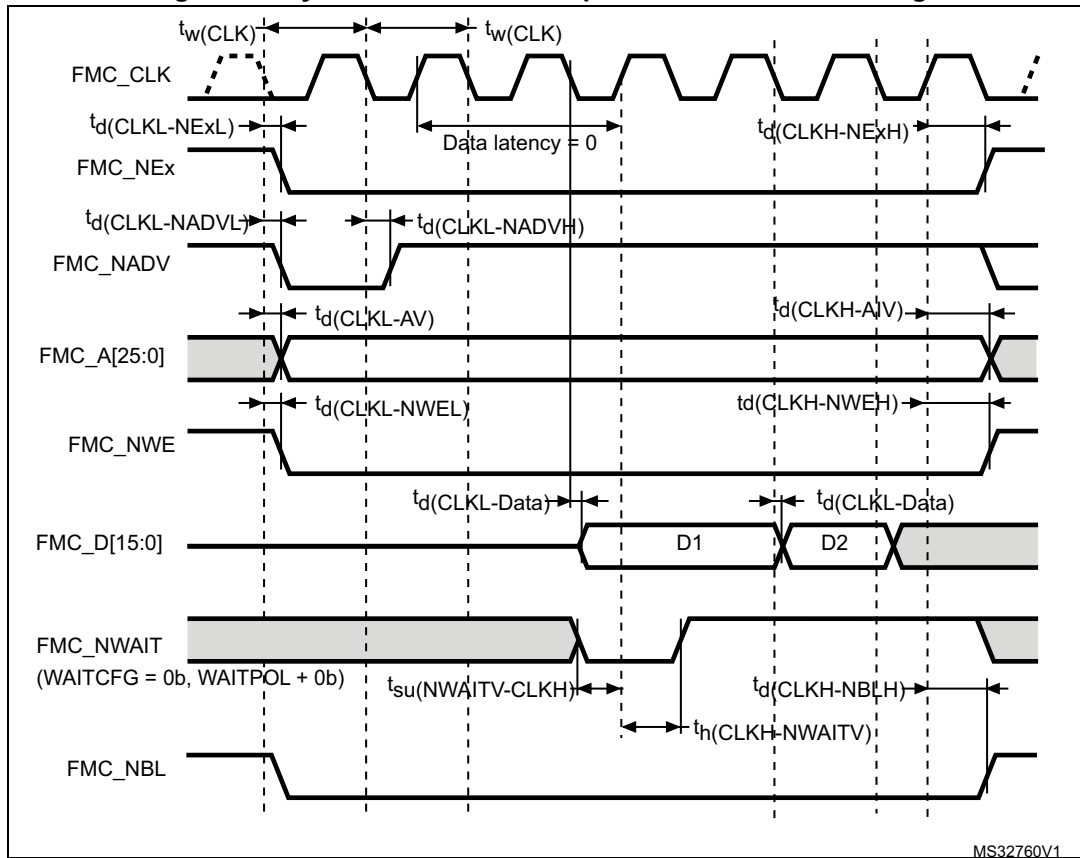
Table 76. Synchronous non-multiplexed NOR/PSRAM read timings<sup>(1)</sup>

| Symbol                          | Parameter                                     | Min                             | Max | Unit |
|---------------------------------|-----------------------------------------------|---------------------------------|-----|------|
| $t_w(\text{CLK})$               | FMC_CLK period                                | $2T_{\text{fmc\_ker\_ck}} - 1$  | -   | ns   |
| $t(\text{CLKL-NExL})$           | FMC_CLK low to FMC_NEx low ( $x=0..2$ )       | -                               | 2   |      |
| $t_d(\text{CLKH-NExH})$         | FMC_CLK high to FMC_NEx high ( $x=0..2$ )     | $T_{\text{fmc\_ker\_ck}} + 0.5$ | -   |      |
| $t_d(\text{CLKL-NADV})$         | FMC_CLK low to FMC_NADV low                   | -                               | 0.5 |      |
| $t_d(\text{CLKL-NADVH})$        | FMC_CLK low to FMC_NADV high                  | 0                               | -   |      |
| $t_d(\text{CLKL-AV})$           | FMC_CLK low to FMC_Ax valid ( $x=16..25$ )    | -                               | 2   |      |
| $t_d(\text{CLKH-AIV})$          | FMC_CLK high to FMC_Ax invalid ( $x=16..25$ ) | $T_{\text{fmc\_ker\_ck}}$       | -   |      |
| $t_d(\text{CLKL-NOEL})$         | FMC_CLK low to FMC_NOE low                    | -                               | 1.5 |      |
| $t_d(\text{CLKH-NOEH})$         | FMC_CLK high to FMC_NOE high                  | $T_{\text{fmc\_ker\_ck}} + 0.5$ | -   |      |
| $t_{\text{su}}(\text{DV-CLKH})$ | FMC_D[15:0] valid data before FMC_CLK high    | 2                               | -   |      |
| $t_h(\text{CLKH-DV})$           | FMC_D[15:0] valid data after FMC_CLK high     | 1                               | -   |      |
| $t(\text{NWAIT-CLKH})$          | FMC_NWAIT valid before FMC_CLK high           | 2                               | -   |      |
| $t_h(\text{CLKH-NWAIT})$        | FMC_NWAIT valid after FMC_CLK high            | 2                               | -   |      |

1. Guaranteed by characterization results.



Figure 31. Synchronous non-multiplexed PSRAM write timings

Table 77. Synchronous non-multiplexed PSRAM write timings<sup>(1)</sup>

| Symbol                      | Parameter                                  | Min                             | Max | Unit |
|-----------------------------|--------------------------------------------|---------------------------------|-----|------|
| $t_{\text{CLK}}$            | FMC_CLK period                             | $2T_{\text{fmc\_ker\_ck}} - 1$  | -   | ns   |
| $t_{\text{d(CLKL-NExL)}}$   | FMC_CLK low to FMC_NEx low (x=0..2)        | -                               | 2   |      |
| $t_{\text{d(CLKH-NExH)}}$   | FMC_CLK high to FMC_NEx high (x= 0...2)    | $T_{\text{fmc\_ker\_ck}} + 0.5$ | -   |      |
| $t_{\text{d(CLKL-NADV)}}$   | FMC_CLK low to FMC_NADV low                | -                               | 0.5 |      |
| $t_{\text{d(CLKL-NADVH)}}$  | FMC_CLK low to FMC_NADV high               | 0                               | -   |      |
| $t_{\text{d(CLKL-AV)}}$     | FMC_CLK low to FMC_Ax valid (x=16...25)    | -                               | 2   |      |
| $t_{\text{d(CLKH-AIV)}}$    | FMC_CLK high to FMC_Ax invalid (x=16...25) | $T_{\text{fmc\_ker\_ck}}$       | -   |      |
| $t_{\text{d(CLKL-NWE)}}$    | FMC_CLK low to FMC_NWE low                 | -                               | 1.5 |      |
| $t_{\text{d(CLKH-NWEH)}}$   | FMC_CLK high to FMC_NWE high               | $T_{\text{fmc\_ker\_ck}} + 1$   | -   |      |
| $t_{\text{d(CLKL-Data)}}$   | FMC_D[15:0] valid data after FMC_CLK low   | -                               | 3.5 |      |
| $t_{\text{d(CLKL-NBL)}}$    | FMC_CLK low to FMC_NBL low                 | -                               | 2   |      |
| $t_{\text{d(CLKH-NBLH)}}$   | FMC_CLK high to FMC_NBL high               | $T_{\text{fmc\_ker\_ck}} + 1$   | -   |      |
| $t_{\text{su(NWAIT-CLKH)}}$ | FMC_NWAIT valid before FMC_CLK high        | 2                               | -   |      |
| $t_{\text{h(CLKH-NWAIT)}}$  | FMC_NWAIT valid after FMC_CLK high         | 2                               | -   |      |

1. Guaranteed by characterization results.

## NAND controller waveforms and timings

Figure 32 through Figure 35 represent synchronous waveforms, and Table 78 and Table 79 provide the corresponding timings. The results shown in this table are obtained with the following FMC configuration:

- COM.FMC\_SetupTime = 0x01
- COM.FMC\_WaitSetupTime = 0x03
- COM.FMC\_HoldSetupTime = 0x02
- COM.FMC\_HiZSetupTime = 0x01
- ATT.FMC\_SetupTime = 0x01
- ATT.FMC\_WaitSetupTime = 0x03
- ATT.FMC\_HoldSetupTime = 0x02
- ATT.FMC\_HiZSetupTime = 0x01
- Bank = FMC\_Bank\_NAND
- MemoryDataWidth = FMC\_MemoryDataWidth\_16b
- ECC = FMC\_ECC\_Enable
- ECCPageSize = FMC\_ECCPageSize\_512Bytes
- TCLRSetupTime = 0
- TARSetupTime = 0
- $C_L = 30 \text{ pF}$

In all timing tables, the  $T_{\text{fmc\_ker\_ck}}$  is the fmc\_ker\_ck clock period.

**Figure 32. NAND controller waveforms for read access**

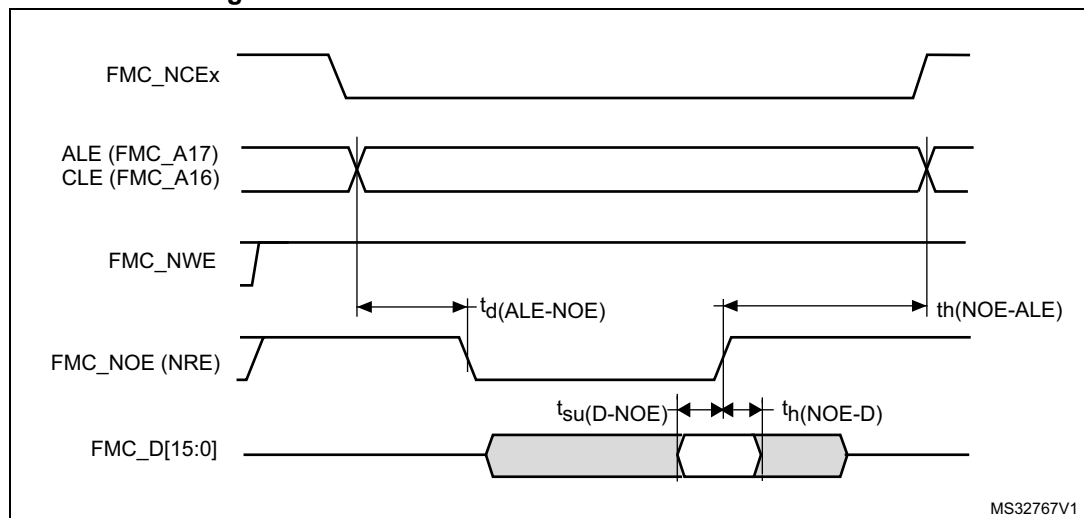


Figure 33. NAND controller waveforms for write access

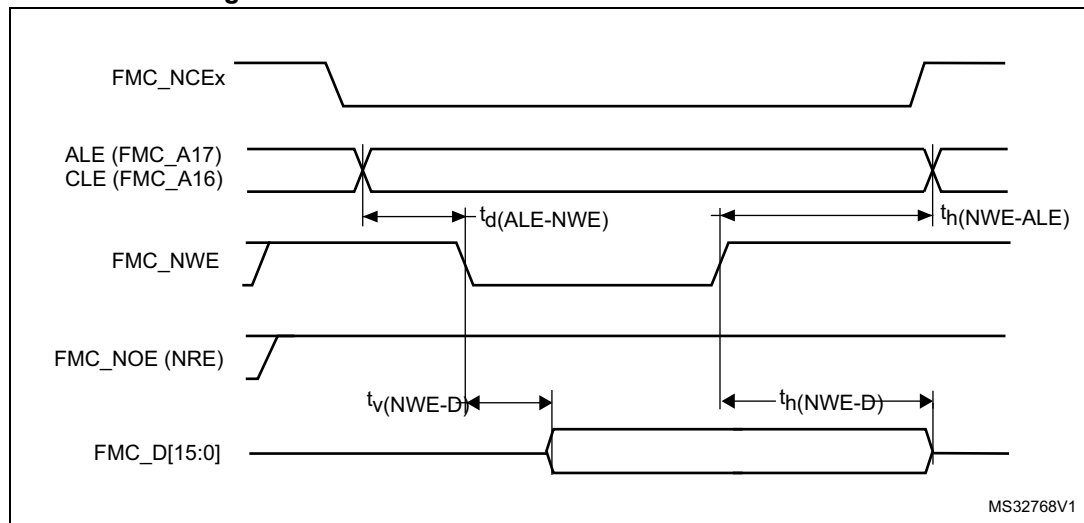


Figure 34. NAND controller waveforms for common memory read access

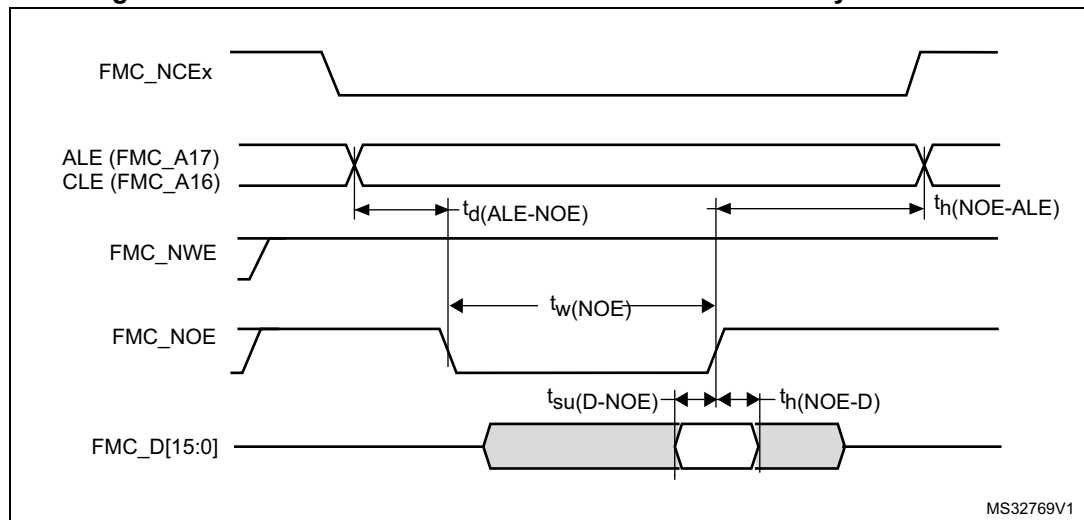
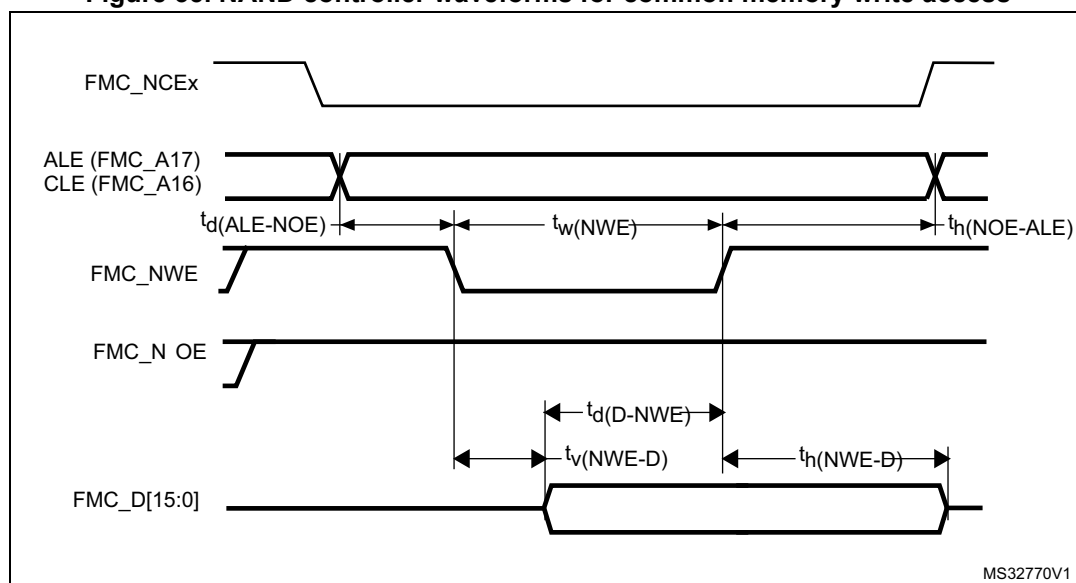


Figure 35. NAND controller waveforms for common memory write access



MS32770V1

Table 78. Switching characteristics for NAND Flash read cycles<sup>(1)</sup>

| Symbol          | Parameter                                  | Min                       | Max                       | Unit |
|-----------------|--------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NOE)}$    | FMC_NOE low width                          | $4T_{fmc\_ker\_ck} - 0.5$ | $4T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su(D-NOE)}$ | FMC_D[15-0] valid data before FMC_NOE high | 8                         | -                         |      |
| $t_h(NOE-D)$    | FMC_D[15-0] valid data after FMC_NOE high  | 0                         | -                         |      |
| $t_d(ALE-NOE)$  | FMC_ALE valid before FMC_NOE low           | -                         | $3T_{fmc\_ker\_ck} + 1$   |      |
| $t_h(NOE-ALE)$  | FMC_NWE high to FMC_ALE invalid            | $4T_{fmc\_ker\_ck} - 2$   | -                         |      |

1. Guaranteed by characterization results.

Table 79. Switching characteristics for NAND Flash write cycles<sup>(1)</sup>

| Symbol         | Parameter                             | Min                       | Max                       | Unit |
|----------------|---------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NWE)}$   | FMC_NWE low width                     | $4T_{fmc\_ker\_ck} - 0.5$ | $4T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_v(NWE-D)$   | FMC_NWE low to FMC_D[15-0] valid      | 0                         | -                         |      |
| $t_h(NWE-D)$   | FMC_NWE high to FMC_D[15-0] invalid   | $2T_{fmc\_ker\_ck} - 0.5$ | -                         |      |
| $t_d(D-NWE)$   | FMC_D[15-0] valid before FMC_NWE high | $5T_{fmc\_ker\_ck} - 1$   | -                         |      |
| $t_d(ALE-NWE)$ | FMC_ALE valid before FMC_NWE low      | -                         | $3T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_h(NWE-ALE)$ | FMC_NWE high to FMC_ALE invalid       | $2T_{fmc\_ker\_ck} - 1$   | -                         |      |

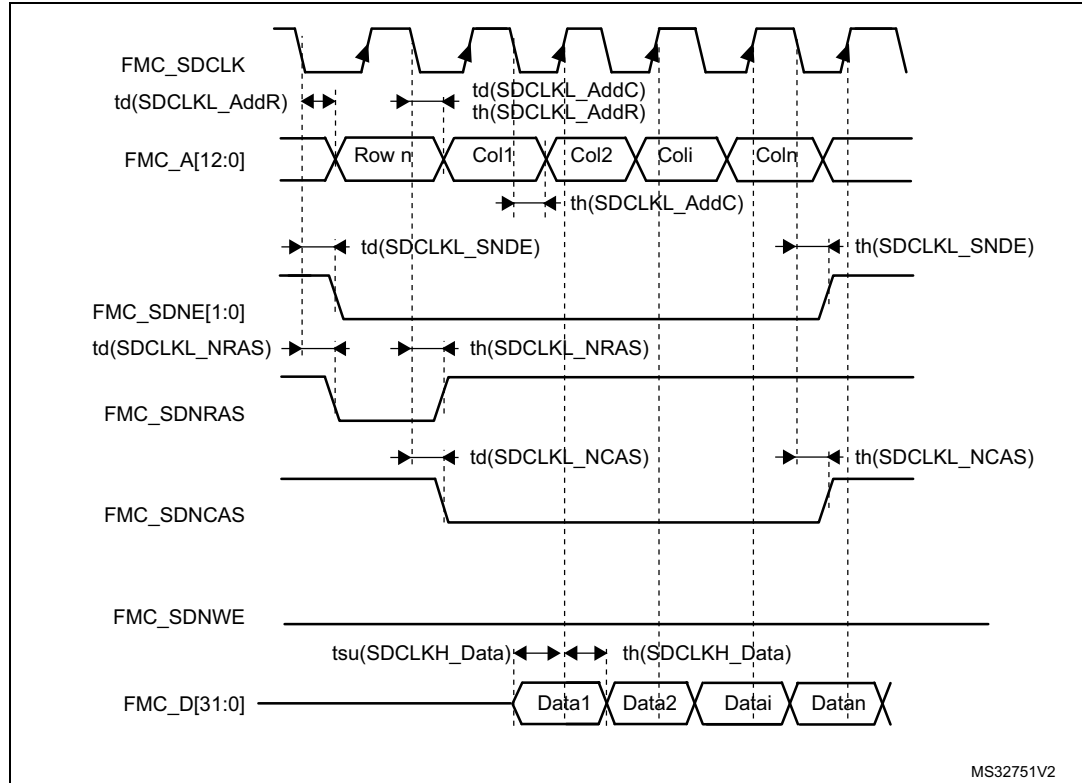
1. Guaranteed by characterization results.

## SDRAM waveforms and timings

In all timing tables, the  $T_{fmc\_ker\_ck}$  is the  $fmc\_ker\_ck$  clock period, with the following FMC\_SDCLK maximum values:

- For  $1.8\text{ V} < V_{DD} < 3.6\text{ V}$ : FMC\_CLK = 100 MHz at 20 pF
- For  $1.62\text{ V} < V_{DD} < 1.8\text{ V}$ , FMC\_CLK = 100 MHz at 30 pF

**Figure 36. SDRAM read access waveforms (CL = 1)**



**Table 80. SDRAM read timings<sup>(1)</sup>**

| Symbol                        | Parameter              | Min                     | Max                       | Unit |
|-------------------------------|------------------------|-------------------------|---------------------------|------|
| $t_w(\text{SDCLK})$           | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su}(\text{SDCLKH\_Data})$ | Data input setup time  | 2                       | -                         |      |
| $t_h(\text{SDCLKH\_Data})$    | Data input hold time   | 1                       | -                         |      |
| $t_d(\text{SDCLKL\_Add})$     | Address valid time     | -                       | 1.5                       |      |
| $t_d(\text{SDCLKL\_SDNE})$    | Chip select valid time | -                       | 1.5                       |      |
| $t_h(\text{SDCLKL\_SDNE})$    | Chip select hold time  | 0.5                     | -                         |      |
| $t_d(\text{SDCLKL\_SDNRAS})$  | SDNRAS valid time      | -                       | 1                         |      |
| $t_h(\text{SDCLKL\_SDNRAS})$  | SDNRAS hold time       | 0.5                     | -                         |      |
| $t_d(\text{SDCLKL\_SDNCAS})$  | SDNCAS valid time      | -                       | 0.5                       |      |
| $t_h(\text{SDCLKL\_SDNCAS})$  | SDNCAS hold time       | 0                       | -                         |      |

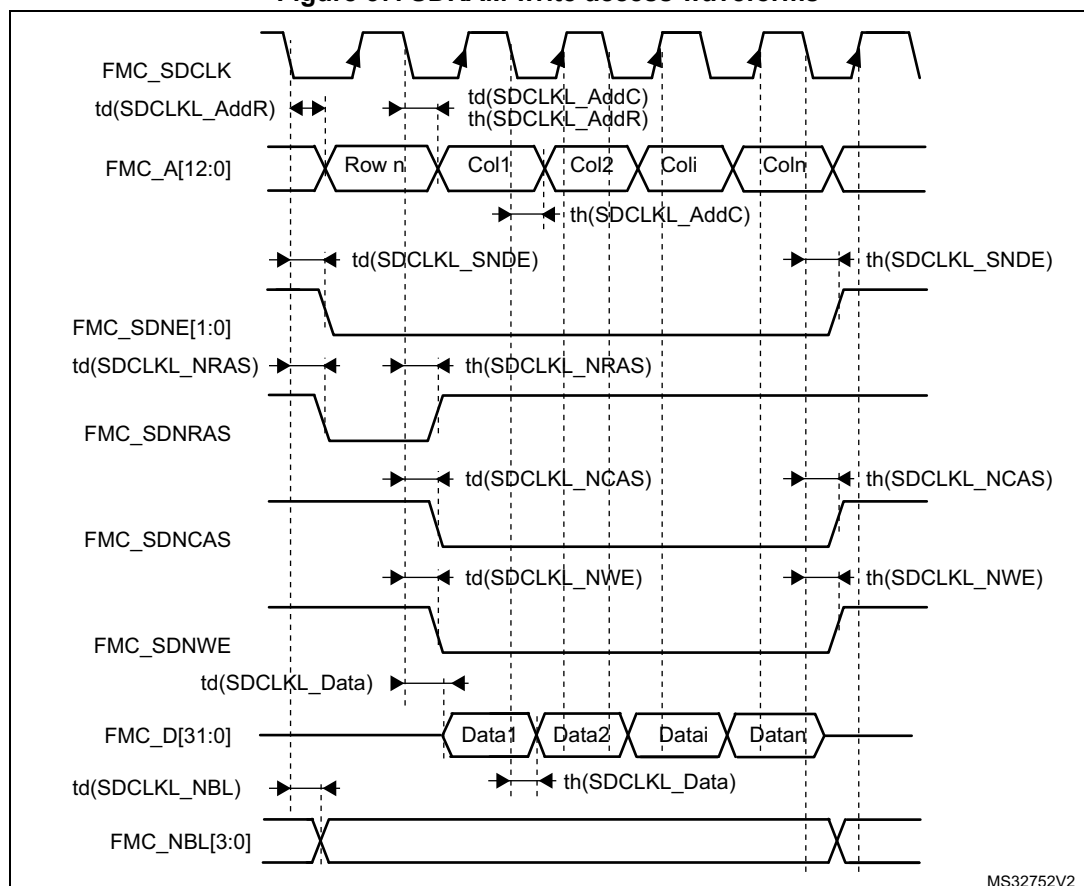
1. Guaranteed by characterization results.

Table 81. LPSDR SDRAM read timings<sup>(1)</sup>

| Symbol                 | Parameter              | Min                     | Max                       | Unit |
|------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{W(SDCLK)}$         | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su(SDCLKH\_Data)}$ | Data input setup time  | 2                       | -                         |      |
| $t_{h(SDCLKH\_Data)}$  | Data input hold time   | 1.5                     | -                         |      |
| $t_d(SDCLKL\_Add)$     | Address valid time     | -                       | 2.5                       |      |
| $t_d(SDCLKL\_SDNE)$    | Chip select valid time | -                       | 2.5                       |      |
| $t_h(SDCLKL\_SDNE)$    | Chip select hold time  | 0                       | -                         |      |
| $t_d(SDCLKL\_SDNRAS)$  | SDNRAS valid time      | -                       | 0.5                       |      |
| $t_h(SDCLKL\_SDNRAS)$  | SDNRAS hold time       | 0                       | -                         |      |
| $t_d(SDCLKL\_SDNCAS)$  | SDNCAS valid time      | -                       | 1.5                       |      |
| $t_h(SDCLKL\_SDNCAS)$  | SDNCAS hold time       | 0                       | -                         |      |

1. Guaranteed by characterization results.

Figure 37. SDRAM write access waveforms



**Table 82. SDRAM write timings<sup>(1)</sup>**

| Symbol                  | Parameter              | Min                     | Max                       | Unit |
|-------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{w(SDCLK)}$          | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{d(SDCLKL\_Data)}$   | Data output valid time | -                       | 3                         |      |
| $t_{h(SDCLKL\_Data)}$   | Data output hold time  | 0                       | -                         |      |
| $t_{d(SDCLKL\_Add)}$    | Address valid time     | -                       | 1.5                       |      |
| $t_{d(SDCLKL\_SDNWE)}$  | SDNWE valid time       | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNWE)}$  | SDNWE hold time        | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNE)}$   | Chip select valid time | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNE)}$   | Chip select hold time  | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNRAS)}$ | SDNRAS valid time      | -                       | 1                         |      |
| $t_{h(SDCLKL\_SDNRAS)}$ | SDNRAS hold time       | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS valid time      | -                       | 1                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS hold time       | 0.5                     | -                         |      |

1. Guaranteed by characterization results.

**Table 83. LPDDR SDRAM write timings<sup>(1)</sup>**

| Symbol                  | Parameter              | Min                     | Max                       | Unit |
|-------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{w(SDCLK)}$          | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{d(SDCLKL\_Data)}$   | Data output valid time | -                       | 2.5                       |      |
| $t_{h(SDCLKL\_Data)}$   | Data output hold time  | 0                       | -                         |      |
| $t_{d(SDCLKL\_Add)}$    | Address valid time     | -                       | 2.5                       |      |
| $t_{d(SDCLKL\_SDNWE)}$  | SDNWE valid time       | -                       | 2.5                       |      |
| $t_{h(SDCLKL\_SDNWE)}$  | SDNWE hold time        | 0                       | -                         |      |
| $t_{d(SDCLKL\_SDNE)}$   | Chip select valid time | -                       | 3                         |      |
| $t_{h(SDCLKL\_SDNE)}$   | Chip select hold time  | 0                       | -                         |      |
| $t_{d(SDCLKL\_SDNRAS)}$ | SDNRAS valid time      | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNRAS)}$ | SDNRAS hold time       | 0                       | -                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS valid time      | -                       | 1.5                       |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS hold time       | 0                       | -                         |      |

1. Guaranteed by characterization results.

### 6.3.18 Quad-SPI interface characteristics

Unless otherwise specified, the parameters given in [Table 84](#) and [Table 85](#) for QUADSPI are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $\text{OSPEEDRy}[1:0] = 11$
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$
- I/O compensation cell enabled
- HSLV activated when  $V_{\text{DD}} \leq 2.7 \text{ V}$

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics.

**Table 84. QUADSPI characteristics in SDR mode<sup>(1)</sup>**

| Symbol                         | Parameter                       | Conditions                                                                           | Min                     | Typ | Max                     | Unit |
|--------------------------------|---------------------------------|--------------------------------------------------------------------------------------|-------------------------|-----|-------------------------|------|
| $F_{\text{ck1}}/T_{\text{CK}}$ | QUADSPI clock frequency         | $2.7 \text{ V} \leq V_{\text{DD}} < 3.6 \text{ V}$<br>$C_{\text{L}} = 20 \text{ pF}$ | -                       | -   | 133                     | MHz  |
|                                |                                 | $1.62 \text{ V} < V_{\text{DD}} < 3.6 \text{ V}$<br>$C_{\text{L}} = 15 \text{ pF}$   | -                       | -   | 100                     |      |
| $t_{\text{w}}(\text{CKH})$     | QUADSPI clock high and low time | -                                                                                    | $T_{\text{CK}}/2 - 0.5$ | -   | $T_{\text{CK}}/2$       | ns   |
| $t_{\text{w}}(\text{CKL})$     |                                 |                                                                                      | $T_{\text{CK}}/2$       | -   | $T_{\text{CK}}/2 + 0.5$ |      |
| $t_{\text{s}}(\text{IN})$      | Data input setup time           | -                                                                                    | 1.5                     | -   | -                       |      |
| $t_{\text{h}}(\text{IN})$      | Data input hold time            |                                                                                      | 2                       | -   | -                       |      |
| $t_{\text{v}}(\text{OUT})$     | Data output valid time          | -                                                                                    | -                       | 1.5 | 2                       |      |
| $t_{\text{h}}(\text{OUT})$     | Data output hold time           | -                                                                                    | 0.5                     | -   | -                       |      |

1. Guaranteed by characterization results.



Table 85. QUADSPI characteristics in DDR mode<sup>(1)</sup>

| Symbol                     | Parameter                       | Conditions                            | Min              | Typ              | Max              | Unit |
|----------------------------|---------------------------------|---------------------------------------|------------------|------------------|------------------|------|
| $F_{CK1/t(CK)}$            | QUADSPI clock frequency         | 2.7 V < $V_{DD}$ < 3.6 V<br>CL=20 pF  | -                | -                | 100              | MHz  |
|                            |                                 | 1.62 V < $V_{DD}$ < 3.6 V<br>CL=15 pF | -                | -                | 100              |      |
| $t_{w(CKH)}$               | QUADSPI clock high and low time | -                                     | $T_{CK}/2 - 0.5$ | -                | $T_{CK}/2$       | ns   |
| $t_{w(CKL)}$               |                                 |                                       | $T_{CK}/2$       | -                | $T_{CK}/2 + 0.5$ |      |
| $t_{sr(IN)}, t_{sf(IN)}$   | Data input setup time           | -                                     | 2                | -                | -                |      |
| $t_{hr(IN)}, t_{hf(IN)}$   | Data input hold time            | -                                     | 2                | -                | -                |      |
| $t_{vr(OUT)}, t_{vf(OUT)}$ | Data output valid time          | DHHC=0                                | -                | 3.5              | 4                |      |
|                            |                                 | DHHC=1<br>Pres=1, 2...                | -                | $T_{CK}/4 + 3.5$ | $T_{CK}/4 + 4$   |      |
| $t_{hr(OUT)}, t_{hf(OUT)}$ | Data output hold time           | DHHC=0                                | 3                | -                | -                |      |
|                            |                                 | DHHC=1<br>Pres=1, 2...                | $T_{CK}/4 + 3$   | -                | -                |      |

1. Guaranteed by characterization results.

Figure 38. Quad-SPI timing diagram - SDR mode

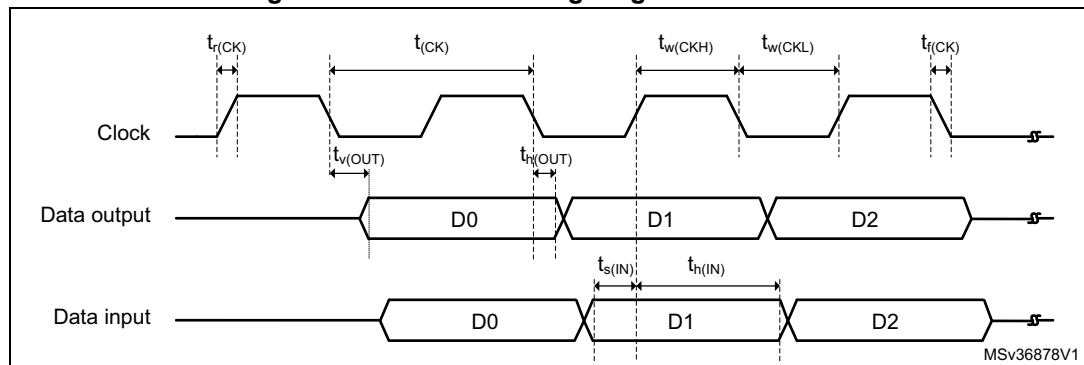
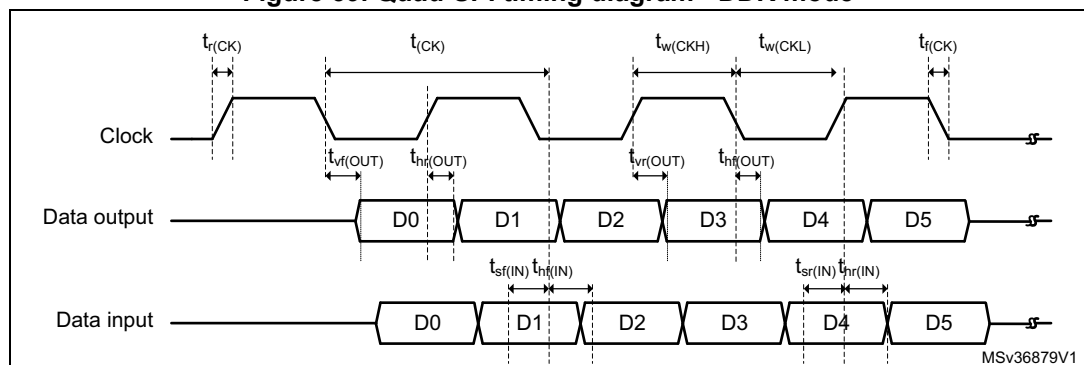


Figure 39. Quad-SPI timing diagram - DDR mode



### 6.3.19 Delay block (DLYB) characteristics

Unless otherwise specified, the parameters given in [Table 87](#) for the delay block are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage summarized in [Table 24: General operating conditions](#).

**Table 86. Dynamics characteristics: Delay Block characteristics<sup>(1)</sup>**

| Symbol            | Parameter     | Conditions | Min  | Typ  | Max  | Unit |
|-------------------|---------------|------------|------|------|------|------|
| $t_{\text{init}}$ | Initial delay | -          | 1400 | 2200 | 2400 | ps   |
| $t_{\Delta}$      | Unit Delay    | -          | 35   | 40   | 45   |      |

1. Guaranteed by characterization results.

### 6.3.20 16-bit ADC characteristics

Unless otherwise specified, the parameters given in [Table 87](#) are derived from tests performed under the ambient temperature,  $f_{\text{PCLK2}}$  frequency and  $V_{\text{DDA}}$  supply voltage conditions summarized in [Table 24: General operating conditions](#).

**Table 87. ADC characteristics<sup>(1)</sup>**

| Symbol            | Parameter                                                                            | Conditions                     |           | Min              | Typ | Max                 | Unit |
|-------------------|--------------------------------------------------------------------------------------|--------------------------------|-----------|------------------|-----|---------------------|------|
| V <sub>DDA</sub>  | Analog power supply                                                                  | -                              |           | 1.62             | -   | 3.6                 | V    |
| V <sub>REF+</sub> | Positive reference voltage                                                           | V <sub>DDA</sub> ≥ 2 V         |           | 2                | -   | V <sub>DDA</sub>    |      |
|                   |                                                                                      | V <sub>DDA</sub> < 2 V         |           | V <sub>DDA</sub> |     |                     |      |
| V <sub>REF-</sub> | Negative reference voltage                                                           | -                              |           | V <sub>SSA</sub> |     |                     |      |
| f <sub>ADC</sub>  | ADC clock frequency                                                                  | 2 V ≤ V <sub>DDA</sub> ≤ 3.3 V | BOOST = 1 | -                | -   | 36                  | MHz  |
|                   |                                                                                      |                                | BOOST = 0 | -                | -   | 20                  |      |
| f <sub>S</sub>    | Sampling rate for Fast channels, BOOST = 1, f <sub>ADC</sub> = 36 MHz <sup>(2)</sup> | 16-bit resolution              |           | -                | -   | 3.60 <sup>(2)</sup> | MSPS |
|                   |                                                                                      | 14-bit resolution              |           | -                | -   | 4.00 <sup>(2)</sup> |      |
|                   |                                                                                      | 12-bit resolution              |           | -                | -   | 4.50 <sup>(2)</sup> |      |
|                   |                                                                                      | 10-bit resolution              |           | -                | -   | 5.00 <sup>(2)</sup> |      |
|                   |                                                                                      | 8-bit resolution               |           |                  |     | 6.00 <sup>(2)</sup> |      |
|                   | Sampling rate for Fast channels, BOOST = 0, f <sub>ADC</sub> = 20 MHz                | 16-bit resolution              |           | -                | -   | 2.00 <sup>(2)</sup> |      |
|                   |                                                                                      | 14-bit resolution              |           | -                | -   | 2.20 <sup>(2)</sup> |      |
|                   |                                                                                      | 12-bit resolution              |           | -                | -   | 2.50 <sup>(2)</sup> |      |
|                   |                                                                                      | 10-bit resolution              |           | -                | -   | 2.80 <sup>(2)</sup> |      |
|                   |                                                                                      | 8-bit resolution               |           |                  |     | 3.30 <sup>(2)</sup> |      |
|                   | Sampling rate for Slow channels, BOOST = 0, f <sub>ADC</sub> = 10 MHz                | 16-bit resolution              |           | -                | -   | 1.00                |      |
|                   |                                                                                      | 14-bit resolution              |           | -                | -   | 1.00                |      |
|                   |                                                                                      | 12-bit resolution              |           | -                | -   | 1.00                |      |
|                   |                                                                                      | 10-bit resolution              |           | -                | -   | 1.00                |      |
|                   |                                                                                      | 8-bit resolution               |           |                  |     | 1.00                |      |

Table 87. ADC characteristics<sup>(1)</sup> (continued)

| Symbol                          | Parameter                                                                                         | Conditions                | Min                                                            | Typ                 | Max                     | Unit               |
|---------------------------------|---------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------|---------------------|-------------------------|--------------------|
| f <sub>TRIG</sub>               | External trigger frequency                                                                        | f <sub>ADC</sub> = 36 MHz | -                                                              | -                   | 3.6                     | MHz                |
|                                 |                                                                                                   | 16-bit resolution         | -                                                              | -                   | 10                      | 1/f <sub>ADC</sub> |
| V <sub>AIN</sub> <sup>(3)</sup> | Conversion voltage range                                                                          | -                         | 0                                                              | -                   | V <sub>REF+</sub>       | V                  |
| V <sub>CMIV</sub>               | Common mode input voltage                                                                         | -                         | V <sub>REF</sub> /2-10%                                        | V <sub>REF</sub> /2 | V <sub>REF</sub> /2+10% |                    |
| R <sub>AIN</sub>                | External input impedance                                                                          | -                         | -                                                              | -                   | 50                      | kΩ                 |
| C <sub>ADC</sub>                | Internal sample and hold capacitor                                                                | -                         | -                                                              | 4                   | -                       | pF                 |
| t <sub>ADCREG_STUP</sub>        | ADC LDO startup time                                                                              | -                         | -                                                              | 5                   | 10                      | μs                 |
| t <sub>STAB</sub>               | ADC power-up time                                                                                 | LDO already started       | 1                                                              |                     |                         | conversion cycle   |
| t <sub>CAL</sub>                | Offset and linearity calibration time                                                             | -                         | 165,010                                                        |                     |                         | 1/f <sub>ADC</sub> |
| t <sub>OFF_CAL</sub>            | Offset calibration time                                                                           | -                         | 1,280                                                          |                     |                         |                    |
| t <sub>LATR</sub>               | Trigger conversion latency for regular and injected channels without aborting the conversion      | CKMODE = 00               | 1.5                                                            | 2                   | 2.5                     |                    |
|                                 |                                                                                                   | CKMODE = 01               | -                                                              | -                   | 2                       |                    |
|                                 |                                                                                                   | CKMODE = 10               |                                                                |                     | 2.25                    |                    |
|                                 |                                                                                                   | CKMODE = 11               |                                                                |                     | 2.125                   |                    |
| t <sub>LATRINJ</sub>            | Trigger conversion latency for regular and injected channels when a regular conversion is aborted | CKMODE = 00               | 2.5                                                            | 3                   | 3.5                     |                    |
|                                 |                                                                                                   | CKMODE = 01               | -                                                              | -                   | 3                       |                    |
|                                 |                                                                                                   | CKMODE = 10               | -                                                              | -                   | 3.25                    |                    |
|                                 |                                                                                                   | CKMODE = 11               | -                                                              | -                   | 3.125                   |                    |
| t <sub>S</sub>                  | Sampling time                                                                                     | -                         | 1.5                                                            | -                   | 640.5                   |                    |
| t <sub>CONV</sub>               | Total conversion time (including sampling time)                                                   | N-bit resolution          | t <sub>S</sub> + 0.5 + N/2<br>(9 to 648 cycles in 14-bit mode) |                     |                         |                    |

1. Guaranteed by design.

2. These values are obtained using the following formula:  $f_{\text{S}} = f_{\text{ADC}} / t_{\text{CONV}}$ , where  $f_{\text{ADC}} = 36 \text{ MHz}$  and  $t_{\text{CONV}} = 1.5 \text{ cycle sampling time} + t_{\text{SAR}} \text{ sampling time}$ . Refer to the product reference manual for the value of  $t_{\text{SAR}}$  depending on resolution.

3. Depending on the package,  $V_{\text{REF+}}$  can be internally connected to  $V_{\text{DDA}}$  and  $V_{\text{REF-}}$  to  $V_{\text{SSA}}$ .

Table 88. ADC accuracy<sup>(1)(2)(3)</sup>

| Symbol               | Parameter                                     | Conditions <sup>(4)</sup> |           | Min | Typ   | Max | Unit |
|----------------------|-----------------------------------------------|---------------------------|-----------|-----|-------|-----|------|
| ET                   | Total unadjusted error                        | Single ended              | BOOST = 1 | -   | ±6    | -   | ±LSB |
|                      |                                               |                           | BOOST = 0 | -   | ±8    | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | ±10   | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | ±16   | -   |      |
| ED                   | Differential linearity error                  | Single ended              | BOOST = 1 | -   | 2     | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 1     | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | 8     | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 2     | -   |      |
| EL                   | Integral linearity error                      | Single ended              | BOOST = 1 | -   | ±6    | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | ±4    | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | ±6    | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | ±4    | -   |      |
| ENOB <sup>(5)</sup>  | Effective number of bits (2 MSPS)             | Single ended              | BOOST = 1 | -   | 11.6  | -   | bits |
|                      |                                               |                           | BOOST = 0 | -   | 12    | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | 13.3  | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 13.5  | -   |      |
| SINAD <sup>(5)</sup> | Signal-to-noise and distortion ratio (2 MSPS) | Single ended              | BOOST = 1 | -   | 71.6  | -   | dB   |
|                      |                                               |                           | BOOST = 0 | -   | 74    | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | 81.83 | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 83    | -   |      |
| SNR <sup>(5)</sup>   | Signal-to-noise ratio (2 MSPS)                | Single ended              | BOOST = 1 | -   | 72    | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 74    | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | 82    | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | 83    | -   |      |
| THD <sup>(5)</sup>   | Total harmonic distortion                     | Single ended              | BOOST = 1 | -   | -78   | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | -80   | -   |      |
|                      |                                               | Differential              | BOOST = 1 | -   | -90   | -   |      |
|                      |                                               |                           | BOOST = 0 | -   | -95   | -   |      |

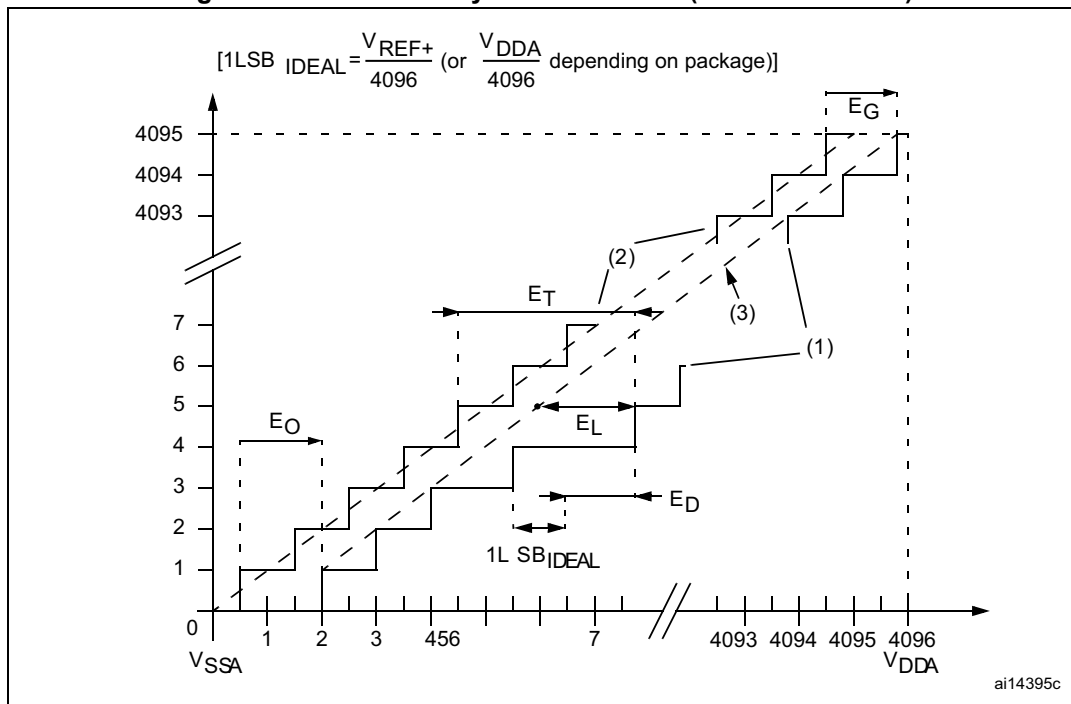
1. Guaranteed by characterization for BGA packages, the values for LQFP packages might differ.
2. ADC DC accuracy values are measured after internal calibration.
3. The above table gives the ADC performance in 16-bit mode.
4. ADC clock frequency  $\leq 36$  MHz,  $2\text{ V} \leq V_{\text{DDA}} \leq 3.3\text{ V}$ ,  $1.6\text{ V} \leq V_{\text{REF}} \leq V_{\text{DDA}}$ , BOOSTEN (for I/O) = 1.
5. ENOB, SINAD, SNR and THD are specified for  $V_{\text{DDA}} = V_{\text{REF}} = 3.3\text{ V}$ .

Note: ADC accuracy vs. negative injection current: injecting a negative current on any analog input pins should be avoided as this significantly reduces the accuracy of the conversion

being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to analog pins which may potentially inject negative currents.

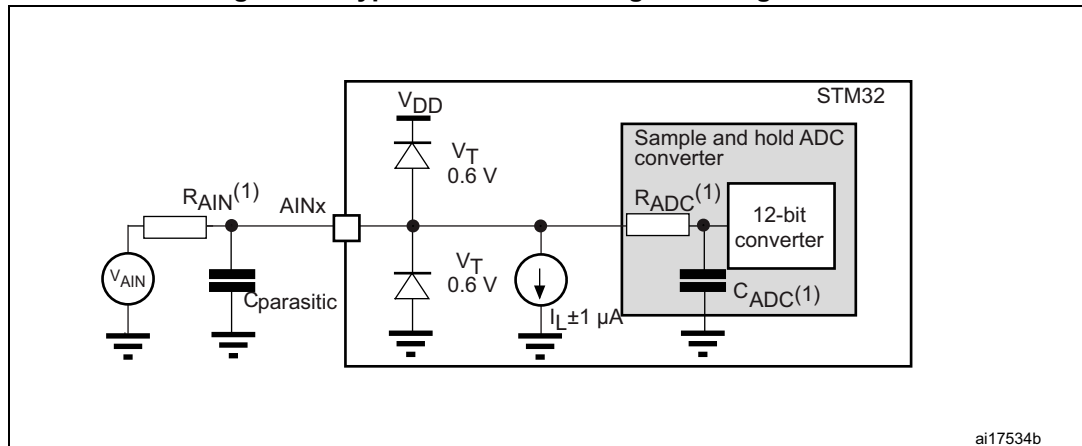
Any positive injection current within the limits specified for  $I_{INJ(PIN)}$  and  $\Sigma I_{INJ(PIN)}$  in [Section 6.3.14](#) does not affect the ADC accuracy.

**Figure 40. ADC accuracy characteristics (12-bit resolution)**



1. Example of an actual transfer curve.
2. Ideal transfer curve.
3. End point correlation line.
4.  $E_T$  = Total Unadjusted Error: maximum deviation between the actual and the ideal transfer curves.  
 $E_O$  = Offset Error: deviation between the first actual transition and the first ideal one.  
 $E_G$  = Gain Error: deviation between the last ideal transition and the last actual one.  
 $E_D$  = Differential Linearity Error: maximum deviation between actual steps and the ideal one.  
 $E_L$  = Integral Linearity Error: maximum deviation between any actual transition and the end point correlation line.

Figure 41. Typical connection diagram using the ADC



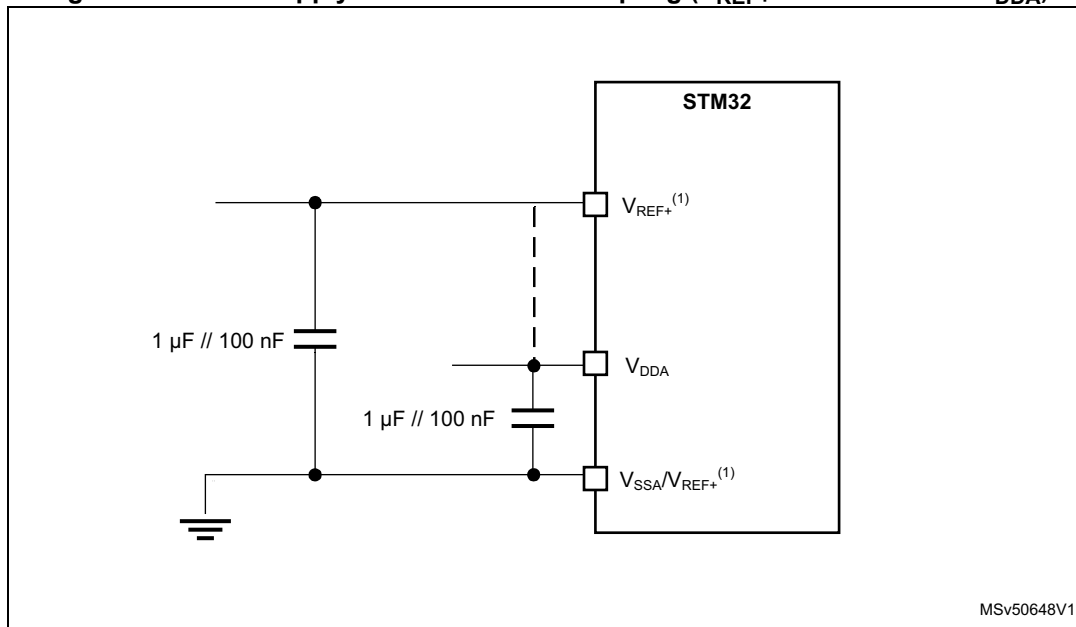
ai17534b

1. Refer to [Table 87](#) for the values of  $R_{AIN}$ ,  $R_{ADC}$  and  $C_{ADC}$ .
2.  $C_{parasitic}$  represents the capacitance of the PCB (dependent on soldering and PCB layout quality) plus the pad capacitance (roughly 5 pF). A high  $C_{parasitic}$  value downgrades conversion accuracy. To remedy this,  $f_{ADC}$  should be reduced.

### General PCB design guidelines

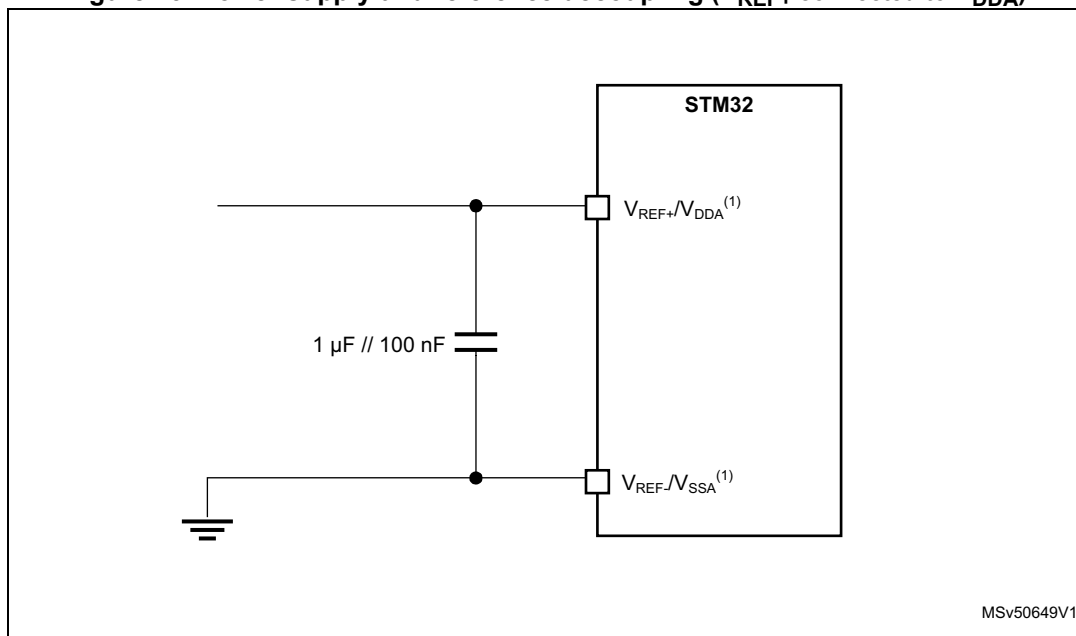
Power supply decoupling should be performed as shown in [Figure 42](#) or [Figure 43](#), depending on whether  $V_{REF+}$  is connected to  $V_{DDA}$  or not. The 100 nF capacitors should be ceramic (good quality). They should be placed as close as possible to the chip.

**Figure 42. Power supply and reference decoupling ( $V_{REF+}$  not connected to  $V_{DDA}$ )**



1.  $V_{REF+}$  input is available on all package whereas the  $V_{REF-}$  is available only on UFBGA176+25 and TFBGA240+25. When  $V_{REF-}$  is not available, it is internally connected to  $V_{DDA}$  and  $V_{SSA}$ .

**Figure 43. Power supply and reference decoupling ( $V_{REF+}$  connected to  $V_{DDA}$ )**



1.  $V_{REF+}$  input is available on all package whereas the  $V_{REF-}$  is available only on UFBGA176+25 and TFBGA240+25. When  $V_{REF-}$  is not available, it is internally connected to  $V_{DDA}$  and  $V_{SSA}$ .

## 6.3.21 DAC electrical characteristics

Table 89. DAC characteristics<sup>(1)</sup>

| Symbol             | Parameter                                                                                                                                                                                                                                            | Conditions                                                                            |                         | Min  | Typ                | Max              | Unit          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------|------|--------------------|------------------|---------------|
| $V_{DDA}$          | Analog supply voltage                                                                                                                                                                                                                                | -                                                                                     |                         | 1.8  | 3.3                | 3.6              | V             |
| $V_{REF+}$         | Positive reference voltage                                                                                                                                                                                                                           | -                                                                                     |                         | 1.80 | -                  | $V_{DDA}$        |               |
| $V_{REF-}$         | Negative reference voltage                                                                                                                                                                                                                           | -                                                                                     |                         | -    | $V_{SSA}$          | -                |               |
| $R_L$              | Resistive Load                                                                                                                                                                                                                                       | DAC output buffer ON                                                                  | connected to $V_{SSA}$  | 5    | -                  | -                | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                       | connected to $V_{DDA}$  | 25   | -                  | -                |               |
| $R_O^{(2)}$        | Output Impedance                                                                                                                                                                                                                                     | DAC output buffer OFF                                                                 |                         | 10.3 | 13                 | 16               |               |
| $R_{BON}$          | Output impedance sample and hold mode, output buffer ON                                                                                                                                                                                              | DAC output buffer ON                                                                  | $V_{DD} = 2.7\text{ V}$ | -    | -                  | 1.6              | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                       | $V_{DD} = 2.0\text{ V}$ | -    | -                  | 2.6              |               |
| $R_{BOFF}$         | Output impedance sample and hold mode, output buffer OFF                                                                                                                                                                                             | DAC output buffer OFF                                                                 | $V_{DD} = 2.7\text{ V}$ | -    | -                  | 17.8             | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                       | $V_{DD} = 2.0\text{ V}$ | -    | -                  | 18.7             |               |
| $C_L^{(2)}$        | Capacitive Load                                                                                                                                                                                                                                      | DAC output buffer OFF                                                                 |                         | -    | -                  | 50               | pF            |
| $C_{SH}^{(2)}$     |                                                                                                                                                                                                                                                      | Sample and Hold mode                                                                  |                         | -    | 0.1                | 1                | $\mu\text{F}$ |
| $V_{DAC\_OUT}$     | Voltage on DAC_OUT output                                                                                                                                                                                                                            | DAC output buffer ON                                                                  |                         | 0.2  | -                  | $V_{REF+} - 0.2$ | V             |
|                    |                                                                                                                                                                                                                                                      | DAC output buffer OFF                                                                 |                         | 0    | -                  | $V_{REF+}$       |               |
| $t_{SETTLING}$     | Settling time (full scale: for a 12-bit code transition between the lowest and the highest input codes when DAC_OUT reaches the final value of $\pm 0.5\text{LSB}$ , $\pm 1\text{LSB}$ , $\pm 2\text{LSB}$ , $\pm 4\text{LSB}$ , $\pm 8\text{LSB}$ ) | Normal mode, DAC output buffer OFF, $\pm 1\text{LSB}$ $C_L = 10\text{ pF}$            |                         | -    | 1.7 <sup>(2)</sup> | 2 <sup>(2)</sup> | $\mu\text{s}$ |
| $t_{WAKEUP}^{(3)}$ | Wakeup time from off state (setting the Enx bit in the DAC Control register) until the $\pm 1\text{LSB}$ final value                                                                                                                                 | Normal mode, DAC output buffer ON, $C_L \leq 50\text{ pF}$ , $R_L = 5\text{ k}\Omega$ |                         | -    | 5                  | 7.5              | $\mu\text{s}$ |
| $V_{offset}^{(2)}$ | Middle code offset for 1 trim code step                                                                                                                                                                                                              | $V_{REF+} = 3.6\text{ V}$                                                             |                         | -    | 850                | -                | $\mu\text{V}$ |
|                    |                                                                                                                                                                                                                                                      | $V_{REF+} = 1.8\text{ V}$                                                             |                         | -    | 425                | -                |               |



Table 89. DAC characteristics<sup>(1)</sup> (continued)

| Symbol         | Parameter                                | Conditions                                                     |                                    | Min | Typ                                 | Max | Unit    |
|----------------|------------------------------------------|----------------------------------------------------------------|------------------------------------|-----|-------------------------------------|-----|---------|
| $I_{DDA}(DAC)$ | DAC quiescent consumption from $V_{DDA}$ | DAC output buffer ON                                           | No load, middle code (0x800)       | -   | 360                                 | -   | $\mu A$ |
|                |                                          |                                                                | No load, worst code (0xF1C)        | -   | 490                                 | -   |         |
|                |                                          | DAC output buffer OFF                                          | No load, middle/worst code (0x800) | -   | 20                                  | -   |         |
|                |                                          | Sample and Hold mode, $C_{SH}=100$ nF                          |                                    | -   | $360 \cdot T_{ON}/(T_{ON}+T_{OFF})$ | -   |         |
| $I_{DDV}(DAC)$ | DAC consumption from $V_{REF+}$          | DAC output buffer ON                                           | No load, middle code (0x800)       | -   | 170                                 | -   |         |
|                |                                          |                                                                | No load, worst code (0xF1C)        | -   | 170                                 | -   |         |
|                |                                          | DAC output buffer OFF                                          | No load, middle/worst code (0x800) | -   | 160                                 | -   |         |
|                |                                          | Sample and Hold mode, Buffer ON, $C_{SH}=100$ nF (worst code)  |                                    | -   | $170 \cdot T_{ON}/(T_{ON}+T_{OFF})$ | -   |         |
|                |                                          | Sample and Hold mode, Buffer OFF, $C_{SH}=100$ nF (worst code) |                                    | -   | $160 \cdot T_{ON}/(T_{ON}+T_{OFF})$ | -   |         |
|                |                                          |                                                                |                                    |     |                                     |     |         |

1. Guaranteed by characterization results.

2. Guaranteed by design.

3. In buffered mode, the output can overshoot above the final value for low input code (starting from the minimum value).

Table 90. DAC accuracy<sup>(1)</sup>

| Symbol | Parameter                                 | Conditions                                                      |                    | Min | Typ     | Max      | Unit |
|--------|-------------------------------------------|-----------------------------------------------------------------|--------------------|-----|---------|----------|------|
| DNL    | Differential non linearity <sup>(2)</sup> | DAC output buffer ON                                            |                    | -   | $\pm 2$ | -        | LSB  |
|        |                                           | DAC output buffer OFF                                           |                    | -   | $\pm 2$ | -        |      |
| INL    | Integral non linearity <sup>(3)</sup>     | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ |                    | -   | $\pm 4$ | -        | LSB  |
|        |                                           | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$               |                    | -   | $\pm 4$ | -        |      |
| Offset | Offset error at code 0x800 <sup>(3)</sup> | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ | $V_{REF+} = 3.6$ V | -   | -       | $\pm 12$ | LSB  |
|        |                                           |                                                                 | $V_{REF+} = 1.8$ V | -   | -       | $\pm 25$ |      |
|        |                                           | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$               |                    | -   | -       | $\pm 8$  |      |

Table 90. DAC accuracy<sup>(1)</sup> (continued)

| Symbol    | Parameter                                            | Conditions                                                                               | Min                | Typ  | Max      | Unit |
|-----------|------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------|------|----------|------|
| Offset1   | Offset error at code 0x001 <sup>(4)</sup>            | DAC output buffer OFF,<br>$C_L \leq 50$ pF, no $R_L$                                     | -                  | -    | $\pm 5$  | LSB  |
| OffsetCal | Offset error at code 0x800 after factory calibration | DAC output buffer ON,<br>$C_L \leq 50$ pF,<br>$R_L \geq 5$ k $\Omega$                    | $V_{REF+} = 3.6$ V | -    | $\pm 5$  | LSB  |
|           |                                                      |                                                                                          | $V_{REF+} = 1.8$ V | -    | $\pm 7$  |      |
| Gain      | Gain error <sup>(5)</sup>                            | DAC output buffer ON, $C_L \leq 50$ pF,<br>$R_L \geq 5$ k $\Omega$                       | -                  | -    | $\pm 1$  | %    |
|           |                                                      | DAC output buffer OFF,<br>$C_L \leq 50$ pF, no $R_L$                                     | -                  | -    | $\pm 1$  |      |
| TUE       | Total unadjusted error                               | DAC output buffer OFF,<br>$C_L \leq 50$ pF, no $R_L$                                     | -                  | -    | $\pm 12$ | LSB  |
| SNR       | Signal-to-noise ratio <sup>(6)</sup>                 | DAC output buffer ON, $C_L \leq 50$ pF,<br>$R_L \geq 5$ k $\Omega$ , 1 kHz, BW = 500 KHz | -                  | 67.8 | -        | dB   |
| SINAD     | Signal-to-noise and distortion ratio <sup>(6)</sup>  | DAC output buffer ON, $C_L \leq 50$ pF,<br>$R_L \geq 5$ k $\Omega$ , 1 kHz               | -                  | 67.5 | -        | dB   |
| ENOB      | Effective number of bits                             | DAC output buffer ON,<br>$C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , 1 kHz               | -                  | 10.9 | -        | bits |

1. Guaranteed by characterization.

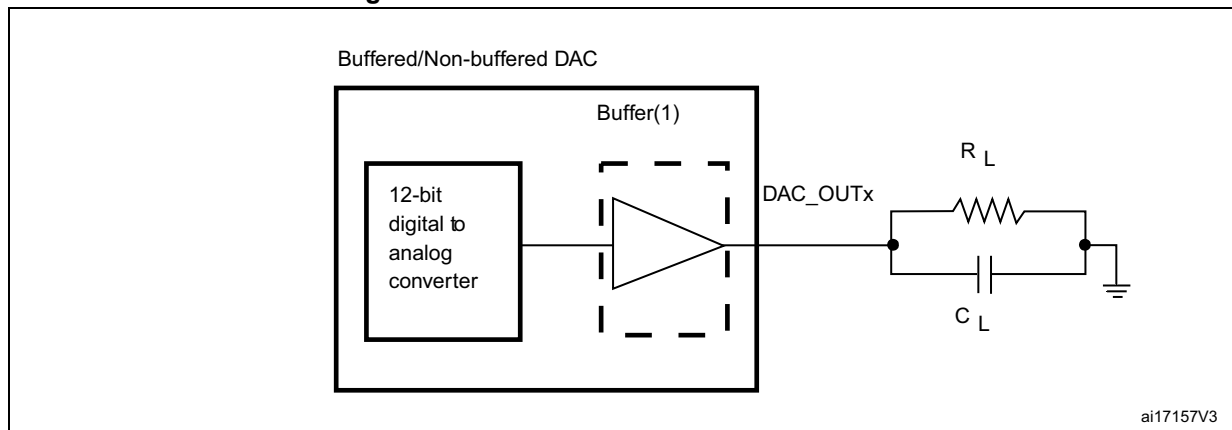
2. Difference between two consecutive codes minus 1 LSB.

3. Difference between the value measured at Code i and the value measured at Code i on a line drawn between Code 0 and last Code 4095.

4. Difference between the value measured at Code (0x001) and the ideal value.

5. Difference between the ideal slope of the transfer function and the measured slope computed from code 0x000 and 0xFFFF when the buffer is OFF, and from code giving 0.2 V and ( $V_{REF+} - 0.2$  V) when the buffer is ON.

6. Signal is -0.5dBFS with  $F_{sampling}=1$  MHz.

**Figure 44. 12-bit buffered /non-buffered DAC**

1. The DAC integrates an output buffer that can be used to reduce the output impedance and to drive external loads directly without the use of an external operational amplifier. The buffer can be bypassed by configuring the BOFFx bit in the DAC\_CR register.

### 6.3.22 Voltage reference buffer characteristics

**Table 91. VREFBUF characteristics<sup>(1)</sup>**

| Symbol            | Parameter                       | Conditions                   |              | Min                       | Typ   | Max       | Unit |
|-------------------|---------------------------------|------------------------------|--------------|---------------------------|-------|-----------|------|
| $V_{DDA}$         | Analog supply voltage           | Normal mode                  | VSCALE = 000 | 2.8                       | 3.3   | 3.6       | V    |
|                   |                                 |                              | VSCALE = 001 | 2.4                       | -     | 3.6       |      |
|                   |                                 |                              | VSCALE = 010 | 2.1                       | -     | 3.6       |      |
|                   |                                 |                              | VSCALE = 011 | 1.8                       | -     | 3.6       |      |
|                   |                                 | Degraded mode                | VSCALE = 000 | 1.62                      | -     | 2.80      |      |
|                   |                                 |                              | VSCALE = 001 | 1.62                      | -     | 2.40      |      |
|                   |                                 |                              | VSCALE = 010 | 1.62                      | -     | 2.10      |      |
|                   |                                 |                              | VSCALE = 011 | 1.62                      | -     | 1.80      |      |
| $V_{REFBUF\_OUT}$ | Voltage Reference Buffer Output | Normal mode                  | VSCALE = 000 | -                         | 2.5   | -         |      |
|                   |                                 |                              | VSCALE = 001 | -                         | 2.048 | -         |      |
|                   |                                 |                              | VSCALE = 010 | -                         | 1.8   | -         |      |
|                   |                                 |                              | VSCALE = 011 | -                         | 1.5   | -         |      |
|                   |                                 | Degraded mode <sup>(2)</sup> | VSCALE = 000 | $V_{DDA} - 150\text{ mV}$ | -     | $V_{DDA}$ |      |
|                   |                                 |                              | VSCALE = 001 | $V_{DDA} - 150\text{ mV}$ | -     | $V_{DDA}$ |      |
|                   |                                 |                              | VSCALE = 010 | $V_{DDA} - 150\text{ mV}$ | -     | $V_{DDA}$ |      |
|                   |                                 |                              | VSCALE = 011 | $V_{DDA} - 150\text{ mV}$ | -     | $V_{DDA}$ |      |
| TRIM              | Trim step resolution            | -                            | -            | -                         | ±0.05 | ±0.2      | %    |
| $C_L$             | Load capacitor                  | -                            | -            | 0.5                       | 1     | 1.50      | uF   |

Table 91. VREFBUF characteristics<sup>(1)</sup> (continued)

| Symbol                    | Parameter                                                                                          | Conditions                        |                            | Min | Typ | Max                                                 | Unit       |
|---------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------|-----|-----|-----------------------------------------------------|------------|
| esr                       | Equivalent Serial Resistor of C <sub>L</sub>                                                       | -                                 | -                          | -   | -   | 2                                                   | Ω          |
| I <sub>load</sub>         | Static load current                                                                                | -                                 | -                          | -   | -   | 4                                                   | mA         |
| I <sub>line_reg</sub>     | Line regulation                                                                                    | 2.8 V ≤ V <sub>DDA</sub> ≤ 3.6 V  | I <sub>load</sub> = 500 μA | -   | 200 | -                                                   | ppm/V      |
|                           |                                                                                                    |                                   | I <sub>load</sub> = 4 mA   | -   | 100 | -                                                   |            |
| I <sub>load_reg</sub>     | Load regulation                                                                                    | 500 μA ≤ I <sub>LOAD</sub> ≤ 4 mA | Normal Mode                | -   | 50  | -                                                   | ppm/<br>mA |
| T <sub>coeff</sub>        | Temperature coefficient                                                                            | -40 °C < T <sub>J</sub> < +125 °C | -                          | -   | -   | T <sub>coeff</sub><br>× V <sub>REFINT</sub><br>+ 75 | ppm/<br>°C |
| PSRR                      | Power supply rejection                                                                             | DC                                | -                          | -   | 60  | -                                                   | dB         |
|                           |                                                                                                    | 100KHz                            | -                          | -   | 40  | -                                                   |            |
| t <sub>START</sub>        | Start-up time                                                                                      | C <sub>L</sub> = 0.5 μF           | -                          | -   | 300 | -                                                   | μs         |
|                           |                                                                                                    | C <sub>L</sub> = 1 μF             | -                          | -   | 500 | -                                                   |            |
|                           |                                                                                                    | C <sub>L</sub> = 1.5 μF           | -                          | -   | 650 | -                                                   |            |
| I <sub>INRUSH</sub>       | Control of maximum DC current drive on V <sub>REFBUF_OUT</sub> during startup phase <sup>(3)</sup> | -                                 |                            | -   | 8   | -                                                   | mA         |
| I <sub>DDA(VREFBUF)</sub> | VREFBUF consumption from V <sub>DDA</sub>                                                          | I <sub>LOAD</sub> = 0 μA          | -                          | -   | 15  | 25                                                  | μA         |
|                           |                                                                                                    | I <sub>LOAD</sub> = 500 μA        | -                          | -   | 16  | 30                                                  |            |
|                           |                                                                                                    | I <sub>LOAD</sub> = 4 mA          | -                          | -   | 32  | 50                                                  |            |

1. Guaranteed by design.

2. In degraded mode, the voltage reference buffer cannot accurately maintain the output voltage (V<sub>DDA</sub>-drop voltage).

3. To properly control VREFBUF I<sub>INRUSH</sub> current during the startup phase and the change of scaling, V<sub>DDA</sub> voltage should be in the range of 1.8 V-3.6 V, 2.1 V-3.6 V, 2.4 V-3.6 V and 2.8 V-3.6 V for VSCALE = 011, 010, 001 and 000, respectively.

### 6.3.23 Temperature sensor characteristics

Table 92. Temperature sensor characteristics

| Symbol                                | Parameter                                      | Min | Typ  | Max  | Unit  |
|---------------------------------------|------------------------------------------------|-----|------|------|-------|
| T <sub>L</sub> <sup>(1)</sup>         | V <sub>SENSE</sub> linearity with temperature  | -   | -    | 3    | °C    |
| Avg_Slope <sup>(2)</sup>              | Average slope                                  | -   | 2    | -    | mV/°C |
| V <sub>30</sub> <sup>(3)</sup>        | Voltage at 30°C ± 5 °C                         | -   | 0.62 | -    | V     |
| t <sub>start_run</sub> <sup>(1)</sup> | Startup time in Run mode (buffer startup)      | -   | -    | 25.2 | μs    |
| t <sub>S_temp</sub> <sup>(1)</sup>    | ADC sampling time when reading the temperature | 9   | -    | -    |       |
| I <sub>sens</sub> <sup>(1)</sup>      | Sensor consumption                             | -   | 0.18 | 0.31 | μA    |
| I <sub>sensbuf</sub> <sup>(1)</sup>   | Sensor buffer consumption                      | -   | 3.8  | 6.5  |       |

1. Guaranteed by design.

2. Guaranteed by characterization.
3. Measured at  $V_{DDA} = 3.3 \text{ V} \pm 10 \text{ mV}$ . The  $V_{30}$  ADC conversion result is stored in the TS\_CAL1 byte.

**Table 93. Temperature sensor calibration values**

| Symbol  | Parameter                                                                     | Memory address            |
|---------|-------------------------------------------------------------------------------|---------------------------|
| TS_CAL1 | Temperature sensor raw data acquired value at 30 °C, $V_{DDA}=3.3 \text{ V}$  | 0x1FF1 E820 - 0x1FF1 E821 |
| TS_CAL2 | Temperature sensor raw data acquired value at 110 °C, $V_{DDA}=3.3 \text{ V}$ | 0x1FF1 E840 - 0x1FF1 E841 |

### 6.3.24 Temperature and $V_{BAT}$ monitoring

**Table 94.  $V_{BAT}$  monitoring characteristics**

| Symbol              | Parameter                                      | Min | Typ  | Max | Unit          |
|---------------------|------------------------------------------------|-----|------|-----|---------------|
| R                   | Resistor bridge for $V_{BAT}$                  | -   | 26   | -   | K $\Omega$    |
| Q                   | Ratio on $V_{BAT}$ measurement                 | -   | 4    | -   | -             |
| $Er^{(1)}$          | Error on Q                                     | -10 | -    | +10 | %             |
| $t_{S\_vbat}^{(1)}$ | ADC sampling time when reading $V_{BAT}$ input | 9   | -    | -   | $\mu\text{s}$ |
| $V_{BAThigh}$       | High supply monitoring                         | -   | 3.55 | -   | V             |
| $V_{BATlow}$        | Low supply monitoring                          | -   | 1.36 | -   |               |

1. Guaranteed by design.

**Table 95.  $V_{BAT}$  charging characteristics**

| Symbol   | Parameter                 | Condition          | Min | Typ | Max | Unit       |
|----------|---------------------------|--------------------|-----|-----|-----|------------|
| $R_{BC}$ | Battery charging resistor | VBRS in PWR_CR3= 0 | -   | 5   | -   | K $\Omega$ |
|          |                           | VBRS in PWR_CR3= 1 |     | 1.5 | -   |            |

**Table 96. Temperature monitoring characteristics**

| Symbol               | Parameter                   | Min | Typ | Max | Unit |
|----------------------|-----------------------------|-----|-----|-----|------|
| TEMP <sub>high</sub> | High temperature monitoring | -   | 117 | -   | °C   |
| TEMP <sub>low</sub>  | Low temperature monitoring  | -   | -25 | -   |      |

### 6.3.25 Voltage booster for analog switch

Table 97. Voltage booster for analog switch characteristics<sup>(1)</sup>

| Symbol          | Parameter            | Condition                                     | Min  | Typ | Max | Unit    |
|-----------------|----------------------|-----------------------------------------------|------|-----|-----|---------|
| $V_{DD}$        | Supply voltage       | -                                             | 1.62 | 2-6 | 3.6 | V       |
| $t_{SU(BOOST)}$ | Booster startup time | -                                             | -    | -   | 50  | $\mu s$ |
| $I_{DD(BOOST)}$ | Booster consumption  | $1.62\text{ V} \leq V_{DD} \leq 2.7\text{ V}$ | -    | -   | 125 | $\mu A$ |
|                 |                      | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$        | -    | -   | 250 |         |

1. Guaranteed by characterization results.

### 6.3.26 Comparator characteristics

Table 98. COMP characteristics<sup>(1)</sup>

| Symbol                         | Parameter                                                                         | Conditions                |                                             | Min                          | Typ | Max              | Unit |
|--------------------------------|-----------------------------------------------------------------------------------|---------------------------|---------------------------------------------|------------------------------|-----|------------------|------|
| V <sub>DDA</sub>               | Analog supply voltage                                                             | -                         |                                             | 1.62                         | 3.3 | 3.6              | V    |
| V <sub>IN</sub>                | Comparator input voltage range                                                    | -                         |                                             | 0                            | -   | V <sub>DDA</sub> |      |
| V <sub>BG</sub> <sup>(2)</sup> | Scaler input voltage                                                              | -                         |                                             | Refer to V <sub>REFINT</sub> |     |                  |      |
| V <sub>SC</sub>                | Scaler offset voltage                                                             | -                         |                                             | -                            | ±5  | ±10              |      |
| I <sub>DDA(SCALER)</sub>       | Scaler static consumption from V <sub>DDA</sub>                                   | BRG_EN=0 (bridge disable) |                                             | -                            | 0.2 | 0.3              | µA   |
|                                |                                                                                   | BRG_EN=1 (bridge enable)  |                                             | -                            | 0.8 | 1                |      |
| t <sub>START_SCALER</sub>      | Scaler startup time                                                               | -                         |                                             | -                            | 140 | 250              | µs   |
| t <sub>START</sub>             | Comparator startup time to reach propagation delay specification                  | High-speed mode           |                                             | -                            | 2   | 5                | µs   |
|                                |                                                                                   | Medium mode               |                                             | -                            | 5   | 20               |      |
|                                |                                                                                   | Ultra-low-power mode      |                                             | -                            | 15  | 80               |      |
| t <sub>D</sub>                 | Propagation delay for 200 mV step with 100 mV overdrive                           | High-speed mode           |                                             | -                            | 50  | 80               | ns   |
|                                |                                                                                   | Medium mode               |                                             | -                            | 0.5 | 1.2              | µs   |
|                                |                                                                                   | Ultra-low-power mode      |                                             | -                            | 2.5 | 7                |      |
|                                | Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs | High-speed mode           |                                             | -                            | 50  | 120              | ns   |
|                                |                                                                                   | Medium mode               |                                             | -                            | 0.5 | 1.2              | µs   |
|                                |                                                                                   | Ultra-low-power mode      |                                             | -                            | 2.5 | 7                |      |
| V <sub>offset</sub>            | Comparator offset error                                                           | Full common mode range    |                                             | -                            | ±5  | ±20              | mV   |
| V <sub>hys</sub>               | Comparator hysteresis                                                             | No hysteresis             |                                             | -                            | 0   | -                | mV   |
|                                |                                                                                   | Low hysteresis            |                                             | -                            | 10  | -                |      |
|                                |                                                                                   | Medium hysteresis         |                                             | -                            | 20  | -                |      |
|                                |                                                                                   | High hysteresis           |                                             | -                            | 30  | -                |      |
| I <sub>DDA(COMP)</sub>         | Comparator consumption from V <sub>DDA</sub>                                      | Ultra-low-power mode      | Static                                      | -                            | 400 | 600              | nA   |
|                                |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -                            | 800 | -                |      |
|                                |                                                                                   | Medium mode               | Static                                      | -                            | 5   | 7                | µA   |
|                                |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -                            | 6   | -                |      |
|                                |                                                                                   | High-speed mode           | Static                                      | -                            | 70  | 100              |      |
|                                |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -                            | 75  | -                |      |

1. Guaranteed by design, unless otherwise specified.

2. Refer to [Table 28: Embedded reference voltage](#).

## 6.3.27 Operational amplifier characteristics

Table 99. OPAMP characteristics<sup>(1)</sup>

| Symbol                       | Parameter                                                       | Conditions                                                                                               | Min | Typ  | Max       | Unit  |
|------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----|------|-----------|-------|
| $V_{DDA}$                    | Analog supply voltage Range                                     | -                                                                                                        | 2   | 3.3  | 3.6       | V     |
| CMIR                         | Common Mode Input Range                                         | -                                                                                                        | 0   | -    | $V_{DDA}$ |       |
| $V_{I\text{OFFSET}}$         | Input offset voltage                                            | 25°C, no load on output                                                                                  | -   | -    | ±1.5      | mV    |
|                              |                                                                 | All voltages and temperature, no load                                                                    | -   | -    | ±2.5      |       |
| $\Delta V_{I\text{OFFSET}}$  | Input offset voltage drift                                      | -                                                                                                        | -   | ±3.0 | -         | µV/°C |
| TRIMOFFSETP<br>TRIMLPOFFSETP | Offset trim step at low common input voltage (0.1* $V_{DDA}$ )  | -                                                                                                        | -   | 1.1  | 1.5       | mV    |
| TRIMOFFSETN<br>TRIMLPOFFSETN | Offset trim step at high common input voltage (0.9* $V_{DDA}$ ) | -                                                                                                        | -   | 1.1  | 1.5       |       |
| $I_{LOAD}$                   | Drive current                                                   | -                                                                                                        | -   | -    | 500       | µA    |
| $I_{LOAD\_PGA}$              | Drive current in PGA mode                                       | -                                                                                                        | -   | -    | 270       |       |
| $C_{LOAD}$                   | Capacitive load                                                 | -                                                                                                        | -   | -    | 50        | pF    |
| CMRR                         | Common mode rejection ratio                                     | -                                                                                                        | -   | 80   | -         | dB    |
| PSRR                         | Power supply rejection ratio                                    | $C_{LOAD} \leq 50\text{pf}$ /<br>$R_{LOAD} \geq 4\text{ k}\Omega^{(2)}$ at 1 kHz,<br>$V_{com}=V_{DDA}/2$ | 50  | 66   | -         | dB    |
| GBW                          | Gain bandwidth for high supply range                            | -                                                                                                        | 4   | 7.3  | 12.3      | MHz   |
| SR                           | Slew rate (from 10% and 90% of output voltage)                  | Normal mode                                                                                              | -   | 3    | -         | V/µs  |
|                              |                                                                 | High-speed mode                                                                                          | -   | 30   | -         |       |
| AO                           | Open loop gain                                                  | -                                                                                                        | 59  | 90   | 129       | dB    |
| $\phi_m$                     | Phase margin                                                    | -                                                                                                        | -   | 55   | -         | °     |
| GM                           | Gain margin                                                     | -                                                                                                        | -   | 12   | -         | dB    |



Table 99. OPAMP characteristics<sup>(1)</sup> (continued)

| Symbol        | Parameter                                                                 | Conditions                                                           |                                                                                                        | Min                        | Typ    | Max | Unit                       |
|---------------|---------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|--------|-----|----------------------------|
| $V_{OHSAT}$   | High saturation voltage                                                   | $I_{load} = \max$ or $R_{LOAD} = \min^{(2)}$ ,<br>Input at $V_{DDA}$ |                                                                                                        | $V_{DDA} - 100 \text{ mV}$ | -      | -   | mV                         |
| $V_{OLSAT}$   | Low saturation voltage                                                    | $I_{load} = \max$ or $R_{LOAD} = \min^{(2)}$ ,<br>Input at 0 V       |                                                                                                        | -                          | -      | 100 |                            |
| $t_{WAKEUP}$  | Wake up time from OFF state                                               | Normal mode                                                          | $C_{LOAD} \leq 50 \text{ pf}$ ,<br>$R_{LOAD} \geq 4 \text{ k}\Omega^{(2)}$ ,<br>follower configuration | -                          | 0.8    | 3.2 | $\mu\text{s}$              |
|               |                                                                           | High speed                                                           | $C_{LOAD} \leq 50 \text{ pf}$ ,<br>$R_{LOAD} \geq 4 \text{ k}\Omega^{(2)}$ ,<br>follower configuration | -                          | 0.9    | 2.8 |                            |
| PGA gain      | Non inverting gain value                                                  | -                                                                    |                                                                                                        | -                          | 2      | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | 4      | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | 8      | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | 16     | -   | -                          |
|               | Inverting gain value                                                      | -                                                                    |                                                                                                        | -                          | -1     | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | -3     | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | -7     | -   | -                          |
|               |                                                                           | -                                                                    |                                                                                                        | -                          | -15    | -   | -                          |
| $R_{network}$ | R2/R1 internal resistance values in non-inverting PGA mode <sup>(3)</sup> | PGA Gain=2                                                           |                                                                                                        | -                          | 10/10  | -   | k $\Omega$ /<br>k $\Omega$ |
|               |                                                                           | PGA Gain=4                                                           |                                                                                                        | -                          | 30/10  | -   |                            |
|               |                                                                           | PGA Gain=8                                                           |                                                                                                        | -                          | 70/10  | -   |                            |
|               |                                                                           | PGA Gain=16                                                          |                                                                                                        | -                          | 150/10 | -   |                            |
|               | R2/R1 internal resistance values in inverting PGA mode <sup>(3)</sup>     | PGA Gain=-1                                                          |                                                                                                        | -                          | 10/10  | -   |                            |
|               |                                                                           | PGA Gain=-3                                                          |                                                                                                        | -                          | 30/10  | -   |                            |
|               |                                                                           | PGA Gain=-7                                                          |                                                                                                        | -                          | 70/10  | -   |                            |
|               |                                                                           | PGA Gain=-15                                                         |                                                                                                        | -                          | 150/10 | -   |                            |
| Delta R       | Resistance variation (R1 or R2)                                           | -                                                                    |                                                                                                        | -15                        | -      | 15  | %                          |
| PGA BW        | PGA bandwidth for different non inverting gain                            | Gain=2                                                               |                                                                                                        | -                          | GBW/2  | -   | MHz                        |
|               |                                                                           | Gain=4                                                               |                                                                                                        | -                          | GBW/4  | -   |                            |
|               |                                                                           | Gain=8                                                               |                                                                                                        | -                          | GBW/8  | -   |                            |
|               |                                                                           | Gain=16                                                              |                                                                                                        | -                          | GBW/16 | -   |                            |

Table 99. OPAMP characteristics<sup>(1)</sup> (continued)

| Symbol                  | Parameter                               | Conditions      |                                   | Min | Typ | Max  | Unit   |
|-------------------------|-----------------------------------------|-----------------|-----------------------------------|-----|-----|------|--------|
| en                      | Voltage noise density                   | at 1 KHz        | output loaded with 4 kΩ           | -   | 140 | -    | nV/√Hz |
|                         |                                         | at 10 KHz       |                                   | -   | 55  | -    |        |
| I <sub>DDA(OPAMP)</sub> | OPAMP consumption from V <sub>DDA</sub> | Normal mode     | no Load, quiescent mode, follower | -   | 570 | 1000 | μA     |
|                         |                                         | High-speed mode |                                   | -   | 610 | 1200 |        |

1. Guaranteed by design, unless otherwise specified.
2. R<sub>LOAD</sub> is the resistive load connected to VSSA or to VDDA.
3. R2 is the internal resistance between the OPAMP output and the OPAMP inverting input. R1 is the internal resistance between the OPAMP inverting input and ground. PGA gain = 1 + R2/R1.

### 6.3.28 Digital filter for Sigma-Delta Modulators (DFSDM) characteristics

Unless otherwise specified, the parameters given in [Table 100](#) for DFSDM are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 10$
- Capacitive load  $C = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (DFSDMx\_CKINx, DFSDMx\_DATINx, DFSDMx\_CKOUT for DFSDMx).

**Table 100. DFSDM measured timing 1.62-3.6 V<sup>(1)</sup>**

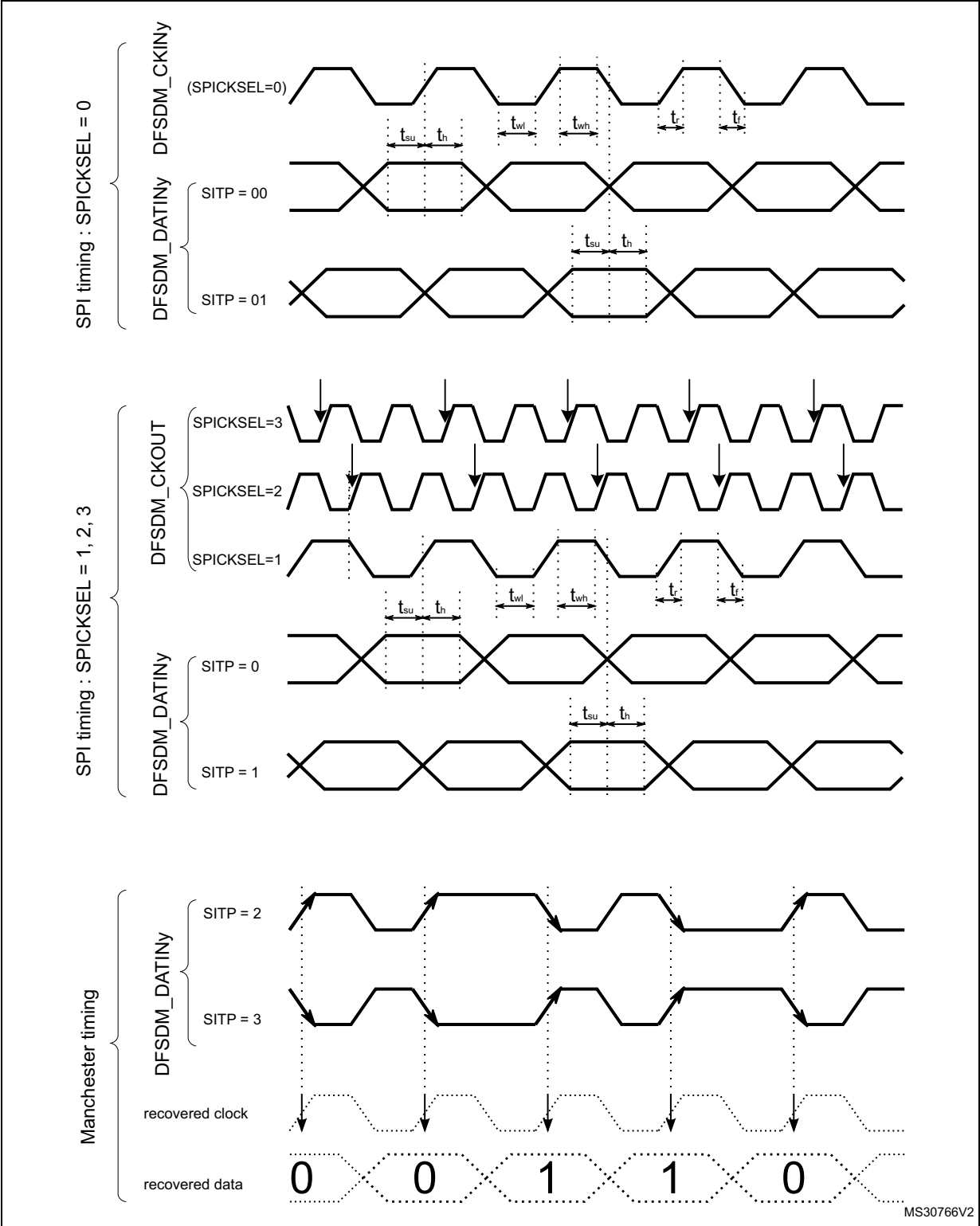
| Symbol                         | Parameter                         | Conditions                                                                                                        | Min | Typ | Max                        | Unit |
|--------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|-----|-----|----------------------------|------|
| $f_{DFSDMCLK}$                 | DFSDM clock                       | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$                                                                           | -   | -   | 133                        | MHz  |
| $f_{CKIN}$<br>( $1/T_{CKIN}$ ) | Input clock frequency             | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | -   | -   | 20<br>( $f_{DFSDMCLK}/4$ ) |      |
|                                |                                   | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$2.7 < V_{DD} < 3.6\text{ V}$           | -   | -   | 20<br>( $f_{DFSDMCLK}/4$ ) |      |
|                                |                                   | SPI mode (SITP[1:0]=0,1),<br>Internal clock mode<br>(SPICKSEL[1:0]≠0),<br>$1.62 < V_{DD} < 3.6\text{ V}$          | -   | -   | 20<br>( $f_{DFSDMCLK}/4$ ) |      |
|                                |                                   | SPI mode (SITP[1:0]=0,1),<br>Internal clock mode<br>(SPICKSEL[1:0]≠0),<br>$2.7 < V_{DD} < 3.6\text{ V}$           | -   | -   | 20<br>( $f_{DFSDMCLK}/4$ ) |      |
| $f_{CKOUT}$                    | Output clock frequency            | $1.62 < V_{DD} < 3.6\text{ V}$                                                                                    | -   | -   | 20                         |      |
| DuCy <sub>CKOUT</sub>          | Output clock frequency duty cycle | $1.62 < V_{DD} < 3.6\text{ V}$                                                                                    | 45  | 50  | 55                         | %    |

Table 100. DFSDM measured timing 1.62-3.6 V<sup>(1)</sup> (continued)

| Symbol                           | Parameter                                       | Conditions                                                                                                   | Min                           | Typ          | Max                           | Unit |
|----------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|-------------------------------|------|
| $t_{wh(CKIN)}$<br>$t_{wl(CKIN)}$ | Input clock high and low time                   | SPI mode (SITP[1:0]=0,1),<br>External clock mode (SPICKSEL[1:0]=0),<br>1.62 < V <sub>DD</sub> < 3.6 V        | $T_{CKIN}/2 - 0.5$            | $T_{CKIN}/2$ | -                             | ns   |
| $t_{su}$                         | Data input setup time                           | SPI mode (SITP[1:0]=0,1),<br>External clock mode (SPICKSEL[1:0]=0),<br>1.62 < V <sub>DD</sub> < 3.6 V        | 4                             | -            | -                             |      |
| $t_h$                            | Data input hold time                            | SPI mode (SITP[1:0]=0,1),<br>External clock mode (SPICKSEL[1:0]=0),<br>1.62 < V <sub>DD</sub> < 3.6 V        | 0.5                           | -            | -                             |      |
| $T_{Manchester}$                 | Manchester data period (recovered clock period) | Manchester mode (SITP[1:0]=2,3),<br>Internal clock mode (SPICKSEL[1:0]≠0),<br>1.62 < V <sub>DD</sub> < 3.6 V | $(CKOUTDIV+1) * T_{DFSDMCLK}$ | -            | $(2*CKOUTDIV) * T_{DFSDMCLK}$ |      |

1. Guaranteed by characterization results.

Figure 45. Channel transceiver timing diagrams



### 6.3.29 Camera interface (DCMI) timing specifications

Unless otherwise specified, the parameters given in [Table 101](#) for DCMI are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage summarized in [Table 24: General operating conditions](#), with the following configuration:

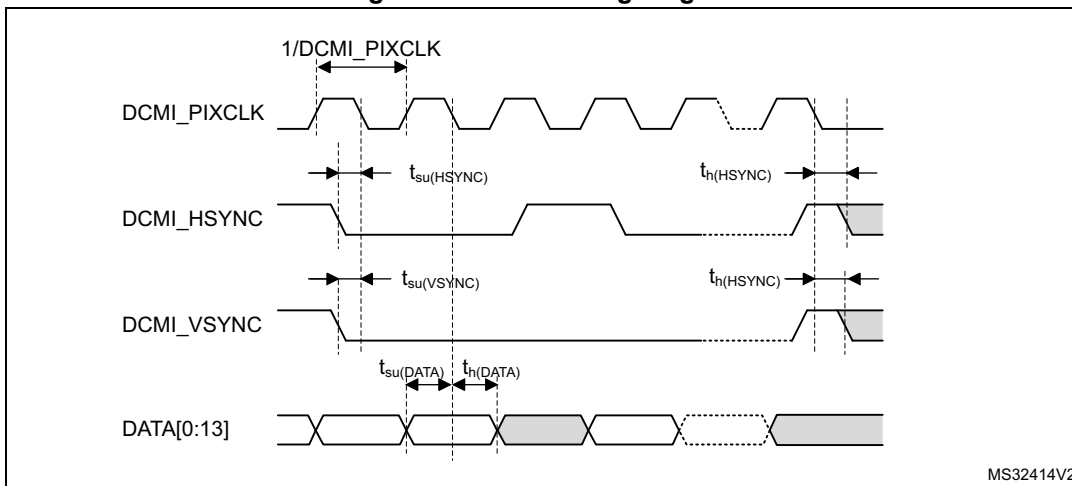
- DCMI\_PIXCLK polarity: falling
- DCMI\_VSYNC and DCMI\_HSYNC polarity: high
- Data formats: 14 bits
- Capacitive load  $C=30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$

**Table 101. DCMI characteristics<sup>(1)</sup>**

| Symbol                                                         | Parameter                                            | Min | Max | Unit |
|----------------------------------------------------------------|------------------------------------------------------|-----|-----|------|
| -                                                              | Frequency ratio DCMI_PIXCLK/ $f_{\text{rcc\_c\_ck}}$ | -   | 0.4 | -    |
| DCMI_PIXCLK                                                    | Pixel clock input                                    | -   | 80  | MHz  |
| $D_{\text{Pixel}}$                                             | Pixel clock input duty cycle                         | 30  | 70  | %    |
| $t_{\text{su}}(\text{DATA})$                                   | Data input setup time                                | 1   | -   | ns   |
| $t_{\text{h}}(\text{DATA})$                                    | Data input hold time                                 | 1   | -   |      |
| $t_{\text{su}}(\text{HSYNC})$<br>$t_{\text{su}}(\text{VSYNC})$ | DCMI_HSYNC/DCMI_VSYNC input setup time               | 1.5 | -   |      |
| $t_{\text{h}}(\text{HSYNC})$<br>$t_{\text{h}}(\text{VSYNC})$   | DCMI_HSYNC/DCMI_VSYNC input hold time                | 1   | -   |      |

1. Guaranteed by characterization results.

**Figure 46. DCMI timing diagram**



### 6.3.30 LCD-TFT controller (LTDC) characteristics

Unless otherwise specified, the parameters given in [Table 102](#) for LCD-TFT are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage summarized in [Table 24: General operating conditions](#), with the following configuration:

- LCD\_CLK polarity: high
- LCD\_DE polarity: low
- LCD\_VSYNC and LCD\_HSYNC polarity: high
- Pixel formats: 24 bits
- Output speed is set to OSPEEDRy[1:0] = 11
- Capacitive load C=30 pF
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$
- I/O compensation cell enabled

**Table 102. LTDC characteristics <sup>(1)</sup>**

| Symbol                                                                                        | Parameter                        | Conditions                                               | Min                              | Max                              | Unit |
|-----------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------|----------------------------------|----------------------------------|------|
| $f_{\text{CLK}}$                                                                              | LTDC clock output frequency      | $2.7\text{ V} < V_{\text{DD}} < 3.6\text{ V}$ ,<br>20 pF | -                                | 150                              | MHz  |
|                                                                                               |                                  | $2.7\text{ V} < V_{\text{DD}} < 3.6\text{ V}$            | -                                | 133                              |      |
|                                                                                               |                                  | $1.62\text{ V} < V_{\text{DD}} < 3.6\text{ V}$           | -                                | 90                               |      |
| $D_{\text{CLK}}$                                                                              | LTDC clock output duty cycle     | -                                                        | 45                               | 55                               | %    |
| $t_{\text{w}}(\text{CLKH})$ ,<br>$t_{\text{w}}(\text{CLKL})$                                  | Clock High time, low time        |                                                          | $t_{\text{w}}(\text{CLK})/2-0.5$ | $t_{\text{w}}(\text{CLK})/2+0.5$ | ns   |
| $t_{\text{v}}(\text{DATA})$                                                                   | Data output valid time           |                                                          | -                                | 0.5                              |      |
| $t_{\text{h}}(\text{DATA})$                                                                   | Data output hold time            |                                                          | 0                                | -                                |      |
| $t_{\text{v}}(\text{HSYNC})$ ,<br>$t_{\text{v}}(\text{VSYNC})$ ,<br>$t_{\text{v}}(\text{DE})$ | HSYNC/VSYNC/DE output valid time |                                                          | -                                | 0.5                              |      |
| $t_{\text{h}}(\text{HSYNC})$ ,<br>$t_{\text{h}}(\text{VSYNC})$ ,<br>$t_{\text{h}}(\text{DE})$ | HSYNC/VSYNC/DE output hold time  |                                                          | 0.5                              | -                                |      |

1. Guaranteed by characterization results.

Figure 47. LCD-TFT horizontal timing diagram

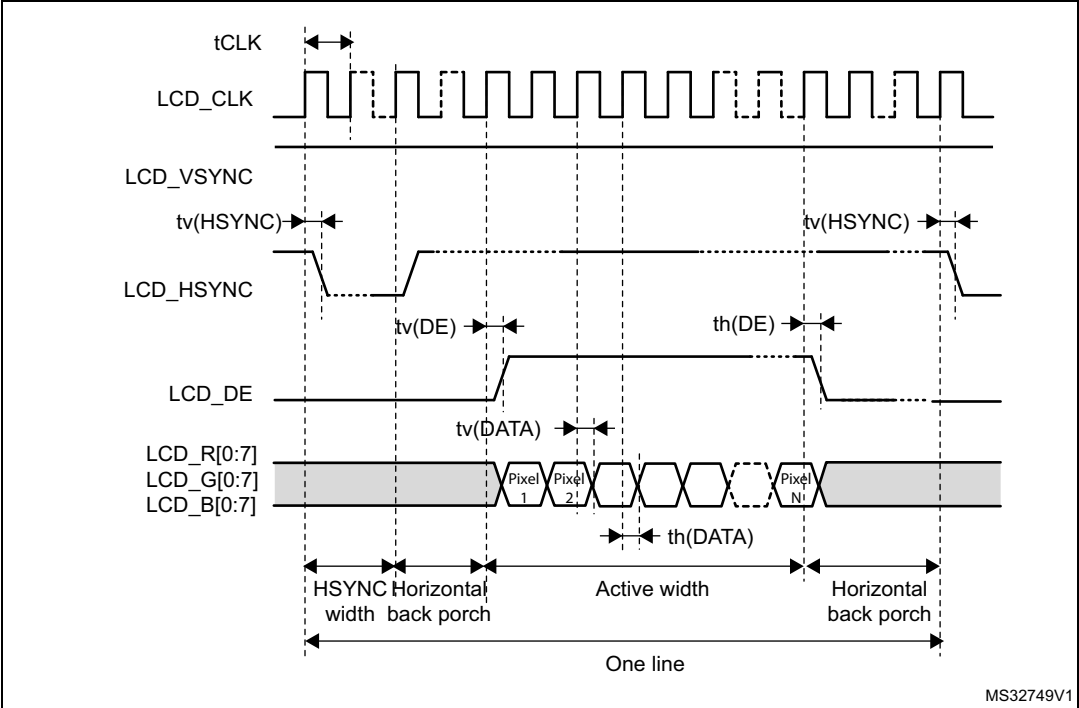
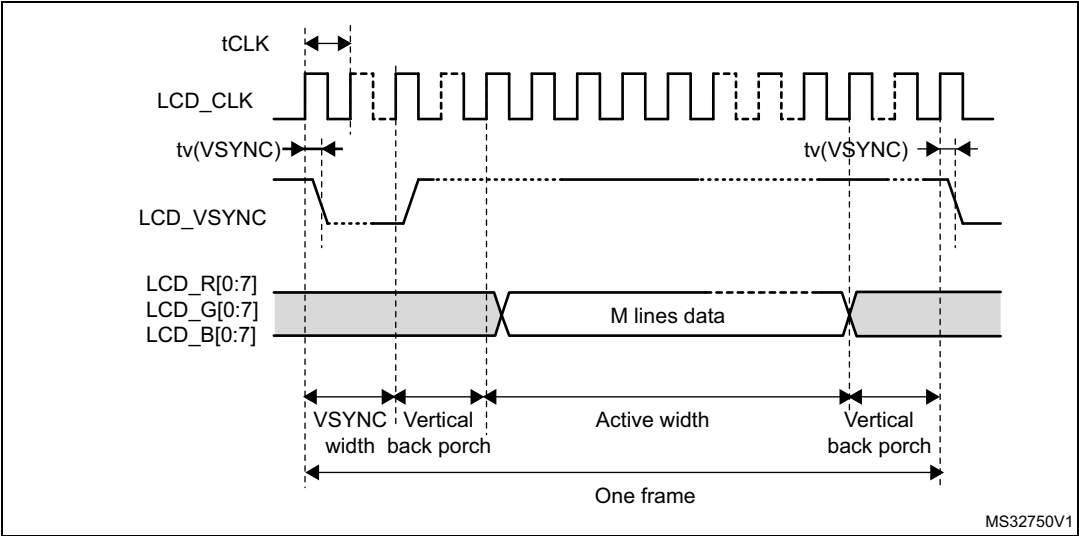


Figure 48. LCD-TFT vertical timing diagram





### 6.3.31 Timer characteristics

The parameters given in [Table 103](#) are guaranteed by design.

Refer to [Section 6.3.15: I/O port characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

**Table 103. TIMx characteristics<sup>(1)(2)</sup>**

| Symbol                    | Parameter                                    | Conditions <sup>(3)</sup>                                              | Min | Max                    | Unit                 |
|---------------------------|----------------------------------------------|------------------------------------------------------------------------|-----|------------------------|----------------------|
| $t_{\text{res(TIM)}}$     | Timer resolution time                        | AHB/APBx prescaler=1 or 2 or 4, $f_{\text{TIMxCLK}} = 200 \text{ MHz}$ | 1   | -                      | $t_{\text{TIMxCLK}}$ |
|                           |                                              | AHB/APBx prescaler>4, $f_{\text{TIMxCLK}} = 100 \text{ MHz}$           | 1   | -                      | $t_{\text{TIMxCLK}}$ |
| $f_{\text{EXT}}$          | Timer external clock frequency on CH1 to CH4 | $f_{\text{TIMxCLK}} = 200 \text{ MHz}$                                 | 0   | $f_{\text{TIMxCLK}}/2$ | MHz                  |
| $\text{Res}_{\text{TIM}}$ | Timer resolution                             |                                                                        | -   | 16/32                  | bit                  |
| $t_{\text{MAX\_COUNT}}$   | Maximum possible count with 32-bit counter   | -                                                                      | -   | $65536 \times 65536$   | $t_{\text{TIMxCLK}}$ |

1. TIMx is used as a general term to refer to the TIM1 to TIM17 timers.
2. Guaranteed by design.
3. The maximum timer frequency on APB1 or APB2 is up to 200 MHz, by setting the TIMPRE bit in the RCC\_CFGR register, if APBx prescaler is 1 or 2 or 4, then  $\text{TIMxCLK} = \text{rcc\_hclk1}$ , otherwise  $\text{TIMxCLK} = 4 \times F_{\text{rcc\_pclkx\_d2}}$ .

### 6.3.32 Communications interfaces

#### I<sup>2</sup>C interface characteristics

The I<sup>2</sup>C interface meets the timings requirements of the I<sup>2</sup>C-bus specification and user manual revision 03 for:

- Standard-mode (Sm): with a bit rate up to 100 kbit/s
- Fast-mode (Fm): with a bit rate up to 400 kbit/s.
- Fast-mode Plus (Fm+): with a bit rate up to 1Mbit/s.

The I<sup>2</sup>C timings requirements are guaranteed by design when the I2C peripheral is properly configured (refer to RM0433 reference manual) and when the i2c\_ker\_ck frequency is greater than the minimum shown in the table below:

**Table 104. Minimum i2c\_ker\_ck frequency in all I<sup>2</sup>C modes**

| Symbol    | Parameter        | Condition      |                            | Min | Unit |
|-----------|------------------|----------------|----------------------------|-----|------|
| f(I2CCLK) | I2CCLK frequency | Standard-mode  |                            | 2   | MHz  |
|           |                  | Fast-mode      | Analog filter ON<br>DNF=0  | 8   |      |
|           |                  |                | Analog filter OFF<br>DNF=1 | 9   |      |
|           |                  | Fast-mode Plus | Analog filter ON<br>DNF=0  | 17  |      |
|           |                  |                | Analog filter OFF<br>DNF=1 | 16  |      |

The SDA and SCL I/O requirements are met with the following restrictions:

- The SDA and SCL I/O pins are not “true” open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V<sub>DD</sub> is disabled, but still present.
- The 20 mA output drive requirement in Fast-mode Plus is not supported. This limits the maximum load C<sub>load</sub> supported in Fm+, which is given by these formulas:

$$t_{r(SDA/SCL)} = 0.8473 \times R_p \times C_{load}$$

$$R_{p(min)} = (V_{DD} - V_{OL(max)}) / I_{OL(max)}$$

Where R<sub>p</sub> is the I2C lines pull-up. Refer to [Section 6.3.15: I/O port characteristics](#) for the I2C I/Os characteristics.

All I<sup>2</sup>C SDA and SCL I/Os embed an analog filter. Refer to [Table 105](#) for the analog filter characteristics:

**Table 105. I2C analog filter characteristics<sup>(1)</sup>**

| Symbol          | Parameter                                                              | Min               | Max                | Unit |
|-----------------|------------------------------------------------------------------------|-------------------|--------------------|------|
| t <sub>AF</sub> | Maximum pulse width of spikes that are suppressed by the analog filter | 50 <sup>(2)</sup> | 260 <sup>(3)</sup> | ns   |

1. Guaranteed by design.
2. Spikes with widths below t<sub>AF(min)</sub> are filtered.
3. Spikes with widths above t<sub>AF(max)</sub> are not filtered.

### SPI interface characteristics

Unless otherwise specified, the parameters given in [Table 106](#) for the SPI interface are derived from tests performed under the ambient temperature,  $f_{\text{PCLKx}}$  frequency and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $\text{OSPEEDRy}[1:0] = 11$
- Capacitive load  $C = 30 \text{ pF}$
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$
- I/O compensation cell enabled
- HSLV activated when  $V_{\text{DD}} \leq 2.7 \text{ V}$

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI).

**Table 106. SPI dynamic characteristics<sup>(1)</sup>**

| Symbol                                         | Parameter             | Conditions                                                                                   | Min                   | Typ               | Max                   | Unit |
|------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------|-----------------------|-------------------|-----------------------|------|
| $f_{\text{SCK}}$<br>$1/t_{\text{c(SCK)}}$      | SPI clock frequency   | Master mode<br>$1.62 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$                        | -                     | -                 | 90                    | MHz  |
|                                                |                       | Master mode<br>$2.7 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$<br>SPI1,2,3             |                       |                   | 133                   |      |
|                                                |                       | Master mode<br>$2.7 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$<br>SPI4,5,6             |                       |                   | 100                   |      |
|                                                |                       | Slave receiver mode<br>$1.62 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$<br>SPI1,2,3    |                       |                   | 150                   |      |
|                                                |                       | Slave receiver mode<br>$1.62 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$<br>SPI4,5,6    |                       |                   | 100                   |      |
|                                                |                       | Slave mode transmitter/full duplex<br>$2.7 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$  |                       |                   | 31                    |      |
|                                                |                       | Slave mode transmitter/full duplex<br>$1.62 \text{ V} \leq V_{\text{DD}} \leq 3.6 \text{ V}$ |                       |                   | 25                    |      |
| $t_{\text{su(NSS)}}$                           | NSS setup time        | Slave mode                                                                                   | 2                     | -                 | -                     | ns   |
| $t_{\text{h(NSS)}}$                            | NSS hold time         |                                                                                              | 1                     | -                 | -                     |      |
| $t_{\text{w(SCKH)}}$ ,<br>$t_{\text{w(SCKL)}}$ | SCK high and low time | Master mode                                                                                  | $T_{\text{PLCK}} - 2$ | $T_{\text{PLCK}}$ | $T_{\text{PLCK}} + 2$ |      |

Table 106. SPI dynamic characteristics<sup>(1)</sup> (continued)

| Symbol        | Parameter                | Conditions                                                | Min | Typ  | Max | Unit |
|---------------|--------------------------|-----------------------------------------------------------|-----|------|-----|------|
| $t_{su(MI)}$  | Data input setup time    | Master mode                                               | 1   | -    | -   | ns   |
| $t_{su(SI)}$  |                          | Slave mode                                                | 2   | -    | -   |      |
| $t_{h(MI)}$   | Data input hold time     | Master mode                                               | 2   | -    | -   |      |
| $t_{h(SI)}$   |                          | Slave mode                                                | 1   | -    | -   |      |
| $t_{a(SO)}$   | Data output access time  | Slave mode                                                | 9   | 13   | 27  |      |
| $t_{dis(SO)}$ | Data output disable time | Slave mode                                                | 0   | 1    | 5   |      |
| $t_{v(SO)}$   | Data output valid time   | Slave mode, $2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -   | 11.5 | 16  |      |
|               |                          | Slave mode $1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -   | 13   | 20  |      |
| $t_{v(MO)}$   |                          | Master mode                                               | -   | 1    | 3   |      |
| $t_{h(SO)}$   | Data output hold time    | Slave mode, $1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ | 9   | -    | -   |      |
| $t_{h(MO)}$   |                          | Master mode                                               | 0   | -    | -   |      |

1. Guaranteed by characterization results.

Figure 49. SPI timing diagram - slave mode and CPHA = 0

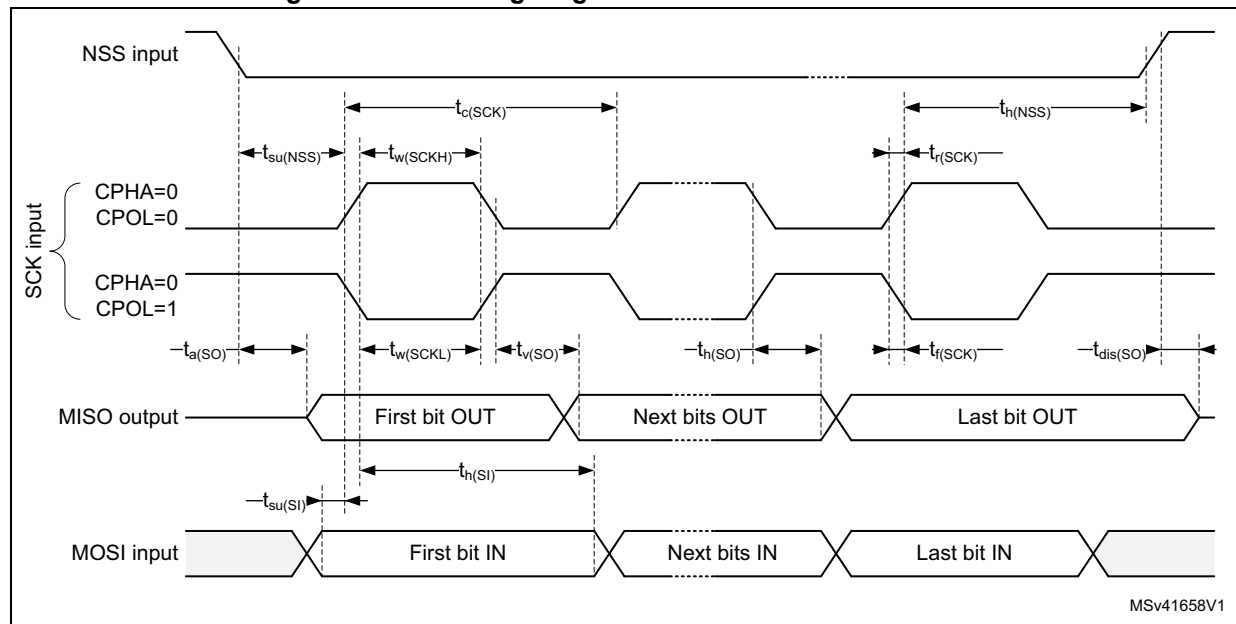
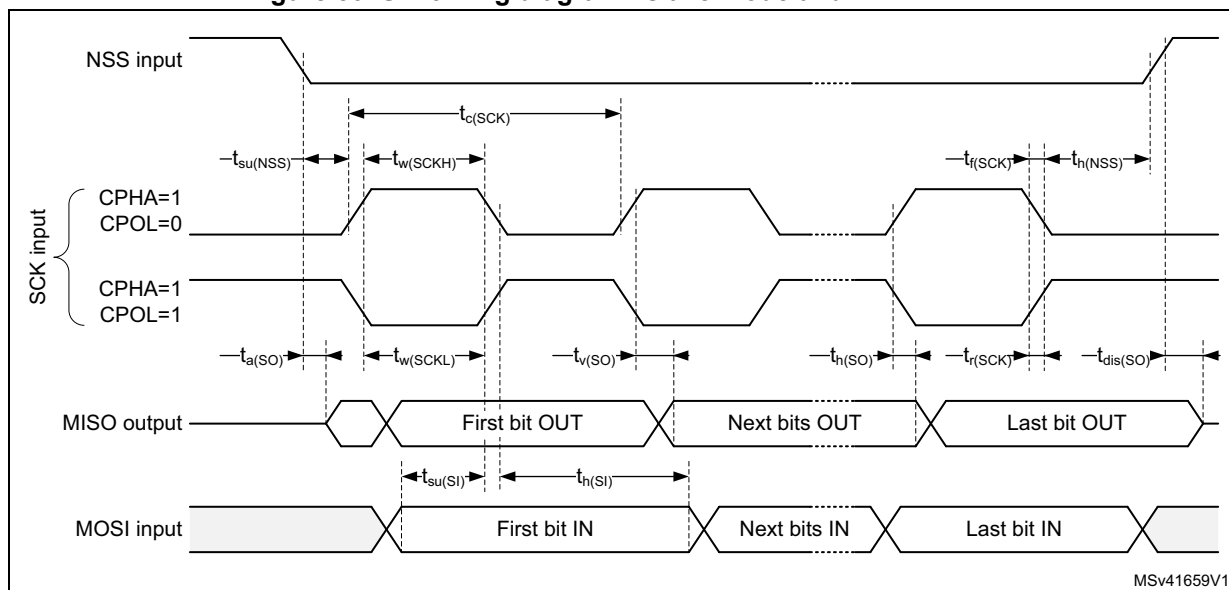
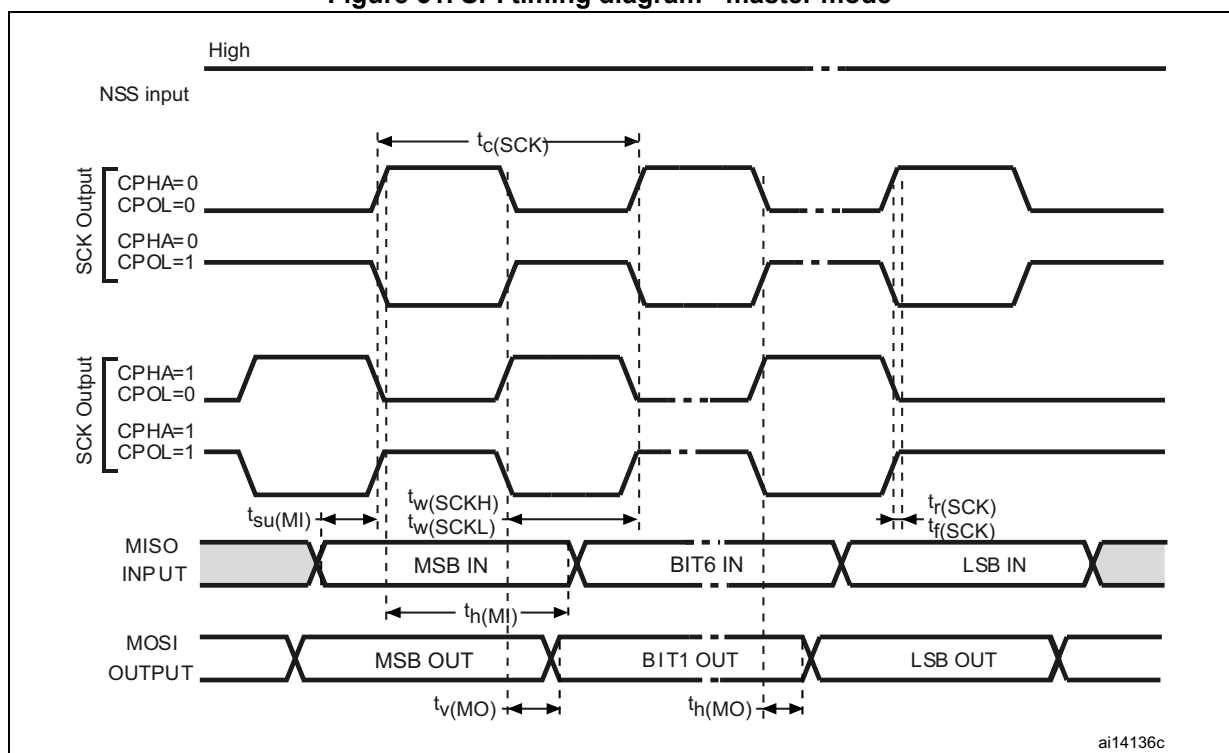


Figure 50. SPI timing diagram - slave mode and CPHA = 1<sup>(1)</sup>

1. Measurement points are done at  $0.5V_{DD}$  and with external  $C_L = 30$  pF.

Figure 51. SPI timing diagram - master mode<sup>(1)</sup>

1. Measurement points are done at  $0.5V_{DD}$  and with external  $C_L = 30$  pF.

## I<sup>2</sup>S interface characteristics

Unless otherwise specified, the parameters given in [Table 107](#) for the I<sup>2</sup>S interface are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

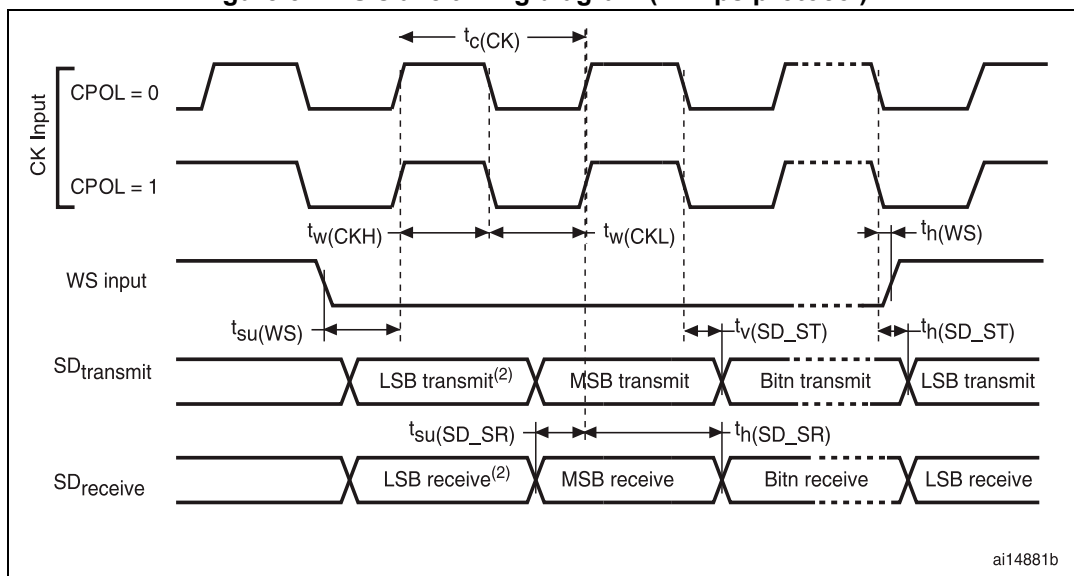
- Output speed is set to  $OSPEEDRy[1:0] = 10$
- Capacitive load  $C = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- I/O compensation cell enabled

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (CK, SD, WS).

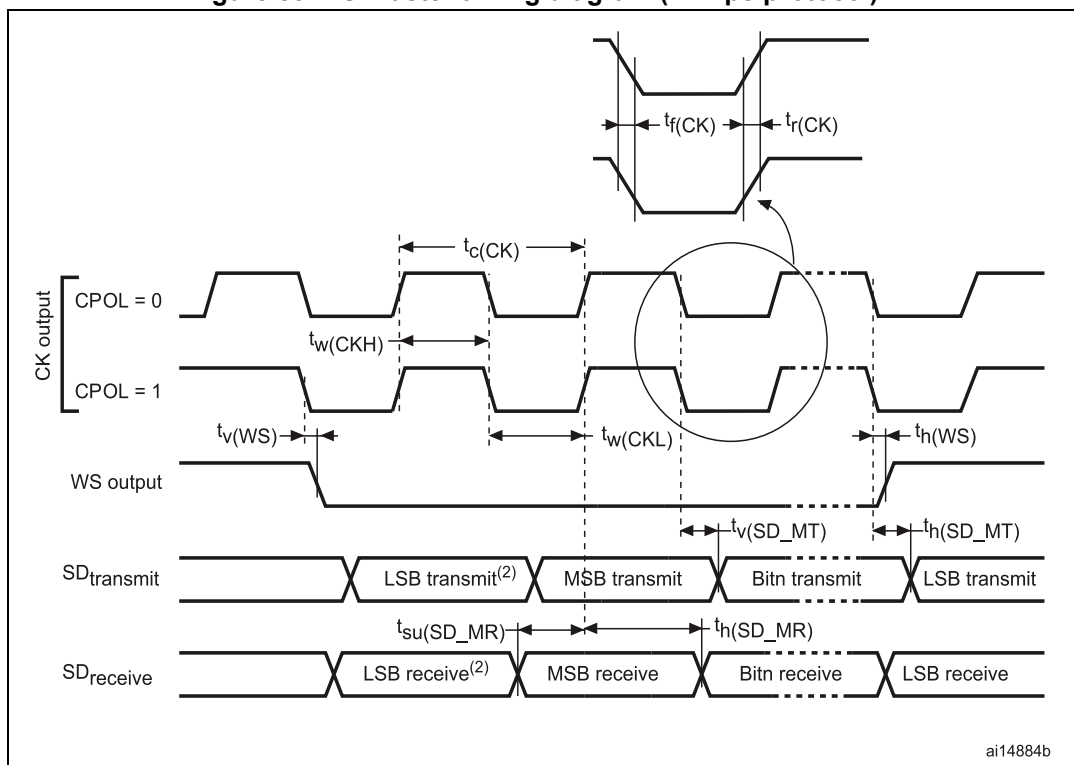
**Table 107. I<sup>2</sup>S dynamic characteristics<sup>(1)</sup>**

| Symbol           | Parameter              | Conditions                             | Min    | Max      | Unit |
|------------------|------------------------|----------------------------------------|--------|----------|------|
| $f_{MCK}$        | I2S Main clock output  | -                                      | 256x8K | $256F_S$ | MHz  |
| $f_{CK}$         | I2S clock frequency    | Master data                            | -      | $64F_S$  | MHz  |
|                  |                        | Slave data                             | -      | $64F_S$  |      |
| $t_{v(WS)}$      | WS valid time          | Master mode                            | -      | 3.5      | ns   |
| $t_{h(WS)}$      | WS hold time           | Master mode                            | 0      | -        |      |
| $t_{su(WS)}$     | WS setup time          | Slave mode                             | 1      | -        |      |
| $t_{h(WS)}$      | WS hold time           | Slave mode                             | 1      | -        |      |
| $t_{su(SD\_MR)}$ | Data input setup time  | Master receiver                        | 1      | -        |      |
| $t_{su(SD\_SR)}$ |                        | Slave receiver                         | 1      | -        |      |
| $t_{h(SD\_MR)}$  | Data input hold time   | Master receiver                        | 4      | -        |      |
| $t_{h(SD\_SR)}$  |                        | Slave receiver                         | 2      | -        |      |
| $t_{v(SD\_ST)}$  | Data output valid time | Slave transmitter (after enable edge)  | -      | 20       |      |
| $t_{v(SD\_MT)}$  |                        | Master transmitter (after enable edge) | -      | 3        |      |
| $t_{h(SD\_ST)}$  | Data output hold time  | Slave transmitter (after enable edge)  | 9      | -        |      |
| $t_{h(SD\_MT)}$  |                        | Master transmitter (after enable edge) | 0      | -        |      |

1. Guaranteed by characterization results.

Figure 52. I<sup>2</sup>S slave timing diagram (Philips protocol)<sup>(1)</sup>

1. LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

Figure 53. I<sup>2</sup>S master timing diagram (Philips protocol)<sup>(1)</sup>

1. LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

### SAI characteristics

Unless otherwise specified, the parameters given in [Table 108](#) for SAI are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and VDD supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load C=30 pF
- Measurement points are performed at CMOS levels:  $0.5V_{DD}$

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (SCK,SD,WS).

**Table 108. SAI characteristics<sup>(1)</sup>**

| Symbol              | Parameter                          | Conditions                                                             | Min      | Max                   | Unit |
|---------------------|------------------------------------|------------------------------------------------------------------------|----------|-----------------------|------|
| $f_{MCK}$           | SAI Main clock output              | -                                                                      | 256 x 8K | 256xFs                | MHz  |
| $F_{CK}$            | SAI clock frequency <sup>(2)</sup> | Master data: 32 bits                                                   | -        | 128xFs <sup>(3)</sup> | MHz  |
|                     |                                    | Slave data: 32 bits                                                    | -        | 128xFs                |      |
| $t_{v(FS)}$         | FS valid time                      | Master mode<br>$2.7 \leq V_{DD} \leq 3.6V$                             | -        | 15                    | ns   |
|                     |                                    | Master mode<br>$1.71 \leq V_{DD} \leq 3.6V$                            | -        | 20                    |      |
| $t_{su(FS)}$        | FS setup time                      | Slave mode                                                             | 7        | -                     |      |
| $t_{h(FS)}$         | FS hold time                       | Master mode                                                            | 1        | -                     |      |
|                     |                                    | Slave mode                                                             | 1        | -                     |      |
| $t_{su(SD\_A\_MR)}$ | Data input setup time              | Master receiver                                                        | 0.5      | -                     |      |
| $t_{su(SD\_B\_SR)}$ |                                    | Slave receiver                                                         | 1        | -                     |      |
| $t_{h(SD\_A\_MR)}$  | Data input hold time               | Master receiver                                                        | 3.5      | -                     |      |
| $t_{h(SD\_B\_SR)}$  |                                    | Slave receiver                                                         | 2        | -                     |      |
| $t_{v(SD\_B\_ST)}$  | Data output valid time             | Slave transmitter (after enable edge)<br>$2.7 \leq V_{DD} \leq 3.6V$   | -        | 17                    | ns   |
|                     |                                    | Slave transmitter (after enable edge)<br>$1.62 \leq V_{DD} \leq 3.6V$  | -        | 20                    |      |
| $t_{h(SD\_B\_ST)}$  | Data output hold time              | Slave transmitter (after enable edge)                                  | 7        | -                     |      |
| $t_{v(SD\_A\_MT)}$  | Data output valid time             | Master transmitter (after enable edge)<br>$2.7 \leq V_{DD} \leq 3.6V$  | -        | 17                    |      |
|                     |                                    | Master transmitter (after enable edge)<br>$1.62 \leq V_{DD} \leq 3.6V$ | -        | 20                    |      |
| $t_{h(SD\_A\_MT)}$  | Data output hold time              | Master transmitter (after enable edge)                                 | 7.55     | -                     |      |

1. Guaranteed by characterization results.

2. APB clock frequency must be at least twice SAI clock frequency.

3. With  $F_S=192$  kHz.



Figure 54. SAI master timing waveforms

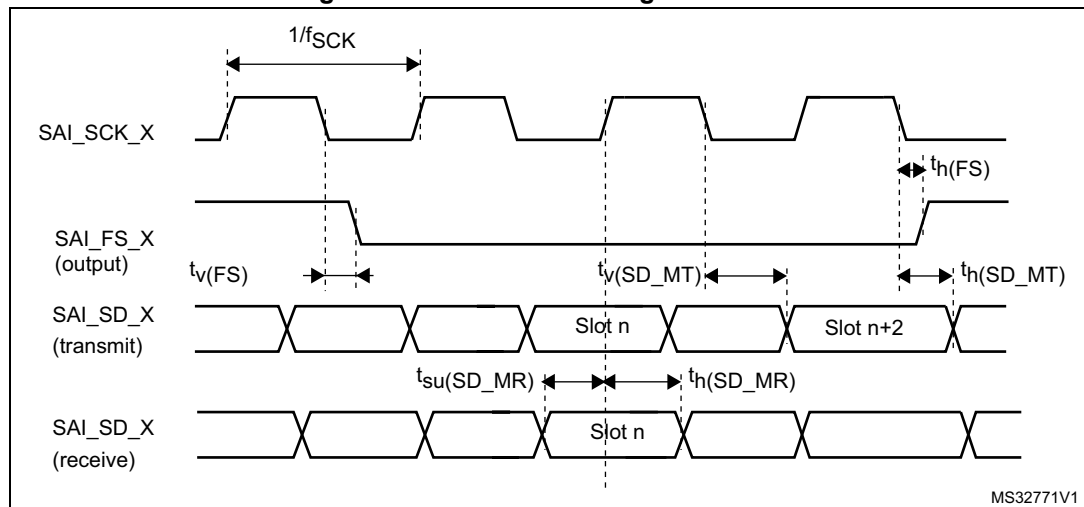
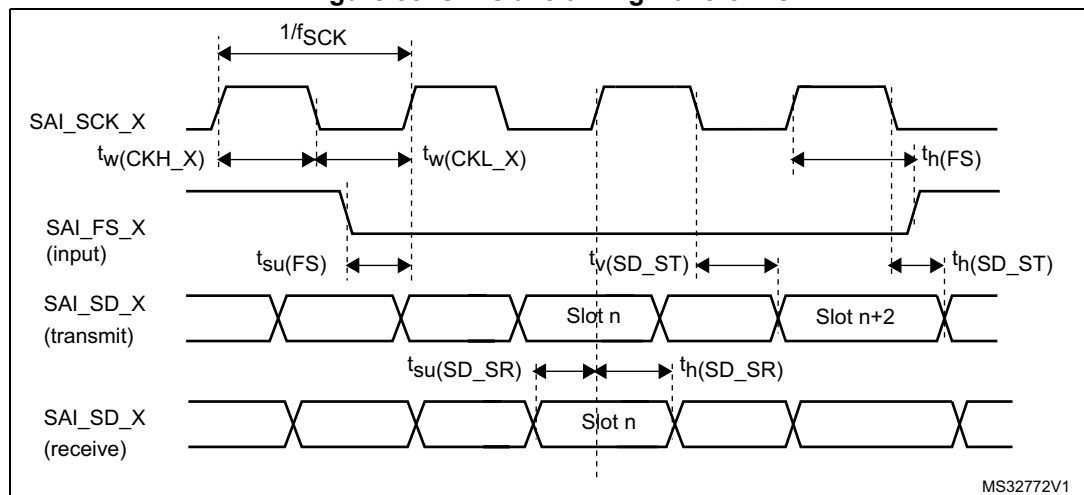


Figure 55. SAI slave timing waveforms



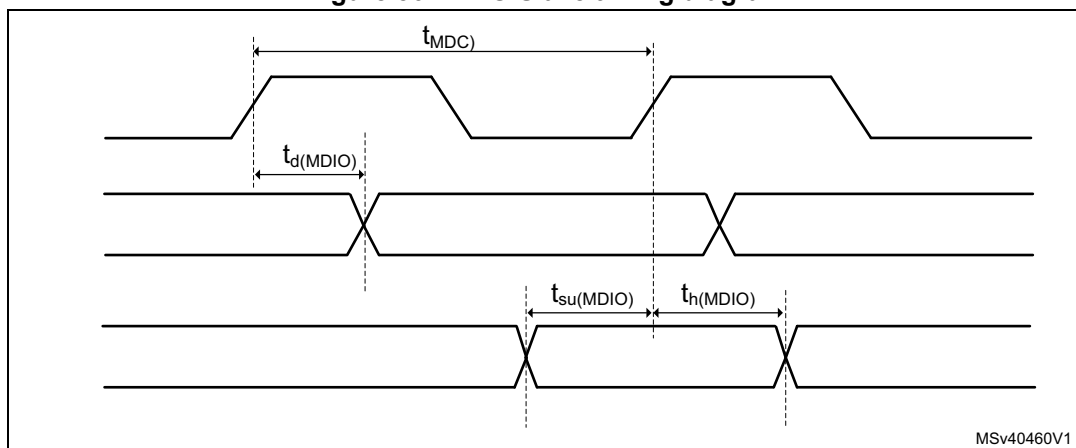
## MDIO characteristics

Table 109. MDIO Slave timing parameters

| Symbol         | Parameter                                             | Min | Typ | Max | Unit |
|----------------|-------------------------------------------------------|-----|-----|-----|------|
| $F_{sDC}$      | Management data clock                                 | -   | -   | 40  | MHz  |
| $t_{d(MDIO)}$  | Management data input/output <b>output valid time</b> | 7   | 8   | 20  | ns   |
| $t_{su(MDIO)}$ | Management data input/output <b>setup time</b>        | 4   | -   | -   |      |
| $t_{h(MDIO)}$  | Management data input/output <b>hold time</b>         | 1   | -   | -   |      |

The MDIO controller is mapped on APB2 domain. The frequency of the APB bus should at least 1.5 times the MDC frequency:  $F_{PCLK2} \geq 1.5 * F_{MDC}$ .

Figure 56. MDIO Slave timing diagram



### SD/SDIO MMC card host interface (SDMMC) characteristics

Unless otherwise specified, the parameters given in [Table 110](#) for the SDIO/MMC interface are derived from tests performed under the ambient temperature,  $f_{PCLK2}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 11$
- Capacitive load  $C = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- I/O compensation cell enabled
- HSLV activated when  $V_{DD} \leq 2.7$  V

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

**Table 110. Dynamic characteristics: SD / MMC characteristics,  $V_{DD}=2.7V$  to  $3.6V^{(1)(2)}$**

| Symbol                                                          | Parameter                             | Conditions               | Min | Typ  | Max | Unit |
|-----------------------------------------------------------------|---------------------------------------|--------------------------|-----|------|-----|------|
| f <sub>PP</sub>                                                 | Clock frequency in data transfer mode | -                        | 0   | -    | 125 | MHz  |
| t <sub>W(CKL)</sub>                                             | Clock low time                        | f <sub>PP</sub> =50 MHz  | 9.5 | 10.5 | -   | ns   |
| t <sub>W(CKH)</sub>                                             | Clock high time                       |                          | 8.5 | 9.5  | -   |      |
| CMD, D inputs (referenced to CK) in MMC and SD HS/SDR/DDR mode  |                                       |                          |     |      |     |      |
| t <sub>ISU</sub>                                                | Input setup time HS                   | f <sub>PP</sub> ≥ 50 MHz | 2   | -    | -   | ns   |
| t <sub>IH</sub>                                                 | Input hold time HS                    |                          | 1.5 | -    | -   |      |
| t <sub>IDW</sub> <sup>(3)</sup>                                 | Input valid window (variable window)  |                          | 3   | -    | -   |      |
| CMD, D outputs (referenced to CK) in MMC and SD HS/SDR/DDR mode |                                       |                          |     |      |     |      |
| t <sub>OV</sub>                                                 | Output valid time HS                  | f <sub>PP</sub> ≥ 50 MHz | -   | 3.5  | 5   | ns   |
| t <sub>OH</sub>                                                 | Output hold time HS                   |                          | 2   | -    | -   |      |

**Table 110. Dynamic characteristics: SD / MMC characteristics,  $V_{DD}=2.7V$  to  $3.6V^{(1)(2)}$** 

| Symbol                                               | Parameter                    | Conditions              | Min | Typ | Max | Unit |
|------------------------------------------------------|------------------------------|-------------------------|-----|-----|-----|------|
| CMD, D inputs (referenced to CK) in SD default mode  |                              |                         |     |     |     |      |
| t <sub>ISUD</sub>                                    | Input setup time SD          | f <sub>PP</sub> =25 MHz | 2   | -   | -   | ns   |
| t <sub>IHD</sub>                                     | Input hold time SD           |                         | 1.5 | -   | -   |      |
| CMD, D outputs (referenced to CK) in SD default mode |                              |                         |     |     |     |      |
| t <sub>OVD</sub>                                     | Output valid default time SD | f <sub>PP</sub> =25 MHz | -   | 1   | 2   | ns   |
| t <sub>OHD</sub>                                     | Output hold default time SD  |                         | 0   | -   | -   |      |

1. Guaranteed by characterization results.
2. Above 100 MHz,  $C_L = 20 \text{ pF}$ .
3. The minimum window of time where the data needs to be stable for proper sampling in tuning mode.

**Table 111. Dynamic characteristics: eMMC characteristics,  $V_{DD}=1.71V$  to  $1.9V^{(1)(2)}$** 

| Symbol                                         | Parameter                             | Conditions               | Min | Typ  | Max | Unit |
|------------------------------------------------|---------------------------------------|--------------------------|-----|------|-----|------|
| f <sub>PP</sub>                                | Clock frequency in data transfer mode | -                        | 0   | -    | 120 | MHz  |
| t <sub>W(CKL)</sub>                            | Clock low time                        | f <sub>PP</sub> =50 MHz  | 9.5 | 10.5 | -   | ns   |
| t <sub>W(CKH)</sub>                            | Clock high time                       |                          | 8.5 | 9.5  | -   |      |
| CMD, D inputs (referenced to CK) in eMMC mode  |                                       |                          |     |      |     |      |
| t <sub>ISU</sub>                               | Input setup time HS                   | f <sub>PP</sub> ≥ 50 MHz | 1.5 | -    | -   | ns   |
| t <sub>IH</sub>                                | Input hold time HS                    |                          | 2   | -    | -   |      |
| t <sub>IDW</sub> <sup>(3)</sup>                | Input valid window (variable window)  |                          | 3.5 | -    | -   |      |
| CMD, D outputs (referenced to CK) in eMMC mode |                                       |                          |     |      |     |      |
| t <sub>OV</sub>                                | Output valid time HS                  | f <sub>PP</sub> ≥ 50 MHz | -   | 5    | 7   | ns   |
| t <sub>OH</sub>                                | Output hold time HS                   |                          | 3   | -    | -   |      |

1. Guaranteed by characterization results.
2.  $C_L = 20 \text{ pF}$ .
3. The minimum window of time where the data needs to be stable for proper sampling in tuning mode.

Figure 57. SDIO high-speed mode

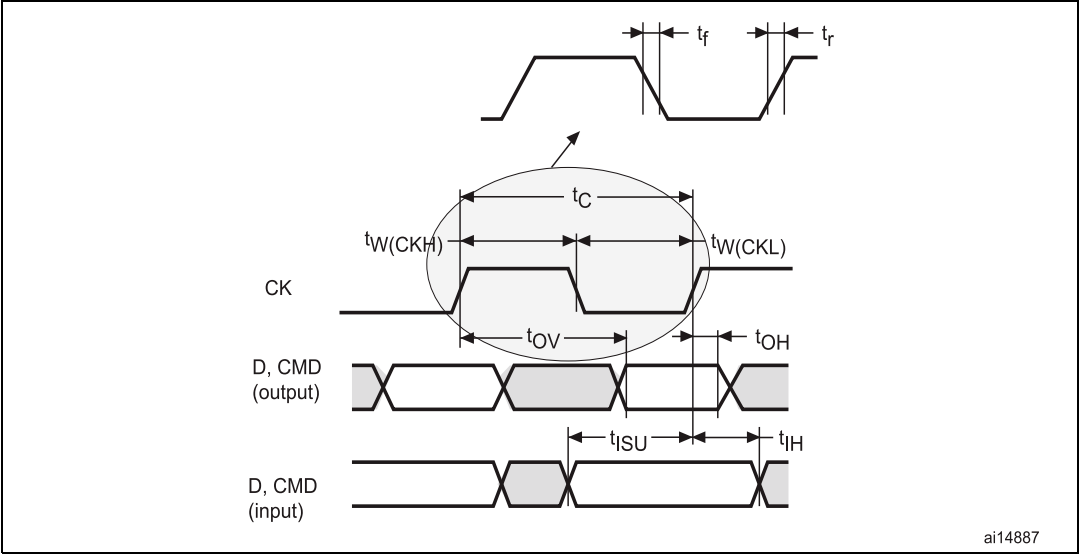


Figure 58. SD default mode

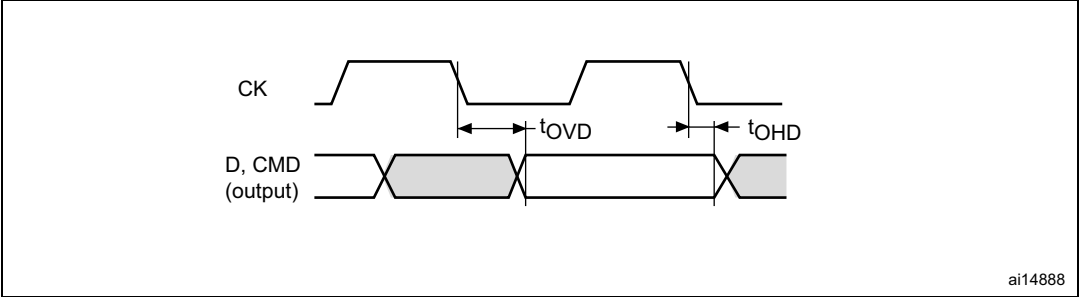
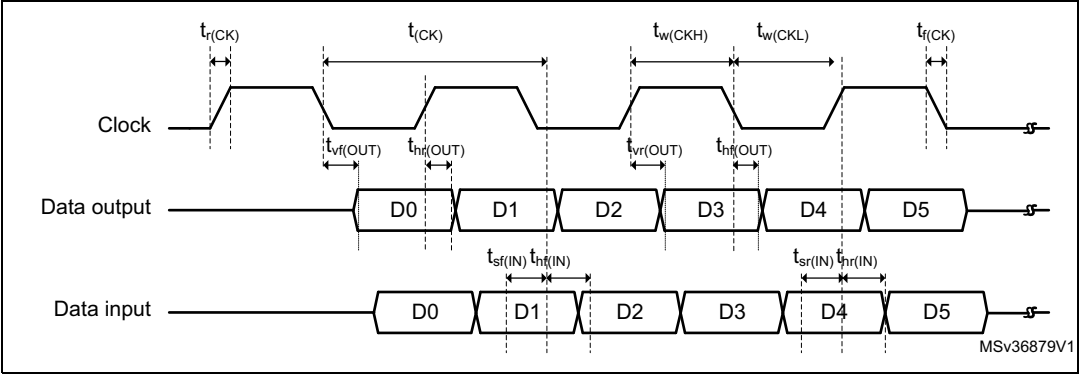


Figure 59. DDR mode



### CAN (controller area network) interface

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (FDCANx\_TX and FDCANx\_RX).

**USB OTG\_FS characteristics**

The USB interface is fully compliant with the USB specification version 2.0 and is USB-IF certified (for Full-speed device operation).

**Table 112. USB OTG\_FS electrical characteristics**

| Symbol       | Parameter                                      | Condition           | Min                | Typ  | Max  | Unit     |
|--------------|------------------------------------------------|---------------------|--------------------|------|------|----------|
| $V_{DD3USB}$ | USB transceiver operating voltage              | -                   | 3.0 <sup>(1)</sup> | -    | 3.6  | V        |
| $R_{PUI}$    | Embedded USB_DP pull-up value during idle      | -                   | 900                | 1250 | 1600 | $\Omega$ |
| $R_{PUR}$    | Embedded USB_DP pull-up value during reception | -                   | 1400               | 2300 | 3200 |          |
| $Z_{DRV}$    | Output driver impedance <sup>(2)</sup>         | Driver high and low | 28                 | 36   | 44   |          |

1. The USB functionality is ensured down to 2.7 V but not the full USB electrical characteristics which are degraded in the 2.7 to 3.0 V voltage range.
2. No external termination series resistors are required on USB\_DP (D+) and USB\_DM (D-); the matching impedance is already included in the embedded driver.

**USB OTG\_HS characteristics**

Unless otherwise specified, the parameters given in [Table 113](#) for ULPI are derived from tests performed under the ambient temperature,  $f_{rcc\_c\_ck}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 11
- Capacitive load  $C = 20$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$ .

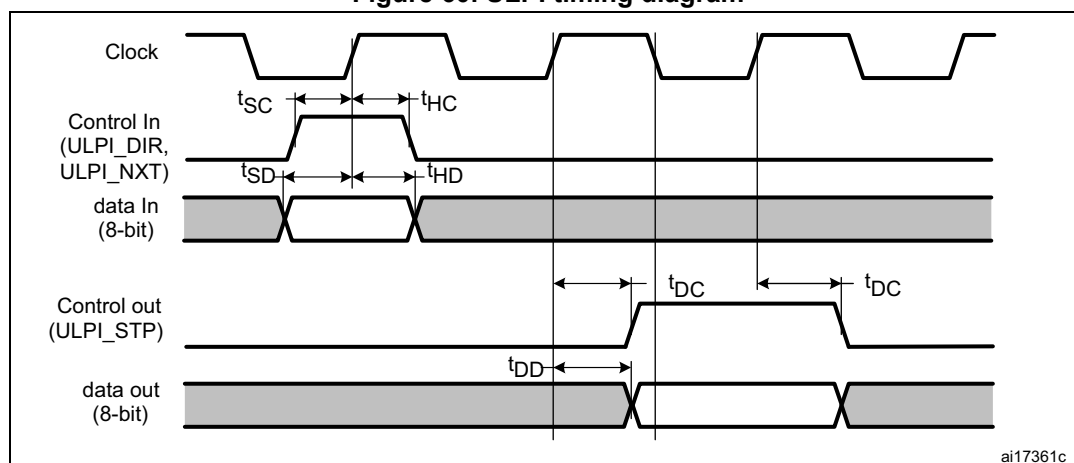
Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

**Table 113. Dynamic characteristics: USB ULPI<sup>(1)</sup>**

| Symbol          | Parameter                                  | Conditions                                                       | Min | Typ | Max | Unit |
|-----------------|--------------------------------------------|------------------------------------------------------------------|-----|-----|-----|------|
| $t_{SC}$        | Control in (ULPI_DIR, ULPI_NXT) setup time | -                                                                | 0.5 | -   | -   | ns   |
| $t_{HC}$        | Control in (ULPI_DIR, ULPI_NXT) hold time  | -                                                                | 6.5 | -   | -   |      |
| $t_{SD}$        | Data in setup time                         | -                                                                | 2.5 | -   | -   |      |
| $t_{HD}$        | Data in hold time                          | -                                                                | 0   | -   | -   |      |
| $t_{DC}/t_{DD}$ | Data/control output delay                  | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$ ,<br>$C_L = 20\text{ pF}$ | -   | 6.5 | 8.5 |      |
|                 |                                            | -                                                                | -   | 6.5 | 13  |      |
|                 |                                            | $1.7\text{ V} < V_{DD} < 3.6\text{ V}$ ,<br>$C_L = 15\text{ pF}$ | -   | -   | -   |      |

1. Guaranteed by characterization results.

Figure 60. ULPI timing diagram



## Ethernet characteristics

Unless otherwise specified, the parameters given in [Table 114](#), [Table 115](#) and [Table 116](#) for SMI, RMII and MII are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $\text{OSPEEDRy}[1:0] = 10$
- Capacitive load  $C = 20 \text{ pF}$
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$ .

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

[Table 114](#) gives the list of Ethernet MAC signals for the SMI and [Figure 61](#) shows the corresponding timing diagram.

**Table 114. Dynamics characteristics: Ethernet MAC signals for SMI<sup>(1)</sup>**

| Symbol                | Parameter               | Min | Typ | Max | Unit |
|-----------------------|-------------------------|-----|-----|-----|------|
| $t_{\text{MDC}}$      | MDC cycle time(2.5 MHz) | 400 | 400 | 403 | ns   |
| $T_{\text{d(MDIO)}}$  | Write data valid time   | 1   | 1.5 | 3   |      |
| $t_{\text{su(MDIO)}}$ | Read data setup time    | 8   | -   | -   |      |
| $t_{\text{h(MDIO)}}$  | Read data hold time     | 0   | -   | -   |      |

1. Guaranteed by characterization results.

Figure 61. Ethernet SMI timing diagram

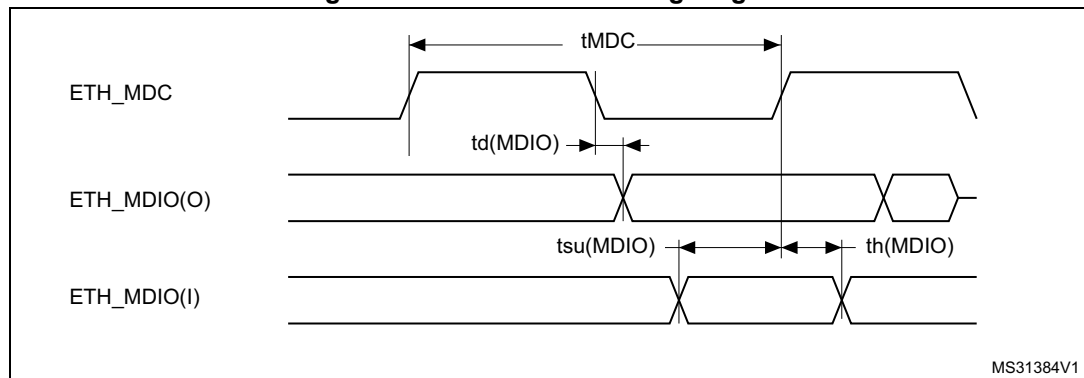


Table 115 gives the list of Ethernet MAC signals for the RMII and Figure 62 shows the corresponding timing diagram.

Table 115. Dynamics characteristics: Ethernet MAC signals for RMII<sup>(1)</sup>

| Symbol        | Parameter                        | Min | Typ | Max  | Unit |
|---------------|----------------------------------|-----|-----|------|------|
| $t_{su}(RXD)$ | Receive data setup time          | 2   | -   | -    | ns   |
| $t_{ih}(RXD)$ | Receive data hold time           | 3   | -   | -    |      |
| $t_{su}(CRS)$ | Carrier sense setup time         | 2.5 | -   | -    |      |
| $t_{ih}(CRS)$ | Carrier sense hold time          | 2   | -   | -    |      |
| $t_d(TXEN)$   | Transmit enable valid delay time | 4   | 4.5 | 7    |      |
| $t_d(TXD)$    | Transmit data valid delay time   | 7   | 7.5 | 11.5 |      |

1. Guaranteed by characterization results.

Figure 62. Ethernet RMII timing diagram

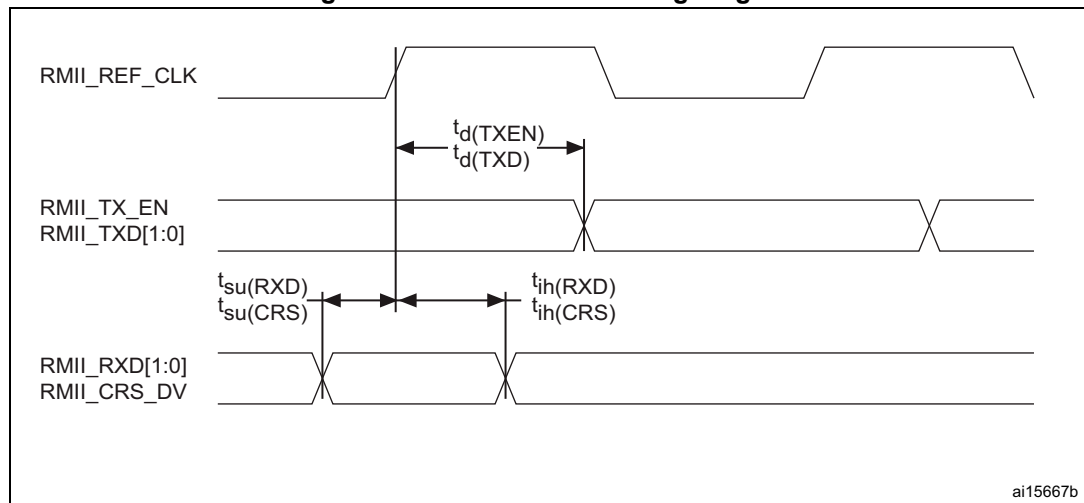


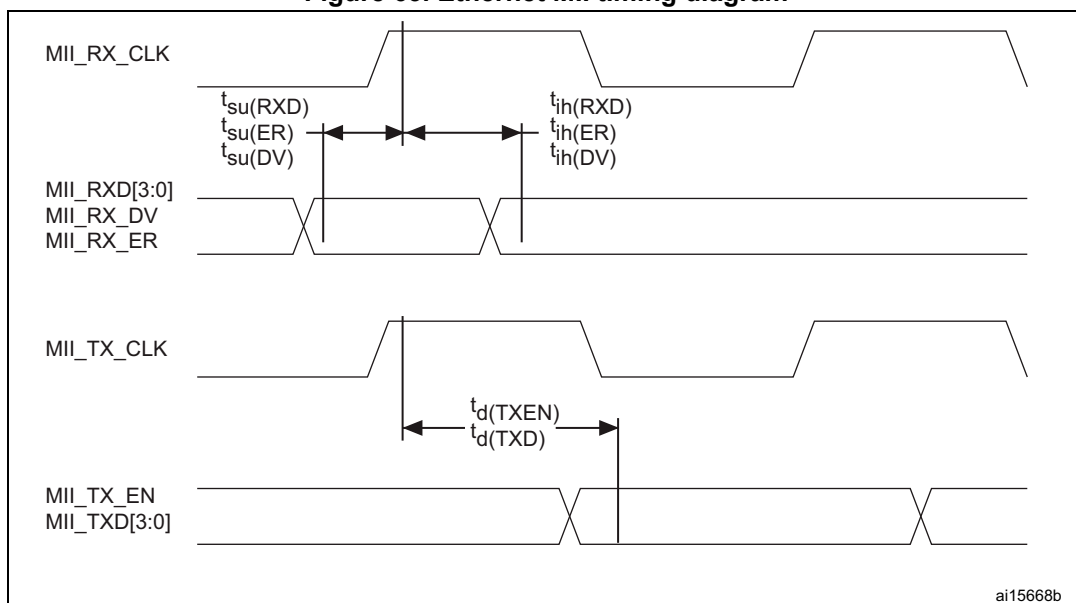
Table 116 gives the list of Ethernet MAC signals for MII and Figure 63 shows the corresponding timing diagram.

Table 116. Dynamics characteristics: Ethernet MAC signals for MII<sup>(1)</sup>

| Symbol        | Parameter                        | Min | Typ | Max | Unit |
|---------------|----------------------------------|-----|-----|-----|------|
| $t_{su}(RXD)$ | Receive data setup time          | 2   | -   | -   | ns   |
| $t_{ih}(RXD)$ | Receive data hold time           | 3   | -   | -   |      |
| $t_{su}(DV)$  | Data valid setup time            | 1.5 | -   | -   |      |
| $t_{ih}(DV)$  | Data valid hold time             | 1   | -   | -   |      |
| $t_{su}(ER)$  | Error setup time                 | 1.5 | -   | -   |      |
| $t_{ih}(ER)$  | Error hold time                  | 0.5 | -   | -   |      |
| $t_d(TXEN)$   | Transmit enable valid delay time | 4.5 | 6.5 | 11  |      |
| $t_d(TXD)$    | Transmit data valid delay time   | 7   | 7.5 | 15  |      |

1. Guaranteed by characterization results.

Figure 63. Ethernet MII timing diagram



### 6.3.33 JTAG/SWD interface characteristics

Unless otherwise specified, the parameters given in [Table 117](#) and [Table 118](#) for JTAG/SWD are derived from tests performed under the ambient temperature,  $f_{rcc\_c\_ck}$  frequency and  $V_{DD}$  supply voltage summarized in [Table 24: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 0x10$
- Capacitive load  $C=30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$

Refer to [Section 6.3.15: I/O port characteristics](#) for more details on the input/output characteristics.



Table 117. Dynamics JTAG characteristics<sup>(1)</sup>

| Symbol            | Parameter                | Conditions                              | Min | Typ | Max  | Unit |
|-------------------|--------------------------|-----------------------------------------|-----|-----|------|------|
| $F_{pp}$          | $T_{CK}$ clock frequency | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$  | -   | -   | 37   | MHz  |
| $1/t_{c(TCK)}$    |                          | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | -   | -   | 27.5 |      |
| $t_{i_{su}(TMS)}$ | TMS input setup time     | -                                       | 2   | -   | -    | ns   |
| $t_{i_h(TMS)}$    | TMS input hold time      | -                                       | 1   | -   | -    |      |
| $t_{i_{su}(TDI)}$ | TDI input setup time     | -                                       | 1.5 | -   | -    |      |
| $t_{i_h(TDI)}$    | TDI input hold time      | -                                       | 1   | -   | -    |      |
| $t_{ov}(TDO)$     | TDO output valid time    | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$  | -   | 8   | 13.5 |      |
|                   |                          | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | -   | 8   | 18   |      |
| $t_{oh}(TDO)$     | TDO output hold time     | -                                       | 7   | -   | -    |      |

1. Guaranteed by characterization results.

Table 118. Dynamics SWD characteristics<sup>(1)</sup>

| Symbol              | Parameter               | Conditions                              | Min | Typ | Max  | Unit |
|---------------------|-------------------------|-----------------------------------------|-----|-----|------|------|
| $F_{pp}$            | SWCLK clock frequency   | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$  | -   | -   | 71   | MHz  |
| $1/t_{c(SWCLK)}$    |                         | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | -   | -   | 55.5 |      |
| $t_{i_{su}(SWDIO)}$ | SWDIO input setup time  | -                                       | 2.5 | -   | -    | ns   |
| $t_{i_h(SWDIO)}$    | SWDIO input hold time   | -                                       | 1   | -   | -    |      |
| $t_{ov}(SWDIO)$     | SWDIO output valid time | $2.7\text{ V} < V_{DD} < 3.6\text{ V}$  | -   | 8.5 | 14   |      |
|                     |                         | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | -   | 8.5 | 18   |      |
| $t_{oh}(SWDIO)$     | SWDIO output hold time  | -                                       | 8   | -   | -    |      |

1. Guaranteed by characterization results.

Figure 64. JTAG timing diagram

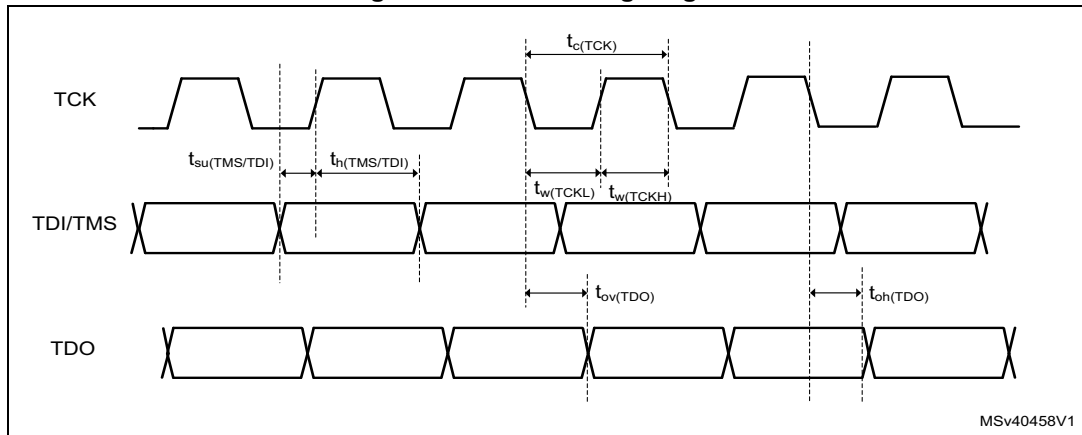
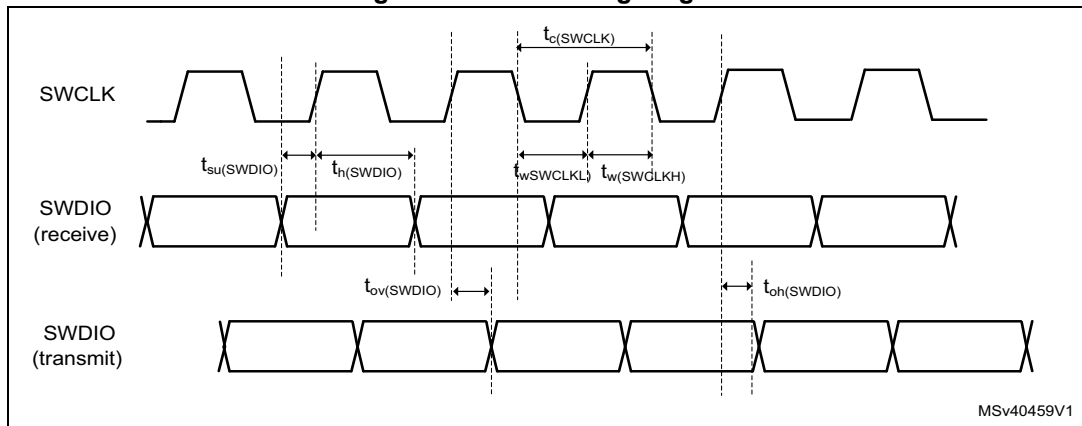


Figure 65. SWD timing diagram



## 7 Electrical characteristics (rev V)

### 7.1 Parameter conditions

Unless otherwise specified, all voltages are referenced to  $V_{SS}$ .

#### 7.1.1 Minimum and maximum values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of junction temperature, supply voltage and frequencies by tests in production on 100% of the devices with an junction temperature at  $T_J = 25\text{ }^{\circ}\text{C}$  and  $T_J = T_{Jmax}$  (given by the selected temperature range).

Data based on characterization results, design simulation and/or technology characteristics are indicated in the table footnotes. Based on characterization, the minimum and maximum values refer to sample tests and represent the mean value plus or minus three times the standard deviation ( $\text{mean} \pm 3\sigma$ ).

#### 7.1.2 Typical values

Unless otherwise specified, typical data are based on  $T_J = 25\text{ }^{\circ}\text{C}$ ,  $V_{DD} = 3.3\text{ V}$  (for the  $1.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  voltage range). They are given only as design guidelines and are not tested.

Typical ADC accuracy values are determined by characterization of a batch of samples from a standard diffusion lot over the full temperature range, where 95% of the devices have an error less than or equal to the value indicated ( $\text{mean} \pm 2\sigma$ ).

#### 7.1.3 Typical curves

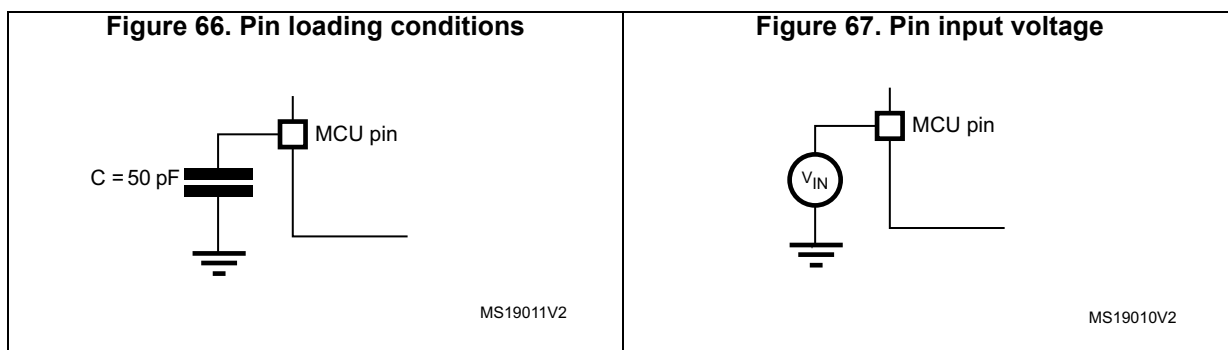
Unless otherwise specified, all typical curves are given only as design guidelines and are not tested.

#### 7.1.4 Loading capacitor

The loading conditions used for pin parameter measurement are shown in [Figure 66](#).

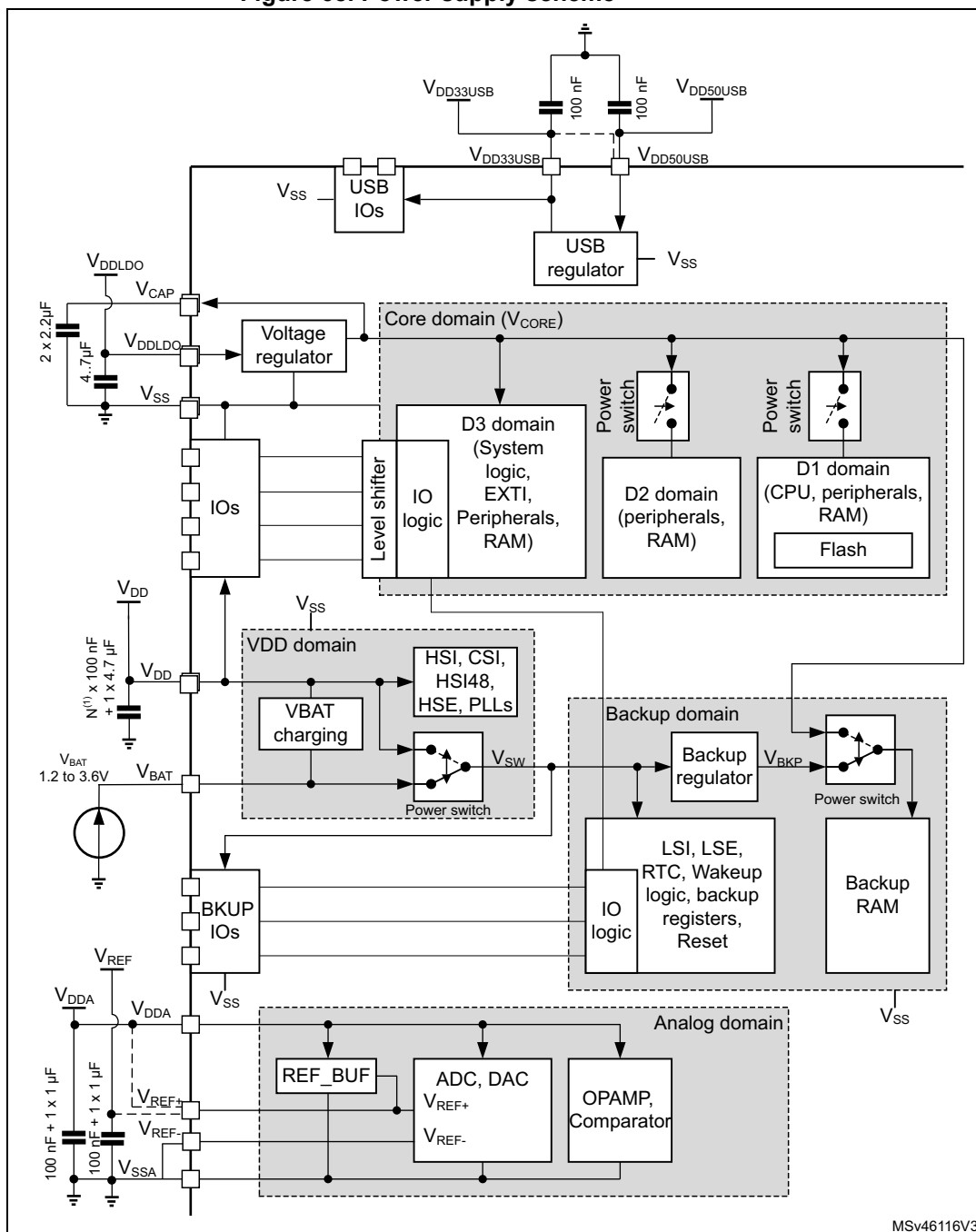
#### 7.1.5 Pin input voltage

The input voltage measurement on a pin of the device is described in [Figure 67](#).



### 7.1.6 Power supply scheme

Figure 68. Power supply scheme



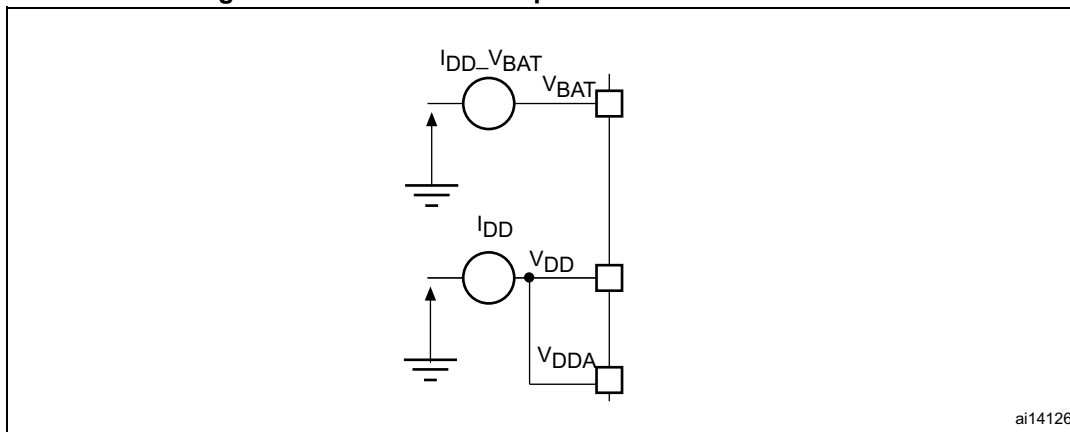
1. N corresponds to the number of VDD pins available on the package.
2. A tolerance of  $\pm 20\%$  is acceptable on decoupling capacitors.

**Caution:** Each power supply pair ( $V_{DD}/V_{SS}$ ,  $V_{DDA}/V_{SSA}$  ...) must be decoupled with filtering ceramic capacitors as shown above. These capacitors must be placed as close as possible to, or below, the appropriate pins on the underside of the PCB to ensure good operation of the

device. It is not recommended to remove filtering capacitors to reduce PCB size or cost. This might cause incorrect operation of the device.

### 7.1.7 Current consumption measurement

Figure 69. Current consumption measurement scheme



## 7.2 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in [Table 119: Voltage characteristics](#), [Table 120: Current characteristics](#), and [Table 121: Thermal characteristics](#) may cause permanent damage to the device. These are stress ratings only and the functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 119. Voltage characteristics <sup>(1)</sup>

| Symbols            | Ratings                                                                                                  | Min          | Max                                                                | Unit |
|--------------------|----------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------|------|
| $V_{DDX} - V_{SS}$ | External main supply voltage (including $V_{DD}$ , $V_{DDLDO}$ , $V_{DDA}$ , $V_{DD33USB}$ , $V_{BAT}$ ) | -0.3         | 4.0                                                                | V    |
| $V_{IN}^{(2)}$     | Input voltage on FT_XXX pins                                                                             | $V_{SS}-0.3$ | $\text{Min}(V_{DD}, V_{DDA}, V_{DD33USB}, V_{BAT}) + 4.0^{(3)(4)}$ | V    |
|                    | Input voltage on TT_XX pins                                                                              | $V_{SS}-0.3$ | 4.0                                                                | V    |
|                    | Input voltage on BOOT0 pin                                                                               | $V_{SS}$     | 9.0                                                                | V    |
|                    | Input voltage on any other pins                                                                          | $V_{SS}-0.3$ | 4.0                                                                | V    |
| $ \Delta V_{DDX} $ | Variations between different $V_{DDX}$ power pins of the same domain                                     | -            | 50                                                                 | mV   |
| $ V_{SSx}-V_{SS} $ | Variations between all the different ground pins                                                         | -            | 50                                                                 | mV   |

1. All main power ( $V_{DD}$ ,  $V_{DDA}$ ,  $V_{DD33USB}$ ,  $V_{BAT}$ ) and ground ( $V_{SS}$ ,  $V_{SSA}$ ) pins must always be connected to the external power supply, in the permitted range.
2.  $V_{IN}$  maximum must always be respected. Refer to [Table 156: I/O current injection susceptibility](#) for the maximum allowed injected current values.
3. This formula has to be applied on power supplies related to the IO structure described by the pin definition table.
4. To sustain a voltage higher than 4V the internal pull-up/pull-down resistors must be disabled.

Table 120. Current characteristics

| Symbols                 | Ratings                                                                         | Max   | Unit |
|-------------------------|---------------------------------------------------------------------------------|-------|------|
| $\Sigma I_{V_{DD}}$     | Total current into sum of all $V_{DD}$ power lines (source) <sup>(1)</sup>      | 620   | mA   |
| $\Sigma I_{V_{SS}}$     | Total current out of sum of all $V_{SS}$ ground lines (sink) <sup>(1)</sup>     | 620   |      |
| $I_{V_{DD}}$            | Maximum current into each $V_{DD}$ power pin (source) <sup>(1)</sup>            | 100   |      |
| $I_{V_{SS}}$            | Maximum current out of each $V_{SS}$ ground pin (sink) <sup>(1)</sup>           | 100   |      |
| $I_{IO}$                | Output current sunk by any I/O and control pin                                  | 20    |      |
| $\Sigma I_{(PIN)}$      | Total output current sunk by sum of all I/Os and control pins <sup>(2)</sup>    | 140   |      |
|                         | Total output current sourced by sum of all I/Os and control pins <sup>(2)</sup> | 140   |      |
| $I_{INJ(PIN)}^{(3)(4)}$ | Injected current on FT_xxx, TT_xx, RST and B pins except PA4, PA5               | -5/+0 |      |
|                         | Injected current on PA4, PA5                                                    | -0/0  |      |
| $\Sigma I_{INJ(PIN)}$   | Total injected current (sum of all I/Os and control pins) <sup>(5)</sup>        | ±25   |      |

1. All main power ( $V_{DD}$ ,  $V_{DDA}$ ,  $V_{DD33USB}$ ) and ground ( $V_{SS}$ ,  $V_{SSA}$ ) pins must always be connected to the external power supplies, in the permitted range.
2. This current consumption must be correctly distributed over all I/Os and control pins. The total output current must not be sunk/sourced between two consecutive power supply pins referring to high pin count QFP packages.
3. Positive injection is not possible on these I/Os and does not occur for input voltages lower than the specified maximum value.
4. A positive injection is induced by  $V_{IN} > V_{DD}$  while a negative injection is induced by  $V_{IN} < V_{SS}$ .  $I_{INJ(PIN)}$  must never be exceeded. Refer also to [Table 119: Voltage characteristics](#) for the maximum allowed input voltage values.
5. When several inputs are submitted to a current injection, the maximum  $\Sigma I_{INJ(PIN)}$  is the absolute sum of the positive and negative injected currents (instantaneous values).

Table 121. Thermal characteristics

| Symbol    | Ratings                      | Value        | Unit |
|-----------|------------------------------|--------------|------|
| $T_{STG}$ | Storage temperature range    | - 65 to +150 | °C   |
| $T_J$     | Maximum junction temperature | 125          |      |

## 7.3 Operating conditions

### 7.3.1 General operating conditions

Table 122. General operating conditions

| Symbol               | Parameter                                                                                                   | Operating conditions                                        | Min                 | Typ  | Max                                                                                            | Unit |
|----------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------|------|------------------------------------------------------------------------------------------------|------|
| V <sub>DD</sub>      | Standard operating voltage                                                                                  | -                                                           | 1.62 <sup>(1)</sup> | -    | 3.6                                                                                            | V    |
| V <sub>DDLDO</sub>   | Supply voltage for the internal regulator                                                                   | V <sub>DDLDO</sub> ≤ V <sub>DD</sub>                        | 1.62 <sup>(1)</sup> | -    | 3.6                                                                                            |      |
| V <sub>DD33USB</sub> | Standard operating voltage, USB domain                                                                      | USB used                                                    | 3.0                 | -    | 3.6                                                                                            | V    |
|                      |                                                                                                             | USB not used                                                | 0                   | -    | 3.6                                                                                            |      |
| V <sub>DDA</sub>     | Analog operating voltage                                                                                    | ADC or COMP used                                            | 1.62                | -    | 3.6                                                                                            |      |
|                      |                                                                                                             | DAC used                                                    | 1.8                 | -    |                                                                                                |      |
|                      |                                                                                                             | OPAMP used                                                  | 2.0                 | -    |                                                                                                |      |
|                      |                                                                                                             | VREFBUF used                                                | 1.8                 | -    |                                                                                                |      |
|                      |                                                                                                             | ADC, DAC, OPAMP, COMP, VREFBUF not used                     | 0                   | -    |                                                                                                |      |
| V <sub>IN</sub>      | I/O Input voltage                                                                                           | TT_xx I/O                                                   | -0.3                | -    | V <sub>DD</sub> +0.3                                                                           |      |
|                      |                                                                                                             | BOOT0                                                       | 0                   | -    | 9                                                                                              |      |
|                      |                                                                                                             | All I/O except BOOT0 and TT_xx                              | -0.3                | -    | Min(V <sub>DD</sub> , V <sub>DDA</sub> , V <sub>DD33USB</sub> ) +3.6V < 5.5V <sup>(2)(3)</sup> |      |
| V <sub>CORE</sub>    | Internal regulator ON (LDO)                                                                                 | VOS3 (max frequency 200 MHz)                                | 0.95                | 1.0  | 1.26                                                                                           | V    |
|                      |                                                                                                             | VOS2 (max frequency 300 MHz)                                | 1.05                | 1.10 | 1.26                                                                                           |      |
|                      |                                                                                                             | VOS1 (max frequency 400 MHz)                                | 1.15                | 1.20 | 1.26                                                                                           |      |
|                      |                                                                                                             | VOS0 <sup>(4)</sup> (max frequency 480 MHz <sup>(5)</sup> ) | 1.26                | 1.35 | 1.40                                                                                           |      |
|                      | Regulator OFF: external V <sub>CORE</sub> voltage must be supplied from external regulator on two VCAP pins | VOS3 (max frequency 200 MHz)                                | 0.98                | 1.03 | 1.26                                                                                           |      |
|                      |                                                                                                             | VOS2 (max frequency 300 MHz)                                | 1.08                | 1.13 | 1.26                                                                                           |      |
|                      |                                                                                                             | VOS1 (max frequency 400 MHz)                                | 1.17                | 1.23 | 1.26                                                                                           |      |
|                      |                                                                                                             | VOS0 (max frequency 480 MHz <sup>(5)</sup> )                | 1.37                | 1.38 | 1.40                                                                                           |      |

Table 122. General operating conditions (continued)

| Symbol            | Parameter                                                                            | Operating conditions                 | Min | Typ | Max                | Unit               |
|-------------------|--------------------------------------------------------------------------------------|--------------------------------------|-----|-----|--------------------|--------------------|
| $f_{\text{CPU}}$  | Arm® Cortex®-M7 clock frequency                                                      | VOS3                                 | -   | -   | 200                | MHz                |
|                   |                                                                                      | VOS2                                 | -   | -   | 300                |                    |
|                   |                                                                                      | VOS1                                 | -   | -   | 400                |                    |
|                   |                                                                                      | VOS0                                 | -   | -   | 480 <sup>(5)</sup> |                    |
| $f_{\text{HCLK}}$ | AHB clock frequency                                                                  | VOS3                                 | -   | -   | 100                |                    |
|                   |                                                                                      | VOS2                                 | -   | -   | 150                |                    |
|                   |                                                                                      | VOS1                                 | -   | -   | 200                |                    |
|                   |                                                                                      | VOS0                                 | -   | -   | 240 <sup>(5)</sup> |                    |
| $f_{\text{PCLK}}$ | APB clock frequency                                                                  | VOS3                                 | -   | -   | 50 <sup>(6)</sup>  |                    |
|                   |                                                                                      | VOS2                                 | -   | -   | 75                 |                    |
|                   |                                                                                      | VOS1                                 | -   | -   | 100                |                    |
|                   |                                                                                      | VOS0                                 | -   | -   | 120 <sup>(5)</sup> |                    |
| $P_{\text{D}}$    | Power dissipation at $T_{\text{A}} = 85^{\circ}\text{C}$ for suffix 6 <sup>(7)</sup> | TFBGA240+25                          | -   | -   | 1093               | mW                 |
|                   |                                                                                      | LQFP208                              | -   | -   | 943                |                    |
|                   |                                                                                      | LQFP176                              | -   | -   | 930                |                    |
|                   |                                                                                      | UFBGA176+25                          | -   | -   | 1070               |                    |
|                   |                                                                                      | UFBGA169                             | -   | -   | 1061               |                    |
|                   |                                                                                      | LQFP144                              | -   | -   | 915                |                    |
|                   |                                                                                      | LQFP100                              | -   | -   | 889                |                    |
|                   |                                                                                      | TFBGA100                             | -   | -   | 1018               |                    |
| $T_{\text{A}}$    | Ambient temperature for the suffix 6 version                                         | Maximum power dissipation            |     | -40 | 85                 | $^{\circ}\text{C}$ |
|                   |                                                                                      | Low-power dissipation <sup>(8)</sup> |     | -40 | 105                |                    |
|                   | Ambient temperature for the suffix 3 version                                         | Maximum power dissipation            |     | -40 | 125                |                    |
|                   |                                                                                      | Low-power dissipation <sup>(5)</sup> |     | -40 | 130                |                    |
| $T_{\text{J}}$    | Junction temperature range                                                           | Suffix 6 version                     |     | -40 | 125                | $^{\circ}\text{C}$ |

1. When RESET is released functionality is guaranteed down to  $V_{\text{BOR0}}$  min
2. This formula has to be applied on power supplies related to the IO structure described by the pin definition table.
3. For operation with voltage higher than Min ( $V_{\text{DD}}$ ,  $V_{\text{DDA}}$ ,  $V_{\text{DD3USB}}$ ) +0.3V, the internal Pull-up and Pull-Down resistors must be disabled.
4. VOS0 is available only when the LDO regulator is ON.
5.  $T_{\text{Jmax}} = 105^{\circ}\text{C}$ .
6. Maximum APB clock frequency when at least one peripheral is enabled.
7. If  $T_{\text{A}}$  is lower, higher  $P_{\text{D}}$  values are allowed as long as  $T_{\text{J}}$  does not exceed  $T_{\text{Jmax}}$  (see [Section 8.9: Thermal characteristics](#)).
8. In low-power dissipation state,  $T_{\text{A}}$  can be extended to this range as long as  $T_{\text{J}}$  does not exceed  $T_{\text{Jmax}}$  (see [Section 8.9: Thermal characteristics](#)).

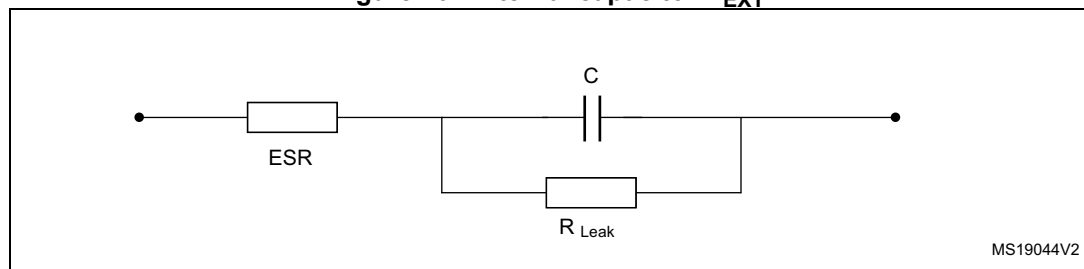


Table 123. Supply voltage and maximum frequency configuration

| Power scale | V <sub>CORE</sub> source | Max T <sub>J</sub> (°C) | Max frequency (MHz) | Min V <sub>DD</sub> (V) |
|-------------|--------------------------|-------------------------|---------------------|-------------------------|
| VOS0        | LDO                      | 105                     | 480                 | 1.7                     |
| VOS1        | LDO                      | 125                     | 400                 | 1.62                    |
| VOS2        | LDO                      | 125                     | 300                 | 1.62                    |
| VOS3        | LDO                      | 125                     | 200                 | 1.62                    |
| SVOS4       | LDO                      | 105                     | N/A                 | 1.62                    |
| SVOS5       | LDO                      | 105                     | N/A                 | 1.62                    |

### 7.3.2 VCAP external capacitor

Stabilization for the main regulator is achieved by connecting an external capacitor C<sub>EXT</sub> to the VCAP pin. C<sub>EXT</sub> is specified in [Table 124](#). Two external capacitors can be connected to VCAP pins.

Figure 70. External capacitor C<sub>EXT</sub>

1. Legend: ESR is the equivalent series resistance.

Table 124. VCAP operating conditions<sup>(1)</sup>

| Symbol           | Parameter                         | Conditions            |
|------------------|-----------------------------------|-----------------------|
| C <sub>EXT</sub> | Capacitance of external capacitor | 2.2 μF <sup>(2)</sup> |
| ESR              | ESR of external capacitor         | < 100 mΩ              |

1. When bypassing the voltage regulator, the two 2.2 μF V<sub>CAP</sub> capacitors are not required and should be replaced by two 100 nF decoupling capacitors.
2. This value corresponds to C<sub>EXT</sub> typical value. A variation of +/-20% is tolerated.

### 7.3.3 Operating conditions at power-up / power-down

Subject to general operating conditions for  $T_A$ .

**Table 125. Operating conditions at power-up / power-down (regulator ON)**

| Symbol       | Parameter                  | Min | Max      | Unit            |
|--------------|----------------------------|-----|----------|-----------------|
| $t_{VDD}$    | $V_{DD}$ rise time rate    | 0   | $\infty$ | $\mu\text{s/V}$ |
|              | $V_{DD}$ fall time rate    | 10  | $\infty$ |                 |
| $t_{VDDA}$   | $V_{DDA}$ rise time rate   | 0   | $\infty$ |                 |
|              | $V_{DDA}$ fall time rate   | 10  | $\infty$ |                 |
| $t_{VDDUSB}$ | $V_{DDUSB}$ rise time rate | 0   | $\infty$ |                 |
|              | $V_{DDUSB}$ fall time rate | 10  | $\infty$ |                 |

### 7.3.4 Embedded reset and power control block characteristics

The parameters given in [Table 126](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

**Table 126. Reset and power control block characteristics**

| Symbol                   | Parameter                                                          | Conditions                 | Min  | Typ  | Max   | Unit    |
|--------------------------|--------------------------------------------------------------------|----------------------------|------|------|-------|---------|
| $t_{RSTEMPO}^{(1)}$      | Reset temporization after BOR0 released                            | -                          | -    | 377  | -     | $\mu s$ |
| $V_{BOR0}$               | Brown-out reset threshold 0                                        | Rising edge <sup>(1)</sup> | 1.62 | 1.67 | 1.71  | V       |
|                          |                                                                    | Falling edge               | 1.58 | 1.62 | 1.68  |         |
| $V_{BOR1}$               | Brown-out reset threshold 1                                        | Rising edge                | 2.04 | 2.10 | 2.15  |         |
|                          |                                                                    | Falling edge               | 1.95 | 2.00 | 2.06  |         |
| $V_{BOR2}$               | Brown-out reset threshold 2                                        | Rising edge                | 2.34 | 2.41 | 2.47  |         |
|                          |                                                                    | Falling edge               | 2.25 | 2.31 | 2.37  |         |
| $V_{BOR3}$               | Brown-out reset threshold 3                                        | Rising edge                | 2.63 | 2.70 | 2.78  |         |
|                          |                                                                    | Falling edge               | 2.54 | 2.61 | 2.68  |         |
| $V_{PVD0}$               | Programmable Voltage Detector threshold 0                          | Rising edge                | 1.90 | 1.96 | 2.01  |         |
|                          |                                                                    | Falling edge               | 1.81 | 1.86 | 1.91  |         |
| $V_{PVD1}$               | Programmable Voltage Detector threshold 1                          | Rising edge                | 2.05 | 2.10 | 2.16  |         |
|                          |                                                                    | Falling edge               | 1.96 | 2.01 | 2.06  |         |
| $V_{PVD2}$               | Programmable Voltage Detector threshold 2                          | Rising edge                | 2.19 | 2.26 | 2.32  |         |
|                          |                                                                    | Falling edge               | 2.10 | 2.15 | 2.21  |         |
| $V_{PVD3}$               | Programmable Voltage Detector threshold 3                          | Rising edge                | 2.35 | 2.41 | 2.47  |         |
|                          |                                                                    | Falling edge               | 2.25 | 2.31 | 2.37  |         |
| $V_{PVD4}$               | Programmable Voltage Detector threshold 4                          | Rising edge                | 2.49 | 2.56 | 2.62  |         |
|                          |                                                                    | Falling edge               | 2.39 | 2.45 | 2.51  |         |
| $V_{PVD5}$               | Programmable Voltage Detector threshold 5                          | Rising edge                | 2.64 | 2.71 | 2.78  |         |
|                          |                                                                    | Falling edge               | 2.55 | 2.61 | 2.68  |         |
| $V_{PVD6}$               | Programmable Voltage Detector threshold 6                          | Rising edge                | 2.78 | 2.86 | 2.94  |         |
|                          |                                                                    | Falling edge in Run mode   | 2.69 | 2.76 | 2.83  |         |
| $V_{hyst\_BOR\_PVD}$     | Hysteresis voltage of BOR (unless BOR0) and PVD                    | Hysteresis in Run mode     | -    | 100  | -     | mV      |
| $I_{DD\_BOR\_PVD}^{(1)}$ | BOR <sup>(2)</sup> (unless BOR0) and PVD consumption from $V_{DD}$ | -                          | -    |      | 0.630 | $\mu A$ |

Table 126. Reset and power control block characteristics (continued)

| Symbol           | Parameter                                         | Conditions      | Min  | Typ  | Max  | Unit    |
|------------------|---------------------------------------------------|-----------------|------|------|------|---------|
| $V_{AVM\_0}$     | Analog voltage detector for $V_{DDA}$ threshold 0 | Rising edge     | 1.66 | 1.71 | 1.76 | V       |
|                  |                                                   | Falling edge    | 1.56 | 1.61 | 1.66 |         |
| $V_{AVM\_1}$     | Analog voltage detector for $V_{DDA}$ threshold 1 | Rising edge     | 2.06 | 2.12 | 2.19 |         |
|                  |                                                   | Falling edge    | 1.96 | 2.02 | 2.08 |         |
| $V_{AVM\_2}$     | Analog voltage detector for $V_{DDA}$ threshold 2 | Rising edge     | 2.42 | 2.50 | 2.58 |         |
|                  |                                                   | Falling edge    | 2.35 | 2.42 | 2.49 |         |
| $V_{AVM\_3}$     | Analog voltage detector for $V_{DDA}$ threshold 3 | Rising edge     | 2.74 | 2.83 | 2.91 |         |
|                  |                                                   | Falling edge    | 2.64 | 2.72 | 2.80 |         |
| $V_{hyst\_VDDA}$ | Hysteresis of $V_{DDA}$ voltage detector          | -               | -    | 100  | -    | mV      |
| $I_{DD\_PVM}$    | PVM consumption from $V_{DD(1)}$                  | -               | -    | -    | 0.25 | $\mu$ A |
| $I_{DD\_VDDA}$   | Voltage detector consumption on $V_{DDA}^{(1)}$   | Resistor bridge | -    | -    | 2.5  | $\mu$ A |

1. Guaranteed by design.

2. BOR0 is enabled in all modes and its consumption is therefore included in the supply current characteristics tables (refer to [Section 7.3.6: Supply current characteristics](#)).

### 7.3.5 Embedded reference voltage

The parameters given in [Table 127](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

Table 127. Embedded reference voltage

| Symbol                    | Parameter                                                           | Conditions                                                                     | Min   | Typ   | Max   | Unit                    |
|---------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|-------|-------|-------|-------------------------|
| $V_{REFINT}$              | Internal reference voltages                                         | $-40^{\circ}\text{C} < T_J < 125^{\circ}\text{C}$ ,<br>$V_{DD} = 3.3\text{ V}$ | 1.180 | 1.216 | 1.255 | V                       |
| $t_{S\_vrefint}^{(1)(2)}$ | ADC sampling time when reading the internal reference voltage       | -                                                                              | 4.3   | -     | -     | $\mu$ s                 |
| $t_{S\_vbat}^{(1)(2)}$    | VBAT sampling time when reading the internal VBAT reference voltage | -                                                                              | 9     | -     | -     |                         |
| $I_{refbuf}^{(2)}$        | Reference Buffer consumption for ADC                                | $V_{DDA}=3.3\text{ V}$                                                         | 9     | 13.5  | 23    | $\mu$ A                 |
| $\Delta V_{REFINT}^{(2)}$ | Internal reference voltage spread over the temperature range        | $-40^{\circ}\text{C} < T_J < 125^{\circ}\text{C}$                              | -     | 5     | 15    | mV                      |
| $T_{coeff}^{(2)}$         | Average temperature coefficient                                     | Average temperature coefficient                                                | -     | 20    | 70    | ppm/ $^{\circ}\text{C}$ |
| $V_{DDcoeff}^{(2)}$       | Average Voltage coefficient                                         | $3.0\text{V} < V_{DD} < 3.6\text{V}$                                           | -     | 10    | 1370  | ppm/V                   |

Table 127. Embedded reference voltage (continued)

| Symbol                   | Parameter             | Conditions | Min | Typ | Max | Unit                     |
|--------------------------|-----------------------|------------|-----|-----|-----|--------------------------|
| V <sub>REFINT_DIV1</sub> | 1/4 reference voltage | -          | -   | 25  | -   | %<br>V <sub>REFINT</sub> |
| V <sub>REFINT_DIV2</sub> | 1/2 reference voltage | -          | -   | 50  | -   |                          |
| V <sub>REFINT_DIV3</sub> | 3/4 reference voltage | -          | -   | 75  | -   |                          |

1. The shortest sampling time for the application can be determined by multiple iterations.
2. Guaranteed by design.

Table 128. Internal reference voltage calibration values

| Symbol                 | Parameter                                                           | Memory address      |
|------------------------|---------------------------------------------------------------------|---------------------|
| V <sub>REFIN_CAL</sub> | Raw data acquired at temperature of 30 °C, V <sub>DDA</sub> = 3.3 V | 1FF1E860 - 1FF1E861 |

### 7.3.6 Supply current characteristics

The current consumption is a function of several parameters and factors such as the operating voltage, ambient temperature, I/O pin loading, device software configuration, operating frequencies, I/O pin switching rate, program location in memory and executed binary code.

The current consumption is measured as described in [Figure 69: Current consumption measurement scheme](#).

All the run-mode current consumption measurements given in this section are performed with a CoreMark code.

#### Typical and maximum current consumption

The MCU is placed under the following conditions:

- All I/O pins are in analog input mode.
- All peripherals are disabled except when explicitly mentioned.
- The Flash memory access time is adjusted with the minimum wait states number, depending on the f<sub>ACLK</sub> frequency (refer to the table “Number of wait states according to CPU clock (f<sub>ICC\_CCK</sub>) frequency and V<sub>CORE</sub> range” available in the reference manual).
- When the peripherals are enabled, the AHB clock frequency is the CPU frequency divided by 2 and the APB clock frequency is AHB clock frequency divided by 2.

The parameters given in the below tables are derived from tests performed under ambient temperature and supply voltage conditions summarized in [Table 122: General operating conditions](#).

Table 129. Typical and maximum current consumption in Run mode, code with data processing running from ITCM, LDO regulator ON<sup>(1)</sup>

| Symbol          | Parameter                        | Conditions                     |      | f <sub>HCLK</sub><br>(MHz) | Typ | Max <sup>(2)</sup> |             |              |              | Unit |
|-----------------|----------------------------------|--------------------------------|------|----------------------------|-----|--------------------|-------------|--------------|--------------|------|
|                 |                                  |                                |      |                            |     | Tj=25<br>°C        | Tj=85<br>°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub> | Supply<br>current in<br>Run mode | All<br>peripherals<br>disabled | VOS0 | 480                        | 148 | 226                | 307         | 390          | -            | mA   |
|                 |                                  |                                |      | 400                        | 125 | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS1 | 400                        | 110 | 168                | 230         | 296          | 384          |      |
|                 |                                  |                                |      | 300                        | 84  | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS2 | 300                        | 76  | 114                | 170         | 224          | 297          |      |
|                 |                                  |                                |      | 216                        | 56  | 88                 | 152         | 205          | 278          |      |
|                 |                                  |                                |      | 200                        | 53  | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS3 | 200                        | 47  | 71                 | 121         | 164          | 223          |      |
|                 |                                  |                                |      | 180                        | 43  | 64                 | 116         | 159          | 218          |      |
|                 |                                  |                                |      | 168                        | 40  | 63                 | 115         | 158          | 217          |      |
|                 |                                  |                                |      | 144                        | 35  | 55                 | 109         | 153          | 212          |      |
|                 |                                  |                                |      | 60                         | 16  | 36                 | 92          | 135          | 194          |      |
|                 |                                  |                                |      | 25                         | 12  | 24                 | 83          | 126          | 185          |      |
|                 |                                  |                                |      | 25                         | 12  | 24                 | 83          | 126          | 185          |      |
|                 |                                  | All<br>peripherals<br>enabled  | VOS0 | 480                        | 226 | 348                | 439         | 550          | -            |      |
|                 |                                  |                                |      | 400                        | 190 | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS1 | 400                        | 167 | 256                | 327         | 416          | 536          |      |
|                 |                                  |                                |      | 300                        | 135 | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS2 | 300                        | 122 | 183                | 248         | 320          | 419          |      |
|                 |                                  |                                |      | 200                        | 85  | -                  | -           | -            | -            |      |
|                 |                                  |                                | VOS3 | 200                        | 76  | 116                | 174         | 233          | 313          |      |
|                 |                                  |                                |      | 200                        | 76  | 116                | 174         | 233          | 313          |      |

1. Data are in DTCM for best computation performance, the cache has no influence on consumption in this case.

2. Guaranteed by characterization results, unless otherwise specified.

**Table 130. Typical and maximum current consumption in Run mode, code with data processing running from Flash memory, cache ON, LDO regulator ON**

| Symbol          | Parameter                  | Conditions               |      | f <sub>HCLK</sub><br>(MHz) | Typ  | Max <sup>(1)</sup>       |                          |                           |                           | Unit |
|-----------------|----------------------------|--------------------------|------|----------------------------|------|--------------------------|--------------------------|---------------------------|---------------------------|------|
|                 |                            |                          |      |                            |      | T <sub>J</sub> =25<br>°C | T <sub>J</sub> =85<br>°C | T <sub>J</sub> =105<br>°C | T <sub>J</sub> =125<br>°C |      |
| I <sub>DD</sub> | Supply current in Run mode | All peripherals disabled | VOS0 | 480                        | 110  | 222                      | 304                      | 388                       | -                         | mA   |
|                 |                            |                          |      | 400                        | 91   | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS1 | 400                        | 80   | 162                      | 228                      | 294                       | 381                       |      |
|                 |                            |                          |      | 300                        | 61.5 | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS2 | 216                        | 55   | 111                      | 168                      | 222                       | 294                       |      |
|                 |                            |                          |      | 200                        | 38.5 | -                        | -                        | -                         | -                         |      |
|                 |                            | All peripherals enabled  | VOS3 | 200                        | 34.5 | 69                       | 120                      | 163                       | 222                       |      |
|                 |                            |                          |      | VOS0                       | 480  | 220                      | 342                      | 436                       | 546                       |      |
|                 |                            |                          | 400  |                            | 195  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS1 | 400                        | 175  | 264                      | 336                      | 424                       | 544                       |      |
|                 |                            |                          |      | 300                        | 135  | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS2 | 300                        | 120  | 180                      | 246                      | 318                       | 418                       |      |
|                 |                            |                          |      | 200                        | 83   | -                        | -                        | -                         | -                         |      |
|                 |                            |                          | VOS3 | 200                        | 75   | 114                      | 173                      | 232                       | 312                       |      |

1. Guaranteed by characterization results, unless otherwise specified.

**Table 131. Typical and maximum current consumption in Run mode, code with data processing running from Flash memory, cache OFF, LDO regulator ON**

| Symbol          | Parameter                        | Conditions                     |      | f <sub>HCLK</sub><br>(MHz) | Typ | Max <sup>(1)</sup> |         |              |              | Unit |
|-----------------|----------------------------------|--------------------------------|------|----------------------------|-----|--------------------|---------|--------------|--------------|------|
|                 |                                  |                                |      |                            |     | Tj=25°C            | Tj=85°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub> | Supply<br>current in<br>Run mode | All<br>peripherals<br>disabled | VOS0 | 480                        | 87  | 157                | 259     | 342          | 453          | mA   |
|                 |                                  |                                | VOS1 | 400                        | 73  | 123                | 201     | 267          | 355          |      |
|                 |                                  |                                | VOS2 | 300                        | 52  | 85                 | 150     | 204          | 277          |      |
|                 |                                  |                                | VOS3 | 200                        | 34  | 54                 | 109     | 152          | 212          |      |
|                 |                                  | All<br>peripherals<br>enabled  | VOS0 | 480                        | 168 | 276                | 390     | 504          | 658          |      |
|                 |                                  |                                | VOS1 | 400                        | 135 | 224                | 308     | 397          | 519          |      |
|                 |                                  |                                | VOS2 | 300                        | 100 | 154                | 228     | 301          | 401          |      |
|                 |                                  |                                | VOS3 | 200                        | 70  | 103                | 167     | 226          | 307          |      |

1. Guaranteed by characterization results, unless otherwise specified.

**Table 132. Typical and maximum current consumption batch acquisition mode, LDO regulator ON**

| Symbol          | Parameter                                            | Conditions                                 |      | f <sub>HCLK</sub><br>(MHz) | Typ | Max <sup>(1)</sup> |         |              |              | Unit |
|-----------------|------------------------------------------------------|--------------------------------------------|------|----------------------------|-----|--------------------|---------|--------------|--------------|------|
|                 |                                                      |                                            |      |                            |     | Tj=25°C            | Tj=85°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub> | Supply<br>current in<br>batch<br>acquisition<br>mode | D1<br>Standby,<br>D2<br>Standby,<br>D3 Run | VOS3 | 64                         | 2.7 | 4.7                | 12.9    | 19.0         | 27.5         | mA   |
|                 |                                                      |                                            |      | 8                          | 1.1 | -                  | -       | -            | -            |      |
|                 |                                                      | D1 Stop,<br>D2 Stop,<br>D3 Run             | VOS3 | 64                         | 5.4 | 18.4               | 83.7    | 132.6        | 202.4        |      |
|                 |                                                      |                                            |      | 8                          | 3.8 | -                  | -       | -            | -            |      |

1. Guaranteed by characterization results, unless otherwise specified.

**Table 133. Typical and maximum current consumption in Stop, LDO regulator ON**

| Symbol                 | Parameter                             | Conditions                |       | Typ  | Max <sup>(1)</sup> |         |          |          | Unit |
|------------------------|---------------------------------------|---------------------------|-------|------|--------------------|---------|----------|----------|------|
|                        |                                       |                           |       |      | Tj=25°C            | Tj=85°C | Tj=105°C | Tj=125°C |      |
| I <sub>DD (Stop)</sub> | D1 Stop,<br>D2 Stop,<br>D3 Stop       | Flash memory OFF, no IWDG | SVOS5 | 1.27 | 6.3                | 42.5    | 72.0     | -        | mA   |
|                        |                                       |                           | SVOS4 | 1.96 | 9.4                | 57.4    | 94.6     | -        |      |
|                        |                                       |                           | SVOS3 | 2.78 | 13.8               | 75.9    | 121.3    | 183.8    |      |
|                        |                                       | Flash memory ON, no IWDG  | SVOS5 | 1.27 | 6.3                | 42.5    | 72.0     | -        |      |
|                        |                                       |                           | SVOS4 | 2.25 | 9.8                | 57.9    | 95.2     | -        |      |
|                        |                                       |                           | SVOS3 | 3.07 | 14.1               | 76.4    | 122.0    | 184.8    |      |
|                        | D1 Stop,<br>D2 Standby,<br>D3 Stop    | Flash memory OFF, no IWDG | SVOS5 | 0.91 | 4.6                | 30.4    | 51.2     | -        |      |
|                        |                                       |                           | SVOS4 | 1.42 | 6.8                | 41.1    | 67.3     | -        |      |
|                        |                                       |                           | SVOS3 | 2.02 | 10.0               | 54.4    | 86.6     | 130.0    |      |
|                        |                                       | Flash memory ON, no IWDG  | SVOS5 | 0.91 | 4.6                | 30.4    | 51.2     | -        |      |
|                        |                                       |                           | SVOS4 | 1.70 | 7.2                | 41.5    | 67.9     | -        |      |
|                        |                                       |                           | SVOS3 | 2.31 | 10.3               | 54.9    | 87.1     | 130.8    |      |
|                        | D1 Standby,<br>D2 Stop,<br>D3 Stop    | Flash memory OFF, no IWDG | SVOS5 | 0.49 | 2.4                | 16.5    | 28.0     | -        |      |
|                        |                                       |                           | SVOS4 | 0.76 | 3.6                | 22.2    | 36.6     | -        |      |
|                        |                                       |                           | SVOS3 | 1.10 | 5.3                | 29.3    | 46.9     | 71.2     |      |
|                        | D1 Standby,<br>D2 Standby,<br>D3 Stop |                           | SVOS5 | 0.15 | 0.7                | 4.3     | 7.3      | -        |      |
|                        |                                       |                           | SVOS4 | 0.22 | 1.0                | 5.8     | 9.6      | -        |      |
|                        |                                       |                           | SVOS3 | 0.35 | 1.5                | 7.8     | 12.3     | 18.6     |      |

1. Guaranteed by characterization results, unless otherwise specified.



Table 134. Typical and maximum current consumption in Sleep mode, LDO regulator

| Symbol                  | Parameter                          | Conditions                     |      | f <sub>HCLK</sub><br>(MHz) | Typ   | Max <sup>(1)</sup> |             |              |              | Unit |
|-------------------------|------------------------------------|--------------------------------|------|----------------------------|-------|--------------------|-------------|--------------|--------------|------|
|                         |                                    |                                |      |                            |       | Tj=25<br>°C        | Tj=85<br>°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub> (Sleep) | Supply<br>current in<br>Sleep mode | All<br>peripherals<br>disabled | VOS0 | 480                        | 50.7  | 96.3               | 253.4       | 366.1        | -            | mA   |
|                         |                                    |                                |      | 400                        | 43.4  | 87.8               | 245.5       | 357.9        | -            |      |
|                         |                                    |                                | VOS1 | 400                        | 35.3  | 66.5               | 181.3       | 265.8        | 379.6        |      |
|                         |                                    |                                |      | 300                        | 27.9  | -                  | -           | -            | -            |      |
|                         |                                    |                                | VOS2 | 300                        | 24.6  | 47.3               | 139.1       | 207.3        | 300.4        |      |
|                         |                                    |                                |      | 200                        | 18.8  | -                  | -           | -            | -            |      |
|                         |                                    |                                | VOS3 | 200                        | 16.5  | 33.6               | 106.4       | 160.9        | 236.1        |      |
|                         |                                    | All<br>peripherals<br>enabled  | VOS0 | 480                        | 136.0 | 194.7              | 348.5       | 464.4        | -            |      |
|                         |                                    |                                |      | 400                        | 115.0 | 169.0              | 325.9       | 441.7        | -            |      |
|                         |                                    |                                | VOS1 | 400                        | 97.7  | 138.2              | 251.3       | 338.4        | 456.4        |      |
|                         |                                    |                                |      | 300                        | 74.9  | -                  | -           | -            | -            |      |
|                         |                                    |                                | VOS2 | 300                        | 67.3  | 95.8               | 187.6       | 257.9        | 354.1        |      |
|                         |                                    |                                |      | 200                        | 52.8  | -                  | -           | -            | -            |      |
|                         |                                    |                                | VOS3 | 200                        | 47.1  | 69.3               | 141.4       | 197.7        | 275.1        |      |

1. Guaranteed by characterization results, unless otherwise specified.

Table 135. Typical and maximum current consumption in Standby

| Symbol                       | Parameter                               | Conditions     |                   | Typ    |       |      |       | Max <sup>(1)</sup> |             |              |              | Unit |
|------------------------------|-----------------------------------------|----------------|-------------------|--------|-------|------|-------|--------------------|-------------|--------------|--------------|------|
|                              |                                         |                |                   | 1.62 V | 2.4 V | 3 V  | 3.3 V | 3 V                |             |              |              |      |
|                              |                                         | Backup<br>SRAM | RTC<br>and<br>LSE |        |       |      |       | Tj=25<br>°C        | Tj=85<br>°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub><br>(Standby) | Supply<br>current in<br>Standby<br>mode | OFF            | OFF               | 1,92   | 1,95  | 2,06 | 2,16  | 4                  | 18          | 40           | 90           | µA   |
|                              |                                         | ON             | OFF               | 3,33   | 3,44  | 3,6  | 3,79  | 8.2                | 47          | 83           | 141          |      |
|                              |                                         | OFF            | ON                | 2,43   | 2,57  | 2,77 | 2,95  | -                  | -           | -            | -            |      |
|                              |                                         | ON             | ON                | 3,82   | 4,05  | 4,31 | 4,55  | -                  | -           | -            | -            |      |

1. Guaranteed by characterization results, unless otherwise specified.

Table 136. Typical and maximum current consumption in V<sub>BAT</sub> mode

| Symbol                    | Parameter                                     | Conditions     |                   | Typ   |      |      |       | Max <sup>(1)</sup> |             |              |              | Unit |
|---------------------------|-----------------------------------------------|----------------|-------------------|-------|------|------|-------|--------------------|-------------|--------------|--------------|------|
|                           |                                               | Backup<br>SRAM | RTC<br>and<br>LSE | 1.2 V | 2 V  | 3 V  | 3.4 V | 3 V                |             |              |              |      |
|                           |                                               |                |                   |       |      |      |       | Tj=25<br>°C        | Tj=85<br>°C | Tj=105<br>°C | Tj=125<br>°C |      |
| I <sub>DD</sub><br>(VBAT) | Supply<br>current in<br>V <sub>BAT</sub> mode | OFF            | OFF               | 0,02  | 0,02 | 0,03 | 0,05  | 0,5                | 4,1         | 10           | 24           | µA   |
|                           |                                               | ON             | OFF               | 1,33  | 1,45 | 1,58 | 1,7   | 4,4                | 22          | 48           | 87           |      |
|                           |                                               | OFF            | ON                | 0,46  | 0,57 | 0,75 | 0,87  | -                  | -           | -            | -            |      |
|                           |                                               | ON             | ON                | 1,77  | 2    | 2,3  | 2,5   | -                  | -           | -            | -            |      |

1. Guaranteed by characterization results, unless otherwise specified.

### I/O system current consumption

The current consumption of the I/O system has two components: static and dynamic.

#### I/O static current consumption

All the I/Os used as inputs with pull-up generate a current consumption when the pin is externally held low. The value of this current consumption can be simply computed by using the pull-up/pull-down resistors values given in [Table 157: I/O static characteristics](#).

For the output pins, any external pull-down or external load must also be considered to estimate the current consumption.

An additional I/O current consumption is due to I/Os configured as inputs if an intermediate voltage level is externally applied. This current consumption is caused by the input Schmitt trigger circuits used to discriminate the input value. Unless this specific configuration is required by the application, this supply current consumption can be avoided by configuring these I/Os in analog mode. This is notably the case of ADC input pins which should be configured as analog inputs.

**Caution:** Any floating input pin can also settle to an intermediate voltage level or switch inadvertently, as a result of external electromagnetic noise. To avoid a current consumption related to floating pins, they must either be configured in analog mode, or forced internally to a definite digital value. This can be done either by using pull-up/down resistors or by configuring the pins in output mode.

**I/O dynamic current consumption**

In addition to the internal peripheral current consumption (see [Table 137: Peripheral current consumption in Run mode](#)), the I/Os used by an application also contribute to the current consumption. When an I/O pin switches, it uses the current from the MCU supply voltage to supply the I/O pin circuitry and to charge/discharge the capacitive load (internal or external) connected to the pin:

$$I_{SW} = V_{DDx} \times f_{SW} \times C_L$$

where

$I_{SW}$  is the current sunk by a switching I/O to charge/discharge the capacitive load

$V_{DDx}$  is the MCU supply voltage

$f_{SW}$  is the I/O switching frequency

$C_L$  is the total capacitance seen by the I/O pin:  $C = C_{INT} + C_{EXT}$

The test pin is configured in push-pull output mode and is toggled by software at a fixed frequency.

**On-chip peripheral current consumption**

The MCU is placed under the following conditions:

- At startup, all I/O pins are in analog input configuration.
- All peripherals are disabled unless otherwise mentioned.
- The I/O compensation cell is enabled.
- $f_{rcc\_c\_ck}$  is the CPU clock.  $f_{PCLK} = f_{rcc\_c\_ck}/4$ , and  $f_{HCLK} = f_{rcc\_c\_ck}/2$ .  
The given value is calculated by measuring the difference of current consumption
  - with all peripherals clocked off
  - with only one peripheral clocked on
  - $f_{rcc\_c\_ck} = 400$  MHz (Scale 1),  $f_{rcc\_c\_ck} = 300$  MHz (Scale 2),  
 $f_{rcc\_c\_ck} = 200$  MHz (Scale 3)
- The ambient operating temperature is 25 °C and  $V_{DD}=3.3$  V.

Table 137. Peripheral current consumption in Run mode

| Bus  | Peripheral         | VOS0       | VOS1       | VOS2       | VOS3       | Unit                     |
|------|--------------------|------------|------------|------------|------------|--------------------------|
| AHB3 | MDMA               | 4.6        | 3.8        | 3.4        | 3.2        | $\mu\text{A}/\text{MHz}$ |
|      | DMA2D              | 2.9        | 2.4        | 2.1        | 1.9        |                          |
|      | JPGDEC             | 4.1        | 3.7        | 3.4        | 3.1        |                          |
|      | FLASH              | 17.0       | 15.0       | 14.0       | 12.0       |                          |
|      | FMC registers      | 0.9        | 1.1        | 0.9        | 0.8        |                          |
|      | FMC kernel         | 7.0        | 6.1        | 5.6        | 5.0        |                          |
|      | QUADSPI registers  | 1.5        | 1.5        | 1.4        | 1.3        |                          |
|      | QSPI kernel        | 1.0        | 0.9        | 0.8        | 0.7        |                          |
|      | SDMMC1 registers   | 8.2        | 7.2        | 6.7        | 6.0        |                          |
|      | SDMMC1 kernel      | 1.3        | 1.2        | 0.9        | 0.9        |                          |
|      | DTCM1              | 7.9        | 6.8        | 6.0        | 5.3        |                          |
|      | DTCM2              | 8.3        | 7.2        | 6.4        | 5.7        |                          |
|      | ITCM               | 7.0        | 6.3        | 5.6        | 5.1        |                          |
|      | D1SRAM1            | 13.0       | 11.0       | 9.9        | 8.7        |                          |
|      | AHB3 bridge        | 35.0       | 32.0       | 29.0       | 26.0       |                          |
|      | <b>Total AHB3</b>  | <b>120</b> | <b>106</b> | <b>96</b>  | <b>86</b>  |                          |
| AHB1 | DMA1               | 54.0       | 48.0       | 41.0       | 37.0       |                          |
|      | DMA2               | 55.0       | 49.0       | 42.0       | 37.0       |                          |
|      | ADC12 registers    | 4.5        | 4.1        | 3.7        | 3.3        |                          |
|      | ADC12 kernel       | 1.0        | 0.7        | 0.4        | 0.6        |                          |
|      | ART accelerator    | 4.1        | 3.7        | 3.2        | 2.9        |                          |
|      | ETH1MAC            | 17.0       | 15.0       | 14.0       | 12.0       |                          |
|      | ETH1TX             | 0.1        | 0.1        | 0.1        | 0.1        |                          |
|      | ETH1RX             | 0.1        | 0.1        | 0.1        | 0.1        |                          |
|      | USB1 OTG registers | 23.0       | 21.0       | 19.0       | 17.0       |                          |
|      | USB1 OTG kernel    | 8.2        | 0.5        | 8.3        | 8.2        |                          |
|      | USB1 ULPI          | 0.1        | 0.1        | 0.1        | 0.1        |                          |
|      | USB2 OTG registers | 21.0       | 19.0       | 17.0       | 15.0       |                          |
|      | USB2 OTG kernel    | 8.5        | 0.4        | 8.6        | 8.3        |                          |
|      | USB2 ULPI          | 23.0       | 19.0       | 20.0       | 19.0       |                          |
|      | AHB1 bridge        | 0.1        | 0.1        | 0.1        | 0.1        |                          |
|      | <b>Total AHB1</b>  | <b>220</b> | <b>181</b> | <b>178</b> | <b>161</b> |                          |

Table 137. Peripheral current consumption in Run mode (continued)

| Bus  | Peripheral        | VOS0      | VOS1      | VOS2      | VOS3      | Unit                     |
|------|-------------------|-----------|-----------|-----------|-----------|--------------------------|
| AHB2 | DCMI              | 2.1       | 1.9       | 1.8       | 1.6       | $\mu\text{A}/\text{MHz}$ |
|      | RNG registers     | 1.7       | 2.0       | 1.3       | 1.2       |                          |
|      | RNG kernel        | 11.0      | 0.1       | 9.7       | 9.4       |                          |
|      | SDMMC2 registers  | 47.0      | 41.0      | 37.0      | 34.0      |                          |
|      | SDMMC2 kernel     | 1.7       | 1.2       | 1.1       | 1.0       |                          |
|      | D2SRAM1           | 5.7       | 4.9       | 4.4       | 3.9       |                          |
|      | D2SRAM2           | 5.2       | 4.5       | 4.0       | 3.5       |                          |
|      | D2SRAM3           | 4.1       | 3.6       | 3.2       | 2.8       |                          |
|      | AHB2 bridge       | 0.1       | 0.1       | 0.1       | 0.1       |                          |
|      | <b>Total AHB2</b> | <b>79</b> | <b>60</b> | <b>63</b> | <b>58</b> |                          |
| AHB4 | GPIOA             | 1.5       | 1.3       | 1.3       | 1.1       |                          |
|      | GPIOB             | 1.2       | 1.0       | 1.0       | 0.9       |                          |
|      | GPIOC             | 0.8       | 0.7       | 0.7       | 0.6       |                          |
|      | GPIOD             | 1.1       | 1.0       | 1.0       | 0.9       |                          |
|      | GPIOE             | 0.7       | 0.7       | 0.7       | 0.6       |                          |
|      | GPIOF             | 0.8       | 0.8       | 0.7       | 0.6       |                          |
|      | GPIOG             | 0.9       | 0.8       | 0.8       | 0.7       |                          |
|      | GPIOH             | 1.1       | 1.0       | 1.0       | 0.9       |                          |
|      | GPIOI             | 0.9       | 0.9       | 0.8       | 0.7       |                          |
|      | GPIOJ             | 0.8       | 0.8       | 0.7       | 0.7       |                          |
|      | GPIOK             | 0.7       | 0.8       | 0.7       | 0.6       |                          |
|      | CRC               | 0.4       | 0.5       | 0.4       | 0.3       |                          |
|      | BDMA              | 6.6       | 5.9       | 5.3       | 4.8       |                          |
|      | ADC3 registers    | 1.7       | 1.5       | 1.2       | 1.2       |                          |
|      | ADC3 kernel       | 0.4       | 0.3       | 0.5       | 0.2       |                          |
|      | BKPRAM            | 2.3       | 1.9       | 1.7       | 1.5       |                          |
|      | AHB4 bridge       | 0.1       | 0.1       | 0.1       | 0.1       |                          |
|      | <b>Total AHB4</b> | <b>22</b> | <b>20</b> | <b>19</b> | <b>16</b> |                          |
| APB3 | WWDG1             | 0.7       | 0.5       | 0.5       | 0.2       | $\mu\text{A}/\text{MHz}$ |
|      | LCD-TFT           | 81.0      | 36.0      | 33.0      | 30.0      |                          |
|      | APB3 bridge       | 0.3       | 0.2       | 0.1       | 0.1       |                          |
|      | <b>Total APB3</b> | <b>87</b> | <b>41</b> | <b>38</b> | <b>34</b> |                          |

Table 137. Peripheral current consumption in Run mode (continued)

| Bus  | Peripheral         | VOS0 | VOS1 | VOS2 | VOS3 | Unit              |
|------|--------------------|------|------|------|------|-------------------|
| APB1 | TIM2               | 7.7  | 3.6  | 3.3  | 3.0  | $\mu\text{A/MHz}$ |
|      | TIM3               | 6.7  | 3.2  | 3.0  | 2.7  |                   |
|      | TIM4               | 6.3  | 3.1  | 2.8  | 2.5  |                   |
|      | TIM5               | 7.4  | 3.5  | 3.2  | 2.8  |                   |
|      | TIM6               | 1.4  | 0.7  | 0.8  | 0.6  |                   |
|      | TIM7               | 1.4  | 0.7  | 0.7  | 0.6  |                   |
|      | TIM12              | 3.2  | 1.5  | 1.5  | 1.3  |                   |
|      | TIM13              | 2.3  | 1.1  | 1.1  | 0.9  |                   |
|      | TIM14              | 2.1  | 1.1  | 1.1  | 0.9  |                   |
|      | LPTIM1 registers   | 0.7  | 0.5  | 0.8  | 0.7  |                   |
|      | LPTIM1 kernel      | 2.4  | 2.3  | 1.9  | 1.7  |                   |
|      | WWDG2              | 0.6  | 0.5  | 0.5  | 0.4  |                   |
|      | SPI2 registers     | 2.0  | 1.8  | 1.7  | 1.4  |                   |
|      | SPI2 kernel        | 0.8  | 0.6  | 0.5  | 0.6  |                   |
|      | SPI3 registers     | 1.8  | 1.6  | 1.6  | 1.3  |                   |
|      | SPI3 kernel        | 0.7  | 0.9  | 0.7  | 0.7  |                   |
|      | SPDIFRX1 registers | 0.5  | 0.7  | 0.7  | 0.6  |                   |
|      | SPDIFRX1 kernel    | 3.5  | 2.8  | 2.4  | 2.2  |                   |
|      | USART2 registers   | 1.9  | 1.7  | 1.4  | 1.3  |                   |
|      | USART2 kernel      | 4.3  | 3.9  | 3.6  | 3.2  |                   |
|      | USART3 registers   | 1.9  | 1.7  | 1.4  | 1.3  |                   |
|      | USART3 kernel      | 4.4  | 3.9  | 3.5  | 3.2  |                   |
|      | UART4 registers    | 1.7  | 1.5  | 1.4  | 1.4  |                   |
|      | UART4 kernel       | 3.9  | 3.4  | 3.1  | 2.8  |                   |
|      | UART5 registers    | 1.6  | 1.4  | 1.4  | 1.3  |                   |
|      | UART5 kernel       | 3.8  | 3.4  | 3.0  | 2.7  |                   |
|      | I2C1 registers     | 1.1  | 0.8  | 0.9  | 0.8  |                   |
|      | I2C1 kernel        | 2.5  | 2.3  | 2.0  | 1.9  |                   |
|      | I2C2 registers     | 1.0  | 0.8  | 0.9  | 0.8  |                   |

Table 137. Peripheral current consumption in Run mode (continued)

| Bus                 | Peripheral         | VOS0       | VOS1       | VOS2       | VOS3      | Unit                     |
|---------------------|--------------------|------------|------------|------------|-----------|--------------------------|
| APB1<br>(continued) | I2C2 kernel        | 2.3        | 2.2        | 1.9        | 1.7       | $\mu\text{A}/\text{MHz}$ |
|                     | I2C3 registers     | 0.8        | 1.0        | 0.8        | 0.8       |                          |
|                     | I2C3 kernel        | 2.4        | 1.9        | 1.8        | 1.6       |                          |
|                     | HDMI-CEC registers | 0.7        | 0.5        | 0.6        | 0.5       |                          |
|                     | HDMI-CEC kernel    | 0.1        | 0.1        | 3.2        | 0.1       |                          |
|                     | DAC12              | 3.6        | 1.3        | 1.2        | 1.0       |                          |
|                     | USART7 registers   | 1.8        | 1.8        | 1.6        | 1.4       |                          |
|                     | USART7 kernel      | 4.0        | 3.3        | 3.0        | 2.8       |                          |
|                     | USART8 registers   | 2.0        | 1.6        | 1.6        | 1.4       |                          |
|                     | USART8 kernel      | 3.9        | 3.4        | 3.1        | 2.8       |                          |
|                     | CRS                | 6.4        | 5.5        | 5.0        | 4.5       |                          |
|                     | SWPMI registers    | 2.7        | 2.4        | 2.3        | 1.9       |                          |
|                     | SWPMI kernel       | 0.1        | 0.1        | 0.1        | 0.1       |                          |
|                     | OPAMP              | 0.2        | 0.3        | 0.3        | 0.2       |                          |
|                     | MDIO               | 3.3        | 2.9        | 2.6        | 2.3       |                          |
|                     | FDCAN registers    | 19.0       | 17.0       | 15.0       | 13.0      |                          |
|                     | FDCAN kernel       | 9.1        | 7.9        | 6.9        | 6.4       |                          |
|                     | APB1 bridge        | 0.1        | 0.1        | 0.1        | 0.1       |                          |
|                     | <b>Total APB1</b>  | <b>142</b> | <b>108</b> | <b>102</b> | <b>88</b> |                          |
| APB2                | TIM1               | 11.0       | 5.0        | 4.5        | 4.0       | $\mu\text{A}/\text{MHz}$ |
|                     | TIM8               | 10.0       | 4.7        | 4.3        | 3.8       |                          |
|                     | USART1 registers   | 3.6        | 2.5        | 2.7        | 2.9       |                          |
|                     | USART1 kernel      | 0.1        | 0.1        | 0.1        | 0.1       |                          |
|                     | USART6 registers   | 4.5        | 3.0        | 3.1        | 3.4       |                          |
|                     | USART6 kernel      | 0.1        | 0.1        | 0.1        | 0.1       |                          |
|                     | SPI1 registers     | 2.0        | 1.7        | 1.6        | 1.4       |                          |
|                     | SPI1 kernel        | 0.9        | 0.8        | 0.7        | 0.6       |                          |
|                     | SPI4 registers     | 2.1        | 1.7        | 1.6        | 1.5       |                          |
|                     | SPI4 kernel        | 0.6        | 0.5        | 0.5        | 0.3       |                          |
|                     | TIM15              | 5.5        | 2.5        | 2.3        | 2.1       |                          |
|                     | TIM16              | 4.1        | 2.0        | 1.8        | 1.7       |                          |
|                     | TIM17              | 4.1        | 1.9        | 1.8        | 1.6       |                          |
|                     | SPI5 registers     | 2.0        | 1.8        | 1.6        | 1.3       |                          |
|                     | SPI5 kernel        | 0.5        | 0.4        | 0.4        | 0.5       |                          |
|                     | SAI1 registers     | 1.3        | 1.1        | 1.1        | 1.0       |                          |

Table 137. Peripheral current consumption in Run mode (continued)

| Bus                 | Peripheral        | VOS0       | VOS1        | VOS2        | VOS3        | Unit                     |
|---------------------|-------------------|------------|-------------|-------------|-------------|--------------------------|
| APB2<br>(continued) | SAI1 kernel       | 1.4        | 1.1         | 1.0         | 0.8         | $\mu\text{A}/\text{MHz}$ |
|                     | SAI2 registers    | 1.5        | 1.3         | 1.2         | 1.0         |                          |
|                     | SAI2 kernel       | 1.1        | 1.0         | 0.9         | 0.9         |                          |
|                     | SAI3 registers    | 1.6        | 1.3         | 1.1         | 1.0         |                          |
|                     | SAI3 kernel       | 1.1        | 1.2         | 1.1         | 0.9         |                          |
|                     | DFSDM1 registers  | 6.5        | 5.8         | 5.2         | 4.7         |                          |
|                     | DFSDM1 kernel     | 0.3        | 0.2         | 0.2         | 0.4         |                          |
|                     | HRTIM             | 84.0       | 39.0        | 35.0        | 32.0        |                          |
|                     | APB2 bridge       | 0.2        | 0.1         | 0.1         | 0.2         |                          |
|                     | <b>Total APB2</b> | <b>150</b> | <b>81</b>   | <b>74</b>   | <b>68</b>   |                          |
| APB4                | SYSCFG            | 0.9        | 1.0         | 0.7         | 0.8         |                          |
|                     | LPUART1 registers | 1.1        | 1.3         | 1.0         | 0.8         |                          |
|                     | LPUART1 kernel    | 2.9        | 2.2         | 2.2         | 2.1         |                          |
|                     | SPI6 registers    | 1.8        | 1.6         | 1.4         | 1.3         |                          |
|                     | SPI6 kernel       | 0.4        | 0.4         | 0.5         | 0.3         |                          |
|                     | I2C4 registers    | 0.9        | 0.7         | 0.7         | 0.4         |                          |
|                     | I2C4 kernel       | 2.2        | 2.1         | 1.9         | 1.8         |                          |
|                     | LPTIM2 registers  | 0.8        | 0.6         | 0.7         | 0.5         |                          |
|                     | LPTIM2 kernel     | 2.3        | 2.1         | 1.8         | 1.4         |                          |
|                     | LPTIM3 registers  | 0.7        | 0.7         | 0.7         | 0.4         |                          |
|                     | LPTIM3 kernel     | 2.1        | 1.7         | 1.6         | 1.5         |                          |
|                     | LPTIM4 registers  | 0.8        | 0.4         | 0.6         | 0.4         |                          |
|                     | LPTIM4 kernel     | 2.2        | 2.0         | 1.7         | 1.5         |                          |
|                     | LPTIM5 registers  | 0.5        | 0.4         | 0.6         | 0.4         |                          |
|                     | LPTIM5 kernel     | 2.0        | 1.8         | 1.5         | 1.2         |                          |
|                     | COMP12            | 0.6        | 0.4         | 0.5         | 0.2         |                          |
|                     | VREF              | 0.4        | 0.2         | 0.2         | 0.1         |                          |
|                     | RTC               | 1.1        | 0.9         | 1.0         | 0.6         |                          |
|                     | SAI4 registers    | 1.7        | 1.4         | 1.3         | 1.0         |                          |
|                     | SAI4 kernel       | 2.0        | 2.0         | 1.8         | 1.6         |                          |
|                     | APB4 bridge       | 0.1        | 0.1         | 0.1         | 0.1         |                          |
|                     | <b>Total APB4</b> | <b>28</b>  | <b>24.4</b> | <b>22.4</b> | <b>18.9</b> |                          |



### 7.3.7 Wakeup time from low-power modes

The wakeup times given in [Table 138](#) are measured starting from the wakeup event trigger up to the first instruction executed by the CPU:

- For Stop or Sleep modes: the wakeup event is WFE.
- WKUP (PC1) pin is used to wakeup from Standby, Stop and Sleep modes.

All timings are derived from tests performed under ambient temperature and  $V_{DD}=3.3$  V.

**Table 138. Low-power mode wakeup timings**

| Symbol                    | Parameter                            | Conditions                                | Typ <sup>(1)</sup> | Max <sup>(1)</sup> | Unit             |
|---------------------------|--------------------------------------|-------------------------------------------|--------------------|--------------------|------------------|
| $t_{WUSLEEP}^{(2)}$       | Wakeup from Sleep                    | -                                         | 9                  | 10                 | CPU clock cycles |
| $t_{WUSTOP}^{(2)}$        | Wakeup from Stop                     | VOS3, HSI, Flash memory in normal mode    | 4.4                | 5.6                | $\mu s$          |
|                           |                                      | VOS3, HSI, Flash memory in low-power mode | 12                 | 15                 |                  |
|                           |                                      | VOS4, HSI, Flash memory in normal mode    | 15                 | 20                 |                  |
|                           |                                      | VOS4, HSI, Flash memory in low-power mode | 23                 | 28                 |                  |
|                           |                                      | VOS5, HSI, Flash memory in normal mode    | 39                 | 71                 |                  |
|                           |                                      | VOS5, HSI, Flash memory in low-power mode | 39                 | 47                 |                  |
|                           |                                      | VOS3, CSI, Flash memory in normal mode    | 30                 | 37                 |                  |
|                           |                                      | VOS3, CSI, Flash memory in low power mode | 36                 | 50                 |                  |
|                           |                                      | VOS4, CSI, Flash memory in normal mode    | 38                 | 48                 |                  |
|                           |                                      | VOS4, CSI, Flash memory in low-power mode | 47                 | 61                 |                  |
|                           |                                      | VOS5, CSI, Flash memory in normal mode    | 68                 | 75                 |                  |
|                           |                                      | VOS5, CSI, Flash memory in low-power mode | 68                 | 77                 |                  |
| $t_{WUSTOP\_KERON}^{(2)}$ | Wakeup from Stop, clock kept running | VOS3, HSI, Flash memory in normal mode    | 2.6                | 3.4                | $\mu s$          |
|                           |                                      | VOS3, CSI, Flash memory in normal mode    | 26                 | 36                 |                  |
| $t_{WUSTDBY}^{(2)}$       | Wakeup from Standby mode             | -                                         | 390                | 500                |                  |

1. Guaranteed by characterization results.

2. The wakeup times are measured from the wakeup event to the point in which the application code reads the first instruction.

### 7.3.8 External clock source characteristics

#### High-speed external user clock generated from an external source

In bypass mode the HSE oscillator is switched off and the input pin is a standard I/O.

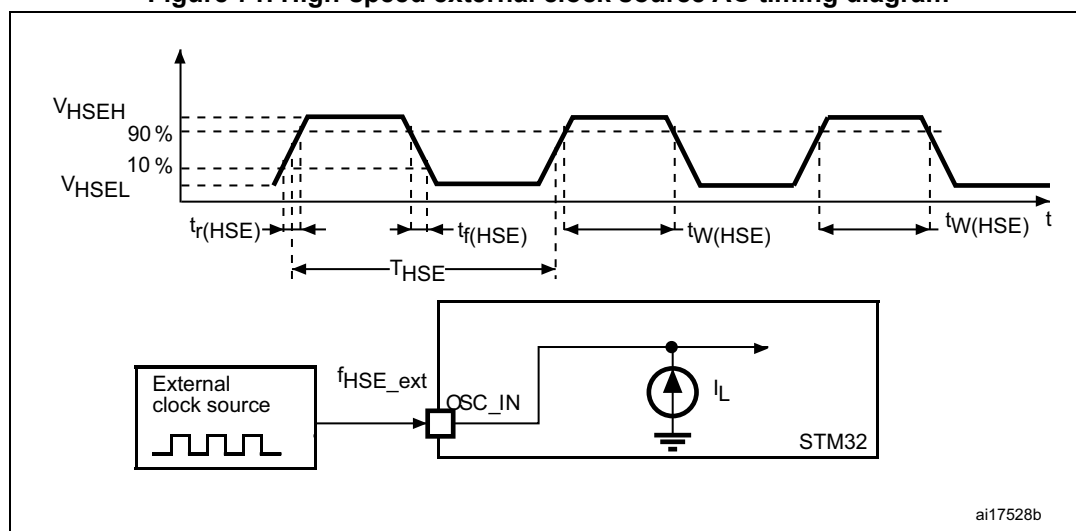
The external clock signal has to respect the [Table 157: I/O static characteristics](#). However, the recommended clock input waveform is shown in [Figure 71](#).

**Table 139. High-speed external user clock characteristics<sup>(1)</sup>**

| Symbol                                | Parameter                            | Min         | Typ | Max         | Unit |
|---------------------------------------|--------------------------------------|-------------|-----|-------------|------|
| $f_{HSE\_ext}$                        | User external clock source frequency | 4           | 25  | 50          | MHz  |
| $V_{SW}$<br>( $V_{HSEH} - V_{HSEL}$ ) | OSC_IN amplitude                     | $0.7V_{DD}$ | -   | $V_{DD}$    | V    |
| $V_{DC}$                              | OSC_IN input voltage                 | $V_{SS}$    | -   | $0.3V_{SS}$ |      |
| $t_{W(HSE)}$                          | OSC_IN high or low time              | 7           | -   | -           | ns   |

1. Guaranteed by design.

**Figure 71. High-speed external clock source AC timing diagram**



ai17528b

### Low-speed external user clock generated from an external source

In bypass mode the LSE oscillator is switched off and the input pin is a standard I/O. The external clock signal has to respect the [Table 157: I/O static characteristics](#). However, the recommended clock input waveform is shown in [Figure 72](#).

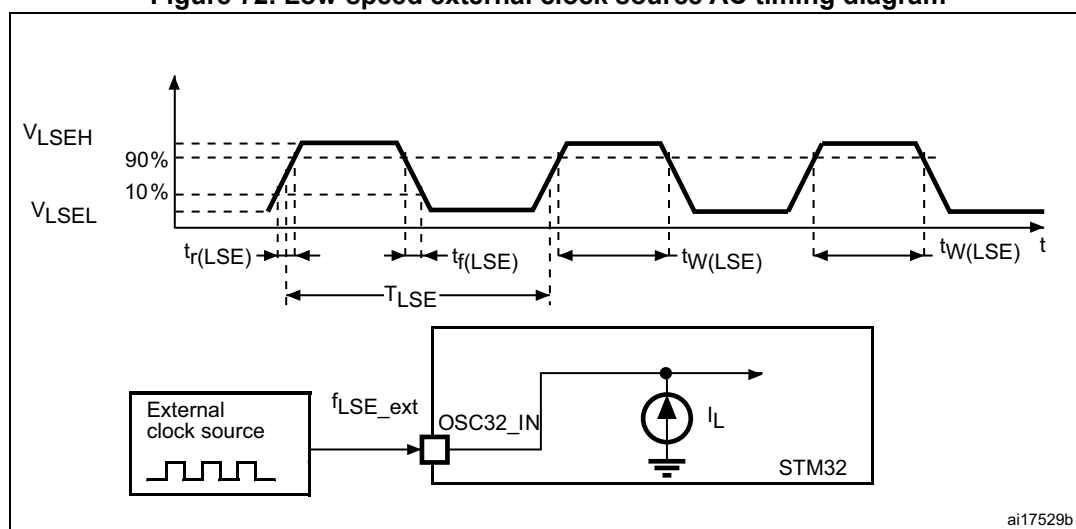
**Table 140. Low-speed external user clock characteristics<sup>(1)</sup>**

| Symbol                         | Parameter                             | Conditions | Min             | Typ    | Max             | Unit |
|--------------------------------|---------------------------------------|------------|-----------------|--------|-----------------|------|
| $f_{LSE\_ext}$                 | User external clock source frequency  | -          | -               | 32.768 | 1000            | kHz  |
| $V_{LSEH}$                     | OSC32_IN input pin high level voltage | -          | $0.7 V_{DDIOx}$ | -      | $V_{DDIOx}$     | V    |
| $V_{LSEL}$                     | OSC32_IN input pin low level voltage  | -          | $V_{SS}$        | -      | $0.3 V_{DDIOx}$ |      |
| $t_{w(LSEH)}$<br>$t_{w(LSEL)}$ | OSC32_IN high or low time             | -          | 250             | -      | -               | ns   |

1. Guaranteed by design.

**Note:** For information on selecting the crystal, refer to the application note AN2867 “Oscillator design guide for ST microcontrollers” available from the ST website [www.st.com](http://www.st.com).

**Figure 72. Low-speed external clock source AC timing diagram**



### High-speed external clock generated from a crystal/ceramic resonator

The high-speed external (HSE) clock can be supplied with a 4 to 48 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in [Table 141](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

**Table 141. 4-48 MHz HSE oscillator characteristics<sup>(1)</sup>**

| Symbol                           | Parameter                   | Operating conditions <sup>(2)</sup>                                           | Min | Typ  | Max | Unit |
|----------------------------------|-----------------------------|-------------------------------------------------------------------------------|-----|------|-----|------|
| F                                | Oscillator frequency        | -                                                                             | 4   | -    | 48  | MHz  |
| R <sub>F</sub>                   | Feedback resistor           | -                                                                             | -   | 200  | -   | kΩ   |
| I <sub>DD(HSE)</sub>             | HSE current consumption     | During startup <sup>(3)</sup>                                                 | -   | -    | 4   | mA   |
|                                  |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF@4MHz      | -   | 0.35 | -   |      |
|                                  |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 8 MHz  | -   | 0.40 | -   |      |
|                                  |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 16 MHz | -   | 0.45 | -   |      |
|                                  |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 32 MHz | -   | 0.65 | -   |      |
|                                  |                             | V <sub>DD</sub> =3 V, R <sub>m</sub> =30 Ω<br>C <sub>L</sub> =10 pF at 48 MHz | -   | 0.95 | -   |      |
| G <sub>m<sub>crit</sub>max</sub> | Maximum critical crystal gm | Startup                                                                       | -   | -    | 1.5 | mA/V |
| t <sub>SU</sub> <sup>(4)</sup>   | Start-up time               | V <sub>DD</sub> is stabilized                                                 | -   | 2    | -   | ms   |

1. Guaranteed by design.

2. Resonator characteristics given by the crystal/ceramic resonator manufacturer.

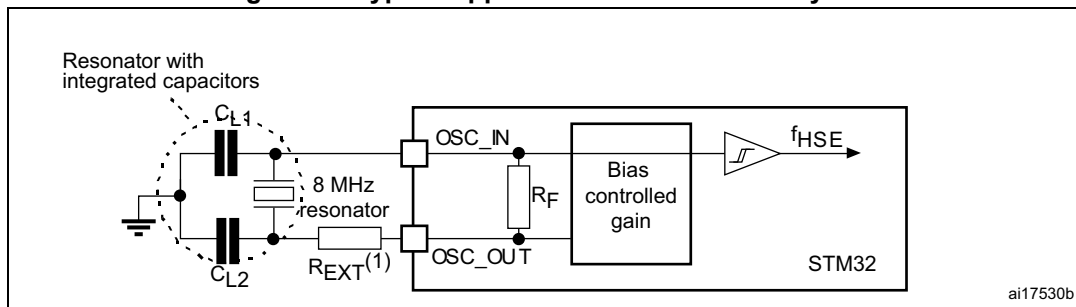
3. This consumption level occurs during the first 2/3 of the t<sub>SU(HSE)</sub> startup time.

4. t<sub>SU(HSE)</sub> is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

For C<sub>L1</sub> and C<sub>L2</sub>, it is recommended to use high-quality external ceramic capacitors in the 5 pF to 25 pF range (typical), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator (see [Figure 73](#)). C<sub>L1</sub> and C<sub>L2</sub> are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C<sub>L1</sub> and C<sub>L2</sub>. The PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing C<sub>L1</sub> and C<sub>L2</sub>.

**Note:** For information on selecting the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website [www.st.com](http://www.st.com).

Figure 73. Typical application with an 8 MHz crystal



1.  $R_{EXT}$  value depends on the crystal characteristics.

### Low-speed external clock generated from a crystal/ceramic resonator

The low-speed external (LSE) clock can be supplied with a 32.768 kHz crystal/ceramic resonator oscillator. All the information given in this paragraph are based on characterization results obtained with typical external components specified in [Table 142](#). In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details on the resonator characteristics (frequency, package, accuracy).

Table 142. Low-speed external user clock characteristics<sup>(1)</sup>

| Symbol            | Parameter                   | Operating conditions <sup>(2)</sup>               | Min | Typ    | Max  | Unit      |
|-------------------|-----------------------------|---------------------------------------------------|-----|--------|------|-----------|
| F                 | Oscillator frequency        | -                                                 | -   | 32.768 | -    | kHz       |
| $I_{DD}$          | LSE current consumption     | LSEDRV[1:0] = 00,<br>Low drive capability         | -   | 290    | -    | nA        |
|                   |                             | LSEDRV[1:0] = 01,<br>Medium Low drive capability  | -   | 390    | -    |           |
|                   |                             | LSEDRV[1:0] = 10,<br>Medium high drive capability | -   | 550    | -    |           |
|                   |                             | LSEDRV[1:0] = 11,<br>High drive capability        | -   | 900    | -    |           |
| $G_{m_{critmax}}$ | Maximum critical crystal gm | LSEDRV[1:0] = 00,<br>Low drive capability         | -   | -      | 0.5  | $\mu A/V$ |
|                   |                             | LSEDRV[1:0] = 01,<br>Medium Low drive capability  | -   | -      | 0.75 |           |
|                   |                             | LSEDRV[1:0] = 10,<br>Medium high drive capability | -   | -      | 1.7  |           |
|                   |                             | LSEDRV[1:0] = 11,<br>High drive capability        | -   | -      | 2.7  |           |
| $t_{SU}^{(3)}$    | Startup time                | VDD is stabilized                                 | -   | 2      | -    | s         |

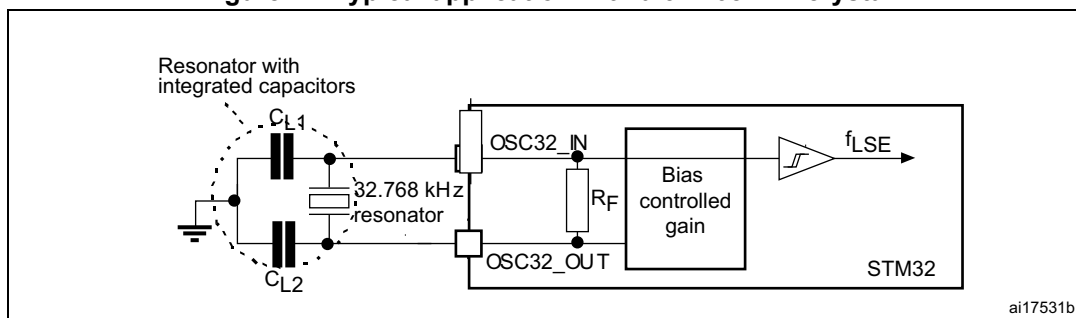
1. Guaranteed by design.

2. Refer to the note and caution paragraphs below the table, and to the application note AN2867 "Oscillator design guide for ST microcontrollers.

3.  $t_{SU}$  is the startup time measured from the moment it is enabled (by software) to a stabilized 32.768k Hz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

**Note:** For information on selecting the crystal, refer to the application note AN2867 “Oscillator design guide for ST microcontrollers” available from the ST website [www.st.com](http://www.st.com).

**Figure 74. Typical application with a 32.768 kHz crystal**



1. An external resistor is not required between OSC32\_IN and OSC32\_OUT and it is forbidden to add one.

### 7.3.9 Internal clock source characteristics

The parameters given in [Table 143](#) to [Table 146](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

#### 48 MHz high-speed internal RC oscillator (HSI48)

**Table 143. HSI48 oscillator characteristics**

| Symbol                              | Parameter                                                                  | Conditions                                                  | Min                 | Typ        | Max                 | Unit          |
|-------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|---------------------|------------|---------------------|---------------|
| $f_{\text{HSI48}}$                  | HSI48 frequency                                                            | $V_{DD}=3.3\text{ V}$ ,<br>$T_J=30\text{ }^{\circ}\text{C}$ | 47.5 <sup>(1)</sup> | 48         | 48.5 <sup>(1)</sup> | MHz           |
| TRIM <sup>(2)</sup>                 | USER trimming step                                                         | -                                                           | -                   | 0.175      | -                   | %             |
| USER TRIM COVERAGE <sup>(3)</sup>   | USER TRIMMING Coverage                                                     | $\pm 32$ steps                                              | $\pm 4.79$          | $\pm 5.60$ | -                   | %             |
| DuCy(HSI48) <sup>(2)</sup>          | Duty Cycle                                                                 | -                                                           | 45                  | -          | 55                  | %             |
| ACCHSI48_REL <sup>(3)(4)</sup>      | Accuracy of the HSI48 oscillator over temperature (factory calibrated)     | $T_J=-40$ to $125\text{ }^{\circ}\text{C}$                  | -4.5                | -          | 3.5                 | %             |
| $\Delta V_{DD}(\text{HSI48})^{(3)}$ | HSI48 oscillator frequency drift with $V_{DD}$ <sup>(5)</sup>              | $V_{DD}=3$ to $3.6\text{ V}$                                | -                   | 0.025      | 0.05                | %             |
|                                     |                                                                            | $V_{DD}=1.62\text{ V}$ to $3.6\text{ V}$                    | -                   | 0.05       | 0.1                 |               |
| $t_{\text{su}}(\text{HSI48})^{(2)}$ | HSI48 oscillator start-up time                                             | -                                                           | -                   | 2.1        | 4.0                 | $\mu\text{s}$ |
| $I_{DD}(\text{HSI48})^{(2)}$        | HSI48 oscillator power consumption                                         | -                                                           | -                   | 350        | 400                 | $\mu\text{A}$ |
| $N_T$ jitter                        | Next transition jitter<br>Accumulated jitter on 28 cycles <sup>(6)</sup>   | -                                                           | -                   | $\pm 0.15$ | -                   | ns            |
| $P_T$ jitter                        | Paired transition jitter<br>Accumulated jitter on 56 cycles <sup>(6)</sup> | -                                                           | -                   | $\pm 0.25$ | -                   | ns            |

1. Guaranteed by test in production.

2. Guaranteed by design.

3. Guaranteed by characterization.

4.  $\Delta f_{\text{HSI}} = \text{ACCHSI48\_REL} + \Delta V_{DD}$ .

5. These values are obtained by using the formula:  $(\text{Freq}(3.6\text{V}) - \text{Freq}(3.0\text{V})) / \text{Freq}(3.0\text{V})$  or  $(\text{Freq}(3.6\text{V}) - \text{Freq}(1.62\text{V})) / \text{Freq}(1.62\text{V})$ .
6. Jitter measurements are performed without clock source activated in parallel.

### 64 MHz high-speed internal RC oscillator (HSI)

**Table 144. HSI oscillator characteristics<sup>(1)</sup>**

| Symbol                             | Parameter                                                                | Conditions                                                                 | Min                 | Typ   | Max                 | Unit          |
|------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------|-------|---------------------|---------------|
| $f_{\text{HSI}}$                   | HSI frequency                                                            | $V_{\text{DD}}=3.3\text{ V}$ , $T_{\text{J}}=30\text{ }^{\circ}\text{C}$   | 63.7 <sup>(2)</sup> | 64    | 64.3 <sup>(2)</sup> | MHz           |
| TRIM                               | HSI user trimming step                                                   | Trimming is not a multiple of 32                                           | -                   | 0.24  | 0.32                | %             |
|                                    |                                                                          | Trimming is 128, 256 and 384                                               | -5.2                | -1.8  | -                   |               |
|                                    |                                                                          | Trimming is 64, 192, 320 and 448                                           | -1.4                | -0.8  | -                   |               |
|                                    |                                                                          | Other trimming are a multiple of 32 (not including multiple of 64 and 128) | -0.6                | -0.25 | -                   |               |
| DuCy(HSI)                          | Duty Cycle                                                               | -                                                                          | 45                  | -     | 55                  | %             |
| $\Delta_{\text{VDD}}(\text{HSI})$  | HSI oscillator frequency drift over $V_{\text{DD}}$ (reference is 3.3 V) | $V_{\text{DD}}=1.62\text{ to }3.6\text{ V}$                                | -0.12               | -     | 0.03                | %             |
| $\Delta_{\text{TEMP}}(\text{HSI})$ | HSI oscillator frequency drift over temperature (reference is 64 MHz)    | $T_{\text{J}}=-20\text{ to }105\text{ }^{\circ}\text{C}$                   | -1 <sup>(3)</sup>   | -     | 1 <sup>(3)</sup>    | %             |
|                                    |                                                                          | $T_{\text{J}}=-40\text{ to }T_{\text{Jmax}}\text{ }^{\circ}\text{C}$       | -2 <sup>(3)</sup>   | -     | 1 <sup>(3)</sup>    |               |
| $t_{\text{su}}(\text{HSI})$        | HSI oscillator start-up time                                             | -                                                                          | -                   | 1.4   | 2                   | $\mu\text{s}$ |
| $t_{\text{stab}}(\text{HSI})$      | HSI oscillator stabilization time                                        | at 1% of target frequency                                                  | -                   | 4     | 8                   | $\mu\text{s}$ |
| $I_{\text{DD}}(\text{HSI})$        | HSI oscillator power consumption                                         | -                                                                          | -                   | 300   | 400                 | $\mu\text{A}$ |

1. Guaranteed by design unless otherwise specified.
2. Guaranteed by test in production.
3. Guaranteed by characterization.

### 4 MHz low-power internal RC oscillator (CSI)

**Table 145. CSI oscillator characteristics<sup>(1)</sup>**

| Symbol                             | Parameter                                           | Conditions                                                               | Min                 | Typ                 | Max                 | Unit |
|------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------|---------------------|---------------------|---------------------|------|
| $f_{\text{CSI}}$                   | CSI frequency                                       | $V_{\text{DD}}=3.3\text{ V}$ , $T_{\text{J}}=30\text{ }^{\circ}\text{C}$ | 3.96 <sup>(2)</sup> | 4                   | 4.04 <sup>(2)</sup> | MHz  |
| TRIM                               | Trimming step                                       | -                                                                        | -                   | 0.35                | -                   | %    |
| DuCy(CSI)                          | Duty Cycle                                          | -                                                                        | 45                  | -                   | 55                  | %    |
| $\Delta_{\text{TEMP}}(\text{CSI})$ | CSI oscillator frequency drift over temperature     | $T_{\text{J}} = 0\text{ to }85\text{ }^{\circ}\text{C}$                  | -                   | -3.7 <sup>(3)</sup> | 4.5 <sup>(3)</sup>  | %    |
|                                    |                                                     | $T_{\text{J}} = -40\text{ to }125\text{ }^{\circ}\text{C}$               | -                   | -11 <sup>(3)</sup>  | 7.5 <sup>(3)</sup>  |      |
| $\Delta_{\text{VDD}}(\text{CSI})$  | CSI oscillator frequency drift over $V_{\text{DD}}$ | $V_{\text{DD}} = 1.62\text{ to }3.6\text{ V}$                            | -                   | -0.06               | 0.06                | %    |

Table 145. CSI oscillator characteristics<sup>(1)</sup> (continued)

| Symbol          | Parameter                                                            | Conditions | Min | Typ | Max | Unit    |
|-----------------|----------------------------------------------------------------------|------------|-----|-----|-----|---------|
| $t_{su(CSI)}$   | CSI oscillator startup time                                          | -          | -   | 1   | 2   | $\mu s$ |
| $t_{stab(CSI)}$ | CSI oscillator stabilization time (to reach $\pm 3\%$ of $f_{CSI}$ ) | -          | -   | -   | 4   | cycle   |
| $I_{DD(CSI)}$   | CSI oscillator power consumption                                     | -          | -   | 23  | 30  | $\mu A$ |

1. Guaranteed by design.
2. Guaranteed by test in production.
3. Guaranteed by characterization.

### Low-speed internal (LSI) RC oscillator

Table 146. LSI oscillator characteristics

| Symbol                | Parameter                                             | Conditions                                                                                 | Min                  | Typ | Max                 | Unit    |
|-----------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------|-----|---------------------|---------|
| $f_{LSI}$             | LSI frequency                                         | $V_{DD} = 3.3\text{ V}$ , $T_J = 25\text{ }^{\circ}\text{C}$                               | 31.4 <sup>(1)</sup>  | 32  | 32.6 <sup>(1)</sup> | kHz     |
|                       |                                                       | $T_J = -40\text{ to }110\text{ }^{\circ}\text{C}$ , $V_{DD} = 1.62\text{ to }3.6\text{ V}$ | 29.76 <sup>(2)</sup> | -   | 33.6 <sup>(2)</sup> |         |
|                       |                                                       | $T_J = -40\text{ to }125\text{ }^{\circ}\text{C}$ , $V_{DD} = 1.62\text{ to }3.6\text{ V}$ | 29.4                 | -   | 33.6                |         |
| $t_{su(LSI)}^{(3)}$   | LSI oscillator startup time                           | -                                                                                          | -                    | 80  | 130                 | $\mu s$ |
| $t_{stab(LSI)}^{(3)}$ | LSI oscillator stabilization time (5% of final value) | -                                                                                          | -                    | 120 | 170                 |         |
| $I_{DD(LSI)}^{(3)}$   | LSI oscillator power consumption                      | -                                                                                          | -                    | 130 | 280                 | nA      |

1. Guaranteed by test in production.
2. Guaranteed by characterization results.
3. Guaranteed by design.



### 7.3.10 PLL characteristics

The parameters given in [Table 147](#) are derived from tests performed under temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

**Table 147. PLL characteristics (wide VCO frequency range)<sup>(1)</sup>**

| Symbol              | Parameter                            | Conditions                              | Min              | Typ               | Max                | Unit     |
|---------------------|--------------------------------------|-----------------------------------------|------------------|-------------------|--------------------|----------|
| $f_{PLL\_IN}$       | PLL input clock                      | -                                       | 2                | -                 | 16                 | MHz      |
|                     | PLL input clock duty cycle           | -                                       | 10               | -                 | 90                 | %        |
| $f_{PLL\_P\_OUT}$   | PLL multiplier output clock P        | VOS0                                    | 1.5              | -                 | 480 <sup>(2)</sup> | MHz      |
|                     |                                      | VOS1                                    | 1.5              | -                 | 400 <sup>(2)</sup> |          |
|                     |                                      | VOS2                                    | 1.5              | -                 | 300 <sup>(2)</sup> |          |
|                     |                                      | VOS3                                    | 1.5              | -                 | 200 <sup>(2)</sup> |          |
| $f_{VCO\_OUT}$      | PLL VCO output                       | -                                       | 192              | -                 | 960                |          |
| $t_{LOCK}$          | PLL lock time                        | Normal mode                             | -                | 50 <sup>(3)</sup> | 150 <sup>(3)</sup> | $\mu s$  |
|                     |                                      | Sigma-delta mode<br>(CKIN $\geq$ 8 MHz) | -                | 58 <sup>(3)</sup> | 166 <sup>(3)</sup> |          |
| Jitter              | Cycle-to-cycle jitter <sup>(4)</sup> | -                                       | VCO =<br>192 MHz | -                 | 134                | $\pm ps$ |
|                     |                                      |                                         | VCO =<br>200 MHz | -                 | 134                |          |
|                     |                                      |                                         | VCO =<br>400 MHz | -                 | 76                 |          |
|                     |                                      |                                         | VCO =<br>800 MHz | -                 | 39                 |          |
|                     | Long term jitter                     | Normal mode                             | VCO =<br>800 MHz | -                 | $\pm 0.7$          | %        |
|                     |                                      | Sigma-delta<br>mode (CKIN =<br>16 MHz)  | VCO =<br>800 MHz | -                 | $\pm 0.8$          |          |
| $I_{DD(PLL)}^{(3)}$ | PLL power consumption on $V_{DD}$    | VCO freq =<br>836 MHz                   | $V_{DDA}$        | -                 | 590                | $\mu A$  |
|                     |                                      |                                         | $V_{CORE}$       | -                 | 720                |          |
|                     |                                      | VCO freq =<br>192 MHz                   | $V_{DDA}$        | -                 | 180                |          |
|                     |                                      |                                         | $V_{CORE}$       | -                 | 280                |          |

1. Guaranteed by design unless otherwise specified.

2. This value must be limited to the maximum frequency due to the product limitation (480 MHz for VOS0, 400 MHz for VOS1, 300 MHz for VOS2, 200 MHz for VOS3).

3. Guaranteed by characterization results.

4. Integer mode only.

Table 148. PLL characteristics (medium VCO frequency range)<sup>(1)</sup>

| Symbol                | Parameter                                | Conditions                    |               | Min       | Typ               | Max                | Unit |
|-----------------------|------------------------------------------|-------------------------------|---------------|-----------|-------------------|--------------------|------|
| f <sub>PLL_IN</sub>   | PLL input clock                          | -                             |               | 1         | -                 | 2                  | MHz  |
|                       | PLL input clock duty cycle               | -                             |               | 10        | -                 | 90                 | %    |
| f <sub>PLL_OUT</sub>  | PLL multiplier output clock P, Q, R      | VOS1                          |               | 1.17      | -                 | 210                | MHz  |
|                       |                                          | VOS2                          |               | 1.17      | -                 | 210                |      |
|                       |                                          | VOS3                          |               | 1.17      | -                 | 200                |      |
| f <sub>VCO_OUT</sub>  | PLL VCO output                           | -                             |               | 150       | -                 | 420                |      |
| t <sub>LOCK</sub>     | PLL lock time                            | Normal mode                   |               | -         | 60 <sup>(2)</sup> | 100 <sup>(2)</sup> | μs   |
|                       |                                          | Sigma-delta mode              |               | forbidden |                   |                    |      |
| Jitter                | Cycle-to-cycle jitter <sup>(3)</sup>     | -                             | VCO = 150 MHz | -         | 145               | -                  | ±ps  |
|                       |                                          |                               | VCO = 300 MHz | -         | 91                | -                  |      |
|                       |                                          |                               | VCO = 400 MHz | -         | 64                | -                  |      |
|                       |                                          |                               | VCO = 420 MHz | -         | 63                | -                  |      |
|                       | Period jitter                            | f <sub>PLL_OUT</sub> = 50 MHz | VCO = 150 MHz | -         | 55                | -                  | ±-ps |
|                       |                                          |                               | VCO = 400 MHz | -         | 30                | -                  |      |
|                       | Long term jitter                         | Normal mode                   | VCO = 400 MHz | -         | ±0.3              | -                  | %    |
| I(PLL) <sup>(2)</sup> | PLL power consumption on V <sub>DD</sub> | VCO freq = 420MHz             | VDD           | -         | 440               | 1150               | μA   |
|                       |                                          |                               | VCORE         | -         | 530               | -                  |      |
|                       |                                          | VCO freq = 150MHz             | VDD           | -         | 180               | 500                |      |
|                       |                                          |                               | VCORE         | -         | 200               | -                  |      |

1. Guaranteed by design unless otherwise specified.

2. Guaranteed by characterization results.

3. Integer mode only.

### 7.3.11 Memory characteristics

#### Flash memory

The characteristics are given at  $T_J = -40$  to  $125\text{ }^{\circ}\text{C}$  unless otherwise specified.

The devices are shipped to customers with the Flash memory erased.

**Table 149. Flash memory characteristics**

| Symbol   | Parameter      | Conditions                | Min | Typ  | Max | Unit |
|----------|----------------|---------------------------|-----|------|-----|------|
| $I_{DD}$ | Supply current | Write / Erase 8-bit mode  | -   | 6.5  | -   | mA   |
|          |                | Write / Erase 16-bit mode | -   | 11.5 | -   |      |
|          |                | Write / Erase 32-bit mode | -   | 20   | -   |      |
|          |                | Write / Erase 64-bit mode | -   | 35   | -   |      |

**Table 150. Flash memory programming (single bank configuration nDBANK=1)**

| Symbol           | Parameter                        | Conditions                     | Min <sup>(1)</sup> | Typ | Max <sup>(1)</sup> | Unit          |
|------------------|----------------------------------|--------------------------------|--------------------|-----|--------------------|---------------|
| $t_{prog}$       | Word (266 bits) programming time | Program/erase parallelism x 8  | -                  | 290 | 580 <sup>(2)</sup> | $\mu\text{s}$ |
|                  |                                  | Program/erase parallelism x 16 | -                  | 180 | 360                |               |
|                  |                                  | Program/erase parallelism x 32 | -                  | 130 | 260                |               |
|                  |                                  | Program/erase parallelism x 64 | -                  | 100 | 200                |               |
| $t_{ERASE128KB}$ | Sector (128 KB) erase time       | Program/erase parallelism x 8  | -                  | 2   | 4                  | s             |
|                  |                                  | Program/erase parallelism x 16 | -                  | 1.8 | 3.6                |               |
|                  |                                  | Program/erase parallelism x 32 | -                  |     |                    |               |
| $t_{ME}$         | Mass erase time                  | Program/erase parallelism x 8  | -                  | 13  | 26                 |               |
|                  |                                  | Program/erase parallelism x 16 | -                  | 8   | 16                 |               |
|                  |                                  | Program/erase parallelism x 32 | -                  | 6   | 12                 |               |
|                  |                                  | Program/erase parallelism x 64 | -                  | 5   | 10                 |               |
| $V_{prog}$       | Programming voltage              | Program parallelism x 8        | 1.62               | -   | 3.6                | V             |
|                  |                                  | Program parallelism x 16       |                    |     |                    |               |
|                  |                                  | Program parallelism x 32       |                    |     |                    |               |
|                  |                                  | Program parallelism x 64       | 1.8                | -   | 3.6                |               |

1. Guaranteed by characterization results.

2. The maximum programming time is measured after 10K erase operations.

Table 151. Flash memory endurance and data retention

| Symbol           | Parameter      | Conditions                                          | Value              | Unit    |
|------------------|----------------|-----------------------------------------------------|--------------------|---------|
|                  |                |                                                     | Min <sup>(1)</sup> |         |
| N <sub>END</sub> | Endurance      | T <sub>J</sub> = -40 to +125 °C (6 suffix versions) | 10                 | kcycles |
| t <sub>RET</sub> | Data retention | 1 kcycle at T <sub>A</sub> = 85 °C                  | 30                 | Years   |
|                  |                | 10 kcycles at T <sub>A</sub> = 55 °C                | 20                 |         |

1. Guaranteed by characterization results.

### 7.3.12 EMC characteristics

Susceptibility tests are performed on a sample basis during device characterization.

#### Functional EMS (electromagnetic susceptibility)

While a simple application is executed on the device (toggling 2 LEDs through I/O ports), the device is stressed by two electromagnetic events until a failure occurs. The failure is indicated by the LEDs:

- **Electrostatic discharge (ESD)** (positive and negative) is applied to all device pins until a functional disturbance occurs. This test is compliant with the IEC 61000-4-2 standard.
- **FTB**: A burst of fast transient voltage (positive and negative) is applied to V<sub>DD</sub> and V<sub>SS</sub> through a 100 pF capacitor, until a functional disturbance occurs. This test is compliant with the IEC 61000-4-4 standard.

A device reset allows normal operations to be resumed.

The test results are given in [Table 152](#). They are based on the EMS levels and classes defined in application note AN1709.

Table 152. EMS characteristics

| Symbol            | Parameter                                                                                                                                       | Conditions                                                                                                           | Level/Class |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------|
| V <sub>FESD</sub> | Voltage limits to be applied on any I/O pin to induce a functional disturbance                                                                  | V <sub>DD</sub> = 3.3 V, T <sub>A</sub> = +25 °C, UFBGA240, f <sub>rcc_ck</sub> = 400 MHz, conforms to IEC 61000-4-2 | 3B          |
| V <sub>FTB</sub>  | Fast transient voltage burst limits to be applied through 100 pF on V <sub>DD</sub> and V <sub>SS</sub> pins to induce a functional disturbance |                                                                                                                      | 5A          |

As a consequence, it is recommended to add a serial resistor (1 kΩ) located as close as possible to the MCU to the pins exposed to noise (connected to tracks longer than 50 mm on PCB).

## Designing hardened software to avoid noise problems

EMC characterization and optimization are performed at component level with a typical application environment and simplified MCU software. It should be noted that good EMC performance is highly dependent on the user application and the software in particular.

Therefore it is recommended that the user applies EMC software optimization and prequalification tests in relation with the EMC level requested for his application.

### Software recommendations

The software flowchart must include the management of runaway conditions such as:

- Corrupted program counter
- Unexpected reset
- Critical Data corruption (control registers...)

### Prequalification trials

Most of the common failures (unexpected reset and program counter corruption) can be reproduced by manually forcing a low state on the NRST pin or the Oscillator pins for 1 second.

To complete these trials, ESD stress can be applied directly on the device, over the range of specification values. When unexpected behavior is detected, the software can be hardened to prevent unrecoverable errors occurring (see application note AN1015).

## Electromagnetic Interference (EMI)

The electromagnetic field emitted by the device are monitored while a simple application, executing EEMBC code, is running. This emission test is compliant with SAE IEC61967-2 standard which specifies the test board and the pin loading.

**Table 153. EMI characteristics**

| Symbol           | Parameter  | Conditions                                                                                  | Monitored frequency band | Max vs. $[f_{HSE}/f_{CPU}]$ | Unit |
|------------------|------------|---------------------------------------------------------------------------------------------|--------------------------|-----------------------------|------|
|                  |            |                                                                                             |                          | 8/400 MHz                   |      |
| S <sub>EMI</sub> | Peak level | V <sub>DD</sub> = 3.6 V, T <sub>A</sub> = 25 °C, UFBGA240 package, conforming to IEC61967-2 | 0.1 to 30 MHz            | 11                          | dBμV |
|                  |            |                                                                                             | 30 to 130 MHz            | 6                           |      |
|                  |            |                                                                                             | 130 MHz to 1 GHz         | 12                          |      |
|                  |            |                                                                                             | 1 GHz to 2 GHz           | 7                           |      |
|                  |            |                                                                                             | EMI Level                | 2.5                         | -    |

### 7.3.13 Absolute maximum ratings (electrical sensitivity)

Based on three different tests (ESD, LU) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

#### Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse) are applied to the pins of each sample according to each pin combination. This test conforms to the ANSI/ESDA/JEDEC JS-001 and ANSI/ESDA/JEDEC JS-002 standards.

**Table 154. ESD absolute maximum ratings**

| Symbol                | Ratings                                               | Conditions                                                 | Packages | Class | Maximum value <sup>(1)</sup> | Unit |
|-----------------------|-------------------------------------------------------|------------------------------------------------------------|----------|-------|------------------------------|------|
| $V_{\text{ESD(HBM)}}$ | Electrostatic discharge voltage (human body model)    | $T_A = +25\text{ °C}$ conforming to ANSI/ESDA/JEDEC JS-001 | All      | 1C    | 1000                         | V    |
| $V_{\text{ESD(CDM)}}$ | Electrostatic discharge voltage (charge device model) | $T_A = +25\text{ °C}$ conforming to ANSI/ESDA/JEDEC JS-002 | All      | C1    | 250                          |      |

1. Guaranteed by characterization results.

#### Static latchup

Two complementary static tests are required on six parts to assess the latchup performance:

- A supply overvoltage is applied to each power supply pin
- A current injection is applied to each input, output and configurable I/O pin

These tests are compliant with JESD78 IC latchup standard.

**Table 155. Electrical sensitivities**

| Symbol | Parameter            | Conditions                                 | Class      |
|--------|----------------------|--------------------------------------------|------------|
| LU     | Static latchup class | $T_A = +25\text{ °C}$ conforming to JESD78 | II level A |

### 7.3.14 I/O current injection characteristics

As a general rule, a current injection to the I/O pins, due to external voltage below  $V_{SS}$  or above  $V_{DD}$  (for standard, 3.3 V-capable I/O pins) should be avoided during the normal product operation. However, in order to give an indication of the robustness of the microcontroller in cases when an abnormal injection accidentally happens, susceptibility tests are performed on a sample basis during the device characterization.

#### Functional susceptibility to I/O current injection

While a simple application is executed on the device, the device is stressed by injecting current into the I/O pins programmed in floating input mode. While current is injected into the I/O pin, one at a time, the device is checked for functional failures.

The failure is indicated by an out of range parameter: ADC error above a certain limit (higher than 5 LSB TUE), out of conventional limits of induced leakage current on adjacent pins (out of  $-5\ \mu\text{A}/+0\ \mu\text{A}$  range), or other functional failure (for example reset, oscillator frequency deviation).

The following tables are the compilation of the SIC1/SIC2 and functional ESD results.

Negative induced A negative induced leakage current is caused by negative injection and positive induced leakage current by positive injection.

**Table 156. I/O current injection susceptibility<sup>(1)</sup>**

| Symbol    | Description                                                              | Functional susceptibility |                    | Unit |
|-----------|--------------------------------------------------------------------------|---------------------------|--------------------|------|
|           |                                                                          | Negative injection        | Positive injection |      |
| $I_{INJ}$ | PA7, PC5, PG1, PB14, PJ7, PA11, PA12, PA13, PA14, PA15, PJ12, PB4        | 5                         | 0                  | mA   |
|           | PA2, PH2, PH3, PE8, PA6, PA7, PC4, PE7, PE10, PE11                       | 0                         | NA                 |      |
|           | PA0, PA_C, PA1, PA1_C, PC2, PC2_C, PC3, PC3_C, PA4, PA5, PH4, PH5, BOOT0 | 0                         | 0                  |      |
|           | All other I/Os                                                           | 5                         | NA                 |      |

1. Guaranteed by characterization.

### 7.3.15 I/O port characteristics

#### General input/output characteristics

Unless otherwise specified, the parameters given in [Table 157: I/O static characteristics](#) are derived from tests performed under the conditions summarized in [Table 122: General operating conditions](#). All I/Os are CMOS and TTL compliant (except for BOOT0).

**Table 157. I/O static characteristics**

| Symbol           | Parameter                                                | Condition                                                               | Min                       | Typ | Max                      | Unit |
|------------------|----------------------------------------------------------|-------------------------------------------------------------------------|---------------------------|-----|--------------------------|------|
| $V_{IL}$         | I/O input low level voltage except BOOT0                 | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                              | -                         | -   | $0.3V_{DD}^{(1)}$        | V    |
|                  | I/O input low level voltage except BOOT0                 |                                                                         | -                         | -   | $0.4V_{DD} - 0.1^{(2)}$  |      |
|                  | BOOT0 I/O input low level voltage                        |                                                                         | -                         | -   | $0.19V_{DD} + 0.1^{(2)}$ |      |
| $V_{IH}$         | I/O input high level voltage except BOOT0                | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                              | $0.7V_{DD}^{(1)}$         | -   | -                        | V    |
|                  | I/O input high level voltage except BOOT0 <sup>(3)</sup> |                                                                         | $0.47V_{DD} + 0.25^{(2)}$ | -   | -                        |      |
|                  | BOOT0 I/O input high level voltage <sup>(3)</sup>        |                                                                         | $0.17V_{DD} + 0.6^{(2)}$  | -   | -                        |      |
| $V_{HYS}^{(2)}$  | TT_xx, FT_xxx and NRST I/O input hysteresis              | $1.62\text{ V} < V_{DDIOx} < 3.6\text{ V}$                              | -                         | 250 | -                        | mV   |
|                  | BOOT0 I/O input hysteresis                               |                                                                         | -                         | 200 | -                        |      |
| $I_{leak}^{(4)}$ | FT_xx Input leakage current <sup>(2)</sup>               | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                           | -                         | -   | +/-250                   | nA   |
|                  |                                                          | $\text{Max}(V_{DDXXX}) < V_{IN} \leq 5.5\text{ V}$ <sup>(5)(6)(9)</sup> | -                         | -   | 1500                     |      |
|                  | FT_u IO                                                  | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                           | -                         | -   | +/- 350                  |      |
|                  |                                                          | $\text{Max}(V_{DDXXX}) < V_{IN} \leq 5.5\text{ V}$ <sup>(5)(6)(9)</sup> | -                         | -   | 5000 <sup>(7)</sup>      |      |
|                  | TT_xx Input leakage current                              | $0 < V_{IN} \leq \text{Max}(V_{DDXXX})^{(9)}$                           | -                         | -   | +/-250                   |      |
|                  | VPP (BOOT0 alternate function)                           | $0 < V_{IN} \leq V_{DDIOx}$                                             | -                         | -   | 15                       |      |
|                  |                                                          | $V_{DDIOx} < V_{IN} \leq 9\text{ V}$                                    | -                         | -   | 35                       |      |
| $R_{PU}$         | Weak pull-up equivalent resistor <sup>(8)</sup>          | $V_{IN} = V_{SS}$                                                       | 30                        | 40  | 50                       | kΩ   |
| $R_{PD}$         | Weak pull-down equivalent resistor <sup>(8)</sup>        | $V_{IN} = V_{DD}^{(9)}$                                                 | 30                        | 40  | 50                       |      |
| $C_{IO}$         | I/O pin capacitance                                      | -                                                                       | -                         | 5   | -                        | pF   |

1. Compliant with CMOS requirements.

2. Guaranteed by design.

3.  $V_{DDIOx}$  represents  $V_{DDIO1}$ ,  $V_{DDIO2}$  or  $V_{DDIO3}$ .  $V_{DDIOx} = V_{DD}$ .

4. This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula:  $I_{Total\_leak\_max} = 10\text{ }\mu\text{A} + [\text{number of I/Os where } V_{IN} \text{ is applied on the pad}] \times I_{lk}(Max)$ .

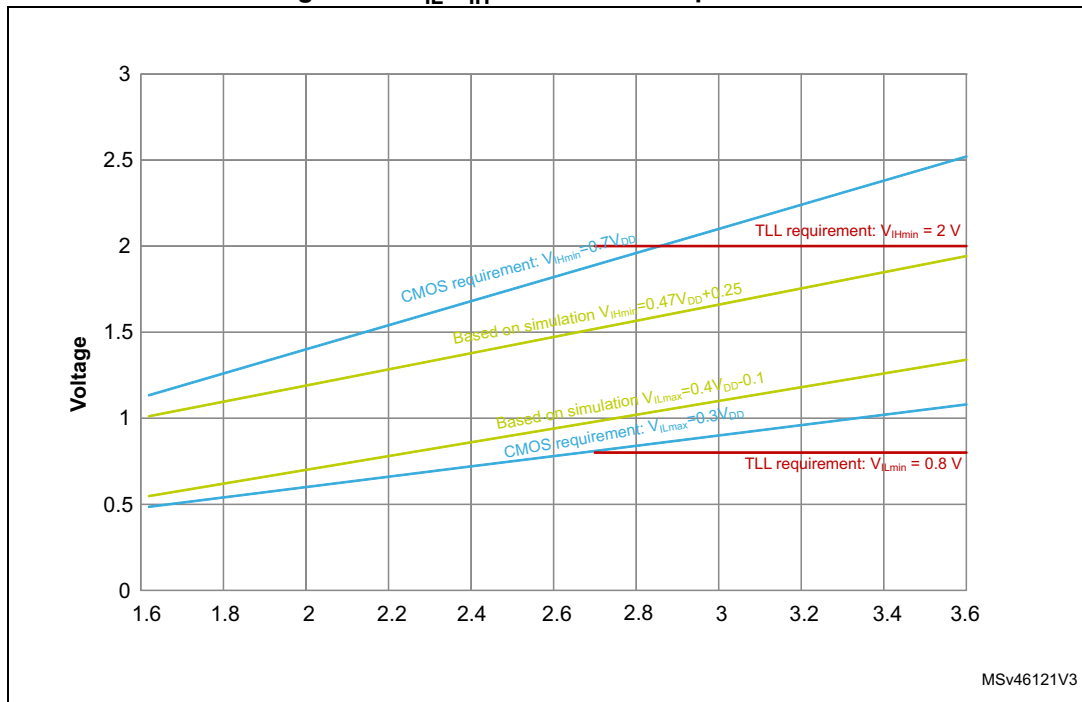
5. All FT\_xx IO except FT\_lu, FT\_u and PC3.



6.  $V_{IN}$  must be less than  $\text{Max}(V_{DDXXX}) + 3.6 \text{ V}$ .
7. To sustain a voltage higher than  $\text{MIN}(V_{DD}, V_{DDA}, V_{DD33USB}) + 0.3 \text{ V}$ , the internal pull-up and pull-down resistors must be disabled.
8. The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).
9.  $\text{Max}(V_{DDXXX})$  is the maximum value of all the I/O supplies.

All I/Os are CMOS and TTL compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements for FT I/Os is shown in [Figure 75](#).

**Figure 75.  $V_{IL}/V_{IH}$  for all I/Os except BOOT0**



### Output driving current

The GPIOs (general purpose input/outputs) can sink or source up to  $\pm 8 \text{ mA}$ , and sink or source up to  $\pm 20 \text{ mA}$  (with a relaxed  $V_{OL}/V_{OH}$ ).

In the user application, the number of I/O pins which can drive current must be limited to respect the absolute maximum rating specified in [Section 7.2](#). In particular:

- The sum of the currents sourced by all the I/Os on  $V_{DD}$ , plus the maximum Run consumption of the MCU sourced on  $V_{DD}$ , cannot exceed the absolute maximum rating  $\Sigma I_{VDD}$  (see [Table 120](#)).
- The sum of the currents sunk by all the I/Os on  $V_{SS}$  plus the maximum Run consumption of the MCU sunk on  $V_{SS}$  cannot exceed the absolute maximum rating  $\Sigma I_{VSS}$  (see [Table 120](#)).

### Output voltage levels

Unless otherwise specified, the parameters given in [Table 158: Output voltage characteristics for all I/Os except PC13, PC14, PC15 and PI8](#) and [Table 159: Output voltage characteristics for PC13, PC14, PC15 and PI8](#) are derived from tests performed under ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#). All I/Os are CMOS and TTL compliant.

**Table 158. Output voltage characteristics for all I/Os except PC13, PC14, PC15 and PI8<sup>(1)</sup>**

| Symbol            | Parameter                                               | Conditions <sup>(3)</sup>                                                                         | Min          | Max | Unit |
|-------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------|-----|------|
| $V_{OL}$          | Output low level voltage                                | CMOS port <sup>(2)</sup><br>$I_{IO}=8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -            | 0.4 | V    |
| $V_{OH}$          | Output high level voltage                               | CMOS port <sup>(2)</sup><br>$I_{IO}=-8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ | $V_{DD}-0.4$ | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | TTL port <sup>(2)</sup><br>$I_{IO}=8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$   | -            | 0.4 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | TTL port <sup>(2)</sup><br>$I_{IO}=-8\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | 2.4          | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | $I_{IO}=20\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 1.3 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | $I_{IO}=-20\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                            | $V_{DD}-1.3$ | -   |      |
| $V_{OL}^{(3)}$    | Output low level voltage                                | $I_{IO}=4\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 0.4 |      |
| $V_{OH}^{(3)}$    | Output high level voltage                               | $I_{IO}=-4\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} < 3.6\text{ V}$                               | $V_{DD}-0.4$ | -   |      |
| $V_{OLFM+}^{(3)}$ | Output low level voltage for an FTf I/O pin in FM+ mode | $I_{IO}=20\text{ mA}$<br>$2.3\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                             | -            | 0.4 |      |
|                   |                                                         | $I_{IO}=10\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                            | -            | 0.4 |      |

1. The I/O current sourced or sunk by the device must always respect the absolute maximum rating specified in [Table 119: Voltage characteristics](#), and the sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings  $\Sigma I_{IO}$ .
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Guaranteed by design.

Table 159. Output voltage characteristics for PC13, PC14, PC15 and PI8<sup>(1)</sup>

| Symbol         | Parameter                 | Conditions <sup>(3)</sup>                                                                         | Min          | Max | Unit |
|----------------|---------------------------|---------------------------------------------------------------------------------------------------|--------------|-----|------|
| $V_{OL}$       | Output low level voltage  | CMOS port <sup>(2)</sup><br>$I_{IO}=3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | -            | 0.4 | V    |
| $V_{OH}$       | Output high level voltage | CMOS port <sup>(2)</sup><br>$I_{IO}=-3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$ | $V_{DD}-0.4$ | -   |      |
| $V_{OL}^{(3)}$ | Output low level voltage  | TTL port <sup>(2)</sup><br>$I_{IO}=3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$   | -            | 0.4 |      |
| $V_{OH}^{(2)}$ | Output high level voltage | TTL port <sup>(2)</sup><br>$I_{IO}=-3\text{ mA}$<br>$2.7\text{ V} \leq V_{DD} \leq 3.6\text{ V}$  | 2.4          | -   |      |
| $V_{OL}^{(2)}$ | Output low level voltage  | $I_{IO}=1.5\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                           | -            | 0.4 |      |
| $V_{OH}^{(2)}$ | Output high level voltage | $I_{IO}=-1.5\text{ mA}$<br>$1.62\text{ V} \leq V_{DD} \leq 3.6\text{ V}$                          | $V_{DD}-0.4$ | -   |      |

1. The I/O current sourced or sunk by the device must always respect the absolute maximum rating specified in [Table 119: Voltage characteristics](#), and the sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always respect the absolute maximum ratings  $\Sigma I_{IO}$ .
2. TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.
3. Guaranteed by design.

**Output buffer timing characteristics (HSLV option disabled)**

The HSLV bit of SYSCFG\_CCCSR register can be used to optimize the I/O speed when the product voltage is below 2.7 V.

**Table 160. Output timing characteristics (HSLV OFF)<sup>(1)(2)</sup>**

| Speed | Symbol           | Parameter                                                                 | conditions                                | Min | Max  | Unit |
|-------|------------------|---------------------------------------------------------------------------|-------------------------------------------|-----|------|------|
| 00    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 12   | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 3    |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 12   |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 3    |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 16   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 4    |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 16.6 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 33.3 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 13.3 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 25   |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 10   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 20   |      |
| 01    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 60   | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 15   |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 80   |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 15   |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 110  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 20   |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 5.2  | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 10   |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 4.2  |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 7.5  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V  | -   | 2.8  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -   | 5.2  |      |

Table 160. Output timing characteristics (HSLV OFF)<sup>(1)(2)</sup> (continued)

| Speed | Symbol           | Parameter                                                                 | conditions                                               | Min | Max | Unit |
|-------|------------------|---------------------------------------------------------------------------|----------------------------------------------------------|-----|-----|------|
| 10    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 85  | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 35  |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 110 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 40  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 166 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 100 |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 3.8 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 6.9 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 2.8 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 5.2 |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 1.8 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 3.3 |      |
| 11    | $F_{\max}^{(3)}$ | Maximum frequency                                                         | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 100 | MHz  |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 50  |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 133 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 66  |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 220 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 85  |      |
|       | $t_r/t_f^{(4)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 3.3 | ns   |
|       |                  |                                                                           | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 6.6 |      |
|       |                  |                                                                           | C=30 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 2.4 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 4.5 |      |
|       |                  |                                                                           | C=10 pF, 2.7 V ≤ V <sub>DD</sub> ≤ 3.6 V <sup>(5)</sup>  | -   | 1.5 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(5)</sup> | -   | 2.7 |      |

1. Guaranteed by design.
2. The frequency of the GPIOs that can be supplied in V<sub>BAT</sub> mode (PC13, PC14, PC15 and PI8) is limited to 2 MHz
3. The maximum frequency is defined with the following conditions:  
 $(t_r + t_f) \leq 2/3 T$   
Skew ≤ 1/20 T  
45% < Duty cycle < 55%
4. The fall and rise times are defined between 90% and 10% and between 10% and 90% of the output waveform, respectively.
5. Compensation system enabled.

## Output buffer timing characteristics (HSLV option enabled)

Table 161. Output timing characteristics (HSLV ON)<sup>(1)</sup>

| Speed | Symbol           | Parameter                                                                 | conditions                                               | Min | Max | Unit |
|-------|------------------|---------------------------------------------------------------------------|----------------------------------------------------------|-----|-----|------|
| 00    | $F_{\max}^{(2)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 10  |      |
|       | $t_r/t_f^{(3)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 11  | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 9   |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 6.6 |      |
| 01    | $F_{\max}^{(2)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 50  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 58  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 66  |      |
|       | $t_r/t_f^{(3)}$  | Output high to low level fall time and output low to high level rise time | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 6.6 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 4.8 |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V                | -   | 3   |      |
| 10    | $F_{\max}^{(2)}$ | Maximum frequency                                                         | C=50 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 55  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 80  |      |
|       |                  |                                                                           | C=10 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 133 |      |
|       | $t_r/t_f^{(3)}$  | Output high to low level fall time and output low to high level rise time | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 5.8 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 4   |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 2.4 |      |
| 11    | $F_{\max}^{(2)}$ | Maximum frequency                                                         | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 60  | MHz  |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 90  |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 175 |      |
|       | $t_r/t_f^{(3)}$  | Output high to low level fall time and output low to high level rise time | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 5.3 | ns   |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 3.6 |      |
|       |                  |                                                                           | C=30 pF, 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V <sup>(4)</sup> | -   | 1.9 |      |

1. Guaranteed by design.

2. The maximum frequency is defined with the following conditions:  
 $(t_r + t_f) \leq 2/3 T$   
 Skew ≤ 1/20 T  
 45% < Duty cycle < 55%

3. The fall and rise times are defined between 90% and 10% and between 10% and 90% of the output waveform, respectively.

4. Compensation system enabled.

### 7.3.16 NRST pin characteristics

The NRST pin input driver uses CMOS technology. It is connected to a permanent pull-up resistor,  $R_{PU}$  (see [Table 157: I/O static characteristics](#)).

Unless otherwise specified, the parameters given in [Table 162](#) are derived from tests performed under the ambient temperature and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

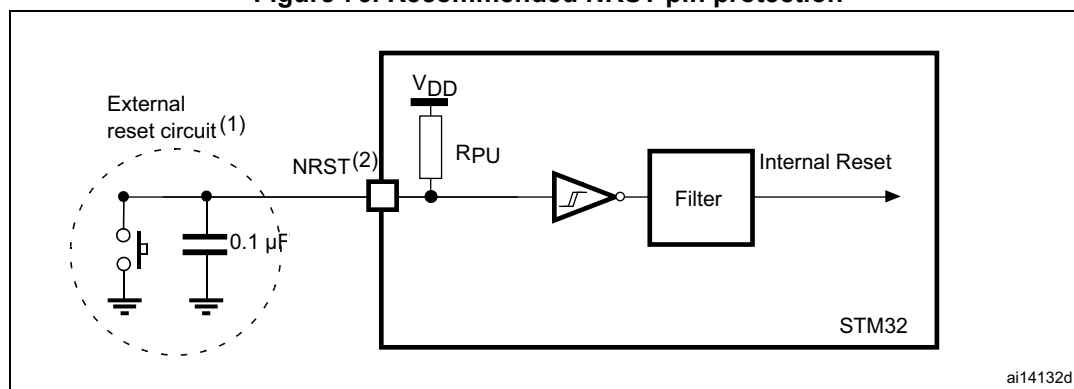
**Table 162. NRST pin characteristics**

| Symbol               | Parameter                                       | Conditions                              | Min  | Typ | Max | Unit       |
|----------------------|-------------------------------------------------|-----------------------------------------|------|-----|-----|------------|
| $R_{PU}^{(2)}$       | Weak pull-up equivalent resistor <sup>(1)</sup> | $V_{IN} = V_{SS}$                       | 30   | 40  | 50  | k $\Omega$ |
| $V_{F(NRST)}^{(2)}$  | NRST Input filtered pulse                       | $1.71\text{ V} < V_{DD} < 3.6\text{ V}$ | -    | -   | 50  | ns         |
| $V_{NF(NRST)}^{(2)}$ | NRST Input not filtered pulse                   | $1.71\text{ V} < V_{DD} < 3.6\text{ V}$ | 300  | -   | -   |            |
|                      |                                                 | $1.62\text{ V} < V_{DD} < 3.6\text{ V}$ | 1000 | -   | -   |            |

1. The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance must be minimum (~10% order).

2. Guaranteed by design.

**Figure 76. Recommended NRST pin protection**



1. The reset network protects the device against parasitic resets.
2. The user must ensure that the level on the NRST pin can go below the  $V_{IL(NRST)}$  max level specified in [Table 157](#). Otherwise the reset is not taken into account by the device.

### 7.3.17 FMC characteristics

Unless otherwise specified, the parameters given in [Table 163](#) to [Table 176](#) for the FMC interface are derived from tests performed under the ambient temperature,  $f_{HCLK}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 11$
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- IO Compensation cell activated.
- HSLV activated when  $V_{DD} \leq 2.7\text{ V}$
- VOS level set to VOS1.

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics.

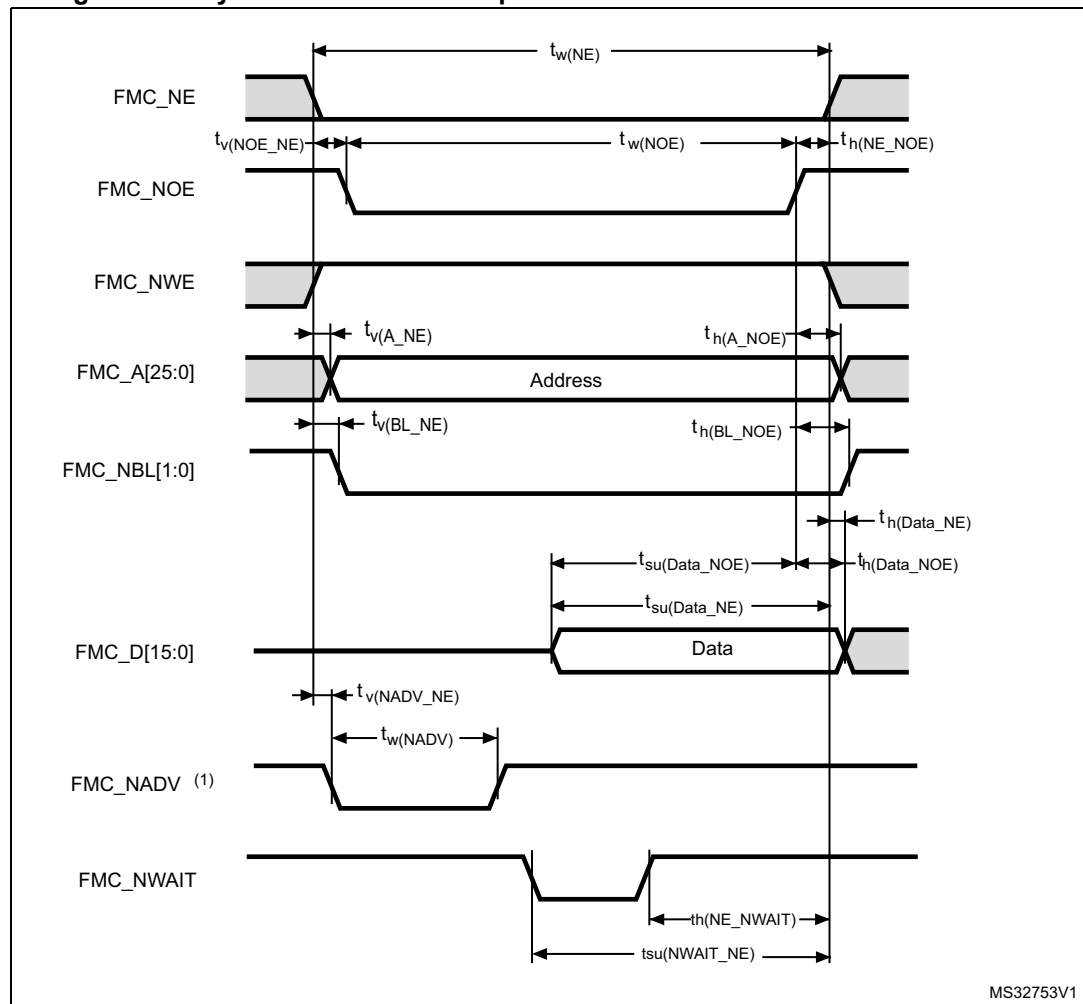
### Asynchronous waveforms and timings

[Figure 77](#) through [Figure 79](#) represent asynchronous waveforms and [Table 163](#) through [Table 170](#) provide the corresponding timings. The results shown in these tables are obtained with the following FMC configuration:

- AddressSetupTime = 0x1
- AddressHoldTime = 0x1
- DataSetupTime = 0x1 (except for asynchronous NWAIT mode , DataSetupTime = 0x5)
- BusTurnAroundDuration = 0x0
- Capacitive load  $C_L = 30 \text{ pF}$

In all timing tables, the  $T_{KERCK}$  is the  $f_{mc\_ker\_ck}$  clock period.

**Figure 77. Asynchronous non-multiplexed SRAM/PSRAM/NOR read waveforms**



1. Mode 2/B, C and D only. In Mode 1, FMC\_NADV is not used.



**Table 163. Asynchronous non-multiplexed SRAM/PSRAM/NOR read timings<sup>(1)</sup>**

| Symbol              | Parameter                             | Min                   | Max                   | Unit |
|---------------------|---------------------------------------|-----------------------|-----------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                       | $3T_{fmc\_ker\_ck}-1$ | $3T_{fmc\_ker\_ck}+1$ | ns   |
| $t_{v(NOE\_NE)}$    | FMC_NEx low to FMC_NOE low            | 0                     | 0.5                   |      |
| $t_{w(NOE)}$        | FMC_NOE low time                      | $2T_{fmc\_ker\_ck}-1$ | $2T_{fmc\_ker\_ck}+1$ |      |
| $t_{h(NE\_NOE)}$    | FMC_NOE high to FMC_NE high hold time | 0                     | -                     |      |
| $t_{v(A\_NE)}$      | FMC_NEx low to FMC_A valid            | -                     | 0.5                   |      |
| $t_{h(A\_NOE)}$     | Address hold time after FMC_NOE high  | 0                     | -                     |      |
| $t_{su(Data\_NE)}$  | Data to FMC_NEx high setup time       | 11                    | -                     |      |
| $t_{su(Data\_NOE)}$ | Data to FMC_NOEx high setup time      | 11                    | -                     |      |
| $t_{h(Data\_NOE)}$  | Data hold time after FMC_NOE high     | 0                     | -                     |      |
| $t_{h(Data\_NE)}$   | Data hold time after FMC_NEx high     | 0                     | -                     |      |
| $t_{v(NADV\_NE)}$   | FMC_NEx low to FMC_NADV low           | -                     | 0                     |      |
| $t_{w(NADV)}$       | FMC_NADV low time                     | -                     | $T_{fmc\_ker\_ck}+1$  |      |

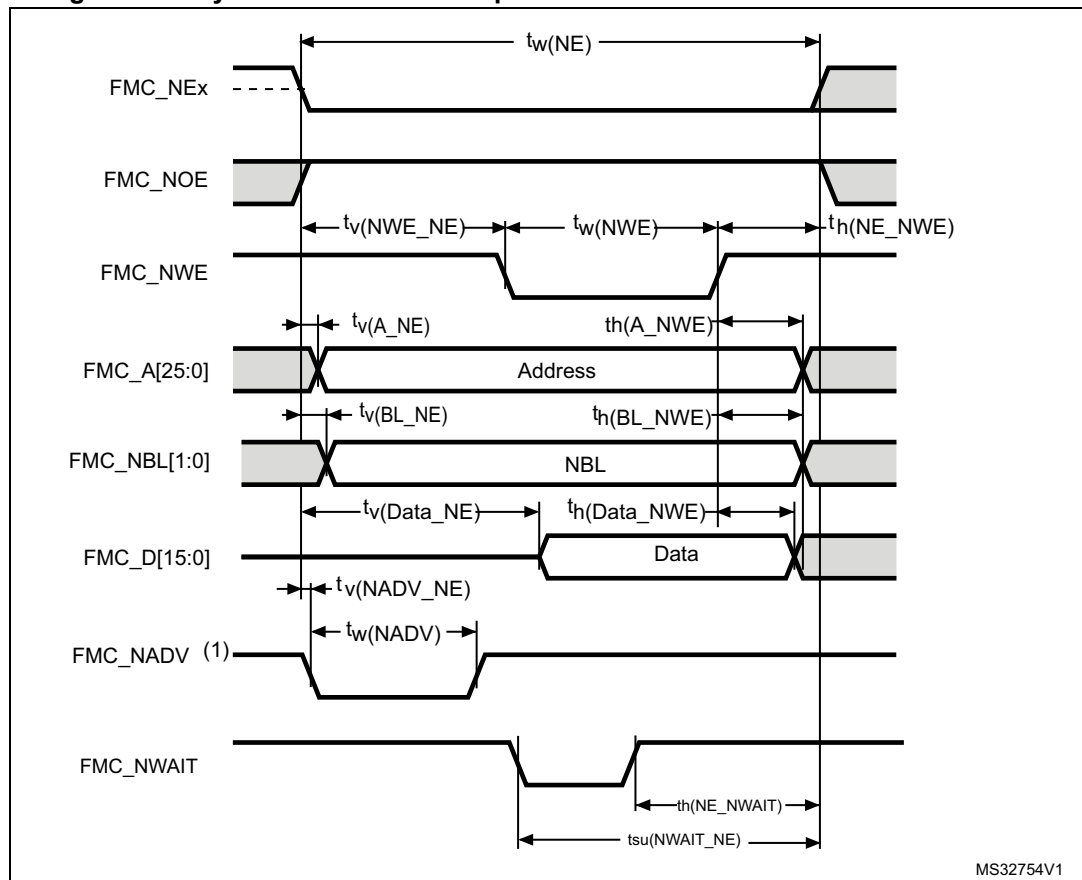
1. Guaranteed by characterization results.

**Table 164. Asynchronous non-multiplexed SRAM/PSRAM/NOR read-NWAIT timings<sup>(1)(2)</sup>**

| Symbol              | Parameter                                 | Min                      | Max                   | Unit |
|---------------------|-------------------------------------------|--------------------------|-----------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $7T_{fmc\_ker\_ck}+1$    | $7T_{fmc\_ker\_ck}+1$ | ns   |
| $t_{w(NOE)}$        | FMC_NOE low time                          | $5T_{fmc\_ker\_ck}-1$    | $5T_{fmc\_ker\_ck}+1$ |      |
| $t_{w(NWAIT)}$      | FMC_NWAIT low time                        | $T_{fmc\_ker\_ck}-0.5$   | -                     |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $4T_{fmc\_ker\_ck}+11$   | -                     |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $3T_{fmc\_ker\_ck}+11.5$ | -                     |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

**Figure 78. Asynchronous non-multiplexed SRAM/PSRAM/NOR write waveforms**

1. Mode 2/B, C and D only. In Mode 1, FMC\_NADV is not used.

**Table 165. Asynchronous non-multiplexed SRAM/PSRAM/NOR write timings<sup>(1)</sup>**

| Symbol             | Parameter                             | Min                      | Max                      | Unit |
|--------------------|---------------------------------------|--------------------------|--------------------------|------|
| $t_{w(NE)}$        | FMC_NE low time                       | $3T_{fmc\_ker\_ck} - 1$  | $3T_{fmc\_ker\_ck}$      | ns   |
| $t_{v(NWE\_NE)}$   | FMC_NEx low to FMC_NWE low            | $T_{fmc\_ker\_ck}$       | $T_{fmc\_ker\_ck} + 1$   |      |
| $t_{w(NWE)}$       | FMC_NWE low time                      | $T_{fmc\_ker\_ck} - 0.5$ | $T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{h(NE\_NWE)}$   | FMC_NWE high to FMC_NE high hold time | $T_{fmc\_ker\_ck}$       | -                        |      |
| $t_{v(A\_NE)}$     | FMC_NEx low to FMC_A valid            | -                        | 2                        |      |
| $t_{h(A\_NWE)}$    | Address hold time after FMC_NWE high  | $T_{fmc\_ker\_ck} - 0.5$ | -                        |      |
| $t_{v(BL\_NE)}$    | FMC_NEx low to FMC_BL valid           | -                        | 0.5                      |      |
| $t_{h(BL\_NWE)}$   | FMC_BL hold time after FMC_NWE high   | $T_{fmc\_ker\_ck} - 0.5$ | -                        |      |
| $t_{v(Data\_NE)}$  | Data to FMC_NEx low to Data valid     | -                        | $T_{fmc\_ker\_ck} + 2.5$ |      |
| $t_{h(Data\_NWE)}$ | Data hold time after FMC_NWE high     | $T_{fmc\_ker\_ck} + 0.5$ | -                        |      |
| $t_{v(NADV\_NE)}$  | FMC_NEx low to FMC_NADV low           | -                        | 0                        |      |
| $t_{w(NADV)}$      | FMC_NADV low time                     | -                        | $T_{fmc\_ker\_ck} + 1$   |      |

1. Guaranteed by characterization results.

**Table 166. Asynchronous non-multiplexed SRAM/PSRAM/NOR write-NWAIT timings<sup>(1)(2)</sup>**

| Symbol              | Parameter                                 | Min                       | Max                       | Unit |
|---------------------|-------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $8T_{fmc\_ker\_ck} - 1$   | $8T_{fmc\_ker\_ck} + 1$   | ns   |
| $t_{w(NWE)}$        | FMC_NWE low time                          | $6T_{fmc\_ker\_ck} - 1.5$ | $6T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $5T_{fmc\_ker\_ck} + 13$  | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $4T_{fmc\_ker\_ck} + 13$  | -                         |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

Figure 79. Asynchronous multiplexed PSRAM/NOR read waveforms

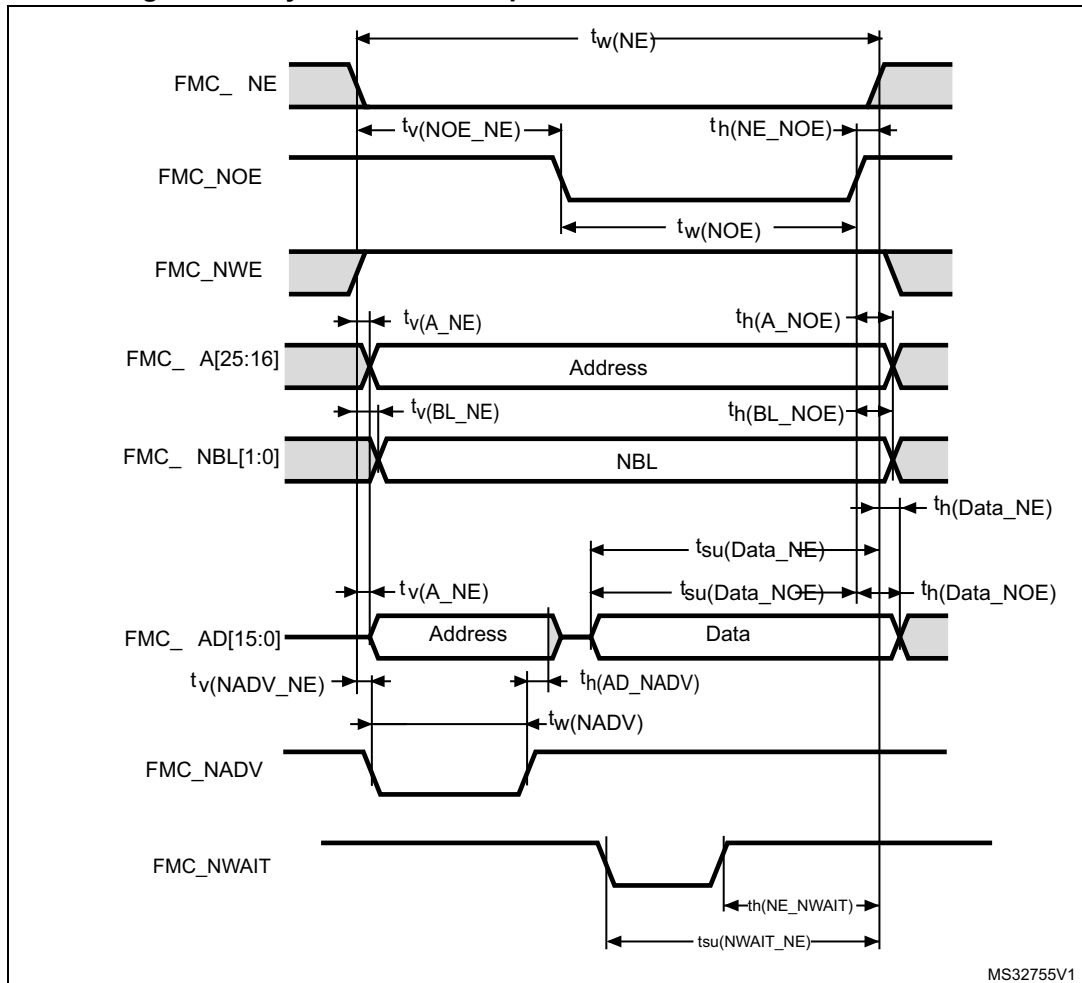


Table 167. Asynchronous multiplexed PSRAM/NOR read timings<sup>(1)</sup>

| Symbol              | Parameter                                           | Min                      | Max                       | Unit |
|---------------------|-----------------------------------------------------|--------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                                     | $4T_{fmc\_ker\_ck} - 1$  | $4T_{fmc\_ker\_ck} + 1$   | ns   |
| $t_{v(NOE\_NE)}$    | FMC_NEx low to FMC_NOE low                          | $2T_{fmc\_ker\_ck}$      | $2T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{w(NOE)}$        | FMC_NOE low time                                    | $T_{fmc\_ker\_ck} - 1$   | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_{h(NE\_NOE)}$    | FMC_NOE high to FMC_NE high hold time               | 0                        | -                         |      |
| $t_{v(A\_NE)}$      | FMC_NEx low to FMC_A valid                          | -                        | 0.5                       |      |
| $t_{v(NADV\_NE)}$   | FMC_NEx low to FMC_NADV low                         | 0                        | 0.5                       |      |
| $t_{w(NADV)}$       | FMC_NADV low time                                   | $T_{fmc\_ker\_ck} - 0.5$ | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_{h(AD\_NADV)}$   | FMC_AD(address) valid hold time after FMC_NADV high | $T_{fmc\_ker\_ck} + 0.5$ | -                         |      |
| $t_{h(A\_NOE)}$     | Address hold time after FMC_NOE high                | $T_{fmc\_ker\_ck} - 0.5$ | -                         |      |
| $t_{su(Data\_NE)}$  | Data to FMC_NEx high setup time                     | 11                       | -                         |      |
| $t_{su(Data\_NOE)}$ | Data to FMC_NOE high setup time                     | 11                       | -                         |      |
| $t_{h(Data\_NE)}$   | Data hold time after FMC_NEx high                   | 0                        | -                         |      |
| $t_{h(Data\_NOE)}$  | Data hold time after FMC_NOE high                   | 0                        | -                         |      |

1. Guaranteed by characterization results.

Table 168. Asynchronous multiplexed PSRAM/NOR read-NWAIT timings<sup>(1)(2)</sup>

| Symbol              | Parameter                                 | Min                        | Max                       | Unit |
|---------------------|-------------------------------------------|----------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $8T_{fmc\_ker\_ck} - 1$    | $8T_{fmc\_ker\_ck}$       | ns   |
| $t_{w(NOE)}$        | FMC_NWE low time                          | $5T_{fmc\_ker\_ck} - 1.5$  | $5T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $4T_{fmc\_ker\_ck} + 11$   | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $3T_{fmc\_ker\_ck} + 11.5$ | -                         |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

**Table 169. Asynchronous multiplexed PSRAM/NOR write timings<sup>(1)</sup>**

| Symbol              | Parameter                                           | Min                       | Max                       | Unit |
|---------------------|-----------------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                                     | $4T_{fmc\_ker\_ck} - 1$   | $4T_{fmc\_ker\_ck}$       | ns   |
| $t_{v(NWE\_NE)}$    | FMC_NEx low to FMC_NWE low                          | $T_{fmc\_ker\_ck} - 1$    | $T_{fmc\_ker\_ck} + 0.5$  |      |
| $t_{w(NWE)}$        | FMC_NWE low time                                    | $2T_{fmc\_ker\_ck} - 0.5$ | $2T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{h(NE\_NWE)}$    | FMC_NWE high to FMC_NE high hold time               | $T_{fmc\_ker\_ck} - 0.5$  | -                         |      |
| $t_{v(A\_NE)}$      | FMC_NEx low to FMC_A valid                          | -                         | 0                         |      |
| $t_{v(NADV\_NE)}$   | FMC_NEx low to FMC_NADV low                         | 0                         | 0.5                       |      |
| $t_{w(NADV)}$       | FMC_NADV low time                                   | $T_{fmc\_ker\_ck}$        | $T_{fmc\_ker\_ck} + 1$    |      |
| $t_{h(AD\_NADV)}$   | FMC_AD(address) valid hold time after FMC_NADV high | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |
| $t_{h(A\_NWE)}$     | Address hold time after FMC_NWE high                | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |
| $t_{h(BL\_NWE)}$    | FMC_BL hold time after FMC_NWE high                 | $T_{fmc\_ker\_ck} - 0.5$  | -                         |      |
| $t_{v(BL\_NE)}$     | FMC_NEx low to FMC_BL valid                         | -                         | 0.5                       |      |
| $t_{v(Data\_NADV)}$ | FMC_NADV high to Data valid                         | -                         | $T_{fmc\_ker\_ck} + 2$    |      |
| $t_{h(Data\_NWE)}$  | Data hold time after FMC_NWE high                   | $T_{fmc\_ker\_ck} + 0.5$  | -                         |      |

1. Guaranteed by characterization results.

**Table 170. Asynchronous multiplexed PSRAM/NOR write-NWAIT timings<sup>(1)(2)</sup>**

| Symbol              | Parameter                                 | Min                        | Max                       | Unit |
|---------------------|-------------------------------------------|----------------------------|---------------------------|------|
| $t_{w(NE)}$         | FMC_NE low time                           | $9T_{fmc\_ker\_ck} - 1$    | $9T_{fmc\_ker\_ck}$       | ns   |
| $t_{w(NWE)}$        | FMC_NWE low time                          | $7T_{fmc\_ker\_ck} - 0.5$  | $7T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_{su(NWAIT\_NE)}$ | FMC_NWAIT valid before FMC_NEx high       | $5T_{fmc\_ker\_ck} + 11$   | -                         |      |
| $t_{h(NE\_NWAIT)}$  | FMC_NEx hold time after FMC_NWAIT invalid | $4T_{fmc\_ker\_ck} + 11.5$ | -                         |      |

1. Guaranteed by characterization results.

2.  $N_{WAIT}$  pulse width is equal to 1 AHB cycle.

## Synchronous waveforms and timings

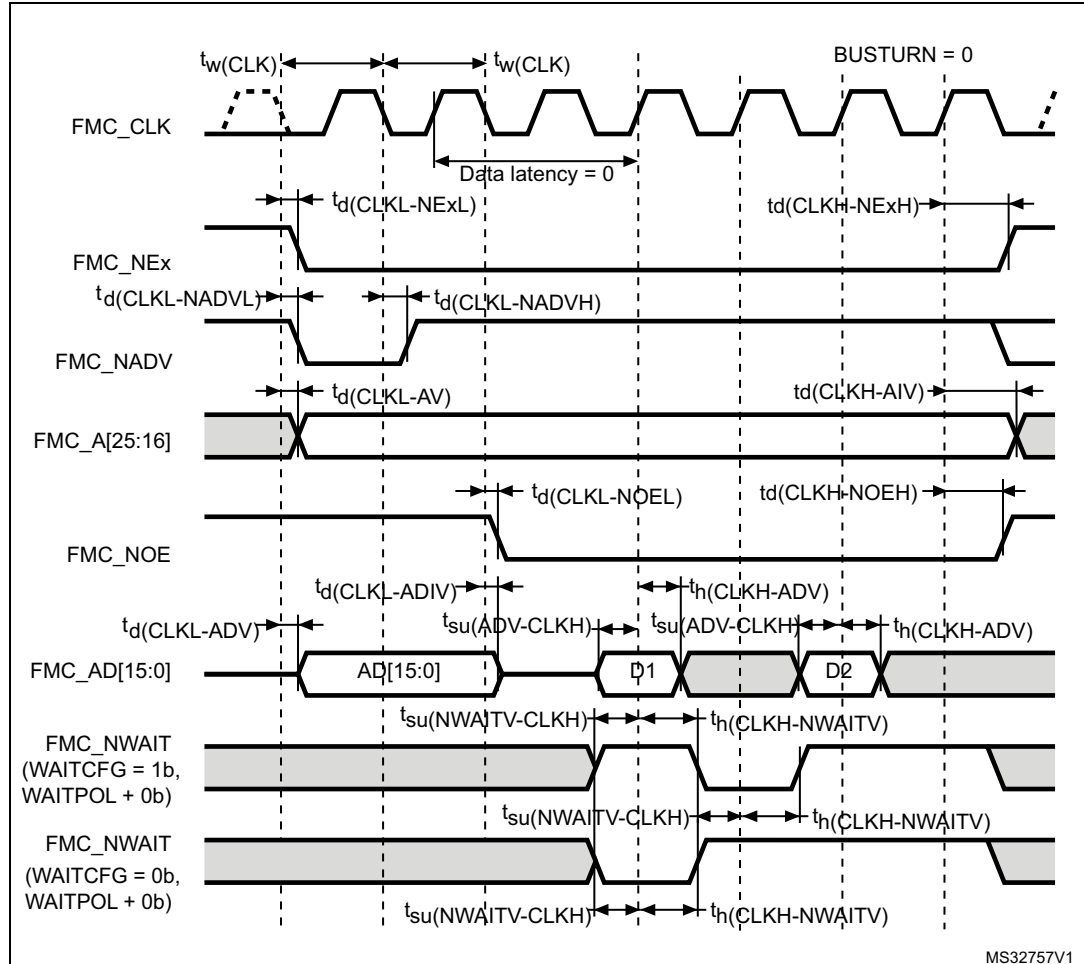
Figure 80 through Figure 83 represent synchronous waveforms and Table 171 through Table 174 provide the corresponding timings. The results shown in these tables are obtained with the following FMC configuration:

- BurstAccessMode = FMC\_BurstAccessMode\_Enable
- MemoryType = FMC\_MemoryType\_CRAM
- WriteBurst = FMC\_WriteBurst\_Enable
- CLKDivision = 1
- DataLatency = 1 for NOR Flash; DataLatency = 0 for PSRAM

In all the timing tables, the  $T_{fmc\_ker\_ck}$  is the  $f_{fmc\_ker\_ck}$  clock period, with the following FMC\_CLK maximum values:

- For  $2.7\text{ V} < V_{DD} < 3.6\text{ V}$ , FMC\_CLK = 125 MHz at 20 pF
- For  $1.8\text{ V} < V_{DD} < 1.9\text{ V}$ , FMC\_CLK = 100 MHz at 20 pF
- For  $1.62\text{ V} < V_{DD} < 1.8\text{ V}$ , FMC\_CLK = 100 MHz at 15 pF

**Figure 80. Synchronous multiplexed NOR/PSRAM read timings**



MS32757V1

Table 171. Synchronous multiplexed NOR/PSRAM read timings<sup>(1)</sup>

| Symbol               | Parameter                                     | Min                    | Max | Unit |
|----------------------|-----------------------------------------------|------------------------|-----|------|
| $t_{w(CLK)}$         | FMC_CLK period                                | $2T_{fmc\_ker\_ck}-1$  | -   | ns   |
| $t_d(CLKL-NExL)$     | FMC_CLK low to FMC_NEx low ( $x=0..2$ )       | -                      | 1   |      |
| $t_d(CLKH-NExH)$     | FMC_CLK high to FMC_NEx high ( $x=0..2$ )     | $T_{fmc\_ker\_ck}+0.5$ | -   |      |
| $t_d(CLKL-NADV_L)$   | FMC_CLK low to FMC_NADV low                   | -                      | 1   |      |
| $t_d(CLKL-NADV_H)$   | FMC_CLK low to FMC_NADV high                  | 0                      | -   |      |
| $t_d(CLKL-AV)$       | FMC_CLK low to FMC_Ax valid ( $x=16..25$ )    | -                      | 2.5 |      |
| $t_d(CLKH-AIV)$      | FMC_CLK high to FMC_Ax invalid ( $x=16..25$ ) | $T_{fmc\_ker\_ck}$     | -   |      |
| $t_d(CLKL-NOEL)$     | FMC_CLK low to FMC_NOE low                    | -                      | 1.5 |      |
| $t_d(CLKH-NOEH)$     | FMC_CLK high to FMC_NOE high                  | $T_{fmc\_ker\_ck}-0.5$ | -   |      |
| $t_d(CLKL-ADV)$      | FMC_CLK low to FMC_AD[15:0] valid             | -                      | 3   |      |
| $t_d(CLKL-ADIV)$     | FMC_CLK low to FMC_AD[15:0] invalid           | 0                      | -   |      |
| $t_{su}(ADV-CLKH)$   | FMC_A/D[15:0] valid data before FMC_CLK high  | 2                      | -   |      |
| $t_h(CLKH-ADV)$      | FMC_A/D[15:0] valid data after FMC_CLK high   | 1                      | -   |      |
| $t_{su}(NWAIT-CLKH)$ | FMC_NWAIT valid before FMC_CLK high           | 2                      | -   |      |
| $t_h(CLKH-NWAIT)$    | FMC_NWAIT valid after FMC_CLK high            | 2                      | -   |      |

1. Guaranteed by characterization results.



### Figure 81. Synchronous multiplexed PSRAM write timings

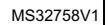


Table 172. Synchronous multiplexed PSRAM write timings<sup>(1)</sup>

| Symbol               | Parameter                                       | Min                      | Max | Unit |
|----------------------|-------------------------------------------------|--------------------------|-----|------|
| $t_{w(CLK)}$         | FMC_CLK period, $V_{DD} = 2.7$ to $3.6$ V       | $2T_{fmc\_ker\_ck} - 1$  | -   | Ns   |
| $t_d(CLKL-NExL)$     | FMC_CLK low to FMC_NEx low ( $x = 0..2$ )       | -                        | 1   |      |
| $t_d(CLKH-NExH)$     | FMC_CLK high to FMC_NEx high ( $x = 0..2$ )     | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_d(CLKL-NADV_L)$   | FMC_CLK low to FMC_NADV low                     | -                        | 1.5 |      |
| $t_d(CLKL-NADV_H)$   | FMC_CLK low to FMC_NADV high                    | 0                        | -   |      |
| $t_d(CLKL-AV)$       | FMC_CLK low to FMC_Ax valid ( $x = 16..25$ )    | -                        | 2   |      |
| $t_d(CLKH-AIV)$      | FMC_CLK high to FMC_Ax invalid ( $x = 16..25$ ) | $T_{fmc\_ker\_ck}$       | -   |      |
| $t_d(CLKL-NWEL)$     | FMC_CLK low to FMC_NWE low                      | -                        | 1.5 |      |
| $t_d(CLKH-NWEH)$     | FMC_CLK high to FMC_NWE high                    | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_d(CLKL-ADV)$      | FMC_CLK low to FMC_AD[15:0] valid               | -                        | 2.5 |      |
| $t_d(CLKL-ADIV)$     | FMC_CLK low to FMC_AD[15:0] invalid             | 0                        | -   |      |
| $t_d(CLKL-DATA)$     | FMC_A/D[15:0] valid data after FMC_CLK low      | -                        | 2.5 |      |
| $t_d(CLKL-NBLL)$     | FMC_CLK low to FMC_NBL low                      | -                        | 2   |      |
| $t_d(CLKH-NBLH)$     | FMC_CLK high to FMC_NBL high                    | $T_{fmc\_ker\_ck} + 0.5$ | -   |      |
| $t_{su}(NWAIT-CLKH)$ | FMC_NWAIT valid before FMC_CLK high             | 2                        | -   |      |
| $t_h(CLKH-NWAIT)$    | FMC_NWAIT valid after FMC_CLK high              | 2                        | -   |      |

1. Guaranteed by characterization results.

Figure 82. Synchronous non-multiplexed NOR/PSRAM read timings

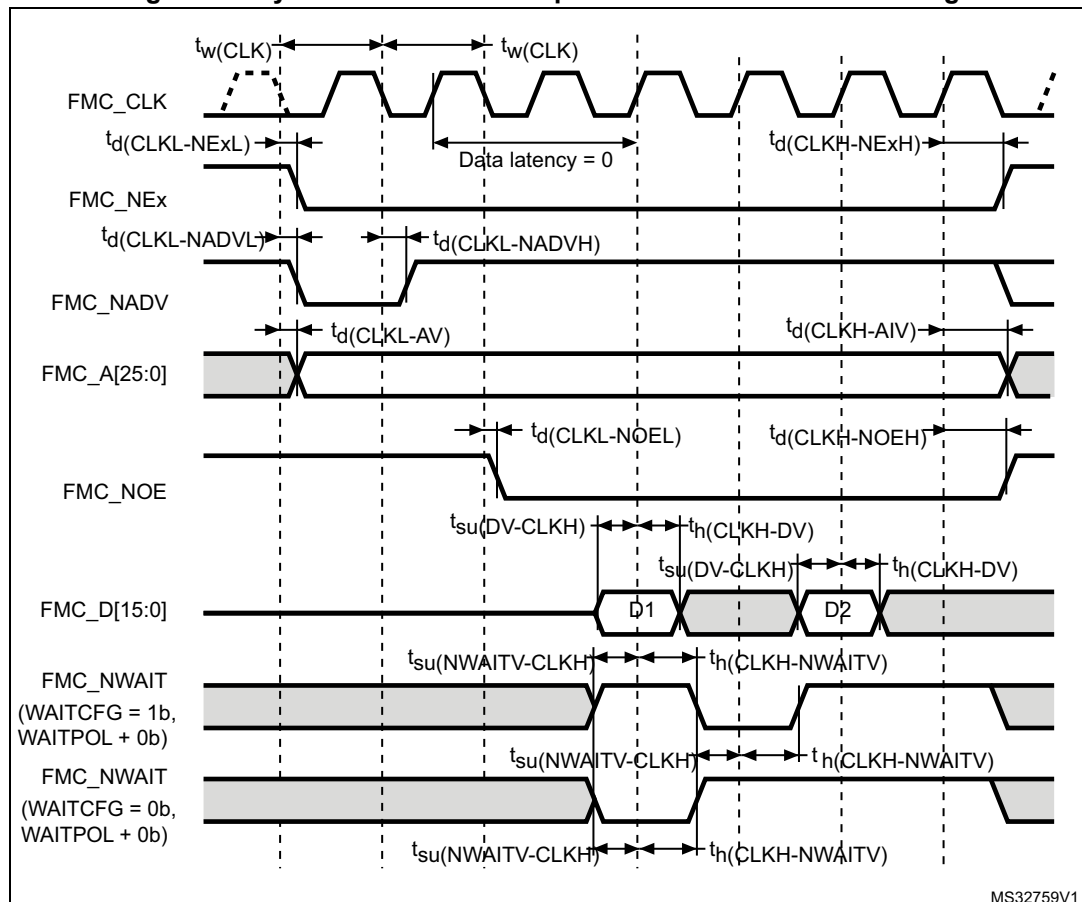


Table 173. Synchronous non-multiplexed NOR/PSRAM read timings<sup>(1)</sup>

| Symbol               | Parameter                                  | Min                     | Max | Unit |
|----------------------|--------------------------------------------|-------------------------|-----|------|
| $t_{w(CLK)}$         | FMC_CLK period                             | $2T_{fmc\_ker\_ck}-1$   | -   | ns   |
| $t_{(CLKL-NExL)}$    | FMC_CLK low to FMC_NEx low (x=0..2)        | -                       | 1   |      |
| $t_{d(CLKH-NExH)}$   | FMC_CLK high to FMC_NEx high (x= 0...2)    | $2T_{fmc\_ker\_ck}+0.5$ | -   |      |
| $t_{d(CLKL-NADV L)}$ | FMC_CLK low to FMC_NADV low                | -                       | 0.5 |      |
| $t_{d(CLKL-NADV H)}$ | FMC_CLK low to FMC_NADV high               | 0                       | -   |      |
| $t_{d(CLKL-AV)}$     | FMC_CLK low to FMC_Ax valid (x=16...25)    | -                       | 2   |      |
| $t_{d(CLKH-AIV)}$    | FMC_CLK high to FMC_Ax invalid (x=16...25) | $2T_{fmc\_ker\_ck}$     | -   |      |
| $t_{d(CLKL-NOEL)}$   | FMC_CLK low to FMC_NOE low                 | -                       | 1.5 |      |
| $t_{d(CLKH-NOEH)}$   | FMC_CLK high to FMC_NOE high               | $2T_{fmc\_ker\_ck}-0.5$ | -   |      |
| $t_{su(DV-CLKH)}$    | FMC_D[15:0] valid data before FMC_CLK high | 2                       | -   |      |
| $t_{h(CLKH-DV)}$     | FMC_D[15:0] valid data after FMC_CLK high  | 1                       | -   |      |
| $t_{(NWAIT-CLKH)}$   | FMC_NWAIT valid before FMC_CLK high        | 2                       | -   |      |
| $t_{h(CLKH-NWAIT)}$  | FMC_NWAIT valid after FMC_CLK high         | 2                       | -   |      |

1. Guaranteed by characterization results.

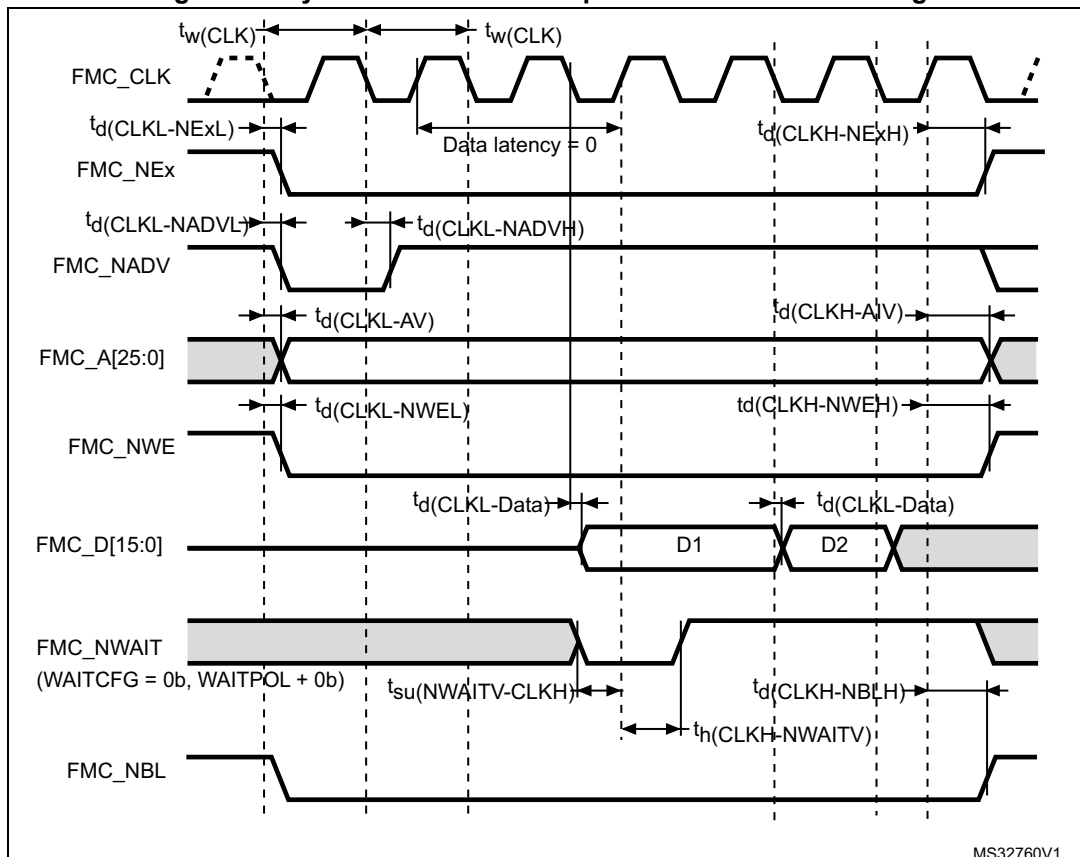
**Figure 83. Synchronous non-multiplexed PSRAM write timings**

Table 174. Synchronous non-multiplexed PSRAM write timings<sup>(1)</sup>

| Symbol                      | Parameter                                  | Min                             | Max | Unit |
|-----------------------------|--------------------------------------------|---------------------------------|-----|------|
| $t_{\text{CLK}}$            | FMC_CLK period                             | $2T_{\text{fmc\_ker\_ck}} - 1$  | -   | ns   |
| $t_{\text{d(CLKL-NExL)}}$   | FMC_CLK low to FMC_NEx low (x=0..2)        | -                               | 2   |      |
| $t_{\text{(CLKH-NExH)}}$    | FMC_CLK high to FMC_NEx high (x= 0...2)    | $T_{\text{fmc\_ker\_ck}} + 0.5$ | -   |      |
| $t_{\text{d(CLKL-NADV)}}$   | FMC_CLK low to FMC_NADV low                | -                               | 0.5 |      |
| $t_{\text{d(CLKL-NADV)H}}$  | FMC_CLK low to FMC_NADV high               | 0                               | -   |      |
| $t_{\text{d(CLKL-AV)}}$     | FMC_CLK low to FMC_Ax valid (x=16...25)    | -                               | 2.  |      |
| $t_{\text{d(CLKH-AIV)}}$    | FMC_CLK high to FMC_Ax invalid (x=16...25) | $T_{\text{fmc\_ker\_ck}}$       | -   |      |
| $t_{\text{d(CLKL-NWEL)}}$   | FMC_CLK low to FMC_NWE low                 | -                               | 1.5 |      |
| $t_{\text{d(CLKH-NWEH)}}$   | FMC_CLK high to FMC_NWE high               | $T_{\text{fmc\_ker\_ck}} + 1$   | -   |      |
| $t_{\text{d(CLKL-Data)}}$   | FMC_D[15:0] valid data after FMC_CLK low   | -                               | 3.5 |      |
| $t_{\text{d(CLKL-NBLL)}}$   | FMC_CLK low to FMC_NBL low                 | -                               | 2   |      |
| $t_{\text{d(CLKH-NBLH)}}$   | FMC_CLK high to FMC_NBL high               | $T_{\text{fmc\_ker\_ck}} + 1$   | -   |      |
| $t_{\text{su(NWAIT-CLKH)}}$ | FMC_NWAIT valid before FMC_CLK high        | 2                               | -   |      |
| $t_{\text{h(CLKH-NWAIT)}}$  | FMC_NWAIT valid after FMC_CLK high         | 2                               | -   |      |

1. Guaranteed by characterization results.

## NAND controller waveforms and timings

Figure 84 through Figure 87 represent synchronous waveforms, and Table 175 and Table 176 provide the corresponding timings. The results shown in this table are obtained with the following FMC configuration:

- COM.FMC\_SetupTime = 0x01
- COM.FMC\_WaitSetupTime = 0x03
- COM.FMC\_HoldSetupTime = 0x02
- COM.FMC\_HiZSetupTime = 0x01
- ATT.FMC\_SetupTime = 0x01
- ATT.FMC\_WaitSetupTime = 0x03
- ATT.FMC\_HoldSetupTime = 0x02
- ATT.FMC\_HiZSetupTime = 0x01
- Bank = FMC\_Bank\_NAND
- MemoryDataWidth = FMC\_MemoryDataWidth\_16b
- ECC = FMC\_ECC\_Enable
- ECCPageSize = FMC\_ECCPageSize\_512Bytes
- TCLRSetupTime = 0
- TARSetupTime = 0
- Capacitive load  $C_L = 30 \text{ pF}$

In all timing tables, the  $T_{\text{fmc\_ker\_ck}}$  is the fmc\_ker\_ck clock period.

**Figure 84. NAND controller waveforms for read access**

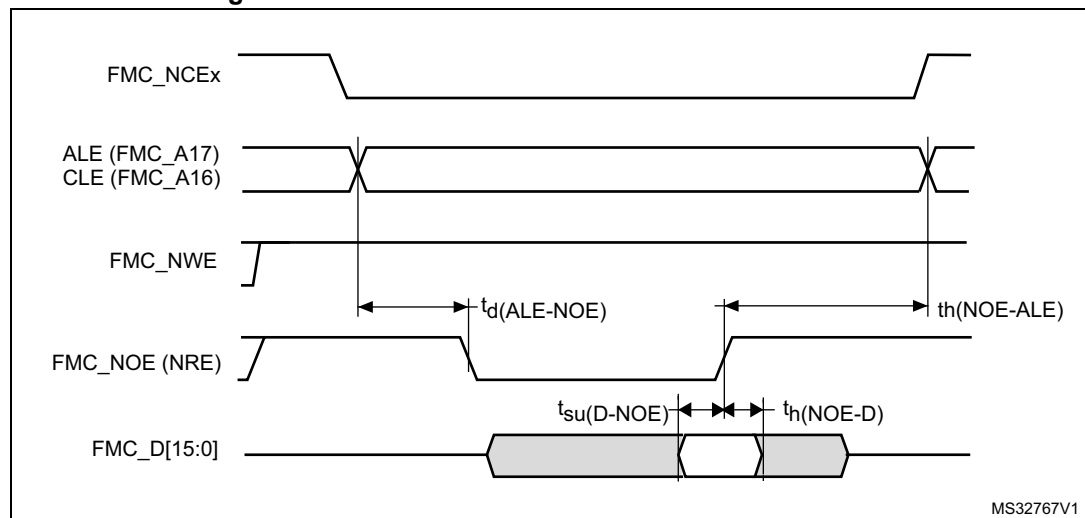


Figure 85. NAND controller waveforms for write access

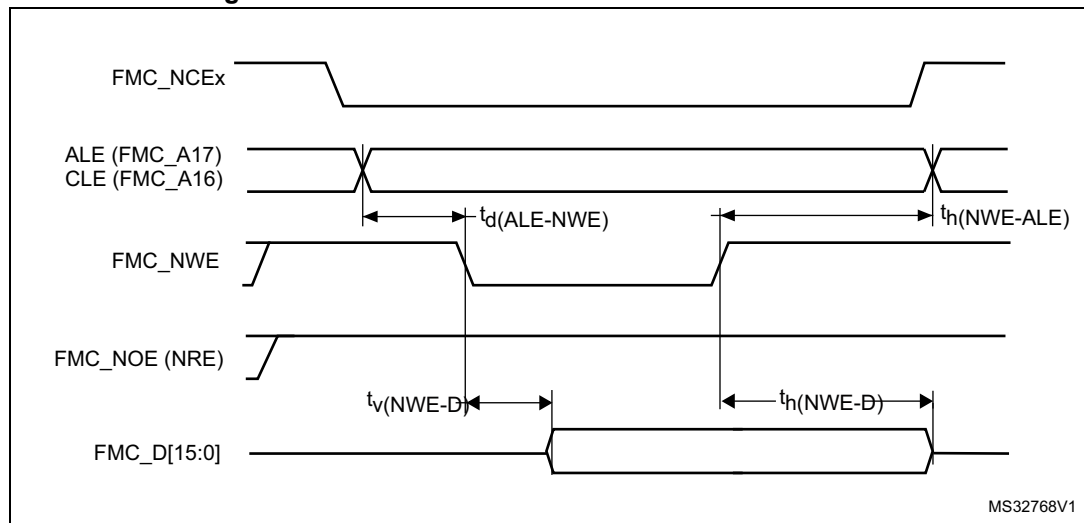
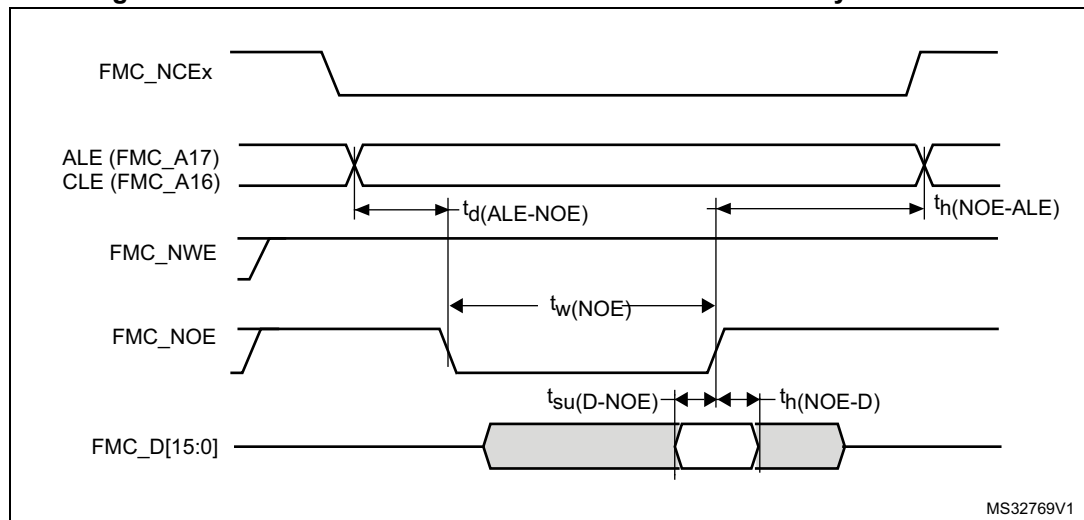
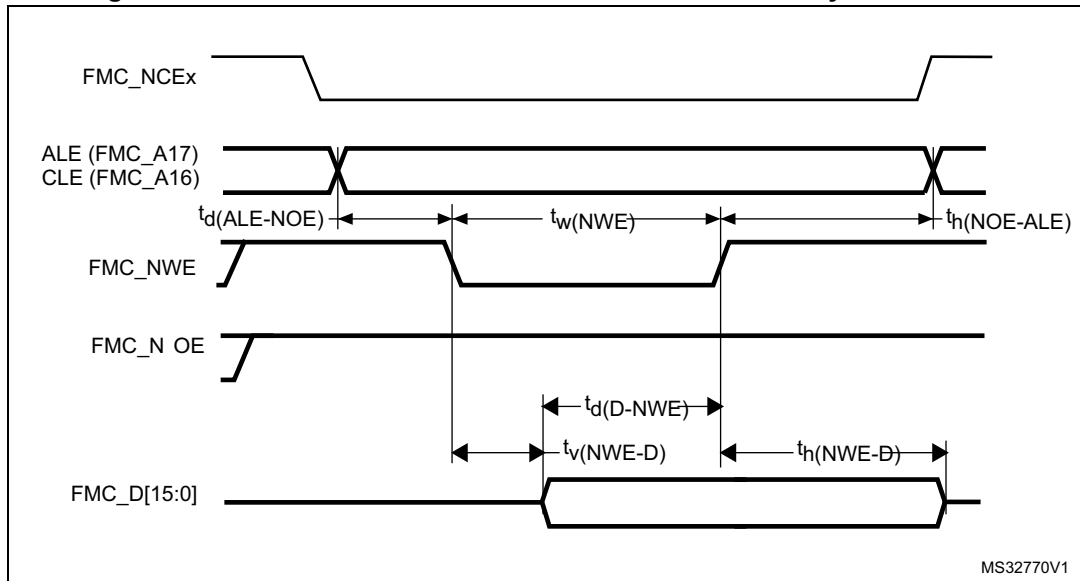


Figure 86. NAND controller waveforms for common memory read access





**Figure 87. NAND controller waveforms for common memory write access****Table 175. Switching characteristics for NAND Flash read cycles<sup>(1)</sup>**

| Symbol          | Parameter                                  | Min                       | Max                       | Unit |
|-----------------|--------------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NOE)}$    | FMC_NOE low width                          | $4T_{fmc\_ker\_ck} - 0.5$ | $4T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su(D-NOE)}$ | FMC_D[15-0] valid data before FMC_NOE high | 8                         | -                         |      |
| $t_h(NOE-D)$    | FMC_D[15-0] valid data after FMC_NOE high  | 0                         | -                         |      |
| $t_d(ALE-NOE)$  | FMC_ALE valid before FMC_NOE low           | -                         | $3T_{fmc\_ker\_ck} + 1$   |      |
| $t_h(NOE-ALE)$  | FMC_NWE high to FMC_ALE invalid            | $4T_{fmc\_ker\_ck} - 2$   | -                         |      |

1. Guaranteed by characterization results.

**Table 176. Switching characteristics for NAND Flash write cycles<sup>(1)</sup>**

| Symbol         | Parameter                             | Min                       | Max                       | Unit |
|----------------|---------------------------------------|---------------------------|---------------------------|------|
| $t_{w(NWE)}$   | FMC_NWE low width                     | $4T_{fmc\_ker\_ck} - 0.5$ | $4T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{v(NWE-D)}$ | FMC_NWE low to FMC_D[15-0] valid      | 0                         | -                         |      |
| $t_h(NWE-D)$   | FMC_NWE high to FMC_D[15-0] invalid   | $2T_{fmc\_ker\_ck} - 0.5$ | -                         |      |
| $t_d(D-NWE)$   | FMC_D[15-0] valid before FMC_NWE high | $5T_{fmc\_ker\_ck} - 1$   | -                         |      |
| $t_d(ALE-NWE)$ | FMC_ALE valid before FMC_NWE low      | -                         | $3T_{fmc\_ker\_ck} + 0.5$ |      |
| $t_h(NWE-ALE)$ | FMC_NWE high to FMC_ALE invalid       | $2T_{fmc\_ker\_ck} - 1$   | -                         |      |

1. Guaranteed by characterization results.

### SDRAM waveforms and timings

In all timing tables, the  $T_{KERCK}$  is the  $fmc\_ker\_ck$  clock period, with the following  $FMC\_SDCLK$  maximum values:

- For  $2.7\text{ V} < V_{DD} < 3.6\text{ V}$ :  $FMC\_CLK = 110\text{ MHz}$  at  $20\text{ pF}$
- For  $1.8\text{ V} < V_{DD} < 1.9\text{ V}$ :  $FMC\_CLK = 100\text{ MHz}$  at  $20\text{ pF}$
- For  $1.62\text{ V} < V_{DD} < 1.8\text{ V}$ ,  $FMC\_CLK = 100\text{ MHz}$  at  $15\text{ pF}$

**Figure 88. SDRAM read access waveforms (CL = 1)**

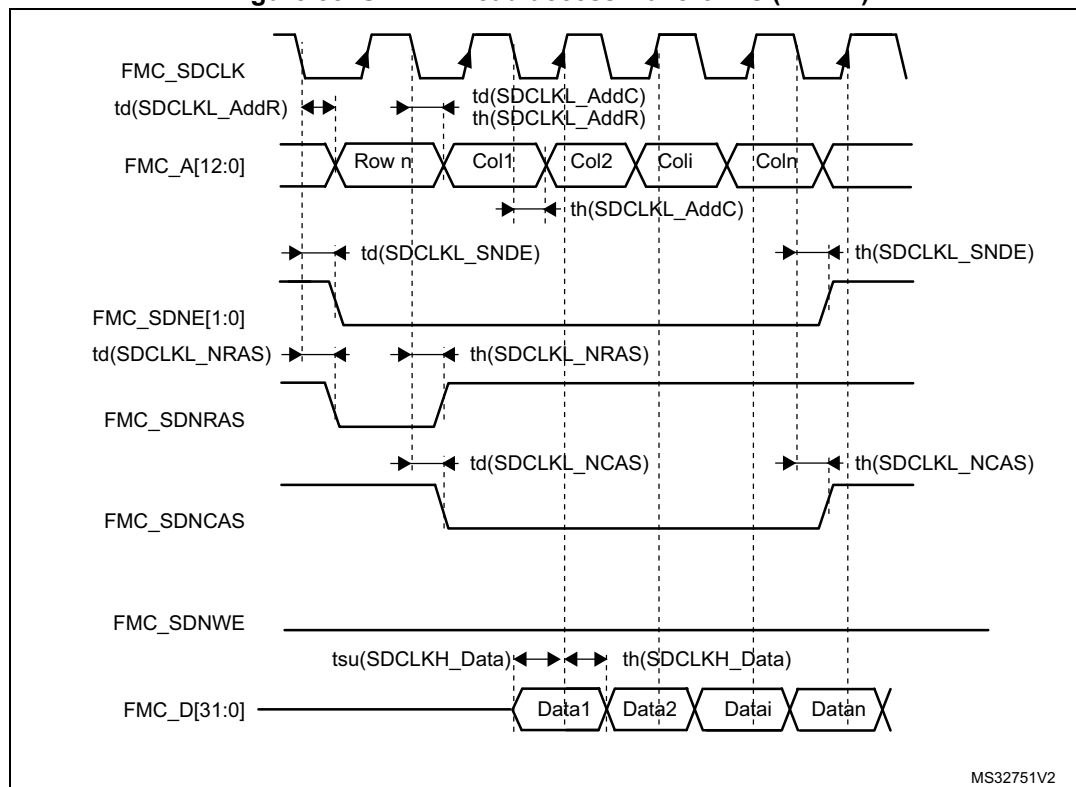


Table 177. SDRAM read timings<sup>(1)</sup>

| Symbol                  | Parameter              | Min                     | Max                       | Unit |
|-------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{w(SDCLK)}$          | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su(SDCLKH\_Data)}$  | Data input setup time  | 2                       | -                         |      |
| $t_{h(SDCLKH\_Data)}$   | Data input hold time   | 1                       | -                         |      |
| $t_{d(SDCLKL\_Add)}$    | Address valid time     | -                       | 1.5                       |      |
| $t_{d(SDCLKL\_SDNE)}$   | Chip select valid time | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNE)}$   | Chip select hold time  | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNRAS)}$ | SDNRAS valid time      | -                       | 1                         |      |
| $t_{h(SDCLKL\_SDNRAS)}$ | SDNRAS hold time       | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS valid time      | -                       | 0.5                       |      |
| $t_{h(SDCLKL\_SDNCAS)}$ | SDNCAS hold time       | 0                       | -                         |      |

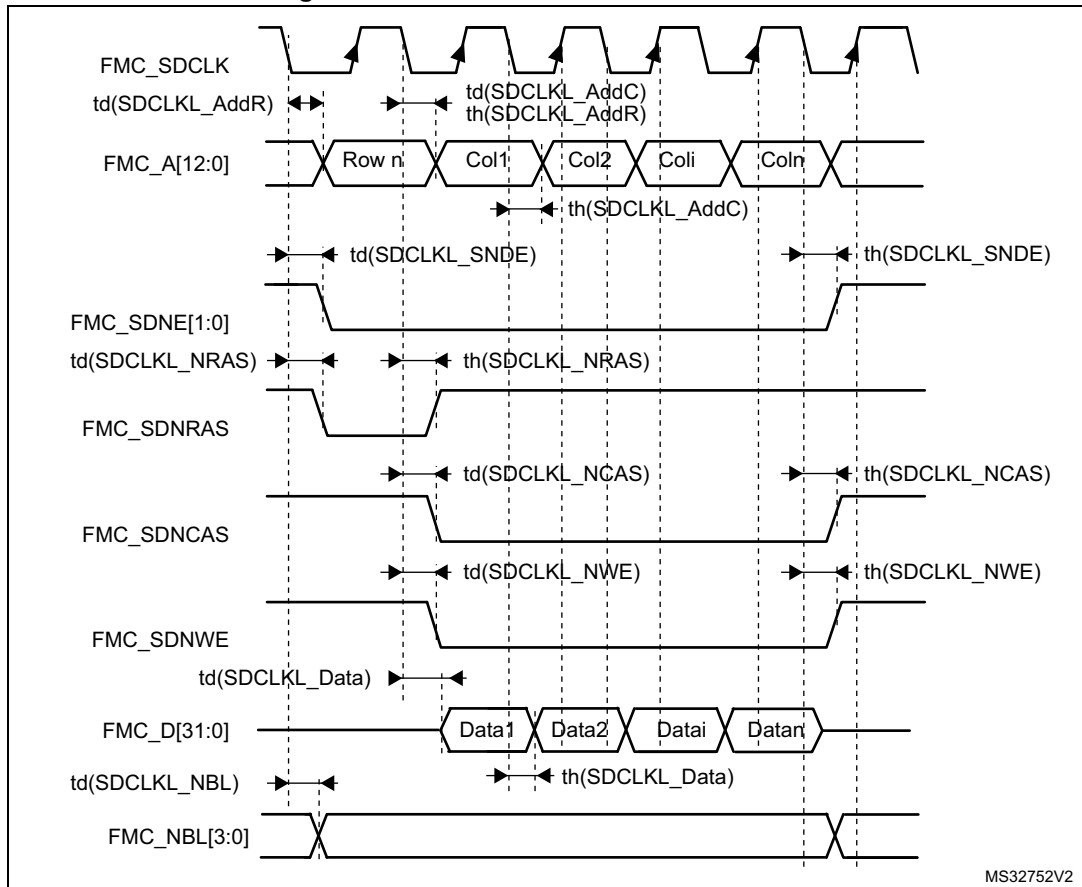
1. Guaranteed by characterization results.

Table 178. LPDDR SDRAM read timings<sup>(1)</sup>

| Symbol                  | Parameter              | Min                     | Max                       | Unit |
|-------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{w(SDCLK)}$          | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{su(SDCLKH\_Data)}$  | Data input setup time  | 2                       | -                         |      |
| $t_{h(SDCLKH\_Data)}$   | Data input hold time   | 1.5                     | -                         |      |
| $t_{d(SDCLKL\_Add)}$    | Address valid time     | -                       | 2.5                       |      |
| $t_{d(SDCLKL\_SDNE)}$   | Chip select valid time | -                       | 2.5                       |      |
| $t_{h(SDCLKL\_SDNE)}$   | Chip select hold time  | 0                       | -                         |      |
| $t_{d(SDCLKL\_SDNRAS)}$ | SDNRAS valid time      | -                       | 0.5                       |      |
| $t_{h(SDCLKL\_SDNRAS)}$ | SDNRAS hold time       | 0                       | -                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS valid time      | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNCAS)}$ | SDNCAS hold time       | 0                       | -                         |      |

1. Guaranteed by characterization results.

Figure 89. SDRAM write access waveforms

Table 179. SDRAM Write timings<sup>(1)</sup>

| Symbol                  | Parameter              | Min                     | Max                       | Unit |
|-------------------------|------------------------|-------------------------|---------------------------|------|
| $t_{w(SDCLK)}$          | FMC_SDCLK period       | $2T_{fmc\_ker\_ck} - 1$ | $2T_{fmc\_ker\_ck} + 0.5$ | ns   |
| $t_{d(SDCLKL\_Data)}$   | Data output valid time | -                       | 1                         |      |
| $t_{h(SDCLKL\_Data)}$   | Data output hold time  | 0                       | -                         |      |
| $t_{d(SDCLKL\_Add)}$    | Address valid time     | -                       | 1.5                       |      |
| $t_{d(SDCLKL\_SDNWE)}$  | SDNWE valid time       | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNWE)}$  | SDNWE hold time        | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNE)}$   | Chip select valid time | -                       | 1.5                       |      |
| $t_{h(SDCLKL\_SDNE)}$   | Chip select hold time  | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNRAS)}$ | SDNRAS valid time      | -                       | 1                         |      |
| $t_{h(SDCLKL\_SDNRAS)}$ | SDNRAS hold time       | 0.5                     | -                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS valid time      | -                       | 1                         |      |
| $t_{d(SDCLKL\_SDNCAS)}$ | SDNCAS hold time       | 0.5                     | -                         |      |

1. Guaranteed by characterization results.

Table 180. LPDDR SDRAM Write timings<sup>(1)</sup>

| Symbol                        | Parameter              | Min                            | Max                              | Unit |
|-------------------------------|------------------------|--------------------------------|----------------------------------|------|
| $t_{w}(\text{SDCLK})$         | FMC_SDCLK period       | $2T_{\text{fmc\_ker\_ck}} - 1$ | $2T_{\text{fmc\_ker\_ck}} + 0.5$ | ns   |
| $t_{d}(\text{SDCLKL\_Data})$  | Data output valid time | -                              | 2.5                              |      |
| $t_{h}(\text{SDCLKL\_Data})$  | Data output hold time  | 0                              | -                                |      |
| $t_{d}(\text{SDCLKL\_Add})$   | Address valid time     | -                              | 2.5                              |      |
| $t_{d}(\text{SDCLKL-SDNWE})$  | SDNWE valid time       | -                              | 2.5                              |      |
| $t_{h}(\text{SDCLKL-SDNWE})$  | SDNWE hold time        | 0                              | -                                |      |
| $t_{d}(\text{SDCLKL-SDNE})$   | Chip select valid time | -                              | 3                                |      |
| $t_{h}(\text{SDCLKL-SDNE})$   | Chip select hold time  | 0                              | -                                |      |
| $t_{d}(\text{SDCLKL-SDNRAS})$ | SDNRAS valid time      | -                              | 1.5                              |      |
| $t_{h}(\text{SDCLKL-SDNRAS})$ | SDNRAS hold time       | 0                              | -                                |      |
| $t_{d}(\text{SDCLKL-SDNCAS})$ | SDNCAS valid time      | -                              | 1.5                              |      |
| $t_{d}(\text{SDCLKL-SDNCAS})$ | SDNCAS hold time       | 0                              | -                                |      |

1. Guaranteed by characterization results.

### 7.3.18 Quad-SPI interface characteristics

Unless otherwise specified, the parameters given in [Table 181](#) and [Table 182](#) for QUADSPI are derived from tests performed under the ambient temperature,  $f_{\text{AHB}}$  frequency and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 11
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$
- IO Compensation cell activated.
- HSLV activated when  $V_{\text{DD}} \leq 2.7 \text{ V}$
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics.

The following table summarizes the parameters measured in SDR mode.

Table 181. QUADSPI characteristics in SDR mode<sup>(1)</sup>

| Symbol                         | Parameter               | Conditions                                           | Min | Typ | Max | Unit |
|--------------------------------|-------------------------|------------------------------------------------------|-----|-----|-----|------|
| $F_{\text{ck1}}/T_{\text{CK}}$ | QUADSPI clock frequency | $2.7 < V_{\text{DD}} < 3.6 \text{ V}$<br>CL = 20 pF  | -   | -   | 133 | MHz  |
|                                |                         | $1.62 < V_{\text{DD}} < 3.6 \text{ V}$<br>CL = 15 pF | -   | -   | 100 |      |

Table 181. QUADSPI characteristics in SDR mode<sup>(1)</sup> (continued)

| Symbol       | Parameter                                     | Conditions                               | Min                            | Typ | Max                              | Unit |
|--------------|-----------------------------------------------|------------------------------------------|--------------------------------|-----|----------------------------------|------|
| $t_{w(CKH)}$ | QUADSPI clock high and low time Even division | PRESCALER[7:0] = $n = 0, 1, 3, 5, \dots$ | $T_{CK}/2 - 0.5$               | -   | $T_{CK}/2$                       | ns   |
| $t_{w(CKL)}$ |                                               |                                          | $T_{CK}/2$                     | -   | $T_{CK}/2 + 0.5$                 |      |
| $t_{s(IN)}$  | QUADSPI clock high and low time Odd division  | PRESCALER[7:0] = $n = 2, 4, 6, 8, \dots$ | $(n/2) * T_{CK} / (n+1) - 0.5$ | -   | $(n/2) * T_{CK} / (n+1)$         |      |
| $t_{h(IN)}$  |                                               |                                          | $(n/2+1) * T_{CK} / (n+1)$     | -   | $(n/2+1) * T_{CK} / (n+1) + 0.5$ |      |
| $t_{v(OUT)}$ | Data input setup time                         | -                                        | 1                              | -   | -                                |      |
| $t_{h(OUT)}$ | Data input hold time                          |                                          | 3.5                            | -   | -                                |      |
| $t_{w(CKH)}$ | Data output valid time                        | -                                        | -                              | 1   | 2                                |      |
| $t_{w(CKL)}$ | Data output hold time                         | -                                        | 0                              | -   | -                                |      |

1. Guaranteed by characterization results.

The following table summarizes the parameters measured in DDR mode.

Table 182. QUADSPI characteristics in DDR mode<sup>(1)</sup>

| Symbol                     | Parameter                                     | Conditions                               | Min                            | Typ            | Max                              | Unit |
|----------------------------|-----------------------------------------------|------------------------------------------|--------------------------------|----------------|----------------------------------|------|
| $F_{ck1}/T_{CK}$           | QUADSPI clock frequency                       | $2.7 < V_{DD} < 3.6$ V<br>CL = 20 pF     | -                              | -              | 100                              | MHz  |
|                            |                                               | $1.62 < V_{DD} < 3.6$ V<br>CL = 15 pF    | -                              | -              | 100                              |      |
| $t_{w(CKH)}$               | QUADSPI clock high and low time Even division | PRESCALER[7:0] = $n = 0, 1, 3, 5, \dots$ | $T_{CK}/2 - 0.5$               | -              | $T_{CK}/2$                       | ns   |
| $t_{w(CKL)}$               |                                               |                                          | $T_{CK}/2$                     | -              | $T_{CK}/2 + 0.5$                 |      |
| $t_{w(CKH)}$               | QUADSPI clock high and low time Odd division  | PRESCALER[7:0] = $n = 2, 4, 6, 8, \dots$ | $(n/2) * T_{CK} / (n+1) - 0.5$ | -              | $(n/2) * T_{CK} / (n+1)$         |      |
| $t_{w(CKL)}$               |                                               |                                          | $(n/2+1) * T_{CK} / (n+1)$     | -              | $(n/2+1) * T_{CK} / (n+1) + 0.5$ |      |
| $t_{sr(IN)}, t_{sf(IN)}$   | Data input setup time                         | -                                        | 1.5                            | -              | -                                |      |
| $t_{hr(IN)}, t_{hf(IN)}$   | Data input hold time                          | -                                        | 3.5                            | -              | -                                |      |
| $t_{vr(OUT)}, t_{vf(OUT)}$ | Data output valid time                        | DHHC=0                                   | -                              | 5              | 6                                |      |
|                            |                                               | DHHC=1<br>PRESCALER[7:0] = 1, 2, ...     | -                              | $T_{CK}/4 + 1$ | $T_{CK}/4 + 2$                   |      |
| $t_{hr(OUT)}, t_{hf(OUT)}$ | Data output hold time                         | DHHC=0                                   | 3                              | -              | -                                |      |
|                            |                                               | DHHC=1<br>PRESCALER[7:0] = 1, 2, ...     | $T_{CK}/4$                     | -              | -                                |      |

1. Guaranteed by characterization results.

Figure 90. Quad-SPI timing diagram - SDR mode

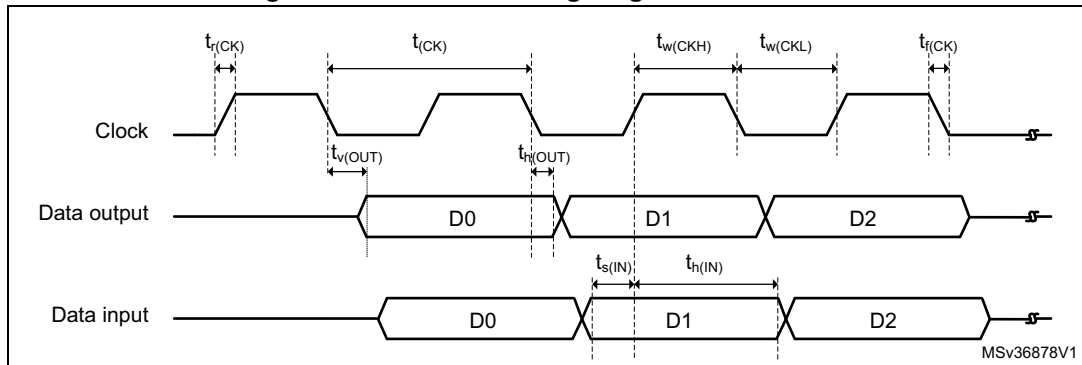
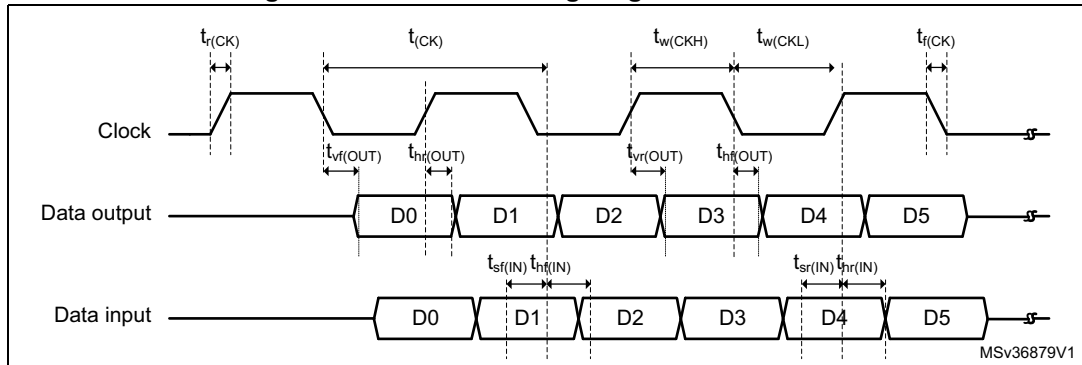


Figure 91. Quad-SPI timing diagram - DDR mode



### 7.3.19 Delay block (DLYB) characteristics

Unless otherwise specified, the parameters given in [Table 183](#) for Delay Block are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and VDD supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

Table 183. Delay Block characteristics

| Symbol            | Parameter     | Conditions | Min  | Typ  | Max  | Unit |
|-------------------|---------------|------------|------|------|------|------|
| $t_{\text{init}}$ | Initial delay | -          | 1400 | 2200 | 2400 | ps   |
| $t_{\Delta}$      | Unit Delay    | -          | 35   | 40   | 45   | -    |

### 7.3.20 16-bit ADC characteristics

Unless otherwise specified, the parameters given in [Table 184](#) are derived from tests performed under the ambient temperature,  $f_{PCLK2}$  frequency and  $V_{DDA}$  supply voltage conditions summarized in [Table 122: General operating conditions](#).

**Table 184. ADC characteristics<sup>(1)(2)</sup>**

| Symbol                          | Parameter                                        | Conditions                                     |                         |                           |           | Min                       | Typ                 | Max                       | Unit               |     |
|---------------------------------|--------------------------------------------------|------------------------------------------------|-------------------------|---------------------------|-----------|---------------------------|---------------------|---------------------------|--------------------|-----|
| V <sub>DDA</sub>                | Analog supply voltage for ADC ON                 | -                                              |                         |                           |           | 1.62                      | -                   | 3.6                       | V                  |     |
| V <sub>REF+</sub>               | Positive reference voltage                       | -                                              |                         |                           |           | 1.62                      | -                   | V <sub>DDA</sub>          | V                  |     |
| V <sub>REF-</sub>               | Negative reference voltage                       | -                                              |                         |                           |           | V <sub>SSA</sub>          |                     |                           | V                  |     |
| f <sub>ADC</sub>                | ADC clock frequency                              | 1.62 V ≤ V <sub>DDA</sub> ≤ 3.6 V              |                         |                           |           | BOOST = 11                | 0.12                | -                         | 50                 | MHz |
|                                 |                                                  |                                                |                         |                           |           | BOOST = 10                | 0.12                | -                         | 25                 |     |
|                                 |                                                  |                                                |                         |                           |           | BOOST = 01                | 0.12                | -                         | 12.5               |     |
|                                 |                                                  |                                                |                         |                           |           | BOOST = 00                | -                   | -                         | 6.25               |     |
| f <sub>s</sub> <sup>(3)</sup>   | Sampling rate for Direct channels <sup>(4)</sup> | Resolution = 16 bits, V <sub>DDA</sub> > 2.5 V | T <sub>J</sub> = 90 °C  | f <sub>ADC</sub> =36 MHz  | SMP = 1.5 | -                         | -                   | 3.60                      | MSps               |     |
|                                 |                                                  | Resolution = 16 bits                           |                         | f <sub>ADC</sub> =37 MHz  | SMP = 2.5 | -                         | -                   | 3.35                      |                    |     |
|                                 |                                                  | Resolution = 14 bits                           | T <sub>J</sub> = 125 °C | f <sub>ADC</sub> = 50 MHz | SMP = 2.5 | -                         | -                   | 5.00                      |                    |     |
|                                 |                                                  | Resolution = 12 bits                           |                         | f <sub>ADC</sub> = 50 MHz | SMP = 2.5 | -                         | -                   | 5.50                      |                    |     |
|                                 |                                                  | Resolution = 10 bits                           |                         | f <sub>ADC</sub> = 50 MHz | SMP = 1.5 | -                         | -                   | 7.10                      |                    |     |
|                                 |                                                  | Resolution = 8 bits                            |                         | f <sub>ADC</sub> = 50 MHz | SMP = 1.5 | -                         | -                   | 8.30                      |                    |     |
|                                 | Sampling rate for Fast channels                  | Resolution = 16 bits, V <sub>DDA</sub> > 2.5 V | T <sub>J</sub> = 90 °C  | f <sub>ADC</sub> =32 MHz  | SMP = 2.5 | -                         | -                   | 2.90                      |                    |     |
|                                 |                                                  | Resolution = 16 bits                           |                         | f <sub>ADC</sub> =31 MHz  | SMP = 2.5 | -                         | -                   | 2.80                      |                    |     |
|                                 |                                                  | Resolution = 14 bits                           | T <sub>J</sub> = 125 °C | f <sub>ADC</sub> = 33 MHz | SMP = 2.5 | -                         | -                   | 3.30                      |                    |     |
|                                 |                                                  | Resolution = 12 bits                           |                         | f <sub>ADC</sub> = 39 MHz | SMP = 2.5 | -                         | -                   | 4.30                      |                    |     |
|                                 |                                                  | Resolution = 10 bits                           |                         | f <sub>ADC</sub> = 48 MHz | SMP = 2.5 | -                         | -                   | 6.00                      |                    |     |
|                                 |                                                  | Resolution = 8 bits                            |                         | f <sub>ADC</sub> = 50 MHz | SMP = 2.5 | -                         | -                   | 7.10                      |                    |     |
|                                 | Sampling rate for Slow channels                  | Resolution = 16 bits                           | T <sub>J</sub> = 90 °C  | f <sub>ADC</sub> = 10 MHz | SMP = 1.5 | -                         | -                   | 1.00                      |                    |     |
|                                 |                                                  | resolution = 14 bits                           |                         |                           |           | -                         | -                   |                           |                    |     |
|                                 |                                                  | resolution = 12 bits                           | T <sub>J</sub> = 125 °C |                           |           | -                         | -                   |                           |                    |     |
|                                 |                                                  | resolution = 10 bits                           |                         |                           |           | -                         | -                   |                           |                    |     |
|                                 |                                                  | resolution = 8 bits                            |                         |                           |           | -                         | -                   |                           |                    |     |
| t <sub>TRIG</sub>               | External trigger period                          | Resolution = 16 bits                           |                         |                           |           | -                         | -                   | 10                        | 1/f <sub>ADC</sub> |     |
| V <sub>AIN</sub> <sup>(5)</sup> | Conversion voltage range                         | -                                              |                         |                           |           | 0                         | -                   | V <sub>REF+</sub>         | V                  |     |
| V <sub>CMIV</sub>               | Common mode input voltage                        | -                                              |                         |                           |           | V <sub>REF</sub> /2 - 10% | V <sub>REF</sub> /2 | V <sub>REF</sub> /2 + 10% | V                  |     |



Table 184. ADC characteristics<sup>(1)(2)</sup> (continued)

| Symbol              | Parameter                                                                          | Conditions                                                |   |   | Min               | Typ | Max   | Unit             |
|---------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|---|---|-------------------|-----|-------|------------------|
| $R_{AIN}^{(6)}$     | External input impedance                                                           | Resolution = 16 bits, $T_J = 125\text{ }^{\circ}\text{C}$ | - | - | -                 | -   | 170   | $\Omega$         |
|                     |                                                                                    | Resolution = 14 bits, $T_J = 125\text{ }^{\circ}\text{C}$ | - | - | -                 | -   | 435   |                  |
|                     |                                                                                    | Resolution = 12 bits, $T_J = 125\text{ }^{\circ}\text{C}$ | - | - | -                 | -   | 1150  |                  |
|                     |                                                                                    | Resolution = 10 bits, $T_J = 125\text{ }^{\circ}\text{C}$ | - | - | -                 | -   | 5650  |                  |
|                     |                                                                                    | Resolution = 8 bits, $T_J = 125\text{ }^{\circ}\text{C}$  | - | - | -                 | -   | 26500 |                  |
| $C_{ADC}$           | Internal sample and hold capacitor                                                 | -                                                         |   |   | -                 | 4   | -     | pF               |
| $t_{ADCVREG\_STUP}$ | ADC LDO startup time                                                               | -                                                         |   |   | -                 | 5   | 10    | us               |
| $t_{STAB}$          | ADC Power-up time                                                                  | LDO already started                                       |   |   | 1                 | -   | -     | conversion cycle |
| $t_{CAL}$           | Offset and linearity calibration time                                              | -                                                         |   |   | 165010            | -   | -     | $1/f_{ADC}$      |
| $t_{OFF\_CAL}$      | Offset calibration time                                                            | -                                                         |   |   | 1280              | -   | -     | $1/f_{ADC}$      |
| $t_{LATR}$          | Trigger conversion latency regular and injected channels without conversion abort  | CKMODE = 00                                               |   |   | 1.5               | 2   | 2.5   | $1/f_{ADC}$      |
|                     |                                                                                    | CKMODE = 01                                               |   |   | -                 | -   | 2.5   |                  |
|                     |                                                                                    | CKMODE = 10                                               |   |   | -                 | -   | 2.5   |                  |
|                     |                                                                                    | CKMODE = 11                                               |   |   | -                 | -   | 2.25  |                  |
| $t_{LATRINJ}$       | Trigger conversion latency regular injected channels aborting a regular conversion | CKMODE = 00                                               |   |   | 2.5               | 3   | 3.5   | $1/f_{ADC}$      |
|                     |                                                                                    | CKMODE = 01                                               |   |   | -                 | -   | 3.5   |                  |
|                     |                                                                                    | CKMODE = 10                                               |   |   | -                 | -   | 3.5   |                  |
|                     |                                                                                    | CKMODE = 11                                               |   |   | -                 | -   | 3.25  |                  |
| $t_S$               | Sampling time                                                                      | -                                                         |   |   | 1.5               | -   | 810.5 | $1/f_{ADC}$      |
| $t_{CONV}$          | Total conversion time (including sampling time)                                    | Resolution = N bits                                       |   |   | $t_S + 0.5 + N/2$ | -   | -     | $1/f_{ADC}$      |

Table 184. ADC characteristics<sup>(1)(2)</sup> (continued)

| Symbol                 | Parameter                                                                      | Conditions                              |   |   | Min | Typ   | Max | Unit    |
|------------------------|--------------------------------------------------------------------------------|-----------------------------------------|---|---|-----|-------|-----|---------|
| $I_{DDA\_D}$<br>(ADC)  | ADC consumption on $V_{DDA}$ , BOOST=11, Differential mode                     | Resolution = 16 bits, $f_{ADC}$ =25 MHz | - | - | -   | 1440  | -   | $\mu A$ |
|                        |                                                                                | Resolution = 14 bits, $f_{ADC}$ =30 MHz | - | - | -   | 1350  | -   |         |
|                        |                                                                                | Resolution = 12 bits, $f_{ADC}$ =40 MHz | - | - | -   | 990   | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=10, Differential mode $f_{ADC}$ =25 MHz   | Resolution = 16 bits                    | - | - | -   | 1080  | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 810   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 585   | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=01, Differential mode $f_{ADC}$ =12.5 MHz | Resolution = 16 bits                    | - | - | -   | 630   | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 432   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 315   | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=00, Differential mode $f_{ADC}$ =6.25 MHz | Resolution = 16 bits                    | - | - | -   | 360   | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 270   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 225   | -   |         |
| $I_{DDA\_SE}$<br>(ADC) | ADC consumption on $V_{DDA}$ , BOOST=11, Single-ended mode                     | Resolution = 16 bits, $f_{ADC}$ =25 MHz | - | - | -   | 720   | -   | $\mu A$ |
|                        |                                                                                | Resolution = 14 bits, $f_{ADC}$ =30 MHz | - | - | -   | 675   | -   |         |
|                        |                                                                                | Resolution = 12 bits, $f_{ADC}$ =40 MHz | - | - | -   | 495   | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=10, Single-ended mode $f_{ADC}$ =25 MHz   | Resolution = 16 bits                    | - | - | -   | 540   | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 405   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 292.5 | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=01, Single-ended mode $f_{ADC}$ =12.5 MHz | Resolution = 16 bits                    | - | - | -   | 315   | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 216   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 157.5 | -   |         |
|                        | ADC consumption on $V_{DDA}$ , BOOST=00, Single-ended mode $f_{ADC}$ =6.25 MHz | Resolution = 16 bits                    | - | - | -   | 180   | -   |         |
|                        |                                                                                | Resolution = 14 bits                    | - | - | -   | 135   | -   |         |
|                        |                                                                                | Resolution = 12 bits                    | - | - | -   | 112.5 | -   |         |
| $I_{DD}$<br>(ADC)      | ADC consumption on $V_{DD}$ , BOOST=11                                         | $f_{ADC}$ =50 MHz                       | - | - | -   | 400   | -   | $\mu A$ |
|                        |                                                                                | $f_{ADC}$ =25 MHz                       | - | - | -   | 220   | -   |         |
|                        |                                                                                | $f_{ADC}$ =12.5 MHz                     | - | - | -   | 180   | -   |         |
|                        |                                                                                | $f_{ADC}$ =6.25 MHz                     | - | - | -   | 120   | -   |         |
|                        |                                                                                | $f_{ADC}$ =3.125 MHz                    | - | - | -   | 80    | -   |         |

1. Guaranteed by design.
2. The voltage booster on ADC switches must be used for  $V_{DDA} < 2.4$  V (embedded I/O switches).
3. These values are valid for UFBGA169 and one ADC. The values for other packages and multiple ADCs may be different.
4. Direct channels are connected to analog I/Os (PA0\_C, PA1\_C, PC2\_C and PC3\_C) to optimize ADC performance.
5. Depending on the package,  $V_{REF+}$  can be internally connected to  $V_{DDA}$  and  $V_{REF-}$  to  $V_{SSA}$ .
6. The tolerance is 10 LSBs for 16-bit resolution, 4 LSBs for 14-bit resolution, and 2 LSBs for 12-bit, 10-bit and 8-bit resolutions.

Table 185. Minimum sampling time vs  $R_{AIN}^{(1)(2)}$ 

| Resolution | RAIN ( $\Omega$ ) | Minimum sampling time (s)      |                              |                              |
|------------|-------------------|--------------------------------|------------------------------|------------------------------|
|            |                   | Direct channels <sup>(3)</sup> | Fast channels <sup>(4)</sup> | Slow channels <sup>(5)</sup> |
| 16 bits    | 47                | 7.37E-08                       | 1.14E-07                     | 1.72E-07                     |
| 14 bits    | 47                | 6.29E-08                       | 9.74E-08                     | 1.55E-07                     |
|            | 68                | 6.84E-08                       | 1.02E-07                     | 1.58E-07                     |
|            | 100               | 7.80E-08                       | 1.12E-07                     | 1.62E-07                     |
|            | 150               | 9.86E-08                       | 1.32E-07                     | 1.80E-07                     |
|            | 220               | 1.32E-07                       | 1.61E-07                     | 2.01E-07                     |
| 12 bits    | 47                | 5.32E-08                       | 8.00E-08                     | 1.29E-07                     |
|            | 68                | 5.74E-08                       | 8.50E-08                     | 1.32E-07                     |
|            | 100               | 6.58E-08                       | 9.31E-08                     | 1.40E-07                     |
|            | 150               | 8.37E-08                       | 1.10E-07                     | 1.51E-07                     |
|            | 220               | 1.11E-07                       | 1.34E-07                     | 1.73E-07                     |
|            | 330               | 1.56E-07                       | 1.78E-07                     | 2.14E-07                     |
|            | 470               | 2.16E-07                       | 2.39E-07                     | 2.68E-07                     |
|            | 680               | 3.01E-07                       | 3.29E-07                     | 3.54E-07                     |
| 10 bits    | 47                | 4.34E-08                       | 6.51E-08                     | 1.08E-07                     |
|            | 68                | 4.68E-08                       | 6.89E-08                     | 1.11E-07                     |
|            | 100               | 5.35E-08                       | 7.55E-08                     | 1.16E-07                     |
|            | 150               | 6.68E-08                       | 8.77E-08                     | 1.26E-07                     |
|            | 220               | 8.80E-08                       | 1.08E-07                     | 1.40E-07                     |
|            | 330               | 1.24E-07                       | 1.43E-07                     | 1.71E-07                     |
|            | 470               | 1.69E-07                       | 1.89E-07                     | 2.13E-07                     |
|            | 680               | 2.38E-07                       | 2.60E-07                     | 2.80E-07                     |
|            | 1000              | 3.45E-07                       | 3.66E-07                     | 3.84E-07                     |
|            | 1500              | 5.15E-07                       | 5.35E-07                     | 5.48E-07                     |
|            | 2200              | 7.42E-07                       | 7.75E-07                     | 7.78E-07                     |
|            | 3300              | 1.10E-06                       | 1.14E-06                     | 1.14E-06                     |

Table 185. Minimum sampling time vs  $R_{AIN}^{(1)(2)}$  (continued)

| Resolution | RAIN ( $\Omega$ ) | Minimum sampling time (s)      |                              |                              |
|------------|-------------------|--------------------------------|------------------------------|------------------------------|
|            |                   | Direct channels <sup>(3)</sup> | Fast channels <sup>(4)</sup> | Slow channels <sup>(5)</sup> |
| 8 bits     | 47                | 3.32E-08                       | 5.10E-08                     | 8.61E-08                     |
|            | 68                | 3.59E-08                       | 5.35E-08                     | 8.83E-08                     |
|            | 100               | 4.10E-08                       | 5.83E-08                     | 9.22E-08                     |
|            | 150               | 5.06E-08                       | 6.76E-08                     | 9.95E-08                     |
|            | 220               | 6.61E-08                       | 8.22E-08                     | 1.11E-07                     |
|            | 330               | 9.17E-08                       | 1.08E-07                     | 1.32E-07                     |
|            | 470               | 1.24E-07                       | 1.40E-07                     | 1.63E-07                     |
|            | 680               | 1.74E-07                       | 1.91E-07                     | 2.12E-07                     |
|            | 1000              | 2.53E-07                       | 2.70E-07                     | 2.85E-07                     |
|            | 1500              | 3.73E-07                       | 3.93E-07                     | 4.05E-07                     |
|            | 2200              | 5.39E-07                       | 5.67E-07                     | 5.75E-07                     |
|            | 3300              | 8.02E-07                       | 8.36E-07                     | 8.38E-07                     |
|            | 4700              | 1.13E-06                       | 1.18E-06                     | 1.18E-06                     |
|            | 6800              | 1.62E-06                       | 1.69E-06                     | 1.68E-06                     |
|            | 10000             | 2.36E-06                       | 2.47E-06                     | 2.45E-06                     |
|            | 15000             | 3.50E-06                       | 3.69E-06                     | 3.65E-06                     |

1. Guaranteed by design.
2. Data valid at up to 125 °C, with a 47 pF PCB capacitor, and  $V_{DDA}=1.6$  V.
3. Direct channels are connected to analog I/Os (PA0\_C, PA1\_C, PC2\_C and PC3\_C) to optimize ADC performance.
4. Fast channels correspond to PC0, PC1, PC2, PC3, PA0, and PA1.
5. Slow channels correspond to all ADC inputs except for the Fast channels.

Table 186. ADC accuracy<sup>(1)(2)</sup>

| Symbol | Parameter                            | Conditions <sup>(3)</sup> |              | Min | Typ     | Max | Unit |
|--------|--------------------------------------|---------------------------|--------------|-----|---------|-----|------|
| ET     | Total undadjusted error              | Direct channel            | Single ended | -   | +10/-20 | -   | LSB  |
|        |                                      |                           | Differential | -   | ±15     | -   |      |
|        |                                      | Fast channel              | Single ended | -   | +10/-20 | -   |      |
|        |                                      |                           | Differential | -   | ±15     | -   |      |
|        |                                      | Slow channel              | Single ended | -   | ±10     | -   |      |
|        |                                      |                           | Differential | -   | ±10     | -   |      |
| EO     | Offset error                         | -                         |              | -   | ±10     | -   |      |
| EG     | Gain error                           | -                         |              | -   | ±15     | -   |      |
| ED     | Differential linearity error         | Single ended              |              | -   | +3/-1   | -   |      |
|        |                                      | Differential              |              | -   | +4.5/-1 | -   |      |
| EL     | Integral linearity error             | Direct channel            | Single ended | -   | ±11     | -   |      |
|        |                                      |                           | Differential | -   | ±7      | -   |      |
|        |                                      | Fast channel              | Single ended | -   | ±13     | -   |      |
|        |                                      |                           | Differential | -   | ±7      | -   |      |
|        |                                      | Slow channel              | Single ended | -   | ±10     | -   |      |
|        |                                      |                           | Differential | -   | ±6      | -   |      |
| ENOB   | Effective number of bits             | Single ended              |              | -   | 12.2    | -   | Bits |
|        |                                      | Differential              |              | -   | 13.2    | -   |      |
| SINAD  | Signal-to-noise and distortion ratio | Single ended              |              | -   | 75.2    | -   | dB   |
|        |                                      | Differential              |              | -   | 81.2    | -   |      |
| SNR    | Signal-to-noise ratio                | Single ended              |              | -   | 77.0    | -   |      |
|        |                                      | Differential              |              | -   | 81.0    | -   |      |
| THD    | Total harmonic distortion            | Single ended              |              | -   | 87      | -   |      |
|        |                                      | Differential              |              | -   | 90      | -   |      |

1. Data guaranteed by characterization for BGA packages. The values for LQFP packages might differ.

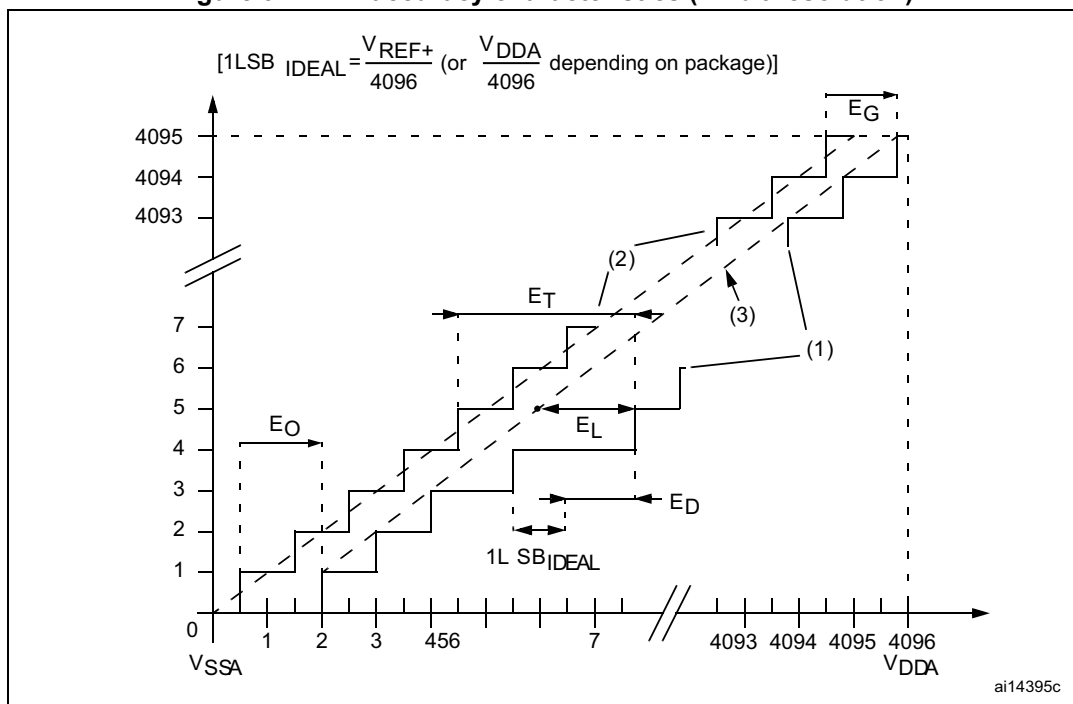
2. ADC DC accuracy values are measured after internal calibration.

3. ADC clock frequency = 25 MHz, ADC resolution = 16 bits,  $V_{DDA}=V_{REF+}=3.3$  V and BOOST=11.

Note: ADC accuracy vs. negative injection current: injecting a negative current on any analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to analog pins which may potentially inject negative currents.

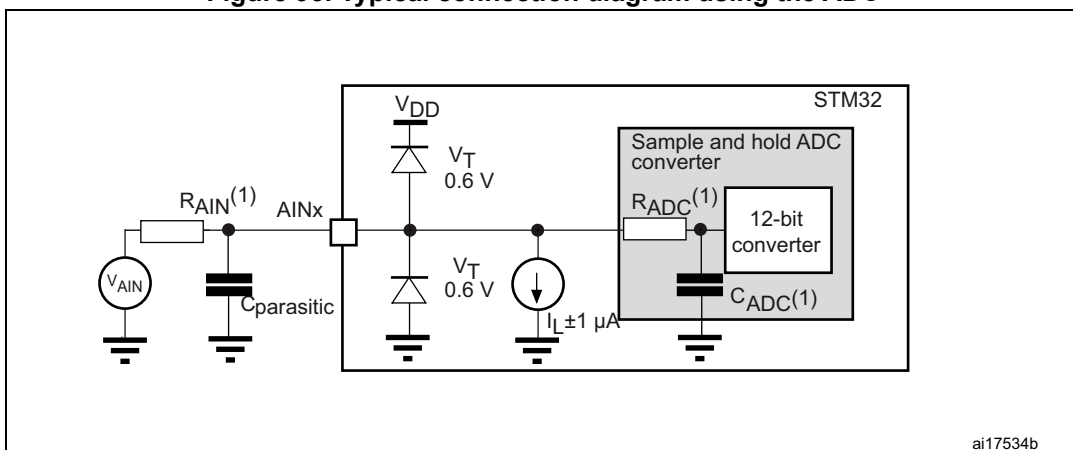
Any positive injection current within the limits specified for  $I_{INJ(PIN)}$  and  $\Sigma I_{INJ(PIN)}$  in [Section 7.3.14](#) does not affect the ADC accuracy.

Figure 92. ADC accuracy characteristics (12-bit resolution)



1. Example of an actual transfer curve.
2. Ideal transfer curve.
3. End point correlation line.
4.  $E_T$  = Total Unadjusted Error: maximum deviation between the actual and the ideal transfer curves.  
 $E_O$  = Offset Error: deviation between the first actual transition and the first ideal one.  
 $E_G$  = Gain Error: deviation between the last ideal transition and the last actual one.  
 $E_D$  = Differential Linearity Error: maximum deviation between actual steps and the ideal one.  
 $E_L$  = Integral Linearity Error: maximum deviation between any actual transition and the end point correlation line.

Figure 93. Typical connection diagram using the ADC

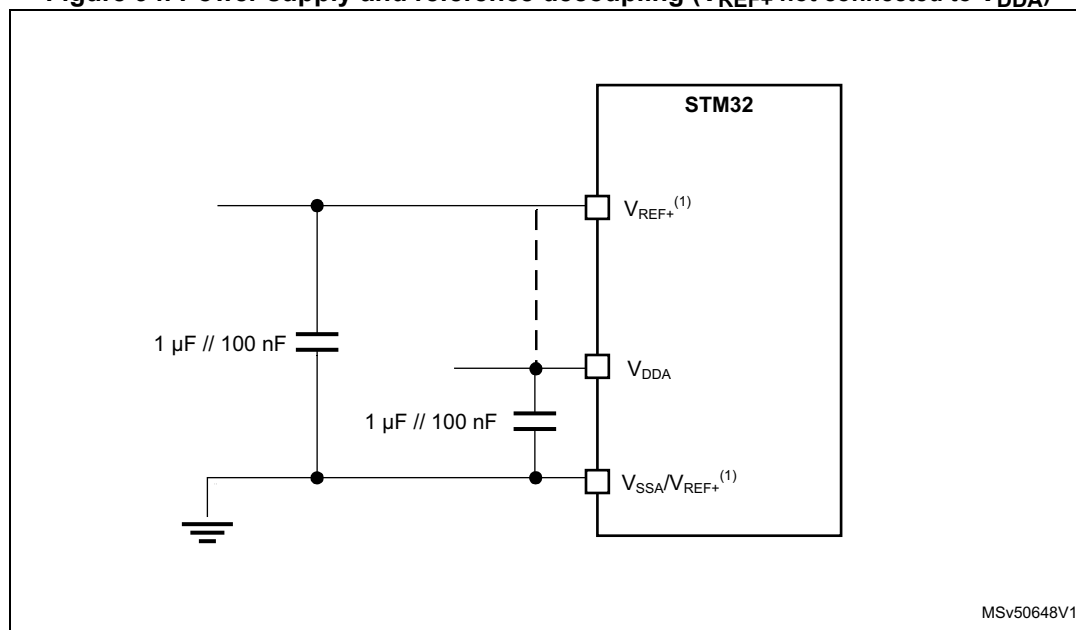


1. Refer to [Table 184](#) for the values of  $R_{AIN}$ ,  $R_{ADC}$  and  $C_{ADC}$ .
2.  $C_{parasitic}$  represents the capacitance of the PCB (dependent on soldering and PCB layout quality) plus the pad capacitance (roughly 5 pF). A high  $C_{parasitic}$  value downgrades conversion accuracy. To remedy this,  $f_{ADC}$  should be reduced.

### General PCB design guidelines

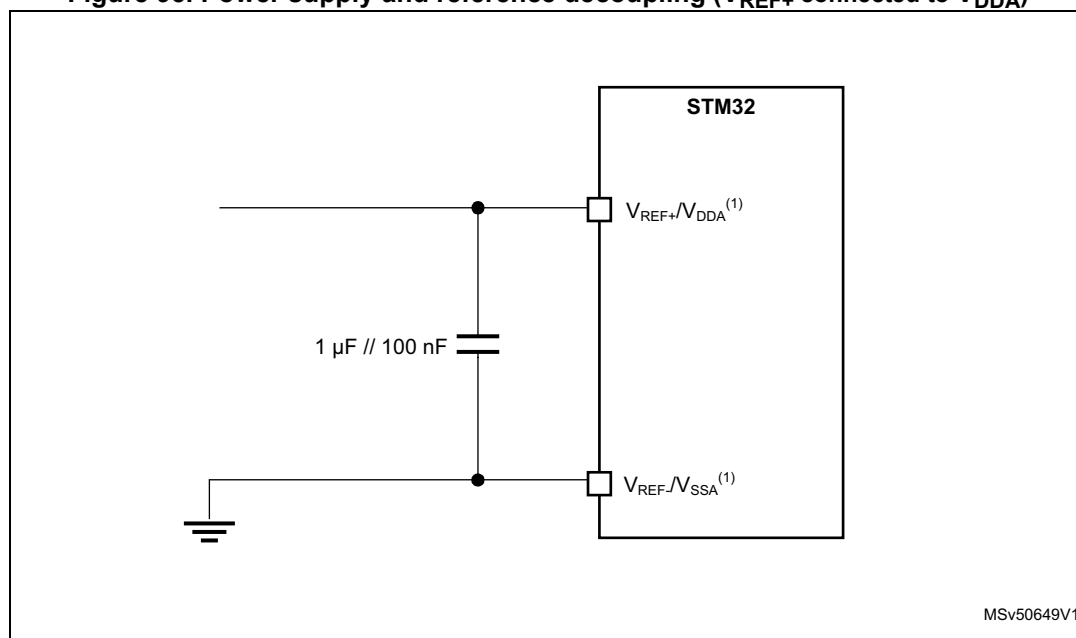
Power supply decoupling should be performed as shown in [Figure 94](#) or [Figure 95](#), depending on whether  $V_{REF+}$  is connected to  $V_{DDA}$  or not. The 100 nF capacitors should be ceramic (good quality). They should be placed as close as possible to the chip.

**Figure 94. Power supply and reference decoupling ( $V_{REF+}$  not connected to  $V_{DDA}$ )**



1.  $V_{REF+}$  input is available on all package whereas the  $V_{REF-}$  is available only on UFBGA176+25 and TFBGA240+25. When  $V_{REF-}$  is not available, it is internally connected to  $V_{DDA}$  and  $V_{SSA}$ .

**Figure 95. Power supply and reference decoupling ( $V_{REF+}$  connected to  $V_{DDA}$ )**



1.  $V_{REF+}$  input is available on all package whereas the  $V_{REF-}$  is available only on UFBGA176+25 and TFBGA240+25. When  $V_{REF-}$  is not available, it is internally connected to  $V_{DDA}$  and  $V_{SSA}$ .

## 7.3.21 DAC characteristics

Table 187. DAC characteristics<sup>(1)(2)</sup>

| Symbol             | Parameter                                                                                                                                                                                                                                            | Conditions                                                                               |                         | Min  | Typ       | Max             | Unit          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------|------|-----------|-----------------|---------------|
| $V_{DDA}$          | Analog supply voltage                                                                                                                                                                                                                                | -                                                                                        |                         | 1.8  | 3.3       | 3.6             | V             |
| $V_{REF+}$         | Positive reference voltage                                                                                                                                                                                                                           | -                                                                                        |                         | 1.80 | -         | $V_{DDA}$       |               |
| $V_{REF-}$         | Negative reference voltage                                                                                                                                                                                                                           | -                                                                                        |                         | -    | $V_{SSA}$ | -               |               |
| $R_L$              | Resistive Load                                                                                                                                                                                                                                       | DAC output buffer ON                                                                     | connected to $V_{SSA}$  | 5    | -         | -               | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | connected to $V_{DDA}$  | 25   | -         | -               |               |
| $R_O$              | Output Impedance                                                                                                                                                                                                                                     | DAC output buffer OFF                                                                    |                         | 10.3 | 13        | 16              |               |
| $R_{BON}$          | Output impedance sample and hold mode, output buffer ON                                                                                                                                                                                              | DAC output buffer ON                                                                     | $V_{DD} = 2.7\text{ V}$ | -    | -         | 1.6             | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $V_{DD} = 2.0\text{ V}$ | -    | -         | 2.6             |               |
| $R_{BOFF}$         | Output impedance sample and hold mode, output buffer OFF                                                                                                                                                                                             | DAC output buffer OFF                                                                    | $V_{DD} = 2.7\text{ V}$ | -    | -         | 17.8            | k $\Omega$    |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $V_{DD} = 2.0\text{ V}$ | -    | -         | 18.7            |               |
| $C_L$              | Capacitive Load                                                                                                                                                                                                                                      | DAC output buffer OFF                                                                    |                         | -    | -         | 50              | pF            |
| $C_{SH}$           |                                                                                                                                                                                                                                                      | Sample and Hold mode                                                                     |                         | -    | 0.1       | 1               | $\mu\text{F}$ |
| $V_{DAC\_OUT}$     | Voltage on DAC_OUT output                                                                                                                                                                                                                            | DAC output buffer ON                                                                     |                         | 0.2  | -         | $V_{DDA} - 0.2$ | V             |
|                    |                                                                                                                                                                                                                                                      | DAC output buffer OFF                                                                    |                         | 0    | -         | $V_{REF+}$      |               |
| $t_{SETTLING}$     | Settling time (full scale: for a 12-bit code transition between the lowest and the highest input codes when DAC_OUT reaches the final value of $\pm 0.5\text{LSB}$ , $\pm 1\text{LSB}$ , $\pm 2\text{LSB}$ , $\pm 4\text{LSB}$ , $\pm 8\text{LSB}$ ) | Normal mode, DAC output buffer ON, $C_L \leq 50\text{ pF}$ , $R_L \geq 5\text{ k}\Omega$ | $\pm 0.5\text{ LSB}$    | -    | 2.05      | -               | $\mu\text{s}$ |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $\pm 1\text{ LSB}$      | -    | 1.97      | -               |               |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $\pm 2\text{ LSB}$      | -    | 1.67      | -               |               |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $\pm 4\text{ LSB}$      | -    | 1.66      | -               |               |
|                    |                                                                                                                                                                                                                                                      |                                                                                          | $\pm 8\text{ LSB}$      | -    | 1.65      | -               |               |
|                    |                                                                                                                                                                                                                                                      | Normal mode, DAC output buffer OFF, $\pm 1\text{LSB}$ $C_L = 10\text{ pF}$               |                         | -    | 1.7       | 2               |               |
| $t_{WAKEUP}^{(3)}$ | Wakeup time from off state (setting the ENx bit in the DAC Control register) until the final value of $\pm 1\text{LSB}$ is reached                                                                                                                   | Normal mode, DAC output buffer ON, $C_L \leq 50\text{ pF}$ , $R_L = 5\text{ k}\Omega$    |                         | -    | 5         | 7.5             | $\mu\text{s}$ |
|                    |                                                                                                                                                                                                                                                      | Normal mode, DAC output buffer OFF, $C_L \leq 10\text{ pF}$                              |                         |      | 2         | 5               |               |
| PSRR               | DC $V_{DDA}$ supply rejection ratio                                                                                                                                                                                                                  | Normal mode, DAC output buffer ON, $C_L \leq 50\text{ pF}$ , $R_L = 5\text{ k}\Omega$    |                         | -    | -80       | -28             | dB            |



Table 187. DAC characteristics<sup>(1)(2)</sup> (continued)

| Symbol                       | Parameter                                                                                                                                                                                            | Conditions                                                                      |                                    | Min | Typ                                                                | Max  | Unit          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------|-----|--------------------------------------------------------------------|------|---------------|
| $t_{\text{SAMP}}$            | Sampling time in Sample and Hold mode<br>$C_L = 100 \text{ nF}$<br>(code transition between the lowest input code and the highest input code when DAC_OUT reaches the $\pm 1\text{LSB}$ final value) | MODE<2:0>_V12=100/101 (BUFFER ON)                                               |                                    | -   | 0.7                                                                | 2.6  | ms            |
|                              |                                                                                                                                                                                                      | MODE<2:0>_V12=110 (BUFFER OFF)                                                  |                                    | -   | 11.5                                                               | 18.7 |               |
|                              |                                                                                                                                                                                                      | MODE<2:0>_V12=111 (INTERNAL BUFFER OFF)                                         |                                    | -   | 0.3                                                                | 0.6  | $\mu\text{s}$ |
| $C_{\text{lint}}$            | Internal sample and hold capacitor                                                                                                                                                                   | -                                                                               |                                    | 1.8 | 2.2                                                                | 2.6  | pF            |
| $t_{\text{TRIM}}$            | Middle code offset trim time                                                                                                                                                                         | Minimum time to verify the each code                                            |                                    | 50  | -                                                                  | -    | $\mu\text{s}$ |
| $V_{\text{offset}}$          | Middle code offset for 1 trim code step                                                                                                                                                              | $V_{\text{REF}+} = 3.6 \text{ V}$                                               |                                    | -   | 850                                                                | -    | $\mu\text{V}$ |
|                              |                                                                                                                                                                                                      | $V_{\text{REF}+} = 1.8 \text{ V}$                                               |                                    | -   | 425                                                                | -    |               |
| $I_{\text{DDA}}(\text{DAC})$ | DAC quiescent consumption from $V_{\text{DDA}}$                                                                                                                                                      | DAC output buffer ON                                                            | No load, middle code (0x800)       | -   | 360                                                                | -    | $\mu\text{A}$ |
|                              |                                                                                                                                                                                                      |                                                                                 | No load, worst code (0xF1C)        | -   | 490                                                                | -    |               |
|                              |                                                                                                                                                                                                      | DAC output buffer OFF                                                           | No load, middle/worst code (0x800) | -   | 20                                                                 | -    |               |
|                              |                                                                                                                                                                                                      | Sample and Hold mode, $C_{\text{SH}} = 100 \text{ nF}$                          |                                    | -   | $360 \cdot T_{\text{ON}} / (T_{\text{ON}} + T_{\text{OFF}})_{(4)}$ | -    |               |
| $I_{\text{DDV}}(\text{DAC})$ | DAC consumption from $V_{\text{REF}+}$                                                                                                                                                               | DAC output buffer ON                                                            | No load, middle code (0x800)       | -   | 170                                                                | -    | $\mu\text{A}$ |
|                              |                                                                                                                                                                                                      |                                                                                 | No load, worst code (0xF1C)        | -   | 170                                                                | -    |               |
|                              |                                                                                                                                                                                                      | DAC output buffer OFF                                                           | No load, middle/worst code (0x800) | -   | 160                                                                | -    |               |
|                              |                                                                                                                                                                                                      | Sample and Hold mode, Buffer ON, $C_{\text{SH}} = 100 \text{ nF}$ (worst code)  |                                    | -   | $170 \cdot T_{\text{ON}} / (T_{\text{ON}} + T_{\text{OFF}})_{(4)}$ | -    |               |
|                              |                                                                                                                                                                                                      | Sample and Hold mode, Buffer OFF, $C_{\text{SH}} = 100 \text{ nF}$ (worst code) |                                    | -   | $160 \cdot T_{\text{ON}} / (T_{\text{ON}} + T_{\text{OFF}})_{(4)}$ | -    |               |
|                              |                                                                                                                                                                                                      |                                                                                 |                                    |     |                                                                    |      |               |

1. Guaranteed by design unless otherwise specified.

2. TBD stands for "to be defined".
3. In buffered mode, the output can overshoot above the final value for low input code (starting from the minimum value).
4.  $T_{ON}$  is the refresh phase duration, while  $T_{OFF}$  is the hold phase duration. Refer to the product reference manual for more details.

Table 188. DAC accuracy<sup>(1)</sup>

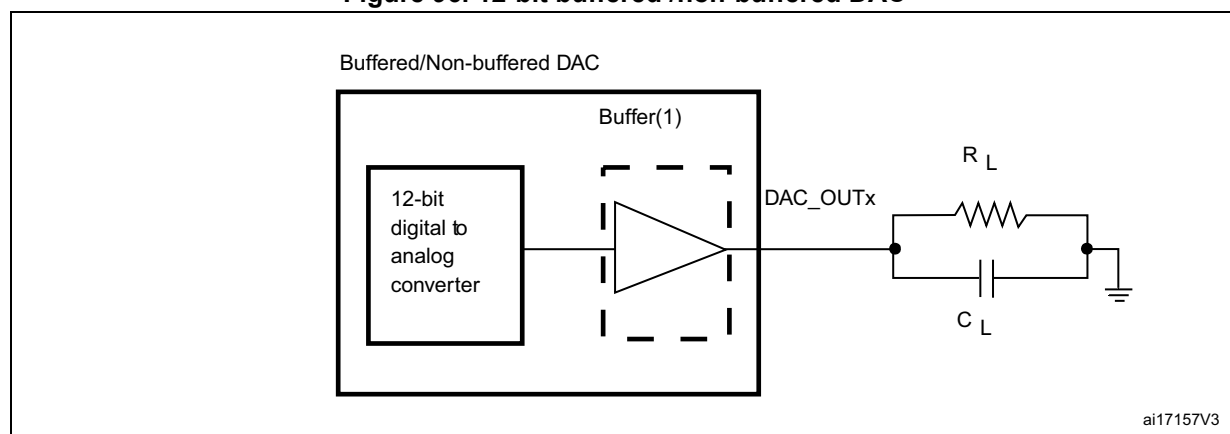
| Symbol    | Parameter                                            | Conditions                                                                            | Min | Typ   | Max      | Unit |
|-----------|------------------------------------------------------|---------------------------------------------------------------------------------------|-----|-------|----------|------|
| DNL       | Differential non linearity <sup>(2)</sup>            | DAC output buffer ON                                                                  | -2  | -     | 2        | LSB  |
|           |                                                      | DAC output buffer OFF                                                                 | -2  | -     | 2        |      |
| -         | Monotonicity                                         | 10 bits                                                                               | -   | -     | -        | -    |
| INL       | Integral non linearity <sup>(3)</sup>                | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$                       | -4  | -     | 4        | LSB  |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$                                     | -4  | -     | 4        |      |
| Offset    | Offset error at code 0x800 <sup>(3)</sup>            | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , $V_{REF+} = 3.6$ V  | -   | -     | $\pm 15$ | LSB  |
|           |                                                      | $V_{REF+} = 1.8$ V                                                                    | -   | -     | $\pm 30$ |      |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$                                     | -   | -     | $\pm 8$  |      |
| Offset1   | Offset error at code 0x001 <sup>(4)</sup>            | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$                                     | -   | -     | $\pm 5$  | LSB  |
| OffsetCal | Offset error at code 0x800 after factory calibration | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , $V_{REF+} = 3.6$ V  | -   | -     | $\pm 6$  | LSB  |
|           |                                                      | $V_{REF+} = 1.8$ V                                                                    | -   | -     | $\pm 7$  |      |
| Gain      | Gain error <sup>(5)</sup>                            | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$                       | -   | -     | $\pm 1$  | %    |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$                                     | -   | -     | $\pm 1$  |      |
| SNR       | Signal-to-noise ratio <sup>(6)</sup>                 | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , 1 kHz, BW = 500 KHz | -   | 67.8  | -        | dB   |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$ , 1 kHz, BW = 500 KHz               | -   | 67.8  | -        |      |
| THD       | Total harmonic distortion <sup>(6)</sup>             | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , 1 kHz               | -   | -78.6 | -        | dB   |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$ , 1 kHz                             | -   | -78.6 | -        |      |
| SINAD     | Signal-to-noise and distortion ratio <sup>(6)</sup>  | DAC output buffer ON, $C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , 1 kHz               | -   | 67.5  | -        | dB   |
|           |                                                      | DAC output buffer OFF, $C_L \leq 50$ pF, no $R_L$ , 1 kHz                             | -   | 67.5  | -        |      |

Table 188. DAC accuracy<sup>(1)</sup> (continued)

| Symbol | Parameter                | Conditions                                                                 | Min | Typ  | Max | Unit |
|--------|--------------------------|----------------------------------------------------------------------------|-----|------|-----|------|
| ENOB   | Effective number of bits | DAC output buffer ON,<br>$C_L \leq 50$ pF, $R_L \geq 5$ k $\Omega$ , 1 kHz | -   | 10.9 | -   | bits |
|        |                          | DAC output buffer OFF,<br>$C_L \leq 50$ pF, no $R_L$ , 1 kHz               | -   | 10.9 | -   |      |

1. Guaranteed by characterization.
2. Difference between two consecutive codes minus 1 LSB.
3. Difference between the value measured at Code  $i$  and the value measured at Code  $i$  on a line drawn between Code 0 and last Code 4095.
4. Difference between the value measured at Code (0x001) and the ideal value.
5. Difference between the ideal slope of the transfer function and the measured slope computed from code 0x000 and 0xFFFF when the buffer is OFF, and from code giving 0.2 V and ( $V_{REF+} - 0.2$  V) when the buffer is ON.
6. Signal is -0.5dBFS with  $F_{sampling}=1$  MHz.

Figure 96. 12-bit buffered /non-buffered DAC



1. The DAC integrates an output buffer that can be used to reduce the output impedance and to drive external loads directly without the use of an external operational amplifier. The buffer can be bypassed by configuring the BOFFx bit in the DAC\_CR register.

## 7.3.22 Voltage reference buffer characteristics

Table 189. VREFBUF characteristics<sup>(1)</sup>

| Symbol            | Parameter                                                         | Conditions                                                       |                           | Min                        | Typ        | Max                               | Unit                     |
|-------------------|-------------------------------------------------------------------|------------------------------------------------------------------|---------------------------|----------------------------|------------|-----------------------------------|--------------------------|
| $V_{DDA}$         | Analog supply voltage                                             | Normal mode                                                      | VSCALE = 000              | 2.8                        | 3.3        | 3.6                               | V                        |
|                   |                                                                   |                                                                  | VSCALE = 001              | 2.4                        | -          | 3.6                               |                          |
|                   |                                                                   |                                                                  | VSCALE = 010              | 2.1                        | -          | 3.6                               |                          |
|                   |                                                                   |                                                                  | VSCALE = 011              | 1.8                        | -          | 3.6                               |                          |
|                   |                                                                   | Degraded mode                                                    | VSCALE = 000              | 1.62                       | -          | 2.80                              |                          |
|                   |                                                                   |                                                                  | VSCALE = 001              | 1.62                       | -          | 2.40                              |                          |
|                   |                                                                   |                                                                  | VSCALE = 010              | 1.62                       | -          | 2.10                              |                          |
|                   |                                                                   |                                                                  | VSCALE = 011              | 1.62                       | -          | 1.80                              |                          |
| $V_{REFBUF\_OUT}$ | Voltage Reference Buffer Output, at 30 °C, $I_{load} = 100 \mu A$ | Normal mode                                                      | VSCALE = 000              | 2.498                      | 2.5        | 2.5035                            | V                        |
|                   |                                                                   |                                                                  | VSCALE = 001              | 2.046                      | 2.049      | 2.052                             |                          |
|                   |                                                                   |                                                                  | VSCALE = 010              | 1.801                      | 1.804      | 1.806                             |                          |
|                   |                                                                   |                                                                  | VSCALE = 011              | 1.4995                     | 1.5015     | 1.504                             |                          |
|                   |                                                                   | Degraded mode <sup>(2)</sup>                                     | VSCALE = 000              | $V_{DDA} - 150 \text{ mV}$ | -          | $V_{DDA}$                         |                          |
|                   |                                                                   |                                                                  | VSCALE = 001              | $V_{DDA} - 150 \text{ mV}$ | -          | $V_{DDA}$                         |                          |
|                   |                                                                   |                                                                  | VSCALE = 010              | $V_{DDA} - 150 \text{ mV}$ | -          | $V_{DDA}$                         |                          |
|                   |                                                                   |                                                                  | VSCALE = 011              | $V_{DDA} - 150 \text{ mV}$ | -          | $V_{DDA}$                         |                          |
| TRIM              | Trim step resolution                                              | -                                                                | -                         | -                          | $\pm 0.05$ | $\pm 0.1$                         | %                        |
| $C_L$             | Load capacitor                                                    | -                                                                | -                         | 0.5                        | 1          | 1.50                              | $\mu F$                  |
| esr               | Equivalent Serial Resistor of $C_L$                               | -                                                                | -                         | -                          | -          | 2                                 | $\Omega$                 |
| $I_{load}$        | Static load current                                               | -                                                                | -                         | -                          | -          | 4                                 | mA                       |
| $I_{line\_reg}$   | Line regulation                                                   | $2.8 \text{ V} \leq V_{DDA} \leq 3.6 \text{ V}$                  | $I_{load} = 500 \mu A$    | -                          | 200        | -                                 | ppm/V                    |
|                   |                                                                   |                                                                  | $I_{load} = 4 \text{ mA}$ | -                          | 100        | -                                 |                          |
| $I_{load\_reg}$   | Load regulation                                                   | $500 \mu A \leq I_{LOAD} \leq 4 \text{ mA}$                      | Normal Mode               | -                          | 50         | -                                 | ppm/<br>mA               |
| $T_{coeff}$       | Temperature coefficient                                           | $-40 \text{ }^\circ\text{C} < T_J < +125 \text{ }^\circ\text{C}$ |                           | -                          | -          | $T_{coeff}$<br>$V_{REFINT} + 100$ | ppm/<br>$^\circ\text{C}$ |
| PSRR              | Power supply rejection                                            | DC                                                               | -                         | -                          | 60         | -                                 | dB                       |
|                   |                                                                   | 100KHz                                                           | -                         | -                          | 40         | -                                 |                          |

Table 189. VREFBUF characteristics<sup>(1)</sup> (continued)

| Symbol                    | Parameter                                                                                          | Conditions                 |   | Min | Typ | Max | Unit |
|---------------------------|----------------------------------------------------------------------------------------------------|----------------------------|---|-----|-----|-----|------|
| t <sub>START</sub>        | Start-up time                                                                                      | C <sub>L</sub> =0.5 μF     | - | -   | 300 | -   | μs   |
|                           |                                                                                                    | C <sub>L</sub> =1 μF       | - | -   | 500 | -   |      |
|                           |                                                                                                    | C <sub>L</sub> =1.5 μF     | - | -   | 650 | -   |      |
| I <sub>INRUSH</sub>       | Control of maximum DC current drive on V <sub>REFBUF_OUT</sub> during startup phase <sup>(3)</sup> | -                          |   | -   | 8   | -   | mA   |
| I <sub>DDA(VREFBUF)</sub> | VREFBUF consumption from V <sub>DDA</sub>                                                          | I <sub>LOAD</sub> = 0 μA   | - | -   | 15  | 25  | μA   |
|                           |                                                                                                    | I <sub>LOAD</sub> = 500 μA | - | -   | 16  | 30  |      |
|                           |                                                                                                    | I <sub>LOAD</sub> = 4 mA   | - | -   | 32  | 50  |      |

1. Guaranteed by design.
2. In degraded mode, the voltage reference buffer cannot accurately maintain the output voltage (V<sub>DDA</sub>-drop voltage).
3. To properly control VREFBUF I<sub>INRUSH</sub> current during the startup phase and the change of scaling, V<sub>DDA</sub> voltage should be in the range of 1.8 V-3.6 V, 2.1 V-3.6 V, 2.4 V-3.6 V and 2.8 V-3.6 V for VSCALE = 011, 010, 001 and 000, respectively.

### 7.3.23 Temperature sensor characteristics

Table 190. Temperature sensor characteristics

| Symbol                              | Parameter                                      | Min | Typ  | Max  | Unit  |
|-------------------------------------|------------------------------------------------|-----|------|------|-------|
| T <sub>L</sub> <sup>(1)</sup>       | V <sub>SENSE</sub> linearity with temperature  | -   | -    | 3    | °C    |
| Avg_Slope <sup>(2)</sup>            | Average slope                                  | -   | 2    | -    | mV/°C |
| V <sub>30</sub> <sup>(3)</sup>      | Voltage at 30°C ± 5 °C                         | -   | 0.62 | -    | V     |
| t <sub>start_run</sub>              | Startup time in Run mode (buffer startup)      | -   | -    | 25.2 | μs    |
| t <sub>S_temp</sub> <sup>(1)</sup>  | ADC sampling time when reading the temperature | 9   | -    | -    |       |
| I <sub>sens</sub> <sup>(1)</sup>    | Sensor consumption                             | -   | 0.18 | 0.31 | μA    |
| I <sub>sensbuf</sub> <sup>(1)</sup> | Sensor buffer consumption                      | -   | 3.8  | 6.5  |       |

1. Guaranteed by design.
2. Guaranteed by characterization.
3. Measured at V<sub>DDA</sub> = 3.3 V ± 10 mV. The V<sub>30</sub> ADC conversion result is stored in the TS\_CAL1 byte.

Table 191. Temperature sensor calibration values

| Symbol  | Parameter                                                                     | Memory address            |
|---------|-------------------------------------------------------------------------------|---------------------------|
| TS_CAL1 | Temperature sensor raw data acquired value at 30 °C, V <sub>DDA</sub> =3.3 V  | 0x1FF1 E820 - 0x1FF1 E821 |
| TS_CAL2 | Temperature sensor raw data acquired value at 110 °C, V <sub>DDA</sub> =3.3 V | 0x1FF1 E840 - 0x1FF1 E841 |

### 7.3.24 Temperature and V<sub>BAT</sub> monitoring

**Table 192. V<sub>BAT</sub> monitoring characteristics**

| Symbol                             | Parameter                                             | Min | Typ  | Max | Unit |
|------------------------------------|-------------------------------------------------------|-----|------|-----|------|
| R                                  | Resistor bridge for V <sub>BAT</sub>                  | -   | 26   | -   | KΩ   |
| Q                                  | Ratio on V <sub>BAT</sub> measurement                 | -   | 4    | -   | -    |
| Er <sup>(1)</sup>                  | Error on Q                                            | -10 | -    | +10 | %    |
| t <sub>S_vbat</sub> <sup>(1)</sup> | ADC sampling time when reading V <sub>BAT</sub> input | 9   | -    | -   | μs   |
| V <sub>BAThigh</sub>               | High supply monitoring                                | -   | 3.55 | -   | V    |
| V <sub>BATlow</sub>                | Low supply monitoring                                 | -   | 1.36 | -   |      |

1. Guaranteed by design.

**Table 193. V<sub>BAT</sub> charging characteristics**

| Symbol          | Parameter                 | Condition          | Min | Typ | Max | Unit |
|-----------------|---------------------------|--------------------|-----|-----|-----|------|
| R <sub>BC</sub> | Battery charging resistor | VBRS in PWR_CR3= 0 | -   | 5   | -   | KΩ   |
|                 |                           | VBRS in PWR_CR3= 1 |     | 1.5 | -   |      |

**Table 194. Temperature monitoring characteristics**

| Symbol               | Parameter                   | Min | Typ | Max | Unit |
|----------------------|-----------------------------|-----|-----|-----|------|
| TEMP <sub>high</sub> | High temperature monitoring | -   | 117 | -   | °C   |
| TEMP <sub>low</sub>  | Low temperature monitoring  | -   | -25 | -   |      |

### 7.3.25 Voltage booster for analog switch

**Table 195. Voltage booster for analog switch characteristics<sup>(1)</sup>**

| Symbol                 | Parameter            | Condition                        | Min  | Typ | Max | Unit |
|------------------------|----------------------|----------------------------------|------|-----|-----|------|
| V <sub>DD</sub>        | Supply voltage       | -                                | 1.62 | 2.6 | 3.6 | V    |
| t <sub>SU(BOOST)</sub> | Booster startup time | -                                | -    | -   | 50  | μs   |
| I <sub>DD(BOOST)</sub> | Booster consumption  | 1.62 V ≤ V <sub>DD</sub> ≤ 2.7 V | -    | -   | 125 | μA   |
|                        |                      | 2.7 V < V <sub>DD</sub> < 3.6 V  | -    | -   | 250 |      |

1. Guaranteed by characterization results.

### 7.3.26 Comparator characteristics

Table 196. COMP characteristics<sup>(1)</sup>

| Symbol                        | Parameter                                                                         | Conditions                |                                             | Min  | Typ | Max              | Unit |
|-------------------------------|-----------------------------------------------------------------------------------|---------------------------|---------------------------------------------|------|-----|------------------|------|
| V <sub>DDA</sub>              | Analog supply voltage                                                             | -                         |                                             | 1.62 | 3.3 | 3.6              | V    |
| V <sub>IN</sub>               | Comparator input voltage range                                                    | -                         |                                             | 0    | -   | V <sub>DDA</sub> |      |
| V <sub>BG</sub>               | Scaler input voltage                                                              | -                         |                                             | (2)  |     |                  |      |
| V <sub>SC</sub>               | Scaler offset voltage                                                             | -                         |                                             | -    | ±5  | ±10              | mV   |
| I <sub>DDA(SCALER)</sub>      | Scaler static consumption from V <sub>DDA</sub>                                   | BRG_EN=0 (bridge disable) |                                             | -    | 0.2 | 0.3              | µA   |
|                               |                                                                                   | BRG_EN=1 (bridge enable)  |                                             | -    | 0.8 | 1                |      |
| t <sub>START_SCALER</sub>     | Scaler startup time                                                               | -                         |                                             | -    | 140 | 250              | µs   |
| t <sub>START</sub>            | Comparator startup time to reach propagation delay specification                  | High-speed mode           |                                             | -    | 2   | 5                | µs   |
|                               |                                                                                   | Medium mode               |                                             | -    | 5   | 20               |      |
|                               |                                                                                   | Ultra-low-power mode      |                                             | -    | 15  | 80               |      |
| t <sub>D</sub> <sup>(3)</sup> | Propagation delay for 200 mV step with 100 mV overdrive                           | High-speed mode           |                                             | -    | 50  | 80               | ns   |
|                               |                                                                                   | Medium mode               |                                             | -    | 0.5 | 1.2              | µs   |
|                               |                                                                                   | Ultra-low-power mode      |                                             | -    | 2.5 | 7                |      |
|                               | Propagation delay for step > 200 mV with 100 mV overdrive only on positive inputs | High-speed mode           |                                             | -    | 50  | 120              | ns   |
|                               |                                                                                   | Medium mode               |                                             | -    | 0.5 | 1.2              | µs   |
|                               |                                                                                   | Ultra-low-power mode      |                                             | -    | 2.5 | 7                |      |
| V <sub>offset</sub>           | Comparator offset error                                                           | Full common mode range    |                                             | -    | ±5  | ±20              | mV   |
| V <sub>hys</sub>              | Comparator hysteresis                                                             | No hysteresis             |                                             | -    | 0   | -                | mV   |
|                               |                                                                                   | Low hysteresis            |                                             | 5    | 10  | 22               |      |
|                               |                                                                                   | Medium hysteresis         |                                             | 8    | 20  | 37               |      |
|                               |                                                                                   | High hysteresis           |                                             | 16   | 30  | 52               |      |
| I <sub>DDA(COMP)</sub>        | Comparator consumption from V <sub>DDA</sub>                                      | Ultra-low-power mode      | Static                                      | -    | 400 | 600              | nA   |
|                               |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -    | 800 | -                |      |
|                               |                                                                                   | Medium mode               | Static                                      | -    | 5   | 7                | µA   |
|                               |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -    | 6   | -                |      |
|                               |                                                                                   | High-speed mode           | Static                                      | -    | 70  | 100              |      |
|                               |                                                                                   |                           | With 50 kHz ±100 mV overdrive square signal | -    | 75  | -                |      |

1. Guaranteed by design, unless otherwise specified.

2. Refer to [Table 127: Embedded reference voltage](#).

3. Guaranteed by characterization results.

### 7.3.27 Operational amplifier characteristics

**Table 197. Operational amplifier characteristics**

| Symbol                       | Parameter                                                       | Conditions                                                                                               | Min                  | Typ  | Max       | Unit  |
|------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------|------|-----------|-------|
| $V_{DDA}$                    | Analog supply voltage Range                                     | -                                                                                                        | 2                    | 3.3  | 3.6       | V     |
| CMIR                         | Common Mode Input Range                                         | -                                                                                                        | 0                    | -    | $V_{DDA}$ |       |
| $V_{I\text{OFFSET}}$         | Input offset voltage                                            | 25°C, no load on output                                                                                  | -                    | -    | ±1.5      | mV    |
|                              |                                                                 | All voltages and temperature, no load                                                                    | -                    | -    | ±2.5      |       |
| $\Delta V_{I\text{OFFSET}}$  | Input offset voltage drift                                      | -                                                                                                        | -                    | ±3.0 | -         | µV/°C |
| TRIMOFFSETP<br>TRIMLPOFFSETP | Offset trim step at low common input voltage (0.1* $V_{DDA}$ )  | -                                                                                                        | -                    | 1.1  | 1.5       | mV    |
| TRIMOFFSETN<br>TRIMLPOFFSETN | Offset trim step at high common input voltage (0.9* $V_{DDA}$ ) | -                                                                                                        | -                    | 1.1  | 1.5       |       |
| $I_{LOAD}$                   | Drive current                                                   | -                                                                                                        | -                    | -    | 500       | µA    |
| $I_{LOAD\_PGA}$              | Drive current in PGA mode                                       | -                                                                                                        | -                    | -    | 270       |       |
| $C_{LOAD}$                   | Capacitive load                                                 | -                                                                                                        | -                    | -    | 50        | pF    |
| CMRR                         | Common mode rejection ratio                                     | -                                                                                                        | -                    | 80   | -         | dB    |
| PSRR                         | Power supply rejection ratio                                    | $C_{LOAD} \leq 50\text{pf}$ /<br>$R_{LOAD} \geq 4\text{ k}\Omega^{(1)}$ at 1 kHz,<br>$V_{com}=V_{DDA}/2$ | 50                   | 66   | -         | dB    |
| GBW                          | Gain bandwidth for high supply range                            | 200 mV ≤ Output dynamic range ≤ $V_{DDA}$ - 200 mV                                                       | 4                    | 7.3  | 12.3      | MHz   |
| SR                           | Slew rate (from 10% and 90% of output voltage)                  | Normal mode                                                                                              | -                    | 3    | -         | V/µs  |
|                              |                                                                 | High-speed mode                                                                                          | -                    | 30   | -         |       |
| AO                           | Open loop gain                                                  | 200 mV ≤ Output dynamic range ≤ $V_{DDA}$ - 200 mV                                                       | 59                   | 90   | 129       | dB    |
| φm                           | Phase margin                                                    | -                                                                                                        | -                    | 55   | -         | °     |
| GM                           | Gain margin                                                     | -                                                                                                        | -                    | 12   | -         | dB    |
| $V_{OHSAT}$                  | High saturation voltage                                         | $I_{load}=\text{max}$ or $R_{LOAD}=\text{min}$ ,<br>Input at $V_{DDA}$                                   | $V_{DDA}$<br>-100 mV | -    | -         | mV    |
| $V_{OLSAT}$                  | Low saturation voltage                                          | $I_{load}=\text{max}$ or $R_{LOAD}=\text{min}$ ,<br>Input at 0 V                                         | -                    | -    | 100       |       |



Table 197. Operational amplifier characteristics (continued)

| Symbol               | Parameter                                                                 | Conditions      |                                                                                                             | Min  | Typ    | Max | Unit                       |
|----------------------|---------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------|------|--------|-----|----------------------------|
| $t_{\text{WAKEUP}}$  | Wake up time from OFF state                                               | Normal mode     | $C_{\text{LOAD}} \leq 50\text{pf}$ ,<br>$R_{\text{LOAD}} \geq 4\text{ k}\Omega$ ,<br>follower configuration | -    | 0.8    | 3.2 | $\mu\text{s}$              |
|                      |                                                                           | High speed mode | $C_{\text{LOAD}} \leq 50\text{pf}$ ,<br>$R_{\text{LOAD}} \geq 4\text{ k}\Omega$ ,<br>follower configuration | -    | 0.9    | 2.8 |                            |
| PGA gain             | Non inverting gain error value                                            | PGA gain = 2    |                                                                                                             | -1   | -      | 1   | %                          |
|                      |                                                                           | PGA gain = 4    |                                                                                                             | -2   | -      | 2   |                            |
|                      |                                                                           | PGA gain = 8    |                                                                                                             | -2.5 | -      | 2.5 |                            |
|                      |                                                                           | PGA gain = 16   |                                                                                                             | -3   | -      | 3   |                            |
|                      | Inverting gain error value                                                | PGA gain = 2    |                                                                                                             | -1   | -      | 1   |                            |
|                      |                                                                           | PGA gain = 4    |                                                                                                             | -1   | -      | 1   |                            |
|                      |                                                                           | PGA gain = 8    |                                                                                                             | -2   | -      | 2   |                            |
|                      |                                                                           | PGA gain = 16   |                                                                                                             | -3   | -      | 3   |                            |
|                      | External non-inverting gain error value                                   | PGA gain = 2    |                                                                                                             | -1   | -      | 1   |                            |
|                      |                                                                           | PGA gain = 4    |                                                                                                             | -3   | -      | 3   |                            |
|                      |                                                                           | PGA gain = 8    |                                                                                                             | -3.5 | -      | 3.5 |                            |
|                      |                                                                           | PGA gain = 16   |                                                                                                             | -4   | -      | 4   |                            |
| $R_{\text{network}}$ | R2/R1 internal resistance values in non-inverting PGA mode <sup>(2)</sup> | PGA Gain=2      |                                                                                                             | -    | 10/10  | -   | k $\Omega$ /<br>k $\Omega$ |
|                      |                                                                           | PGA Gain=4      |                                                                                                             | -    | 30/10  | -   |                            |
|                      |                                                                           | PGA Gain=8      |                                                                                                             | -    | 70/10  | -   |                            |
|                      |                                                                           | PGA Gain=16     |                                                                                                             | -    | 150/10 | -   |                            |
|                      | R2/R1 internal resistance values in inverting PGA mode <sup>(2)</sup>     | PGA Gain = -1   |                                                                                                             | -    | 10/10  | -   |                            |
|                      |                                                                           | PGA Gain = -3   |                                                                                                             | -    | 30/10  | -   |                            |
|                      |                                                                           | PGA Gain = -7   |                                                                                                             | -    | 70/10  | -   |                            |
|                      |                                                                           | PGA Gain = -15  |                                                                                                             | -    | 150/10 | -   |                            |
| Delta R              | Resistance variation (R1 or R2)                                           | -               |                                                                                                             | -15  | -      | 15  | %                          |

Table 197. Operational amplifier characteristics (continued)

| Symbol                  | Parameter                                      | Conditions      | Min | Typ    | Max  | Unit                   |
|-------------------------|------------------------------------------------|-----------------|-----|--------|------|------------------------|
| PGA BW                  | PGA bandwidth for different non inverting gain | Gain=2          | -   | GBW/2  | -    | MHz                    |
|                         |                                                | Gain=4          | -   | GBW/4  | -    |                        |
|                         |                                                | Gain=8          | -   | GBW/8  | -    |                        |
|                         |                                                | Gain=16         | -   | GBW/16 | -    |                        |
|                         | PGA bandwidth for different inverting gain     | Gain = -1       | -   | 5.00   | -    | MHz                    |
|                         |                                                | Gain = -3       | -   | 3.00   | -    |                        |
|                         |                                                | Gain = -7       | -   | 1.50   | -    |                        |
|                         |                                                | Gain = -15      | -   | 0.80   | -    |                        |
| en                      | Voltage noise density                          | at 1 KHz        | -   | 140    | -    | nV/ $\sqrt{\text{Hz}}$ |
|                         |                                                | at 10 KHz       |     | 55     | -    |                        |
| I <sub>DDA(OPAMP)</sub> | OPAMP consumption from V <sub>DDA</sub>        | Normal mode     | -   | 570    | 1000 | $\mu\text{A}$          |
|                         |                                                | High-speed mode | -   | 610    | 1200 |                        |

1. R<sub>LOAD</sub> is the resistive load connected to VSSA or to VDDA.

2. R2 is the internal resistance between the OPAMP output and the OPAMP inverting input. R1 is the internal resistance between the OPAMP inverting input and ground. PGA gain = 1 + R2/R1.

### 7.3.28 Digital filter for Sigma-Delta Modulators (DFSDM) characteristics

Unless otherwise specified, the parameters given in [Table 198](#) for DFSDM are derived from tests performed under the ambient temperature, fPCLKx frequency and supply voltage conditions summarized in [Table 122: General operating conditions](#).

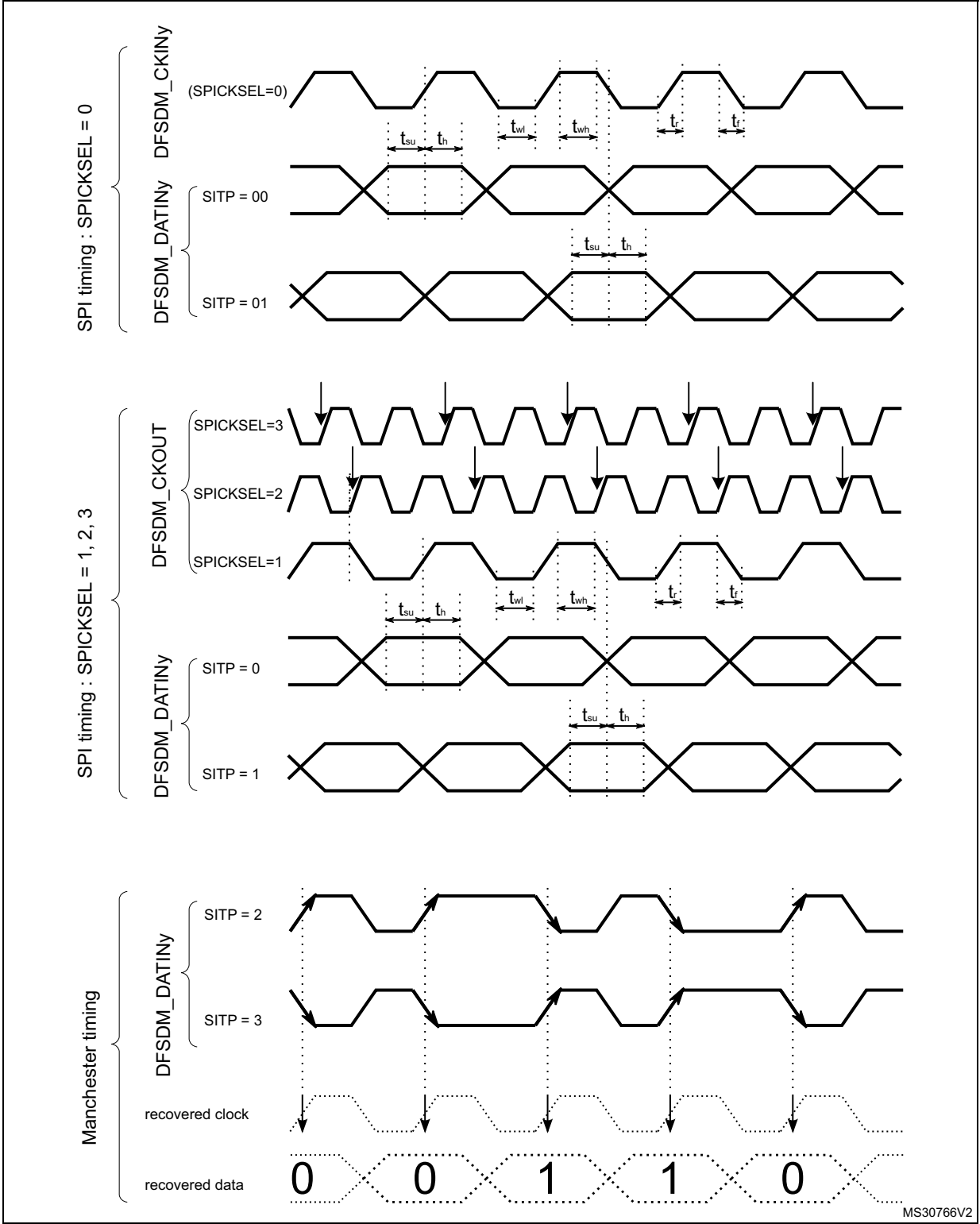
- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load C<sub>L</sub> = 30 pF
- Measurement points are done at CMOS levels: 0.5V<sub>DD</sub>
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (DIFSDM\_CKINx, DFSDM\_DATINx, DFSDM\_CKOUT for DFSDM).

Table 198. DFSDM measured timing 1.62-3.6 V

| Symbol                                                       | Parameter                                       | Conditions                                                                                                              | Min                                             | Typ                 | Max                                                   | Unit |
|--------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------|-------------------------------------------------------|------|
| $f_{\text{DFSDMCLK}}$                                        | DFSDM clock                                     | $1.62 < V_{\text{DD}} < 3.6 \text{ V}$                                                                                  | -                                               | -                   | 133                                                   | MHz  |
| $f_{\text{CKIN}}$<br>( $1/T_{\text{CKIN}}$ )                 | Input clock frequency                           | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$        | -                                               | -                   | 20                                                    |      |
|                                                              |                                                 | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$2.7 < V_{\text{DD}} < 3.6 \text{ V}$         | -                                               | -                   | 20                                                    |      |
|                                                              |                                                 | SPI mode (SITP[1:0]=0,1),<br>Internal clock mode<br>(SPICKSEL[1:0]'0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$        | -                                               | -                   | 20                                                    |      |
|                                                              |                                                 | SPI mode (SITP[1:0]=0,1),<br>Internal clock mode<br>(SPICKSEL[1:0]'0),<br>$2.7 < V_{\text{DD}} < 3.6 \text{ V}$         | -                                               | -                   | 20                                                    |      |
| $f_{\text{CKOUT}}$                                           | Output clock frequency                          | $1.62 < V_{\text{DD}} < 3.6 \text{ V}$                                                                                  | -                                               | -                   | 20                                                    |      |
| $\text{DuCy}_{\text{CKOUT}}$<br>T                            | Output clock frequency duty cycle               | $1.62 < V_{\text{DD}} < 3.6 \text{ V}$                                                                                  | 45                                              | 50                  | 55                                                    | %    |
| $t_{\text{wh}}(\text{CKIN})$<br>$t_{\text{wl}}(\text{CKIN})$ | Input clock high and low time                   | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$        | $T_{\text{CKIN}}/2-0.5$                         | $T_{\text{CKIN}}/2$ | -                                                     | ns   |
| $t_{\text{su}}$                                              | Data input setup time                           | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$        | 1.5                                             | -                   | -                                                     |      |
| $t_{\text{h}}$                                               | Data input hold time                            | SPI mode (SITP[1:0]=0,1),<br>External clock mode<br>(SPICKSEL[1:0]=0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$        | 0.5                                             | -                   | -                                                     |      |
| $T_{\text{Manchester}}$                                      | Manchester data period (recovered clock period) | Manchester mode (SITP[1:0]=2,3),<br>Internal clock mode<br>(SPICKSEL[1:0]'0),<br>$1.62 < V_{\text{DD}} < 3.6 \text{ V}$ | $(\text{CKOUTDIV}+1) \cdot T_{\text{DFSDMCLK}}$ | -                   | $(2 \cdot \text{CKOUTDIV}) \cdot T_{\text{DFSDMCLK}}$ |      |

Figure 97. Channel transceiver timing diagrams



### 7.3.29 Camera interface (DCMI) timing specifications

Unless otherwise specified, the parameters given in [Table 199](#) for DCMI are derived from tests performed under the ambient temperature,  $f_{HCLK}$  frequency and VDD supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

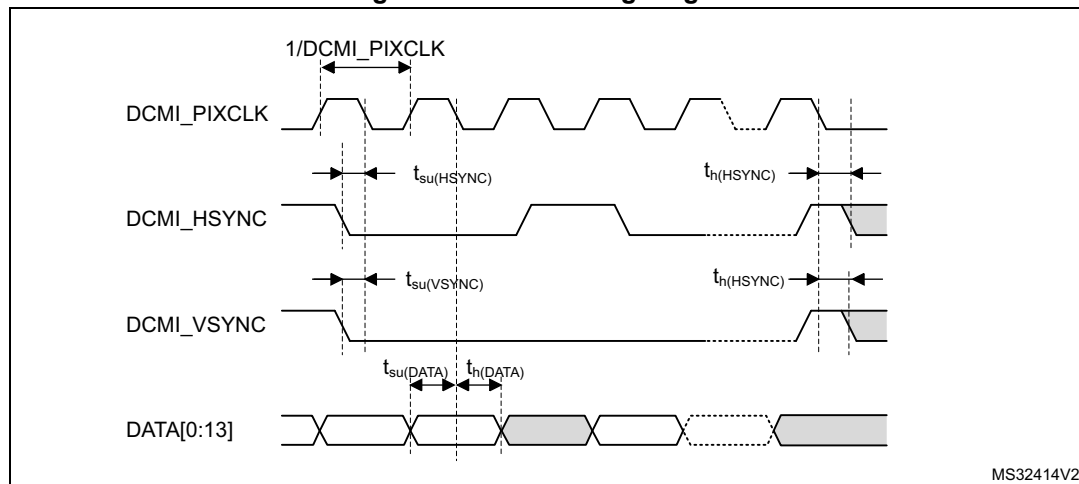
- DCMI\_PIXCLK polarity: falling
- DCMI\_VSYNC and DCMI\_HSYNC polarity: high
- Data formats: 14 bits
- Capacitive load  $C_L=30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- VOS level set to VOS1

**Table 199. DCMI characteristics<sup>(1)</sup>**

| Symbol                               | Parameter                               | Min | Max | Unit |
|--------------------------------------|-----------------------------------------|-----|-----|------|
| -                                    | Frequency ratio DCMI_PIXCLK/ $f_{HCLK}$ | -   | 0.4 | -    |
| DCMI_PIXCLK                          | Pixel Clock input                       | -   | 80  | MHz  |
| $D_{pixel}$                          | Pixel Clock input duty cycle            | 30  | 70  | %    |
| $t_{su}(DATA)$                       | Data input setup time                   | 3   | -   | -    |
| $t_h(DATA)$                          | Data hold time                          | 1   | -   |      |
| $t_{su}(HSYNC)$ ,<br>$t_{su}(VSYNC)$ | DCMI_HSYNC/ DCMI_VSYNC input setup time | 2   | -   | ns   |
| $t_h(HSYNC)$ ,<br>$t_h(VSYNC)$       | DCMI_HSYNC/ DCMI_VSYNC input hold time  | 1   | -   | -    |

1. Guaranteed by characterization results.

**Figure 98. DCMI timing diagram**



### 7.3.30 LCD-TFT controller (LTDC) characteristics

Unless otherwise specified, the parameters given in [Table 200](#) for LCD-TFT are derived from tests performed under the ambient temperature,  $f_{\text{HCLK}}$  frequency and VDD supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

- LCD\_CLK polarity: high
- LCD\_DE polarity: low
- LCD\_VSYNC and LCD\_HSYNC polarity: high
- Pixel formats: 24 bits
- Output speed is set to OSPEEDRy[1:0] = 11
- Capacitive load  $C_L = 30$  pF
- Measurement points are done at CMOS levels: 0.5VDD
- IO Compensation cell activated.
- HSLV activated when  $V_{DD} \leq 2.7$  V
- VOS level set to VOS1

**Table 200. LTDC characteristics<sup>(1)</sup>**

| Symbol                                                                   | Parameter                        |                                | Min                         | Max                         | Unit |
|--------------------------------------------------------------------------|----------------------------------|--------------------------------|-----------------------------|-----------------------------|------|
| $f_{\text{CLK}}$                                                         | LTDC clock output frequency      | $2.7 < V_{DD} < 3.6$ V<br>20pF | -                           | 150                         | MHz  |
|                                                                          |                                  | $2.7 < V_{DD} < 3.6$ V         |                             | 133                         |      |
|                                                                          |                                  | $1.62 < V_{DD} < 3.6$ V        |                             | 90                          |      |
| $D_{\text{CLK}}$                                                         | LTDC clock output duty cycle     |                                | 45                          | 55                          | %    |
| $t_{w(\text{CLKH})}$ ,<br>$t_{w(\text{CLKL})}$                           | Clock High time, low time        |                                | $t_{w(\text{CLK})}/2 - 0.5$ | $t_{w(\text{CLK})}/2 + 0.5$ | -    |
| $t_{v(\text{DATA})}$                                                     | Data output valid time           | $2.7 < V_{DD} < 3.6$ V         | -                           | 0.5                         |      |
| $t_{h(\text{DATA})}$                                                     |                                  | $1.62 < V_{DD} < 3.6$ V        |                             | 5                           |      |
| $t_{v(\text{DATA})}$                                                     | Data output hold time            |                                | 0                           | -                           |      |
| $t_{v(\text{HSYNC})}$ ,<br>$t_{v(\text{VSYNC})}$ ,<br>$t_{v(\text{DE})}$ | HSYNC/VSYNC/DE output valid time | $2.7 < V_{DD} < 3.6$ V         | -                           | 0.5                         |      |
|                                                                          |                                  | $1.62 < V_{DD} < 3.6$ V        | -                           | 5                           |      |
| $t_{h(\text{HSYNC})}$ ,<br>$t_{h(\text{VSYNC})}$ ,<br>$t_{h(\text{DE})}$ | HSYNC/VSYNC/DE output hold time  |                                | 0                           | -                           |      |

1. Guaranteed by characterization results.

Figure 99. LCD-TFT horizontal timing diagram

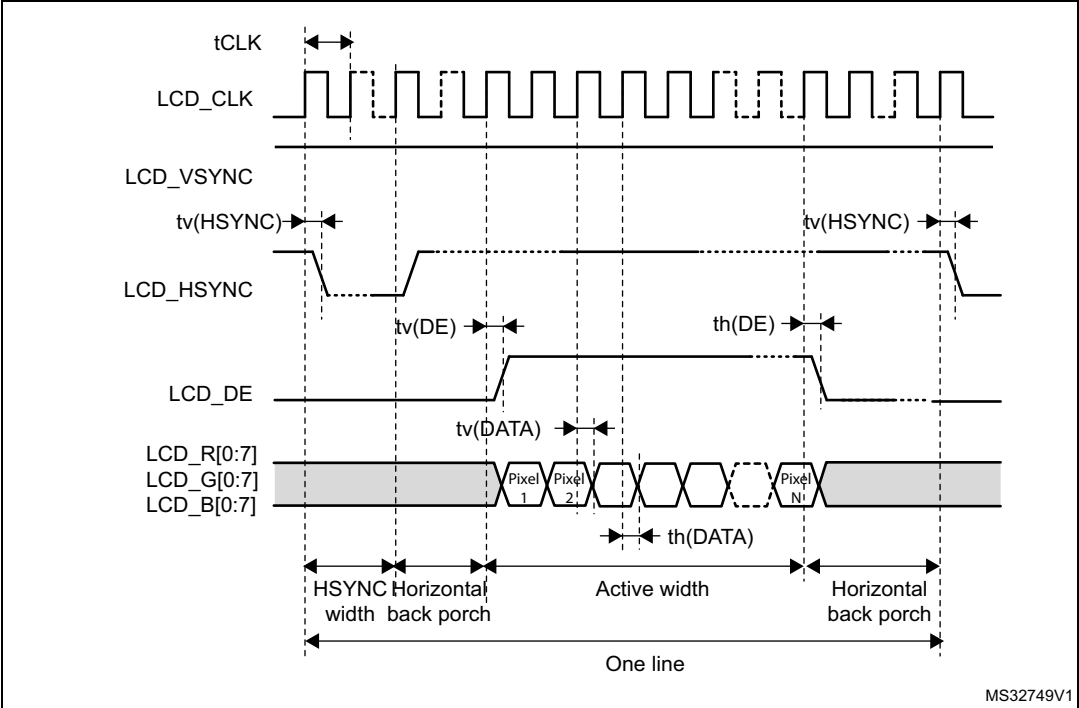
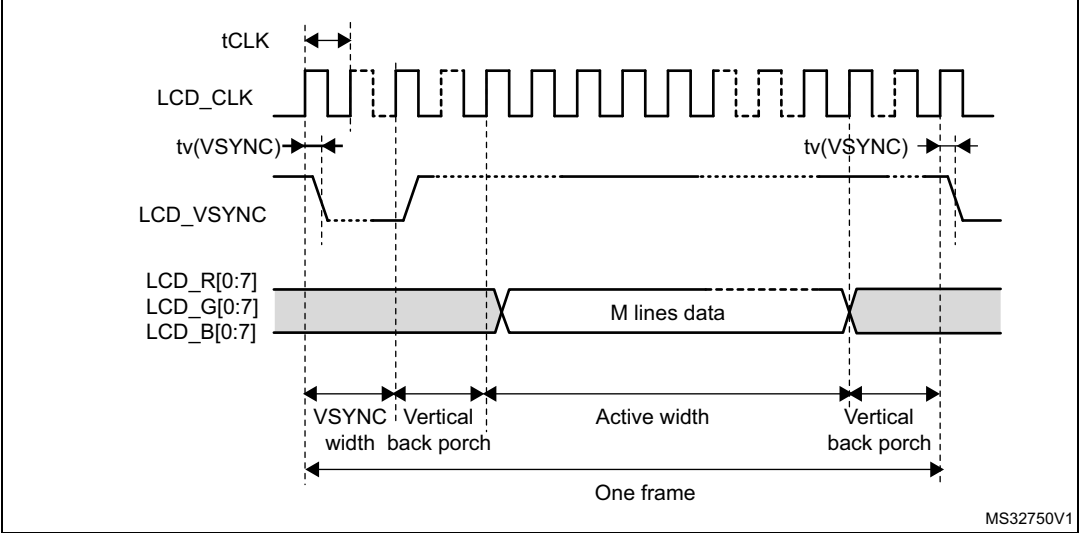


Figure 100. LCD-TFT vertical timing diagram



### 7.3.31 Timer characteristics

The parameters given in [Table 201](#) are guaranteed by design.

Refer to [Section 7.3.15: I/O port characteristics](#) for details on the input/output alternate function characteristics (output compare, input capture, external clock, PWM output).

**Table 201. TIMx characteristics<sup>(1)(2)</sup>**

| Symbol           | Parameter                                    | Conditions <sup>(3)</sup>                                      | Min | Max                  | Unit          |
|------------------|----------------------------------------------|----------------------------------------------------------------|-----|----------------------|---------------|
| $t_{res(TIM)}$   | Timer resolution time                        | AHB/APBx prescaler=1 or 2 or 4, $f_{TIMxCLK} = 240\text{ MHz}$ | 1   | -                    | $t_{TIMxCLK}$ |
|                  |                                              | AHB/APBx prescaler>4, $f_{TIMxCLK} = 120\text{ MHz}$           | 1   | -                    | $t_{TIMxCLK}$ |
| $f_{EXT}$        | Timer external clock frequency on CH1 to CH4 | $f_{TIMxCLK} = 240\text{ MHz}$                                 | 0   | $f_{TIMxCLK}/2$      | MHz           |
| $Res_{TIM}$      | Timer resolution                             |                                                                | -   | 16/32                | bit           |
| $t_{MAX\_COUNT}$ | Maximum possible count with 32-bit counter   | -                                                              | -   | $65536 \times 65536$ | $t_{TIMxCLK}$ |

1. TIMx is used as a general term to refer to the TIM1 to TIM17 timers.
2. Guaranteed by design.
3. The maximum timer frequency on APB1 or APB2 is up to 240 MHz, by setting the TIMPRE bit in the RCC\_CFGR register, if APBx prescaler is 1 or 2 or 4, then  $TIMxCLK = rcc\_hclk1$ , otherwise  $TIMxCLK = 4 \times F_{rcc\_pclkx\_d2}$ .

### 7.3.32 Communication interfaces

#### I<sup>2</sup>C interface characteristics

The I<sup>2</sup>C interface meets the timings requirements of the I2C-bus specification and user manual revision 03 for:

- Standard-mode (Sm): with a bit rate up to 100 kbit/s
- Fast-mode (Fm): with a bit rate up to 400 kbit/s
- Fast-mode Plus (Fm+): with a bit rate up to 1 Mbit/s.

The I<sup>2</sup>C timings requirements are guaranteed by design when the I<sup>2</sup>C peripheral is properly configured (refer to RM0399 reference manual) and when the `i2c_ker_ck` frequency is greater than the minimum shown in the table below:



Table 202. Minimum i2c\_ker\_ck frequency in all I<sup>2</sup>C modes

| Symbol    | Parameter        | Condition      |                            | Min | Unit |
|-----------|------------------|----------------|----------------------------|-----|------|
| f(I2CCLK) | I2CCLK frequency | Standard-mode  | -                          | 2   | MHz  |
|           |                  | Fast-mode      | Analog Filtre ON<br>DNF=0  | 8   |      |
|           |                  |                | Analog Filtre OFF<br>DNF=1 | 9   |      |
|           |                  | Fast-mode Plus | Analog Filtre ON<br>DNF=0  | 17  |      |
|           |                  |                | Analog Filtre OFF<br>DNF=1 | 16  | -    |

The SDA and SCL I/O requirements are met with the following restrictions:

- The SDA and SCL I/O pins are not “true” open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V<sub>DDIOx</sub> is disabled, but still present.
- The 20 mA output drive requirement in Fast-mode Plus is not supported. This limits the maximum load C<sub>Load</sub> supported in Fm+, which is given by these formulas:

$$t_{r(SDA/SCL)} = 0.8473 \times R_P \times C_{Load}$$

$$R_{P(min)} = (V_{DD} - V_{OL(max)}) / I_{OL(max)}$$

Where R<sub>P</sub> is the I2C lines pull-up. Refer to [Section 7.3.15: I/O port characteristics](#) for the I<sup>2</sup>C I/Os characteristics.

All I<sup>2</sup>C SDA and SCL I/Os embed an analog filter. Refer to the table below for the analog filter characteristics:

Table 203. I<sup>2</sup>C analog filter characteristics<sup>(1)</sup>

| Symbol          | Parameter                                                          | Min               | Max               | Unit |
|-----------------|--------------------------------------------------------------------|-------------------|-------------------|------|
| t <sub>AF</sub> | Maximum pulse width of spikes that are suppressed by analog filter | 50 <sup>(2)</sup> | 80 <sup>(3)</sup> | ns   |

- Guaranteed by characterization results.
- Spikes with widths below t<sub>AF(min)</sub> are filtered.
- Spikes with widths above t<sub>AF(max)</sub> are not filtered.

### USART interface characteristics

Unless otherwise specified, the parameters given in [Table 204](#) for USART are derived from tests performed under the ambient temperature, f<sub>PCLKx</sub> frequency and V<sub>DD</sub> supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load C<sub>L</sub> = 30 pF
- Measurement points are done at CMOS levels: 0.5V<sub>DD</sub>
- IO Compensation cell activated.
- VOS level set to VOS1

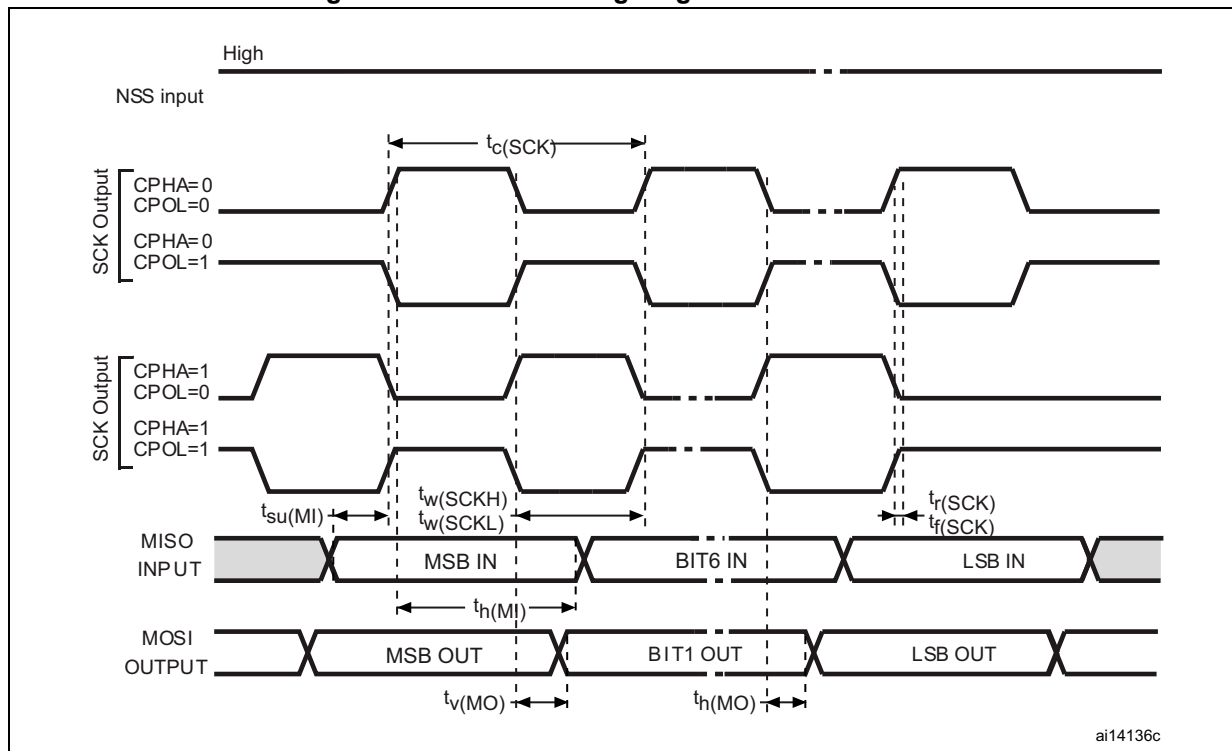
Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (NSS, CK, TX, RX for USART).

**Table 204. USART characteristics<sup>(1)</sup>**

| Symbol                 | Parameter              | Conditions  | Min            | Typ          | Max            | Unit |
|------------------------|------------------------|-------------|----------------|--------------|----------------|------|
| $f_{CK}$               | USART clock frequency  | Master mode | -              | -            | 12.5           | MHz  |
|                        |                        | Slave mode  |                |              | 25             |      |
| $t_{su(NSS)}$          | NSS setup time         | Slave mode  | $t_{ker}+1$    | -            | -              | -    |
| $t_h(NSS)$             | NSS hold time          | Slave mode  | 2              | -            | -              |      |
| $t_w(SCKH), t_w(SCKL)$ | CK high and low time   | Master mode | $1/f_{CK}/2-2$ | $1/f_{CK}/2$ | $1/f_{CK}/2+2$ |      |
| $t_{su(RX)}$           | Data input setup time  | Master mode | $t_{ker}+6$    | -            | -              | ns   |
|                        |                        | Slave mode  | 1.5            | -            | -              |      |
| $t_h(RX)$              | Data input hold time   | Master mode | 0              | -            | -              |      |
|                        |                        | Slave mode  | 1.5            | -            | -              |      |
| $t_v(TX)$              | Data output valid time | Slave mode  | -              | 12           | 20             |      |
|                        |                        | Master mode | -              | 0.5          | 1              |      |
| $t_h(TX)$              | Data output hold time  | Slave mode  | 9              | -            | -              |      |
|                        |                        | Master mode | 0              | -            | -              |      |

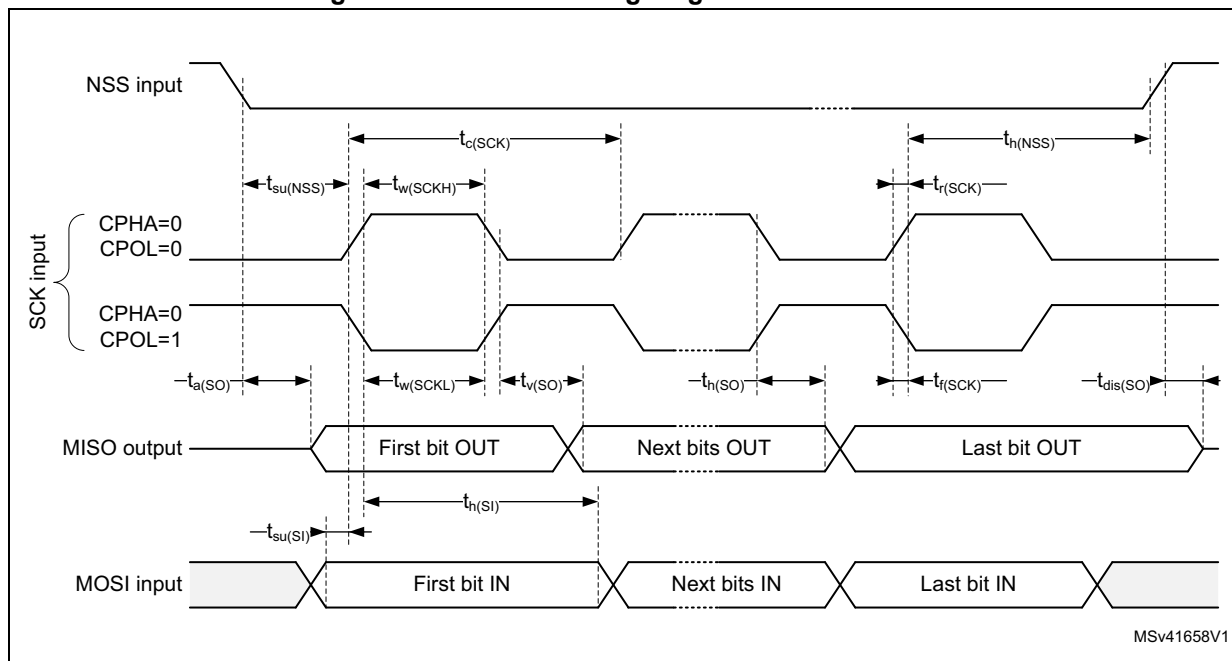
1. Guaranteed by characterization results.

Figure 101. USART timing diagram in Master mode



1. Measurement points are done at  $0.5V_{DD}$  and with external  $C_L = 30 \text{ pF}$ .

Figure 102. USART timing diagram in Slave mode



### SPI interface characteristics

Unless otherwise specified, the parameters given in [Table 205](#) for SPI are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 11$
- Capacitive load  $C_L = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- IO Compensation cell activated.
- HSLV activated when  $V_{DD} \leq 2.7$  V
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI).

**Table 205. SPI characteristics<sup>(1)</sup>**

| Symbol                 | Parameter             | Conditions                                                    | Min          | Typ        | Max          | Unit |
|------------------------|-----------------------|---------------------------------------------------------------|--------------|------------|--------------|------|
| $f_{SCK}$              | SPI clock frequency   | Master mode<br>$1.62 < V_{DD} < 3.6$ V<br>SPI1, 2, 3          | -            | -          | 80           | MHz  |
|                        |                       | Master mode<br>$2.7 < V_{DD} < 3.6$ V<br>SPI1, 2, 3           |              |            | 100          |      |
|                        |                       | Master mode<br>$1.62 < V_{DD} < 3.6$ V<br>SPI4, 5, 6          |              |            | 50           |      |
|                        |                       | Slave receiver mode<br>$1.62 < V_{DD} < 3.6$ V                |              |            | 100          |      |
|                        |                       | Slave mode transmitter/full duplex<br>$2.7 < V_{DD} < 3.6$ V  |              |            | 31           |      |
|                        |                       | Slave mode transmitter/full duplex<br>$1.62 < V_{DD} < 3.6$ V |              |            | 29           |      |
| $t_{su}(NSS)$          | NSS setup time        | Slave mode                                                    | 2            | -          | -            | -    |
| $t_h(NSS)$             | NSS hold time         | Slave mode                                                    | 1            | -          | -            |      |
| $t_w(SCKH), t_w(SCKL)$ | SCK high and low time | Master mode                                                   | $T_{PCLK}-2$ | $T_{PCLK}$ | $T_{PCLK}+2$ |      |

Table 205. SPI characteristics<sup>(1)</sup> (continued)

| Symbol        | Parameter                | Conditions                            | Min | Typ  | Max | Unit |
|---------------|--------------------------|---------------------------------------|-----|------|-----|------|
| $t_{su(MI)}$  | Data input setup time    | Master mode                           | 1   | -    | -   | ns   |
| $t_{su(SI)}$  |                          | Slave mode                            | 1   | -    | -   |      |
| $t_{h(MI)}$   | Data input hold time     | Master mode                           | 4   | -    | -   |      |
| $t_{h(SI)}$   |                          | Slave mode                            | 2   | -    | -   |      |
| $t_{a(SO)}$   | Data output access time  | Slave mode                            | 9   | 13   | 27  |      |
| $t_{dis(SO)}$ | Data output disable time | Slave mode                            | 0   | 1    | 5   |      |
| $t_{v(SO)}$   | Data output valid time   | Slave mode<br>$2.7 < V_{DD} < 3.6$ V  | -   | 12.5 | 16  |      |
|               |                          | Slave mode<br>$1.62 < V_{DD} < 3.6$ V | -   | 12.5 | 17  |      |
| $t_{v(MO)}$   |                          | Master mode                           | -   | 1    | 3   |      |
| $t_{h(SO)}$   | Data output hold time    | Slave mode<br>$1.62 < V_{DD} < 3.6$ V | 10  | -    | -   |      |
| $t_{h(MO)}$   |                          | Master mode                           | 0   | -    | -   |      |

1. Guaranteed by characterization results.

Figure 103. SPI timing diagram - slave mode and CPHA = 0

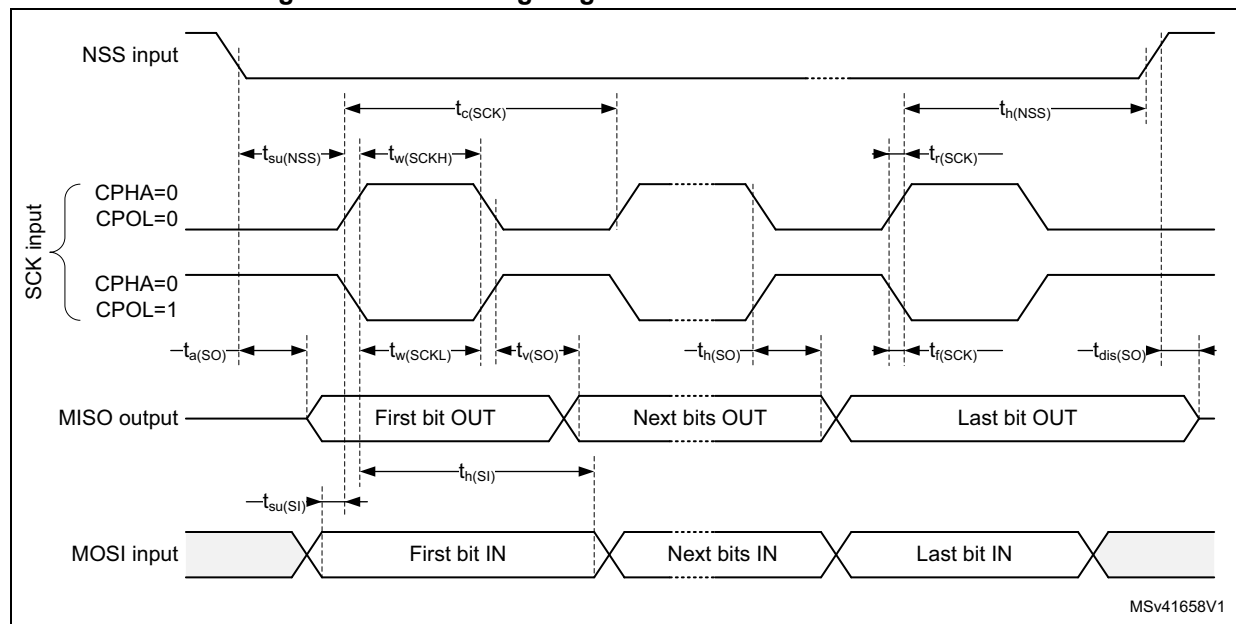
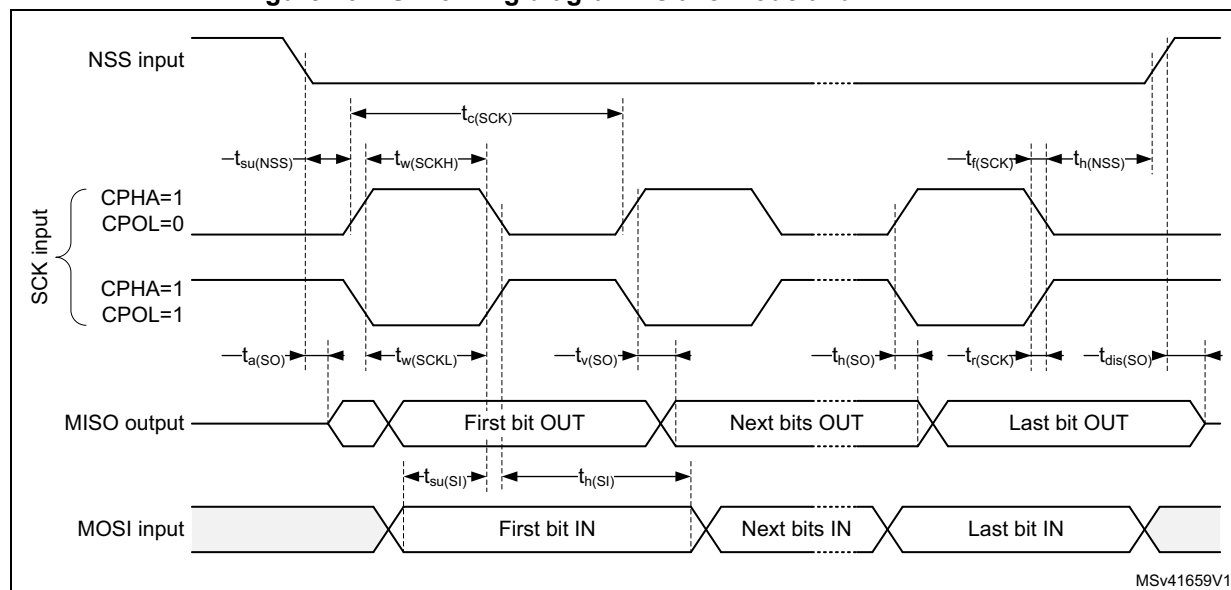
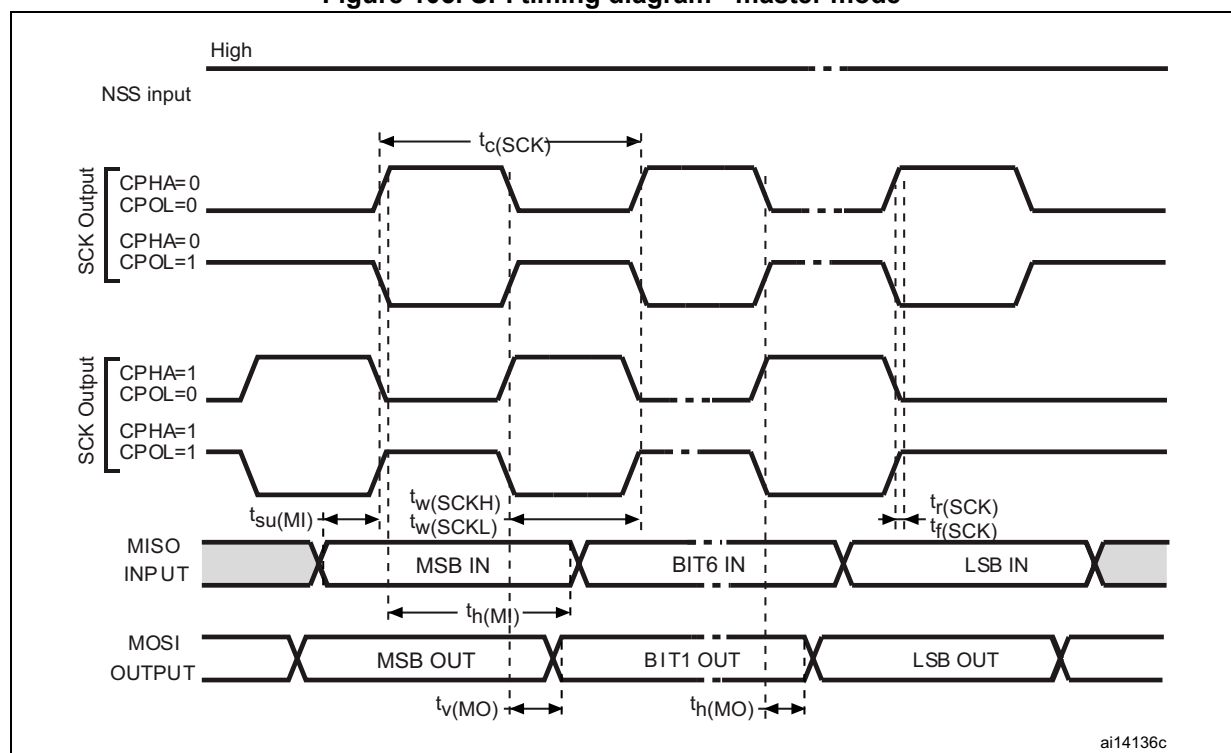


Figure 104. SPI timing diagram - slave mode and CPHA = 1<sup>(1)</sup>

1. Measurement points are done at  $0.5V_{DD}$  and with external  $C_L = 30$  pF.

Figure 105. SPI timing diagram - master mode<sup>(1)</sup>

1. Measurement points are done at  $0.5V_{DD}$  and with external  $C_L = 30$  pF.

## I<sup>2</sup>S Interface characteristics

Unless otherwise specified, the parameters given in [Table 206](#) for I<sup>2</sup>S are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

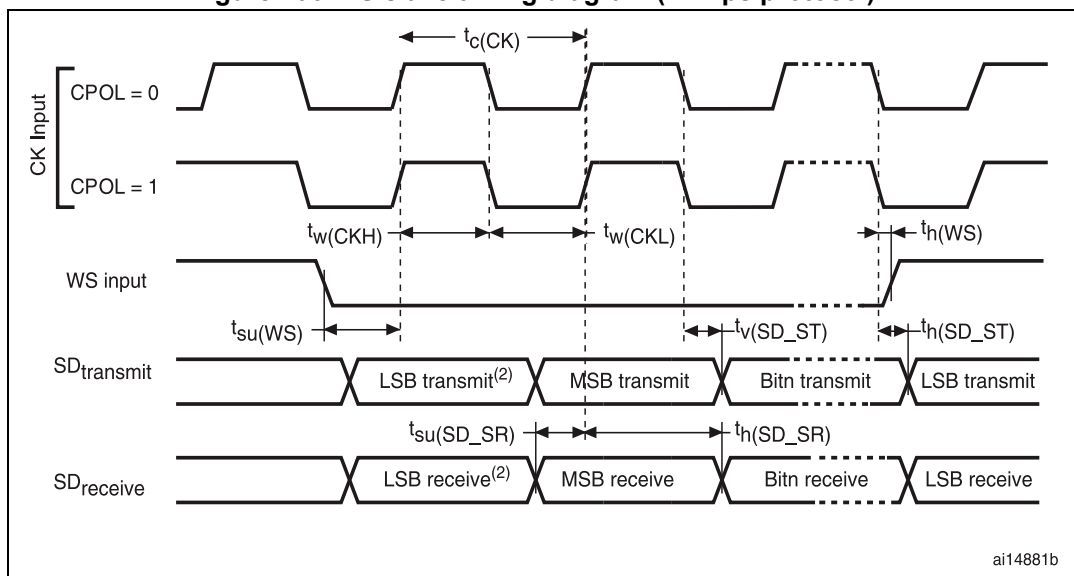
- Output speed is set to  $OSPEEDRy[1:0] = 10$
- Capacitive load  $C_L = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- IO Compensation cell activated.
- HSLV activated when  $V_{DD} \leq 2.7$  V
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (CK,SD,WS).

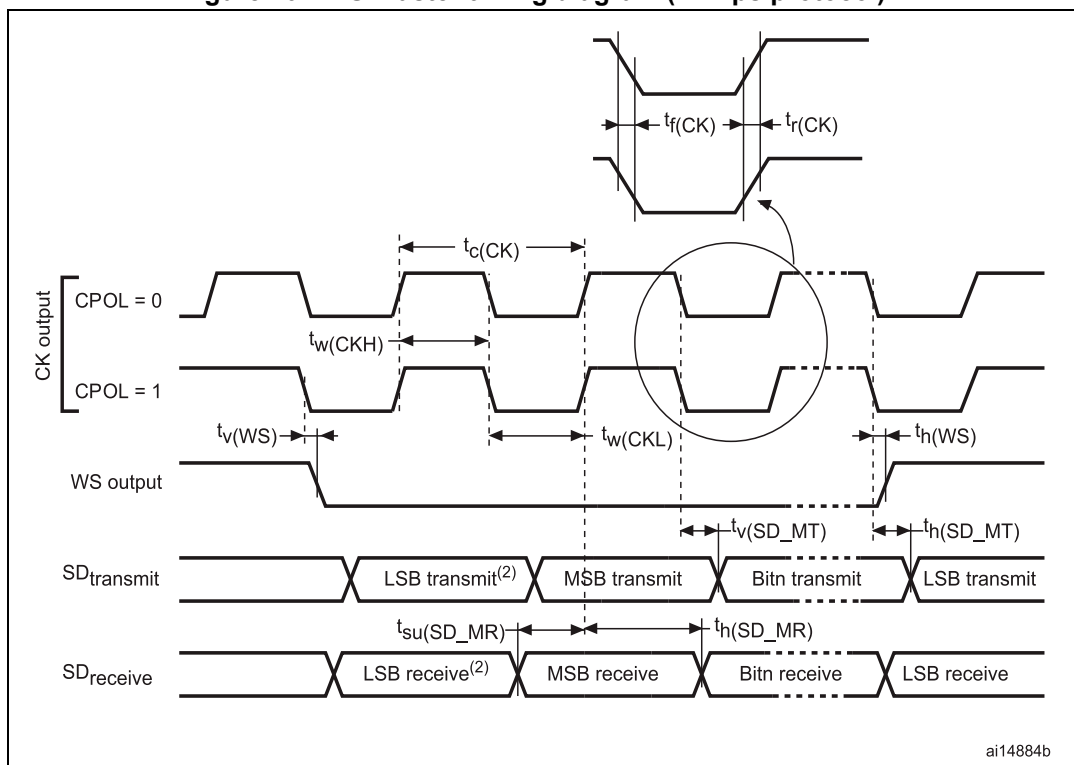
**Table 206. I<sup>2</sup>S dynamic characteristics<sup>(1)</sup>**

| Symbol           | Parameter                          | Conditions                             | Min    | Max      | Unit |
|------------------|------------------------------------|----------------------------------------|--------|----------|------|
| $f_{MCK}$        | I <sup>2</sup> S main clock output | -                                      | 256x8K | $256F_S$ | MHz  |
| $f_{CK}$         | I <sup>2</sup> S clock frequency   | Master data                            | -      | $64F_S$  | MHz  |
|                  |                                    | Slave data                             | -      | $64F_S$  |      |
| $t_{v(WS)}$      | WS valid time                      | Master mode                            | -      | 3        | ns   |
| $t_{h(WS)}$      | WS hold time                       | Master mode                            | 0      | -        |      |
| $t_{su(WS)}$     | WS setup time                      | Slave mode                             | 1      | -        |      |
| $t_{h(WS)}$      | WS hold time                       | Slave mode                             | 1      | -        |      |
| $t_{su(SD\_MR)}$ | Data input setup time              | Master receiver                        | 1      | -        |      |
| $t_{su(SD\_SR)}$ |                                    | Slave receiver                         | 1      | -        |      |
| $t_{h(SD\_MR)}$  | Data input hold time               | Master receiver                        | 4      | -        |      |
| $t_{h(SD\_SR)}$  |                                    | Slave receiver                         | 2      | -        |      |
| $t_{v(SD\_ST)}$  | Data output valid time             | Slave transmitter (after enable edge)  | -      | 17       |      |
| $t_{v(SD\_MT)}$  |                                    | Master transmitter (after enable edge) | -      | 3        |      |
| $t_{h(SD\_ST)}$  | Data output hold time              | Slave transmitter (after enable edge)  | 9      | -        |      |
| $t_{h(SD\_MT)}$  |                                    | Master transmitter (after enable edge) | 0      | -        |      |

1. Guaranteed by characterization results.

Figure 106. I<sup>2</sup>S slave timing diagram (Philips protocol)<sup>(1)</sup>

1. LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.

Figure 107. I<sup>2</sup>S master timing diagram (Philips protocol)<sup>(1)</sup>

1. LSB transmit/receive of the previously transmitted byte. No LSB transmit/receive is sent before the first byte.



### SAI characteristics

Unless otherwise specified, the parameters given in [Table 207](#) for SAI are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and VDD supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 10$
- Capacitive load  $C_L = 30$  pF
- IO Compensation cell activated.
- Measurement points are done at CMOS levels: 0.5VDD
- VOS level set to VOS1.

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output alternate function characteristics (SCK,SD,WS).

**Table 207. SAI characteristics<sup>(1)</sup>**

| Symbol    | Parameter                          | Conditions           | Min    | Max                    | Unit |
|-----------|------------------------------------|----------------------|--------|------------------------|------|
| $f_{MCK}$ | SAI Main clock output              | -                    | 256x8K | $256 \times F_S$       | MHz  |
| $f_{CK}$  | SAI clock frequency <sup>(2)</sup> | Master Data: 32 bits | -      | $128 \times F_S^{(3)}$ |      |
|           |                                    | Slave Data: 32 bits  | -      | $128 \times F_S^{(3)}$ |      |

Table 207. SAI characteristics<sup>(1)</sup> (continued)

| Symbol              | Parameter              | Conditions                                                            | Min | Max | Unit |
|---------------------|------------------------|-----------------------------------------------------------------------|-----|-----|------|
| $t_{V(FS)}$         | $F_S$ valid time       | Master mode<br>$2.7 \leq V_{DD} \leq 3.6$                             | -   | 13  | ns   |
|                     |                        | Master mode<br>$1.62 \leq V_{DD} \leq 3.6$                            | -   | 20  |      |
| $t_{su(FS)}$        | $F_S$ hold time        | Master mode                                                           | 8   | -   |      |
| $t_{h(FS)}$         | $F_S$ setup time       | Slave mode                                                            | 1   | -   |      |
|                     | $F_S$ hold time        | Slave mode                                                            | 1   | -   |      |
| $t_{su(SD\_A\_MR)}$ | Data input setup time  | Master receiver                                                       | 0.5 | -   |      |
| $t_{su(SD\_B\_SR)}$ |                        | Slave receiver                                                        | 1   | -   |      |
| $t_{h(SD\_A\_MR)}$  | Data input hold time   | Master receiver                                                       | 3.5 | -   |      |
| $t_{h(SD\_B\_SR)}$  |                        | Slave receiver                                                        | 2   | -   |      |
| $t_{V(SD\_B\_ST)}$  | Data output valid time | Slave transmitter (after enable edge)<br>$2.7 \leq V_{DD} \leq 3.6$   | -   | 14  |      |
|                     |                        | Slave transmitter (after enable edge)<br>$1.62 \leq V_{DD} \leq 3.6$  | -   | 20  |      |
| $t_{h(SD\_B\_ST)}$  | Data output hold time  | Slave transmitter (after enable edge)                                 | 9   | -   |      |
| $t_{V(SD\_A\_MT)}$  | Data output valid time | Master transmitter (after enable edge)<br>$2.7 \leq V_{DD} \leq 3.6$  | -   | 12  |      |
|                     |                        | Master transmitter (after enable edge)<br>$1.62 \leq V_{DD} \leq 3.6$ | -   | 19  |      |
| $t_{h(SD\_A\_MT)}$  | Data output hold time  | Master transmitter (after enable edge)                                | 7.5 | -   |      |

1. Guaranteed by characterization results.

2. APB clock frequency must be at least twice SAI clock frequency.

3. With  $F_S=192$  kHz.

Figure 108. SAI master timing waveforms

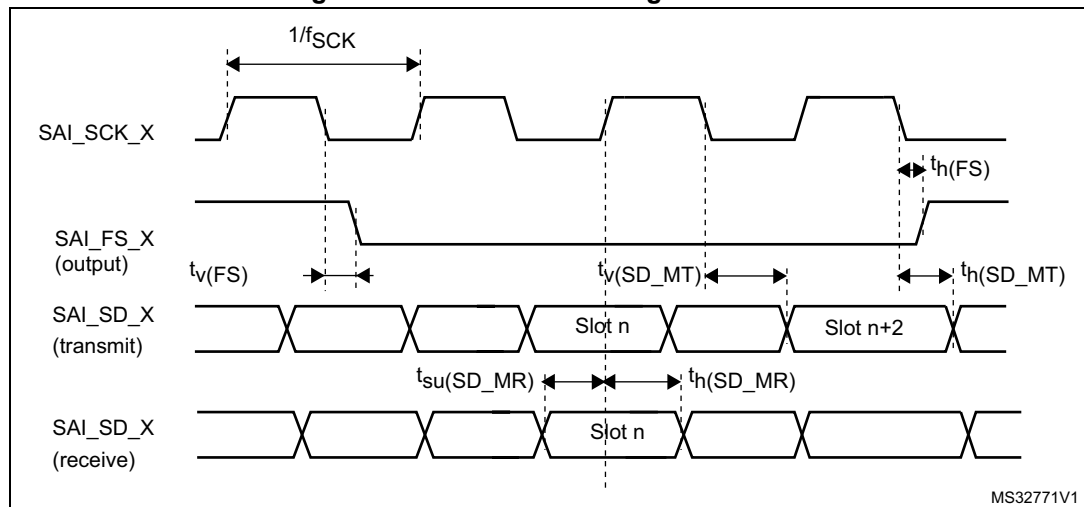
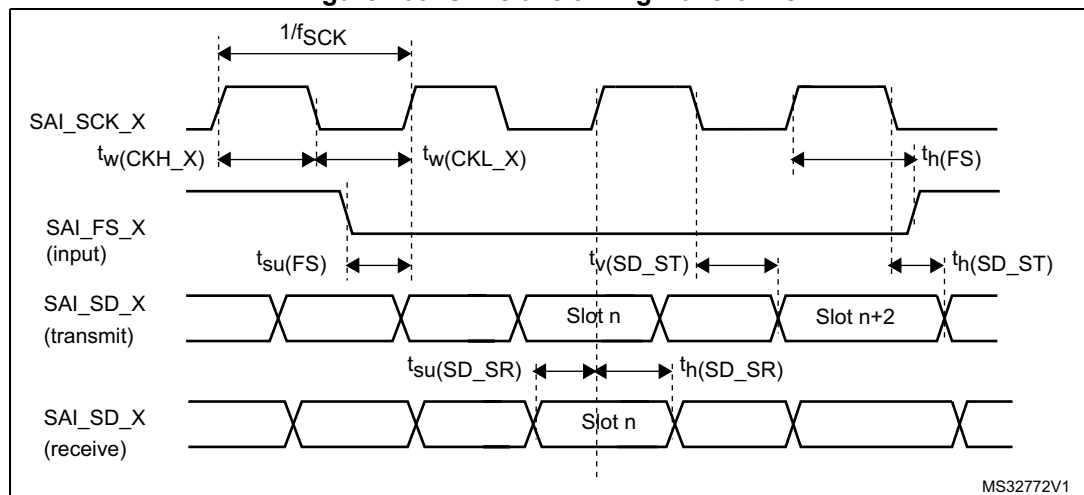


Figure 109. SAI slave timing waveforms

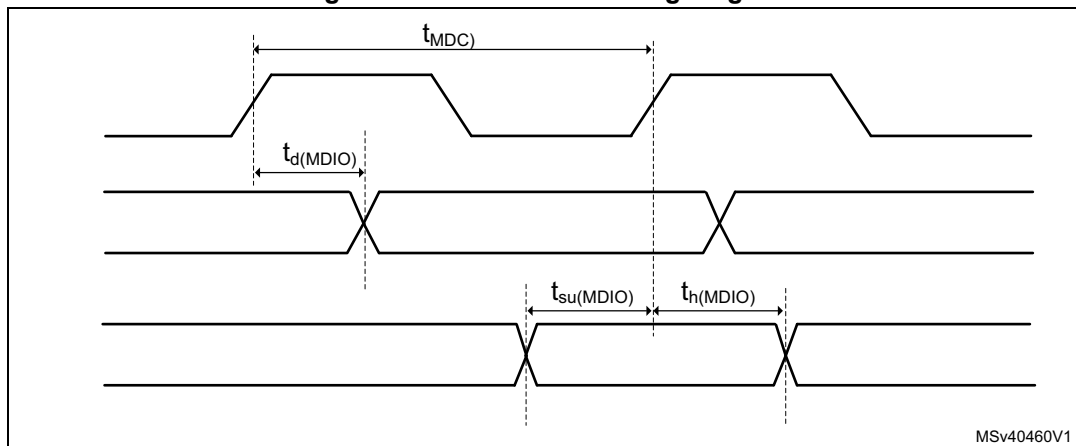


## MDIO characteristics

Table 208. MDIO Slave timing parameters

| Symbol         | Parameter                                      | Min | Typ | Max | Unit |
|----------------|------------------------------------------------|-----|-----|-----|------|
| $F_{MDC}$      | Management Data Clock                          | -   | -   | 30  | MHz  |
| $t_d(MDIO)$    | Management Data Input/output output valid time | 8   | 10  | 19  | ns   |
| $t_{su}(MDIO)$ | Management Data Input/output setup time        | 1   | -   | -   |      |
| $t_h(MDIO)$    | Management Data Input/output hold time         | 1   | -   | -   |      |

Figure 110. MDIO Slave timing diagram



### SD/SDIO MMC card host interface (SDMMC) characteristics

Unless otherwise specified, the parameters given in [Table 209](#) and [Table 210](#) for SDIO are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 0x11$
- Capacitive load  $C_L = 30$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- IO Compensation cell activated.
- HSLV activated when  $V_{DD} \leq 2.7$  V
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

**Table 209. Dynamics characteristics: SD / MMC characteristics,  $V_{DD} = 2.7$  to  $3.6$  V<sup>(1)(2)</sup>**

| Symbol                                                                                                        | Parameter                             | Conditions             | Min | Typ | Max | Unit |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------|-----|-----|-----|------|
| f <sub>PP</sub>                                                                                               | Clock frequency in data transfer mode | -                      | 0   | -   | 133 | MHz  |
| -                                                                                                             | SDIO_CK/fPCLK2 frequency ratio        | -                      | -   | -   | 8/3 | -    |
| t <sub>W(CKL)</sub>                                                                                           | Clock low time                        | f <sub>PP</sub> =52MHz | 8.5 | 9.5 | -   | ns   |
| t <sub>W(CKH)</sub>                                                                                           | Clock high time                       | f <sub>PP</sub> =52MHz | 8.5 | 9.5 | -   |      |
| CMD, D inputs (referenced to CK) in eMMC legacy/SDR/DDR and SD HS/SDR <sup>(3)</sup> /DDR <sup>(3)</sup> mode |                                       |                        |     |     |     |      |
| t <sub>ISU</sub>                                                                                              | Input setup time HS                   | -                      | 1.5 | -   | -   | ns   |
| t <sub>IH</sub>                                                                                               | Input hold time HS                    | -                      | 1.5 | -   | -   |      |
| t <sub>IDW</sub> <sup>(4)</sup>                                                                               | Input valid window (variable window)  | -                      | 3   | -   | -   | -    |
| CMD, D outputs (referenced to CK) in eMMC legacy/SDR/DDR and SD HS/SDR/DDR <sup>(3)</sup> mode                |                                       |                        |     |     |     |      |
| t <sub>OV</sub>                                                                                               | Output valid time HS                  | -                      | -   | 3.5 | 5   | ns   |
| t <sub>OH</sub>                                                                                               | Output hold time HS                   | -                      | 2   | -   | -   |      |

**Table 209. Dynamics characteristics: SD / MMC characteristics,  $V_{DD}=2.7$  to  $3.6$  V<sup>(1)(2)</sup> (continued)**

| Symbol                                               | Parameter                    | Conditions | Min | Typ | Max | Unit |
|------------------------------------------------------|------------------------------|------------|-----|-----|-----|------|
| CMD, D inputs (referenced to CK) in SD default mode  |                              |            |     |     |     |      |
| t <sub>ISUD</sub>                                    | Input setup time SD          | -          | 1.5 |     | -   | ns   |
| t <sub>IHD</sub>                                     | Input hold time SD           | -          | 1.5 |     | -   |      |
| CMD, D outputs (referenced to CK) in SD default mode |                              |            |     |     |     |      |
| t <sub>OVD</sub>                                     | Output valid default time SD | -          | -   | 0.5 | 2   | ns   |
| t <sub>OHD</sub>                                     | Output hold default time SD  | -          | 0   | -   | -   |      |

1. Guaranteed by characterization results.
2. Above 100 MHz,  $C_L = 20$  pF.
3. An external voltage converter is required to support SD 1.8 V.
4. The minimum window of time where the data needs to be stable for proper sampling in tuning mode.

**Table 210. Dynamics characteristics: eMMC characteristics  $V_{DD}=1.71$  V to  $1.9$  V<sup>(1)(2)</sup>**

| Symbol                                         | Parameter                             | Conditions              | Min | Typ | Max | Unit |
|------------------------------------------------|---------------------------------------|-------------------------|-----|-----|-----|------|
| f <sub>PP</sub>                                | Clock frequency in data transfer mode | -                       | 0   | -   | 120 | MHz  |
| -                                              | SDIO_CK/fPCLK2 frequency ratio        | -                       | -   | -   | 8/3 | -    |
| t <sub>W(CKL)</sub>                            | Clock low time                        | f <sub>PP</sub> =52 MHz | 8.5 | 9.5 | -   | ns   |
| t <sub>W(CKH)</sub>                            | Clock high time                       | f <sub>PP</sub> =52 MHz | 8.5 | 9.5 | -   |      |
| CMD, D inputs (referenced to CK) in eMMC mode  |                                       |                         |     |     |     |      |
| t <sub>ISU</sub>                               | Input setup time HS                   | -                       | 1   | -   | -   | ns   |
| t <sub>IH</sub>                                | Input hold time HS                    | -                       | 2.5 | -   | -   |      |
| t <sub>IDW</sub> <sup>(3)</sup>                | Input valid window (variable window)  | -                       | 3.5 | -   | -   |      |
| CMD, D outputs (referenced to CK) in eMMC mode |                                       |                         |     |     |     |      |
| t <sub>OVD</sub>                               | Output valid time HS                  | -                       | -   | 5   | 7   | ns   |
| t <sub>OHD</sub>                               | Output hold time HS                   | -                       | 3   | -   | -   |      |

1. Guaranteed by characterization results.
2.  $C_L = 20$  pF.
3. The minimum window of time where the data needs to be stable for proper sampling in tuning mode.

Figure 111. SDIO high-speed mode

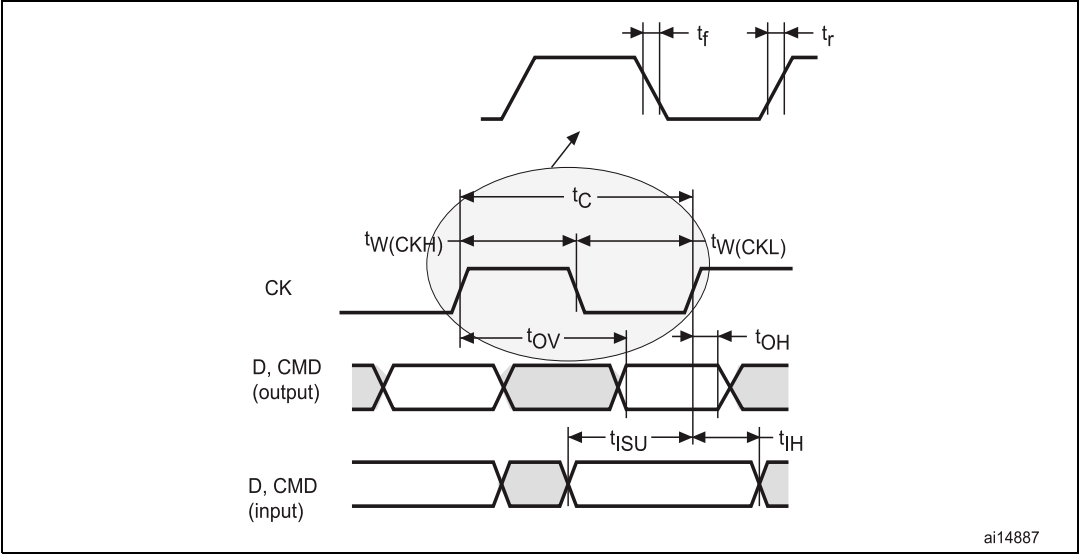


Figure 112. SD default mode

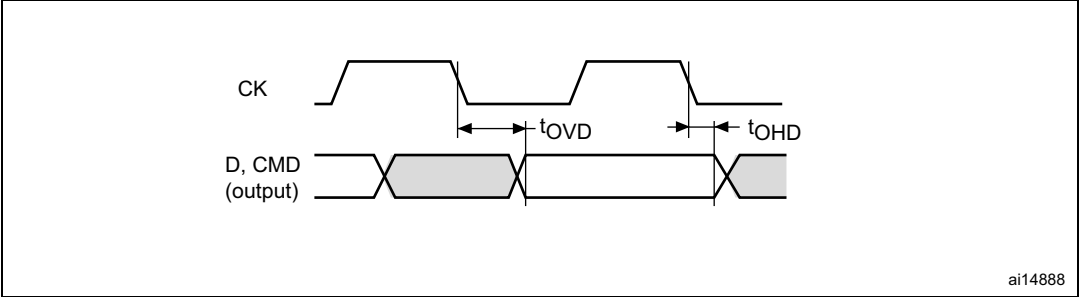
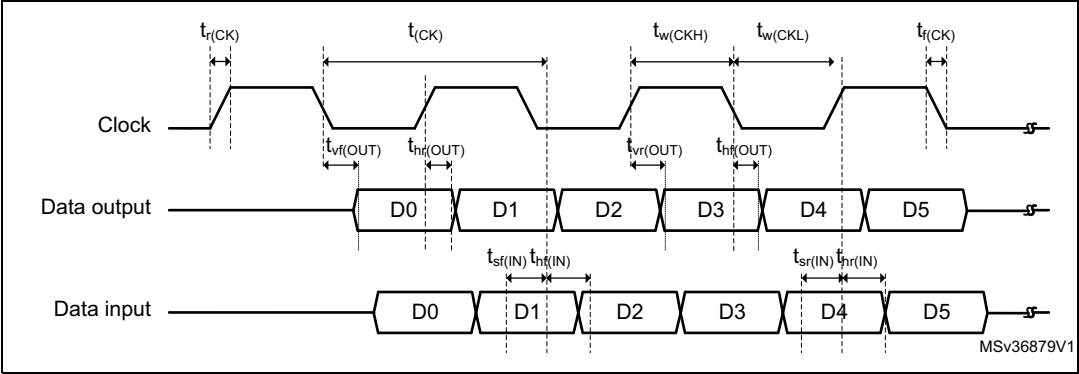


Figure 113. DDR mode



### USB OTG\_HS characteristics

Unless otherwise specified, the parameters given in [Table 211](#) for ULPI are derived from tests performed under the ambient temperature,  $f_{PCLKx}$  frequency and  $V_{DD}$  supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to  $OSPEEDRy[1:0] = 11$
- Capacitive load  $C_L = 20$  pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- IO Compensation cell activated.
- VOS level set to VOS1

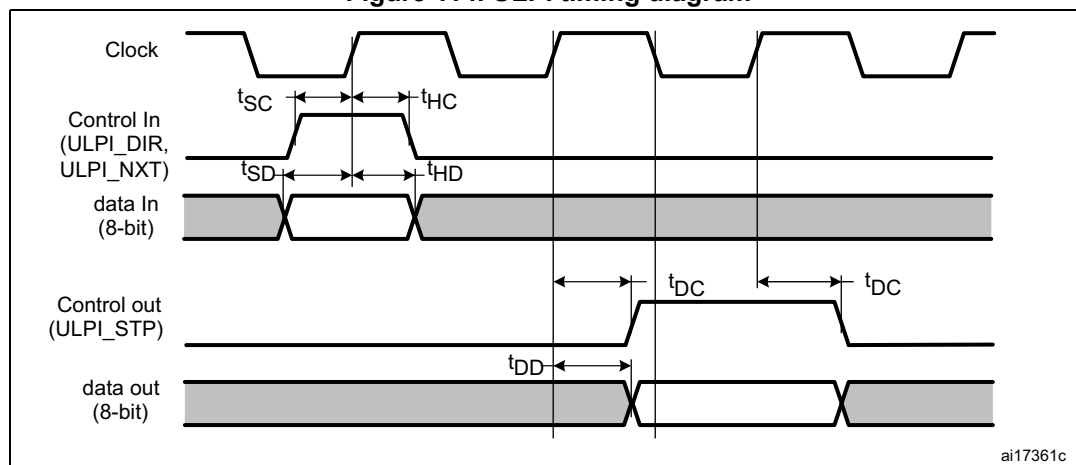
Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output characteristics.

**Table 211. Dynamics characteristics: USB ULPI<sup>(1)</sup>**

| Symbol          | Parameter                                  | Condition                                | Min | Typ | Max | Unit |
|-----------------|--------------------------------------------|------------------------------------------|-----|-----|-----|------|
| $t_{SC}$        | Control in (ULPI_DIR, ULPI_NXT) setup time | -                                        | 2.5 | -   | -   | ns   |
| $t_{HC}$        | Control in (ULPI_DIR, ULPI_NXT) hold time  | -                                        | 2   | -   | -   |      |
| $t_{SD}$        | Data in setup time                         | -                                        | 2.5 | -   | -   |      |
| $t_{HD}$        | Data in hold time                          | -                                        | 0   | -   | -   |      |
| $t_{DC}/t_{DD}$ | Control/Data output delay                  | $2.7 < V_{DD} < 3.6$ V<br>$C_L = 20$ pF  | -   | 9   | 9.5 |      |
|                 |                                            | $1.71 < V_{DD} < 3.6$ V<br>$C_L = 15$ pF | -   | 9   | 14  |      |

1. Guaranteed by characterization results.

**Figure 114. ULPI timing diagram**



## Ethernet interface characteristics

Unless otherwise specified, the parameters given in [Table 212](#), [Table 213](#) and [Table 214](#) for SMI, RMII and MII are derived from tests performed under the ambient temperature,  $f_{\text{rcc\_c\_ck}}$  frequency and  $V_{\text{DD}}$  supply voltage conditions summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- Capacitive load  $C_L = 20$  pF
- Measurement points are done at CMOS levels:  $0.5V_{\text{DD}}$
- IO Compensation cell activated.
- HSLV activated when  $V_{\text{DD}} \leq 2.7$  V
- VOS level set to VOS1

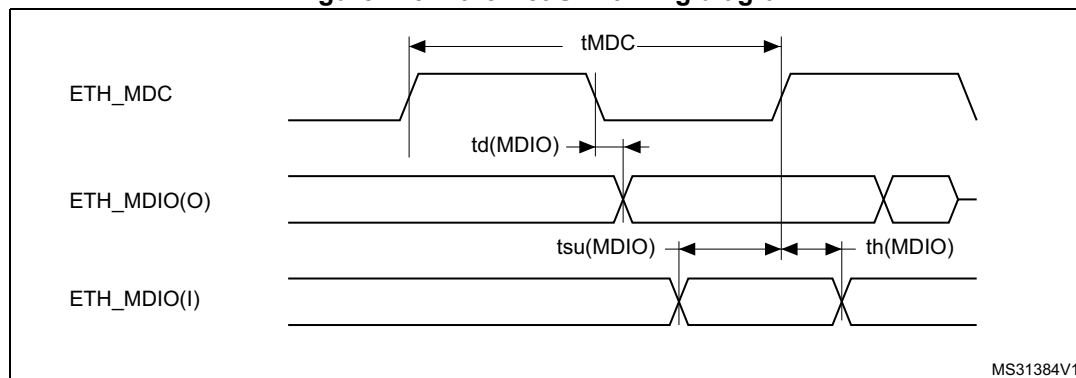
Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output characteristics:

**Table 212. Dynamics characteristics: Ethernet MAC signals for SMI <sup>(1)</sup>**

| Symbol                       | Parameter                | Min  | Typ | Max | Unit |
|------------------------------|--------------------------|------|-----|-----|------|
| $t_{\text{MDC}}$             | MDC cycle time( 2.5 MHz) | 400  | 400 | 403 | ns   |
| $T_{\text{d}}(\text{MDIO})$  | Write data valid time    | 0.5  | 1.5 | 4   |      |
| $t_{\text{su}}(\text{MDIO})$ | Read data setup time     | 12.5 | -   | -   |      |
| $t_{\text{h}}(\text{MDIO})$  | Read data hold time      | 0    | -   | -   |      |

1. Guaranteed by characterization results.

**Figure 115. Ethernet SMI timing diagram**

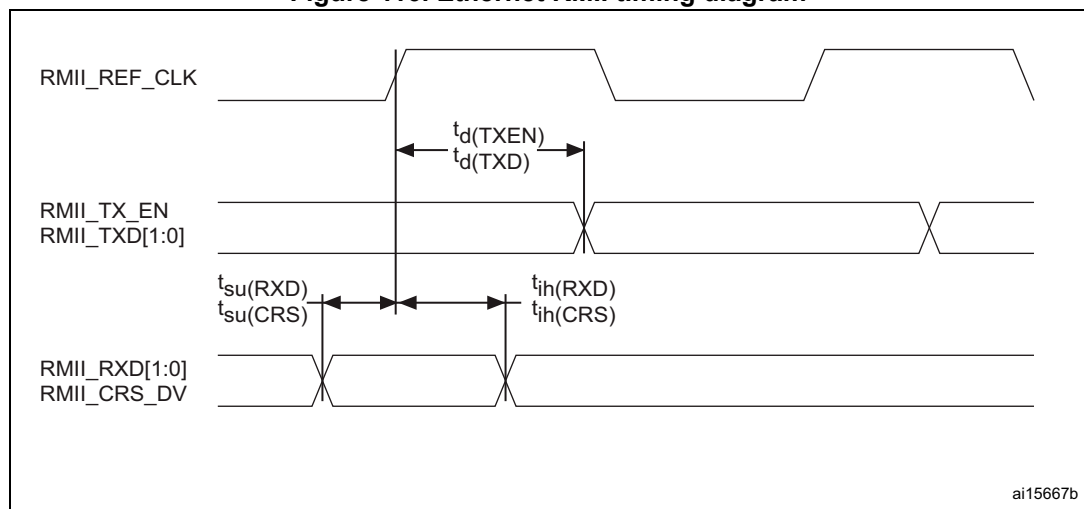




**Table 213. Dynamics characteristics: Ethernet MAC signals for RMII <sup>(1)</sup>**

| Symbol        | Parameter                        | Min | Typ | Max | Unit |
|---------------|----------------------------------|-----|-----|-----|------|
| $t_{su}(RXD)$ | Receive data setup time          | 2   | -   | -   | ns   |
| $t_{ih}(RXD)$ | Receive data hold time           | 2   | -   | -   |      |
| $t_{su}(CRS)$ | Carrier sense setup time         | 1.5 | -   | -   |      |
| $t_{ih}(CRS)$ | Carrier sense hold time          | 1.5 | -   | -   |      |
| $t_d(TXEN)$   | Transmit enable valid delay time | 7   | 8   | 9.5 |      |
| $t_d(TXD)$    | Transmit data valid delay time   | 8   | 9   | 11  |      |

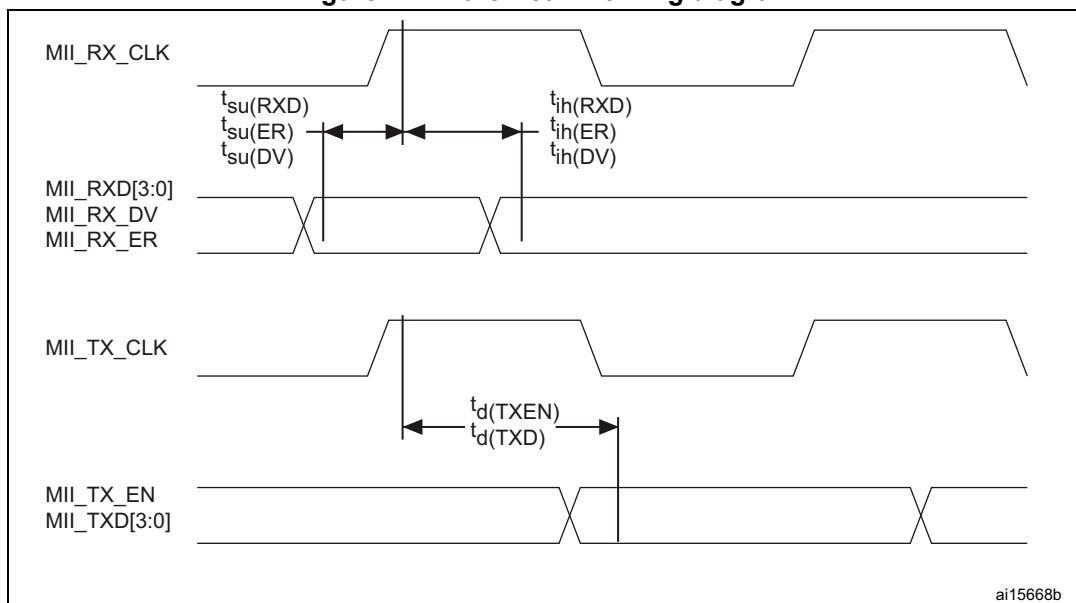
1. Guaranteed by characterization results.

**Figure 116. Ethernet RMII timing diagram****Table 214. Dynamics characteristics: Ethernet MAC signals for MII <sup>(1)</sup>**

| Symbol        | Parameter                        | Min | Typ | Max  | Unit |
|---------------|----------------------------------|-----|-----|------|------|
| $t_{su}(RXD)$ | Receive data setup time          | 2   | -   | -    | ns   |
| $t_{ih}(RXD)$ | Receive data hold time           | 2   | -   | -    |      |
| $t_{su}(DV)$  | Data valid setup time            | 1.5 | -   | -    |      |
| $t_{ih}(DV)$  | Data valid hold time             | 1.5 | -   | -    |      |
| $t_{su}(ER)$  | Error setup time                 | 1.5 | -   | -    |      |
| $t_{ih}(ER)$  | Error hold time                  | 0.5 | -   | -    |      |
| $t_d(TXEN)$   | Transmit enable valid delay time | 9   | 10  | 11   |      |
| $t_d(TXD)$    | Transmit data valid delay time   | 8.5 | 9.5 | 12.5 |      |

1. Guaranteed by characterization results.

Figure 117. Ethernet MII timing diagram



### JTAG/SWD interface characteristics

Unless otherwise specified, the parameters given in [Table 215](#) and [Table 216](#) for JTAG/SWD are derived from tests performed under the ambient temperature,  $f_{rcc\_c\_ck}$  frequency and  $V_{DD}$  supply voltage summarized in [Table 122: General operating conditions](#), with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 0x10
- Capacitive load  $C_L$  = 30 pF
- Measurement points are done at CMOS levels:  $0.5V_{DD}$
- VOS level set to VOS1

Refer to [Section 7.3.15: I/O port characteristics](#) for more details on the input/output characteristics:

Table 215. Dynamics JTAG characteristics

| Symbol            | Parameter                | Conditions             | Min | Typ | Max  | Unit |
|-------------------|--------------------------|------------------------|-----|-----|------|------|
| $F_{pp}$          | $T_{CK}$ clock frequency | $2.7V < V_{DD} < 3.6V$ | -   | -   | 37   | MHz  |
| $1/t_c(TCK)$      |                          | $1.62 < V_{DD} < 3.6V$ | -   | -   | 27.5 |      |
| $t_{i_{su}}(TMS)$ | TMS input setup time     | -                      | 2.5 | -   | -    |      |
| $t_{i_h}(TMS)$    | TMS input hold time      | -                      | 1   | -   | -    |      |
| $t_{i_{su}}(TDI)$ | TDI input setup time     | -                      | 1.5 | -   | -    | -    |
| $t_{i_h}(TDI)$    | TDI input hold time      | -                      | 1   | -   | -    | -    |
| $t_{ov}(TDO)$     | TDO output valid time    | $2.7V < V_{DD} < 3.6V$ | -   | 8   | 13.5 | -    |
|                   |                          | $1.62 < V_{DD} < 3.6V$ | -   | 8   | 18   | -    |
| $t_{oh}(TDO)$     | TDO output hold time     | -                      | 7   | -   | -    | -    |

Table 216. Dynamics SWD characteristics:

| Symbol              | Parameter               | Conditions             | Min | Typ | Max  | Unit |
|---------------------|-------------------------|------------------------|-----|-----|------|------|
| $F_{pp}$            | SWCLK clock frequency   | $2.7V < V_{DD} < 3.6V$ | -   | -   | 71   | MHz  |
| $1/t_c(SWCLK)$      |                         | $1.62 < V_{DD} < 3.6V$ | -   | -   | 52.5 |      |
| $t_{i_{su}}(SWDIO)$ | SWDIO input setup time  | -                      | 2.5 | -   | -    | -    |
| $t_{i_h}(SWDIO)$    | SWDIO input hold time   | -                      | 1   | -   | -    | -    |
| $t_{ov}(SWDIO)$     | SWDIO output valid time | $2.7V < V_{DD} < 3.6V$ | -   | 8.5 | 14   | -    |
|                     |                         | $1.62 < V_{DD} < 3.6V$ | -   | 8.5 | 19   | -    |
| $t_{oh}(SWDIO)$     | SWDIO output hold time  | -                      | 8   | -   | -    | -    |

Figure 118. JTAG timing diagram

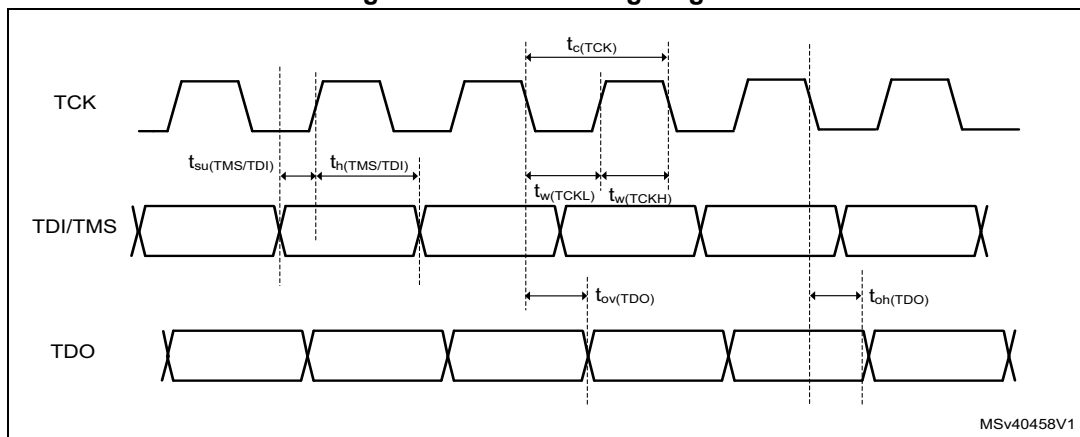
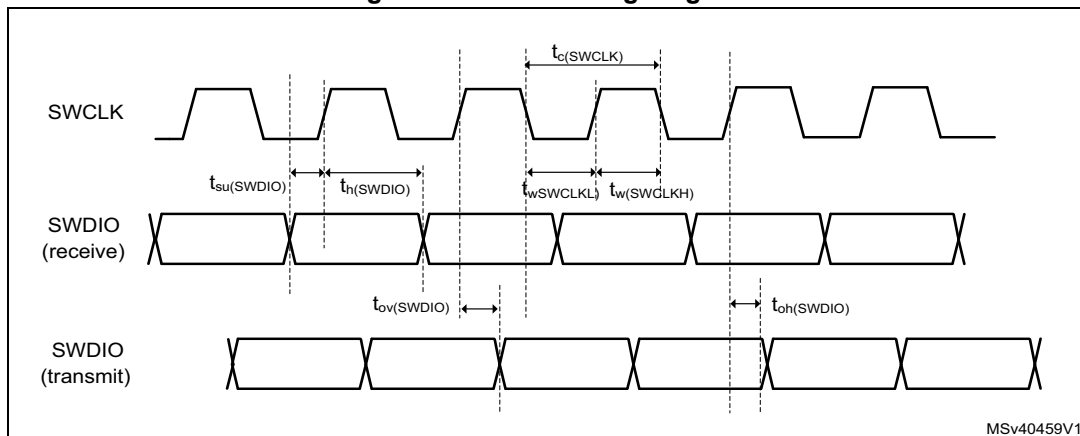


Figure 119. SWD timing diagram



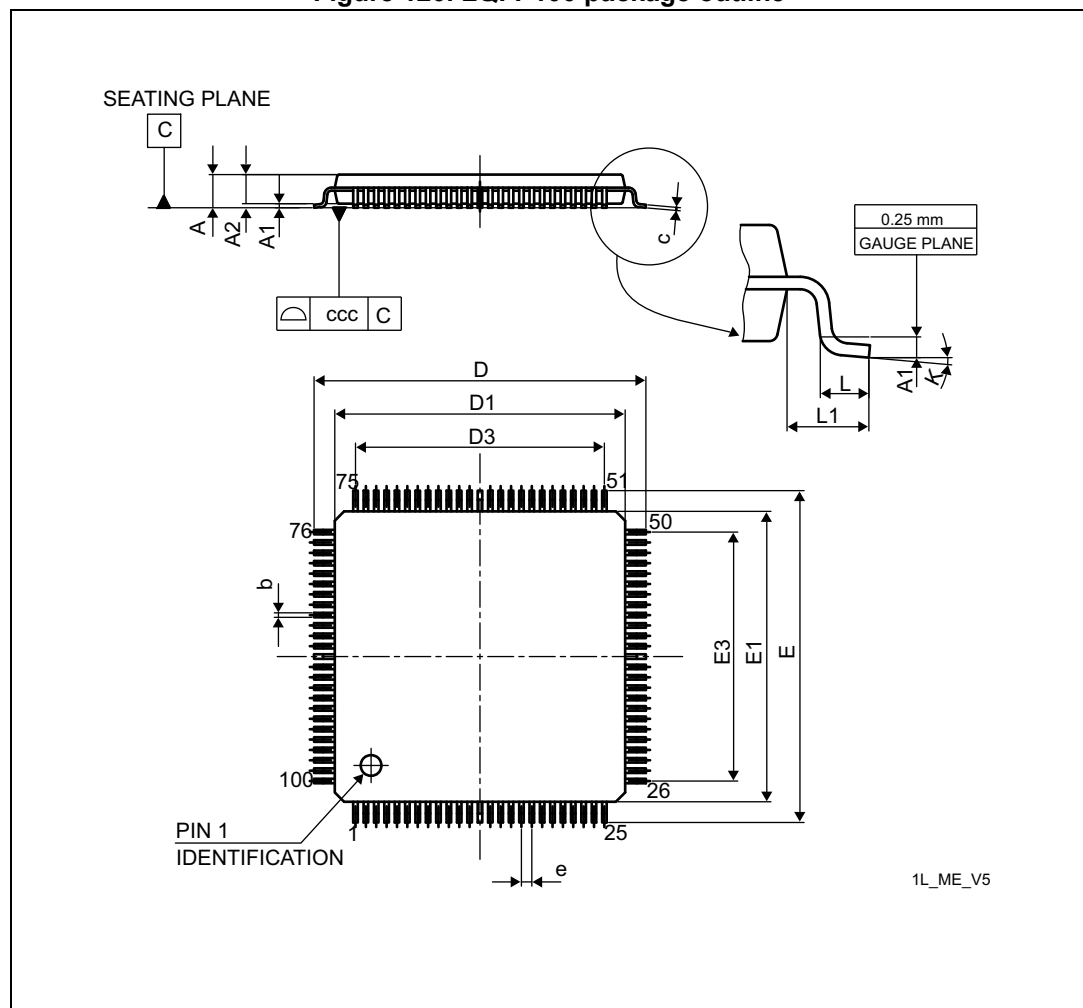
## 8 Package information

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status *are available at* [www.st.com](http://www.st.com). ECOPACK® is an ST trademark.

### 8.1 LQFP100 package information

LQFP100 is a 100-pin, 14 x 14 mm low-profile quad flat package.

Figure 120. LQFP100 package outline



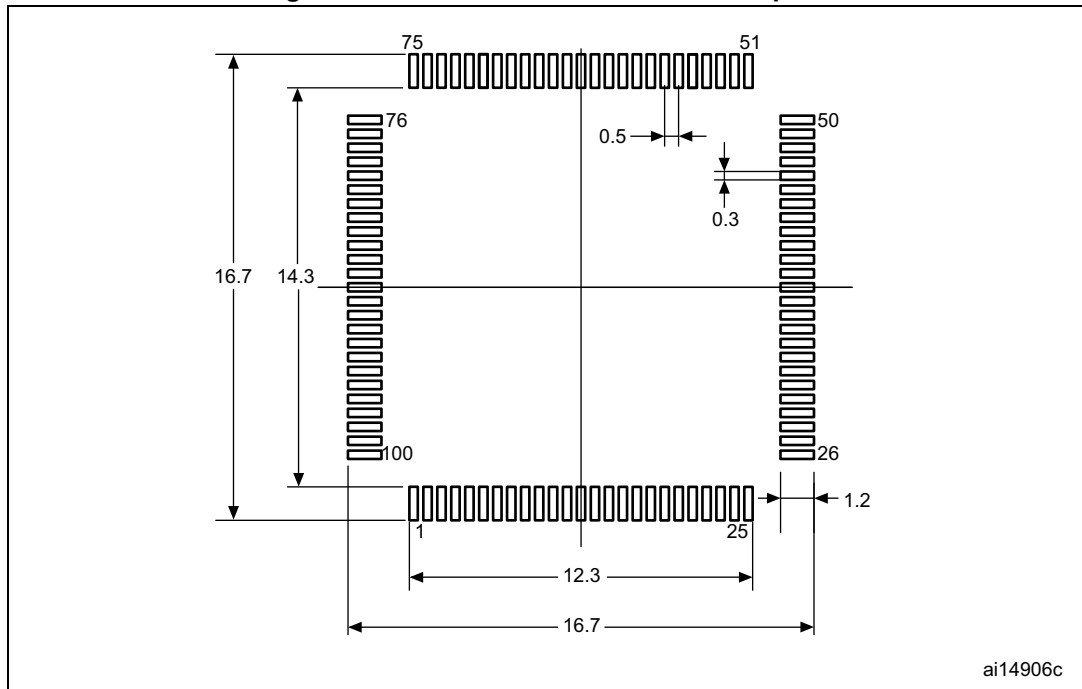
1. Drawing is not to scale.

Table 217. LQPF100 package mechanical data

| Symbol | millimeters |        |        | inches <sup>(1)</sup> |        |        |
|--------|-------------|--------|--------|-----------------------|--------|--------|
|        | Min         | Typ    | Max    | Min                   | Typ    | Max    |
| A      | -           | -      | 1.600  | -                     | -      | 0.0630 |
| A1     | 0.050       | -      | 0.150  | 0.0020                | -      | 0.0059 |
| A2     | 1.350       | 1.400  | 1.450  | 0.0531                | 0.0551 | 0.0571 |
| b      | 0.170       | 0.220  | 0.270  | 0.0067                | 0.0087 | 0.0106 |
| c      | 0.090       | -      | 0.200  | 0.0035                | -      | 0.0079 |
| D      | 15.800      | 16.000 | 16.200 | 0.6220                | 0.6299 | 0.6378 |
| D1     | 13.800      | 14.000 | 14.200 | 0.5433                | 0.5512 | 0.5591 |
| D3     | -           | 12.000 | -      | -                     | 0.4724 | -      |
| E      | 15.800      | 16.000 | 16.200 | 0.6220                | 0.6299 | 0.6378 |
| E1     | 13.800      | 14.000 | 14.200 | 0.5433                | 0.5512 | 0.5591 |
| E3     | -           | 12.000 | -      | -                     | 0.4724 | -      |
| e      | -           | 0.500  | -      | -                     | 0.0197 | -      |
| L      | 0.450       | 0.600  | 0.750  | 0.0177                | 0.0236 | 0.0295 |
| L1     | -           | 1.000  | -      | -                     | 0.0394 | -      |
| k      | 0.0°        | 3.5°   | 7.0°   | 0.0°                  | 3.5°   | 7.0°   |
| ccc    | -           | -      | 0.080  | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 121. LQFP100 recommended footprint



1. Dimensions are expressed in millimeters.

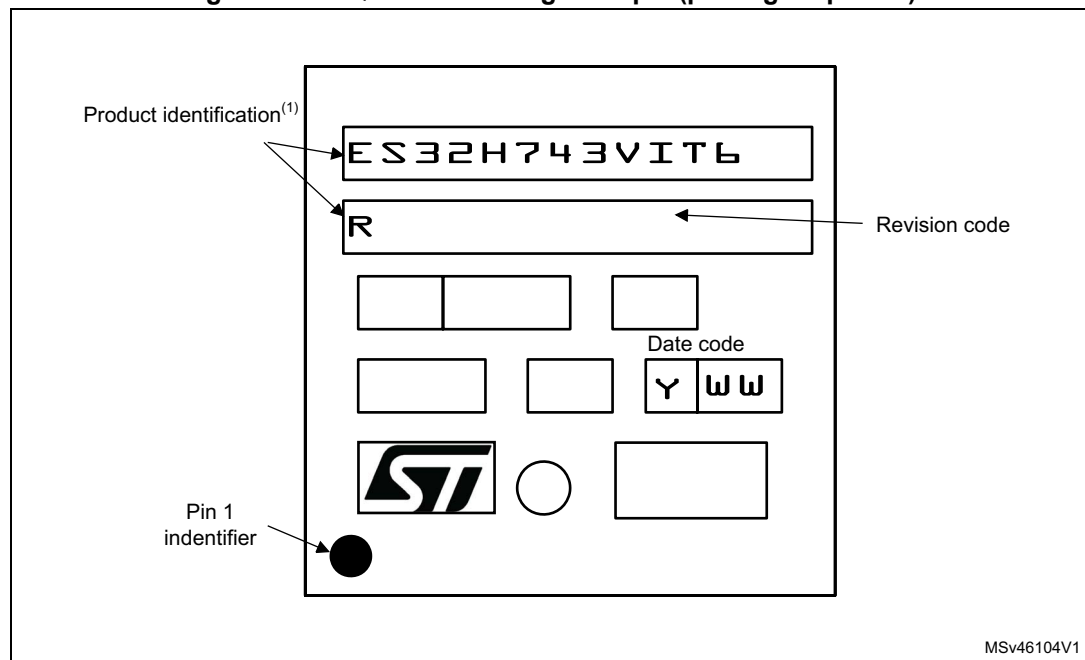
### Device marking for LQFP100

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 122. LQFP100 marking example (package top view)**

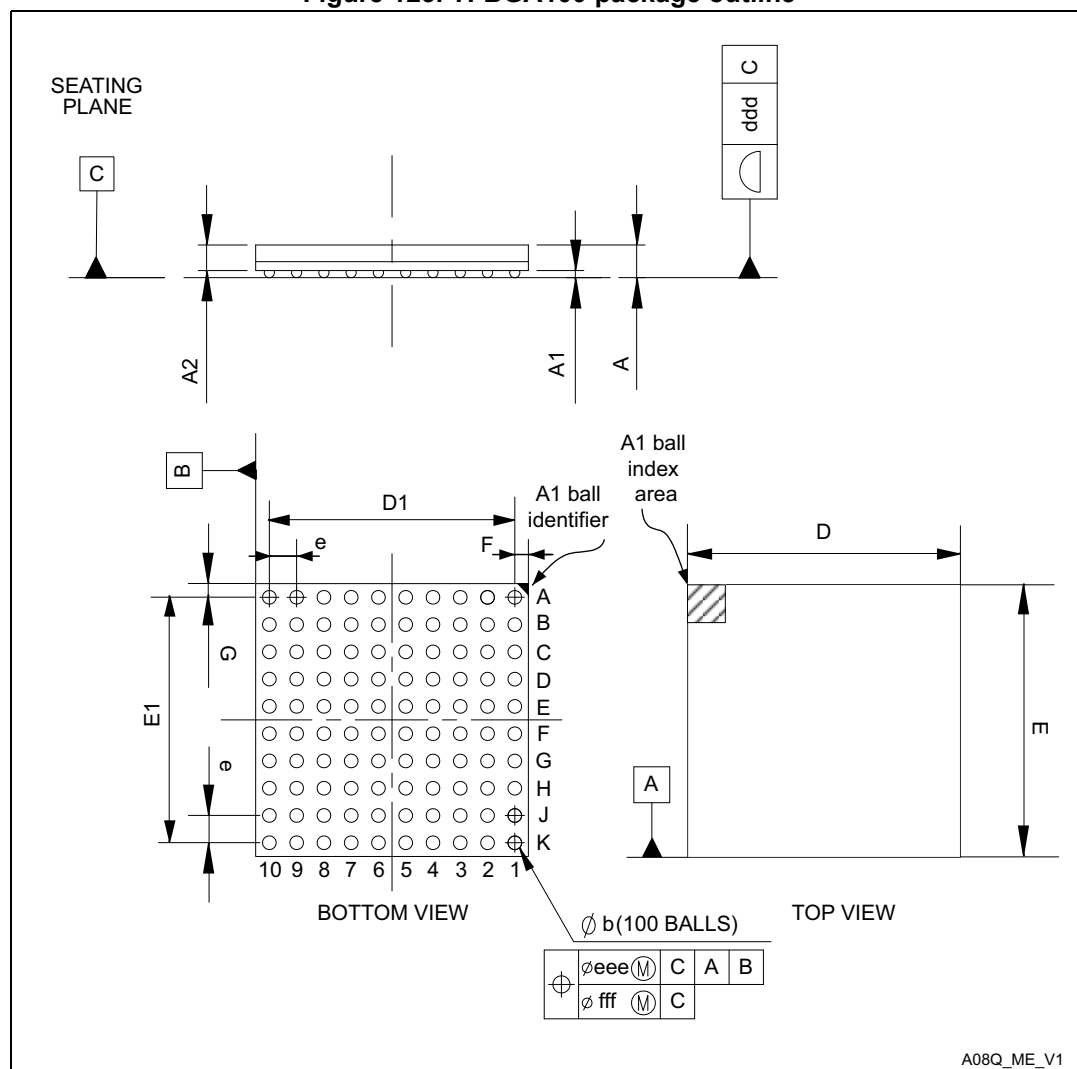


1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

## 8.2 TFBGA100 package information

TFBGA100 is a 100-ball, 8 x 8 mm, 0.8 mm pitch, thin fine-pitch ball grid array package.

Figure 123. TFBGA100 package outline



1. Drawing is not to scale.

Table 218. TFBGA100 package mechanical data

| Symbol | millimeters |       |       | inches <sup>(1)</sup> |        |        |
|--------|-------------|-------|-------|-----------------------|--------|--------|
|        | Min         | Typ   | Max   | Min                   | Typ    | Max    |
| A      | -           | -     | 1.100 | -                     | -      | 0.0433 |
| A1     | 0.150       | -     | -     | 0.0059                | -      | -      |
| A2     | -           | 0.760 | -     | -                     | 0.0299 | -      |
| b      | 0.350       | 0.400 | 0.450 | 0.0138                | 0.0157 | 0.0177 |
| D      | 7.850       | 8.000 | 8.150 | 0.3091                | 0.3150 | 0.3209 |

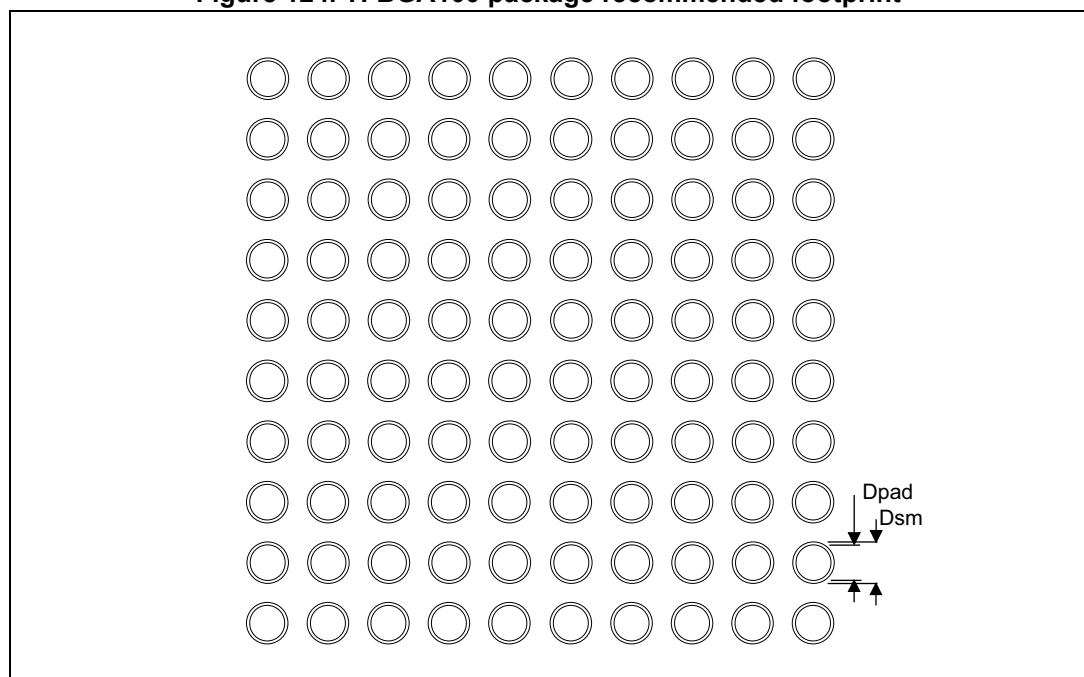


Table 218. TFBGA100 package mechanical data (continued)

| Symbol | millimeters |       |       | inches <sup>(1)</sup> |        |        |
|--------|-------------|-------|-------|-----------------------|--------|--------|
|        | Min         | Typ   | Max   | Min                   | Typ    | Max    |
| D1     | -           | 7.200 |       | -                     | 0.2835 | -      |
| E      | 7.850       | 8.000 | 8.150 | 0.3091                | 0.3150 | 0.3209 |
| E1     | -           | 7.200 | -     | -                     | 0.2835 | -      |
| e      | -           | 0.800 | -     | -                     | 0.0315 | -      |
| F      | -           | 0.400 | -     | -                     | 0.0157 | -      |
| G      | -           | 0.400 | -     | -                     | 0.0157 | -      |
| ddd    | -           | -     | 0.100 | -                     | -      | 0.0039 |
| eee    | -           | -     | 0.150 | -                     | -      | 0.0059 |
| fff    | -           | -     | 0.080 | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 124. TFBGA100 package recommended footprint



1. Dimensions are expressed in millimeters.

Table 219. TFBGA100 recommended PCB design rules (0.8 mm pitch BGA)

| Dimension | Recommended values                                              |
|-----------|-----------------------------------------------------------------|
| Pitch     | 0.8                                                             |
| Dpad      | 0.400 mm                                                        |
| Dsm       | 0.470 mm typ (depends on the soldermask registration tolerance) |

**Table 219. TFBGA100 recommended PCB design rules (0.8 mm pitch BGA) (contin-**

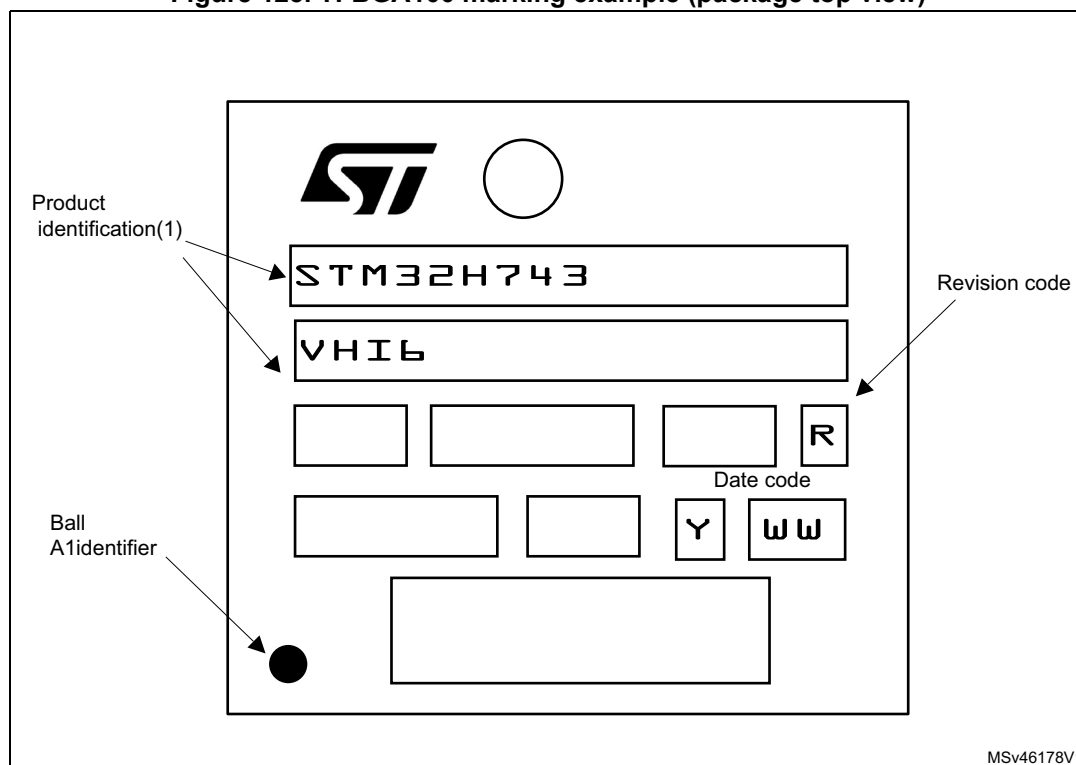
| Dimension         | Recommended values            |
|-------------------|-------------------------------|
| Stencil opening   | 0.400 mm                      |
| Stencil thickness | Between 0.100 mm and 0.125 mm |
| Pad trace width   | 0.120 mm                      |

**Device marking for TFBGA100**

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

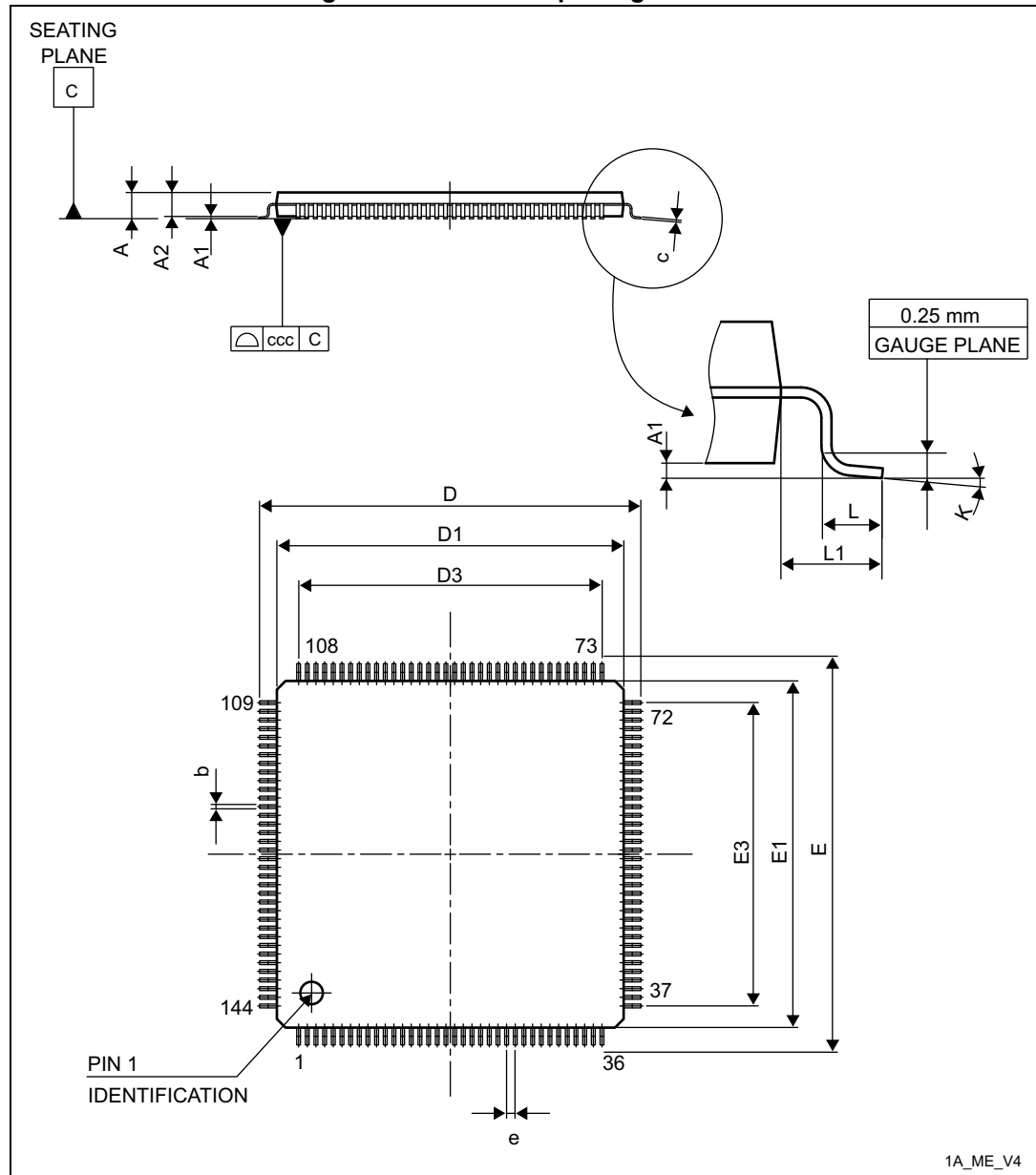
**Figure 125. TFBGA100 marking example (package top view)**

1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

### 8.3 LQFP144 package information

LQFP144 is a 144-pin, 20 x 20 mm low-profile quad flat package.

Figure 126. LQFP144 package outline



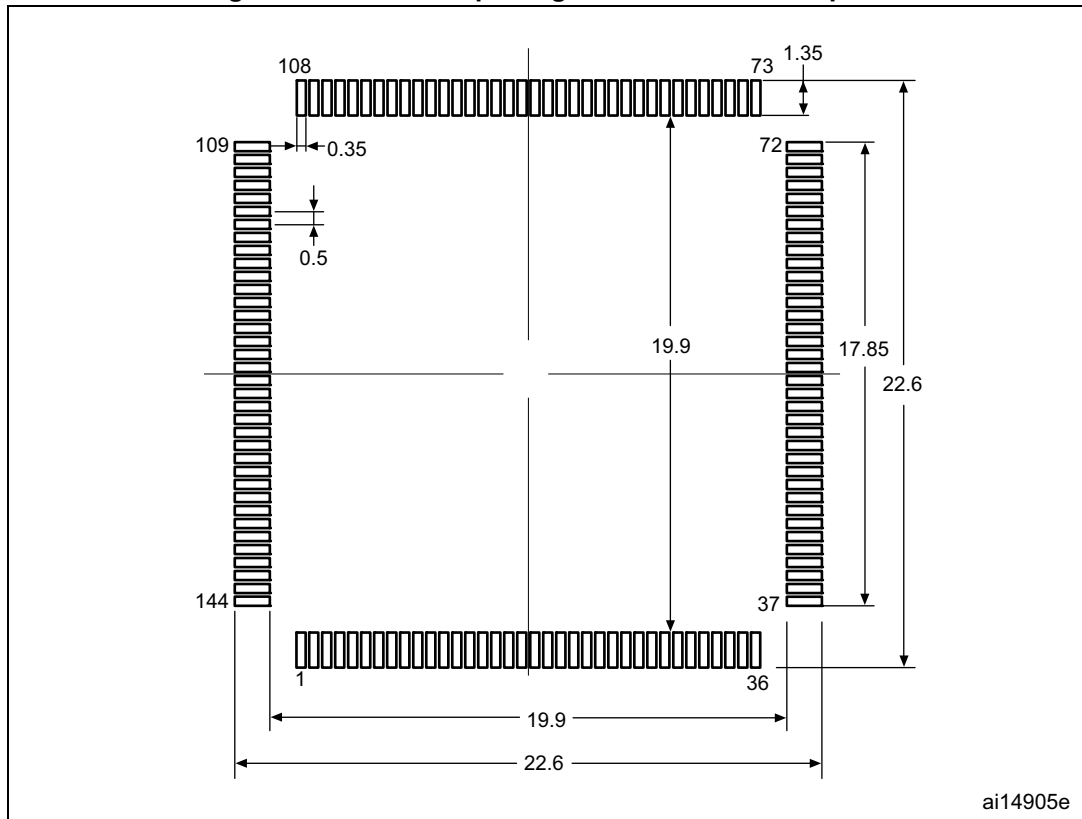
1. Drawing is not to scale.

Table 220. LQFP144 package mechanical data

| Symbol | millimeters |        |        | inches <sup>(1)</sup> |        |        |
|--------|-------------|--------|--------|-----------------------|--------|--------|
|        | Min         | Typ    | Max    | Min                   | Typ    | Max    |
| A      | -           | -      | 1.600  | -                     | -      | 0.0630 |
| A1     | 0.050       | -      | 0.150  | 0.0020                | -      | 0.0059 |
| A2     | 1.350       | 1.400  | 1.450  | 0.0531                | 0.0551 | 0.0571 |
| b      | 0.170       | 0.220  | 0.270  | 0.0067                | 0.0087 | 0.0106 |
| c      | 0.090       | -      | 0.200  | 0.0035                | -      | 0.0079 |
| D      | 21.800      | 22.000 | 22.200 | 0.8583                | 0.8661 | 0.8740 |
| D1     | 19.800      | 20.000 | 20.200 | 0.7795                | 0.7874 | 0.7953 |
| D3     | -           | 17.500 | -      | -                     | 0.6890 | -      |
| E      | 21.800      | 22.000 | 22.200 | 0.8583                | 0.8661 | 0.8740 |
| E1     | 19.800      | 20.000 | 20.200 | 0.7795                | 0.7874 | 0.7953 |
| E3     | -           | 17.500 | -      | -                     | 0.6890 | -      |
| e      | -           | 0.500  | -      | -                     | 0.0197 | -      |
| L      | 0.450       | 0.600  | 0.750  | 0.0177                | 0.0236 | 0.0295 |
| L1     | -           | 1.000  | -      | -                     | 0.0394 | -      |
| k      | 0°          | 3.5°   | 7°     | 0°                    | 3.5°   | 7°     |
| ccc    | -           | -      | 0.080  | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 127. LQFP144 package recommended footprint



1. Dimensions are expressed in millimeters.

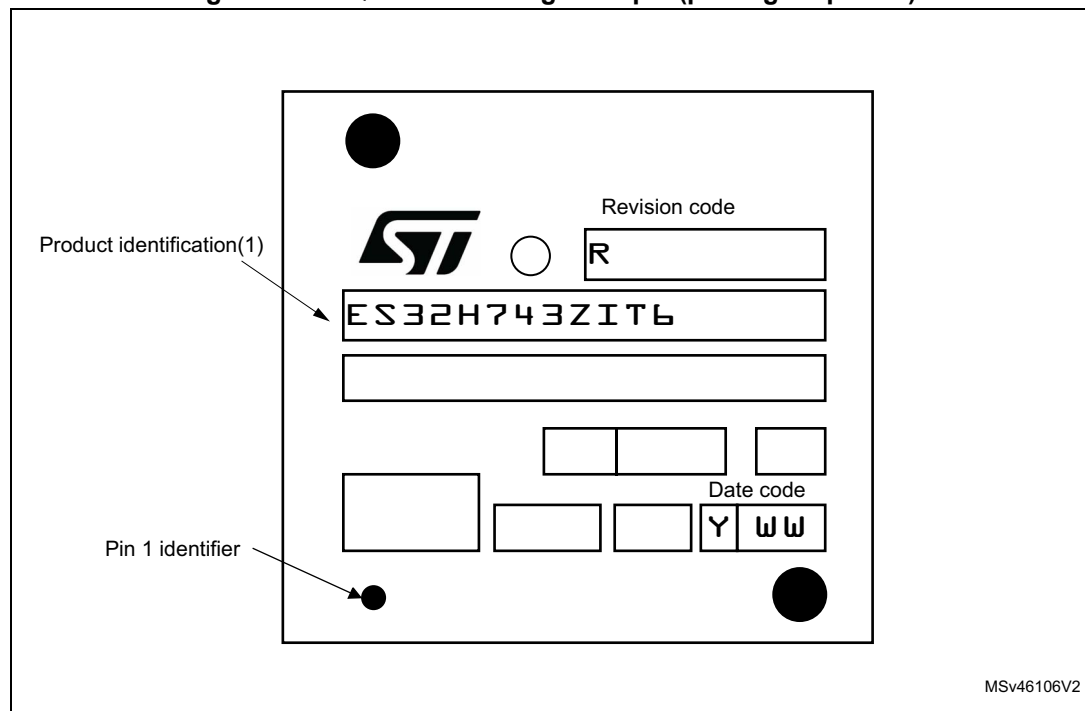
### Device marking for LQFP144

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 128. LQFP144 marking example (package top view)**

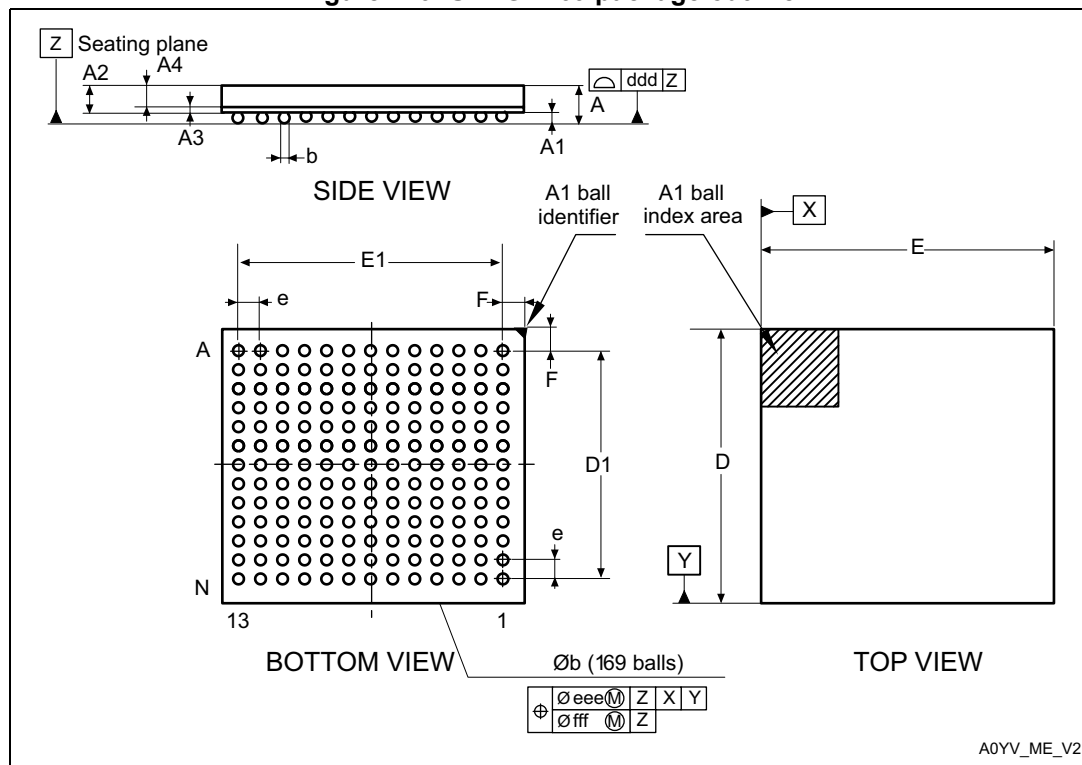


1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

## 8.4 UFBGA169 package information

UFBGA169 is a 169-pin, 7 x 7 mm, 0.50 mm pitch, ultra fine pitch ball grid array package.

Figure 129. UFBGA169 package outline



1. Drawing is not in scale.

Table 221. UFBGA169 package mechanical data

| Symbol | millimeters |       |       | inches <sup>(1)</sup> |        |        |
|--------|-------------|-------|-------|-----------------------|--------|--------|
|        | Min.        | Typ.  | Max.  | Min.                  | Typ.   | Max.   |
| A      | 0.460       | 0.530 | 0.600 | 0.0181                | 0.0209 | 0.0236 |
| A1     | 0.050       | 0.080 | 0.110 | 0.0020                | 0.0031 | 0.0043 |
| A2     | 0.400       | 0.450 | 0.500 | 0.0157                | 0.0177 | 0.0197 |
| A3     | -           | 0.130 | -     | -                     | 0.0051 | -      |
| A4     | 0.270       | 0.320 | 0.370 | 0.0106                | 0.0126 | 0.0146 |
| b      | 0.230       | 0.280 | 0.330 | 0.0091                | 0.0110 | 0.0130 |
| D      | 6.950       | 7.000 | 7.050 | 0.2736                | 0.2756 | 0.2776 |
| D1     | 5.950       | 6.000 | 6.050 | 0.2343                | 0.2362 | 0.2382 |
| E      | 6.950       | 7.000 | 7.050 | 0.2736                | 0.2756 | 0.2776 |
| E1     | 5.950       | 6.000 | 6.050 | 0.2343                | 0.2362 | 0.2382 |
| e      | -           | 0.500 | -     | -                     | 0.0197 | -      |
| F      | 0.450       | 0.500 | 0.550 | 0.0177                | 0.0197 | 0.0217 |

Table 221. UFBGA169 package mechanical data (continued)

| Symbol | millimeters |      |       | inches <sup>(1)</sup> |      |        |
|--------|-------------|------|-------|-----------------------|------|--------|
|        | Min.        | Typ. | Max.  | Min.                  | Typ. | Max.   |
| ddd    | -           | -    | 0.100 | -                     | -    | 0.0039 |
| eee    | -           | -    | 0.150 | -                     | -    | 0.0059 |
| fff    | -           | -    | 0.050 | -                     | -    | 0.0020 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

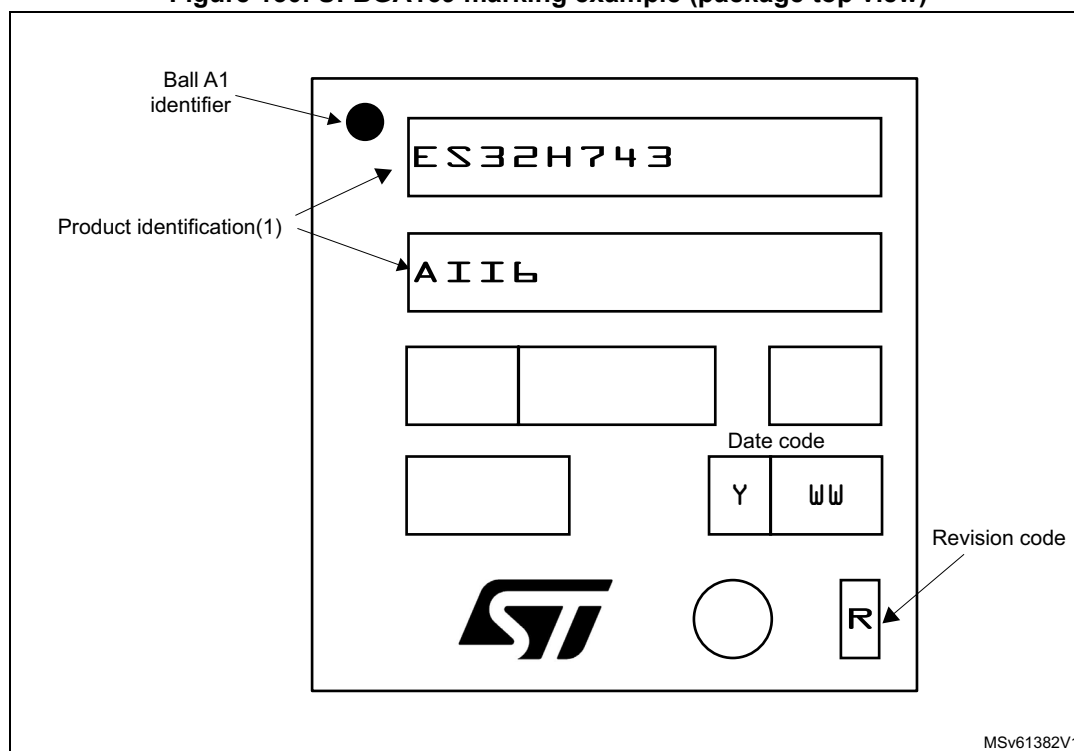
### Device marking for UFBGA169

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

Figure 130. UFBGA169 marking example (package top view)



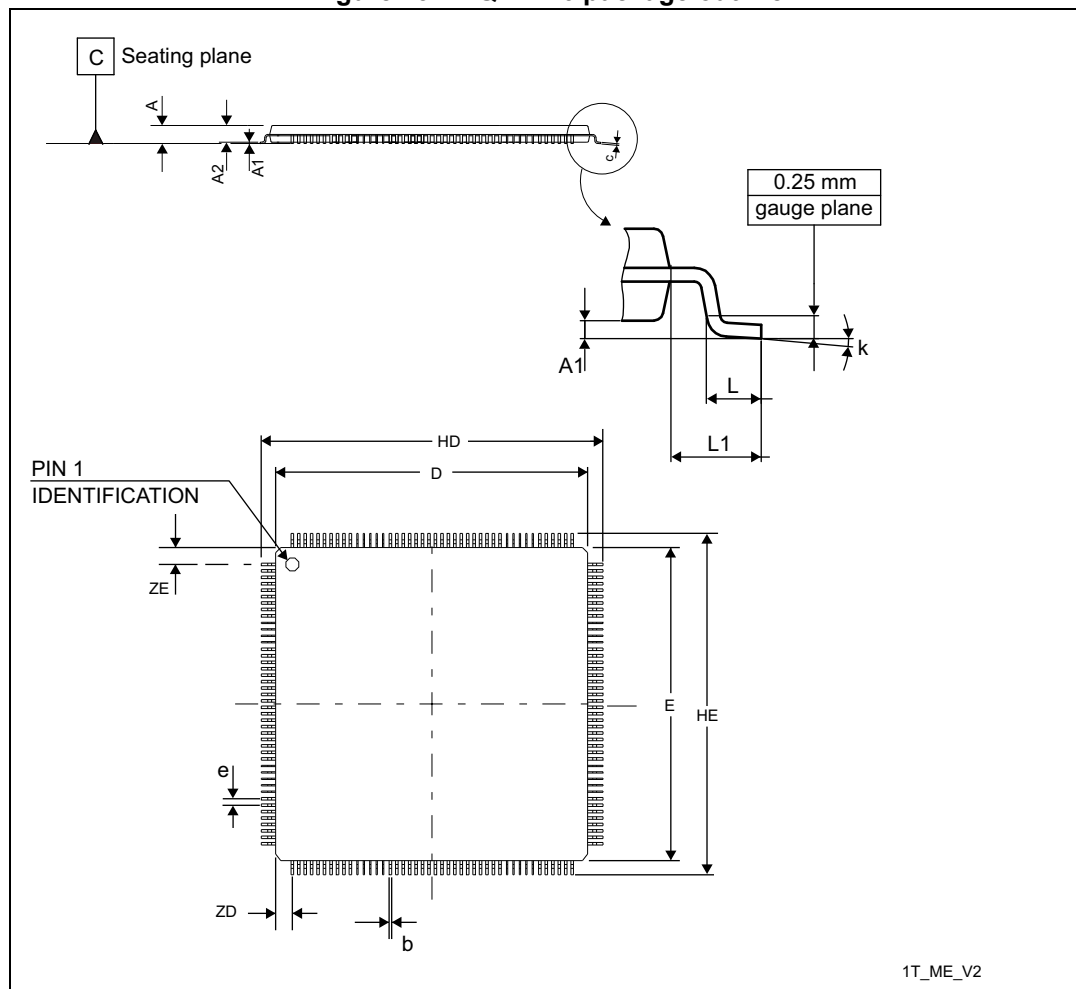
1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.



## 8.5 LQFP176 package information

LQFP176 is a 176-pin, 24 x 24 mm low profile quad flat package.

Figure 131. LQFP176 package outline



1. Drawing is not to scale.

Table 222. LQFP176 package mechanical data

| Ref. | Dimensions  |      |       |                       |      |        |
|------|-------------|------|-------|-----------------------|------|--------|
|      | Millimeters |      |       | Inches <sup>(1)</sup> |      |        |
|      | Min.        | Typ. | Max.  | Min.                  | Typ. | Max.   |
| A    | -           | -    | 1.600 | -                     | -    | 0.0630 |
| A1   | 0.050       | -    | 0.150 | 0.0020                | -    | 0.0059 |
| A2   | 1.350       | -    | 1.450 | 0.0531                | -    | 0.0571 |
| b    | 0.170       | -    | 0.270 | 0.0067                | -    | 0.0106 |
| c    | 0.090       | -    | 0.200 | 0.0035                | -    | 0.0079 |

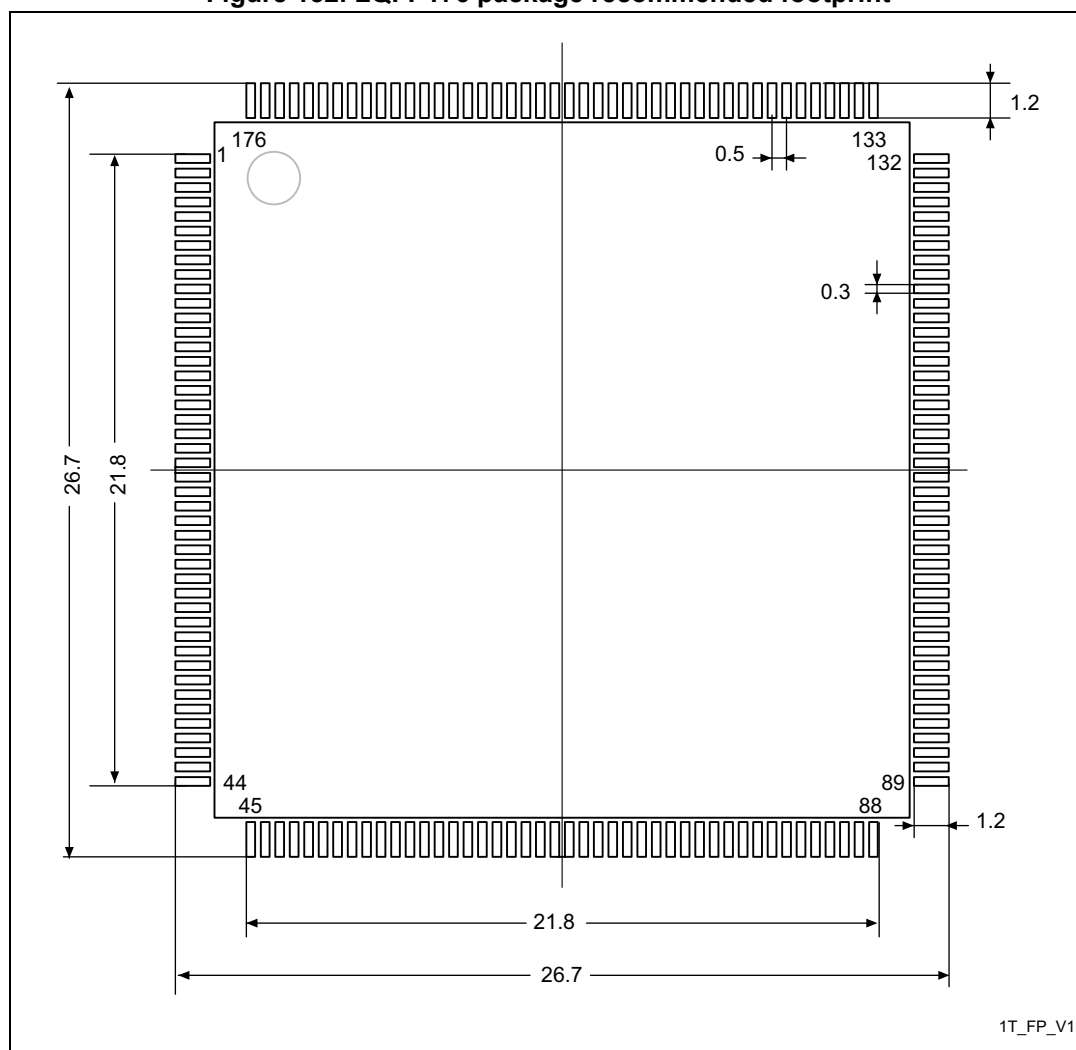
Table 222. LQFP176 package mechanical data (continued)

| Ref.             | Dimensions  |       |        |                       |        |        |
|------------------|-------------|-------|--------|-----------------------|--------|--------|
|                  | Millimeters |       |        | Inches <sup>(1)</sup> |        |        |
|                  | Min.        | Typ.  | Max.   | Min.                  | Typ.   | Max.   |
| D                | 23.900      | -     | 24.100 | 0.9409                | -      | 0.9488 |
| HD               | 25.900      | -     | 26.100 | 1.0197                | -      | 1.0276 |
| ZD               | -           | 1.250 | -      | -                     | 0.0492 | -      |
| E                | 23.900      | -     | 24.100 | 0.9409                | -      | 0.9488 |
| HE               | 25.900      | -     | 26.100 | 1.0197                | -      | 1.0276 |
| ZE               | -           | 1.250 | -      | -                     | 0.0492 | -      |
| e                | -           | 0.500 | -      | -                     | 0.0197 | -      |
| L <sup>(2)</sup> | 0.450       | -     | 0.750  | 0.0177                | -      | 0.0295 |
| L1               | -           | 1.000 | -      | -                     | 0.0394 | -      |
| k                | 0°          | -     | 7°     | 0°                    | -      | 7°     |
| ccc              | -           | -     | 0.080  | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

2. L dimension is measured at gauge plane at 0.25 mm above the seating plane.

Figure 132. LQFP176 package recommended footprint



1. Dimensions are expressed in millimeters.

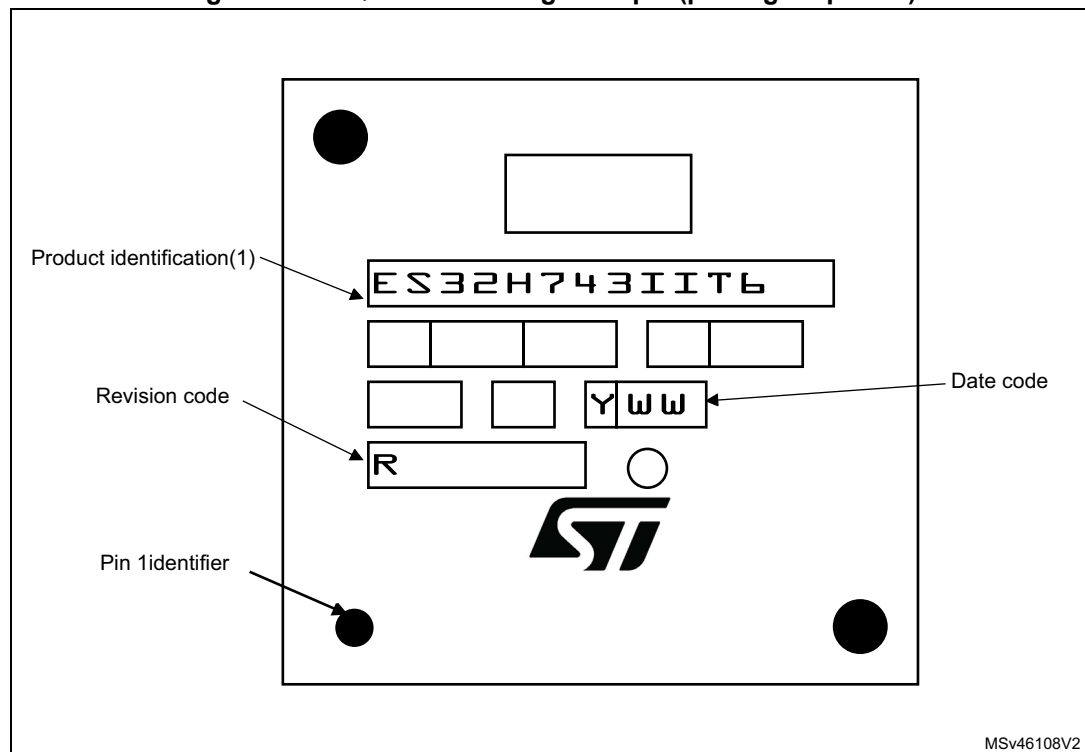
### Device marking for LQFP176

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 133. LQFP176 marking example (package top view)**

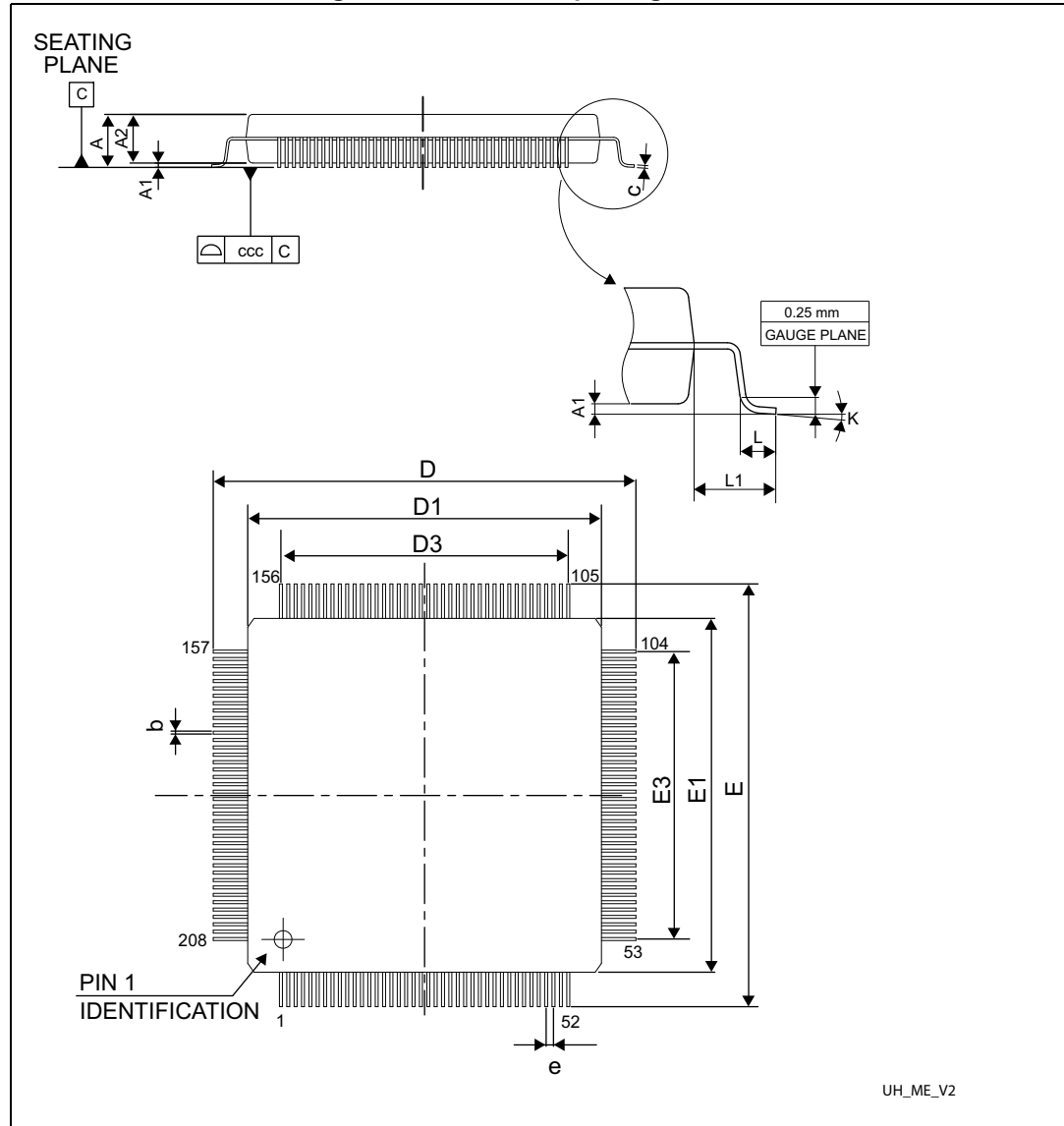


1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

## 8.6 LQFP208 package information

LQFP208 is a 208-pin, 28 x 28 mm low-profile quad flat package.

Figure 134. LQFP208 package outline



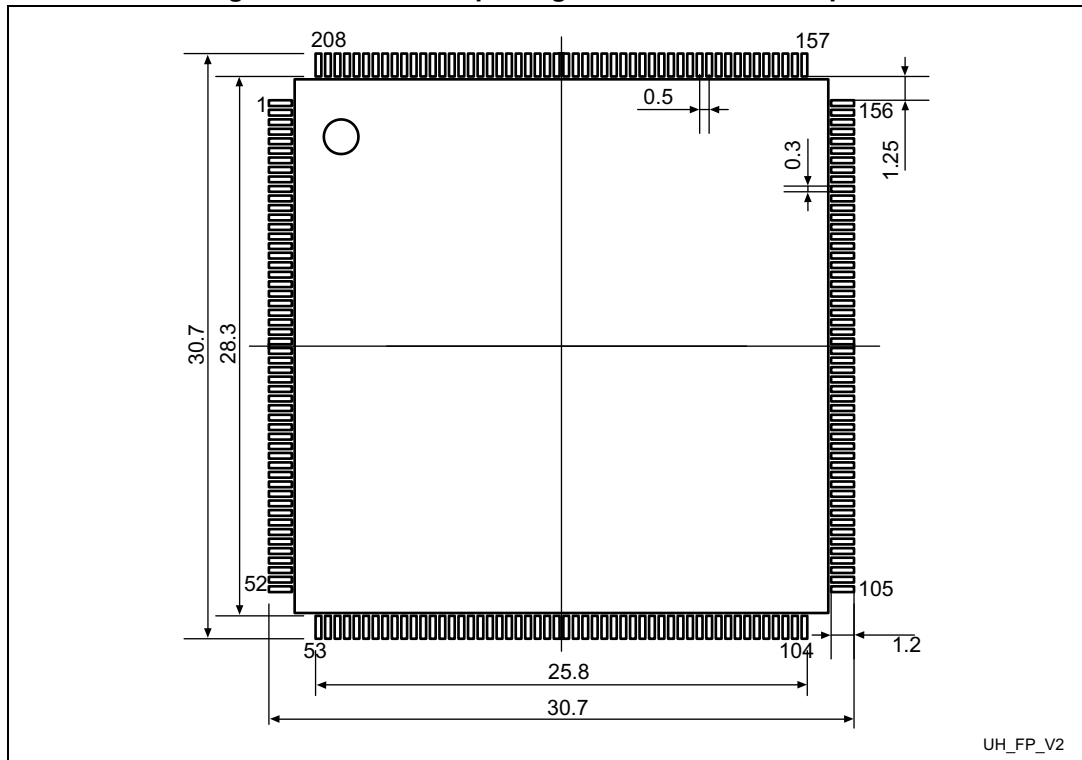
1. Drawing is not to scale.

Table 223. LQFP208 package mechanical data

| Symbol | millimeters |        |        | inches <sup>(1)</sup> |        |        |
|--------|-------------|--------|--------|-----------------------|--------|--------|
|        | Min         | Typ    | Max    | Min                   | Typ    | Max    |
| A      | -           | -      | 1.600  | -                     | -      | 0.0630 |
| A1     | 0.050       | -      | 0.150  | 0.0020                | -      | 0.0059 |
| A2     | 1.350       | 1.400  | 1.450  | 0.0531                | 0.0551 | 0.0571 |
| b      | 0.170       | 0.220  | 0.270  | 0.0067                | 0.0087 | 0.0106 |
| c      | 0.090       | -      | 0.200  | 0.0035                | -      | 0.0079 |
| D      | 29.800      | 30.000 | 30.200 | 1.1811                | 1.1732 | 1.1890 |
| D1     | 27.800      | 28.000 | 28.200 | 1.1024                | 1.0945 | 1.1102 |
| D3     | -           | 25.500 | -      | -                     | 1.0039 | -      |
| E      | 29.800      | 30.000 | 30.200 | 1.1811                | 1.1732 | 1.1890 |
| E1     | 27.800      | 28.000 | 28.200 | 1.1024                | 1.0945 | 1.1102 |
| E3     | -           | 25.500 | -      | -                     | 1.0039 | -      |
| e      | -           | 0.500  | -      | -                     | 0.0197 | -      |
| L      | 0.450       | 0.600  | 0.750  | 0.0177                | 0.0236 | 0.0295 |
| L1     | -           | 1.000  | -      | -                     | 0.0394 | -      |
| k      | 0°          | 3.5°   | 7°     | 0°                    | 3.5°   | 7°     |
| ccc    | -           | -      | 0.080  | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 135. LQFP208 package recommended footprint



1. Dimensions are expressed in millimeters.

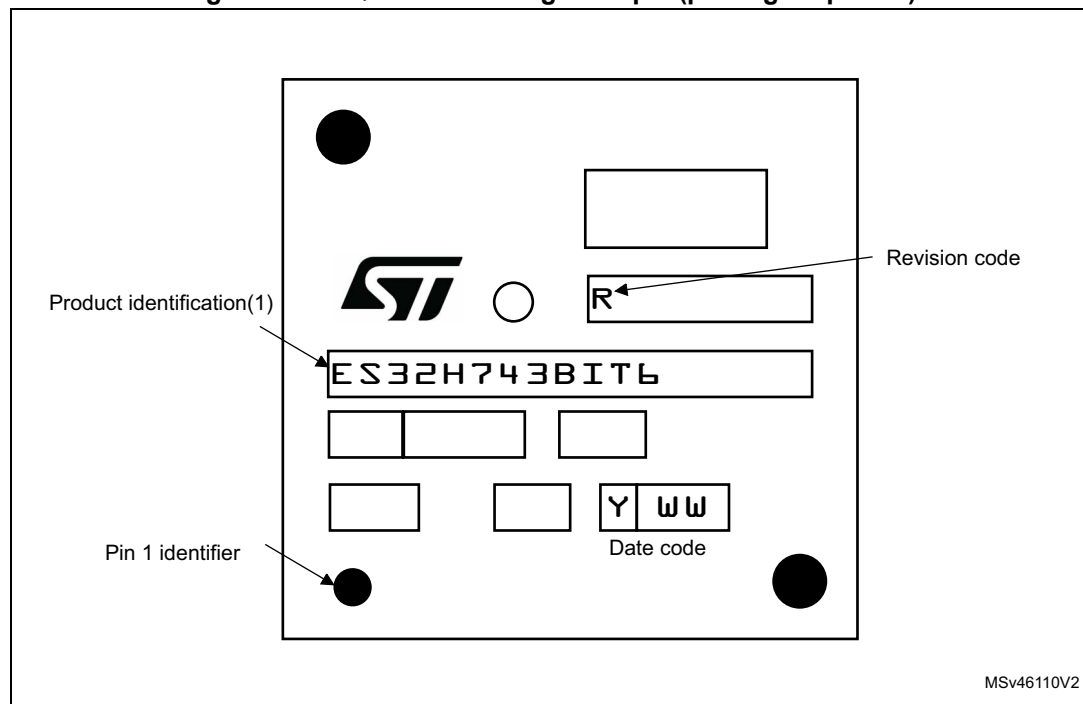
### Device marking for LQFP208

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 136. LQFP208 marking example (package top view)**



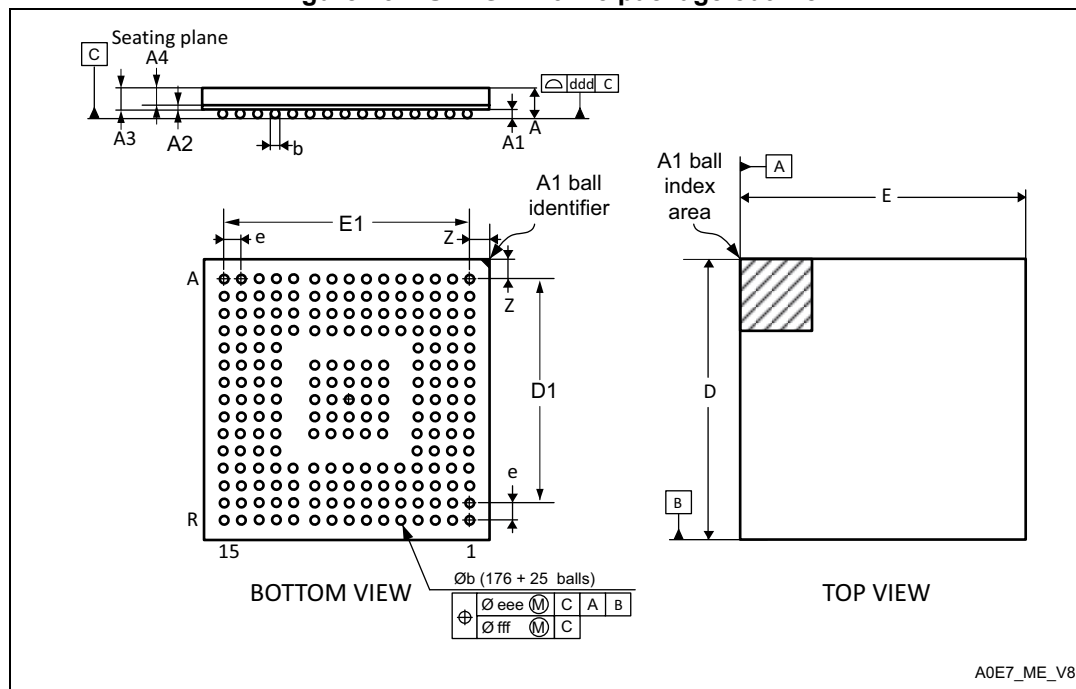
1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.



## 8.7 UFBGA176+25 package information

UFBGA176+25 is a 201-ball, 10 x 10 mm, 0.65 mm pitch, ultra fine pitch ball grid array package.

Figure 137. UFBGA176+25 package outline



1. Drawing is not to scale.

Table 224. UFBGA176+25 package mechanical data

| Symbol | millimeters |        |        | inches <sup>(1)</sup> |        |        |
|--------|-------------|--------|--------|-----------------------|--------|--------|
|        | Min.        | Typ.   | Max.   | Min.                  | Typ.   | Max.   |
| A      | -           | -      | 0.600  | -                     | -      | 0.0236 |
| A1     | -           | -      | 0.110  | -                     | -      | 0.0043 |
| A2     | -           | 0.130  | -      | -                     | 0.0051 | -      |
| A3     | -           | 0.450  | -      | -                     | 0.0177 | -      |
| A4     | -           | 0.320  | -      | -                     | 0.0126 | -      |
| b      | 0.240       | 0.290  | 0.340  | 0.0094                | 0.0114 | 0.0134 |
| D      | 9.850       | 10.000 | 10.150 | 0.3878                | 0.3937 | 0.3996 |
| D1     | -           | 9.100  | -      | -                     | 0.3583 | -      |
| E      | 9.850       | 10.000 | 10.150 | 0.3878                | 0.3937 | 0.3996 |
| E1     | -           | 9.100  | -      | -                     | 0.3583 | -      |
| e      | -           | 0.650  | -      | -                     | 0.0256 | -      |
| Z      | -           | 0.450  | -      | -                     | 0.0177 | -      |
| ddd    | -           | -      | 0.080  | -                     | -      | 0.0031 |

Table 224. UFBGA176+25 package mechanical data (continued)

| Symbol | millimeters |      |       | inches <sup>(1)</sup> |      |        |
|--------|-------------|------|-------|-----------------------|------|--------|
|        | Min.        | Typ. | Max.  | Min.                  | Typ. | Max.   |
| eee    | -           | -    | 0.150 | -                     | -    | 0.0059 |
| fff    | -           | -    | 0.050 | -                     | -    | 0.0020 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 138. UFBGA176+25 package recommended footprint

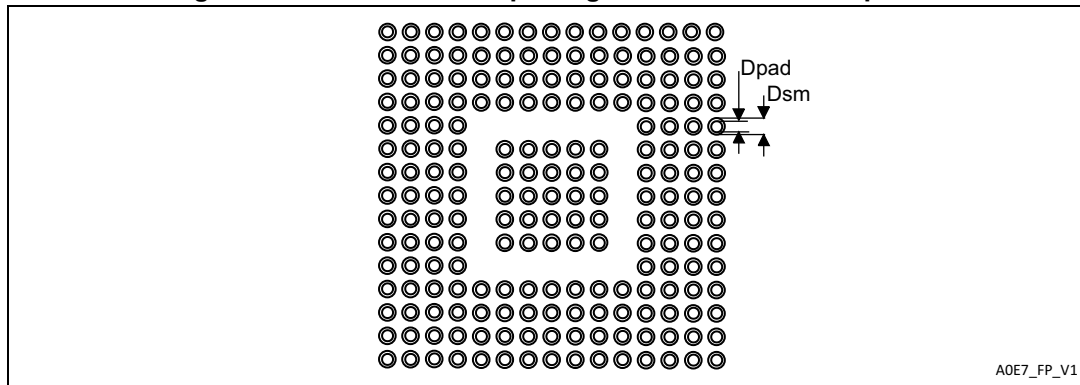


Table 225. UFBGA176+25 recommended PCB design rules (0.65 mm pitch BGA)

| Dimension         | Recommended values                                               |
|-------------------|------------------------------------------------------------------|
| Pitch             | 0.65 mm                                                          |
| Dpad              | 0.300 mm                                                         |
| Dsm               | 0.400 mm typ. (depends on the soldermask registration tolerance) |
| Stencil opening   | 0.300 mm                                                         |
| Stencil thickness | Between 0.100 mm and 0.125 mm                                    |
| Pad trace width   | 0.100 mm                                                         |

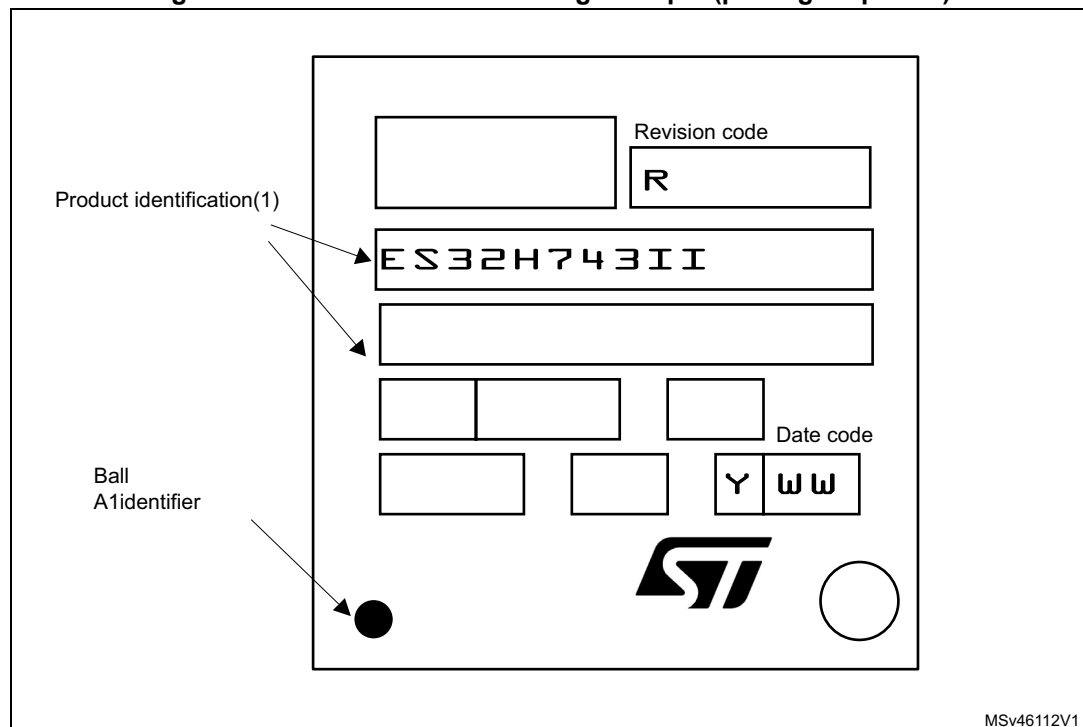
**Device marking for UFBGA176+25**

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 139. UFBGA176+25 marking example (package top view)**

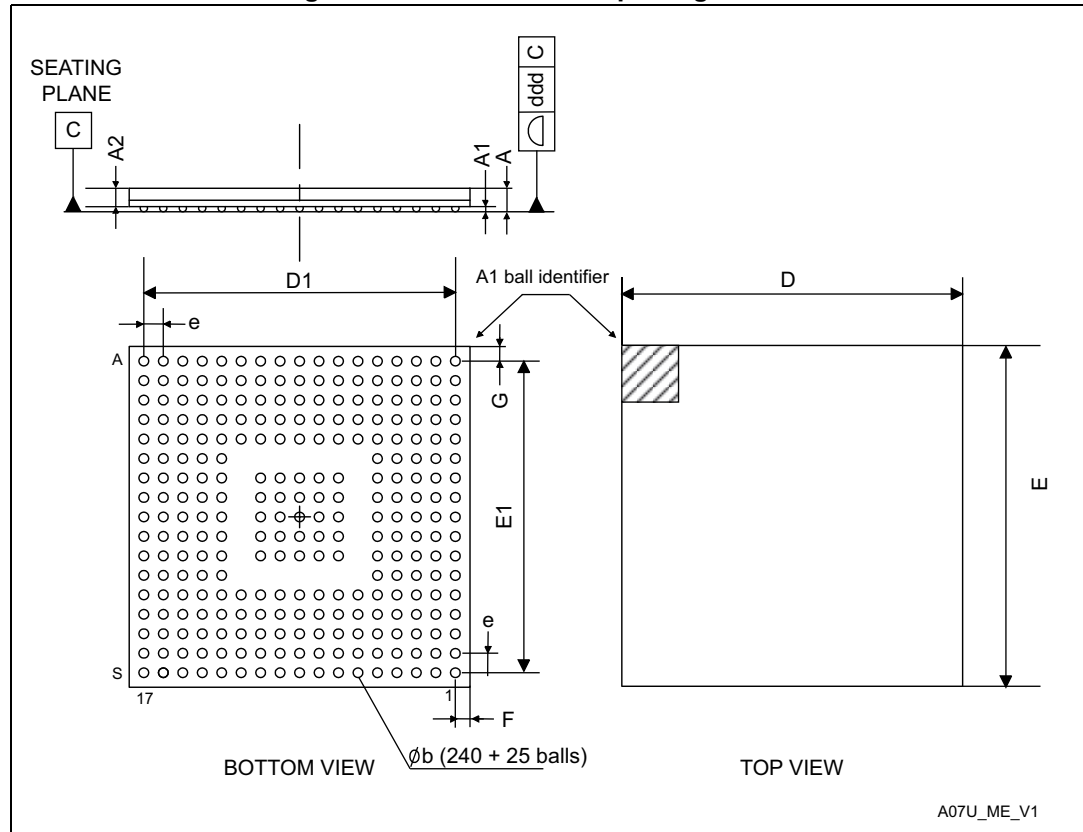


1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

## 8.8 TFBGA240+25 package information

TFBGA240+25 is a 265 ball, 14x14 mm, 0.8 mm pitch, fine pitch ball grid array package.

Figure 140. TFBGA240+25 package outline



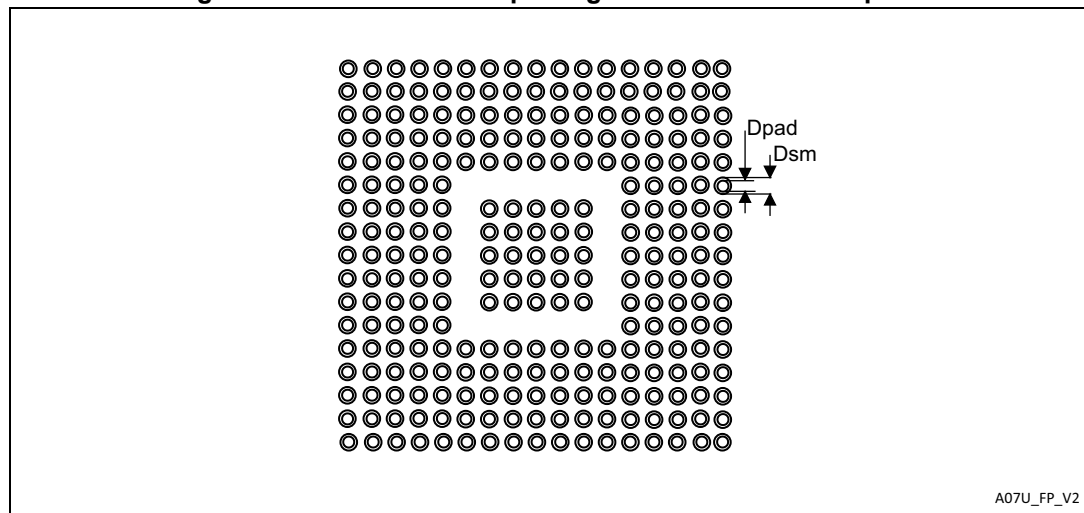
1. Dimensions are expressed in millimeters.

Table 226. TFBG240+25 ball package mechanical data

| Symbol | millimeters |        |        | inches <sup>(1)</sup> |        |        |
|--------|-------------|--------|--------|-----------------------|--------|--------|
|        | Min         | Typ    | Max    | Min                   | Typ    | Max    |
| A      | -           | -      | 1.100  | -                     | -      | 0.0433 |
| A1     | 0.150       | -      | -      | 0.0059                | -      | -      |
| A2     | -           | 0.760  | -      | -                     | 0.0299 | -      |
| b      | 0.350       | 0.400  | 0.450  | 0.0138                | 0.0157 | 0.0177 |
| D      | 13.850      | 14.000 | 14.150 | 0.5453                | 0.5512 | 0.5571 |
| D1     | -           | 12.800 | -      | -                     | 0.5039 | -      |
| E      | 13.850      | 14.000 | 14.150 | 0.5453                | 0.5512 | 0.5571 |
| E1     | -           | 12.800 | -      | -                     | 0.5039 | -      |
| e      | -           | 0.800  | -      | -                     | 0.0315 | -      |
| F      | -           | 0.600  | -      | -                     | 0.0236 | -      |
| G      | -           | 0.600  | -      | -                     | 0.0236 | -      |
| ddd    | -           | -      | 0.100  | -                     | -      | 0.0039 |
| eee    | -           | -      | 0.150  | -                     | -      | 0.0059 |
| fff    | -           | -      | 0.080  | -                     | -      | 0.0031 |

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 141. TFBGA240+25 package recommended footprint



1. Dimensions are expressed in millimeters.

**Table 227. TFBGA240+25 recommended PCB design rules (0.8 mm pitch)**

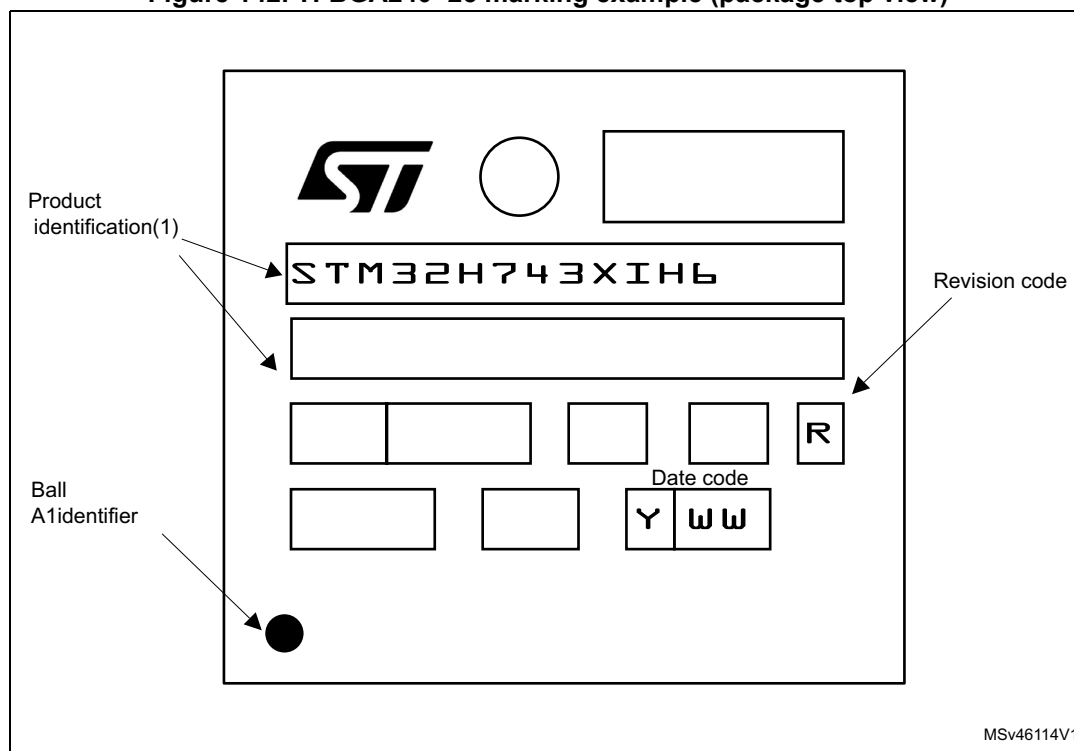
| Dimension         | Recommended values                                               |
|-------------------|------------------------------------------------------------------|
| Pitch             | 0.8 mm                                                           |
| Dpad              | 0.225 mm                                                         |
| Dsm               | 0.290 mm typ. (depends on the soldermask registration tolerance) |
| Stencil opening   | 0.250 mm                                                         |
| Stencil thickness | 0.100 mm                                                         |

**Device marking for TFBGA240+25**

The following figure gives an example of topside marking versus pin 1 position identifier location.

The printed markings may differ depending on the supply chain.

Other optional marking or inset/upset marks, which depend on supply chain operations, are not indicated below.

**Figure 142. TFBGA240+25 marking example (package top view)**

1. Parts marked as “ES”, “E” or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST’s Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

## 8.9 Thermal characteristics

The maximum chip-junction temperature,  $T_J \text{ max}$ , in degrees Celsius, may be calculated using the following equation:

$$T_J \text{ max} = T_A \text{ max} + (P_D \text{ max} \times \Theta_{JA})$$

Where:

- $T_A \text{ max}$  is the maximum ambient temperature in °C,
- $\Theta_{JA}$  is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D \text{ max}$  is the sum of  $P_{INT} \text{ max}$  and  $P_{I/O} \text{ max}$  ( $P_D \text{ max} = P_{INT} \text{ max} + P_{I/O} \text{ max}$ ),
- $P_{INT} \text{ max}$  is the product of  $I_{DD}$  and  $V_{DD}$ , expressed in Watts. This is the maximum chip internal power.

$P_{I/O} \text{ max}$  represents the maximum power dissipation on output pins where:

$$P_{I/O} \text{ max} = \Sigma (V_{OL} \times I_{OL}) + \Sigma (V_{DD} - V_{OH}) \times I_{OH},$$

taking into account the actual  $V_{OL} / I_{OL}$  and  $V_{OH} / I_{OH}$  of the I/Os at low and high level in the application.

**Table 228. Thermal characteristics<sup>(1)</sup>**

| Symbol        | Definition                          | Parameter                                                                             | Value | Unit |
|---------------|-------------------------------------|---------------------------------------------------------------------------------------|-------|------|
| $\Theta_{JA}$ | Thermal resistance junction-ambient | <b>Thermal resistance junction-ambient</b><br>LQFP100 - 14 x 14 mm /0.5 mm pitch      | 45.0  | °C/W |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>TFBGA100 - 8 x 8 mm /0.8 mm pitch       | 39.3  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>LQFP144 - 20 x 20 mm /0.5 mm pitch      | 43.7  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>UFBGA169 - 7 x 7 mm /0.5 mm pitch       | 37.7  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>LQFP176 - 24 x 24 mm /0.5 mm pitch      | 43.0  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>LQFP208 - 28 x 28 mm /0.5 mm pitch      | 42.4  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>UFBGA176+25 - 10 x 10 mm /0.65 mm pitch | 37.4  |      |
|               |                                     | <b>Thermal resistance junction-ambient</b><br>TFBGA240+25 - 14 x 14 mm / 0.8 mm pitch | 36.6  |      |

Table 228. Thermal characteristics<sup>(1)</sup> (continued)

| Symbol        | Definition                        | Parameter                                                                      | Value | Unit |
|---------------|-----------------------------------|--------------------------------------------------------------------------------|-------|------|
| $\Theta_{JC}$ | Thermal resistance junction-case  | Thermal resistance junction-ambient<br>LQFP100 - 14 x 14 mm /0.5 mm pitch      | 11.5  | °C/W |
|               |                                   | Thermal resistance junction-ambient<br>TFBGA100 - 8 x 8 mm /0.8 mm pitch       | 17.1  |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP144 - 20 x 20 mm /0.5 mm pitch      | 11.3  |      |
|               |                                   | Thermal resistance junction-ambient<br>UFBGA169 - 7 x 7 mm /0.5 mm pitch       | TBD   |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP176 - 24 x 24 mm /0.5 mm pitch      | 11.2  |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP208 - 28 x 28 mm /0.5 mm pitch      | 11.1  |      |
|               |                                   | Thermal resistance junction-ambient<br>UFBGA176+25 - 10 x 10 mm /0.65 mm pitch | 23.9  |      |
|               |                                   | Thermal resistance junction-ambient<br>TFBGA240+25 - 14 x 14 mm / 0.8 mm pitch | 7.4   |      |
| $\Theta_{JB}$ | Thermal resistance junction-board | Thermal resistance junction-ambient<br>LQFP100 - 14 x 14 mm /0.5 mm pitch      | 36.3  | °C/W |
|               |                                   | Thermal resistance junction-ambient<br>TFBGA100 - 8 x 8 mm /0.8 mm pitch       | 21.1  |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP144 - 20 x 20 mm /0.5 mm pitch      | 38.3  |      |
|               |                                   | Thermal resistance junction-ambient<br>UFBGA169 - 7 x 7 mm /0.5 mm pitch       | TBD   |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP176 - 24 x 24 mm /0.5 mm pitch      | 39.4  |      |
|               |                                   | Thermal resistance junction-ambient<br>LQFP208 - 28 x 28 mm /0.5 mm pitch      | 40.3  |      |
|               |                                   | Thermal resistance junction-ambient<br>UFBGA176+25 - 10 x 10 mm /0.65 mm pitch | 19.3  |      |
|               |                                   | Thermal resistance junction-ambient<br>TFBGA240+25 - 14 x 14 mm / 0.8 mm pitch | 24.3  |      |

1. TBD stands for "to be defined".

### 8.9.1 Reference document

- JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions - Natural Convection (Still Air). Available from [www.jedec.org](http://www.jedec.org).
- For information on thermal management, refer to application note "Thermal management guidelines for STM32 32-bit Arm Cortex MCUs applications" (AN5036) available from [www.st.com](http://www.st.com).



# 9 Ordering information

|                                                |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
|------------------------------------------------|-------|---|-----|---|---|---|---|----|--|--|--|--|--|--|--|--|
| Example:                                       | STM32 | H | 743 | X | I | T | 6 | TR |  |  |  |  |  |  |  |  |
| Device family                                  |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| STM32 = Arm-based 32-bit microcontroller       |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Product type                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| H = High performance                           |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Device subfamily                               |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| 743 = STM32H7x3                                |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| 742 = STM32H7x2                                |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Pin count                                      |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| V = 100 pins                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Z = 144 pins                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| A = 169 pins                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| I = 176 pins/balls                             |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| B = 208 pins                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| X = 240 balls                                  |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Flash memory size                              |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| G = 1 Mbytes                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| I = 2 Mbytes                                   |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Package                                        |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| T = LQFP ECOPACK®2                             |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| K = UFBGA pitch 0.65 mm ECOPACK®2              |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| I = UFBGA pitch 0.5 mm ECOPACK®2               |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| H = TFBGA ECOPACK®2                            |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Temperature range                              |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| 6 = Industrial temperature range, –40 to 85 °C |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| Packing                                        |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| TR = tape and reel                             |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |
| No character = tray or tube                    |       |   |     |   |   |   |   |    |  |  |  |  |  |  |  |  |

For a list of available options (speed, package, etc.) or for further information on any aspect of this device, please contact your nearest ST sales office.

## 10 Revision history

**Table 229. Document revision history**

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22-Jun-2017 | 1        | Initial release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 27-Sep-2017 | 2        | <p>Updated list of features. Changed datasheet status to “production data”.</p> <p>Added UFBGA169 and TFBGA100 packages and well as notes related their status on cover page and in <a href="#">Table 2: STM32H742xI/G and STM32H743xI/G features and peripheral counts</a>. Differentiated number of GPIOs for each package in <a href="#">Table 2: STM32H742xI/G and STM32H743xI/G features and peripheral counts</a>.</p> <p>Updated <a href="#">Error code correction (ECC)</a> in <a href="#">Section 3.3.2: Embedded SRAM</a>. Change PWR_CR3 into PWR_D3CR in <a href="#">Section 3.5.1: Power supply scheme</a>. Updated <a href="#">Section 3.12: Nested vectored interrupt controller (NVIC)</a>.</p> <p>Added ADC sampling rate values in <a href="#">Section 3.17: Analog-to-digital converters (ADCs)</a>.</p> <p>Added <a href="#">Table 4: DFSDM implementation</a> in <a href="#">Section 3.23: Digital filter for sigma-delta modulators (DFSDM)</a></p> <p>Changed PC2/3 to PC2/3_C and VDD33USB to VDD in <a href="#">Figure 5: LQFP100 pinout</a>.</p> <p>Changed PC2/3 to PC2/3_C in <a href="#">Figure 7: LQFP144 pinout</a>. Changed PC2/3 to PC2/3_C in <a href="#">Figure 9: LQFP176 pinout</a>. Changed PC2/3 to PC2/3_C in <a href="#">Figure 11: LQFP208 pinout</a>.</p> <p><a href="#">Table 9: Pin/ball definition</a>:</p> <ul style="list-style-type: none"> <li>– Modified PA7, PC4, PC5, PB1, PG1, PE7, PE8 and PE9 I/O structure</li> <li>– TFBGA240 +25: removed duplicate occurrence of F1, F2 and P17 pin; added notes related to F1, F2, G2 pin connection; added note on E1, L16, L17, M16, M17, K16, K17, N17.</li> <li>– UFBGA176+25: changed G10 pin name to VSS.</li> <li>– Added note to VREF+ pin.</li> </ul> <p>Added current consumption corresponding to 125 °C ambient temperature in <a href="#">Section 6.3.6: Supply current characteristics</a>. Removed CRYP peripheral from <a href="#">Table 39: Peripheral current consumption in Run mode</a>.</p> <p>Replaced FMC_CLK by FMC_SDCLK in <a href="#">Section : SDRAM waveforms and timings</a>.</p> <p>Changed description of the last five <math>f_s</math> values and updated <math>t_{LATRINJ}</math> in <a href="#">Table 87: ADC characteristics</a>.</p> <p>For TFBGA100, TFBGA240+25 and UFBGA169, updated thermal resistance power-junction in <a href="#">Table 228: Thermal characteristics</a> as well as power dissipation in <a href="#">Table 24: General operating conditions</a>.</p> |
| 23-Oct-2017 | 3        | <p><a href="#">Features</a>:</p> <ul style="list-style-type: none"> <li>– Removed secure firmware upgrade support.</li> <li>– Total current consumption changed to 4 <math>\mu</math>A minimum.</li> </ul> <p>Updated <a href="#">Figure 8: UFBGA169 ballout</a>.</p> <p>Updated dpad and dsm in <a href="#">Table 227: TFBGA240+25 recommended PCB design rules (0.8 mm pitch)</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Table 229. Document revision history

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18-May-2018 | 4        | <p>Updated LSI clock frequency and ADC on cover page. Removed note related to UFBGA169 package.</p> <p>Updated USB OTG interfaces to add crystal-less capability.</p> <p>Updated ADC features on cover page and in <a href="#">Table 2: STM32H742xI/G and STM32H743xI/G features and peripheral counts</a>.</p> <p>Added Arm trademark notice in <a href="#">Section 1: Introduction</a>.</p> <p>Updated <a href="#">Figure 1: STM32H743xI/G block diagram</a>.</p> <p>Updated GPIO default mode in <a href="#">Section 3.8: General-purpose input/outputs (GPIOs)</a>.</p> <p>Added ADC sampling rate values in <a href="#">Section 3.17: Analog-to-digital converters (ADCs)</a>.</p> <p>Updated <a href="#">Section 3.18: Temperature sensor</a>.</p> <p>Updated LCD-TFT FIFO Size in <a href="#">Section 3.25: LCD-TFT controller</a>.</p> <p><a href="#">Section 3.33: Serial peripheral interface (SPI)/inter- integrated sound interfaces (I2S)</a>: changed maximum SPI frequency to 150 Mbits/s.</p> <p>Modified number of bidirectional endpoints in <a href="#">Section 3.40: Universal serial bus on-the-go high-speed (OTG_HS)</a>.</p> <p><a href="#">Table 9: Pin/ball definition</a>: updated PC14 and PC15 function after reset; changed CAN1_TX/RX to FDCAN1_TX/RX and CAN1_TXFD/RXFD to FDCAN1_TXFD_MODE/RXFD_MODE; changed CAN2_TX/RX to FDCAN2_TX/RX and CAN2_TXFD/RXFD to FDCAN2_TXFD_MODE/RXFD_MODE and replaced VCAP1/2/3 and VDDLDO1/2/3 by VCAP and VDDLDO, respectively.</p> <p>Updated PA0, PA13, PA14, PC14 and PC15 pin/ball signals in pinout/ballout schematics.</p> <p>Replaced <math>f_{\text{ACLK}}</math> by <math>f_{\text{rcc\_c\_ck}}</math> in <a href="#">Section : Typical and maximum current consumption</a>.</p> <p>Replaced system clock by CPU clock and <math>f_{\text{ACLK}}</math> by <math>f_{\text{rcc\_c\_ck}}</math> in <a href="#">Section : On-chip peripheral current consumption</a>.</p> <p>Updated Note 2. in <a href="#">Table 27: Reset and power control block characteristics</a>, <a href="#">Table 28: Embedded reference voltage</a>, <a href="#">Table 30: Typical and maximum current consumption in Run mode, code with data processing running from ITCM, regulator ON</a>, <a href="#">Table 31: Typical and maximum current consumption in Run mode, code with data processing running from Flash memory, cache ON, regulator ON</a>, <a href="#">Table 36: Typical and maximum current consumption in Stop mode, regulator ON</a>, <a href="#">Table 37: Typical and maximum current consumption in Standby mode</a> and <a href="#">Table 38: Typical and maximum current consumption in VBAT mode</a>.</p> <p>Added note to <math>f_{\text{LSI}}</math> in <a href="#">Table 49: LSI oscillator characteristics</a>.</p> <p>Updated <a href="#">Figure 22: VIL/VIH for all I/Os except BOOT0</a>.</p> <p>Added note in <a href="#">Table 84: QUADSPI characteristics in SDR mode</a>, <a href="#">Table 85: QUADSPI characteristics in DDR mode</a> and <a href="#">Table 86: Dynamics characteristics: Delay Block characteristics</a>.</p> <p><a href="#">Section 6.3.20: 16-bit ADC characteristics</a>: updated THD conditions in <a href="#">Table 88: ADC accuracy</a>; removed formula to compute <math>R_{\text{AIN}}</math>.</p> <p>Changed decoupling capacitor value to 100 nF in <a href="#">Section : General PCB design guidelines</a>.</p> <p>Added note in <a href="#">Table 89: DAC characteristics</a>, <a href="#">Table 97: Voltage booster for analog switch characteristics</a>, <a href="#">Table 100: DFSDM measured timing 1.62-3.6 V</a>, <a href="#">Table 117: Dynamics JTAG characteristics</a> and <a href="#">Table 118: Dynamics SWD characteristics</a>.</p> <p>Updated <a href="#">Figure 128: LQFP144 marking example (package top view)</a>, <a href="#">Figure 133: LQFP176 marking example (package top view)</a> and <a href="#">Figure 136: LQFP208 marking example (package top view)</a>.</p> <p>Updated TFBGA240+25 package information to final mechanical data.</p> |

Table 229. Document revision history

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13-Jul-2018 | 5        | <p>Added description of power-up and power-down phases in <a href="#">Section 3.5.1: Power supply scheme</a>.</p> <p>Removed ETH_TX_ER from <a href="#">Table 9: Pin/ball definition</a> and <a href="#">Table 10: Port A alternate functions</a> to <a href="#">Table 20: Port K alternate functions</a>.</p> <p>Added note related to decoupling capacitor tolerance below <a href="#">Figure 15: Power supply scheme</a>. Added note 2. related to CEXT in <a href="#">Table 25: VCAP operating conditions</a>.</p> <p>Updated <a href="#">Table 46: HSI48 oscillator characteristics</a>, <a href="#">Table 47: HSI oscillator characteristics</a> and <a href="#">Table 48: CSI oscillator characteristics</a>. Renamed <a href="#">Table 50</a> into "PLL characteristics (wide VCO frequency range)" and updated note 2.. Added <a href="#">Table 51: PLL characteristics (medium VCO frequency range)</a>.</p> <p>Updated <math>T_{\text{coeff}}</math> in <a href="#">Table 91: VREFBUF characteristics</a> and <math>t_{\text{S\_vbat}}</math> in <a href="#">Table 94: V<sub>BAT</sub> monitoring characteristics</a>. Updated <a href="#">Table 99: OPAMP characteristics</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 05-Apr-2019 | 6        | <p>Added STM32H743xG part numbers corresponding to 1 Mbyte of Flash memory as well as STM32H742xI/G part numbers.</p> <p>Changed maximum Arm Core-M7 frequency to 480 MHz.</p> <p><b>Features:</b></p> <ul style="list-style-type: none"> <li>– Changed operational amplifier bandwidth to 7.3 MHz</li> <li>– Updated high-resolution timer to 2.1 ns</li> <li>– Updated low-power consumption feature.</li> <li>– Updated <a href="#">Figure 2: STM32H743xI/G block diagram</a>.</li> </ul> <p>Updated voltage scaling in <a href="#">Section 3.5.1: Power supply scheme</a>. Added VOS0 in <a href="#">Section 3.5.3: Voltage regulator</a>. Updated HSE clock in <a href="#">Section 3.7.1: Clock management</a>.</p> <p>Changed FMC NOR/NAND maximum clock frequency to 100 MHz in <a href="#">Features</a> and <a href="#">Synchronous waveforms and timings</a>.</p> <p>Added note related to VDDLDO in <a href="#">Table 9: Pin/ball definition</a>.</p> <p>Updated <a href="#">Section 6: Electrical characteristics (rev Y)</a>:</p> <ul style="list-style-type: none"> <li>– Added note 2. related to CEXT in <a href="#">Table 25: VCAP operating conditions</a>.</li> <li>– Updated <math>f_{\text{HSI48}}</math> in <a href="#">Table 46: HSI48 oscillator characteristics</a>.</li> <li>– Updated <math>t_{\text{stab}}</math> in <a href="#">Table 47: HSI oscillator characteristics</a>.</li> <li>– Updated <a href="#">Table 60: I/O static characteristics</a> and <a href="#">Figure 22: VIL/VIH for all I/Os except BOOT0</a>.</li> <li>– Added <a href="#">Table 62: Output voltage characteristics for PC13, PC14, PC15 and PI8</a>.</li> <li>– Added note related to PC13, PC14, PC15 and PI8 limited frequency in <a href="#">Table 63: Output timing characteristics (HSLV OFF)</a>.</li> <li>– Updated note 2 below <a href="#">Figure 23: Recommended NRST pin protection</a>.</li> <li>– <a href="#">Table 87: ADC characteristics</a>: updated <math>f_{\text{S}}</math> and added note related to <math>f_{\text{S}}</math> formula; updated <math>t_{\text{CAL}}</math>.</li> <li>– Renamed <a href="#">Section 6.3.24</a> into <i>Temperature and VBAT monitoring</i> and content updated.</li> <li>– Updated <math>f_{\text{DFSDMCLK}}</math> in <a href="#">Table 100: DFSDM measured timing 1.62-3.6 V</a>.</li> </ul> <p>Added <a href="#">Section 7: Electrical characteristics (rev V)</a>.</p> <p>Updated paragraph introducing all package marking schematics to add the new sentence "The printed markings may differ depending on the supply chain". Updated <a href="#">Table 228: Thermal characteristics</a>. Added note related to ECOPACK®2 compliance in <a href="#">Section 9: Ordering information</a>.</p> |

Table 229. Document revision history

| Date        | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25-Apr-2019 | 7        | Updated <a href="#">Figure 1: STM32H742xI/G block diagram</a><br>Updated <a href="#">Figure 2: STM32H743xI/G block diagram</a><br>Updated <a href="#">Table 9: Pin/ball definition</a> .<br>Updated <a href="#">Table 10</a> to <a href="#">Table 20</a> (alternate functions).<br>Updated <a href="#">Table 39: Peripheral current consumption in Run mode</a> .<br>Updated <a href="#">Table 137: Peripheral current consumption in Run mode</a> .<br>Updated <a href="#">Table 184: ADC characteristics</a> .<br>Updated <a href="#">Table 185: Minimum sampling time vs RAIN</a> .<br>Updated <a href="#">Table 186: ADC accuracy</a> . |

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